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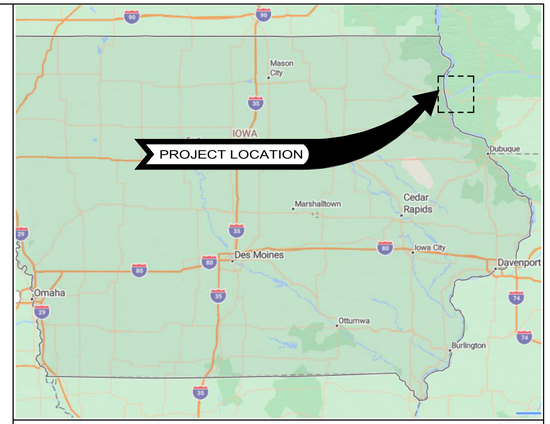
PROPOSED WEST ELEVATION
SHOWN FOR REFERENCE ONLY
SEE SPECIFIC DISCIPLINE SHEETS FOR SCOPE OF WORK

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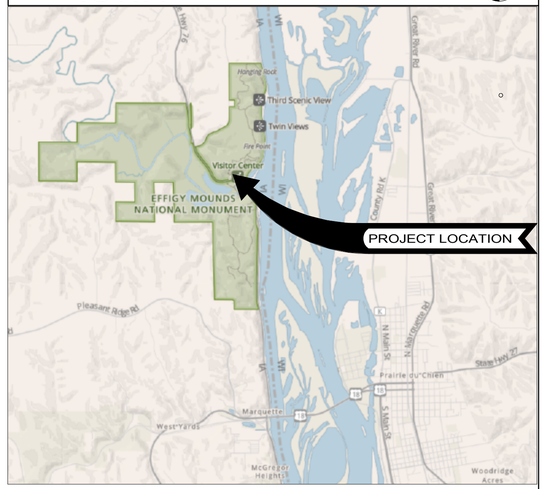

AERIAL IMAGE OF SITE

## CIVIL — CERTIFICATION

I hereby certify that this engineering document was prepared by me or under my direct personal supervision and that I am a duly Licensed Professional Engineer under the laws of the State of Iowa.

Name: EMILY MAGGIO
My license renewal date is: 12/31/2027
Pages or sheets covered by this seal: C SERIES

## LANDSCAPE — CERTIFICATION

I hereby certify that this engineering document was prepared by me or under my direct personal supervision and that I am a duly Licensed Professional Landscape Architect under the laws of the State of Iowa.

Name: LINDSAY MARTIN
My license renewal date is: 06/30/2026
Pages or sheets covered by this seal: L SERIES

## ARCHITECT — CERTIFICATION

I hereby certify that this engineering document was prepared by me or under my direct personal supervision and that I am a duly Licensed Professional Architect under the laws of the State of Iowa.

Name: Elizabeth Hallas
My license renewal date is: 06/30/2028
Pages or sheets covered by this seal: G-SERIES, A-SERIES

## STRUCTURAL — CERTIFICATION

I hereby certify that this engineering document was prepared by me or under my direct personal supervision and that I am a duly Licensed Professional Engineer under the laws of the State of Iowa.

Name: Sean Clark
My license renewal date is: 12/31/2026
Pages or sheets covered by this seal: S-SERIES

## MECH/PLUMB — CERTIFICATION

I hereby certify that this engineering document was prepared by me or under my direct personal supervision and that I am a duly Licensed Professional Engineer under the laws of the State of Iowa.

Name: Denise Dihle
My license renewal date is: 12/31/2027
Pages or sheets covered by this seal: M SERIES, P SERIES

## FIRE PROTEC — CERTIFICATION

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Name: KENTON AIKENS
My license renewal date is: 12/31/2026
Pages or sheets covered by this seal: FA SERIES, FP SERIES

## ELECTRICAL — CERTIFICATION

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Name:
My license renewal date is: 12/31/2026
Pages or sheets covered by this seal: E-SERIES, T-SERIES

# EFFIGY MOUNDS NATIONAL MONUMENT
# REHABILITATE VISITOR CENTER

N170 EFMO318539

OTAK, INC. - 140P2022D0020

| | | | |
|---|---|---|---|
| PRIME / CIVIL / LANDSCAPE: OTAK REDMOND, WA 98052 | ARCHITECT: ANDERSON HALLAS, PC GOLDEN, CO 80401 | MECHANICAL / PLUMBING: 360 ENGINEERING GOLDEN, CO 80403 | FIRE PROTECTION: INTERFACE ENGINEERING, INC PORTLAND, OR 97204 |
| STRUCTURAL: OTAK PORTLAND, OR 97204 | ELECTRICAL: AE DESIGN DENVER, CO 80202 | | |

| Mark | Sheet | REVISION | Date | Initial |
|---|---|---|---|---|
| | | | | |

QUALITY DESIGN CERTIFICATION

- ☐ Prepared in Accordance with design Development (Title I) _____ Drawing No.
- OR
- ☐ Variance from Design Development (Title I) Approved by Superintendent on _____ Date
- OR
- ☐ Construction Drawing Not Preceded by Design Development (Title I)

Project Manager _____ Date

CONSTRUCTION DRAWINGS

UNITED STATES
DEPARTMENT OF THE INTERIOR

NATIONAL PARK SERVICE
DENVER SERVICE CENTER

TITLE OF DRAWING
**REHABILITATE VISITOR CENTER**
LOCATION WITHIN PARK
EFFIGY MOUNDS VISITOR CENTER
NAME OF PARK
EFFIGY MOUNDS NATIONAL MONUMENT

| REGION | COUNTY | STATE |
|---|---|---|
| MIDWEST | ALLAMAKEE | IOWA |

DRAWING NO. 394 194248
PMIS/PKG NO. 318539
SHEET 1 OF 142

7/7/2026 1:22:53 PM Autodesk Docs://2025-110 NPS EFMO Visitor Center DD-CD/2024250 NPS EFMO Visitor Center - v24_BIM360.rvt

# GENERAL ABBREVIATIONS

| | | | |
|---|---|---|---|
| # | POUND(S) or NUMBER | IBC | INTERNATIONAL BUILDING CODE |
| & | AND | ID | INSIDE DIAMETER |
| (S) | SEALANT | INCL | INCLUDING |
| (T) | TEMPERED | INFO | INFORMATION |
| < | ANGLE | INS | INSULATION or INSULATED |
| <E> | EXISTING | INT | INTERIOR |
| <N> | NEW | JT | JOINT |
| @ | AT | LAV | LAVATORY |
| A/V | AUDIO VISUAL | LIN | LINOLEUM |
| ABA | ARCHITECTURAL BARRIERS ACT | LS | LANDSCAPE |
| ABV | ABOVE | MATL | MATERIAL |
| ACM | ASBESTOS CONTAINING MATERIAL | MAX | MAXIMUM |
| ACT | ACOUSTICAL CEILING TILE | MECH | MECHANICAL |
| ADD | ADDENDUM | MEP | MECHANICAL, ELECTRICAL and PLUMBING |
| ADJ | ADJACENT or ADJUSTABLE | MFG | MANUFACTURING |
| AFF | ABOVE FINISHED FLOOR | MFR | MANUFACTURER |
| AHU | AIR HANDLING UNIT | MIN | MINIMUM |
| AL | ALUMINUM | MISC | MISCELLANEOUS |
| ALT | ALTERNATE | MO | MASONRY OPENING |
| APPROX | APPROXIMATE(LY) | MTD | MOUNTED |
| ARCH | ARCHITECTURAL | MTL | METAL |
| ASPH | ASPHALT | N | NORTH |
| BC | BASE CABINET | NA | NOT APPLICABLE |
| BLDG | BUILDING | NIC | NOT IN CONTRACT |
| BLKG | BLOCKING | NO | NUMBER |
| BO | BOTTOM OF | NOM | NOMINAL |
| BR | BACKER ROD | NTS | NOT TO SCALE |
| BTWN | BETWEEN | OC | ON CENTER |
| CAB | CABINET | OD | OUTSIDE DIAMETER |
| CJ | CONTROL JOINT | OH | OPPOSITE HAND |
| CL | CENTERLINE | OPNG | OPENING |
| CLG | CEILING | OPP | OPPOSITE |
| CLR | CLEAR | OSB | ORIENTED STRAND BOARD |
| CMU | CONCRETE MASONRY UNIT | OTE | OPEN TO EXISTING |
| CO | CONTRACTING OFFICER | OTS | OPEN TO STRUCTURE |
| COL | COLUMN | OVH | OVERHANG |
| CONC | CONCRETE | PLAM | PLASTIC LAMINATE |
| CONST | CONSTRUCTION | PLAS | PLASTER |
| CONT | CONTINUOUS | PLUMB | PLUMBING |
| CORR | CORRIDOR | PT | PAINT |
| CPT | CARPET | PTD | PAINTED |
| CT | CERAMIC TILE | PWD | PLYWOOD |
| CTBB | CEMENTITIOUS TILE BACKER BOARD | QT | QUARRY TILE |
| CTR | CENTER | RAD | RADIUS |
| DBL | DOUBLE | RB | RUBBER BASE |
| DEMO | DEMOLITION | RCP | REFLECTED CEILING PLAN |
| DET | DETAIL | RD | ROOF DRAIN |
| DF | DRINKING FOUNTAIN | RE: | REFER TO or REFERENCE |
| DIAM | DIAMETER | REINF | REINFORCED |
| DIM | DIMENSION | RELOC | RELOCATED |
| DN | DOWN | REQ | REQUIRED |
| DS | DOWNSPOUT | REV | REVISE, REVISED or REVISION |
| E | EAST | RM | ROOM |
| EA | EACH | RO | ROUGH OPENING |
| EG | FOR EXAMPLE | RR | RESTROOM(S) |
| ELEC | ELECTRICAL | RS | ROUGH SAWN |
| ELEV | ELEVATION | S | SOUTH |
| EMER | EMERGENCY | SCHED | SCHEDULE |
| ENGR | ENGINEER | SF | SQUARE FEET |
| EP | EPOXY | SHTG | SHEATHING |
| EQ | EQUAL | SIM | SIMILAR |
| EQP | EQUIPMENT | SOG | SLAB ON GRADE |
| ETR | EXISTING TO REMAIN | SPEC | SPECIFICATION(S) |
| EXT | EXTERIOR | SQ | SQUARE |
| FACP | FIRE ALARM CONTROL PANEL | SS | STAINLESS STEEL |
| FD | FLOOR DRAIN | STD | STANDARD |
| FDN | FOUNDATION | STL | STEEL |
| FE | FIRE EXTINGUISHER | STN | STAIN |
| FEC | FIRE EXTINGUISHER CABINET | STRUC | STRUCTURE or STRUCTURAL |
| FIN | FINISH(ED) | T&G | TONGUE AND GROOVE |
| FIXT | FIXTURE | TD | THRESHOLD |
| FLR | FLOOR(ING) | TO | TOP OF |
| FO | FACE OF | TYP | TYPICAL |
| FP | FIRE PROTECTION | UC | UPPER CABINET |
| FRP | FIBERGLASS REINFORCED PANEL(ING) | UON | UNLESS OTHERWISE NOTED |
| FT | FEET | VCT | VINYL COMPOSITION TILE |
| FTG | FOOTING | VERT | VERTICAL |
| FURR | FURRING | VIF | VERIFY IN FIELD |
| GA | GAUGE OR GYPSUM ASSOCIATION | VIN | VINYL |
| GALV | GALVANIZED | VR | VAPOR RETARDER |
| GC | GENERAL CONTRACTOR | VTR | VENT THROUGH ROOF |
| GD | GRADE | W | WEST |
| GL | GLASS or GLAZING | W/ | WITH |
| GND | GROUND | W/O | WITHOUT |
| GRT | GROUT | WC | WATER CLOSET |
| GWB | GYPSUM WALLBOARD | WD | WOOD |
| HAZMAT | HAZARDOUS MATERIAL | WIN | WINDOW |
| HDR | HEADER | WRB | WEATHER RESISTANT BARRIER |
| HDW | HARDWARE | | |
| HM | HOLLOW METAL | | |
| HORIZ | HORIZONTAL | | |
| HSS | HOLLOW STEEL SECTION | | |
| HT | HEIGHT | | |
| HVAC | HEATING, VENTILATION and AIR CONDITIONING | | |

# GENERAL NOTES

1. DO NOT SCALE DRAWINGS. WRITTEN DIMENSIONS GOVERN.
2. THE CONTRACTOR SHALL BE RESPONSIBLE FOR FAMILIARIZING THEMSELF WITH THE CONTRACT DOCUMENTS, VERIFYING FIELD CONDITIONS AND DIMENSIONS, AND CONFIRMING THAT THE WORK MAY BE ACCOMPLISHED AS SHOWN PRIOR TO PROCEEDING WITH CONSTRUCTION. CONTRACTOR SHALL NOTIFY C.O. IN WRITING OF ANY DISCREPANCY WITHIN THE CONTRACT DOCUMENTS AND REQUEST CLARIFICATION PRIOR TO PROCEEDING WITH DESIGN CONSTRUCTION.
3. SHOULD THERE BE ANY QUESTIONS CONCERNING THE CONTRACT DOCUMENTS, EXISTING CONDITIONS, AND/OR DESIGN INTENT, THE CONTRACTOR SHALL BE RESPONSIBLE FOR OBTAINING A CLARIFICATION FROM THE C.O. PRIOR TO PROCEEDING WITH THE WORK, OR RELATED WORK IN QUESTION.
4. THE CONTRACT DOCUMENTS IN THEIR ENTIRETY ARE THE RESPONSIBILITY OF ALL TRADES. WHERE REQUIREMENTS ARE SHOWN IN ONE SECTION OF THE SPECIFICATIONS OR DRAWINGS BUT NOT ANOTHER, THE CONTRACTOR IS NOT RELIEVED FROM PROVIDING COMPLETELY FINISHED AND PROPERLY FUNCTIONING SYSTEMS.
5. THE CONTRACTOR SHALL BE RESPONSIBLE FOR FAMILIARIZING THEMSELVES WITH THE PROJECT SCOPE OF WORK, SCHEDULE, AND DEADLINES. THE CONTRACTOR SHALL BE RESPONSIBLE FOR ADVISING THE CONTRACTING OFFICER OF ALL ITEMS REQUIRING A LONG LEAD TIME UPON NOTICE TO PROCEED THAT WILL AFFECT THE SCHEDULE. ORDER CONFIRMATIONS AND DELIVERY DATES FOR THE ITEMS IN QUESTION SHALL BE INCLUDED IN THE CONSTRUCTION SCHEDULE AND SUBMITTED TO THE CONTRACTING OFFICER.
6. ALL WORK TO BE PERFORMED TO APPLICABLE BUILDING CODES (SEE CODE SUMMARY ON SHEET G110 FOR LIST OF APPLICABLE CODES).
7. THE CONTRACTOR SHALL BE FULLY RESPONSIBLE FOR THE LAYOUT AND COORDINATION OF DIMENSIONS IN THE FIELD THAT ESTABLISH WORK POINTS FOR THEIR WORK AND VARIOUS TRADES.
8. "ALIGN" AS USED IN THESE DOCUMENTS SHALL MEAN TO ACCURATELY LOCATE FINISHED FACES IN THE SAME PLANE.
9. DIMENSIONS ARE TO FACE OF <N> STUD WHERE INTERFACING WITH <N> FRAMING AND TO FACE OF FINISH WHERE INTERFACING WITH <E> WALLS, UNLESS NOTED OTHERWISE. ALL DIMENSIONS INDICATING REQUIRED CLEARANCES ARE TO FACE OF FINISH UNLESS OTHERWISE NOTED.
10. TYPICAL (TYP) OR SIMILAR (SIM) NOTATION USED IN THESE DOCUMENTS SHALL MEAN THAT THE CONDITIONS ARE THE SAME OR REPRESENTATIVE FOR ALL SIMILAR CONDITIONS, UNLESS NOTED OTHERWISE. ACTUAL CONDITIONS MAY VARY, BUT DO NOT REVOKE THE INTENT OF THE CONFIGURATION INDICATED. DETAILS ARE USUALLY KEYED AND NOTED "TYPICAL" OR "SIMILAR" ONLY ONCE WITH THE FIRST OCCURRENCE AND ARE REPRESENTATIVE FOR ALL SIMILAR CONDITIONS THROUGHOUT, UNLESS NOTED OTHERWISE.
11. INSTALL ALL MANUFACTURED ITEMS, MATERIALS, AND EQUIPMENT IN STRICT ACCORDANCE WITH THE MANUFACTURER'S RECOMMENDED SPECIFICATIONS, EXCEPT WHERE THE CONTRACT DOCUMENTS ARE MORE STRINGENT.
12. ANY MISCELLANEOUS ITEMS OR MATERIALS NOT SPECIFICALLY NOTED BUT REQUIRED FOR PROPER INSTALLATION SHALL BE FURNISHED AND INSTALLED BY THE GENERAL CONTRACTOR.
13. THE CONTRACTOR SHALL FURNISH TO THE CO ALL WARRANTIES AND GUARANTEES REQUIRED AT THE CONCLUSION OF THE PROJECT.
14. CONTRACTOR SHALL BE RESPONSIBLE FOR ALL HOOK UPS / UTILITY CONNECTIONS, ETC. TO TEMPORARY TRAILERS.
15. IF DISCREPANCIES OCCUR BETWEEN THE DRAWINGS AND THE SPECIFICATIONS, NOTIFY THE CONTRACTING OFFICER FOR RESOLUTION.
16. THE CONSTRUCTION SITE IS TO BE KEPT CLEAN AND FREE OF DEBRIS. THE CONTRACTOR IS RESPONSIBLE FOR ALL PHASES OF SECURING, HANDLING, TRANSPORTING AND DISPOSING OF DEBRIS.
17. THE CONTRACTORS SHALL NOT SHUT OFF OR INTERRUPT ANY SYSTEMS OR SERVICES WITHOUT FIRST COORDINATING WITH THE CONTRACTING OFFICER.
18. THE CONTRACTOR SHALL COORDINATE WITH THE CONTRACTING OFFICER TO IDENTIFY AND PROTECT EXISTING STRUCTURE,SYSTEMS, UTILITIES, SURFACES AND FINISHES WHICH ARE TO REMAIN. ALL SYSTEMS DAMAGED DURING CONSTRUCTION ARE TO BE REPAIRED BY THE CONTRACTOR AT THEIR EXPENSE.
19. DELIVERY AND STORAGE OF CONSTRUCTION MATERIALS AND EQUIPMENT (STAGING AREAS) IS TO BE COORDINATED WITH THE CONTRACTING OFFICER.
20. PROVIDE ELECTROLYTIC PROTECTION / ISOLATION BETWEEN ALL DISSIMILAR METALS, WHERE THEY OCCUR TO PREVENT ELECTROLYTIC REACTION AND/OR CORROSION.

# OPTIONS

**BASE STRUCTURAL SAFETY AND FOUNDATION IMPROVEMENTS**
REMOVE EXISTING PLANTERS, SALVAGE STONE VENEER, EXCAVATE FOUNDATIONS AND NW CRAWLSPACE, PROVIDE NEW FOUNDATION WALL, COLUMN EXTENSIONS, HELICAL ANCHORS, RETAINING WALL AT NW CORNER, FOUNDATION WATERPROOFING, DRAINAGE MAT, AND PERIMETER DRAINS. REPAIR DAMAGED RAFTERS AT FASCIA AND CONNECTIONS TO ROOF BEAMS. REPAIR DAMAGED CONCRETE FLOOR PLANKS. WATER SERVICE WORK.

**BASE COLLECTIONS MOVE**
MOVE COLLECTIONS TO TEMPORARY SPACE AND THEN MOVE BACK.

**OPTION A: HVAC & FIRE SUPPRESSION**
REMOVE EXISTING HVAC AND FIRE SUPPRESSION, INSTALL NEW HVAC AND FIRE SUPPRESSION AND SECURITY/FIRE ALARM.
IF THIS CLIN IS PERFORMED WITHOUT THE INTERIOR WORK, IT ASSUMES ALL DUCTING IS IN ATTIC AND CEILING REMOVAL FOR BASEMENT LEVEL FIRE ALARM CONDUIT IS NOT REPAIRED. INSTALL NEW WOOD CATWALK AND REPLACE EXISTING ATTIC HATCH WITH NEW IN NEW LOCATION. REPLACE EXISTING ATTIC INSULATION WITH NEW.

**OPTION B EXTERIOR ENVELOPE REHABILITATION**
REMOVE EXTERIOR SIDING AND ROOFING, REFRAME NEW WINDOW OPENINGS, REPAIR ROOF SHEATHING, REPLACE EXISTING ROOFING WITH NEW AND INSTALL ROOF ACCESSORIES (SNOW GUARDS AND ROOF VENTS), REHABILITATE EXISTING BRICK. INSTALL NEW EXTERIOR WEATHER BARRIER, INSULATION, ALUMINUM CLADDING, METAL PANELS, EXTERIOR WINDOWS, EXTERIOR DOORS, EXTERIOR SOFFIT, EXTERIOR COLUMNS (INCLUDING CONCRETE PLINTHS), AND REHAB EXISTING RAILING. INCLUDES FINALIZING SITE WORK, SUCH AS INSTALLING THE FLAG POLE AND OTHER SITE FURNISHINGS.

**OPTION C INTERIOR ACCESSIBILITY COMPLIANCE, PLUMBING, & ELECTRICAL**
PERFORM HAZARDOUS MATERIAL ABATEMENT. REMOVE INTERIOR FINISHES AND DEMO SELECT INTERIOR WALLS FOR INTERIOR RECONFIGURATION. INSTALL NEW WALLS AT RECONFIGURED RESTROOMS AND OFFICE SPACES, INSTALL NEW TRANSACTION DESK, LIGHT FIXTURES, ELECTRICAL/DATA CONDUIT, CABLING, LOBBY MONITORS, PLUMBING AND PLUMBING FIXTURES. INSTALL FINISHES (EPOXY, SOLID SURFACE PANELING, CARPET, RUBBER FLOORING, RUBBER TREADS/RISERS), TOILET ACCESSORIES, INTERIOR DOORS, INTERIOR WINDOWS, ACT WHERE INDICATED. PAINT ALL WALLS, SELECT CEILINGS, RETAINED INTERIOR DOORS, AND STAIN EXISTING LOBBY WOOD CEILING. PROVIDE TABLES AND CHAIRS IN NEW BREAKROOM AND MEETING ROOM.

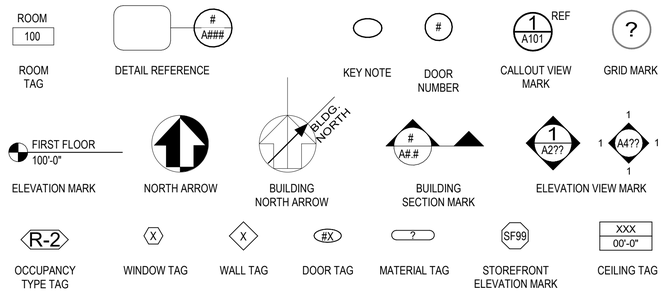
**OPTION D LOW VOLTAGE FIRE ALARM UPDATE**
NEW CABLING FOR LOW VOLTAGE FIRE ALARM IMPROVEMENTS".

**OPTION E LIGHTNING PROTECTION**
INSTALL NEW LIGHTNING PROTECTION SYSTEM INCLUDING DISCREET CABLING.

**OPTION F MEETING ROOM/AUDITORIUM AV & EXHIBIT AREA RENOVATION**
INSTALL NEW OPERABLE PARTITION, OVERHEAD STEEL BEAM TO SUPPORT PARTITION, WINDOW SHADES, AV, ASSISTED LISTENING SYSTEM, AND AUDITORIUM FURNISHINGS.

**OPTION G EXPAND DECK AT NE CORNER**
SALVAGE EXISTING GUARDRAIL AND SELECTIVE DEMO OF EXISTING DECK. INSTALL NEW STEEL COLUMNS, BEAMS, METAL DECK, CIP CONCRETE, ADDITIONAL TRAFFIC COATING, GUARDRAIL, AND DUAL VIEWER.

# SYMBOLS

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| 9 | GC 000 | CIVIL ABBREVIATIONS AND LEGEND |
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| 30 | A 142 | MAIN LEVEL RCP |
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| 34 | A 202 | BUILDING ELEVATIONS |
| 35 | A 301 | BUILDING SECTIONS |
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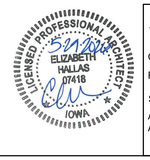


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# CODE ANALYSIS

REFER TO INDIVIDUAL DISCIPLINE SHEETS FOR ADDITIONAL CODE INFORMATION

PROJECT NUMBER: 2025110
PROJECT NAME: EFFIGY MOUNDS NATIONAL MONUMENT REHABILITATE VISITOR CENTER
ADDRESS: 151 IA-76, HARPERS FERRY, IOWA 52146
AHJ: NATIONAL PARK SERVICE
STRUCTURAL FIRE AHJ: NATIONAL PARK SERVICE, MIDWEST REGION
CLIMATE ZONE: 5A PER 2024 IECC

**APPLICABLE CODES AND STANDARDS**
- 2024 INTERNATIONAL BUILDING CODE (IBC)
- 2024 INTERNATIONAL EXISTING BUILDING CODE (IEBC)
- 2024 INTERNATIONAL ENERGY CONSERVATION CODE (IECC)
- 2024 INTERNATIONAL MECHANICAL CODE (IMC)
- 2024 INTERNATIONAL PLUMBING CODE (IMC)
- 2024 INTERNATIONAL FUEL GAS CODE (IFGC)
- 2024 INTERNATIONAL FIRE CODE (IFC)
- 2023 NATIONAL ELECTRICAL CODE (NEC)
- NFPA (NATIONAL FIRE PROTECTION ASSOCIATION) 101 LIFE SAFETY
- NFPA 13 STANDARD FOR THE INSTALLATION OF SPRINKLER SYSTEMS
- ASHRAE 90.1-2016 ENERGY STANDARD FOR SITES AND BUILDINGS EXCEPT LOW-RISE RESIDENTIAL BUILDINGS
- ARCHITECTURAL BARRIERS ACT ACCESSIBILITY STANDARDS (ABAAS)
- 2017 ICC ANSI A117.1-2017 ACCESSIBLE AND USABLE BUILDINGS AND FACILITIES
- DIRECTOR'S ORDER 58: STRUCTURAL FIRE MANAGEMENT

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**IEBC CLASSIFICATION**
ALTERATION LEVEL 3 PER IEBC §604 - WORK AREA COMPLIANCE PATH
THE WORK EXCEEDS 50% OF THE ENTIRE BUILDING AREA

ALL <N> CONSTRUCTION TO COMPLY WITH IBC AND IECC

---

**USE & OCCUPANCY CLASSIFICATION - IBC CHAPTER 3**
EXISTING OCCUPANCY:
- EXISTING BUILDING IS AND WILL CONTINUE TO BE A VISITOR CENTER

OCCUPANCY CLASSIFICATION:
- A-3* ASSEMBLY (AUDITORIUM, LOBBY, EXHIBIT SPACE) PER IBC §303.4
- B BUSINESS* (STAFF OFFICES) PER IBC §304.1
- S-1 STORAGE (COLLECTION STORAGE) PER IBC §311.3

*THE BUILDING IS CATEGORIZED AS AN A-3 OCCUPANCY, WHICH IS THE MOST RESTRICTIVE AND WILL HAVE NONSEPARATED OCCUPANCIES.

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**GENERAL BUILDING HEIGHTS & AREAS - IBC CHAPTER 5**
BUILDING HEIGHT §504

ALLOWABLE BUILDING HEIGHT (TABLE 504.3), AND NUMBER OF STORIES (TABLE 504.4)

| VB CONSTRUCTION TYPE (SPRINKLERED) | | ALLOWABLE | ACTUAL |
|---|---|---|---|
| A-3 OCCUPANCY | BUILDING HEIGHT | 60' | 34'-6" |
| | NUMBER OF STORIES | 2 STORIES * | 2 STORIES |
| B OCCUPANCY | BUILDING HEIGHT | 60' | |
| | NUMBER OF STORIES | 3 STORIES | 2 STORIES |
| S-1 OCCUPANCY | BUILDING HEIGHT | 60' | |
| | NUMBER OF STORIES | 2 STORIES | 2 STORIES |

ALLOWABLE BUILDING AREA (TABLE 506.2)

| OCCUPANCY GROUP, CONSTRUCTION TYPE VB, SPRINKLERED (SM) | ALLOWABLE BUILDING AREA (TOTAL AREA) | ACTUAL BUILDING AREA (PER FLOOR) EXISTING | PROPOSED |
|---|---|---|---|
| A-3 OCCUPANCY | 57,000 SF* | 5,419 SF | 5,419 SF (MAIN FLOOR) |
| (MORE STRINGENT THAN B & S-1 OCCUPANCIES) | | 3,728 SF | 4,399F (BASEMENT) |
| | | **9,147 SF** | **9,818 SF (TOTAL AREA)** |

*ALLOWABLE AREA = 24,000 (S1) + (6,000 (NS) * 0.75) = 28,500 SF PER STORY * 2 STORIES = 57,000 SF
PROPOSED WORK DOES NOT MODIFY BUILDING HEIGHT, OR NUMBER OF STORIES.

NON-SEPARATED OCCUPANCIES PER IBC §508.3.2
PROJECT SHALL UTILIZE THE MOST RESTRICTIVE OCCUPANCY (A3) MAXIMUM ON ALLOWABLE BUILDING AREA, HEIGHT AND NUMBER OF STORIES. PROPOSED WORK DOES NOT MODIFY BUILDING HEIGHT, OR NUMBER OF STORIES.

SEPARATION OF OCCUPANCIES PER 508.3.3
NO SEPARATION REQUIRED BETWEEN UNSEPARATED OCCUPANCIES.

INCIDENTAL USES PER IBC §509
THE MECHANICAL ROOM DOES NOT CONTAIN ANY SINGLE PIECE OF EQUIPMENT THAT IS OVER 400,000 BTU PER HOUR INPUT. THERE ARE NO INCIDENTAL USES IN THIS PROJECT THAT REQUIRE A FIRE RATED SEPARATION.

---

**TYPES OF CONSTRUCTION - CHAPTER 6**
CONSTRUCTION CLASSIFICATION: VB PER IBC §602.5
PROPOSED WORK DOES NOT MODIFY BUILDING HEIGHT, OR NUMBER OF STORIES.

FIRE-RESISTANCE RATING REQUIREMENTS FOR BUILDING ELEMENTS (HOURS) PER TABLE 601

| | TYPE VB | |
|---|---|---|
| BUILDING ELEMENT | REQUIRED | PROVIDED |
| Primary structural frame [f] (see Section 202) | 0 | 0 |
| Bearing walls | | |
| EXT [e, f] | 0 | 0 |
| Interior | 0 | 0 |
| Nonbearing walls and partitions EXT | Re: Table 705.5 | NA |
| Nonbearing walls and partitions INT[d] | 0 | 0 |
| Floor construction and secondary members (see Section 202) | 0 | 0 |
| Roof construction and secondary members (see Section 202) | 0 | 0 |

**FIRE AND SMOKE PROTECTION FEATURES - CHAPTER 7**

FIRE-RESISTANCE RATING REQUIREMENTS FOR EXT WALLS BASED ON FIRE SEPARATION DISTANCE PER TABLE 705.5

| FIRE SEPARATION DISTANCE = X (feet) | TYPE OF CONSTRUCTION | OCCUPANCY GROUP REQUIRED | PROVIDED |
|---|---|---|---|
| X < 5 [b] | ALL | 1 | NA |
| 5 ≤ X < 10 | IA<br>OTHERS | 1 | NA |
| 10 ≤ X < 30 | IA, IB<br>IIB, VB<br>OTHERS | 1<br>0<br>1 | NA |
| X ≥ 30 | ALL | 0 | 0 |

THE PROPOSED WORK COMPLIES WITH TABLE 705.8 OPENINGS. THERE ARE NO OTHER STRUCTURES WITHIN 30' OF THE <E> BUILDING AND THEREFORE NO LIMIT TO THE NUMBER OF UNPROTECTED OPENINGS.

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**INTERIOR FINISHES - CHAPTER 8**
THE INTERIOR WALL AND CEILING FINISHES IN ROOMS AND CORRIDORS SHALL BE TESTED BY AN APPROVED TESTING AGENCY AND SHALL COMPLY WITH IBC §803 AND TABLE 803.13

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**FIRE PROTECTIVE SYSTEMS - CHAPTER 9**

AUTOMATIC SPRINKLER SYSTEMS PER §903.3.1.1
NFPA 13 AUTOMATIC SPRINKLER SYSTEMS WILL BE INSTALLED THROUGHOUT THE BUILDING, INCLUDING THE FOLLOWING SYSTEMS FOR SPECIFIC AREAS:

WET SPRINKLER SYSTEM - TEMPERED AREAS OF THE BUILDING PER IBC §903.3.1.1
DRY SPRINKLER SYSTEM - UNTEMPERED ATTIC SPACE PER IBC §903.3.1.1
PRE-ACTION CLEAN-AGENT SPRINKLER SYSTEM - COLLECTIONS STORAGE AREA PER IBC §903.3.1.1
CLEAN AGENT FIRE SUPPRESSION SYSTEM – COLLECTIONS STORAGE AREA PER IBC §904.10.

SPRINKLERS SHALL BE OMITTED FROM BELOW THE EAVES (EXTERIOR PROJECTIONS) AS THE ALL FINISH IS NON COMBUSTIBLE AND THE CONCEALED SPACE IS FULLY SPRINKED PER NFPA 13 §9.2.3.2.

PORTABLE FIRE EXTINGUISHERS TO BE PROVIDED PER IBC §906
- PROVIDE CLASS A FIRE HAZARD (LOW HAZARD OCCUPANCY) PORTABLE FIRE EXTINGUISHERS IN THE ASSEMBLY AND BUSINESS OCCUPANCIES OF THE BUILDING.
- PROVIDE ORDINARY HAZARD OCCUPANCY PORTABLE FIRE EXTINGUISHERS IN THE COLLECTIONS STORAGE OCCUPANCY OF THE BUILDING.
- PORTABLE FIRE EXTINGUISHERS SHALL BE INSTALLED IN CONSPICUOUS LOCATIONS ALONG NORMAL PATHS OF TRAVEL IN ACCORDANCE WITH §906.5.
- MAXIMUM TRAVEL DISTANCE TO ANY FIRE EXTINGUISHER SHALL BE 75 FEET MAX AND 11,250 SF MAX FLOOR AREA PER TABLE 906.3(1)

FIRE ALARM AND DETECTION SYSTEMS TO BE PROVIDED PER §907.2
- REFER TO FIRE ALARM DRAWINGS

FIRE DEPARTMENT CONNECTION PER §912
A FIRE DEPARTMENT CONNECTION WILL BE PROVIDED ON THE WEST ELEVATION OF THE BUILDING ALONG GRID 9. THE KNOX BOX IS TO BE LOCATED ADJACENT AND SOUTH OF THE MAIN ENTRY PER DISCUSSIONS WITH THE AHJ IN ACCORDANCE WITH IFC §506

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**MEANS OF EGRESS - CHAPTER 10**
REFER TO SHEET G111 FOR MAIN LEVEL AND BASEMENT LEVEL CODE PLANS

HEADROOM PER IBC §1003.3.1
80" MINIMUM OVER ANY WALKING SURFACE

DESIGN OCCUPANT LOAD - TABLE 1004.5

| FUNCTION OF SPACE: | ASSEMBLY | BUSINESS | EXHIBIT SPACE | STORAGE |
|---|---|---|---|---|
| FLOOR AREA PER OCCUPANT: | 15 SF NET | 150 SF GROSS | 30 SF NET | 300 SF GROSS |

REFER TO THE OCCUPANT TABLE ON SHEET G111 CODE ANALYSIS CODE PLAN

MEANS OF EGRESS SIZING PER IBC §1005
WHERE MULTIPLE MEANS OF EGRESS ARE REQUIRED, THEY SHALL BE SIZED SUCH THAT IN THE EVENT ONE OF THE MEANS OF EGRESS IS LOST, THEN THE REMAINING MEANS OF EGRESS SHALL NOT REDUCE THE AVAILABLE CAPACITY TO LESS THAN 50 PERCENT OF THE REQUIRED CAPACITY.

EACH LEVEL IS PROVIDED WITH AT LEAST TWO EXITS FROM THAT SAME LEVEL. ADDITIONALLY, ASSEMBLY SPACES HAVE TWO EXITS EACH. THE EXISTING STAIRWAY IS NOT USED AS A MEANS OF EGRESS.

EGRESS COMPONENT SIZING OF DOORS (.2" PER OCCUPANT)
**MAIN LEVEL** TOTAL OCCUPANTS 159 X .2" = 31.8" WIDTH REQUIRED (DEFAULTS TO 32")
36" EGRESS WIDTH PROVIDED
**BASEMENT LEVEL** TOTAL OCCUPANTS 16 X .2" = 3.2" WIDTH REQUIRED (DEFAULTS TO 32")
36" EGRESS WIDTH PROVIDED

DOOR ENCROACHMENT: WHEN FULLY OPEN, DOORS SHALL NOT REDUCE THE REQUIRED MEANS OF EGRESS WIDTH BY MORE THAN 7". DOORS IN ANY POSITION SHALL NOT REDUCE THE REQUIRED WIDTH BY MORE THAN ONE-HALF.

NUMBER OF EXITS PER IBC §1006.3.3
MAIN LEVEL OCCUPANT LOAD = 159
REQUIRED NUMBER OF EXITS = 2 PER §1006.3.3
PROVIDED NUMBER OF EXITS = 5
- LOBBY = 2 EXITS TO EXTERIOR
- AUDITORIUM = 1 EXIT TO EXTERIOR (2 ROOM EXITS PROVIDED)
- EXHIBIT SPACE = 1 EXIT TO EXTERIOR (2 ROOM EXITS PROVIDED)
- OFFICE AREA = 1 EXIT TO EXTERIOR (2 ROOM EXITS PROVIDED)

BASEMENT LEVEL OCCUPANT LOAD = 15
REQUIRED NUMBER OF EXITS = 2 PER §1006.2.1 (COMMON PATH OF TRAVEL EXCEEDS ALLOWABLE W/ 1 EXIT)
PROVIDED NUMBER OF EXITS = 2

EXIT AND EXIT ACCESS DOORWAY CONFIGURATION PER IBC §1007
WHERE TWO EXITS ARE REQUIRED, THEY SHALL BE PLACED A DISTANCE APART EQUAL TO NOT LESS THAN ONE-THIRD OF THE LENGTH OF THE MAXIMUM OVERALL DIAGONAL DIMENSION OF THE BUILDING OR AREA TO BE SERVED MEASURED IN A STRAIGHT LINE BETWEEN THEM.

MAIN LEVEL
REQUIRED MIN EXIT SEPARATION DISTANCE: 119'-4" / 3 = 40'-0"
EXIT SEPARATION PROVIDED: 42'-10 1/2"

BASEMENT LEVEL
REQUIRED MIN EXIT SEPARATION DISTANCE: 99'-9" / 3 = 33'-3"
EXIT SEPARATION PROVIDED: 51'-11 7/8"

ACCESSIBLE MEANS OF EGRESS PER IBC §1009
ACCESSIBLE SPACES SHALL BE PROVIDED WITH NOT LESS THAN ONE ACCESSIBLE MEANS OF EGRESS. WHERE MORE THAN ONE MEANS OF EGRESS ARE REQUIRED FROM ANY ACCESSIBLE SPACE, EACH ACCESSIBLE PORTION OF THE SPACE SHALL BE SERVED BY NOT LESS THAN TWO ACCESSIBLE MEANS OF EGRESS PER SECTION 1009.1.

ACCESSIBLE MEANS OF EGRESS PROVIDED = 6
MAIN LEVEL = 4
BASEMENT LEVEL = 2*

*BASEMENT LEVEL EXITS ARE PROVIDED WITH AN EXTERIOR AREA FOR ASSISTED RESCUE PER §1009.7 DUE TO ABSENCE OF A CONTINUOUS ACCESSIBLE PATHWAY TO A PUBLIC WAY.

CONTINUITY PER IBC §1009.2
EACH REQUIRED ACCESSIBLE MEANS OF EGRESS SHALL BE CONTINUOUS TO A PUBLIC WAY. REFER TO NEXT SECTION BELOW FOR RESOLUTION OF EXITS WITHOUT A CONTINUOUS PATH TO A PUBLIC WAY.

EXTERIOR AREA FOR ASSISTED RESCUE IBC §1009.7
EXCEPTION: FIRE-RATED WALL ASSEMBLY BETWEEN THE EXTERIOR AREA FOR ASSISTED RESCUE & BUILDING INTERIOR IS NOT REQUIRED WHEN THE BUILDING IS PROTECTED WITH AN AUTOMATIC SPRINKLER SYSTEM.

SIZE OF DOORS PER §1010.1
MINIMUM CLEAR WIDTH: 32"
BUILDING DOORS COMPLY WITH MINIMUM CLEAR DOOR WIDTH OF 32"

EXIT ACCESS TRAVEL DISTANCE - TABLE 1017.2
A OCCUPANCY, SPRINKLERED: 250'-0"
MAIN LEVEL - ACTUAL MAXIMUM TRAVEL DISTANCE 45'-6"
BASEMENT LEVEL - ACTUAL MAXIMUM TRAVEL DISTANCE 66'-6"

**ACCESSIBILITY - CHAPTER 11**
ACCESSIBLE ROUTE PER IBC §1104

ACCESSIBLE ENTRANCES PER IBC §1105.1
PUBLIC ENTRANCES: AT LEAST 60 PERCENT OF ALL PUBLIC ENTRANCES SHALL BE ACCESSIBLE. PROJECT PROVIDES 100% OF PUBLIC ENTRANCES THAT ARE ACCESSIBLE BUILDING ENTRIES, RE: G111 CODE PLANS.

SIGNAGE PER IBC §1112
SIGNS: REQUIRED ACCESSIBLE ELEMENTS SHALL BE IDENTIFIED BY THE INTERNATIONAL SYMBOL OF ACCESSIBILITY AT THE LOCATIONS LISTED IN SECTION 1112.1

SIGNS IDENTIFYING TOILETS: SIGNAGE INDICATING TOILETS SHALL BE PROVIDED IN ACCORDANCE WITH SECTION PER SECTION 1112.2

DIRECTIONAL SIGNAGE: DIRECTIONAL SIGNAGE INDICATING THE ROUTE TO THE NEAREST LIKE ACCESSIBLE ELEMENT SHALL INCLUDE THE INTERNATIONAL SYMBOL OF ACCESSIBILITY AND BE PROVIDED AT THE LOCATIONS LISTED IN SECTION 1112.3

OTHER SIGNS: SIGNAGE INDICATING SPECIAL ACCESSIBILITY PROVISIONS SHALL BE PROVIDED IN EACH DOOR TO AN EXIT PASSAGEWAY, EXTERIOR AREAS FOR ASSISTED RESCUE, AND EXIT DISCHARGE IN ACCORDANCE WITH SECTION 1112.4 AND 1009.11. AT INTERIOR RAMPS SIGNAGE SHALL BE PROVIDED IN ACCORDANCE WITH 1023.9. AT TWO WAY COMMUNICATION SYSTEMS SIGNAGE SHALL BE PROVIDED IN ACCORDANCE WITH 1009.8.2.

GENERAL ACCESSIBILITY NOTES
THE DETAILS AND NOTES PROVIDED WITHIN THESE DOCUMENTS REPRESENT ONLY A PORTION OF THE REQUIREMENTS SET FORTH BY FEDERAL LAW, STATE LAW, CITY AND COUNTY LAW AND THE BUILDING CODE FOR ACCESSIBILITY REQUIREMENTS.

CONTRACTORS MUST REVIEW ALL ACCESSIBILITY REQUIREMENTS AS PRESENTED IN THESE CONSTRUCTION DOCUMENTS AND THE REFERENCES HEREIN PRIOR TO STARTING ANY CONSTRUCTION.

FOR SITE ACCESSIBILITY REQUIREMENTS AND ALL SITE ACCESSIBILITY DETAILING RE: CIVIL

**PLUMBING FIXTURES - CHAPTER 29**
§2902.1 - MINIMUM NUMBER OF FIXTURES SHALL BE PROVIDED FOR THE TYPE OF OCCUPANCY AND IN THE MINIMUM NUMBERS PER TABLE 2902.1

WATER CLOSETS:
ASSEMBLY OCCUPANCY: (147 TOTAL BUSINESS OCCUPANTS = 73 MEN & 73 WOMEN)
1 PER 125 FOR MEN & 1 PER 65 FOR WOMEN
REQUIRED = 1 MEN & 2 WOMEN = 3 TOTAL

BUSINESS OCCUPANCY**: (17 TOTAL BUSINESS OCCUPANTS)
1 PER 25 FOR FIRST 50 AND 1 PER 50 FOR THE REMAINDER EXCEEDING 50
REQUIRED = 1

STORAGE OCCUPANCY: (10 TOTAL STORAGE OCCUPANTS)
1 PER 100
REQUIRED = 1

TOTAL REQUIRED = 5 WATER CLOSETS
7 TOTAL WATER CLOSETS PROVIDED (3 MEN, 3 WOMEN & 2 SINGLE-USER RESTROOMS)
1 URINAL PROVIDED IN THE RESTROOM FOR MEN

LAVATORIES:
ASSEMBLY OCCUPANCY: (147 TOTAL BUSINESS OCCUPANTS)
1 PER 200
REQUIRED = 1

BUSINESS OCCUPANCY**: (17 TOTAL BUSINESS OCCUPANTS)
1 PER 40 FOR THE FIRST 80 AND 1 PER 80 FOR THE REMAINDER EXCEEDING 80
REQUIRED = 1

STORAGE OCCUPANCY: (10 TOTAL STORAGE OCCUPANTS)
1 PER 100
REQUIRED = 1

TOTAL REQUIRED = 3
6 TOTAL LAVATORIES PROVIDED (2 FOR MEN, 2 FOR WOMEN, & 2 IN SINGLE-USER RESTROOMS)

DRINKING FOUNTAINS:
ASSEMBLY OCCUPANCY: (147 TOTAL BUSINESS OCCUPANTS)
1 PER 500
REQUIRED = 1

BUSINESS OCCUPANCY**: (17 TOTAL BUSINESS OCCUPANTS)
1 PER 100
REQUIRED = 1

STORAGE OCCUPANCY: (10 TOTAL STORAGE OCCUPANTS)
1 PER 1000
REQUIRED = 1

TOTAL REQUIRED: 3
4 DRINKING FOUNTAINS PROVIDED (2 ABAAS & 2 NON-ABAAS)

SERVICE SINKS:
SERVICE SINKS REQUIRED = 3 (1 FOR EACH OCCUPANCY TYPE)
SERVICE SINKS PROVIDED = 2

* SINGLE-USER RESTROOMS: 2024 IPC FOR S-1 AND B OCCUPANCIES DOES NOT DISTINGUISH A FIXTURE NEED DIFFERENCE BETWEEN MALE AND FEMALE, PERMITTING THE USE OF GENDER NEUTRAL RESTROOMS FOR FULFILLMENT OF FIXTURE REQUIREMENTS.

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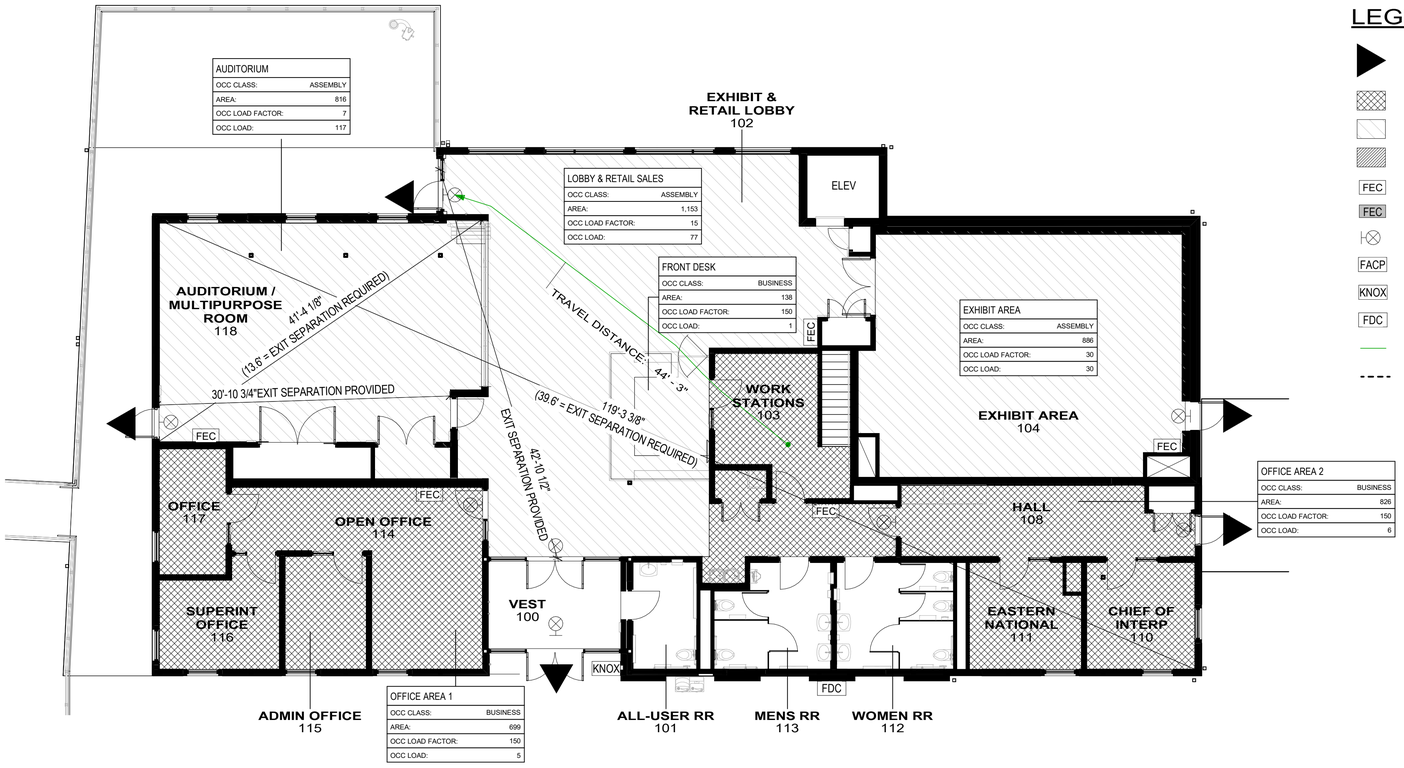
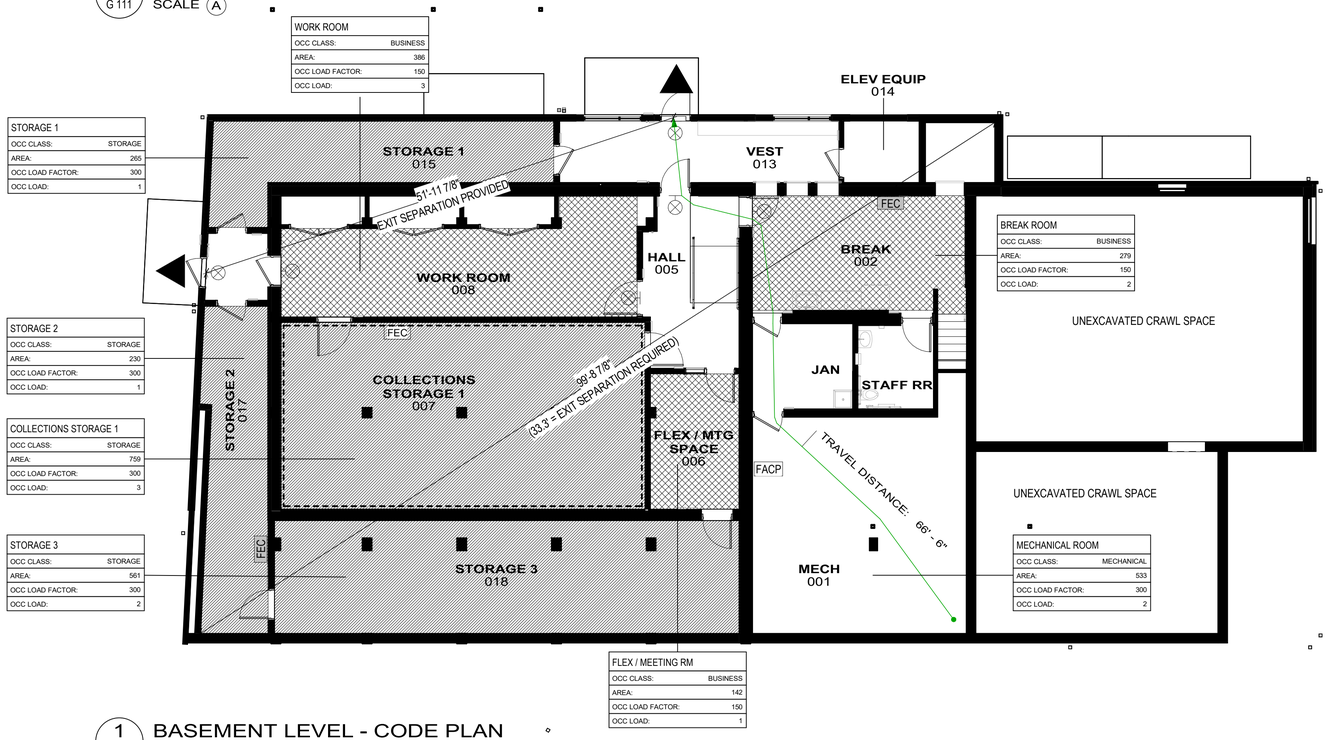
| A/E FIRM: | DESIGNED: | SUB SHEET NO. | TITLE OF SHEET | DRAWING NO. |
| PRIME: OTAK, INC. REDMOND, WA | LB/CM | G 110 | CODE ANALYSIS | 394 194248 |
| SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | CADD LB/MM | | | PMIS/PKG NO. 318539 |
| | TECH REVIEW: ACN / EMH | | EFFIGY MOUNDS NATIONAL MONUMENT | SHEET 3 OF 142 |
| | DATE: 05/29/2026 | | | |




## 2 MAIN LEVEL - CODE PLAN
G 111 SCALE A




## 1 BASEMENT LEVEL - CODE PLAN
G 111 SCALE A

## LEGEND

- BUILDING EGRESS DOOR
- BUSINESS OCCUPANCY CLASSIFICATION
- ASSEMBLY OCCUPANCY CLASSIFICATION
- STORAGE OCCUPANCY CLASSIFICATION
- FEC — <N> SEMI-RECESSED FIRE EXTINGUISHER CABINET
- FEC — <N> SURFACE MOUNT FIRE EXTINGUISHER CABINET
- EXIT SIGNAGE RE: ELEC
- FACP — FIRE ALARM CONTROL PANEL RE: ELEC
- KNOX — KNOX BOX
- FDC — FIRE DEPARTMENT CONNECTION (FDC)
- PATH OF TRAVEL
- <N> 1-HR RATED ASSEMBLY, SEAL ALL PENETRATIONS

OCCUPANT LOAD CALCULATIONS

| Level | Occ Function | Area | Occ Classification | Occ Load Factor | Occ Load |
|---|---|---|---|---|---|
| MAIN LEVEL | AUDITORIUM | 816 SF | ASSEMBLY | 7 | 55 |
| MAIN LEVEL | EXHIBIT AREA | 886 SF | ASSEMBLY | 30 | 30 |
| MAIN LEVEL | LOBBY & RETAIL SALES | 1,153 SF | ASSEMBLY | 15 | 62 |
| MAIN LEVEL | OFFICE AREA 1 | 699 SF | BUSINESS | 150 | 5 |
| MAIN LEVEL | OFFICE AREA 2 | 826 SF | BUSINESS | 150 | 6 |
| MAIN LEVEL | FRONT DESK | 138 SF | BUSINESS | 150 | 1 |
| | | | | | |
| BASEMENT LEVEL | WORK ROOM | 386 SF | BUSINESS | 150 | 3 |
| BASEMENT LEVEL | FLEX / MEETING RM | 142 SF | BUSINESS | 150 | 1 |
| BASEMENT LEVEL | BREAK ROOM | 279 SF | BUSINESS | 150 | 2 |
| BASEMENT LEVEL | MECHANICAL ROOM | 533 SF | MECHANICAL | 300 | 2 |
| BASEMENT LEVEL | COLLECTIONS STORAGE 1 | 759 SF | STORAGE | 300 | 3 |
| BASEMENT LEVEL | STORAGE 1 | 265 SF | STORAGE | 300 | 1 |
| BASEMENT LEVEL | STORAGE 2 | 230 SF | STORAGE | 300 | 1 |
| BASEMENT LEVEL | STORAGE 3 | 561 SF | STORAGE | 300 | 2 |
| TOTAL BUILDING OCCUPANT LOAD | | | | | 174 |

ACCESSIBILITY PLANS ON SHEET G400 FURNITURE PLAN & G401 ACCESSIBILITY PLAN

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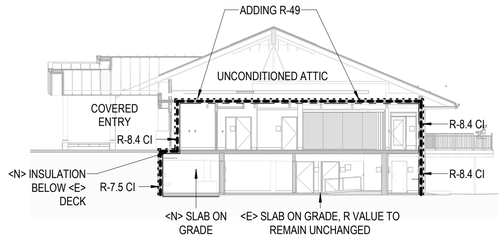
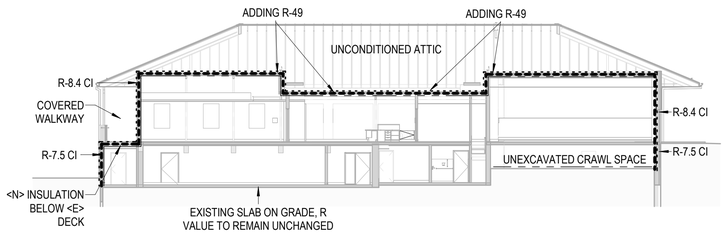


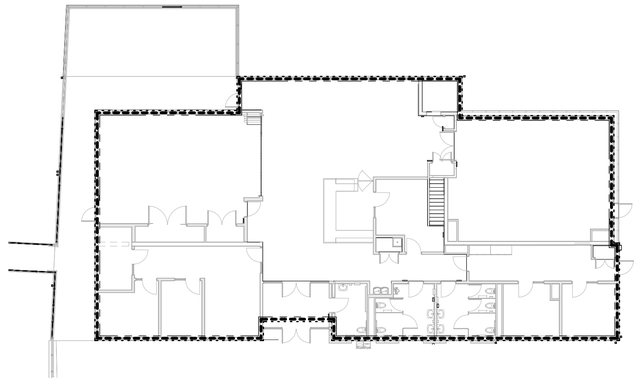

## 2 G 300 E/W BUILDING SECTION - THERMAL ENVELOPE

SCALE (A)



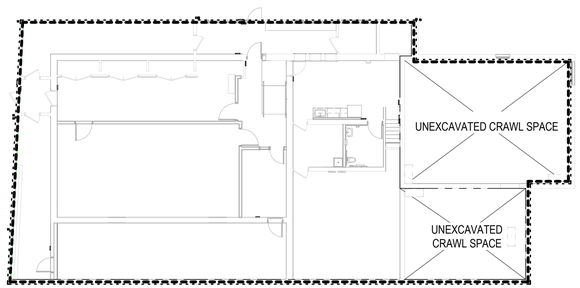

## 1 G 300 N/S BUILDING SECTION - THERMAL ENVELOPE

SCALE (A)



## 4 G 300 MAIN LEVEL FLOOR PLAN - THERMAL ENVELOPE

SCALE (A)




## 3 G 300 BASEMENT FLOOR PLAN - THERMAL ENVELOPE

SCALE (A)

# ENERGY CODE SUMMARY

| | | | | |
|---|---|---|---|---|
| APPLICABLE CODES | 2024 INTERNATIONAL ENERGY CONSERVATION CODE<br>2024 INTERNATIONAL EXISTING BUILDING CODE | | | |
| CLIMATE ZONE | 5A PROJECT LOCATION: HARPER'S FERRY, IOWA | | | |
| COMPLIANCE PATH | PRESCRIPTIVE PATH: R-VALUE METHOD | | | |
| TABLE C402.1.2 | OPAQUE THERMAL ENVELOPE MINIMUM R-VALUE (REFER TO ASSEMBLIES (G500) FOR CONFIGURATIONS) | | | |
| | ASSEMBLY LOCATION | REQUIRED | PROVIDED | INSULATION TYPE |
| | ROOF - ATTIC<br>WALLS - WOOD FRAMED - CAVITY<br>WALLS - WOOD FRAMED - CONTINUOUS<br>WALLS - BELOW GRADE<br>UNHEATED SLAB ON GRADE | R-49<br>R-13<br>R-7.5 CI<br>R-7.5 CI FOR 24"<br>R-15 FOR 24" BELOW | R-49<br>R-21<br>R-8.4 CI<br>R-7.5CI FOR 24"<br>N/A* | MINERAL WOOL BATT<br>MINERAL WOOL BATT<br>MINERAL WOOL SEMI-RIGID<br>XPS<br>N/A* |
| | *PROPOSED SLAB-ON-GRADE IS MORE THAN 24" BELOW GRADE AND THEREFORE NSULATION IS NOT REQUIRED. | | | |
| | OPAQUE DOORS | MAX U-FACTOR | PROVIDED | |
| | NON-SWINGING OPAQUE DOORS<br>SWINGING DOORS | U-0.31<br>U-0.37 | REFER TO SPECIFCATIONS | |
| TABLE C402.5 | BUILDING ENVELOPE FENESTRATION MAXIMUM U-FACTOR & SHGC REQUIREMENTS | | | |
| | | MAX U-FACTOR | PROVIDED | |
| | FIXED FENESTRATION<br>OPERABLE FENESTRATION<br>DOOR ENTRANCES | U-0.34<br>U-0.45<br>U-0.63 | REFER TO SPECIFICATIONS | |
| | FENESTRATION MAXIMUM SOLAR HEAT GAIN COEFFICIENT (SHGC) | | | |
| | | FIXED | OPERABLE | PROVIDED |
| | PF < 0.2<br>0.2 < PF < 0.5<br>PF > 0.5 | U-0.38<br>U-0.46<br>U-0.61 | U-0.33<br>U-0.40<br>U-0.53 | REFER TO SPECIFICATIONS |
| | PF (PROJECTION FACTOR) = A/B<br>A = DISTANCE MEASURED HORIZONTALLY FROM GLAZING TO EDGE OF OVERHANG<br>B = DISTANCE MEASURED VERTICALLY FROM BOTTOM OF GLAZING TO UNDERSIDE OF OVERHANG | | | |
| TABLE C402.6.2.1 | WHOLE BUILDING TEST METHOD AND REPORTING:<br>BUILDING THERMAL ENVELOPE TESTING IS THE RESPONSIBILITY OF THE GENERAL CONTRACTOR. SINCE THE BUILDING IS LESS THAN 10,000SF, THE ENTIRE BUILDING THERMAL ENVELOPE IS PERMITTED TO BE TESTED IN ACCORDANCE WITH ASTM E779, ASTME3158, ASTM E1827, OR AN EQUIVALENT APPROVED METHOD. | | | |
| TABLE C402.6.2.3 | NOT APPLICABLE SINCE SECTION C402.6.2.1 APPLIES | | | |
| TABLE C406 | ADDITIONAL EFFICIENCY PACKAGES | | | |
| | REQUIREMENT | PROVIDED | C503.6 ADDITIONAL EFFICIENCY CREDIT REQUIREMENTS FOR ALTERATIONS | |
| | 48 POINTS TOTAL FOR BASE ENERGY CREDITS<br><br>0 POINTS TOTAL FOR RENEWABLE CREDITS PER C503.6 | 89 POINTS TOTAL<br><br>0 POINTS TOTAL | ALTERATIONS THAT ARE SUBSTANTIAL IMPROVEMENTS SHALL COMPLY WITH MEASURES FROM SECTION C406.2, C406.3 OR BOTH TO EARN THE NUMBER OF REQUIRED CREDITS SPECIFIED IN TABLE 406.1.1 (1) SHALL BE BASED ON BUILDING OCCUPANCY GROUP AND CLIMATE ZONE. | |
| APPENDIX CB | SOLAR READY ZONE COMMERCIAL | | | |
| | PV READY REQUIREMENTS ONLY APPLY TO NEW CONSTRUCTION AND ADDITIONS TO EXISTING BUILDING. PV READY REQUIRMENTS DO NOT APPLY TO THIS PROJECT. SEE C502.3.8 & CB101.1 | | | |
| C501.2 | EXISTING BUILDINGS | | | |
| APPLICABLE SECTIONS | C503.2 BUILDING THERMAL ENVELOPE<br>C503.2.1 ROOF, CEILING & ATTIC ALTERATIONS<br>C503.3.4 ABOVE-GRADE WALL ALTERATIONS<br>C503.2.5 FLOOR ALTERATIONS<br>C503.2.6 BELOW-GRADE WALL ALTERATIONS | | | |

# BUILDING ENVELOPE LEGEND

- THERMAL ENVELOPE
- AIR BARRIER
- VAPOR RETARDER

C402.6.1
ALTERED BUILDING THERMAL ENVELOPE ASSEMBLIES SHALL BE PROVIDED WITH AN AIR BARRIER IN ACCORDANCE WITH C402.6. SUCH AIR BARRIER NEED NOT BE CONTINUOUS WITH UNALTERED PORTIONS OF THE BUILDING THERMAL ENVELOPE. TESTING REQUIREMENTS OF SECTION C402.6.1.2 SHALL NOT BE REQUIRED.

# ADDITIONAL EFFICIENCY CREDITS IECC SECTION C406.2

| OCCUPANCY GROUP CALCULATIONS | SF | % OF SF |
|---|---|---|
| A-3, other | 5203 | 68% |
| B BUSINESS | 2470 | 32% |
| TOTAL | 7673 | 100% |

| TABLE C406.1.1(1)ENERGY CREDIT REQUIREMENTS BY BUILDING OCCUPANCY GROUP | CLIMATE ZONE | | | | | | | | | | | | | | | | | | | OCCUPANCY GROUP MULTIPLIER | WEIGHTED ENERGY CREDIT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| BUILDING OCCUPANCY GROUP | OA | 0B | IA | 1B | 2A | 2B | 3A | 3B | 3C | 4A | 4B | 4C | 5A | 5B | 5C | 6A | 6B | 7 | 8 | | |
| B | | | | | | | | | | | | | 71 | | | | | | | 0.32 | 23 |
| All other | | | | | | | | | | | | | 37 | | | | | | | 0.68 | 25 |
| | | | | | | | | | | | | | | | | | | | | SUBTOTAL | 48 |

| CREDITS<br>TABLE C406.2(4)BASE ENERGY CREDITS FOR GROUP B OCCUPANCIES | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ID | ENERGY CREDIT MEASURE | SECTION | CLIMATE ZONE | | | | | | | | | | | | | | | | | | |
| | | | OA | 0B | IA | 1B | 2A | 2B | 3A | 3B | 3C | 4A | 4B | 4C | 5A | 5B | 5C | 6A | 6B | 7 | 8 |
| H02 | Heating efficiency | C406.2.2.2 | | | | | | | | | | | | | 21 | | | | | | |
| H03 | Cooling efficiency | C406.2.2.3 | | | | | | | | | | | | | 2 | | | | | | |
| W04 | SHW pipe insulation | C406.2.3.2 | | | | | | | | | | | | | 2 | | | | | | |
| W06 | Thermostatic bal. valves | C406.2.3.3 | | | | | | | | | | | | | 5 | | | | | | |
| L03 | Increase occp. sensor | C406.2.5.3 | | | | | | | | | | | | | 5 | | | | | | |
| L06 | Light power reduction | C406.2.5.6 | | | | | | | | | | | | | 22 | | | | | | |
| | | | | | | | | | | | | | | | TOTAL | 57.00 | | | | | |
| | | | | | | | | | | | | | | | ADJUSTED TOTAL | 18 | | | | | |

Total multipied by % of SF and rounded to nearest whole number: 57 x 0.32=

| TABLE C406.2(6 )BASE ENERGY CREDITS FOR GROUP ALL OTHER OCCUPANCIES | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ID | ENERGY CREDIT MEASURE | SECTION | CLIMATE ZONE | | | | | | | | | | | | | | | | | | |
| | | | OA | 0B | IA | 1B | 2A | 2B | 3A | 3B | 3C | 4A | 4B | 4C | 5A | 5B | 5C | 6A | 6B | 7 | 8 |
| H02 | Heating efficiency | C406.2.2.2 | | | | | | | | | | | | | 63 | | | | | | |
| W03 | Efficient gas water heater | C406.2.3.1c | | | | | | | | | | | | | 18 | | | | | | |
| W04 | SHW pipe insulation | C406.2.3.2 | | | | | | | | | | | | | 2 | | | | | | |
| W06 | Thermostatic bal. valves | C406.2.3.3 | | | | | | | | | | | | | 1 | | | | | | |
| L03 | Increase occp. sensor | C406.2.5.3 | | | | | | | | | | | | | 4 | | | | | | |
| L06 | Light power reduction | C406.2.5.6 | | | | | | | | | | | | | 16 | | | | | | |
| | | | | | | | | | | | | | | | TOTAL | 104.00 | | | | | |
| | | | | | | | | | | | | | | | ADJUSTED TOTAL | 71 | | | | | |

Total multipied by % of SF and rounded to nearest whole number: 104 X 0.68 =

| COMBINED TOTALS | 89 |
|---|---|
| NEEDED = | 48 |

# IECC CREDIT NOTES

1. REFERENCE MECH FOR H02 AND H03 CREDITS
2. REFERENCE PLUMB FOR W03, W04 AND W06 CREDITS
3. REFERENCE ELEC FOR L03 AND L06 CREDITS

SCALE: (A) 16 0 16 32
SCALE OF FEET

LICENSED PROFESSIONAL ARCHITECT
ELIZABETH HALLAS 07418
IOWA

| A/E FIRM: | DESIGNED: | SUB SHEET NO. | TITLE OF SHEET | DRAWING NO. |
|---|---|---|---|---|
| PRIME:<br>OTAK, INC.<br>REDMOND, WA<br>SUBCONTRACTOR<br>ANDERSON HALLAS ARCHITECTS<br>GOLDEN, CO | LB/CM<br>CADD<br>LB/MM<br>TECH REVIEW:<br>ACN / EMH<br>DATE:<br>05/29/2026 | G 300 | ENERGY CODE ANALYSIS<br>EFFIGY MOUNDS NATIONAL MONUMENT | 394 194248<br>PMIS/PKG NO. 318539<br>SHEET 5 OF 142 |

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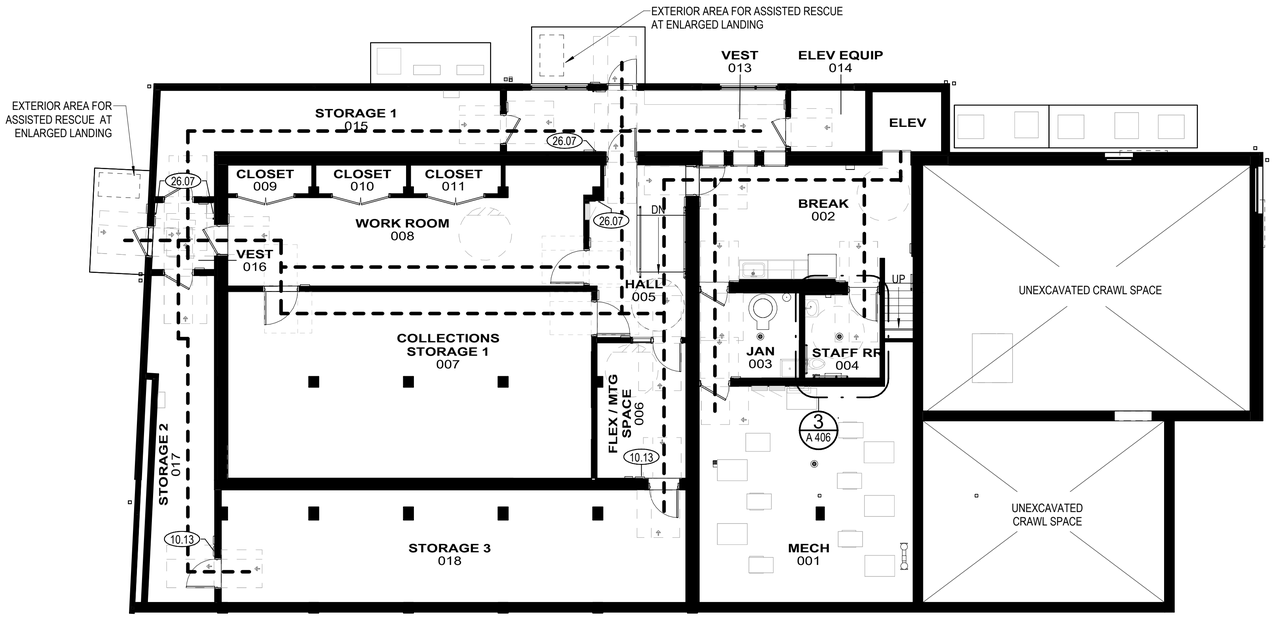
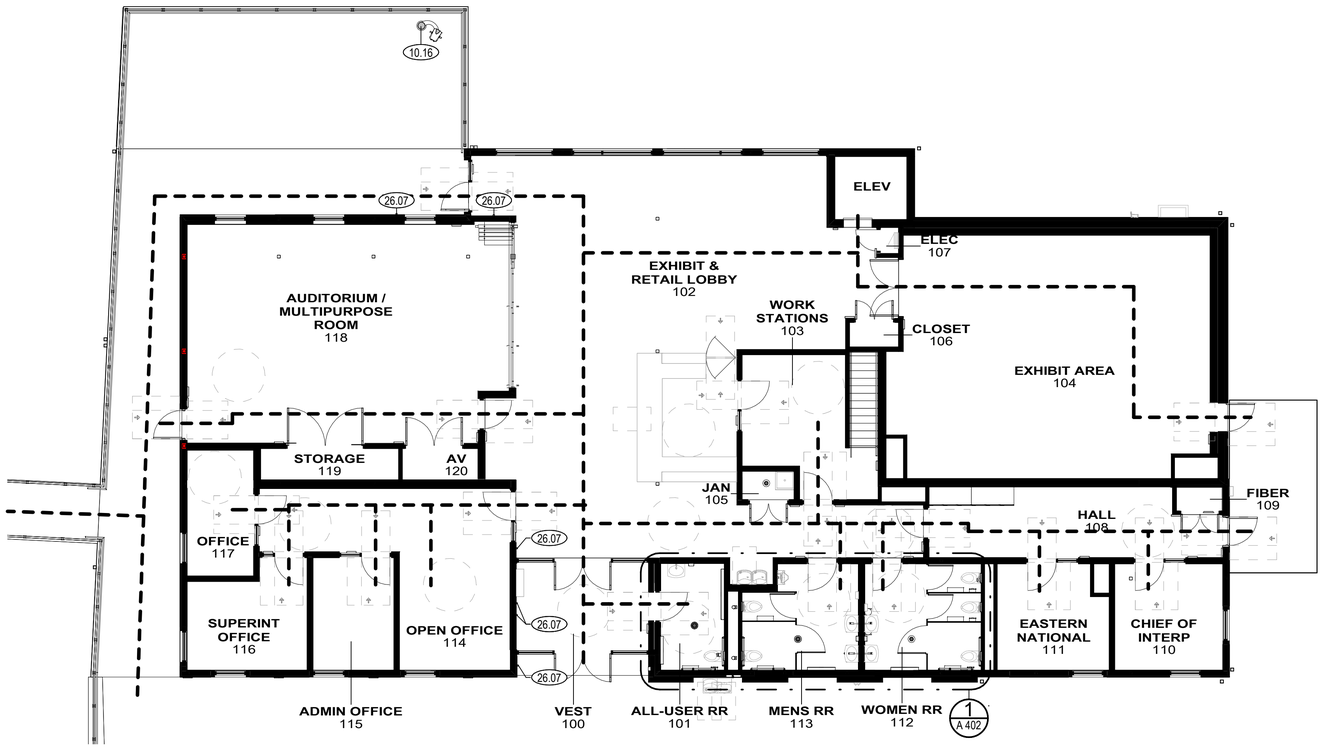



2 G 400 BASEMENT LEVEL - ACCESSIBILITY PLAN
SCALE A




1 G 400 BASEMENT LEVEL - ACCESSIBILITY PLAN
SCALE A

# GENERAL NOTES

1. REFER TO WALL TYPES FOR ASSEMBLY INFORMATION.
2. REFER TO INDEX FOR SYMBOL INFORMATION.
3. REFER TO INTERIOR ELEVATIONS FOR ADDITIONAL INFORMATION ON WALL MOUNTED ELEMENTS.
4. ALL PLAN DIMENSIONS LOCATING EXTERIOR ASSEMBLIES ARE TO GRID LINE OR EXTERIOR OF STUD, UNLESS NOTED OTHERWISE.
5. ALL GWB IN 'WET' AREAS TO BE MOISTURE AND IMPACT RESISTANT.
6. DOOR DIMS ARE TO CENTERLINE UNLESS STOREFRONT OR UON.
7. PATCH AND REPAIR ALL FINISHES IMPACTED BY DEMO AND SYSTEMS INSTALL. MATCH ADJ FINISH UON.
8. PAINT ALL EXPOSED GWB UON.

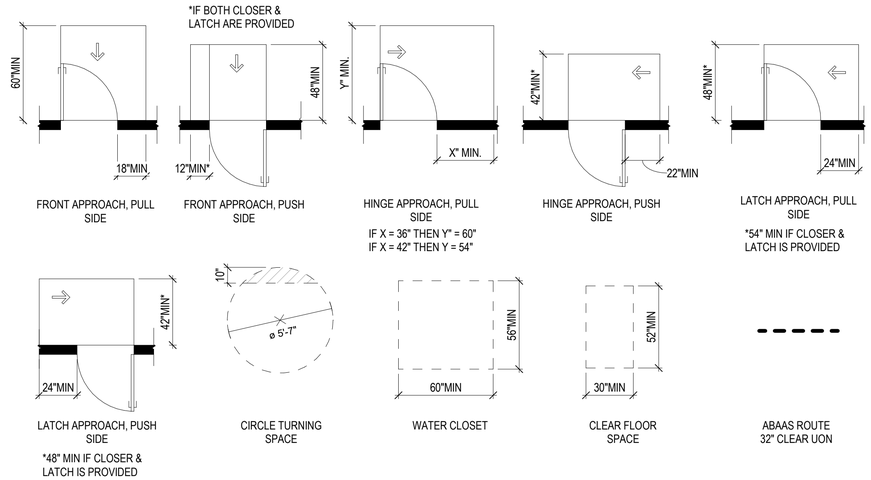
# KEYNOTES

| | |
|---|---|
| 10.13 | OPTION C: <N> SIGNAGE TO IDENTIFY STORAGE 3 HAS LOW HEAD CLEARANCE WITH ELEMENTS LESS THAN 6'-10" AFF |
| 10.16 | OPTION G: <N> ACCESSIBLE PREFINISHED MTL DUAL VIEWING MACHINE |
| 26.07 | OPTION C: <N> ABAAS DOOR PUSH BUTTON RE: ELEC |

# ACCESSIBLE DOOR LEGEND

ANSI A117.1 - SECTION 404

⇐ = DOOR APPROACH DIRECTION




FRONT APPROACH, PULL SIDE

FRONT APPROACH, PUSH SIDE

HINGE APPROACH, PULL SIDE
IF X = 36" THEN Y" = 60"
IF X = 42" THEN Y = 54"

HINGE APPROACH, PUSH SIDE

LATCH APPROACH, PULL SIDE
*54" MIN IF CLOSER & LATCH IS PROVIDED

LATCH APPROACH, PUSH SIDE
*48" MIN IF CLOSER & LATCH IS PROVIDED

CIRCLE TURNING SPACE

WATER CLOSET

CLEAR FLOOR SPACE

ABAAS ROUTE
32" CLEAR UON

SCALE: A 8 0 8 16
SCALE OF FEET

LICENSED PROFESSIONAL ARCHITECT ELIZABETH HALLAS 07418 IOWA

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA<br>SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM<br>DRAWN: LB/MM<br>TECH REVIEW: ACN / EMH<br>DATE: 05/29/2026 | SUB SHEET NO. G 400 | TITLE OF SHEET: ACCESSIBILITY PLANS<br>EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248<br>PMIS/PKG NO. 318539<br>SHEET 6 OF 142 |

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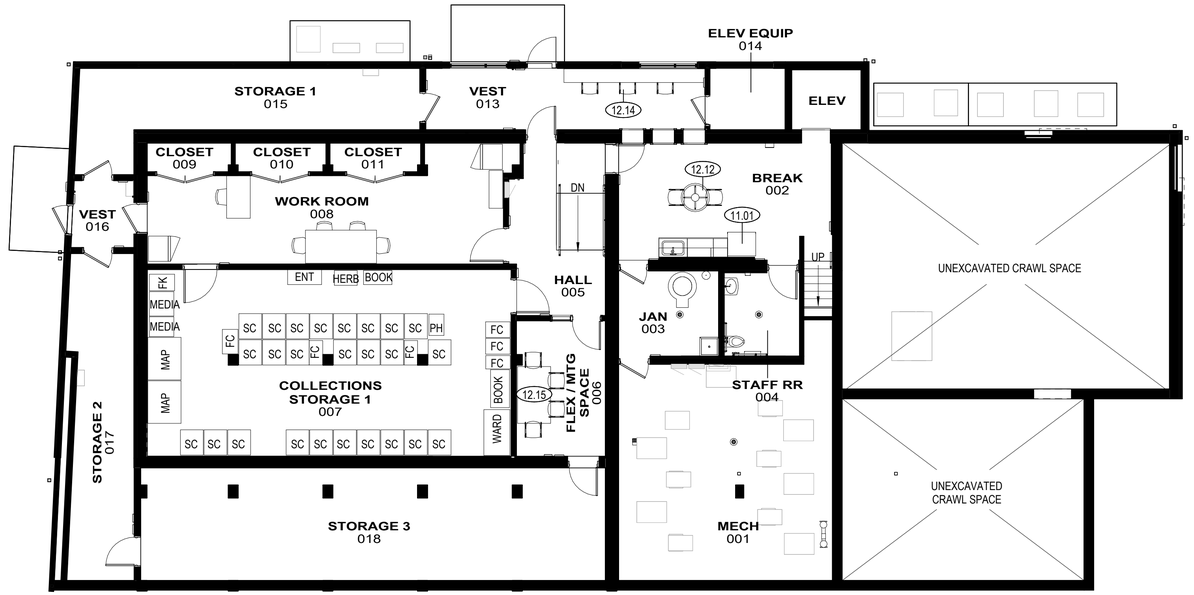
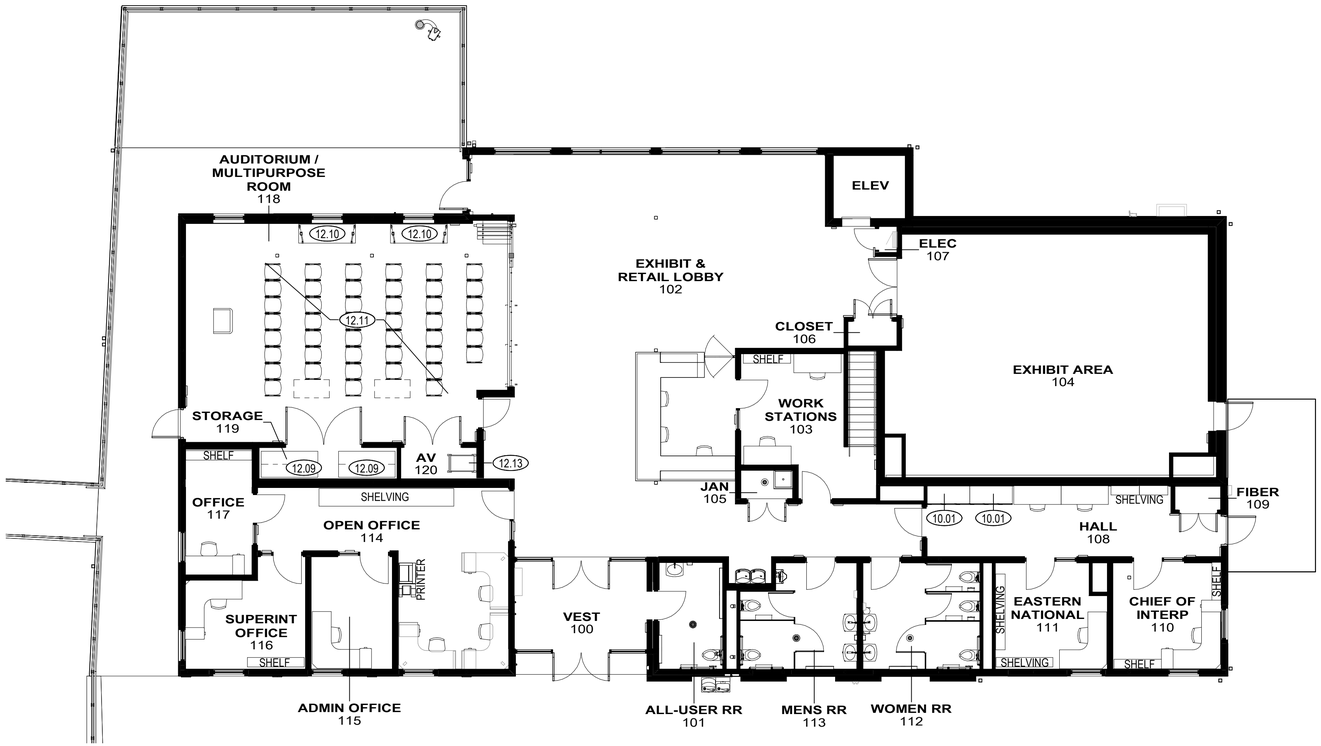



2 G 401 BASEMENT LEVEL - FURNISHING PLAN

SCALE (A)




1 G 401 MAIN LEVEL - FURNISHING PLAN

SCALE (A)

# GENERAL NOTES

1. REFER TO WALL TYPES FOR ASSEMBLY INFORMATION.
2. REFER TO INDEX FOR SYMBOL INFORMATION.
3. REFER TO INTERIOR ELEVATIONS FOR ADDITIONAL INFORMATION ON WALL MOUNTED ELEMENTS.
4. CONTRACTOR TO RELOCATE AND STORE EXISTING FURNITURE TO BE REINSTALLED. COORDINATE FURNITURE TO REMAIN WITH CO.
5. FURNITURE SHOWN IS FOR REFERENCE ONLY AND IS NOT INCLUDED IN THE CONTRACT UON. ALL SHELVING TO BE FURNITURE SOLUTION AND SHOWN FOR REFERENCE ONLY.

# KEYNOTES

| | |
|---|---|
| 10.01 | OPTION C: <N> TWO TIERED EXTRA WIDE WD LOCKERS, LOWER LOCKERS SHALL BE PROVIDED WITH AN ADJUSTABLE SHELF LESS THAN 15" AFF IN COMPLIANCE WITH ABAAS. |
| 11.01 | OPTION C: <N> REFRIGERATOR WITH INTEGRAL ICE MAKER AND WATER DISPENSER. |
| 12.09 | OPTION F: PREFINISHED MTL 30" X 60" NESTING TABLES (X4) WITH P-LAM SURFACE AND CASTER FEET, PROVIDED BY GC |
| 12.10 | OPTION F: PREFINISHED MTL 18" X 60" NESTING TABLES (X2)WITH P-LAM SURFACE AND GLIDE FEET, PROVIDED BY GC |
| 12.11 | OPTION F: STACKING CHAIRS (X50) WITH INTEGRAL CUSHIONED SEATS, PROVIDED BY GC |
| 12.12 | OPTION C: PREFINISHED MTL 36" DIAM CAFÉ STYLE TABLE WITH SOLID WD SURFACE AND GLIDE FEET, PROVIDE STACKING CHAIRS (X2) WITH INTEGRAL CUSHIONED SEATS, PROVIDED BY GC |
| 12.13 | OPTION F: PREFINISHED MTL STACKING DOLLY/CART, PROVIDED BY GC |
| 12.14 | OPTION C: STACKING CHAIRS (X3) WITH INTEGRAL CUSHIONED SEATS, PROVIDED BY GC |
| 12.15 | OPTION C: PREFINISHED MTL 42" X 72" CONFERENCE TABLE WITH P-LAM SURFACE AND GLIDE FEET, PROVIDE STACKING CHAIRS (X4) WITH INTEGRAL CUSHIONED SEATS, PROVIDED BY GC |

SCALE: (A) 8 0 8 16 SCALE OF FEET

LICENSED PROFESSIONAL ARCHITECT ELIZABETH HALLAS 07418 IOWA

| | | | | | |
|---|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM DRAWN: LB/MM TECH REVIEW: ACN / EMH DATE: 05/29/2026 | SUB SHEET NO. G 401 | TITLE OF SHEET FURNITURE PLANS EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 7 OF 142 | |

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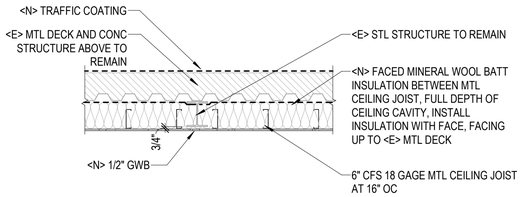
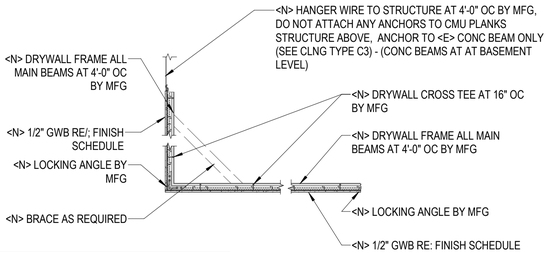
# CEILING TYPES ASSEMBLIES

**CEILING**

**C1**

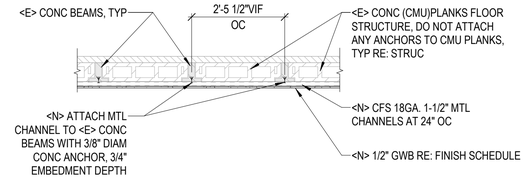



**C2**

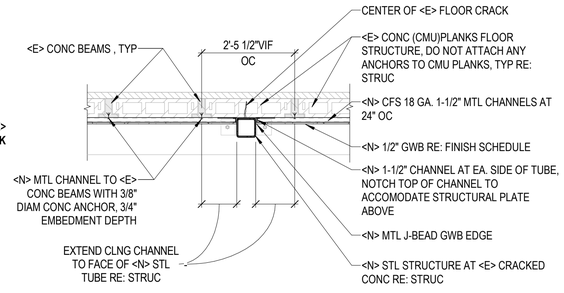



NOTE: STRUC TEE CEILING FRAMING SYSTEM BOD: ARMSTRONG SHORTSPAN DRYWALL FRAMING SYSTEM

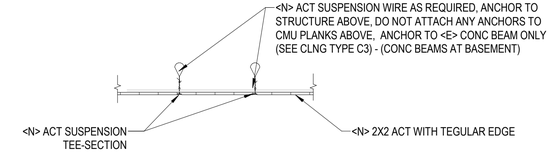
**C3**




**C3.1**
**REPAIR AT <E> FLOOR CRACK**




**C4**




# WALL TYPES ASSEMBLIES

**EXTERIOR WALLS**

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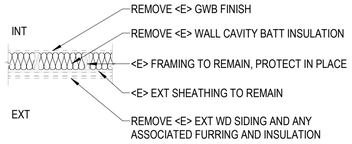


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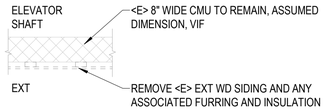


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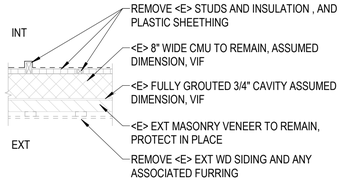


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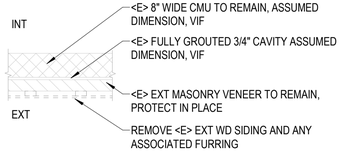


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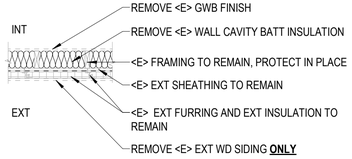


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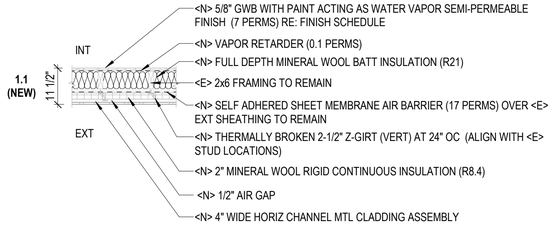
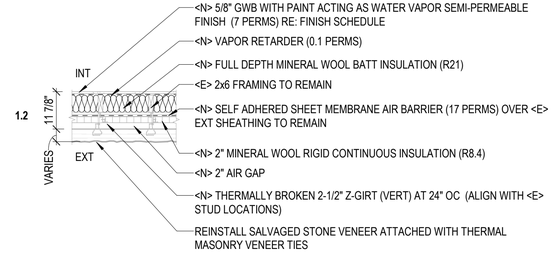



**1.2**

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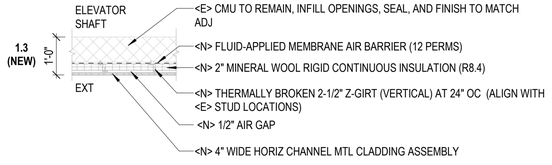


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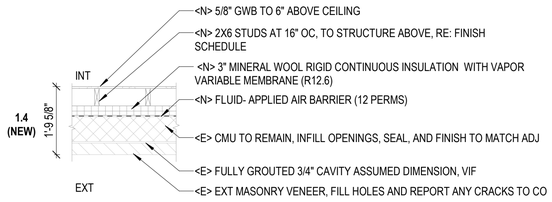
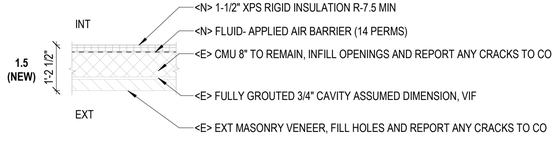
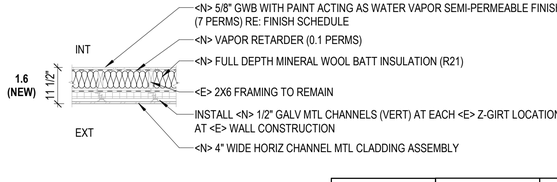



**1.5 (NEW)**




**1.6 (NEW)**



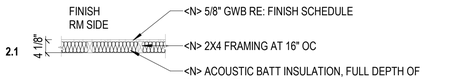

# OPTION NOTES

OPTIONS ON THIS SHEET ARE:
OPTION B - EXTERIOR SCOPE, FROM EXISTING BUILDING SHEATHING OUT PLUS REVISIONS TO FENESTRATION.
OPTION C - INTERIOR SCOPE, FROM EXISTING BUILDING SHEATHING IN.
RE: G001 FOR MORE INFORMATION.

**INTERIOR WALLS**

**NOTE:**
AT PARTITION WALLS IN OFFICES, EXTEND STUDS TO STRUCTURE ABOVE AND BRACE AS REQUIRED, EXTEND GWB FINISH TO 6" MIN ABOVE ACT SYSTEM.

**2.1**

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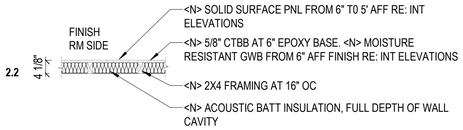


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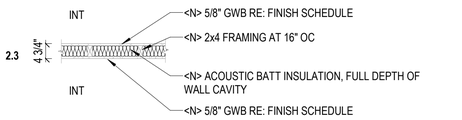


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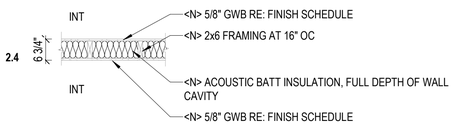


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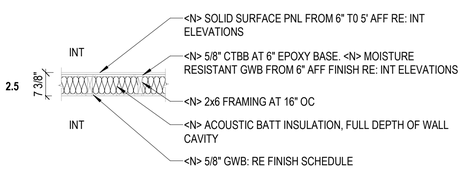
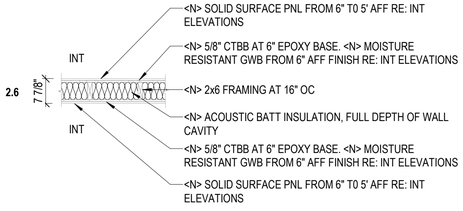



**2.6**

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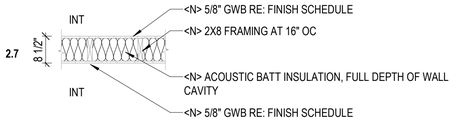


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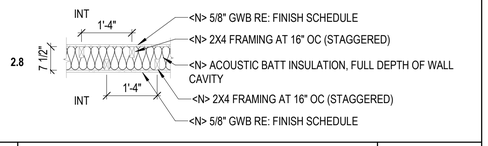



7/7/2026 1:23:12 PM Autodesk Docs://2025-110 NPS EFMO Visitor Center DD-CD/2024250 NPS EFMO Visitor Center - v24_BIM360.rvt

LICENSED PROFESSIONAL ARCHITECT
ELIZABETH HALLAS
07418
IOWA

| | | | |
|---|---|---|---|
| A/E FIRM: | DESIGNED: LB/CM | SUB SHEET NO. | TITLE OF SHEET |
| PRIME: OTAK, INC. REDMOND, WA | CADD LB/MM | G 500 | ASSEMBLIES |
| SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | TECH REVIEW: ACN / EMH | | EFFIGY MOUNDS NATIONAL MONUMENT |
| | DATE: 05/29/2026 | | DRAWING NO. 394 194248 / PMIS/PKG NO. 318539 / SHEET 8 OF 142 |

# LEGEND

| EXISTING | DESCRIPTION | PROPOSED |
|---|---|---|
| △ | SURVEY CONTROL POINT | |
| ◯ | SEWER MANHOLE | |
| ▥ | CATCH BASIN | |
| RD | ROOF DRAIN | |
| ∽ | LINE CONTINUES | |
| H | FIRE HYDRANT | |
| ⋈ | WATER VALVE | |
| ⌓ | AUTO SPRINKLER | |
| ⊸ | UTILITY POLE | |
| ⟵ | GUY ANCHOR | |
| Ⓜ | ELECTRIC METER | |
| Ⓙ | ELECTRIC JUNCTION BOX | |
| Ⓜ | GAS METER | |
| Ⓙ | COMMUNICATION JUNCTION BOX | |
| ♿ | HANDICAPPED PARKING | |
| FP | FLAG POLE | |
| ⊓ | SIGN | |
| • | BOLLARD | |
| ⌓ | AUTO SPRINKLER | |
| ❀ | DECIDUOUS TREE | |
| ✳ | EVERGREEN TREE | |
| × 680.55 | SPOT ELEVATION | |
| ⁘ | CONCRETE SURFACE | |
| ▭ | BITUMINOUS SURFACE | |
| ▨ | RETAINING WALLS | |
| ▩ | LANDSCAPE WOODCHIPS | |
| ▦ | PAVER SURFACE | |

# LEGEND (CONTINUED)

| EXISTING | DESCRIPTION | PROPOSED |
|---|---|---|
| W | UNDERGROUND WATER LINE | W |
| UE | UNDERGROUND ELECTRIC LINE | |
| UT | UNDERGROUND PHONE LINE | |
| S S | UNDERGROUND SEWER LINE | |
| OHL | OVERHEAD LINE | |
| | CULVERT (VARIOUS SIZES) | |
| | BRUSH LINE | |
| 50 | MAJOR (5') CONTOUR LINE | 51 |
| 51 | MINOR (1') CONTOUR LINE | 50 |
| | CEMENT CONCRETE PAVEMENT REMOVAL | |
| | ASPHALT PAVEMENT REMOVAL | |
| | SAWCUT | |
| | LIMITS OF CONSTRUCTION | |
| | SALVAGE STONE WALLS | |
| | REMOVE WATER LINE | |
| | CEMENT CONCRETE WALKWAY | |
| | STRAW WATTLE | |
| | PERFORATED, CORRUGATED FOUNDATION DRAIN | SD |
| | CORRUGATED PVC PIPE | SD |
| | GRADING LIMITS | |
| | DRAINAGE SWALE | |
| | DETECTABLE WARNING SURFACE | |
| | STONE SEAT WALLS | |
| | NEW PLANTER | |
| | CRUSHED ROCK STRIP | |
| | FREE STANDING BENCH | |

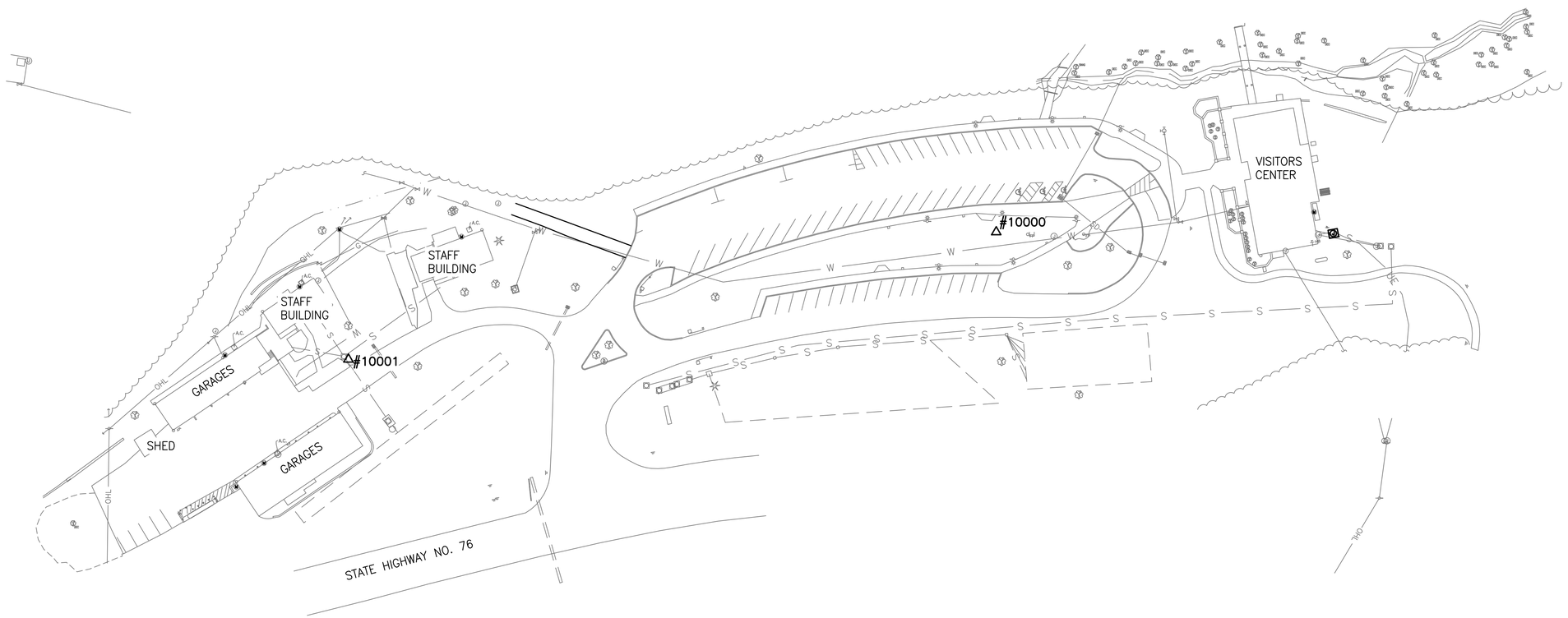
# ABBREVIATIONS

| | | |
|---|---|---|
| ABA | = | ARCHITECTURAL BARRIERS ACT |
| ABAAS | = | ARCHITECTURAL BARRIERS ACT ACCESSIBILITY STANDARD |
| AC | = | ASPHALT CONCRETE |
| AP | = | ANGLE POINT |
| ASPH | = | ASPHALT |
| BOW/BW | = | BOTTOM OF WALL |
| BTM | = | BOTTOM |
| BVCS | = | BEGIN VERTICAL CURVE STATION |
| BVCE | = | BEGIN VERTICAL CURVE ELEVATION |
| CIP | = | CAST–IN–PLACE |
| CLK | = | CHAIN LINK |
| CLIN | = | CONTRACT LINE ITEM NUMBER |
| CMP | = | CORRUGATED METAL PIPE |
| CO | = | CONTRACTING OFFICER |
| CONC | = | CONCRETE |
| CP | = | CONTROL POINT |
| CPP | = | CORRUGATED POLYETHYLENE PIPE |
| DEC | = | DECIDUOUS |
| DI | = | DUCTILE IRON |
| DIA | = | DIAMETER |
| E | = | EASTING |
| EOP | = | EDGE OF PAVEMENT |
| EVCS | = | END VERTICAL CURVE STATION |
| EVCE | = | END VERTICAL CURVE ELEVATION |
| EVER | = | EVERGREEN |
| ELEV | = | ELEVATION |
| EX/EXIST | = | EXISTING |
| FFE | = | FINISHED FLOOR ELEVATION |
| FG | = | FINISHED GRADE |
| FL | = | FLOW LINE |
| FOC | = | FACE OF CURB |
| GB | = | GRADE BREAK |
| HMA | = | HOT MIX ASPHALT |
| HORIZ | = | HORIZONTAL |
| IE | = | INVERT ELEVATION |
| INFO | = | INFORMATION |
| ISA | = | INTERNATIONAL SYMBOL OF ACCESSIBILITY |
| LF | = | LINEAR FEET |
| LVC | = | LENGTH OF VERTICAL CURVE |
| OC | = | ON CENTER |
| N | = | NORTHING |
| NAVD | = | NORTH AMERICAN VERTICAL DATUM |
| NIC | = | NOT IN CONTRACT |
| PC | = | POINT OF CURVATURE |
| PL | = | PARKING LOT |
| PT | = | POINT OF TANGENCY |
| PT | = | POINT |
| PT | = | PRESSURE TREATED |
| PVC | = | POLYVINYL CHLORIDE |
| PVI | = | POINT OF VERTICAL INTERSECTION |
| R | = | RADIUS |
| RCP | = | REINFORCED CONCRETE PIPE |
| RP | = | RADIUS POINT |
| REBAR | = | STEEL REINFORCING ROD |
| S | = | SLOPE |
| SD | = | STORM DRAIN |
| SSCO | = | SANITARY SEWER CLEANOUT |
| SSMH | = | SANITARY SEWER MANHOLE |
| STA | = | STATION |
| STD | = | STANDARD |
| TBC | = | TOP BACK OF CURB |
| TESC | = | TEMPORARY EROSION & SEDIMENT CONTROL |
| T.O.C. | = | TOP OF CONCRETE |
| TBD | = | TO BE DETERMINED |
| TW/T.O.W. | = | TOP OF WALL |
| TYP. | = | TYPICAL |
| UON | = | UNLESS OTHERWISE NOTED |
| VERT | = | VERTICAL |
| VCP | = | VITRIFIED CLAY PIPE |

LICENSED PROFESSIONAL ENGINEER
EMILY A. MAGGIO
P30462
IOWA

| | | | | |
|---|---|---|---|---|
| AE FIRM<br>PRIME:<br>OTAK INC.<br>REDMOND, WA<br><br>SUBCONTRACTOR:<br>ANDERSON HALLAS ARCHITECTS<br>GOLDEN, CO | DESIGNED: EM, LM, CS<br>CADD: EM, CS, ZS<br>TECH REVIEW: EM, KK, LM<br>DATE: 5/29/2026 | SUB SHEET NO.<br>GC000 | TITLE OF SHEET<br>CIVIL ABBREVIATIONS AND LEGEND<br><br>EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 / 194248<br>PMIS/PKG NO. 318539<br>SHEET 9 OF 142 |

Plotted: May 22, 2026 - 2:50pm lindsey.martin O:\PROJECT\33200\332784E\04 CAD\xCAD\EFMO\318539\DRAWINGS\CIVIL\CIVIL ABBREVIATIONS AND LEGEND.dwg Layout Name: GC000




## NOTES

A) THIS SURVEY MADE TIES TO "NEW" FEATURES AND CONDITIONS WITHIN THE LIMITS OF THE PRIOR TOPOGRAPHIC SURVEY. THESE AREAS ARE OUTSIDE OF THE ADDITIONAL SURVEY SCOPE AND WERE TIED AND NOTED FOR REFERENCE ONLY.

B) THE TREE INFORMATION SHOWN HEREON WAS COLLECTED DURING THE FIELD SURVEY BY NON-FORESTRY TRAINED ANDERSON ENGINEERING OF MINNESOTA SURVEY PERSONNEL. TREE SIZES ARE ESTIMATES AND LOCATIONS ARE ACCURATE TO PLUS OR MINUS THREE FEET.

C) NO TITLE WORK WAS PROVIDED FOR THE PREPARATION OF THIS SURVEY TO VERIFY THE LEGAL DESCRIPTION OR THE EXISTENCE OF ANY EASEMENTS OR ENCUMBRANCES.

D) THE LOCATION AND EXTENT OF UNDERGROUND UTILITIES, IF SHOWN, ARE BASED UPON EXISTING DRAWINGS PROVIDED BY THE UTILITY COMPANIES, ABOVE GROUND EVIDENCE AND IOWA ONE CALL MARKINGS. THERE IS NO GUARANTEE AS TO THE ACCURACY OR THE COMPLETENESS OF THIS INFORMATION. THE SIZE AND LOCATION SHOULD BE CONSIDERED APPROXIMATE. ADDITIONAL UNDERGROUND UTILITIES MAY BE PRESENT. VERIFICATION OF THE EXISTENCE AND LOCATION OF ALL UTILITIES SHOULD BE OBTAINED FROM THE UTILITY OWNERS PRIOR TO ANY PLANNING OR DESIGN. IN ACCORDANCE WITH IOWA STATUTE, THE LOCATION OF UTILITIES SHALL BE CONFIRMED PRIOR TO ANY DEMOLITION OR CONSTRUCTION.

THIS DRAWING ONLY REFLECTS AND VERIFIES THE FILED SURVEY DATA OF THOSE FEATURES AND CONDITIONS PRESENT WITHIN THE ORIGINAL SURVEY LIMITS AS OF DEC 12, 2021 AND WITHIN THE ADDITIONAL SURVEY LIMITS AS OF JULY 29, 2024

HOR. DATUM: IOWA STATE PLANE NORTH ZONE NAD83(2011)(EPOCH 2010.00).

VERT. DATUM: NAVD 88

## CONTROL

CONTROL POINTS USED:

POINT NO. 10000
N: 3,868,265.08
E: 5,539,212.14
ELEV: 660.86
DESC: 1/2" IRON ROD WITH ORANGE CAP STAMPED "CONTROL"

POINT NO. 10001
N: 3,868,162.81
E: 5,538,692.17
ELEV: 679.200
DESC: 1/2" IRON ROD WITH ORANGE CAP STAMPED "CONTROL"

50' 0' 50' 100'
SCALE OF FEET

EMILY A. MAGGIO
LICENSED PROFESSIONAL ENGINEER
P30462
IOWA

AE FIRM

PRIME:
OTAK INC.
REDMOND, WA

SUBCONTRACTOR:
ANDERSON HALLAS ARCHITECTS
GOLDEN, CO

DESIGNED: EM, LM, CS
DRAWN: EM, CS, ZS
TECH REVIEW: EM, KK, LM
DATE: 5/29/2026

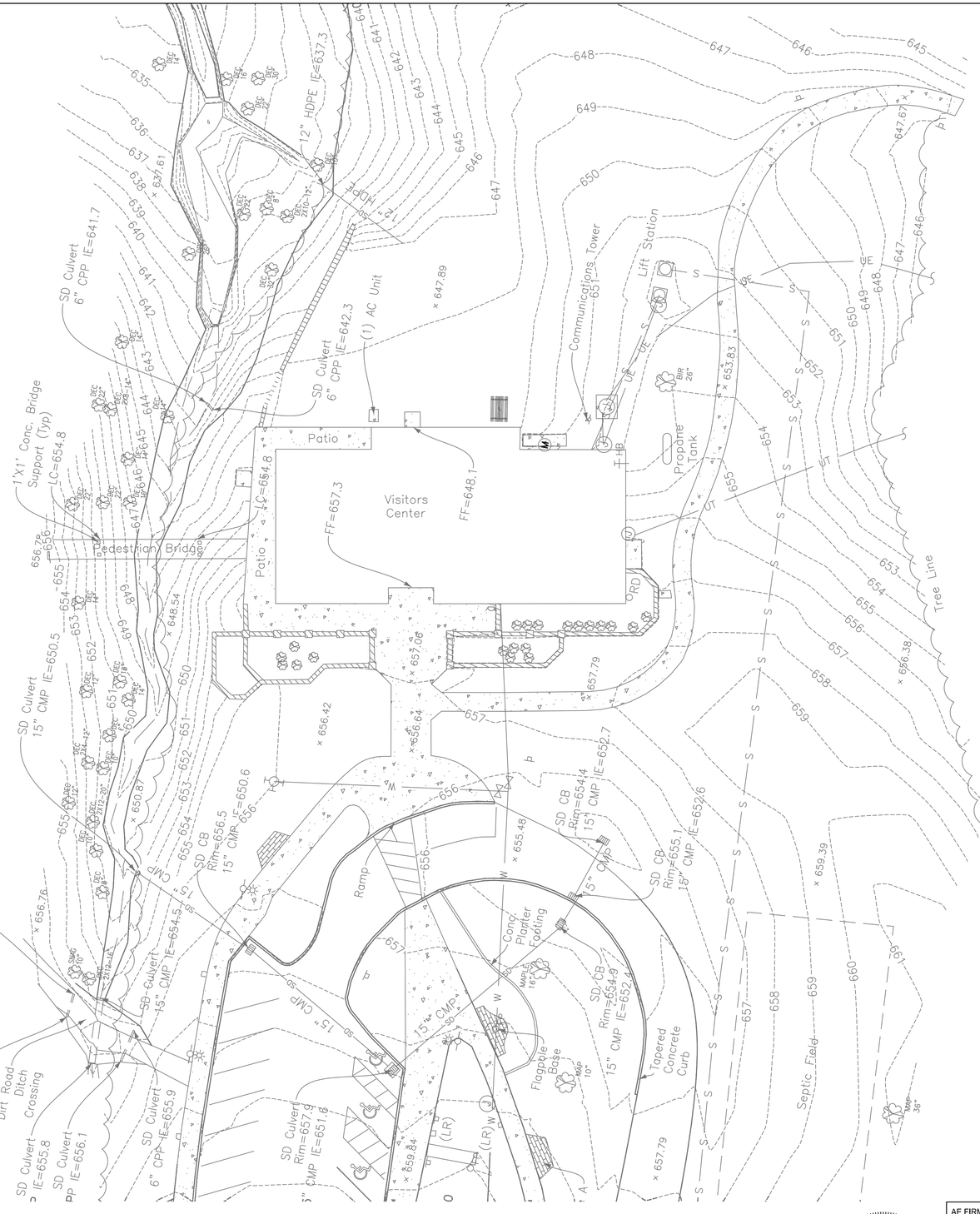
SUB SHEET NO. GC100

TITLE OF SHEET
SURVEY CONTROL PLAN

EFFIGY MOUNDS
NATIONAL MONUMENT

DRAWING NO. 394 194248
PMIS/PKG NO. 318539
SHEET 10 OF 142

Plotted: May 22, 2026 - 2:59pm lindsay.martin O:\PROJECT\332000\332784E\04 CAD\ACAD\ETMP\318539\DRAWINGS\CIVIL\SURVEY CNTRL.dwg Layout Name: GC100


Visitors Center
Patio
Patio
FF=657.3
FF=648.1
LC=654.8
LC=654.8
Pedestrian Bridge
1'X1' Conc. Bridge Support (Typ)
(1) AC Unit
Communications Tower
Lift Station
Propane Tank
Tree Line
OHL
Ramp
Conc. Planter Footing
Flagpole Base
Tapered Concrete Curb
Septic Field
SD Culvert 6" CPP IE=641.7
SD Culvert 6" CPP IE=642.3
12" HDPE IE=637.3
SD Culvert 15" CMP IE=650.5
SD Culvert 6" CPP IE=656.2
Dirt Road Ditch Crossing
SD Culvert IE=655.8
SD Culvert IE=656.1
SD Culvert 6" CPP IE=655.9
SD Culvert 6" CPP IE=655.7
SD Culvert Rim=657.9 CMP IE=651.6
SD CB Rim=656.5 15" CMP=656.6
SD CB Rim=654.14 15" CMP IE=652.7
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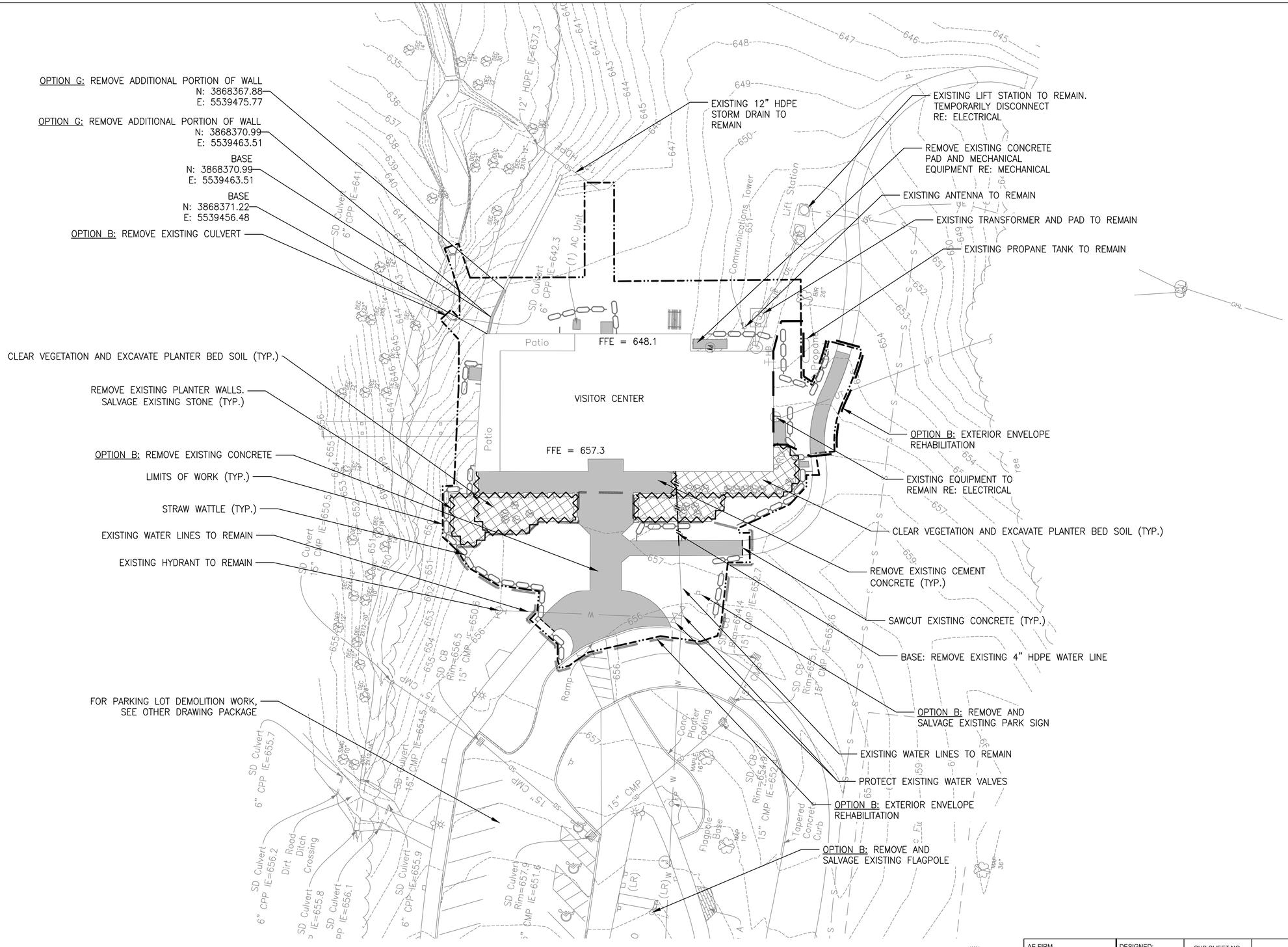
SD CB Rim=654.9 15" CMP IE=652.4
15" CMP
15" CMP
15" CMP




## LEGEND

- REMOVE EXISTING CONCRETE
- REMOVE EXISTING PLANTER BED
- SALVAGE STONE WALLS
- SAWCUT EXISTING CONCRETE
- WATTLE
- LIMITS OF WORK
- REMOVE EXISTING UTILITY LINE

## TESC/DEMOLITION NOTES

1. CONTRACTOR SHALL BE RESPONSIBLE FOR MAINTAINING EXISTING DRAINAGE THROUGH THE SITE DURING CONSTRUCTION.
2. PROTECT EXISTING UTILITIES, SIGNAGE, AND FEATURES IN PLACE NOT CALLED OUT FOR REMOVAL OR RELOCATION.
3. CONTRACTOR SHALL PROVIDE EROSION CONTROL MAINTENANCE DURING CONSTRUCTION AND SHALL REMOVE EROSION CONTROL MEASURES AFTER CONSTRUCTION.
4. CONTRACTOR IS RESPONSIBLE FOR VERIFYING LOCATION OF UNDERGROUND UTILITIES.
5. TESC MEASURES SHOWN ARE APPROXIMATE AND SHALL BE FIELD LOCATED AND APPROVED TO ACCOMMODATE SITE CONDITIONS AND WORK SCHEDULE.
6. REMOVE AND SALVAGE ALL EXTERIOR AMENITIES SUCH AS BENCHES, BIKE RACKS, STANCHION, PICNIC TABLES, AND TRASH RECEPTACLES WITHIN LIMITS OF WORK.

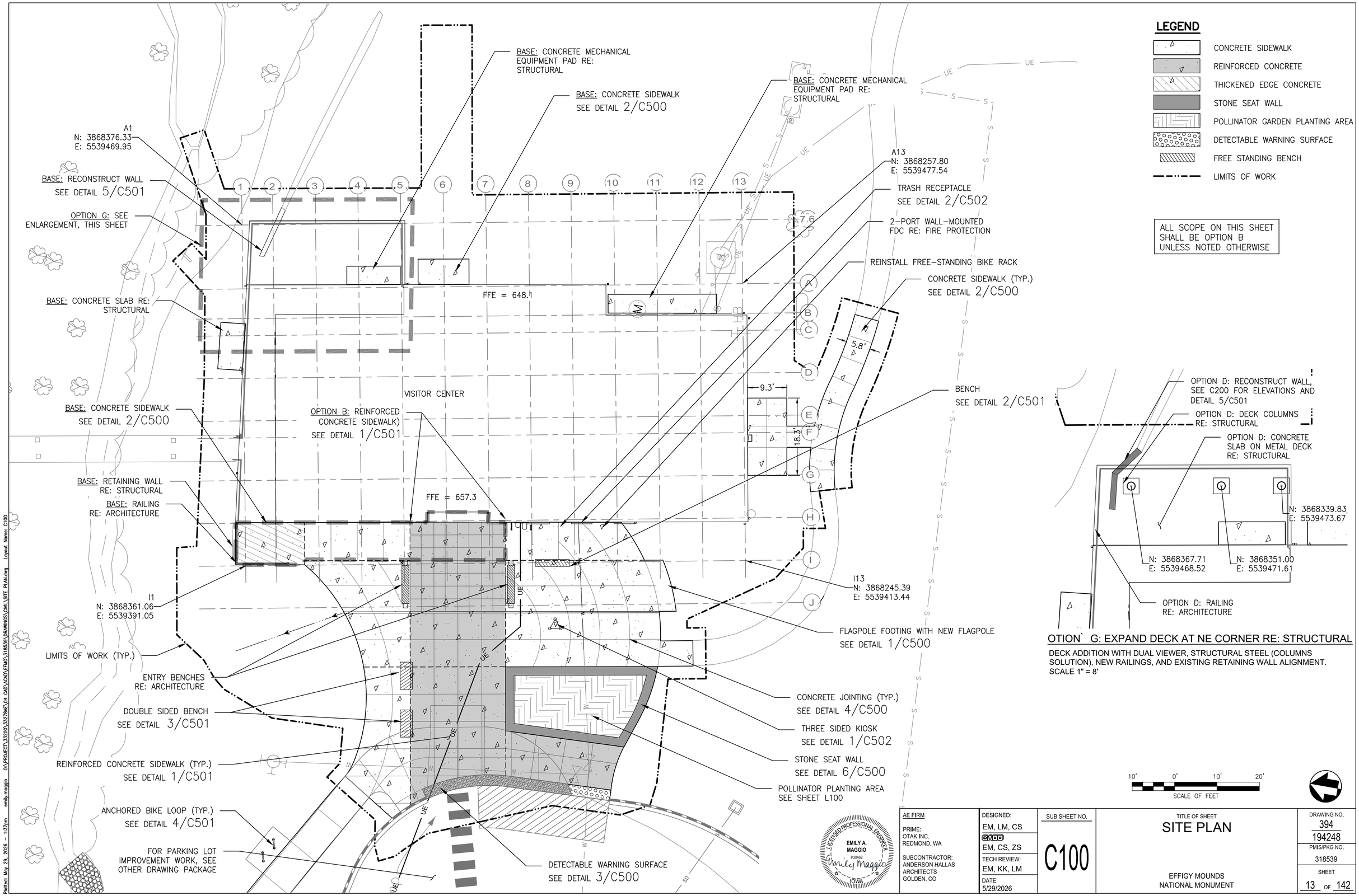
ALL SCOPE ON THIS SHEET SHALL BE BASE SCOPE UNLESS NOTED OTHERWISE

20' 0' 20' 40'
SCALE OF FEET

Plotted: May. 29, 2026 - 1:33pm emily.maggio O:\PROJECT\33200\33278AE\04 CAD\ACAD\EFMO\318539\DRAWINGS\CIVIL\DEMO.dwg Layout Name: C200

EMILY A. MAGGIO
P30462
IOWA
LICENSED PROFESSIONAL ENGINEER

| | | | | |
|---|---|---|---|---|
| AE FIRM<br>PRIME:<br>OTAK INC.<br>REDMOND, WA<br><br>SUBCONTRACTOR:<br>ANDERSON HALLAS ARCHITECTS<br>GOLDEN, CO | DESIGNED:<br>EM, LM, CS<br>DRAWN:<br>EM, CS, ZS<br>TECH REVIEW:<br>EM, KK, LM<br>DATE:<br>5/29/2026 | SUB SHEET NO.<br>CD100 | TITLE OF SHEET<br>TESC & DEMOLITION PLAN<br><br>EFFIGY MOUNDS<br>NATIONAL MONUMENT | DRAWING NO.<br>394<br>194248<br>PMIS/PKG NO.<br>318539<br>SHEET<br>12 OF 142 |




Plotted: May 29, 2026 - 1:37pm emily.maggio O:\PROJECT\33200\332784\04 CAD\ACAD\CIVIL\185391(DRAWINGS)\CIVIL\SITE PLAN.dwg Layout Name: C100

EMILY A. MAGGIO P30462 LICENSED PROFESSIONAL ENGINEER IOWA

| | | | | |
|---|---|---|---|---|
| AE FIRM<br>PRIME:<br>OTAK INC.<br>REDMOND, WA<br><br>SUBCONTRACTOR:<br>ANDERSON HALLAS ARCHITECTS<br>GOLDEN, CO | DESIGNED:<br>EM, LM, CS<br>CADD<br>EM, CS, ZS<br>TECH REVIEW:<br>EM, KK, LM<br>DATE:<br>5/29/2026 | SUB SHEET NO.<br>C100 | TITLE OF SHEET<br>SITE PLAN<br><br>EFFIGY MOUNDS<br>NATIONAL MONUMENT | DRAWING NO.<br>394<br>194248<br>PMIS/PKG NO.<br>318539<br>SHEET<br>13 OF 142 |

BASE:
OUTFLOW
IE=634.51
N: 3868390.27
E: 5539486.32

BASE: 6" DIAMETER PVC AT 11.2%, SEE PROFILE THIS SHEET

BASE: GRADE AT 3:1 MAX

BASE: CONNECT TO 6" DIA. PVC PIPE IE=638.16

BASE SCOPE

CONNECT FOUNDATION DRAIN TO WALL FOUNDATION DRAIN AT 646.84'

RETAINING WALL WITH SALVAGE STONE VENEER, MAX 4' HIGH RE: STRUCTURAL

WALL FOUNDATION DRAIN AT 0.5%

DRAIN PIPE AT 2.0%

SWALE AT 7.8%

GRADING LIMITS (TYP.)

OPTION G: RECONSTRUCT WALL

BASE SCOPE

FFE = 648.1

VISITOR CENTER

BASE SCOPE

FFE = 657.3

BASE SCOPE

BASE: 4" PERFORATED CORRUGATED FOUNDATION DRAIN AT 0.5% MIN.
HIGHPOINT = 647.33
N: 3868253.34
E: 5539425.07

BEGIN TRENCH DRAIN
IE = 656.82
N: 3868263.22
E: 5539397.83

4" TRENCH DRAIN

ROOF DRAIN PLAN AND PROFILE SEE SHEET C202

ACCESSIBLE ROUTE (TYP.)

END DRAIN PIPE
IE = 655.01
N: 3868359.74
E: 5539370.87

BEGIN DRAIN PIPE
IE = 656.07
N: 3868336.25
E: 5539384.38

FOR CONTINUATION OF ACCESSIBLE ROUTE, SEE OTHER DRAWING PACKAGE

1.5% (TYP.)

1.5%

1.0%

EX. 4.2%

EX. 1.2%

1.2%

0.4%

0.5%

0.4%

1.0%

1.5%

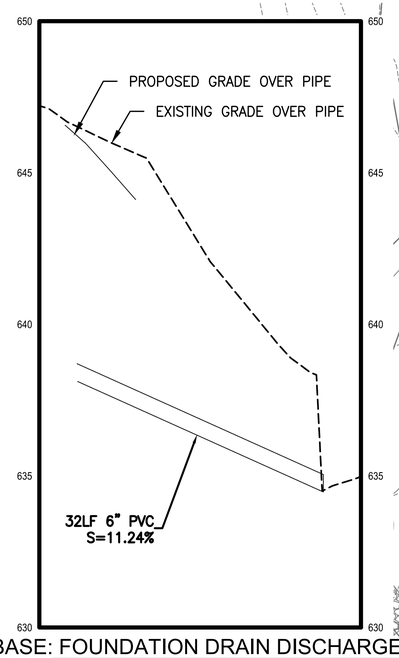
## LEGEND

- DRAINAGE SWALE
- RD — ROOF DRAIN
- FD — CORRUGATED PVC FOUNDATION DRAIN
- ACCESSIBLE ROUTE
- GRADING LIMITS
- FLOW DIRECTION
- AXX — LAYOUT CONTROL POINT

## GRADING NOTES

1. LANDINGS, ACCESSIBLE PARKING STALLS, AND PARKING ACCESS AISLES SHALL NOT EXCEED 1.5% SLOPE IN ANY DIRECTION.
2. ABA WALKWAYS SHALL NOT EXCEED A 4.5% MAXIMUM RUNNING SLOPE AND A 1.5% MAXIMUM CROSS SLOPE.
3. WHERE EXCAVATION OCCURS FOR FOUNDATION DRAINS AND ROOF DRAINS, BACKFILLED AREAS SHALL BE GRADED TO MATCH EXISTING EXISTING SURROUNDING GRADES. ALL EXISTING DRAINAGE PATTERNS SHALL BE MAINTAINED AND DIRECTED AWAY FROM BUILDINGS.
4. SEE SHEET C202 FOR ROOF DRAIN PLAN AND PROFILE.

ALL SCOPE ON THIS SHEET SHALL BE OPTION B UNLESS NOTED OTHERWISE




BASE: FOUNDATION DRAIN DISCHARGE

10' 0' 10' 20'
SCALE OF FEET

LICENSED PROFESSIONAL ENGINEER
EMILY A. MAGGIO
P30462
IOWA

| | | | |
|---|---|---|---|
| AE FIRM<br>PRIME:<br>OTAK INC.<br>REDMOND, WA<br><br>SUBCONTRACTOR:<br>ANDERSON HALLAS ARCHITECTS<br>GOLDEN, CO | DESIGNED:<br>EM, LM, CS<br>CADD<br>EM, CS, ZS<br>TECH REVIEW:<br>EM, KK, LM<br>DATE:<br>5/29/2026 | SUB SHEET NO.<br>C200 | TITLE OF SHEET<br>GRADING AND DRAINAGE PLAN<br><br>EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO.<br>394<br>194248<br>PMIS/PKG NO.<br>318539<br>SHEET<br>14 OF 142 |

Plotted: Jul 09, 2026 - 2:35pm Zach.Schneider O:\PROJECT\13200\13278\E\04 CAD\ACAD\CFM01\318539\DRAWINGS\CIVIL\GRADING AND DRAINAGE.dwg Layout Name: C200

| LAYOUT POINTS | | | | |
|---|---|---|---|---|
| POINT | NORTHING | EASTING | DESCRIPTION | ELEV |
| A1 | 3868377.12 | 5539446.50 | EDGE OF CONC PAD – BASE | 648.00’ |
| A2 | 3868375.88 | 5539435.57 | EDGE OF CONC PAD – BASE | 648.16’ |
| A3 | 3868372.72 | 5539456.27 | GRADING POINT – BASE | 646.89’ |
| A4 | 3868371.67 | 5539470.40 | BOTTOM OF WALL – OPTION G | 645.06’ |
| A5 | 3868371.32 | 5539447.16 | EDGE OF CONC PAD – BASE | 648.06’ |
| A6 | 3868370.99 | 5539463.51 | BOTTOM OF WALL – OPTION G | 646.70’ |
| A7 | 3868370.63 | 5539470.12 | TOP OF WALL – OPTION G | 648.00’ |
| A8 | 3868370.08 | 5539436.23 | EDGE OF CONC PAD – BASE | 648.22’ |
| A9 | 3868369.88 | 5539463.62 | TOP OF WALL – OPTION G | 648.00’ |
| A10 | 3868366.91 | 5539475.13 | TOP OF WALL – OPTION G | 648.00’ |
| A11 | 3868365.24 | 5539400.64 | EDGE OF SIDEWALK – BASE SCOPE LIMITS | 657.16’ |
| A12 | 3868363.46 | 5539390.87 | EDGE OF SIDEWALK – BASE SCOPE LIMITS | 656.98’ |
| A13 | 3868355.93 | 5539402.35 | EDGE OF SIDEWALK – BASE | 657.29’ |
| A14 | 3868352.09 | 5539392.96 | EDGE OF SIDEWALK – BASE | 657.16’ |
| A15 | 3868350.29 | 5539464.83 | EDGE OF CONC PAD – BASE | 648.55’ |
| A16 | 3868349.51 | 5539460.57 | EDGE OF CONC PAD – BASE | 648.10’ |
| A17 | 3868349.24 | 5539403.59 | EDGE OF SIDEWALK – BASE | 657.29’ |
| A18 | 3868347.43 | 5539393.78 | EDGE OF SIDEWALK – BASE | 657.19’ |
| A19 | 3868337.92 | 5539467.11 | EDGE OF CONC PAD – BASE | 648.55’ |
| A20 | 3868337.14 | 5539462.87 | EDGE OF CONC PAD – BASE | 648.10’ |
| A21 | 3868329.06 | 5539346.46 | EDGE OF SIDEWALK | 657.64’ |
| A22 | 3868322.64 | 5539342.52 | EDGE OF SIDEWALK | 657.71’ |
| A23 | 3868320.75 | 5539408.84 | EDGE OF SIDEWALK – BASE | 657.30’ |
| A24 | 3868315.99 | 5539338.44 | EDGE OF SIDEWALK | 657.79’ |
| A25 | 3868310.63 | 5539346.70 | WARNING SURFACE | 657.54’ |
| A26 | 3868309.20 | 5539344.65 | WARNING SURFACE | 657.58’ |
| A27 | 3868306.52 | 5539411.46 | EDGE OF SIDEWALK – BASE | 657.30’ |
| A28 | 3868297.91 | 5539349.84 | WARNING SURFACE | 657.34’ |
| A29 | 3868296.50 | 5539378.62 | BOTTOM OF WALL | 657.16’ |
| A30 | 3868293.77 | 5539363.78 | BOTTOM OF WALL | 657.37’ |

| POINT | NORTHING | EASTING | DESCRIPTION | ELEV |
|---|---|---|---|---|
| A31 | 3868289.68 | 5539471.32 | GRADING POINT – BASE | 648.97’ |
| A32 | 3868288.62 | 5539464.37 | GRADING POINT – BASE | 649.74’ |
| A33 | 3868288.53 | 5539469.36 | EDGE OF CONC PAD – BASE | 650.90’ |
| A34 | 3868287.73 | 5539465.02 | EDGE OF CONC PAD – BASE | 650.94’ |
| A35 | 3868275.84 | 5539417.04 | EDGE OF SIDEWALK | 657.30’ |
| A36 | 3868278.58 | 5539471.07 | EDGE OF CONC PAD – BASE | 650.95’ |
| A37 | 3868266.87 | 5539354.48 | WARNING SURFACE | 657.78’ |
| A38 | 3868266.76 | 5539351.99 | WARNING SURFACE | 657.81’ |
| A39 | 3868278.17 | 5539471.15 | EDGE OF CONC PAD – BASE | 652.26’ |
| A40 | 3868265.69 | 5539365.04 | BOTTOM OF WALL | 657.65’ |
| A41 | 3868268.84 | 5539409.23 | EDGE OF SIDEWALK | 657.18’ |
| A42 | 3868263.37 | 5539474.00 | EDGE OF CONC PAD – BASE | 652.26’ |
| A43 | 3868263.27 | 5539398.00 | EDGE OF SIDEWALK | 657.14’ |
| A44 | 3868262.55 | 5539469.57 | EDGE OF CONC PAD – BASE | 652.31’ |
| A45 | 3868261.70 | 5539391.18 | EDGE OF SIDEWALK | 657.25’ |
| A46 | 3868261.26 | 5539385.13 | BOTTOM OF WALL | 657.34’ |
| A47 | 3868264.56 | 5539410.03 | EDGE OF SIDEWALK | 657.24’ |
| A48 | 3868259.31 | 5539398.73 | EDGE OF SIDEWALK | 657.18’ |
| A49 | 3868257.16 | 5539352.67 | GRADING POINT | 658.30’ |
| A50 | 3868255.43 | 5539470.46 | EDGE OF BLDG | 653.10’ |
| A51 | 3868251.82 | 5539451.20 | EDGE OF SIDEWALK | 657.30’ |
| A52 | 3868248.45 | 5539433.26 | EDGE OF SIDEWALK | 657.30’ |
| A53 | 3868242.69 | 5539452.87 | EDGE OF SIDEWALK | 657.16’ |
| A54 | 3868241.49 | 5539446.36 | EDGE OF SIDEWALK | 657.16’ |
| A55 | 3868240.59 | 5539441.44 | EDGE OF SIDEWALK | 657.16’ |
| A56 | 3868239.40 | 5539434.92 | EDGE OF SIDEWALK | 657.16’ |
| A57 | 3868235.07 | 5539447.53 | EDGE OF SIDEWALK | 656.90’ |
| A58 | 3868235.00 | 5539442.46 | EDGE OF SIDEWALK | 656.95’ |
| A59 | 3868233.19 | 5539432.14 | EDGE OF SIDEWALK | 656.98’ |
| A60 | 3868230.43 | 5539475.13 | EDGE OF SIDEWALK | 654.86’ |

| POINT | NORTHING | EASTING | DESCRIPTION | ELEV |
|---|---|---|---|---|
| A61 | 3868227.62 | 5539433.76 | EDGE OF SIDEWALK | 657.01’ |
| A62 | 3868224.66 | 5539474.61 | EDGE OF SIDEWALK | 654.91’ |
| A63 | 3868229.21 | 5539442.85 | EDGE OF SIDEWALK | 656.91’ |
| A64 | 3868229.25 | 5539447.88 | EDGE OF SIDEWALK | 656.81’ |
| A65 | 3868227.80 | 5539457.04 | EDGE OF SIDEWALK | 656.07’ |
| A66 | 3868233.40 | 5539458.56 | EDGE OF SIDEWALK | 656.04’ |
| A67 | 3868249.97 | 5539441.41 | EDGE OF SIDEWALK | 657.30’ |
| A68 | 3868249.50 | 5539441.50 | EDGE OF SIDEWALK | 657.29’ |
| A69 | 3868249.68 | 5539442.49 | EDGE OF SIDEWALK | 657.29’ |
| A70 | 3868250.16 | 5539442.42 | EDGE OF SIDEWALK | 657.30’ |
| A71 | 3868332.99 | 5539463.59 | EDGE OF CONCRETE – BASE | 648.10’ |
| A72 | 3868334.08 | 5539469.49 | EDGE OF CONCRETE – BASE | 648.24’ |
| A73 | 3868322.28 | 5539471.66 | EDGE OF CONCRETE – BASE | 648.07’ |
| A74 | 3868321.19 | 5539465.76 | EDGE OF CONCRETE – BASE | 648.10’ |
| A75 | 3868327.74 | 5539349.99 | EDGE OF SIDEWALK | 657.58’ |
| A76 | 3868327.99 | 5539367.83 | EDGE OF SIDEWALK | 657.28’ |
| A77 | 3868336.25 | 5539384.38 | EDGE OF SIDEWALK–LOW POINT | 656.97’ |
| A78 | 3868321.42 | 5539387.11 | LOW POINT | 656.99’ |
| A79 | 3868298.87 | 5539391.26 | LOW POINT | 657.03’ |
| A80 | 3868301.05 | 5539403.30 | BASE SCOPE LIMITS | 657.19’ |
| A81 | 3868302.68 | 5539412.16 | BASE SCOPE LIMITS | 657.30’ |

ALL SCOPE ON THIS SHEET SHALL BE OPTION B UNLESS NOTED OTHERWISE

Plotted: May. 22, 2026 – 2:54pm lindsey.martin O:\PROJECT_13200_13270HE\04 CAD\ACAD\EFMO\318539\DRAWINGS\CIVIL\GRADING AND DRAINAGE.dwg Layout Name: C201

AE FIRM

PRIME:
OTAK INC.
REDMOND, WA

SUBCONTRACTOR:
ANDERSON HALLAS ARCHITECTS
GOLDEN, CO

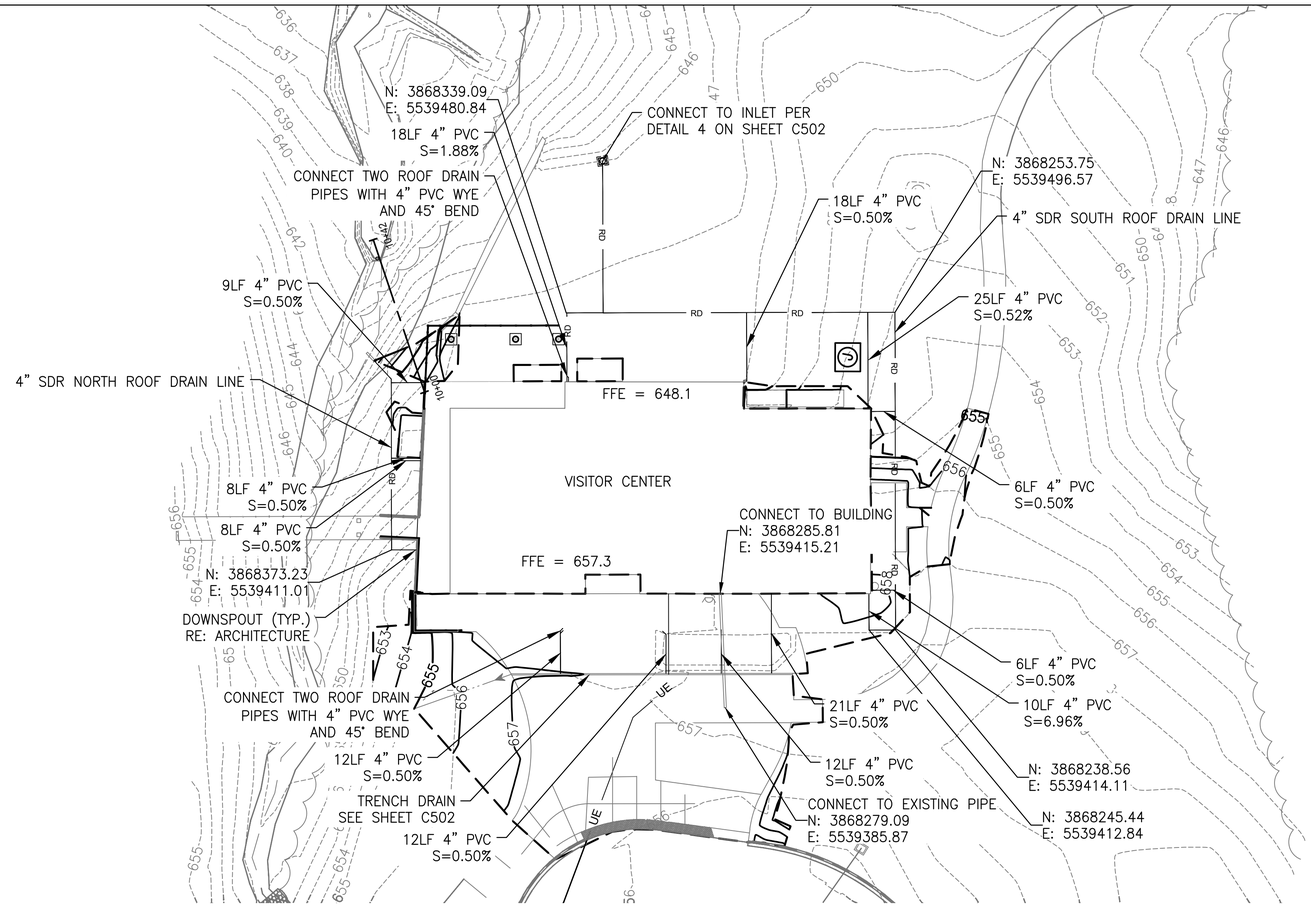
DESIGNED: EM, LM, CS
DRAWN: EM, CS, ZS
TECH REVIEW: EM, KK, LM
DATE: 5/29/2026

SUB SHEET NO. C201

TITLE OF SHEET
GRADING POINTS

EFFIGY MOUNDS
NATIONAL MONUMENT

DRAWING NO. 394 / 194248
PMIS/PKG NO. 318539
SHEET 15 OF 142



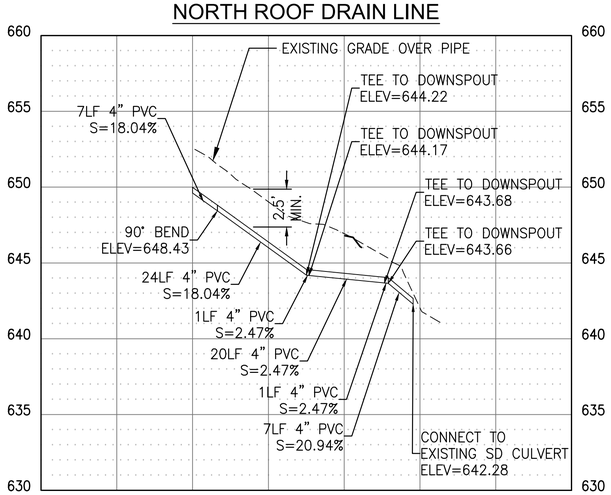

**LEGEND**

- RD — ROOF DRAIN
- GRADING LIMITS
- DOWNSPOUT

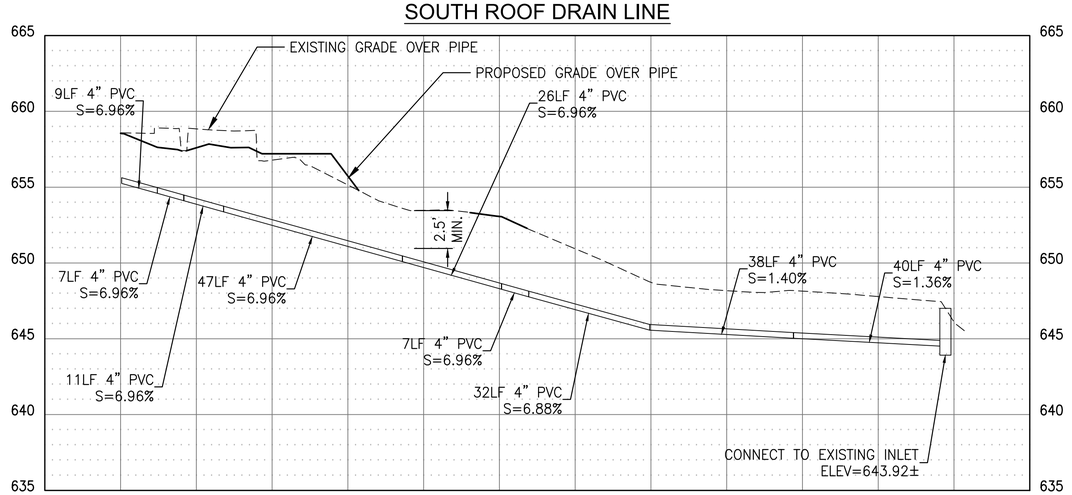
NOTE
SEE 7/C500 FOR DOWNSPOUT & ROOF DRAINAGE CONNECTION DETAIL

ALL SCOPE ON THIS SHEET SHALL BE OPTION B

## NORTH ROOF DRAIN LINE




## SOUTH ROOF DRAIN LINE

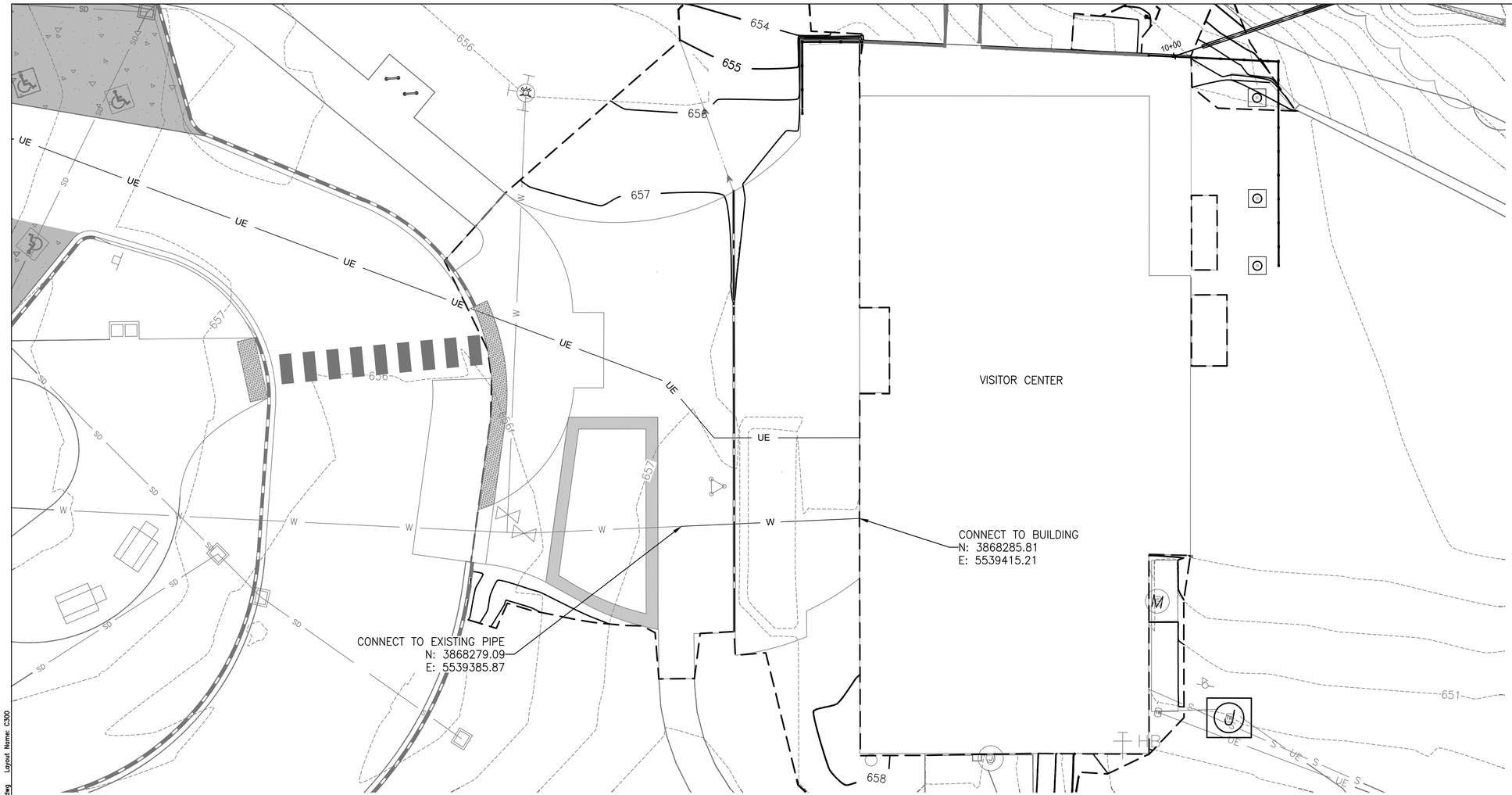


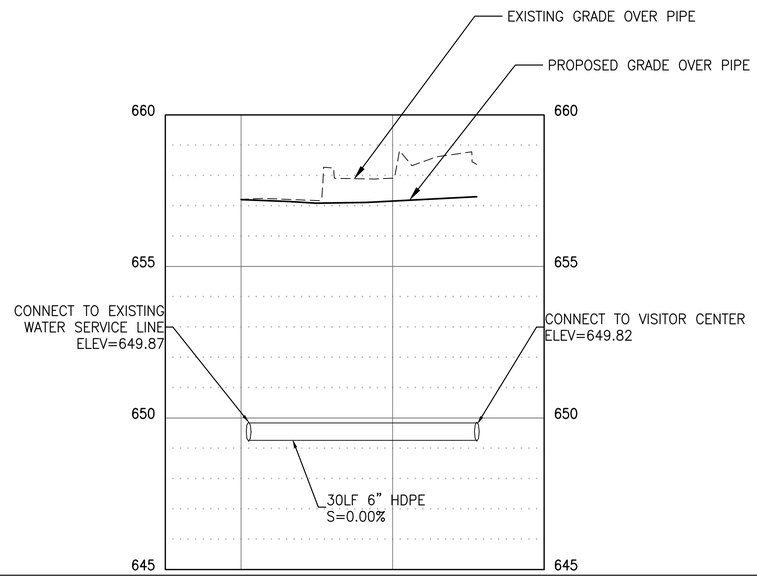

HORZ. 20' 0' 20' 40'
VERT. 5' 0' 5' 10'
SCALE OF FEET

EMILY A. MAGGIO — LICENSED PROFESSIONAL ENGINEER — P30462 — IOWA

| | | | | |
|---|---|---|---|---|
| AE FIRM<br>PRIME:<br>OTAK INC.<br>REDMOND, WA<br><br>SUBCONTRACTOR:<br>ANDERSON HALLAS ARCHITECTS<br>GOLDEN, CO | DESIGNED: EM, LM, CS<br>DRAWN: EM, CS, ZS<br>TECH REVIEW: EM, KK, LM<br>DATE: 5/29/2026 | SUB SHEET NO.<br>C202 | TITLE OF SHEET<br>ROOF DRAIN PLAN AND PROFILE<br><br>EFFIGY MOUNDS<br>NATIONAL MONUMENT | DRAWING NO. 394<br>194248<br>PMIS/PKG NO. 318539<br>SHEET 16 OF 142 |

Plotted: May 22, 2026 - 2:53pm lindsey.martin O:\PROJECT\332000\332784E\04 CAD\ACAD\ETMP\318539\DRAWINGS\CIVIL\ROOF DRAIN.dwg Layout Name: C202




## LEGEND

W — 6" HDPE WATER SERVICE LINE

GRADING LIMITS

NOTE
1. SEE 5/C500 FOR WATER SERVICE LINE TRENCH DETAIL.
2. SEE SPECIFICATION SECTION 33 14 16 FOR WATER LINE MATERIAL REQUIREMENTS.

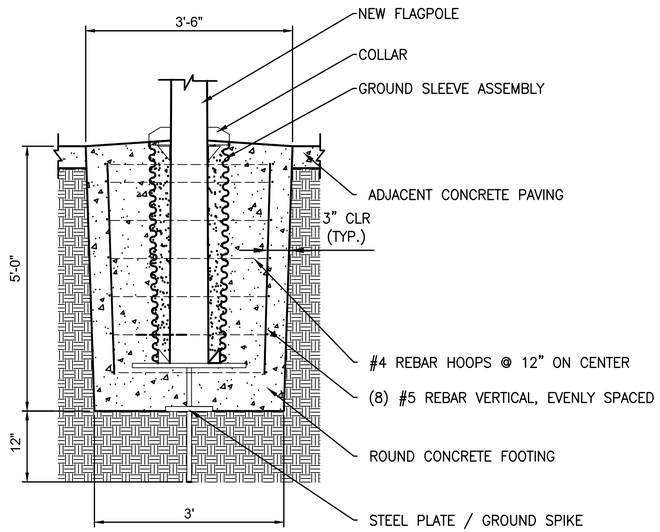
ALL SCOPE ON THIS SHEET SHALL BE BASE SCOPE

HORZ. 10' 0' 10' 20'
VERT. 2.5' 0' 2.5' 5'
SCALE OF FEET

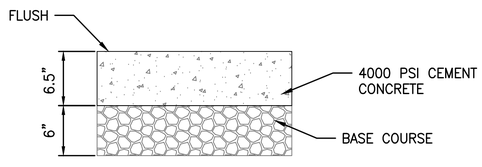
EMILY A. MAGGIO — LICENSED PROFESSIONAL ENGINEER — P30462 — IOWA

| | | | | |
|---|---|---|---|---|
| AE FIRM<br>PRIME:<br>OTAK INC.<br>REDMOND, WA<br><br>SUBCONTRACTOR:<br>ANDERSON HALLAS ARCHITECTS<br>GOLDEN, CO | DESIGNED:<br>EM, LM, CS<br>CADD<br>EM, CS, ZS<br>TECH REVIEW:<br>EM, KK, LM<br>DATE:<br>5/29/2026 | SUB SHEET NO.<br>C300 | TITLE OF SHEET<br>WATER PLAN AND PROFILE<br><br>EFFIGY MOUNDS<br>NATIONAL MONUMENT | DRAWING NO.<br>394<br>194248<br>PMIS/PKG NO.<br>318539<br>SHEET<br>17 OF 142 |

Plotted: May 22, 2026 - 2:53pm lindsay.martin O:\PROJECT\332001\332784E\04 CAD\ACAD\EFMO\318539\DRAWINGS\CIVIL\MOSF DRAW.dwg Layout Name: C300

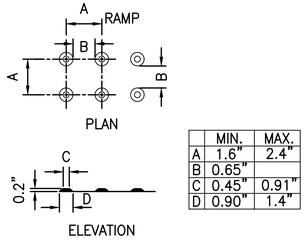



1 C500 FLAGPOLE FOOTING
NTS




NOTES:
1. COLOR: NATURAL COLOR
2. FINISH: MEDIUM BROOM

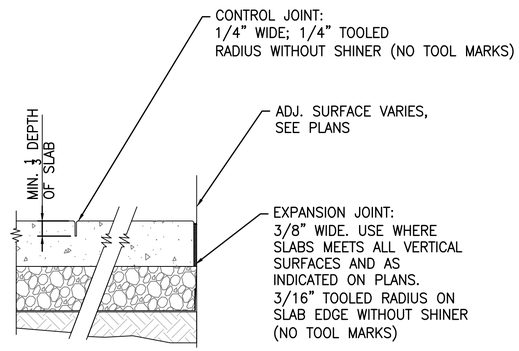
2 C500 BASE & OPTION B: CONCRETE SIDEWALK
NTS



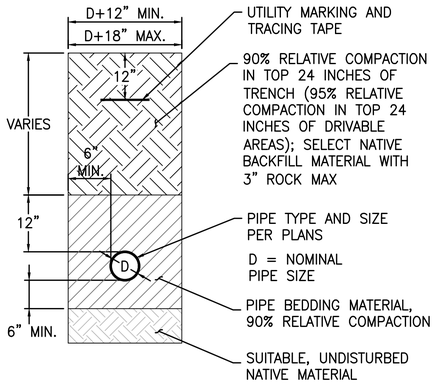

| | MIN. | MAX. |
|---|---|---|
| A | 1.6" | 2.4" |
| B | 0.65" | |
| C | 0.45" | 0.91" |
| D | 0.90" | 1.4" |

DETECTABLE WARNING SURFACE
3 C500 NTS

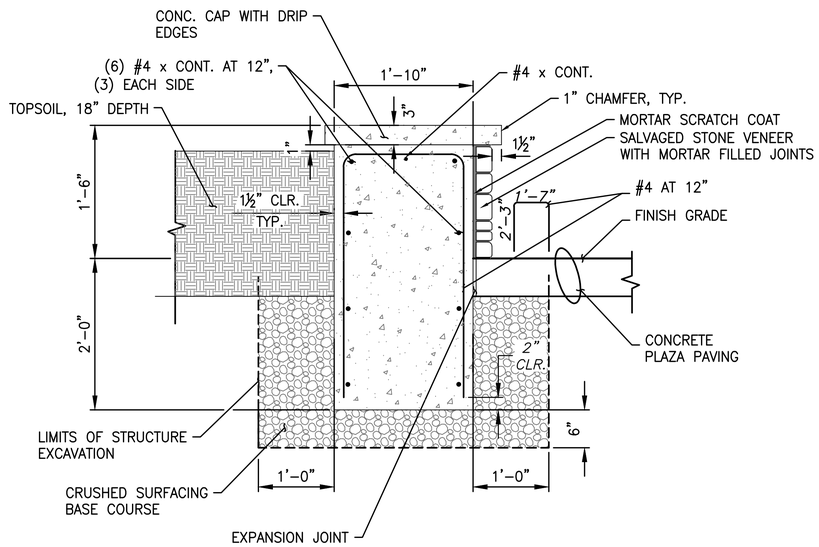
NOTE: SEE PLANS FOR CONCRETE SURFACE TREATMENTS



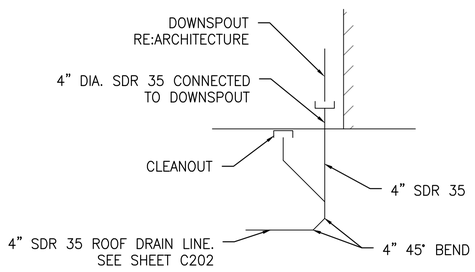

4 C500 BASE & OPTION B: CONCRETE JOINTING
NTS




5 C500 BASE: WATER SERVICE LINE TRENCH
NTS




6 C500 STONE SEAT WALL
NTS



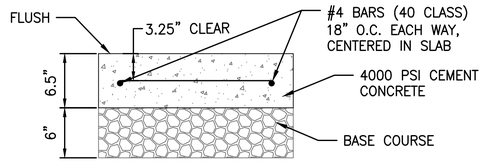

7 C500 DOWNSPOUT & ROOF DRAINAGE CONNECTION
NTS

ALL SCOPE ON THIS SHEET SHALL BE OPTION B UNLESS NOTED OTHERWISE

EMILY A. MAGGIO
LICENSED PROFESSIONAL ENGINEER
25462
IOWA

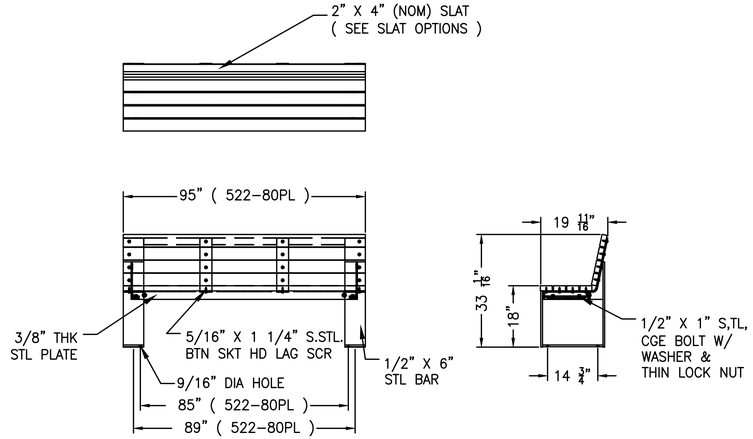
| | | | | |
|---|---|---|---|---|
| AE FIRM<br>PRIME:<br>OTAK INC.<br>REDMOND, WA<br><br>SUBCONTRACTOR:<br>ANDERSON HALLAS ARCHITECTS<br>GOLDEN, CO | DESIGNED:<br>EM, LM, CS<br>DRAWN:<br>EM, CS, ZS<br>TECH REVIEW:<br>EM, KK, LM<br>DATE:<br>5/29/2026 | SUB SHEET NO.<br>C500 | TITLE OF SHEET<br>CIVIL DETAILS I<br><br>EFFIGY MOUNDS<br>NATIONAL MONUMENT | DRAWING NO.<br>394<br>194248<br>PMIS/PKG NO.<br>318539<br>SHEET<br>18 OF 142 |

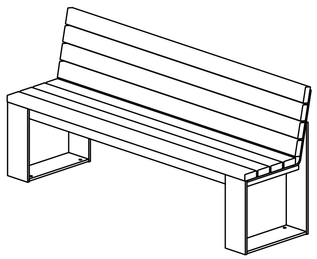
Plotted: May 22, 2026 - 2:53pm lindsay.martin O:\PROJECT\332000\332784E\04 CAD\ACAD\EFMV\318539\DRAWINGS\CIVIL\DETAILS.dwg Layout Name: C500




NOTES:
1. COLOR: NATURAL COLOR
2. FINISH: MEDIUM BROOM

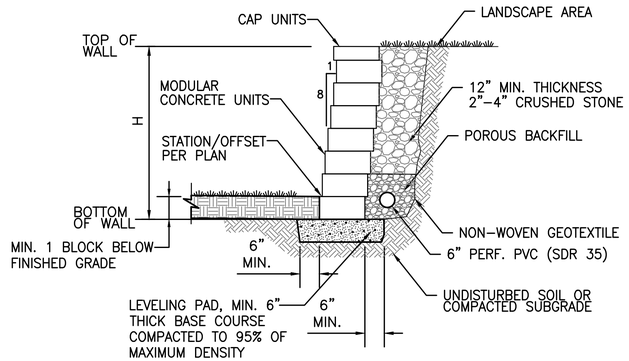
## 1 C501 REINFORCED CONCRETE SIDEWALK
NTS



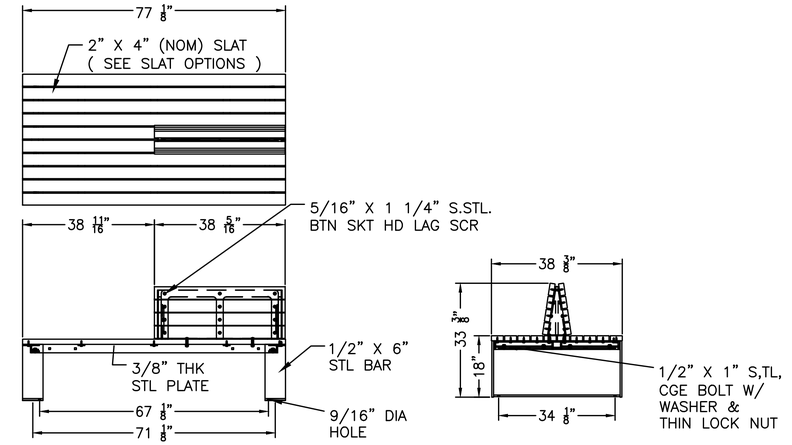



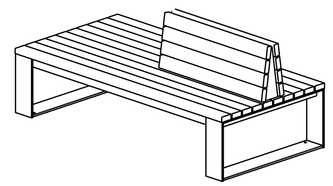
NOTES:
1. ALL STL. MEMBERS COATED W/ ZINC RICH EPOXY THEN FINISHED: MEDIUM BROOM W/ POLYESTER POWDER COATING.
2. DO NOT ANCHOR BENCHES TO CONCRETE SIDEWALK.
3. ANTIQUE MAHOGANY WOOD GRAIN RECYCLED PLASTIC.

## 2 C501 BENCH
NTS



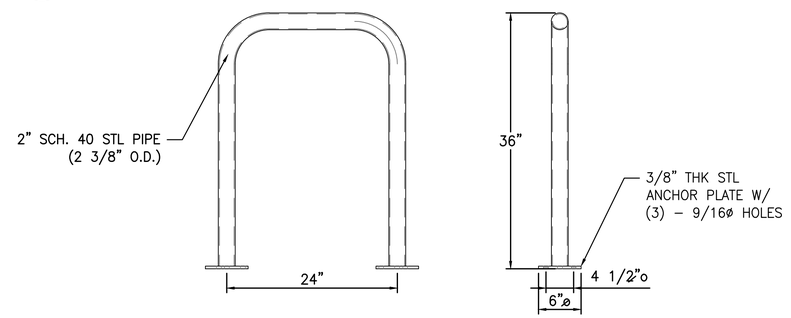

## 5 C501 BASE: BLOCK WALL
NTS



NOTES:
1. ALL STL. MEMBERS COATED W/ ZINC RICH EPOXY THEN FINISHED W/POLYESTER POWDER COATING.
2. DO NOT ANCHOR BENCHES TO CONCRETE SIDEWALK.
3. ANTIQUE MAHOGANY WOOD GRAIN RECYCLED PLASTIC.

## 3 C501 DOUBLE SIDED BENCH
NTS




NOTES:
1. ALL STL. MEMBERS COATED W/ ZINC RICH EPOXY THEN FINISHED W/ POLYESTER POWDER COATING (COLOR: TEXTURED BLACK).
2. SURFACE MOUNT BIKE LOOP WITH 1/2" X 3 3/4" EXPANSION ANCHOR BOLTS
3. EXPANSION ANCHOR BOLT INSTALLATION TO AVOID SLAB REBAR, TYPICAL FOR ALL POST INSTALLED CONNECTIONS.
4. INSTALL BOTH BIKE LOOPS PARALLEL AND EQUIDISTANT.

## 4 C501 ANCHORED BIKE LOOP
NTS

ALL SCOPE ON THIS SHEET SHALL BE OPTION B UNLESS NOTED OTHERWISE

LICENSED PROFESSIONAL ENGINEER
EMILY A. MAGGIO
P30462
IOWA

AE FIRM

PRIME:
OTAK INC.
REDMOND, WA

SUBCONTRACTOR:
ANDERSON HALLAS ARCHITECTS
GOLDEN, CO

DESIGNED: EM, LM, CS
CADD: EM, CS, ZS
TECH REVIEW: EM, KK, LM
DATE: 5/29/2026

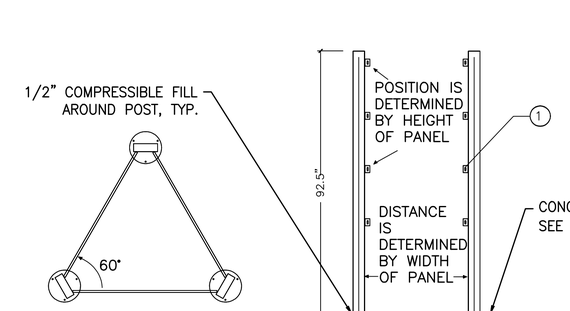
SUB SHEET NO. C501

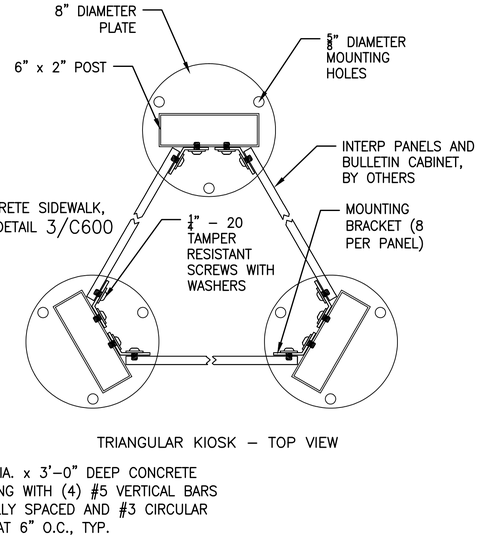
TITLE OF SHEET
CIVIL DETAILS II

EFFIGY MOUNDS
NATIONAL MONUMENT

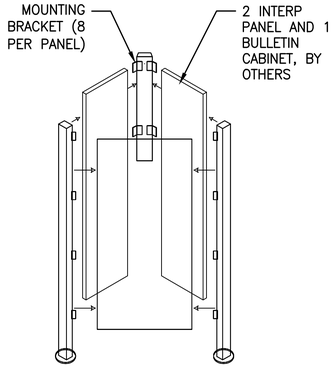
DRAWING NO. 394 194248
PMIS/PKG NO. 318539
SHEET 19 OF 142

Plotted: May. 22, 2026 - 2:51pm lindsey.martin O:\PROJECT\32000\332784E\04 CAD\ACAD\EFMO\318539\DRAWINGS\CIVIL\DETAILS.dwg Layout Name: C501

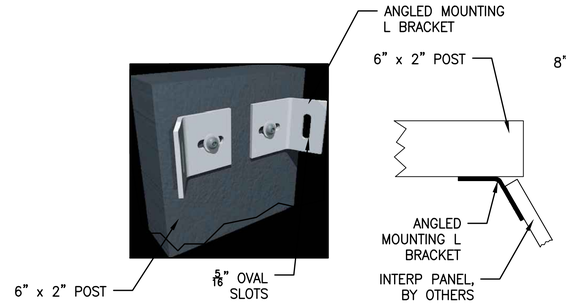


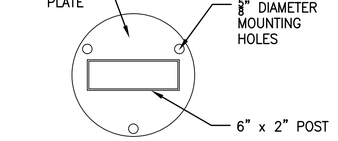

TRIANGULAR KIOSK – TOP VIEW




MULTIPLE POST ASSEMBLY

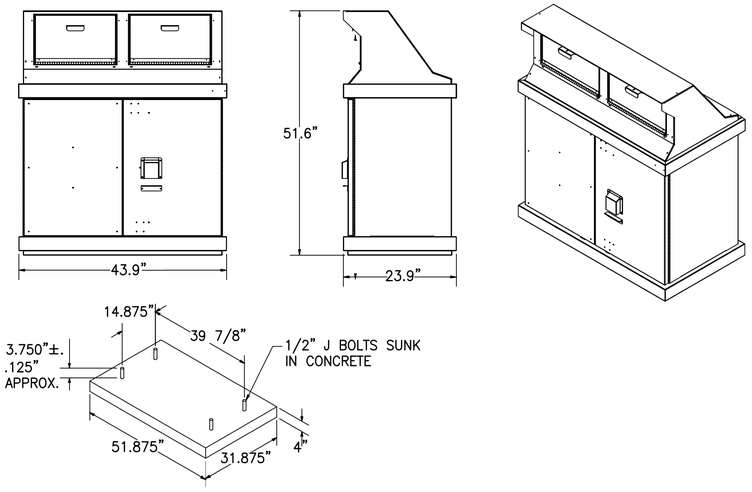



(1) ANGLED STEEL MOUNTING L SHAPED BRACKET



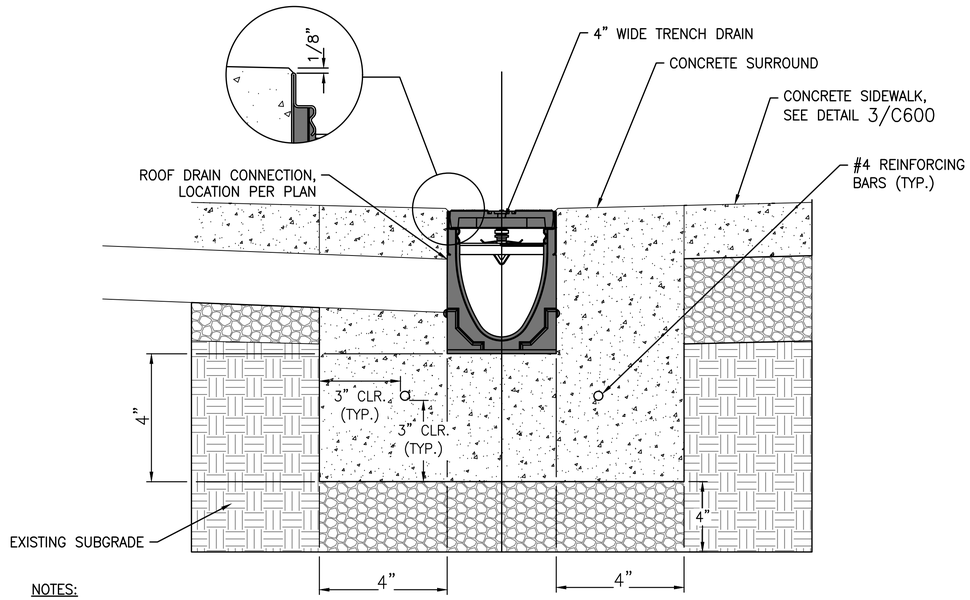

(2) 8" DIAMETER BASE PLATE (6"x2") POST)

## 1 C502 THREE SIDED KIOSK
NTS




## 2 C502 TRASH RECEPTACLE
NTS

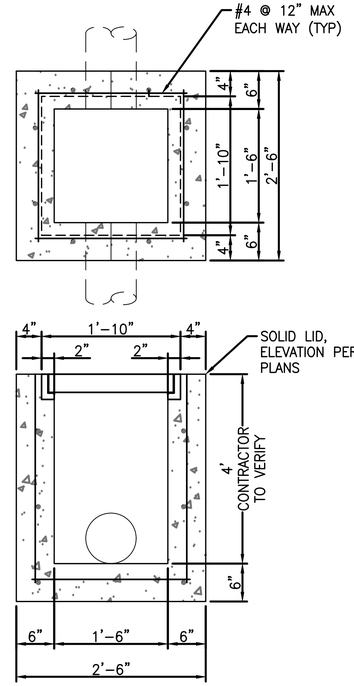
ALL SCOPE ON THIS SHEET SHALL BE OPTION B




NOTES:

1. IT IS NECESSARY TO ENSURE MINIMUM DIMENSIONS SHOWN ARE SUITABLE FOR EXISTING GROUND CONDITIONS.
2. MINIMUM CONCRETE STRENGTH OF 4,000 PSI IS RECOMMENDED. CONCRETE SHALL BE VIBRATED TO ELIMINATE AIR POCKETS.
3. EXPANSION AND CONTRACTION CONTROL JOINTS AND REINFORCEMENT ARE RECOMMENDED TO PROTECT CHANNEL AND CONCRETE SURROUND.
4. THE FINISHED LEVEL OF THE CONCRETE SURROUND MUST BE APPROX. 1/8" ABOVE THE TOP OF THE CHANNEL EDGE.
5. CONCRETE BASE THICKNESS SHOULD MATCH SLAB THICKNESS.

## 3 C502 TRENCH DRAIN
NTS

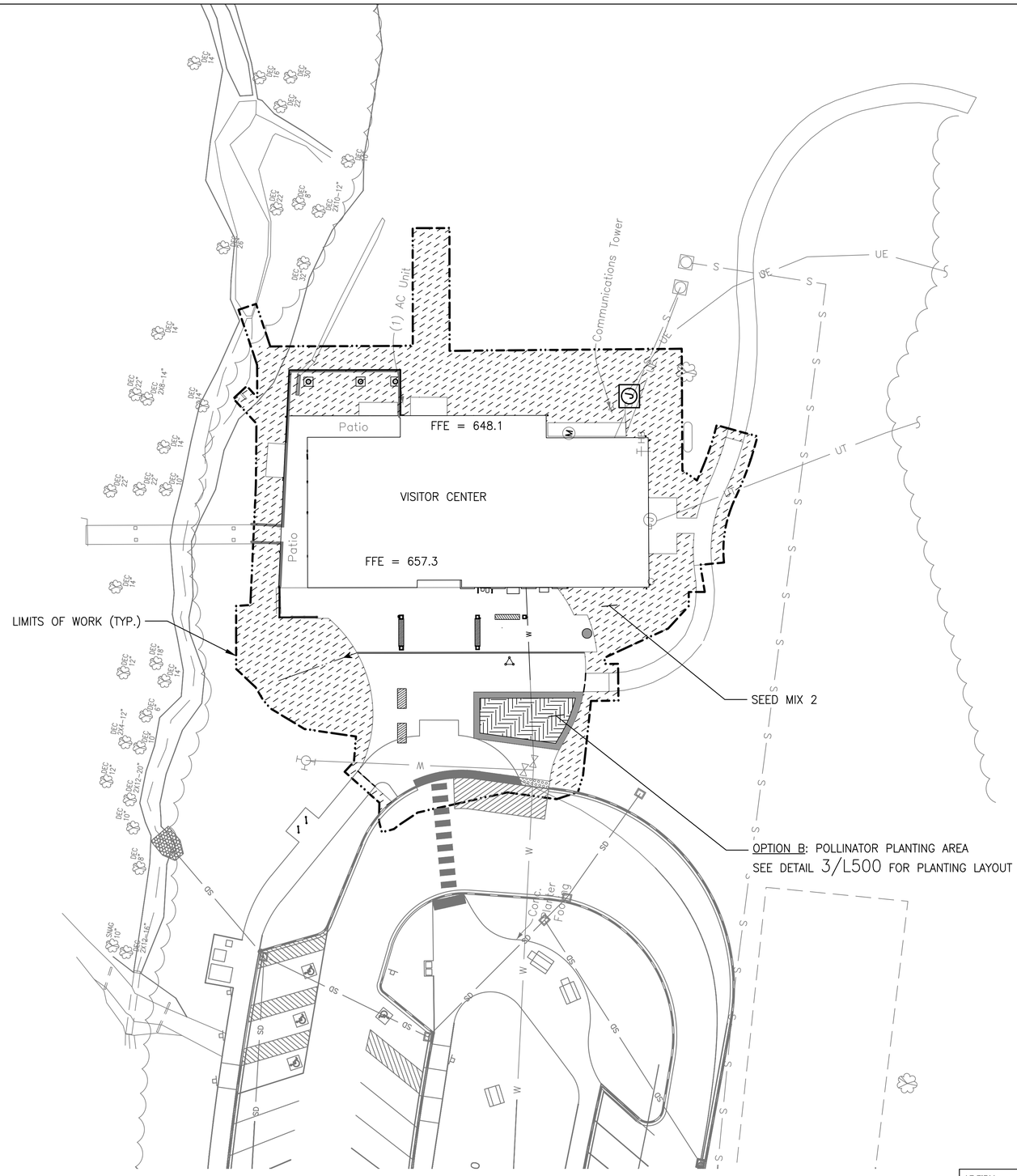



## 4 C502 INLET WITH CLOSED LID
NTS

EMILY A. MAGGIO — LICENSED PROFESSIONAL ENGINEER — P30462 — IOWA

| | | | | |
|---|---|---|---|---|
| AE FIRM<br>PRIME:<br>OTAK INC.<br>REDMOND, WA<br><br>SUBCONTRACTOR:<br>ANDERSON HALLAS ARCHITECTS<br>GOLDEN, CO | DESIGNED:<br>EM, LM, CS<br>CADD<br>EM, CS, ZS<br>TECH REVIEW:<br>EM, KK, LM<br>DATE:<br>5/29/2026 | SUB SHEET NO.<br>C502 | TITLE OF SHEET<br>CIVIL DETAILS III<br><br>EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO.<br>394<br>194248<br>PMIS/PKG NO.<br>318539<br>SHEET<br>20 OF 142 |

Plotted: May 22, 2026 – 2:51pm lindsay.martin O:\PROJECT\332001\332784E\04 CAD\ACAD\EFMO\318539\DRAWINGS\CIVIL\DETAILS.dwg Layout Name: C502




## LEGEND

- LIMITS OF WORK
- POLLINATOR PLANTING AREA
- SEED MIX 2

## GENERAL NOTES

1. ALL PLANT MATERIAL TO BE PROVIDED BY NPS.

ALL SCOPE ON THIS SHEET SHALL BE BASE SCOPE UNLESS NOTED OTHERWISE

20' 0' 20' 40'
SCALE OF FEET

STATE OF IOWA — LINDSAY MARTIN — LANDSCAPE ARCHITECT — No. LA00675 — REGISTERED LANDSCAPE ARCHITECT
5/29/2026

| | | | |
|---|---|---|---|
| AE FIRM<br>PRIME:<br>OTAK INC.<br>REDMOND, WA<br><br>SUBCONTRACTOR:<br>ANDERSON HALLAS ARCHITECTS<br>GOLDEN, CO | DESIGNED:<br>KB, MS, LM<br>DRAWN:<br>LM, RC<br>TECH REVIEW:<br>LM<br>DATE:<br>5/29/2026 | SUB SHEET NO.<br>L100 | TITLE OF SHEET<br>LANDSCAPE PLAN<br><br>EFFIGY MOUNDS<br>NATIONAL MONUMENT | DRAWING NO.<br>394<br>194248<br>PMIS/PKG NO.<br>318539<br>SHEET<br>21 OF 142 |

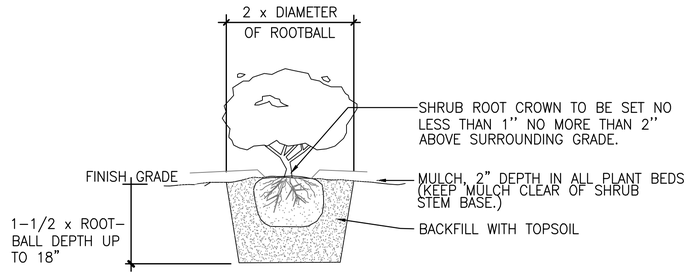
Plotted: May. 22, 2026 - 2:52pm lindsey.martin O:\PROJECT_13200_13270\94E\04 CAD\ACAD\EFMO_318539\DRAWINGS\CIVIL\LANDSCAPE PLAN.dwg Layout Name: LANDSCAPE PLAN

## PLANTING SCHEDULE

| | | |
|---|---|---|
| | POLLINATOR PLANTING AREA<br>SEE DETAIL 3 ON THIS SHEET FOR PLANTING LAYOUT | 378 SF |
| | BASE: SEED MIX 2<br>NATIVE TURF REPLACEMENT–MOWED FREQUENTLY | 7,052 SF |

POLLINATOR PLANTING AREA SCHEDULE

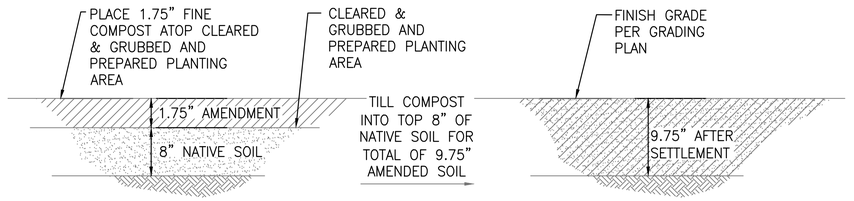
| SHRUBS | QTY | BOTANICAL / COMMON NAME | CONT |
|---|---|---|---|
| | 14 | ANTENNARIA PLANTAGINIFOLIA / PUSSYTOES | 3 GAL |
| | 17 | ARALIA NUDICAULIS / WILD SARSAPARILLA | 3 GAL |
| | 4 | ARALIA RACEMOSA / AMERICAN SPIKENARD | 3 GAL |
| | 17 | ASCLEPIAS TUBEROSA / BUTTERFLY MILKWEED | 3 GAL |
| | 5 | CEANOTHUS AMERICANUS / NEW JERSEY TEA | 5 GAL |
| | 11 | FRAGARIA VIRGINIANA / WILD STRAWBERRY | 3 GAL |
| | 7 | OSMUNDA CLAYTONIANA / INTERRUPTED FERN | 5 GAL |
| | 11 | RUDBECKIA HIRTA / BLACK–EYED SUSAN | 5 GAL |
| | 6 | SOLIDAGO SPECIOSA / SHOWY GOLDENROD | 2 GAL. |




SECTION

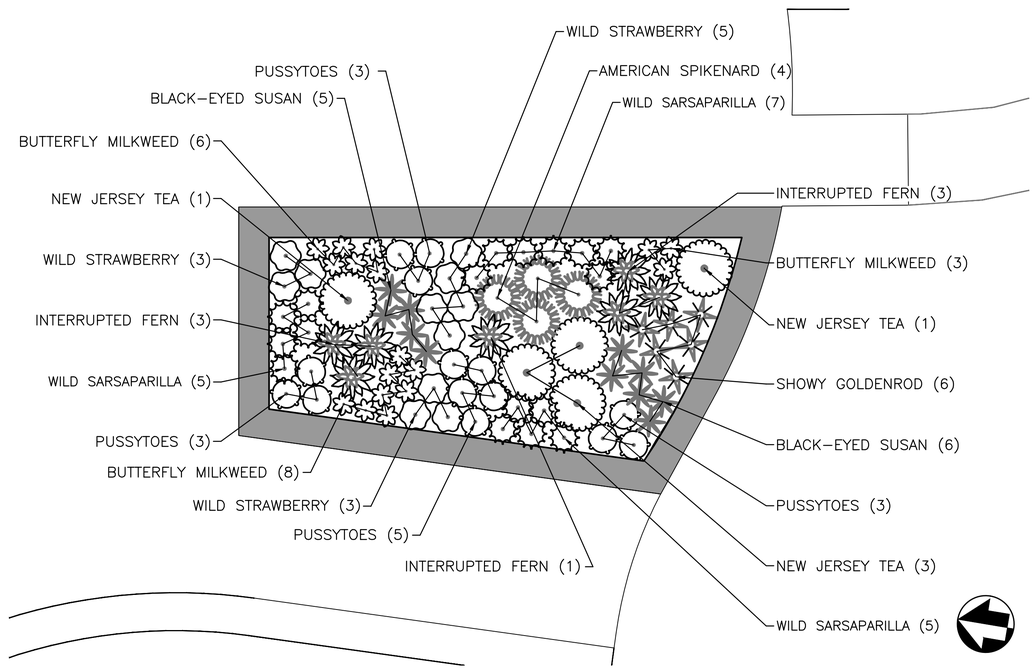
### 1 L500 SHRUB PLANTING
NTS

ALL SCOPE ON THIS SHEET SHALL BE OPTION B UNLESS NOTED OTHERWISE




NOTE: ALL AREAS SHOWN FOR SEEDING AND PLANTING SHALL BE PREPARED ACCORDING TO THIS DETAIL PRIOR TO PLANTING.

### 2 L500 BASE & OPTION B: SOIL PREPARATION
NTS




NOTES

1. PLANT MATERIAL TO BE PROVIDED BY NPS.
2. SEE DETAIL 1 ON THIS SHEET FOR SHRUB PLANTING.
3. SEE DETAIL 2 ON THIS SHEET FOR SOIL PREPARATION.

### 3 L500 POLLINATOR PLANTING AREA LAYOUT
SCALE: 1"= 5'


Plotted: May 22, 2026 – 2:52pm lindsey.martin G:\PROJECT\13200\13279AC\04 CAD\ACAD\GTMO\318539\DRAWINGS\CIVIL\LANDSCAPE DETAILS.dwg Layout Name: LANDSCAPE DETAILS



STATE OF IOWA LINDSAY MARTIN LANDSCAPE ARCHITECT No. LA00075 PROFESSIONAL LANDSCAPE ARCHITECT
5/29/2026

AE FIRM
PRIME:
OTAK INC.
REDMOND, WA

SUBCONTRACTOR:
ANDERSON HALLAS ARCHITECTS
GOLDEN, CO

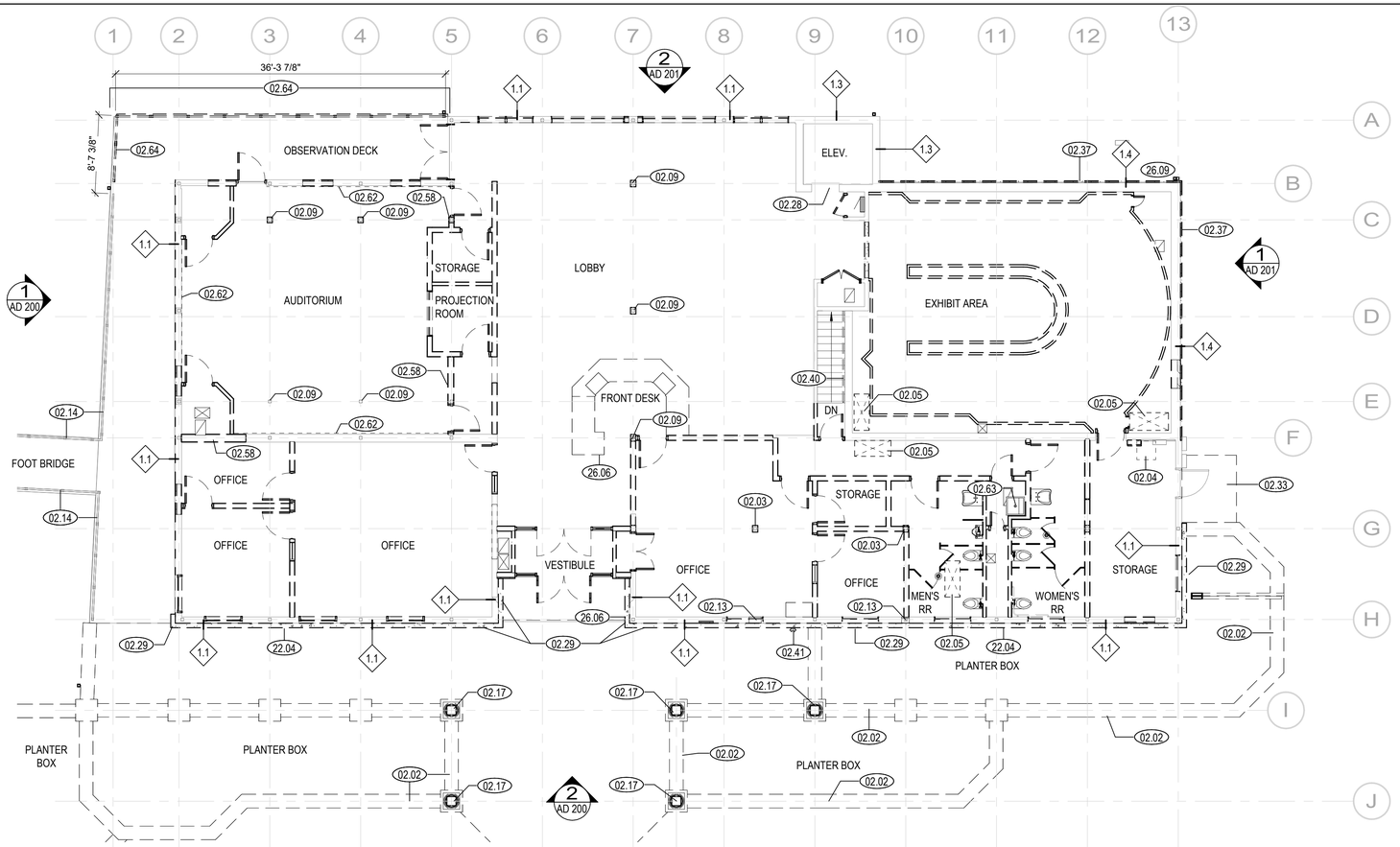

| | |
|---|---|
| DESIGNED: | KB, MS, LM |
| CADD | LM, RC |
| TECH REVIEW: | LM |
| DATE: | 5/29/2026 |

SUB SHEET NO. L500

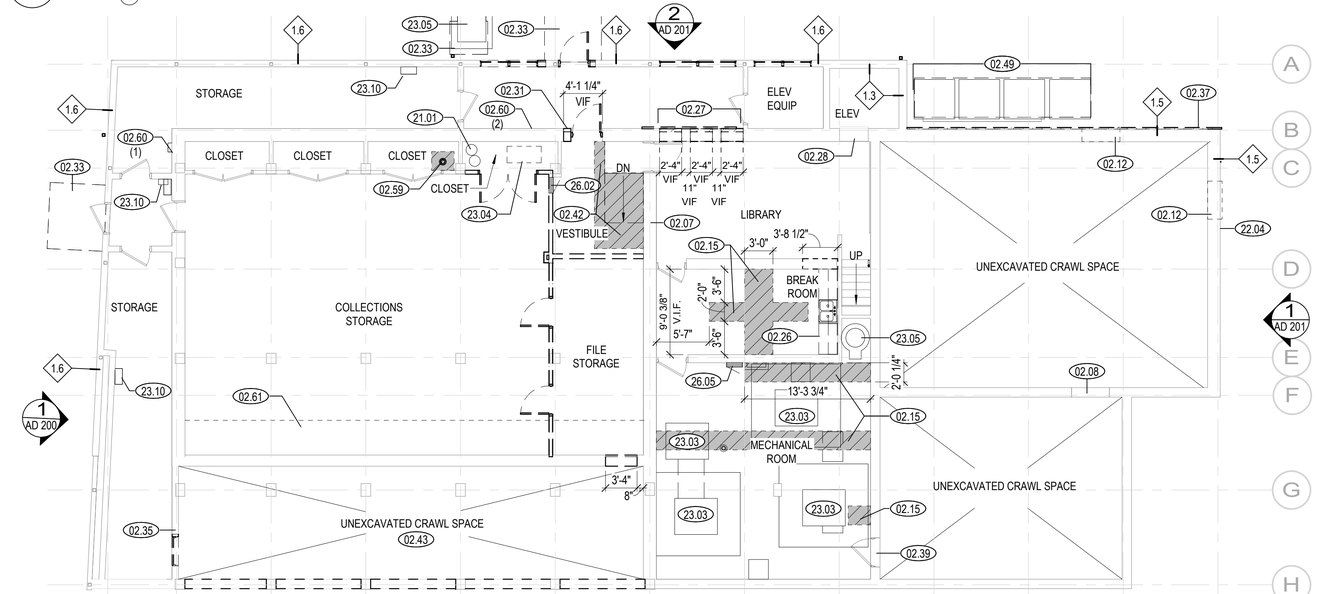
TITLE OF SHEET
LANDSCAPE DETAILS

EFFIGY MOUNDS
NATIONAL MONUMENT

| | |
|---|---|
| DRAWING NO. | 394<br>194248 |
| PMIS/PKG NO. | 318539 |
| SHEET | 22 OF 142 |




## 2 MAIN LEVEL - DEMOLITION PLAN
2 AD 100 SCALE A




## 1 BASEMENT LEVEL - DEMOLITION PLAN
1 AD 100 SCALE A

## DEMOLITION PLAN LEGEND

- <E> ELEMENT TO REMAIN
- <E> ELEMENT TO BE REMOVED
- <E> DOOR AND FRAME TO BE REMOVED
- <E> WALL TO BE REMOVED
- <E> WALL TO REMAIN
- X.X WALL TAG RE: G500 FOR (DEMO) WALL ASSEMBLIES
- SAWCUT AND REMOVE PORTION OF <E> CONCRETE SLAB ON GRADE. REMOVE APPROX. 2'-0" OF GRADE BELOW <E> CONCRETE SLAB, RE: TO PLUMBING FOR <N> PLUMBING LINES, AND DRAIN LOCATIONS

## GENERAL NOTES

1. DRAWINGS ARE REPRESENTATIVE OF SCOPE AND RANGE OF DEMOLITION REQUIRED BUT ARE NOT INTENDED TO BE INCLUSIVE OF ALL NECESSARY WORK.
2. VERIFY ALL DIMENSIONS IN FIELD. GC SHALL BE RESPONSIBLE FOR PROTECTING ALL ELEMENTS TO REMAIN.
3. RE: CIVIL, LAND, STRUC, MECH, PLUMB, ELEC, AND FIRE PROTECTION FOR MORE INFO.
4. CONTRACTOR SHALL HOLD A PRE-DEMOLITION SITE WALK WITH CO.
5. ALL <E> ELEMENTS ARE SHOWN BASED ON AVAILABLE INFO, ON NON-DESTRUCTIVE FIELD OBSERVATION, AND PREVIOUS DRAWINGS PROVIDED BY NPS. CONTRACTOR TO FIELD VERIFY ALL <E> CONSTRUCTION AND NOTIFY CO OF ANY DISCREPANCY BETWEEN CONTRACT DOCUMENTS AND ACTUAL FIELD CONDITIONS AND REQUEST CLARIFICATION PRIOR TO PROCEEDING WITH CONSTRUCTION.
6. DIMS SHOWN FOR DEMO ARE APPROXIMATE. VIF EXTENT OF DEMO REQUIRED TO ACHIEVE <N> WORK. EXPECT THAT <E> BUILDING ELEMENTS MAY NOT BE PLUMB, LEVEL OR SQUARE.
7. FF: MAIN LEVEL 100'-0" =USGS XXX' +/- AT EXT DOORS.
8. FOR ALL ITEMS TO BE REMOVED AND SALVAGED FOR REINSTALLATION, PROVIDE TEMPORARY, SECURE, WEATHERPROOF STORAGE UNTIL REINSTALLATION.
9. STOP WORK AND NOTIFY CO IF ANY ADDITIONAL SUSPECTED HAZARDOUS MATERIAL IS FOUND DURING DEMO. REFER TO THE HAZARDOUS MATERIALS REPORT IN SPEC APPENDICES.
10. RE: FINISH PLAN (A600) FOR ADDITIONAL INFO ON <E> FINISHES TO REMAIN.
11. ALL <E> INSULATION SHALL BE REMOVED FROM THE ATTIC RE: ATTIC PLAN AND WALL SECTIONS.
12. REMOVE ALL <E> NON-STRUCTURAL CEILINGS, INCLUDING METAL SOFFIT AT EXTERIOR. STRUCTURAL ELEMENTS SUCH AS CONCRETE PLANKS AND WOOD DECKING SHALL REMAIN, PROTECTED FROM DEMOLITION ACTIVITIES.

## KEYNOTES

| | |
|---|---|
| 02.02 | REMOVE <E> PLANTERS RE: CIVIL |
| 02.03 | OPTION C: REMOVE <E> COLUMN FROM THIS LOCATION RE: STRUC |
| 02.04 | <E> CLING ATTIC ACCESS TO REMAIN |
| 02.05 | OPTION A: <N> OPENING IN FLOOR FOR MECHANICAL CHASE RE: STRUC |
| 02.07 | OPTION C: REMOVE <E> CONC RAMP |
| 02.08 | <E> ACCESS OPENING TO REMAIN, PROTECT IN PLACE |
| 02.09 | OPTION C: REMOVE GWB WRAP, WD BLOCKING, AND BOLT ATTACHMENT FROM <E> HSS COLUMN, PROTECT IN PLACE |
| 02.12 | OPTION A: <N> OPENING IN <E> WALL FOR MECH LOUVER RE: MECH |
| 02.13 | OPTION C: DEMO PORTION OF <E> WALL FOR INSTALLATION OF <N> STL STRUC COLUMNS RE: STRUC |
| 02.14 | OPTION B: <E> MTL RAILING TO REMAIN. PROTECT IN PLACE. PREP FOR REFINISH AND REPAINT |
| 02.15 | OPTION C: REMOVE PORTION OF <E> CONC SLAB TO ALLOW ACCESS FOR <N> PLUMBING LINES RE: STRUC & PLUMB |
| 02.17 | OPTION B: REMOVE <E> WD COLUMN WRAP, SUPPORT <E> WD BEAM ABOVE FOR INSTALLATION OF <N> STL COLUMNS RE: STRUC. SALVAGE STONE VENEER AT BASE OF COLUMN FOR REUSE |
| 02.26 | OPTION C: REMOVE <E> CASEWORK, COUNTERTOP, AND PLUMB FIXTURE RE: PLUMB |
| 02.27 | OPTION C: SELECTIVELY DEMO <E> CMU WALL TO LOCATE ORIGINAL WINDOW OPENINGS |
| 02.28 | <E> ELEVATOR DOOR TO REMAIN, PROTECT IN PLACE |
| 02.29 | AT <E> EXT WALL REMOVE AND SALVAGE STONE WAINSCOT FINISH AND STONE CAP. REMOVE MORTAR FROM SALVAGED STONE AND PREP FOR REUSE. |
| 02.31 | OPTION C: REMOVE PORTION OF <E> MASONRY WALL AT LATCH SIDE OF REMOVED DOOR RE: STRUC |
| 02.33 | OPTION B: REMOVE <E> CONC SLAB ON GRADE |
| 02.35 | OPTION C: REMOVE <E> ACCESS PANEL AND REMOVE PORTION OF <E> MASONRY WALL TO PROVIDE OPENING FOR <N> DOOR RE: STRUC |
| 02.37 | OPTION B: REMOVE <E> EXT WD SIDING, FURRING, AND ATTACHMENTS |
| 02.39 | <E> ACCESS OPENING WITH ACCESS PANEL +/- 36" AFF TO REMAIN. PROTECT IN PLACE |
| 02.40 | OPTION C: REMOVE <E> WD HANDRAIL |
| 02.41 | OPTION A: <E> SINGLE HEAD FIRE DEPARTMENT CONNECTION TO BE REMOVED; PIPING TO REMAIN, PROTECT IN PLACE RE: FIRE |
| 02.42 | OPTION C: REMOVE <E> CONCRETE LANDING AND RAMP |
| 02.43 | REMOVE <E> GRADE FROM ENTIRE UNEXCAVATED CRAWL SPACE DOWN TO 2'-0 MIN. BELOW LOWER LEVEL. RE: STRUC |
| 02.49 | OPTION A: <E> EQUIP TO BE REMOVED, REMOVE HOUSEKEEPING PAD RE: MECH |
| 02.58 | OPTION C: REMOVE PORTION OF <E> WALL TO UNDERSIDE OF 1959 ROOF CONSTRUCTION, STL BEAMS AT 1959 ROOF CONSTRUCTION TO REMAIN. |
| 02.59 | OPTION C: REMOVE <E> PIPE SUB UP AT FLOOR TO BELOW <E> CONC SLAB, CAP OF <E> PIPE AND PATCH CONC FLOOR TO MATCH ADJACENT |
| 02.60 | OPTION A: REMOVE <E> AIR TRANSFER GRILLE (3) LOCATIONS |
| 02.61 | OPTION A: REMOVE <E> MECH CHASE ENCLOSURE ABOVE FLOOR TO STRUCTURE ABOVE. RE: MECH FOR REMOVAL OF ENCLOSED EQUIPMENT |
| 02.62 | OPTION F: REMOVE <E> FABRIC WALL FINISH AND WD TRIM |
| 02.63 | OPTION C: REMOVE <E> MOP SINK RE: PLUMB |
| 02.64 | OPTION G: REMOVE <E> MTL RAILING |
| 21.01 | OPTION A: <E> CLEAN AGENT TANK TO REMAIN, PROTECT IN PLACE RE: FIRE PROTECTION |
| 22.04 | OPTION C: <N> HOSE BIB RE: PLUMB |
| 23.03 | OPTION A: <E> EQUIPMENT TO BE REMOVED, REMOVE HOUSEKEEPING PAD RE: MECH |
| 23.04 | OPTION A: <E> FURNACE TO BE REMOVED RE: MECH |
| 23.05 | OPTION A: <E> EQUIPMENT TO BE REMOVED RE: MECH |
| 23.10 | OPTION A: <E> WALL HEATER TO REMAIN, RE: MECH |
| 26.02 | OPTION C: <E> ELEC PANEL TO BE SALVAGED FOR REINSTALLATION RE: ELEC |
| 26.05 | OPTION C: <E> ELEC PANEL TO REMAIN, PROTECT IN PLACE RE: ELEC |
| 26.06 | OPTION C: <E> ABAAS DOOR PUSH BUTTON TO BE REMOVED RE: ELEC |
| 26.09 | OPTION C: <E> ELEC PANEL TO BE REMOVED RE: ELEC |

SCALE: A 8 0 8 16
SCALE OF FEET

LICENSED PROFESSIONAL ARCHITECT ELIZABETH HALLAS 07418 IOWA

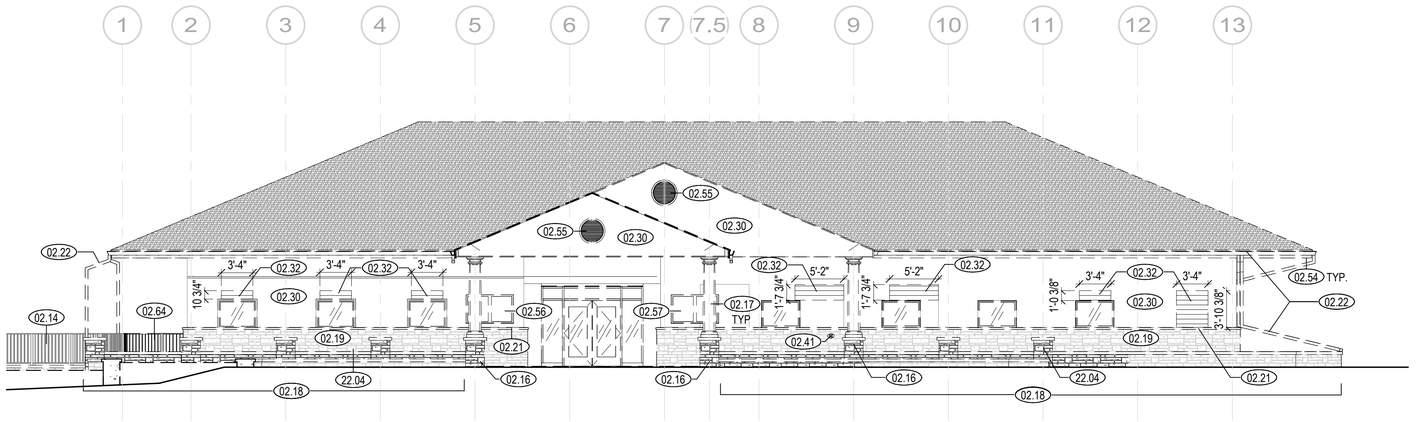
| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM DATED LB/MM TECH REVIEW: ACN / EMH DATE: 05/29/2026 | SUB SHEET NO. AD 100 | TITLE OF SHEET DEMOLITION FLOOR PLANS EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 23 OF 142 |

7/7/2026 1:21:14 PM Autodesk Docs://2025-110 NPS EFMO Visitor Center DD-CD/2024250 NPS EFMO Visitor Center - v24_BIM360.rvt

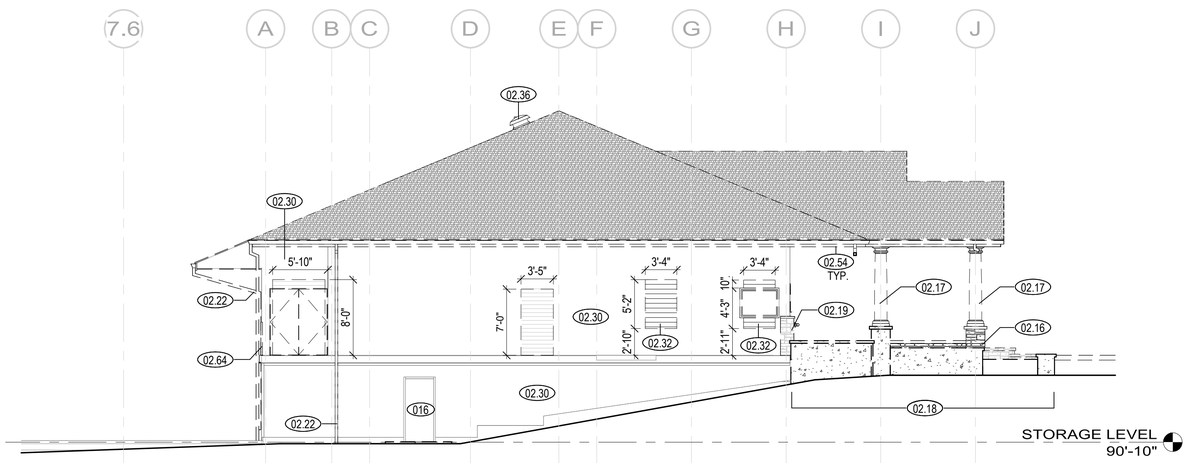
7/7/2026 1:21:18 PM Autodesk Docs://2025-110 NPS EFMO Visitor Center DD-CD/2024250 NPS EFMO Visitor Center - v24_BIM360.rvt

# GENERAL NOTES

1. DRAWINGS ARE REPRESENTATIVE OF SCOPE AND RANGE OF DEMOLITION REQUIRED BUT ARE NOT INTENDED TO BE INCLUSIVE OF ALL NECESSARY WORK.
2. VERIFY ALL DIMENSIONS IN FIELD. GC SHALL BE RESPONSIBLE FOR PROTECTING ALL ELEMENTS TO REMAIN.
3. RE: CIVIL, LAND, STRUC, MECH, PLUMB, ELEC, AND FIRE PROTECTION FOR MORE INFO.
4. CONTRACTOR SHALL HOLD A PRE-DEMOLITION SITE WALK WITH CO.
5. ALL <E> ELEMENTS ARE SHOWN BASED ON AVAILABLE INFO, ON NON-DESTRUCTIVE FIELD OBSERVATION, AND PREVIOUS DRAWINGS PROVIDED BY NPS. CONTRACTOR TO FIELD VERIFY ALL <E> CONSTRUCTION AND NOTIFY CO OF ANY DISCREPANCY BETWEEN CONTRACT DOCUMENTS AND ACTUAL FIELD CONDITIONS AND REQUEST CLARIFICATION PRIOR TO PROCEEDING WITH CONSTRUCTION.
6. DIMS SHOWN FOR DEMO ARE APPROXIMATE. VIF EXTENT OF DEMO REQUIRED TO ACHIEVE <N> WORK. EXPECT THAT <E> BUILDING ELEMENTS MAY NOT BE PLUMB, LEVEL OR SQUARE.
7. FF: MAIN LEVEL 100'-0" =USGS XXX' +/- AT EXT DOORS.
8. FOR ALL ITEMS TO BE REMOVED AND SALVAGED FOR REINSTALLATION, PROVIDE TEMPORARY, SECURE, WEATHERPROOF STORAGE UNTIL REINSTALLATION.
9. STOP WORK AND NOTIFY CO IF ANY ADDITIONAL SUSPECTED HAZARDOUS MATERIAL IS FOUND DURING DEMO. REFER TO THE HAZARDOUS MATERIALS REPORT IN SPEC APPENDICES.
10. RE: FINISH PLAN (A600) FOR ADDITIONAL INFO ON <E> FINISHES TO REMAIN.
11. ALL <E> INSULATION SHALL BE REMOVED FROM THE ATTIC RE: ATTIC PLAN AND WALL SECTIONS.
12. REMOVE ALL <E> NON-STRUCTURAL CEILINGS, INCLUDING METAL SOFFIT AT EXTERIOR. STRUCTURAL ELEMENTS SUCH AS CONCRETE PLANKS AND WOOD DECKING SHALL REMAIN, PROTECTED FROM DEMOLITION ACTIVITIES.




2 AD 200 WEST ELEVATION-DEMOLITION
SCALE A




1 AD 200 NORTH ELEVATION-DEMOLITION
SCALE A

# KEYNOTES

| | |
|---|---|
| 02.14 | OPTION B: <E> MTL RAILING TO REMAIN. PROTECT IN PLACE. PREP FOR REFINISH AND REPAINT |
| 02.16 | AT <E> STONE AT COLUMN BASE REMOVE AND SALVAGE STONE WAINSCOT FINISH AND STONE CAP. REMOVE MORTAR FROM SALVAGED STONE AND PREP FOR REUSE. |
| 02.17 | OPTION B: REMOVE <E> WD COLUMN WRAP, SUPPORT <E> WD BEAM ABOVE FOR INSTALLATION OF <N> STL COLUMNS RE: STRUC. SALVAGE STONE VENEER AT BASE OF COLUMN FOR REUSE |
| 02.18 | REMOVE AND SALVAGE <E> STONE SITE WALL RE: LAND & CIVIL |
| 02.19 | REMOVE AND SALVAGE <E> STONE BASE AT EXT WALL |
| 02.21 | REMOVE AND SALVAGE <E> STONE WATER TABLE TRIM |
| 02.22 | OPTION B: REMOVE <E> MTL GUTTER AND DOWNSPOUTS, TYP |
| 02.30 | OPTION B: AT <E> EXT WALL REMOVE <E> EXT FINISH DOWN TO <E> EXT SHEATHING (SHEATHING AND WALL STUDS TO REMAIN), REMOVE AND REPLACE IN KIND ANY DAMAGED SHEATHING. |
| 02.32 | OPTION B: REMOVE PORTION OF <E> WALL FOR <N> OPENING RE: <N> FLOOR PLAN & ELEVATIONS |
| 02.36 | OPTION B: REMOVE <E> MTL ROOF VENT |
| 02.41 | OPTION A: <E> SINGLE HEAD FIRE DEPARTMENT CONNECTION TO BE REMOVED; PIPING TO REMAIN, PROTECT IN PLACE RE: FIRE |
| 02.54 | OPTION B: REMOVE <E> MTL PANEL SOFFIT |
| 02.55 | OPTION B: REMOVE <E> CIRCULAR LOUVER FROM GABLE |
| 02.56 | OPTION B: REMOVE <E> RECESSED BULLETIN BOARD AND RETURN TO PARK |
| 02.57 | OPTION B: REMOVE <E> RECESSED BULLETIN BOARD FOR REINSTALLATION |
| 02.64 | OPTION G: REMOVE <E> MTL RAILING |
| 22.04 | OPTION C: <N> HOSE BIB RE: PLUMB |

# DEMOLITION ELEVATION LEGEND

- <E> ELEMENT TO REMAIN
- <E> ELEMENT TO BE REMOVED
- <E> WALL TO BE REMOVED
- <E> WALL TO REMAIN
- <E> WALL TO BE REMOVED
- <E> STONE SITE WALL AND BASE WALL TO BE REMOVED AND SALVAGED
- <E> ASPHALT SHINGLE ROOFING TO BE REMOVED AND SALVAGED

SCALE: A 8 0 8 16
SCALE OF FEET

LICENSED PROFESSIONAL ARCHITECT ELIZABETH HALLAS 07418 IOWA

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA<br>SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM<br>CADD: LB/MM<br>TECH REVIEW: ACN / EMH<br>DATE: 05/29/2026 | SUB SHEET NO. AD 200 | TITLE OF SHEET: DEMOLITION ELEVATIONS<br>EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248<br>PMIS/PKG NO. 318539<br>SHEET 24 OF 142 |

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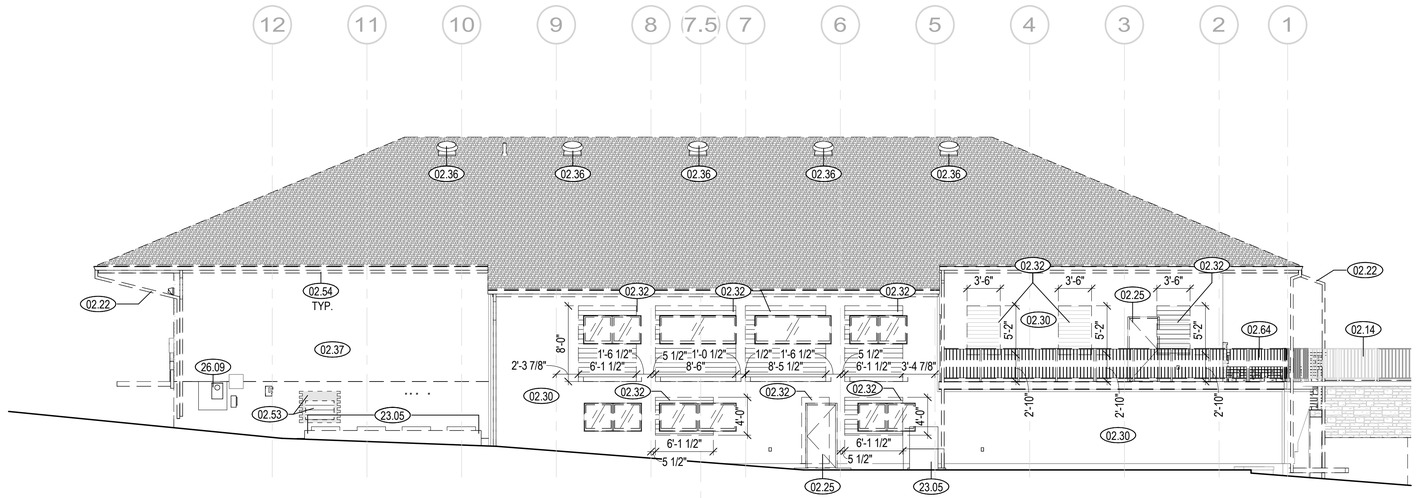
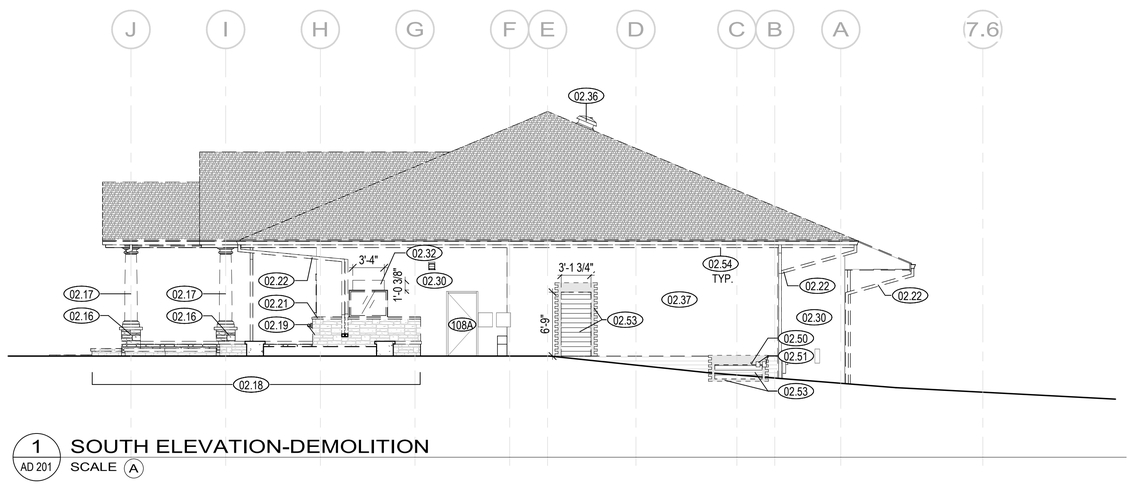



2 AD 201 EAST ELEVATION-DEMOLITION
SCALE A




1 AD 201 SOUTH ELEVATION-DEMOLITION
SCALE A

# GENERAL NOTES

1. DRAWINGS ARE REPRESENTATIVE OF SCOPE AND RANGE OF DEMOLITION REQUIRED BUT ARE NOT INTENDED TO BE INCLUSIVE OF ALL NECESSARY WORK.
2. VERIFY ALL DIMENSIONS IN FIELD. GC SHALL BE RESPONSIBLE FOR PROTECTING ALL ELEMENTS TO REMAIN.
3. RE: CIVIL, LAND, STRUC, MECH, PLUMB, ELEC, AND FIRE PROTECTION FOR MORE INFO.
4. CONTRACTOR SHALL HOLD A PRE-DEMOLITION SITE WALK WITH CO.
5. ALL <E> ELEMENTS ARE SHOWN BASED ON AVAILABLE INFO, ON NON-DESTRUCTIVE FIELD OBSERVATION, AND PREVIOUS DRAWINGS PROVIDED BY NPS. CONTRACTOR TO FIELD VERIFY ALL <E> CONSTRUCTION AND NOTIFY CO OF ANY DISCREPANCY BETWEEN CONTRACT DOCUMENTS AND ACTUAL FIELD CONDITIONS AND REQUEST CLARIFICATION PRIOR TO PROCEEDING WITH CONSTRUCTION.
6. DIMS SHOWN FOR DEMO ARE APPROXIMATE. VIF EXTENT OF DEMO REQUIRED TO ACHIEVE <N> WORK. EXPECT THAT <E> BUILDING ELEMENTS MAY NOT BE PLUMB, LEVEL OR SQUARE.
7. FF: MAIN LEVEL 100'-0" =USGS XXX' +/- AT EXT DOORS.
8. FOR ALL ITEMS TO BE REMOVED AND SALVAGED FOR REINSTALLATION, PROVIDE TEMPORARY, SECURE, WEATHERPROOF STORAGE UNTIL REINSTALLATION.
9. STOP WORK AND NOTIFY CO IF ANY ADDITIONAL SUSPECTED HAZARDOUS MATERIAL IS FOUND DURING DEMO. REFER TO THE HAZARDOUS MATERIALS REPORT IN SPEC APPENDICES.
10. RE: FINISH PLAN (A600) FOR ADDITIONAL INFO ON <E> FINISHES TO REMAIN.
11. ALL <E> INSULATION SHALL BE REMOVED FROM THE ATTIC RE: ATTIC PLAN AND WALL SECTIONS.
12. REMOVE ALL <E> NON-STRUCTURAL CEILINGS, INCLUDING METAL SOFFIT AT EXTERIOR. STRUCTURAL ELEMENTS SUCH AS CONCRETE PLANKS AND WOOD DECKING SHALL REMAIN, PROTECTED FROM DEMOLITION ACTIVITIES.

# KEYNOTES

| | |
|---|---|
| 02.14 | OPTION B: <E> MTL RAILING TO REMAIN. PROTECT IN PLACE. PREP FOR REFINISH AND REPAINT |
| 02.16 | AT <E> STONE AT COLUMN BASE REMOVE AND SALVAGE STONE WAINSCOT FINISH AND STONE CAP. REMOVE MORTAR FROM SALVAGED STONE AND PREP FOR REUSE. |
| 02.17 | OPTION B: REMOVE <E> WD COLUMN WRAP, SUPPORT <E> WD BEAM ABOVE FOR INSTALLATION OF <N> STL COLUMNS RE: STRUC. SALVAGE STONE VENEER AT BASE OF COLUMN FOR REUSE |
| 02.18 | REMOVE AND SALVAGE <E> STONE SITE WALL RE: LAND & CIVIL |
| 02.19 | REMOVE AND SALVAGE <E> STONE BASE AT EXT WALL |
| 02.21 | REMOVE AND SALVAGE <E> STONE WATER TABLE TRIM |
| 02.22 | OPTION B: REMOVE <E> MTL GUTTER AND DOWNSPOUTS, TYP |
| 02.25 | OPTION B: REMOVE <E> DOOR AND FRAME |
| 02.30 | OPTION B: AT <E> EXT WALL REMOVE <E> EXT FINISH DOWN TO <E> EXT SHEATHING (SHEATHING AND WALL STUDS TO REMAIN), REMOVE AND REPLACE IN KIND ANY DAMAGED SHEATHING. |
| 02.32 | OPTION B: REMOVE PORTION OF <E> WALL FOR <N> OPENING RE: <N> FLOOR PLAN & ELEVATIONS |
| 02.36 | OPTION B: REMOVE <E> MTL ROOF VENT |
| 02.37 | OPTION B: REMOVE <E> EXT WD SIDING, FURRING, AND ATTACHMENTS |
| 02.50 | OPTION C: REMOVE AND SALVAGE <E> HOSE BIB AND PIPING. RELOCATED TO <N> LOCATION RE: PLUMB |
| 02.51 | OPTION C: REMOVE <E> ELECTRICAL OUTLETS AND ASSOCIATED (INT CONDUIT) RE: ELEC |
| 02.53 | OPTION A: AT <E> MASONRY WALL, REMOVE <E> MASONRY AS SHOWN IN GRAY HATCH (REMOVE FULL MASONRY UNITS) AT EACH SIDE AND TOP OF <N> DOOR / MECH LOUVER PENETRATION, DO NOT SAW CUT <E> FINISH MASONRY, SAW CUT BACK-UP CMU WALL FOR <N> DOOR / MECH PENETRATIONS RE: STRUC & MECH |
| 02.54 | OPTION B: REMOVE <E> MTL PANEL SOFFIT |
| 02.64 | OPTION G: REMOVE <E> MTL RAILING |
| 23.05 | OPTION A: <E> EQUIPMENT TO BE REMOVED RE: MECH |
| 26.09 | OPTION C: <E> ELEC PANEL TO BE REMOVED RE: ELEC |

# DEMOLITION ELEVATION LEGEND

- <E> ELEMENT TO REMAIN
- <E> ELEMENT TO BE REMOVED
- <E> WALL TO BE REMOVED
- <E> WALL TO REMAIN
- <E> WALL TO BE REMOVED
- <E> STONE SITE WALL AND BASE WALL TO BE REMOVED AND SALVAGED
- <E> ASPHALT SHINGLE ROOFING TO BE REMOVED AND SALVAGED
- <E> MASONRY TO BE MODIFIED RE: KEYNOTE 02.53

SCALE: A 8 0 8 16
SCALE OF FEET

LICENSED PROFESSIONAL ARCHITECT ELIZABETH HALLAS 07418 IOWA

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|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM DRAWN: LB/MM TECH REVIEW: ACN / EMH DATE: 05/29/2026 | SUB SHEET NO. AD 201 | TITLE OF SHEET DEMOLITION ELEVATIONS EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 25 OF 142 |

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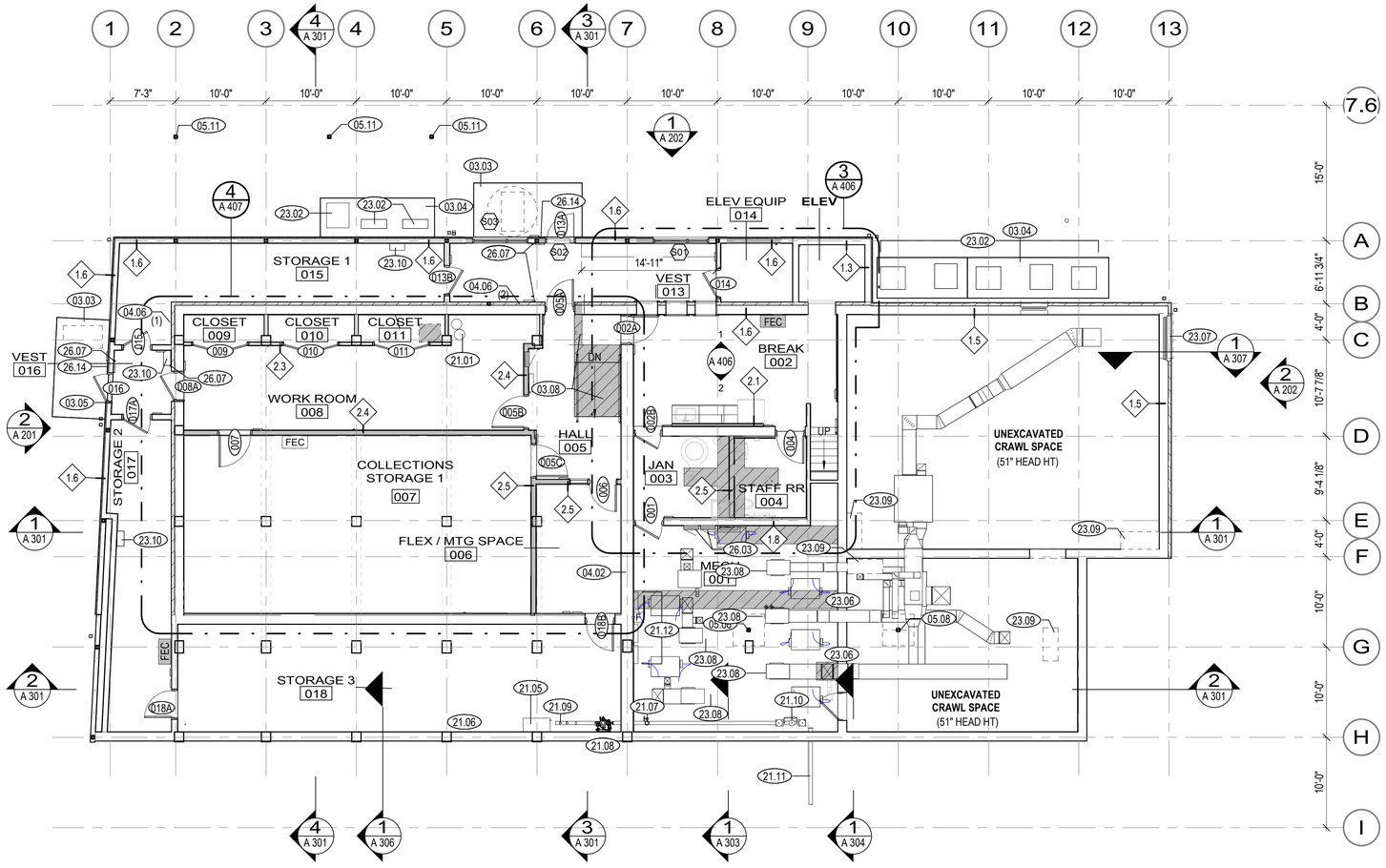


1 A 101 BASEMENT LEVEL - NEW FLOOR PLAN
SCALE A

## GENERAL NOTES

1. RE: CIVIL, LAND, STRUC, MECH, PLUMB, ELEC, AND FIRE PROTECTION FOR MORE INFO.
2. FF: MAIN LEVEL 100'-0" = NAVD88 657.3' +/- AT EXT DOORS.
3. ALL PLAN DIMS LOCATING EXT ASSEMBLIES ARE TO GRID LINE OR FACE OF FINISH, UNO. <E> WALLS WITHOUT WALL TAGS SHALL REMAIN.
4. VERIFY EQUIP ROUGH-IN DIMS WITH MFR FOR ALL EQUIP INCLUDING <E>, REUSED, OR FURNISHED BY CO.
5. PROVIDE ADEQUATE BLOCKING, BACKING, OR STRUC SUPPORT AS REQ'D FOR PROPER INSTALLATION OF ALL MOUNTED ASSEMBLIES, INCLUDING ALL ATTACHED EQUIP, ACCESSORIES (CONTRACTOR FURNISHED ITEMS), PLUMB FIXTURES, MILLWORK, AND CASEWORK.
6. FLOOR DRAIN LOCATIONS SHOWN ARE APPROXIMATE. VIF EXACT LOCATIONS PER PLUMB WITH STRUC COORDINATION. SLOPE CONC TO DRAIN AS INDICATED.
7. ALL CONDUIT & PIPING TO BE ROUTED WITHIN WALLS OR FURRING WALLS UON. CONTRACTOR TO COORDINATE IN FIELD ALL EXPOSED CONDUIT AND PIPING ROUTING WITH CO.
8. <N> PATHS AND SITE ELEMENTS ARE SHOWN FOR REFERENCE ONLY RE: CIVIL & LAND FOR SITE ELEMENTS AND CONTINUATION OF ABA ROUTES.
9. DOORS AND STOREFRONTS OCCUPY <E> FRAMED OPENINGS UON. VIF DIMS IN FIELD. ALIGN ALL <N> ROUGH OPENINGS WITH <E> MASONRY JOINTS UON: RE:STOREFRONT & DOOR SCHEDULE FOR DIMS AND MORE INFO.
10. PATCH AND REPAIR ALL FINISHES IMPACTED BY SYSTEMS AND DEMO WORK RE: FINISH SCHEDULE UON. MATCH ADJ IN MATERIAL, COLOR, TEXTURE AND FINISH.
11. FURNITURE SHOWN IS FOR REFERENCE ONLY AND IS NOT INCLUDED IN THE CONTRACT UON.
12. AT UNDERSIDE OF CONC FLOOR / ROOF STRUCTURES, **DO NOT** ANCHOR **ANY** ELEMENTS (DUCTING, PIPING, CONDUIT, ETC.) INTO THE <E> CMU PLANK FLOOR / ROOF STRUCTURE, ALL ATTACHMENTS SHALL BE INTO THE <E> CONC BEAMS SPACED AT 30" OC, RE: STRUC, MECH, PLUMB, AND FIRE. WHERE <N> SURFACE MOUNTED LIGHTING FIXTURES ARE PROVIDED, COORDINATE THESE LOCATIONS FOR INSTALLATION AND TO BE LOCATED ON <E> CONC BEAMS.

## KEYNOTES

| | |
|---|---|
| 03.03 | OPTION B: <N> EXT CONC SLAB ON GRADE LANDING RE: MECH & STRUC |
| 03.04 | OPTION A: <N> CONC PAD FOR <N> MECH UNITS RE: MECH & STRUC |
| 03.05 | OPTION C: INFILL <E> METAL DOOR FRAME WITH MORTAR AT BOTTOM TO CLOSE GAP AT THE BOTTOM AT BOTH SIDES OF DOOR |
| 03.08 | OPTION C: INSTALL <N> CONC LANDING AND ABAAS COMPLIANT RAMP |
| 04.02 | OPTION A: INFILL <E> OPENING (11X28 APPROXIMATE SIZE, VIF) IN CMU. SEAL, AND FINISH TO MATCH ADJ |
| 04.06 | OPTION A: AT REMOVED AIR TRANSFER GRILL (3) LOCATIONS. IN-FILL <E> OPENING WITH <N> MASONRY TO MATCH ADJ, KEY IN <N> MASONRY INTO <E> MASONRY AND PATCH CMU WALL ON OTHER SIDE TO MATCH |
| 05.08 | OPTION C: <N> HSS COLUMN, PROVIDE RIGID INSULATION AT EXT SIDE OF COLUMN BETWEEN SHEATHING AND STL RE: STRUC |
| 05.11 | OPTION G: <N> STL COLUMN RE: STRUC |
| 21.01 | OPTION A: <E> CLEAN AGENT TANK TO REMAIN, PROTECT IN PLACE RE: FIRE PROTECTION |
| 21.05 | OPTION A: <N> NITROGEN GENERATION PANEL RE: FIRE PROTECTION |
| 21.06 | OPTION A: <N> AIR COMPRESSOR RE: FIRE PROTECTION |
| 21.07 | OPTION A: <N> WET RISER RE: FIRE PROTECTION |
| 21.08 | OPTION A: <N> DRY RISER RE: FIRE PROTECTION |
| 21.09 | OPTION A: <N> PREACTION RISER RE: FIRE PROTECTION |
| 21.10 | OPTION A: <N> BACKFLOW PREVENTER RE: FIRE PROTECTION |
| 21.11 | OPTION A: <N> INCOMING 6" FIRE MAIN RE: FIRE PROTECTION |
| 21.12 | OPTION A: <N> FIRE ALARM PANEL RE: FIRE PROTECTION |
| 23.02 | OPTION A: <N> CONDENSING UNIT RE: MECH |
| 23.06 | OPTION A: <N> MECH VENT, REUSE <E> PENETRATION RE: MECH |
| 23.07 | OPTION A: <N> AIR LOUVER RE: DTLS 3, 4 / A506 & MECH |
| 23.08 | OPTION A: <N> MECH AIR HANDLING UNIT (AHU) RE:MECH |
| 23.09 | OPTION A: <N> MECH CHASE RE: MECH |
| 23.10 | OPTION A: <E> WALL HEATER TO REMAIN, RE: MECH |
| 26.03 | OPTION C: <N> ELEC PANEL RE: ELEC |
| 26.07 | OPTION C: <N> ABAAS DOOR PUSH BUTTON RE: ELEC |
| 26.14 | OPTION C: <N> WEATHERPROOF TWO WAY COMMUNICATION RE:ELEC |

## NEW FLOOR PLAN LEGEND

- <E> WALL TO REMAIN
- <N> WALL RE: WALL TYPES
- <E> ELEMENT TO REMAIN
- <N> ELEMENT TO BE PROVIDED
- FEC <N> SEMI-RECESSED FIRE EXTINGUISHER CABINET
- FEC <N> SURFACE MOUNT FIRE EXTINGUISHER CABINET
- INFILL TRENCHED AREA AFTER INSTALLATION OF <N> PLUMBING LINES, AND DRAINS WITH GRAVEL TO UNDERSIDE OF <E> CONCRETE SLAB, INFILL <N> CONCRETE TO MATCH <E> CONCRETE SLAB, PROVIDE AND INSTALL #4 REBAR DOWEL AT EA. SIDE OF TRENCH AT 24" O.C. (STAGGERED) ENSURE A SMOOTH TRANSITION BETWEEN <E> AND <N> CONCRETE FINISH, RE: PLUMBING AND STRUCTURAL FOR ADDITIONAL INFORMATION

SCALE: A 8 0 8 16
SCALE OF FEET

LICENSED PROFESSIONAL ARCHITECT ELIZABETH HALLAS 07418 IOWA

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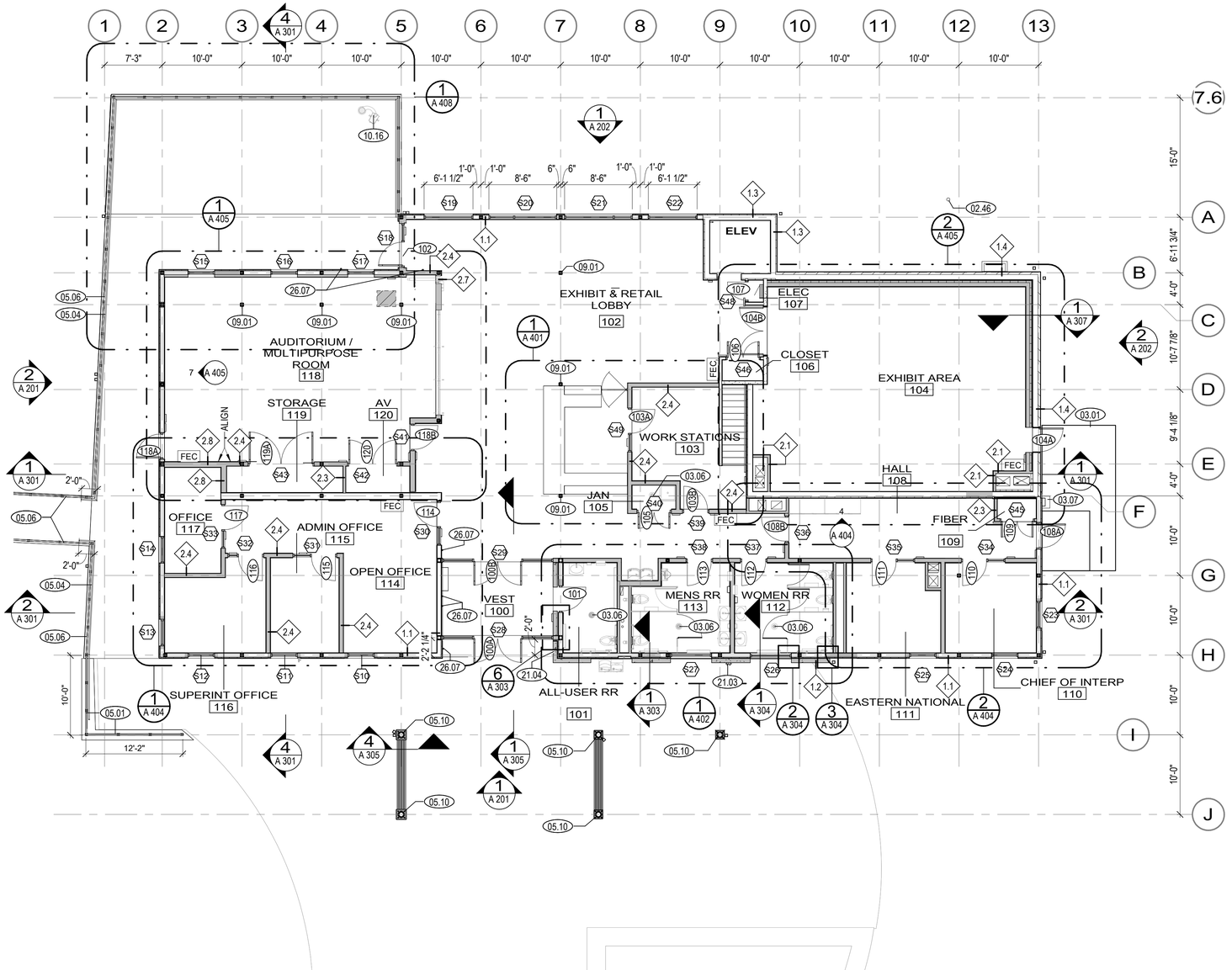
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM DRAWN: LB/MM TECH REVIEW: ACN / EMH DATE: 05/29/2026 | SUB SHEET NO. A 101 | TITLE OF SHEET BASEMENT LEVEL FLOOR PLAN EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 26 OF 142 |




1 A 102 MAIN LEVEL - NEW FLOOR PLAN
SCALE A

SCALE: A 8 0 8 16
SCALE OF FEET

# GENERAL NOTES

1. RE: CIVIL, LAND, STRUC, MECH, PLUMB, ELEC, AND FIRE PROTECTION FOR MORE INFO.
2. FF: MAIN LEVEL 100'-0" = NAVD88 657.3' +/- AT EXT DOORS.
3. ALL PLAN DIMS LOCATING EXT ASSEMBLIES ARE TO GRID LINE OR FACE OF FINISH, UNO. <E> WALLS WITHOUT WALL TAGS SHALL REMAIN.
4. VERIFY EQUIP ROUGH-IN DIMS WITH MFR FOR ALL EQUIP INCLUDING <E>, REUSED, OR FURNISHED BY CO.
5. PROVIDE ADEQUATE BLOCKING, BACKING, OR STRUC SUPPORT AS REQ'D FOR PROPER INSTALLATION OF ALL MOUNTED ASSEMBLIES, INCLUDING ALL ATTACHED EQUIP, ACCESSORIES (CONTRACTOR FURNISHED ITEMS), PLUMB FIXTURES, MILLWORK, AND CASEWORK.
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7. ALL CONDUIT & PIPING TO BE ROUTED WITHIN WALLS OR FURRING WALLS UON. CONTRACTOR TO COORDINATE IN FIELD ALL EXPOSED CONDUIT AND PIPING ROUTING WITH CO.
8. <N> PATHS AND SITE ELEMENTS ARE SHOWN FOR REFERENCE ONLY RE: CIVIL & LAND FOR SITE ELEMENTS AND CONTINUATION OF ABA ROUTES.
9. DOORS AND STOREFRONTS OCCUPY <E> FRAMED OPENINGS UON. VIF DIMS IN FIELD. ALIGN ALL <N> ROUGH OPENINGS WITH <E> MASONRY JOINTS UON: RE:STOREFRONT & DOOR SCHEDULE FOR DIMS AND MORE INFO.
10. PATCH AND REPAIR ALL FINISHES IMPACTED BY SYSTEMS AND DEMO WORK RE: FINISH SCHEDULE UON. MATCH ADJ IN MATERIAL, COLOR, TEXTURE AND FINISH.
11. FURNITURE SHOWN IS FOR REFERENCE ONLY AND IS NOT INCLUDED IN THE CONTRACT UON.
12. AT UNDERSIDE OF CONC FLOOR / ROOF STRUCTURES, **DO NOT** ANCHOR **ANY** ELEMENTS (DUCTING, PIPING, CONDUIT, ETC.) INTO THE <E> CMU PLANK FLOOR / ROOF STRUCTURE. ALL ATTACHMENTS SHALL BE INTO THE <E> CONC BEAMS SPACED AT 30" OC, RE: STRUC, MECH, PLUMB, AND FIRE. WHERE <N> SURFACE MOUNTED LIGHTING FIXTURES ARE PROVIDED, COORDINATE THESE LOCATIONS FOR INSTALLATION AND TO BE LOCATED ON <E> CONC BEAMS.

# KEYNOTES

| | |
|---|---|
| 02.46 | |
| 03.01 | OPTION B: EXTEND EXT CONC LANDING TO ACCOMMODATE <N> EXT DOOR RE: CIVIL/LA |
| 03.06 | OPTION C: GRIND <E> CONC TOPPING SLAB TO PROVIDE SLOPE (2% MIN) 2' FROM <N> FLOORDRAINS |
| 03.07 | OPTION B: KNOTCH CONCRETE SLAB TO ALLOW FOR ELEC EQUIPMENT AND CONDUIT TO REMAIN IN PLACE THROUGHOUT CONSTRUCTION RE: ELEC |
| 05.01 | OPTION B: <N> VERTICAL PICKET STYLE MTL RAILING TO MATCH <E> (PT) WITH MTL HANDRAIL |
| 05.04 | OPTION B: <N> 1-1/2" DIA. MTL HANDRAIL AT 36" AFF, TO TOP OF HANDRAIL, ATTACHED TO <E> MTL RAILING |
| 05.06 | OPTION B: PT <E> MTL RAILING |
| 05.10 | OPTION B: <N> HSS COLUMN IN LOCATION OF REMOVED <E> WD COLUMN RE: STRUC |
| 09.01 | OPTION C: PT <E> STL COLUMN RE: FINISH SCHEDULE |
| 10.16 | OPTION G: <N> ACCESSIBLE PREFINISHED MTL DUAL VIEWING MACHINE |
| 21.03 | OPTION A: <N> DOUBLE-HEAD FIRE DEPARTMENT CONNECTION WHERE <E> REMOVED RE: FIRE PROTECTION |
| 21.04 | OPTION A: <N> RECESSED KNOX BOX |
| 26.07 | OPTION C: <N> ABAAS DOOR PUSH BUTTON RE: ELEC |

# NEW FLOOR PLAN LEGEND

- <E> WALL TO REMAIN
- <N> WALL RE: WALL TYPES
- <E> ELEMENT TO REMAIN
- <N> ELEMENT TO BE PROVIDED
- FEC <N> SEMI-RECESSED FIRE EXTINGUISHER CABINET
- FEC <N> SURFACE MOUNT FIRE EXTINGUISHER CABINET

LICENSED PROFESSIONAL ARCHITECT ELIZABETH HALLAS 07418 IOWA

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|---|---|---|---|---|---|
| A/E FIRM: OTAK, INC. REDMOND, WA | DESIGNED: LB/CM | SUB SHEET NO. A 102 | TITLE OF SHEET MAIN LEVEL FLOOR PLAN | DRAWING NO. 394 194248 |
| PRIME: SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DRAWN: LB/MM | | EFFIGY MOUNDS NATIONAL MONUMENT | PMIS/PKG NO. 318539 |
| | TECH REVIEW: ACN / EMH | | | SHEET 27 OF 142 |
| | DATE: 05/29/2026 | | | |

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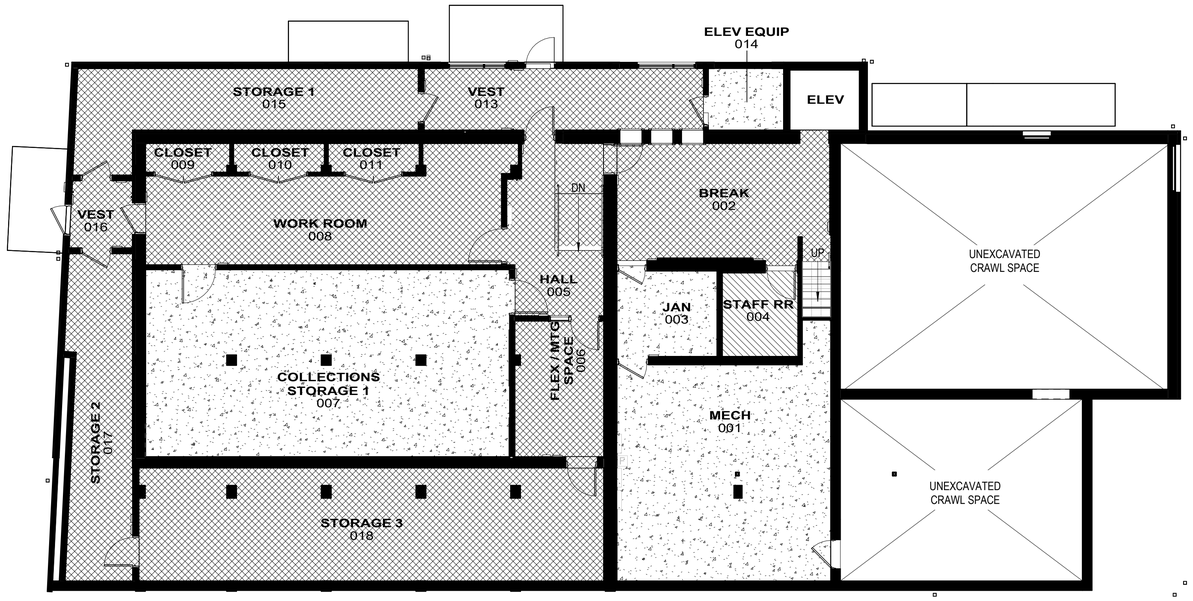
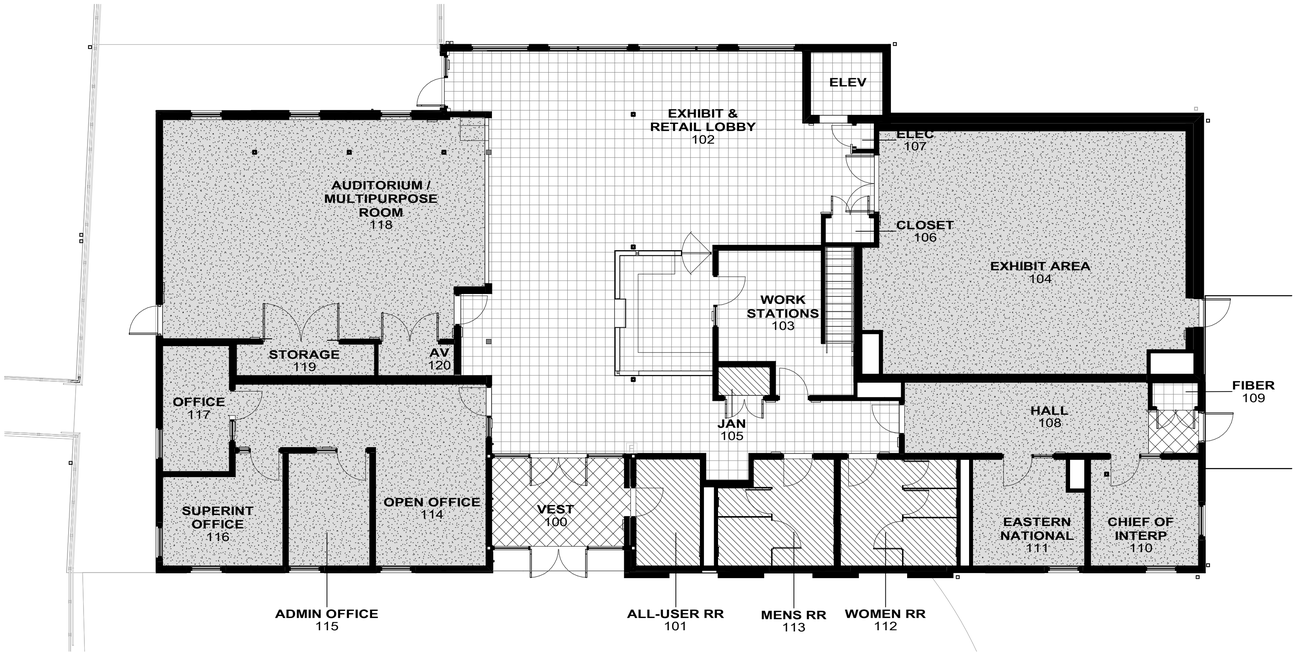



## 2 BASEMENT LEVEL - FINISH PLAN

A 131 SCALE A




## 1 MAIN LEVEL - FINISH PLAN

A 131 SCALE A

## FINISH PLAN LEGEND

- CPT-1 CARPET TILE
- RR-1 RUBBER TILE
- RR-1 ACOUSTIC RUBBER TILE
- SC-SEALED EXISTING CONC
- EF EPOXY FLOORING
- CPT-2 WOG WALK-OFF CPT

## GENERAL NOTES

1. ALL FLOOR FINISHES TO EXTEND UNDER CASEWORK AND EQUIP.
2. FOR ALL FLOOR FINISHES, PREPARE SUBSTRATE PER MFR'S RECOMMENDATIONS.
3. CLEAN <E> CONC. PATCH/ FILL ALL DIVOTS AND CRACKS WIDER THAN 1/4". PREP FOR NEW CLEAR, NON-SLIP SEALER WHERE NOTED ON FINIHS SCHEDULE.
4. PATCH AND REPAIR ALL FINISHES IMPACTED BY DEMO AND SYSTEMS INSTALL. MATCH ADJ FINISH UON.
5. PROVIDE MOLD & MOISTURE RESISTANT GWB AT ALL "WET" AREAS. PROVIDE EPOXY PAINT AT ALL EXPOSED "WET" WALLS. PAINT ALL EXPOSED GWB UON.
6. ABAAS COMPLIANT SS TRANSITIONS BETWEEN ALL FLOORING CHANGES. ALL FLOOR TRANSITIONS TO ALIGN WITH CENTER OF DOOR IN CLOSED POSITION.
7. PROVIDE CORNER GUARDS AT ALL OUTSIDE CORNERS OF GWB WALLS IN BASEMENT LEVEL UON.
8. RE: SPEC FOR PAINT SCHEDULE.
9. ALL EXPOSED GRILLES, VENT COVERS, ACCESS PANELS, ETC TO BE FIELD PAINTED TO BE SAME COLOR AS SURROUNDING WALL UON.
10. TYPICAL THROUGHOUT: NO WALL PNL CUTS TO BE LESS THAN 2" WIDE. CONFIRM/ COORDINATE WITH CO WHEN SMALL CUTS COULD OCCUR PRIOR TO INSTALL.
11. WHERE <N> FLOOR FINISH PROVIDED. <E> FLOOR FINISH TO BE REMOVED.
12. RE: MECH, PLUMB, AND ELEC SETS FOR AREAS WHERE <E> ELEMENTS ARE DEMOD AND REQUIRE PATCHING OF THE FLOOR. FINISH TO MATCH ADJ.
13. AT UNDERSIDE OF CONC FLOOR / ROOF STRUCTURES, **DO NOT** ANCHOR **ANY** ELEMENTS (DUCTING, PIPING, CONDUIT, ETC.) INTO THE <E> CMU PLANK FLOOR / ROOF STRUCTURE. ALL ATTACHMENTS SHALL BE INTO THE <E> CONC BEAMS SPACED AT 30" OC, RE: STRUC, MECH, PLUMB, AND FIRE. WHERE <N> SURFACE MOUNTED LIGHTING FIXTURES ARE PROVIDED, COORDINATE THESE LOCATIONS FOR INSTALLATION AND TO BE LOCATED ON <E> CONC BEAMS.
14. ALL WORK ON THIS SHEET IS OPTION C.

| CODE | FINISH |
|---|---|
| <E> | EXISTING FINISH TO REMAIN |
| OTS | OPEN TO STRUCTURE |
| CPT | CARPET TILE |
| RB | RUBBER BASE |
| SC | SEALED EXISTING CONC |
| GWB | PAINTED GWB |
| PT-1 | FIELD PAINT 1 |
| PT-2 | FIELD PAINT 2 |
| PT-3 | FIELD PAINT 3 (RESTROOMS) |
| ACT | 2X4 ACOUSTIC CEILING TILE (SCORED IN 2X2 PATTERN) |
| EP | EPOXY BASE |
| RR | RUBBER TILE |
| WOG | WALK-OFF CARPET |
| SS | SOLID SURFACE |

## ROOM FINISH SCHEDULE

| NUMBER | NAME | FLOOR FINISH | CEILING FINISH | NORTH WALL FINISH | NORTH WALL BASE | EAST WALL FINISH | EAST WALL BASE | SOUTH WALL FINISH | SOUTH WALL BASE | WEST WALL FINISH | WEST WALL BASE | COMMENTS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 001 | MECH | SC | OTS | CMU, PT-1 | - | CMU, PT-1 | - | CMU, PT-1 | - | CMU, PT-1 | - | |
| 002 | BREAK | RR-2 | GWB, PT-2 | GWB, PT-2 | RB | GWB, PT-2 | RB | GWB, PT-2 | RB | GWB, PT-2 | RB | |
| 003 | JAN | SC | GWB, PT-1 | CMU, PT-1 | - | CMU, PT-1 | - | CMU, PT-1 | - | CMU, PT-1 | - | |
| 004 | STAFF RR | EP | GWB, PT-3 | SS, PT-3 | EP | SS, PT-3 | EP | SS, PT-3 | EP | SS, PT-3 | EP | |
| 005 | HALL | RR-2 | GWB, PT-1 | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | |
| 006 | FLEX / MTG SPACE | RR-2 | GWB, PT-1 | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | |
| 007 | COLLECTIONS STORAGE 1 | SC | GWB, PT-1 | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | |
| 008 | WORK ROOM | RR-2 | GWB, PT-1 | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | |
| 009 | CLOSET | RR-2 | GWB, PT-1 | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | |
| 010 | CLOSET | RR-2 | GWB, PT-1 | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | |
| 011 | CLOSET | RR-2 | GWB, PT-1 | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | |
| 013 | VEST | RR-2 | GWB, PT-1 | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | - | |
| 014 | ELEV EQUIP | SC | OTS | CMU, PT-1 | RB | CMU, PT-1 | RB | CMU, PT-1 | RB | CMU, PT-1 | RB | REPAINT ALL CMU WALLS |
| 015 | STORAGE 1 | RR-2 | GWB, PT-1 | GWB, PT-1 | - | GWB, PT-1 | - | GWB, PT-1 | RB | GWB, PT-1 | -/RB | |
| 016 | VEST | RR-2 | GWB, PT-1 | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | - | GWB, PT-1 | RB | |
| 017 | STORAGE 2 | RR-2 | GWB, PT-1 | CMU, PT-1 | -/RB | CMU, PT-1 | RB | CMU, PT-1 | - | CMU, PT-1 | - | |
| 018 | STORAGE 3 | RR-2 | GWB, PT-1 | CMU, PT-1 | - | CMU, PT-1 | - | GWB, PT-1 | RB | CMU, PT-1 | - | |
| 100 | VEST | WOC | GWB, PT-1 | GWB, PT-1 | WD | GWB, PT-1 | - | GWB, PT-1 | - | GWB, PT-1 | - | |
| 101 | ALL-USER RR | EP | GWB, PT-3 | SS, PT-3 | EP | SS, PT-3 | EP | SS, PT-3 | EP | SS, PT-3 | EP | |
| 102 | EXHIBIT & RETAIL LOBBY | RR-1 | WD/GWB, PT-1 | GWB, PT-1 | WD | GWB, PT-1 | WD | GWB, PT-1 | WD | GWB, PT-1 | WD | |
| 103 | WORK STATIONS | RR-1 | ACT | GWB, PT-2 | WD | GWB, PT-2 | WD | GWB, PT-2 | WD | GWB, PT-2 | WD | |
| 104 | EXHIBIT AREA | CPT | ACT | GWB, PT-1 | WD | GWB, PT-1 | WD | GWB, PT-1 | WD | GWB, PT-1 | WD | |
| 105 | JAN | EP | GWB, PT-1 | GWB, PT-1 | EP | GWB, PT-1 | EP | GWB, PT-1 | EP | GWB, PT-1 | EP | |
| 106 | CLOSET | RR | GWB, PT-1 | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | |
| 107 | ELEC | RR | GWB, PT-1 | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | |
| 108 | HALL | CPT/WOC | ACT | GWB, PT-1 | WD | GWB, PT-1 | WD | GWB, PT-1 | WD | GWB, PT-1 | WD | |
| 109 | FIBER | RR | GWB, PT-1 | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | |
| 110 | CHIEF OF INTERP | CPT | ACT | GWB, PT-1 | WD | GWB, PT-1 | WD | GWB, PT-1 | WD | GWB, PT-1 | WD | |
| 111 | EASTERN NATIONAL | CPT | ACT | GWB, PT-1 | WD | GWB, PT-1 | WD | GWB, PT-1 | WD | GWB, PT-1 | WD | |
| 112 | WOMEN RR | EP | GWB, PT-3 | SS, PT-3 | EP | SS, PT-3 | EP | SS, PT-3 | EP | SS, PT-3 | EP | |
| 113 | MENS RR | EP | GWB, PT-3 | SS, PT-3 | EP | SS, PT-3 | EP | SS, PT-3 | EP | SS, PT-3 | EP | |
| 114 | OPEN OFFICE | CPT | ACT | GWB, PT-1 | WD | GWB, PT-1 | WD | GWB, PT-1 | WD | GWB, PT-1 | WD | |
| 115 | ADMIN OFFICE | CPT | ACT | GWB, PT-1 | WD | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | |
| 116 | SUPERINT OFFICE | CPT | ACT | GWB, PT-1 | WD | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | |
| 117 | OFFICE | CPT | ACT | GWB, PT-1 | WD | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | |
| 118 | AUDITORIUM / MULTIPURPOSE ROOM | CPT | GWB, PT-1 | GWB, PT-1 | WD | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | |
| 119 | STORAGE | CPT | ACT | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | |
| 120 | AV | CPT | ACT | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | GWB, PT-1 | RB | |

**NOTE:**
PROVIDE <N> ELEVATOR CAB FLOORING AND WORKSTATION 103 STAIR FLOORING, FINISH SHALL BE RR-1 RUBBER TILE.

SCALE: A 8 0 8 16
SCALE OF FEET

ELIZABETH HALLAS 07418 LICENSED PROFESSIONAL ARCHITECT IOWA

| | | | |
|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM CADD LB/MM TECH REVIEW: ACN / EMH DATE: 05/29/2026 | SUB SHEET NO. A 131 | TITLE OF SHEET FINISH PLAN EFFIGY MOUNDS NATIONAL MONUMENT |

DRAWING NO. 394 194248
PMIS/PKG NO. 318539
SHEET 28 OF 142

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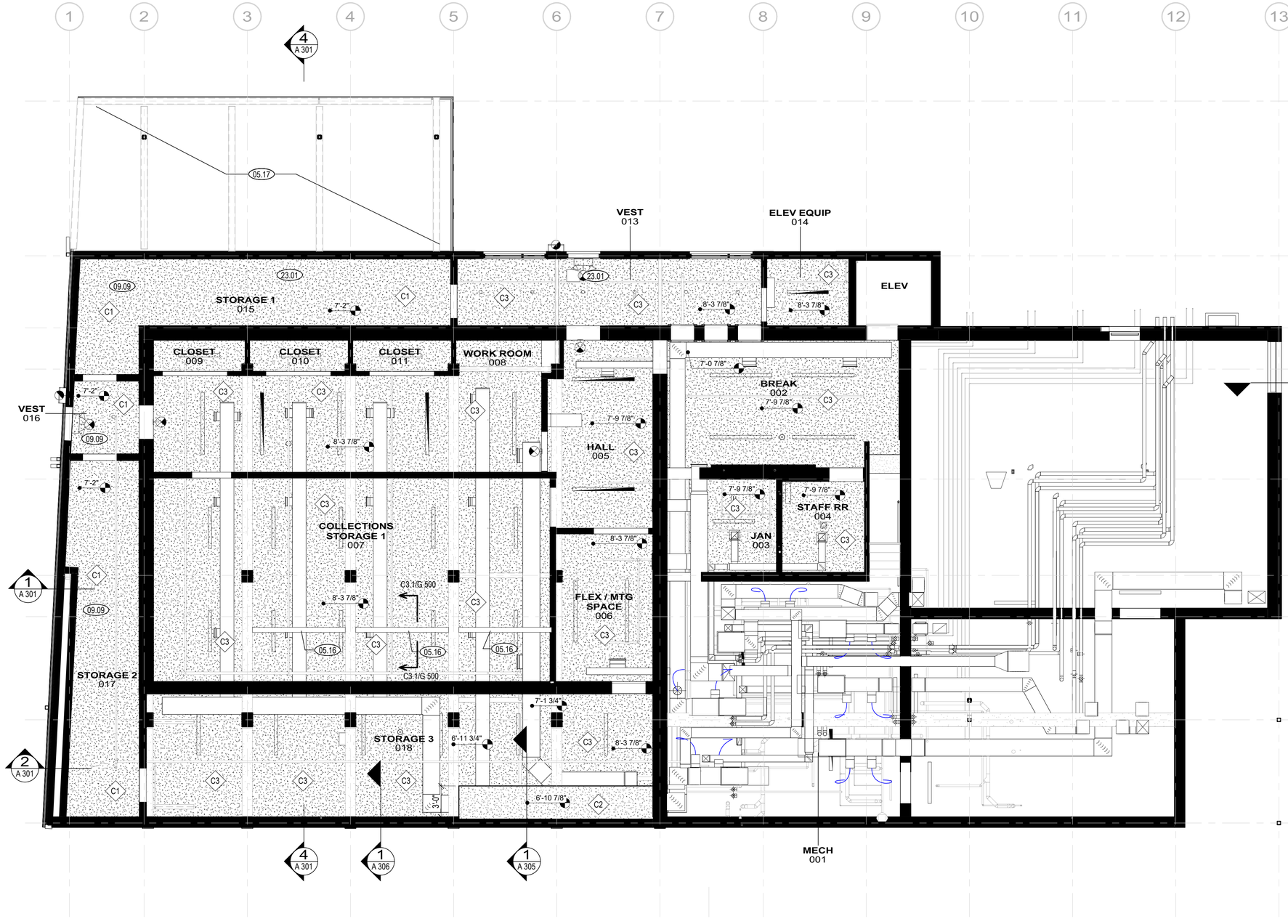


# GENERAL NOTES

1. RE: STRUC, MECH, PLUMB, ELEC, AND FIRE PROTECTION FOR MORE INFO.
2. ALL PLAN DIMS LOCATING EXT ASSEMBLIES ARE TO GRID LINE OR FACE OF FINISH, UNO. <E> WALLS WITHOUT WALL TAGS SHALL REMAIN.
3. VERIFY EQUIP ROUGH-IN DIMS WITH MFR FOR ALL EQUIP INCLUDING <E>, REUSED, OR FURNISHED BY CO.
4. PROVIDE ADEQUATE BLOCKING, BACKING OR STRUC SUPPORT AS REQ'D FOR PROPER INSTALLATION OF ALL MOUNTED ASSEMBLIES, INCLUDING ALL ATTACHED EQUIP, ACCESSORIES (CONTRACTOR FURNISHED ITEMS), PLUMB FIXTURES, MILLWORK, AND CASEWORK.
5. ALL CONDUIT & PIPING TO BE ROUTED WITHIN WALLS OR FURRING WALLS UON. CONTRACTOR TO COORDINATE IN FIELD ALL EXPOSED CONDUIT AND PIPING ROUTING WITH CO.
6. PATCH AND REPAIR ALL FINISHES IMPACTED BY SYSTEMS AND DEMO WORK RE: FINISH SCHEDULE.UON. MATCH ADJ IN MATERIAL, COLOR, TEXTURE AND FINISH.
7. <N> GWB CEILINGS IN THE BASEMENT SHALL BE INSTALLED BELOW A 1-1/2" MTL FURRING CHANNEL.
8. WHERE <N> CEILING FINISH PROVIDED. <E> CEILING FINISH TO BE REMOVED.
9. AT UNDERSIDE OF CONC FLOOR / ROOF STRUCTURES, **DO NOT** ANCHOR **ANY** ELEMENTS (DUCTING, PIPING, CONDUIT, ETC.) INTO THE <E> CMU PLANK FLOOR / ROOF STRUCTURE. ALL ATTACHMENTS SHALL BE INTO THE <E> CONC BEAMS SPACED AT 30" OC, RE: STRUC, MECH, PLUMB, AND FIRE. WHERE <N> SURFACE MOUNTED LIGHTING FIXTURES ARE PROVIDED, COORDINATE THESE LOCATIONS FOR INSTALLATION AND TO BE LOCATED ON <E> CONC BEAMS.

## KEYNOTES

| | |
|---|---|
| 05.16 | LOCATE <E> FLOOR CRACK, INSTALL STUCTURAL STL TUBE CONC PLANK REPAIR, REFER TO CEILING TYPE C3.1 / A500, AND STRUC FOR ADDITIONAL INFORMATION |
| 05.17 | OPTION G: EXPAND DECK AT NORTHEAST CORNER. ADDITION OF CONC OVER MTL DECK, STRUCTURAL STL, <N> MTL RAILINGS. |
| 09.09 | OPTION C: INSTALL FACED MINERAL WOOL BATT INSULATION BETWEEN <N> 6" CFS MTL JOIST AT 16" OC, INSTALL <N> 1/2" GWB RE: ASSEMBLIES C1/G500. |
| 23.01 | OPTION A: <N> MECHANICAL EQUIPMENT RE: MECH |

## LEGEND

- <N> GWB CLNG RE: FINISH SCHED
- <N> 2X4 ACT CLNG RE: FINISH SCHED
- <N> MTL SOFFIT CLADDING
- <N> MTL VENTED SOFFIT CLADDING
- REFINISHED <E> T&G WD CLNG RE: FINISH SCHED
- <N> PWD SHEATHING ABOVE ATTIC FLOOR LANDING
- <N> ATTIC ACCESS PANEL
- <N> DIFFUSER RE: MECH
- <E> 2X2 LIGHT FIXTURE RE: ELEC
- <N> CAN LIGHT RE: ELEC
- <N> WALL MOUNTED LIGHTING RE: ELEC
- <N> CEILING SPEAKER RE: FIRE
- <N> RECESSED SPRINKER (DOWN HEAD) RE: FIRE
- <N> SMOKE DETECTOR RE: FIRE
- <N> EXIT SIGN RE: ELEC
- <N> SECURITY CAMERA RE: ELEC
- <N> MTL ART RAIL

WORK IN THIS LEGEND INCLUDES OPTION A, C, AND G RE: G001 FOR MORE INFO.

1 / A 141 BASEMENT LEVEL - REFLECTED CEILING PLAN
SCALE A

SCALE: A 8 0 8 16
SCALE OF FEET

LICENSED PROFESSIONAL ARCHITECT ELIZABETH HALLAS 07416 IOWA

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|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA<br>SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM<br>DRAFTED: LB/MM<br>TECH REVIEW: ACN / EMH<br>DATE: 05/29/2026 | SUB SHEET NO. A 141 | TITLE OF SHEET: BASEMENT LEVEL RCP<br>EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248<br>PMIS/PKG NO. 318539<br>SHEET 29 OF 142 |

2 A 142 MAIN LEVEL - REFLECTED CEILING PLAN
SCALE (A)

## GENERAL NOTES

1. RE: STRUC, MECH, PLUMB, ELEC, AND FIRE PROTECTION FOR MORE INFO.
2. ALL PLAN DIMS LOCATING EXT ASSEMBLIES ARE TO GRID LINE OR FACE OF FINISH, UNO. <E> WALLS WITHOUT WALL TAGS SHALL REMAIN.
3. VERIFY EQUIP ROUGH-IN DIMS WITH MFR FOR ALL EQUIP INCLUDING <E>, REUSED, OR FURNISHED BY CO.
4. PROVIDE ADEQUATE BLOCKING, BACKING OR STRUC SUPPORT AS REQ'D FOR PROPER INSTALLATION OF ALL MOUNTED ASSEMBLIES, INCLUDING ALL ATTACHED EQUIP, ACCESSORIES (CONTRACTOR FURNISHED ITEMS), PLUMB FIXTURES, MILLWORK, AND CASEWORK.
5. ALL CONDUIT & PIPING TO BE ROUTED WITHIN WALLS OR FURRING WALLS UON. CONTRACTOR TO COORDINATE IN FIELD ALL EXPOSED CONDUIT AND PIPING ROUTING WITH CO.
6. PATCH AND REPAIR ALL FINISHES IMPACTED BY SYSTEMS AND DEMO WORK RE: FINISH SCHEDULE.UON. MATCH ADJ IN MATERIAL, COLOR, TEXTURE AND FINISH.
7. <N> GWB CEILINGS IN THE BASEMENT SHALL BE INSTALLED BELOW A 1-1/2" MTL FURRING CHANNEL.
8. WHERE <N> CEILING FINISH PROVIDED. <E> CEILING FINISH TO BE REMOVED.
9. AT UNDERSIDE OF CONC FLOOR / ROOF STRUCTURES, **DO NOT** ANCHOR **ANY** ELEMENTS (DUCTING, PIPING, CONDUIT, ETC.) INTO THE <E> CMU PLANK FLOOR / ROOF STRUCTURE. ALL ATTACHMENTS SHALL BE INTO THE <E> CONC BEAMS SPACED AT 30" OC, RE: STRUC, MECH, PLUMB, AND FIRE. WHERE <N> SURFACE MOUNTED LIGHTING FIXTURES ARE PROVIDED, COORDINATE THESE LOCATIONS FOR INSTALLATION AND TO BE LOCATED ON <E> CONC BEAMS.

## KEYNOTES

| | |
|---|---|
| 02.20 | OPTION C: REMOVE <E> ACT CEILING GRID AND CEILING ASSEMBLY TO <E> WD PLANK DECK; STRIP, SAND, AND STAIN <E> PLANK DECK, REMOVING ALL UNECESSARY FASTENINGS, FILL HOLES, AND REPORT ANY DAMAGE TO CO. EXPOSED <E> STL BEAMS AND COLUMNS TO BE PAINTED. EXPOSED WD BLOCKING ABOVE STL BEAMS TO BE FINISHED TO MATCH CEILING. CONSOLIDATE ALL <E> SURFACE MOUNTED CONDUIT INTO ONE DISCREET LOCATION, COORDINATE WITH CO. ALL EXPOSED DUCTWORK IN THIS AREA TO BE PAINTED. |
| 05.09 | OPTION B: <N> PREFINISHED MTL 4"x5" DOWNSPOUT RE: BUILDING ELEVATIONS |
| 05.10 | OPTION B: <N> HSS COLUMN IN LOCATION OF REMOVED <E> WD COLUMN RE: STRUC |
| 07.11 | OPTION B: AT EXT SOFFIT INSTALL <N> 6" WIDE PREFINISHED MTL CLADDING, INSTALL 6" WIDE VENTED CLADDING BOARD (3' LONG) AT EVERY 12TH PLANK ALIGNED TO EAVE EDGE UON |
| 07.12 | OPTION B: 6" WIDE VENTED PREFINISHED MTL CLADDING BOARD |
| 08.09 | OPTION C: <N> 24"X36" ACCESS PANEL TO ATTIC TO BE LOCATED IN THE ATTIC FLOOR ABOVE THE <N> ACT CLNG RE: STRUC |
| 09.06 | OPTION C: CLOSE <E> ACCESS TO ATTIC WITH <N> GWB BELOW ATTIC FLOOR |
| 09.12 | OPTION C: AT OFFICES, PROVIDE <N> 24" WIDE ACOUSTIC BATT ABOVE ACT ADJ TO INT PARTITION WALLS, OFFICE SIDE OF PARTITION ONLY |
| 12.06 | OPTION B: SURFACE MOUNT PREFINISHED MTL ART RAIL TO UNDERSIDE OF MTL SOFFIT AT THE TOP INTERSECTION WITH THE EXT WALL. |
| 12.07 | OPTION F: SURFACE MOUNT PREFINISHED MTL ART RAIL TO UNDERSIDE OF GWB AT THE TOP INTERSECTION WITH THE WALLS. |

## LEGEND

- <N> GWB CLNG RE: FINISH SCHED
- <N> 2X4 ACT CLNG RE: FINISH SCHED
- <N> MTL SOFFIT CLADDING
- <N> MTL VENTED SOFFIT CLADDING
- REFINISHED <E> T&G WD CLNG RE: FINISH SCHED
- <N> PWD SHEATHING ABOVE ATTIC FLOOR LANDING
- <N> ATTIC ACCESS PANEL
- <N> DIFFUSER RE: MECH
- <E> 2X2 LIGHT FIXTURE RE: ELEC
- <N> CAN LIGHT RE: ELEC
- <N> WALL MOUNTED LIGHTING RE: ELEC
- <N> CEILING SPEAKER RE: FIRE
- <N> RECESSED SPRINKER (DOWN HEAD) RE: FIRE
- <N> SMOKE DETECTOR RE: FIRE
- <N> EXIT SIGN RE: ELEC
- <N> SECURITY CAMERA RE: ELEC
- <N> MTL ART RAIL

WORK IN THIS LEGEND INCLUDES OPTION A, C, AND G RE: G001 FOR MORE INFO.

SCALE: (A) 4 0 4 8 12
SCALE OF FEET

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA; SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM; CADD: LB/MM; TECH REVIEW: ACN / EMH; DATE: 05/29/2026 | SUB SHEET NO. A 142 | TITLE OF SHEET: MAIN LEVEL RCP; EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248; PMIS/PKG NO. 318539; SHEET 30 OF 142 |

LICENSED PROFESSIONAL ARCHITECT — ELIZABETH HALLAS — 07416 — IOWA

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## ATTIC PLAN LEGEND

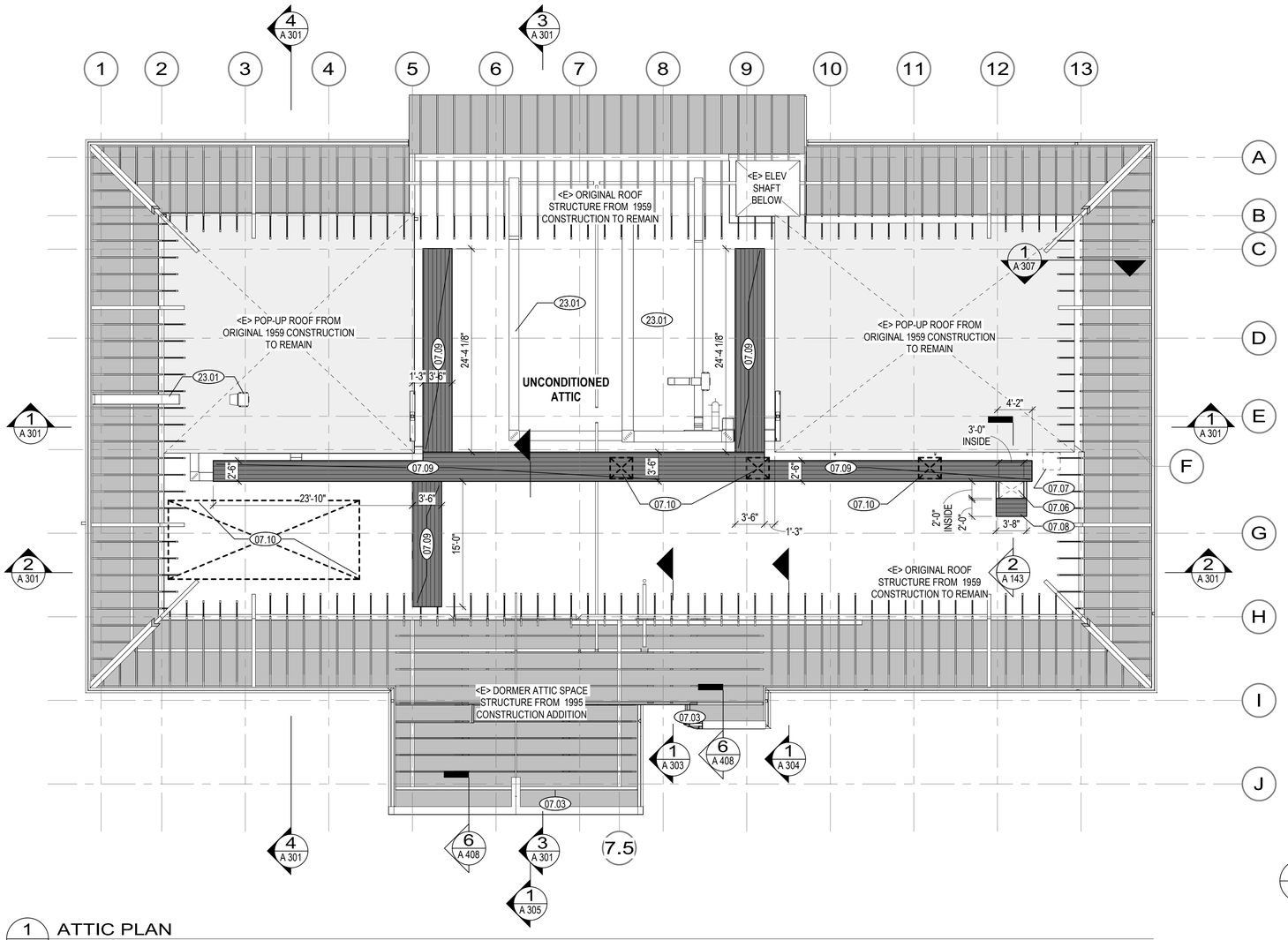
- RAISED AREA OVER AUDITORIUM AND EXHIBITS SPACE BELOW
- <N> RAISED CATWALK WITH 2X PLANK DIRECTION
- <N> MTL SOFFIT BELOW RE: REFLECTED CEILING PLAN
- <E> ROOF STRUCTURE TO REMAIN
- <N> MECHANICAL EQUIPMENT RE: MECH

## GENERAL NOTES

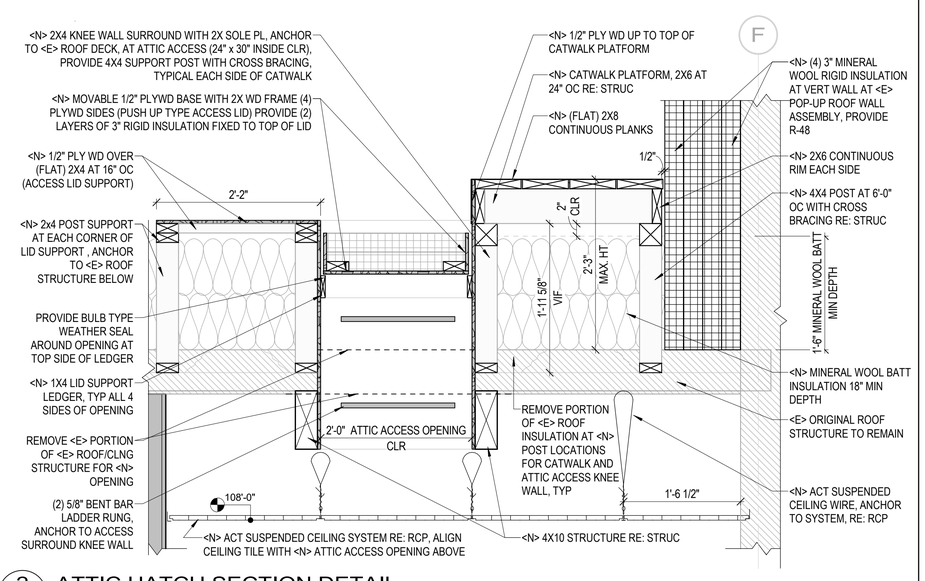
1. RE: CIVIL, LAND, STRUC, MECH, PLUMB, ELEC, AND FIRE PROTECTION FOR MORE INFO.
2. FF: MAIN LEVEL 100'-0" = NAVD88 657.3' +/- AT EXT DOORS.
3. ALL PLAN DIMS LOCATING EXT ASSEMBLIES ARE TO GRID LINE OR FACE OF FINISH, UNO. <E> WALLS WITHOUT WALL TAGS SHALL REMAIN.
4. VERIFY EQUIP ROUGH-IN DIMS WITH MFR FOR ALL EQUIP INCLUDING <E>, REUSED, OR FURNISHED BY CO.
5. PROVIDE ADEQUATE BLOCKING, BACKING, OR STRUC SUPPORT AS REQ'D FOR PROPER INSTALLATION OF ALL MOUNTED ASSEMBLIES, INCLUDING ALL ATTACHED EQUIP, ACCESSORIES (CONTRACTOR FURNISHED ITEMS), PLUMB FIXTURES, MILLWORK, AND CASEWORK.
6. FLOOR DRAIN LOCATIONS SHOWN ARE APPROXIMATE. VIF EXACT LOCATIONS PER PLUMB WITH STRUC COORDINATION. SLOPE CONC TO DRAIN AS INDICATED.
7. ALL CONDUIT & PIPING TO BE ROUTED WITHIN WALLS OR FURRING WALLS UON. CONTRACTOR TO COORDINATE IN FIELD ALL EXPOSED CONDUIT AND PIPING ROUTING WITH CO.
8. <N> PATHS AND SITE ELEMENTS ARE SHOWN FOR REFERENCE ONLY RE: CIVIL & LAND FOR SITE ELEMENTS AND CONTINUATION OF ABA ROUTES.
9. DOORS AND STOREFRONTS OCCUPY <E> FRAMED OPENINGS UON. VIF DIMS IN FIELD. ALIGN ALL <N> ROUGH OPENINGS WITH <E> MASONRY JOINTS UON: RE:STOREFRONT & DOOR SCHEDULE FOR DIMS AND MORE INFO.
10. PATCH AND REPAIR ALL FINISHES IMPACTED BY SYSTEMS AND DEMO WORK RE: FINISH SCHEDULE UON. MATCH ADJ IN MATERIAL, COLOR, TEXTURE AND FINISH.
11. FURNITURE SHOWN IS FOR REFERENCE ONLY AND IS NOT INCLUDED IN THE CONTRACT UON.
12. AT UNDERSIDE OF CONC FLOOR / ROOF STRUCTURES, **DO NOT** ANCHOR **ANY** ELEMENTS (DUCTING, PIPING, CONDUIT, ETC.) INTO THE <E> CMU PLANK FLOOR / ROOF STRUCTURE, ALL ATTACHMENTS SHALL BE INTO THE <E> CONC BEAMS SPACED AT 30" OC, RE: STRUC, MECH, PLUMB, AND FIRE. WHERE <N> SURFACE MOUNTED LIGHTING FIXTURES ARE PROVIDED, COORDINATE THESE LOCATIONS FOR INSTALLATION AND TO BE LOCATED ON <E> CONC BEAMS.

## KEYNOTES

| | |
|---|---|
| 07.03 | OPTION B: 4" WIDE HORIZ CHANNEL MTL CLADDING |
| 07.06 | OPTION B: REMOVE PORTION OF <E> LOWER 1959 ROOF STRUCTURE FOR <N> 24"X36" (CLR) ATTIC ACCESS OPENING, INSTALL <N> 2x4 KNEE WALL SURROUND (4) SIDES, WITH PUSH THRU ACCESS PNL RE: 2/A143 & STRUC FOR <N> STRUCTURE AT OPENING. |
| 07.07 | OPTION B: INFILL <E> ATTIC ACCESS OPENING WITH 2X6 FRAMING, PLYWD TOP (ATTIC SIDE) AND GWB TO MATCH ADJ (CEILING SIDE). |
| 07.08 | OPTIONBC: INSTALL <N> ATTIC ACCESS LID SUPPORT PLATFORM RE: 2/A143 |
| 07.09 | OPTION B: INSTALL <N> CATWALK PLATFORM IN ATTIC, PLATFORM TO BE CONSTRUCTED WITH 2X6 PLANKS OVER 2X8 WD JOIST AT 16" OC. PROVIDE 4X4 WD POST WITH 2X12X12" SQ, BASE PLATE AT 5'-0" OC ON EACH SIDE OF CATWALK, TO SIT ON <E> LOWER 1959 ROOF STRUCTURE RE: 2/A143 |
| 07.10 | OPTION B: CONFIRM OUTLINE LOCATIONS OF 1995 SKYLIGHT BLOCKING INFILL, ENSURE BLOCKING AND INFILL ARE STRUCTURALLY SOUND, IN ADDITION PROVIDE LOCATION FLAGS AT EACH CORNER OF ORIGINAL SKYLIGHT LOCATIONS, FLAGS TO EXTEND ABOVE <N> BATT INSULATION AND CLEARLY INDICATE LOCATIONS OF THE ORIGINAL SKYLIGHT LOCATIONS ONCE SCOPE IS COMPLETE |
| 23.01 | OPTION A: <N> MECHANICAL EQUIPMENT RE: MECH |




1 A 143 ATTIC PLAN
SCALE (A)




2 A 143 ATTIC HATCH SECTION DETAIL
SCALE (B)

SCALE: (B) 1 0 1 2 SCALE OF FEET

SCALE: (A) 8 0 8 16 SCALE OF FEET

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| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM DRAWN: LB/MM TECH REVIEW: ACN / EMH DATE: 05/29/2026 | SUB SHEET NO. A 143 | TITLE OF SHEET ATTIC PLAN EFFIGY MOUNDS NATIONAL MONUMENT |

LICENSED PROFESSIONAL ARCHITECT ELIZABETH HALLAS 07418 IOWA

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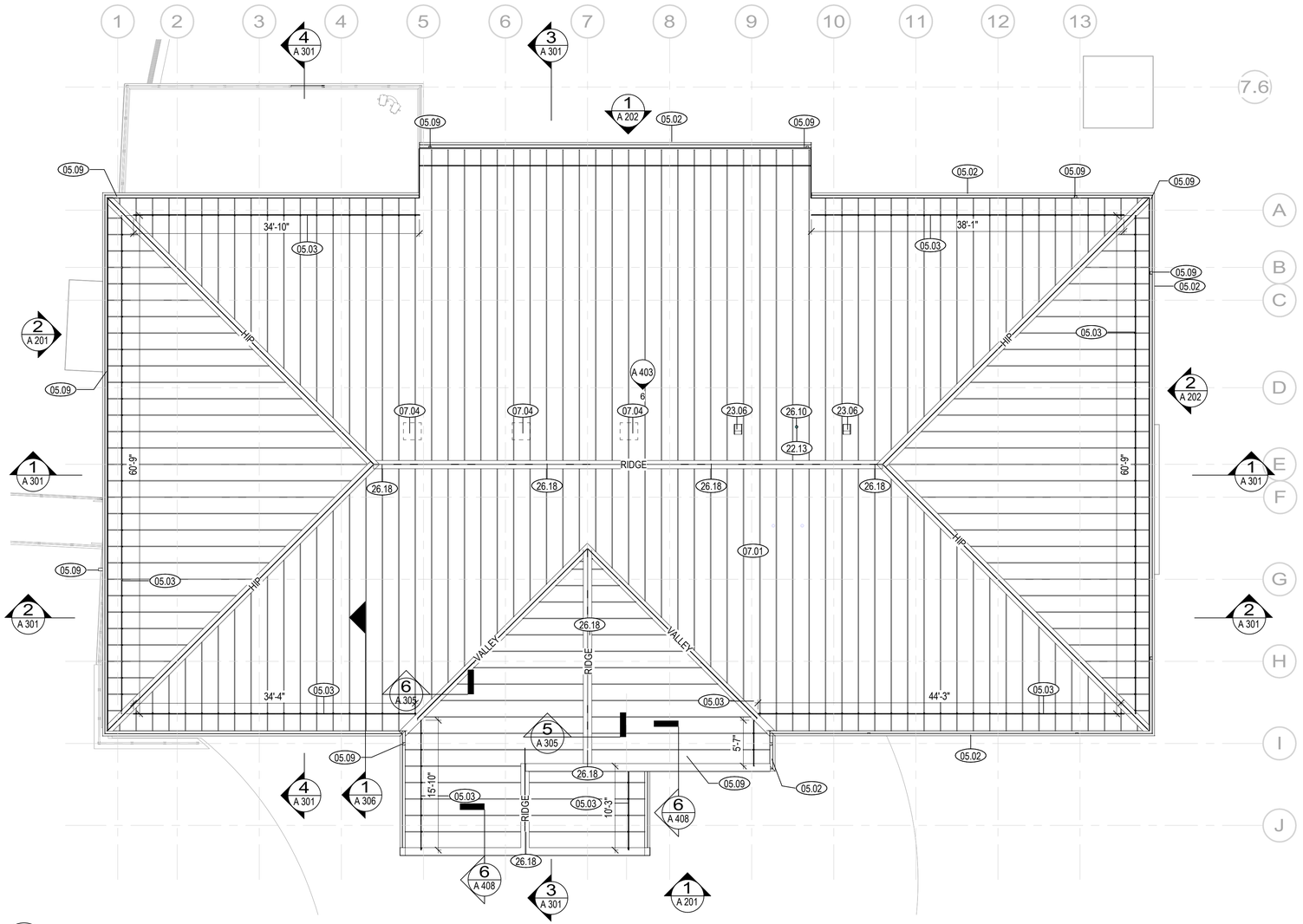
| DRAWING NO. | 394 194248 |
| PMIS/PKG NO. | 318539 |
| SHEET | 31 OF 142 |




1 / A 151 ROOF PLAN
SCALE A

## GENERAL NOTES

1. ROOF SLOPES ARE APPROXIMATE AS MEASURED IN FIELD. MAINTAIN EXISTING ROOF SLOPES.
2. RE: CIVIL, LAND, STRUC, MECH, PLUMB, ELEC, AND FIRE PROTECTION FOR MORE INFO.
3. PROVIDE ADEQUATE BLOCKING, BACKING OR STRUC SUPPORT AS REQ'D FOR PROPER INSTALLATION OF ALL MOUNTED ASSEMBLIES, INCLUDING ALL ATTACHED EQUIP.
4. <N> PATHS AND SITE ELEMENTS ARE SHOWN FOR REFERENCE ONLY RE: CIVIL & LAND FOR SITE ELEMENTS AND CONTINUATION OF ADA ROUTES.
5. PATCH AND REPAIR ALL FINISHES IMPACTED BY SYSTEMS AND DEMO WORK RE: FINISH SCHEDULE UON. MATCH ADJ IN MATERIAL, COLOR, TEXTURE AND FINISH.
6. <N> 18" MINERAL WOOL BATT INSULATION AT ATTIC SIDE OF <E> CEILING

## KEYNOTES

| | |
|---|---|
| 05.02 | OPTION B: <N> PREFINISHED MTL CUSTOM 6" FASCIA-STYLE GUTTER |
| 05.03 | OPTION B: <N> PREFINISHED MTL SNOWGUARD |
| 05.09 | OPTION B: <N> PREFINISHED MTL 4"x5" DOWNSPOUT RE: BUILDING ELEVATIONS |
| 07.01 | OPTION B: PREFINISHED STANDING SEAM MTL ROOF |
| 07.04 | OPTION A: PATCH <E> PLYWD SHEATHING WHERE MECH EQUIP REMOVED IN-KIND |
| 22.13 | OPTION C: <N> STACK VENT, REUSE <E> PENETRATION RE: PLUMB |
| 23.06 | OPTION A: <N> MECH VENT, REUSE <E> PENETRATION RE: MECH |
| 26.10 | |
| 26.18 | OPTION E: <N> LIGHTNING PROTECTION RE: ELEC |

## LEGEND

PREFINISHED STANDING SEAM MTL ROOFING

SCALE: A 8 0 8 16
SCALE OF FEET

LICENSED PROFESSIONAL ARCHITECT ELIZABETH HALLAS 07416 IOWA

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|---|---|---|---|---|---|
| A/E FIRM: | DESIGNED: | SUB SHEET NO. | TITLE OF SHEET | DRAWING NO. | |
| PRIME: OTAK, INC. REDMOND, WA | LB/CM | | ROOF PLAN | 394 / 194248 | |
| SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DRAWN: LB/MM | A 151 | | PMIS/PKG NO. 318539 | |
| | TECH REVIEW: ACN / EMH | | EFFIGY MOUNDS NATIONAL MONUMENT | SHEET 32 OF 142 | |
| | DATE: 05/29/2026 | | | | |

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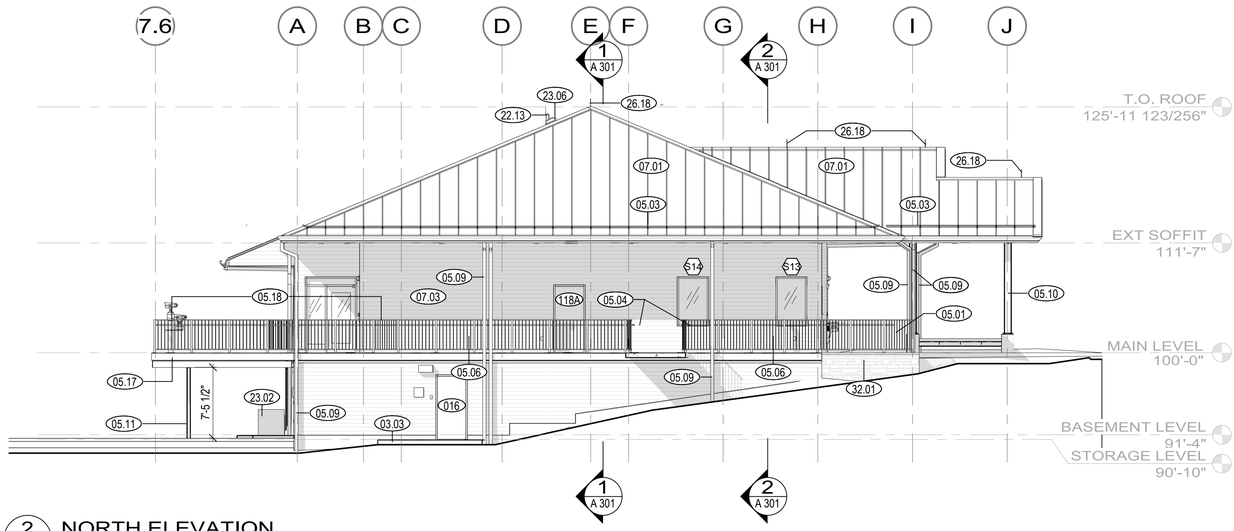
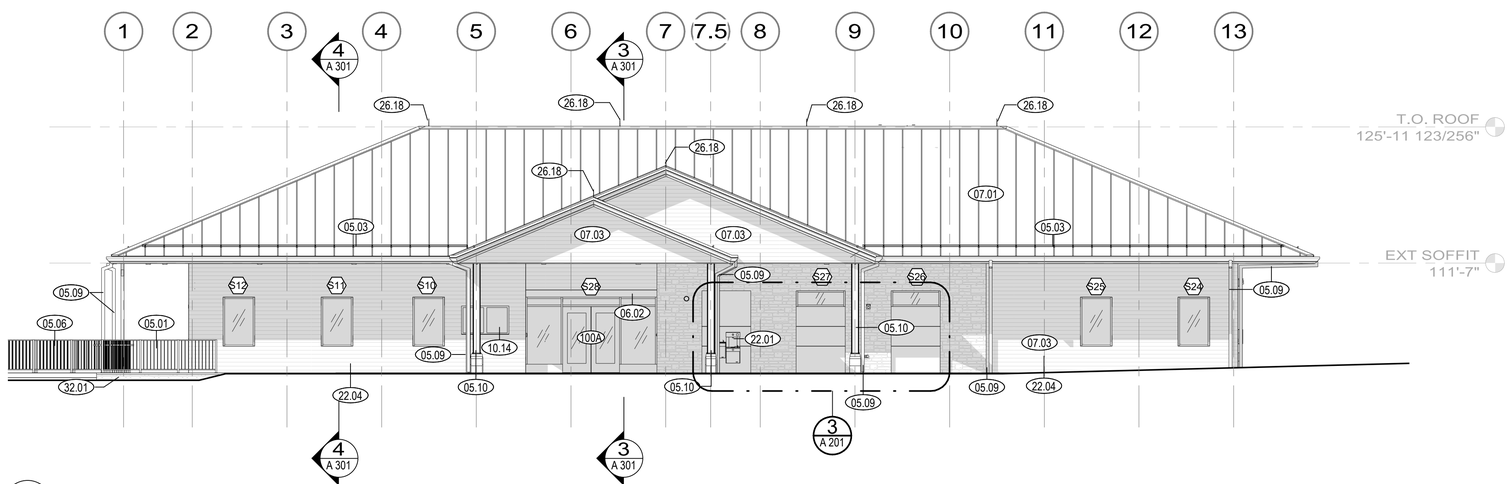



2 A 201 NORTH ELEVATION
SCALE A




1 A 201 WEST ELEVATION
SCALE A

## LEGEND

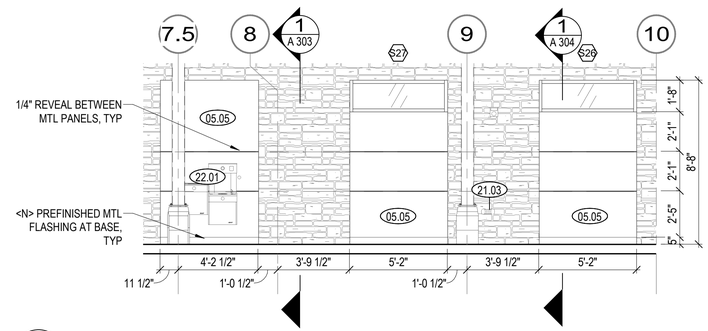
- <N> PREFINISHED STANDING SEAM MTL ROOFING
- <N> CHANNEL MTL CLADDING
- SALVAGED STONE VENEER

# GENERAL NOTES

1. RE: CIVIL, LAND, STRUC, MECH, PLUMB, ELEC, AND FIRE PROTECTION FOR MORE INFO.
2. FF: MAIN LEVEL 100'-0" = NAVD88 657.3' +/- AT EXT DOORS.
3. ALL PLAN DIMS LOCATING EXT ASSEMBLIES ARE TO GRID LINE OR FACE OF FINISH, UNO. <E> WALLS WITHOUT WALL TAGS SHALL REMAIN.
4. VERIFY EQUIP ROUGH-IN DIMS WITH MFR FOR ALL EQUIP INCLUDING <E>, REUSED, OR FURNISHED BY CO.
5. <N> PATHS AND SITE ELEMENTS ARE SHOWN FOR REFERENCE ONLY RE: CIVIL & LAND FOR SITE ELEMENTS AND CONTINUATION OF ABA ROUTES.
6. DOORS AND STOREFRONTS OCCUPY <E> FRAMED OPENINGS UON. VIF DIMS IN FIELD. ALIGN ALL <N> ROUGH OPENINGS WITH <E> MASONRY JOINTS UON RE:STOREFRONT & DOOR SCHEDULE FOR DIMS AND MORE INFO.
7. PATCH AND REPAIR ALL FINISHES IMPACTED BY SYSTEMS AND DEMO WORK RE: FINISH SCHEDULE UON. MATCH ADJ IN MATERIAL, COLOR, TEXTURE AND FINISH.
8. RE: ROOFPLAN FOR GUTTER AND DOWNSPOUT LOCATIONS.

# KEYNOTES

| | |
|---|---|
| 03.03 | OPTION B: <N> EXT CONC SLAB ON GRADE LANDING RE: MECH & STRUC |
| 05.01 | OPTION B: <N> VERTICAL PICKET STYLE MTL RAILING TO MATCH <E> (PT) WITH MTL HANDRAIL |
| 05.03 | OPTION B: <N> PREFINISHED MTL SNOWGUARD |
| 05.04 | OPTION B: <N> 1-1/2" DIA. MTL HANDRAIL AT 36" AFF, TO TOP OF HANDRAIL, ATTACHED TO <E> MTL RAILING |
| 05.05 | OPTION B: <N> PREFINISHED MTL PANEL |
| 05.06 | OPTION B: PT <E> MTL RAILING |
| 05.09 | OPTION B: <N> PREFINISHED MTL 4"x5" DOWNSPOUT RE: BUILDING ELEVATIONS |
| 05.10 | OPTION B: <N> HSS COLUMN IN LOCATION OF REMOVED <E> WD COLUMN RE: STRUC |
| 05.11 | OPTION G: <N> STL COLUMN RE: STRUC |
| 05.17 | OPTION G: EXPAND DECK AT NORTHEAST CORNER. ADDITION OF CONC OVER MTL DECK, STRUCTURAL STL, <N> MTL RAILINGS. |
| 05.18 | OPTION G: <N> VERT PICKET STYLE MTL RAILING TO MATCH <E> (PT) WITH MTL HANDRAIL AT 36" AFF (STN) OPTION G AT EAST DECK EXTENSION ONLY |
| 06.02 | OPTION B: <N> WD HEADER |
| 07.01 | OPTION B: PREFINISHED STANDING SEAM MTL ROOF |
| 07.03 | OPTION B: 4" WIDE HORIZ CHANNEL MTL CLADDING |
| 10.14 | OPTION B: REINSTALLATION SALVAGED BULLETIN BOARD, MAINTAIN INSULATION AND AIR BARRIER OF ASSEMBLY |
| 21.03 | OPTION A: <N> DOUBLE-HEAD FIRE DEPARTMENT CONNECTION WHERE <E> REMOVED RE: FIRE PROTECTION |
| 22.01 | OPTION C: <N> EXT WALL MOUNTED WATER FOUNTAIN WITH BOTTLE FILLER RE: PLUMB |
| 22.04 | OPTION C: <N> HOSE BIB RE: PLUMB |
| 22.13 | OPTION C: <N> STACK VENT, REUSE <E> PENETRATION RE: PLUMB |
| 23.02 | OPTION A: <N> CONDENSING UNIT RE: MECH |
| 23.06 | OPTION A: <N> MECH VENT, REUSE <E> PENETRATION RE: MECH |
| 26.18 | OPTION E: <N> LIGHTNING PROTECTION RE: ELEC |
| 32.01 | <N> STONE RETAINING WALL RE: LAND |




3 A 201 METAL PANEL ALCOVE ELEV
SCALE A

SCALE: A 8 0 8 16
SCALE OF FEET

LICENSED PROFESSIONAL ARCHITECT ELIZABETH HALLAS 07418 IOWA

| | | | | | |
|---|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM DRAWN: LB/MM TECH REVIEW: ACN / EMH DATE: 05/29/2026 | SUB SHEET NO. A 201 | TITLE OF SHEET BUILDING ELEVATIONS EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 33 OF 142 | |

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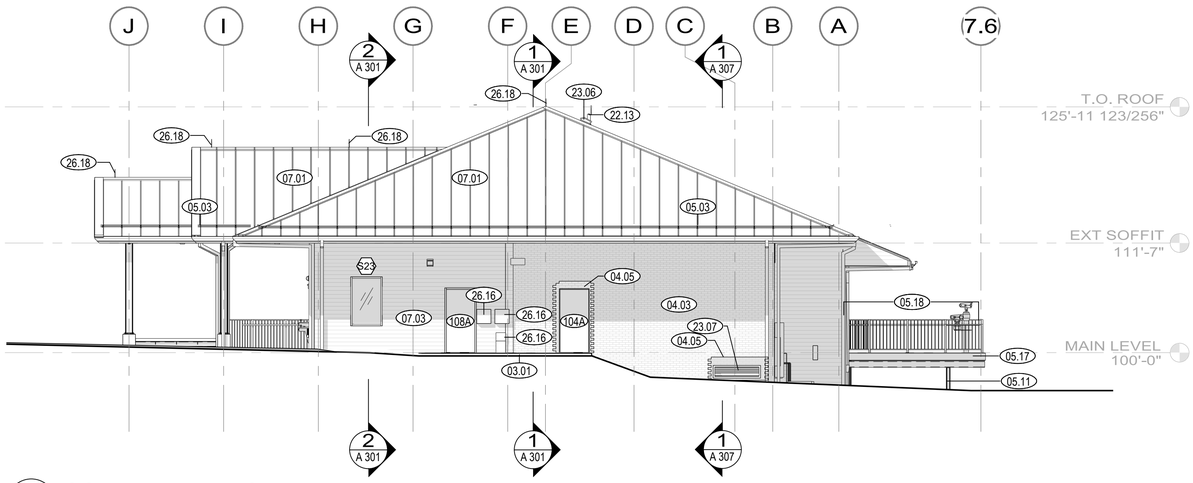
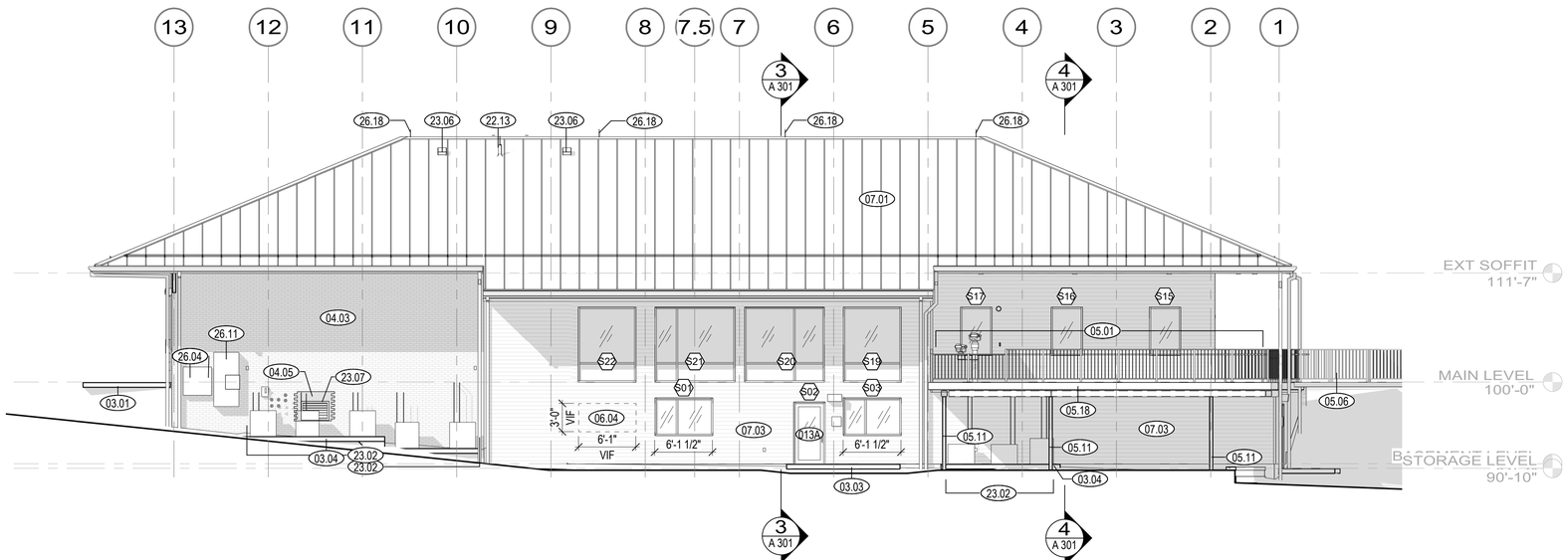



2 A 202 SOUTH ELEVATION
SCALE A




1 A 202 EAST ELEVATION
SCALE A

## LEGEND

<N> PREFINISHED STANDING SEAM MTL ROOFING

<N> CHANNEL MTL CLADDING

## BRICK REHABILITATION GENERAL NOTES

1. ALL BRICK VENEER TO BE CLEANED OF ENVIRONMENTAL AND BIOLOGICAL STAINING. TWO CLEANINGS NEEDED: ONE BEFORE SELECTION OF MATCHING BRICK, AND ONE FINAL CLEANING AFTER OTHER CONSTRUCTION THAT MIGHT DAMAGE OR DIRTY BRICK IS COMPLETED.
2. REPOINT LOOSE OR MISSING MORTAR WITH <N> MORTAR MATCHING <E> IN COLOR, TEXTURE, AND TYPE. REPOINTING SHALL OCCUR AFTER DEMOLITION TO ENSURE IMPACT OF DEMO ACTIVITIES ARE ADDRESSED IN REPOINTING EFFORT. CONTRACTOR TO ASSUME ALLOWANCE OF 25% OF THE <E> BRICK WALLS TO BE REPOINTED ON THE PROJECT.
3. REPAIR CRACKED OR DAMAGED BRICKS PER SPECIFICATIONS. CONTRACTOR TO ASSUME AN ALLOWANCE OF 50 SQUARE INCHES OF PATCHING AND 50 LINEAL INCHES OF CRACK REPAIR ON THE PROJECT.
4. REPLACE INDIVIDUAL BRICKS WHERE DAMAGED BEYOND REPAIR/PATCHING. CONTRACTOR TO ASSUME ALLOWANCE OF 5 REPLACED BRICKS ON THE PROJECT, SEPARATELY FROM THE AREAS OF REINSTALLED SALVAGED BRICK ON ELEVATIONS AND DETAILS.
5. REPORT DAMAGE OVER AND ABOVE THESE ALLOWANCES TO THE CONTRACTING OFFICER PRIOR TO WORK COMMENCEMENT ON THIS SCOPE.

## GENERAL NOTES

1. RE: CIVIL, LAND, STRUC, MECH, PLUMB, ELEC, AND FIRE PROTECTION FOR MORE INFO.
2. FF: MAIN LEVEL 100'-0" = NAVD88 657.3' +/- AT EXT DOORS.
3. ALL PLAN DIMS LOCATING EXT ASSEMBLIES ARE TO GRID LINE OR FACE OF FINISH, UNO. <E> WALLS WITHOUT WALL TAGS SHALL REMAIN.
4. VERIFY EQUIP ROUGH-IN DIMS WITH MFR FOR ALL EQUIP INCLUDING <E>, REUSED, OR FURNISHED BY CO.
5. <N> PATHS AND SITE ELEMENTS ARE SHOWN FOR REFERENCE ONLY RE: CIVIL & LAND FOR SITE ELEMENTS AND CONTINUATION OF ABA ROUTES.
6. DOORS AND STOREFRONTS OCCUPY <E> FRAMED OPENINGS UON. VIF DIMS IN FIELD. ALIGN ALL <N> ROUGH OPENINGS WITH <E> MASONRY JOINTS UON RE:STOREFRONT & DOOR SCHEDULE FOR DIMS AND MORE INFO.
7. PATCH AND REPAIR ALL FINISHES IMPACTED BY SYSTEMS AND DEMO WORK RE: FINISH SCHEDULE UON. MATCH ADJ IN MATERIAL, COLOR, TEXTURE AND FINISH.
8. RE: ROOFPLAN FOR GUTTER AND DOWNSPOUT LOCATIONS.

## KEYNOTES

| | |
|---|---|
| 03.01 | OPTION B: EXTEND EXT CONC LANDING TO ACCOMMODATE <N> EXT DOOR RE: CIVIL/LA |
| 03.03 | OPTION B: <N> EXT CONC SLAB ON GRADE LANDING RE: MECH & STRUC |
| 03.04 | OPTION A: <N> CONC PAD FOR <N> MECH UNITS RE: MECH & STRUC |
| 04.03 | OPTION B: <E> EXT MASONRY VENEER, CLEAN AND REPAIR PER BRICK REHABILITATION GENERAL NOTES |
| 04.05 | OPTION A: INFILL <E> MASONRY WITH SALVAGED MASONRY (SHOWN IN GRAY) AT REMOVED AREAS (FOR <N> OPENING), KEY IN SALVAGED MASONRY INTO <E> MASONRY, PROVIDE <N> GROUT TO MATCH <E> ADJ GROUT AND INSTALL <N> STL LENTILS RE: STRUC. FOR INFORMATION AT MECHANICAL LOUVER LOCATIONS RE: DETAILS 6, 7/A504 |
| 05.01 | OPTION B: <N> VERTICAL PICKET STYLE MTL RAILING TO MATCH <E> (PT) WITH MTL HANDRAIL |
| 05.03 | OPTION B: <N> PREFINISHED MTL SNOWGUARD |
| 05.06 | OPTION B: PT <E> MTL RAILING |
| 05.11 | OPTION G: <N> STL COLUMN RE: STRUC |
| 05.17 | OPTION G: EXPAND DECK AT NORTHEAST CORNER. ADDITION OF CONC OVER MTL DECK, STRUCTURAL STL, <N> MTL RAILINGS. |
| 05.18 | OPTION G: <N> VERT PICKET STYLE MTL RAILING TO MATCH <E> (PT) WITH MTL HANDRAIL AT 36" AFF (STN) OPTION G AT EAST DECK EXTENSION ONLY |
| 06.04 | OPTION B: INFILL <E> WINDOW WITH <N> FRAMING; PROVIDE FULL-DEPTH MINERAL WOOL BATT INSULATION; PROVIDE <N> EXTERIOR SHEATHING, SAME THICKNESS AS <E> RE: ASSEMBLIES |
| 07.01 | OPTION B: PREFINISHED STANDING SEAM MTL ROOF |
| 07.03 | OPTION B: 4" WIDE HORIZ CHANNEL MTL CLADDING |
| 22.13 | OPTION C: <N> STACK VENT, REUSE <E> PENETRATION RE: PLUMB |
| 23.02 | OPTION A: <N> CONDENSING UNIT RE: MECH |
| 23.06 | OPTION A: <N> MECH VENT, REUSE <E> PENETRATION RE: MECH |
| 23.07 | OPTION A: <N> AIR LOUVER RE: DTLS 3, 4 / A506 & MECH |
| 26.04 | OPTION C: <N> ELEC EQUIP RE: ELEC |
| 26.11 | OPTION C: <N> MAIN DISTRIBUTION CENTER PANEL RE: ELEC |
| 26.16 | OPTION C: <E> ELEC EQUIP TO REMAIN RE: ELEC |
| 26.18 | OPTION E: <N> LIGHTNING PROTECTION RE: ELEC |

SCALE: A 8 0 8 16
SCALE OF FEET

LICENSED PROFESSIONAL ARCHITECT ELIZABETH HALLAS 07418 IOWA

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|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM DRAWN: LB/MM TECH REVIEW: ACN / EMH DATE: 05/29/2026 | SUB SHEET NO. A 202 | TITLE OF SHEET: BUILDING ELEVATIONS<br>EFFIGY MOUNDS NATIONAL MONUMENT |

DRAWING NO. 394 / 194248
PMIS/PKG NO. 318539
SHEET 34 OF 142

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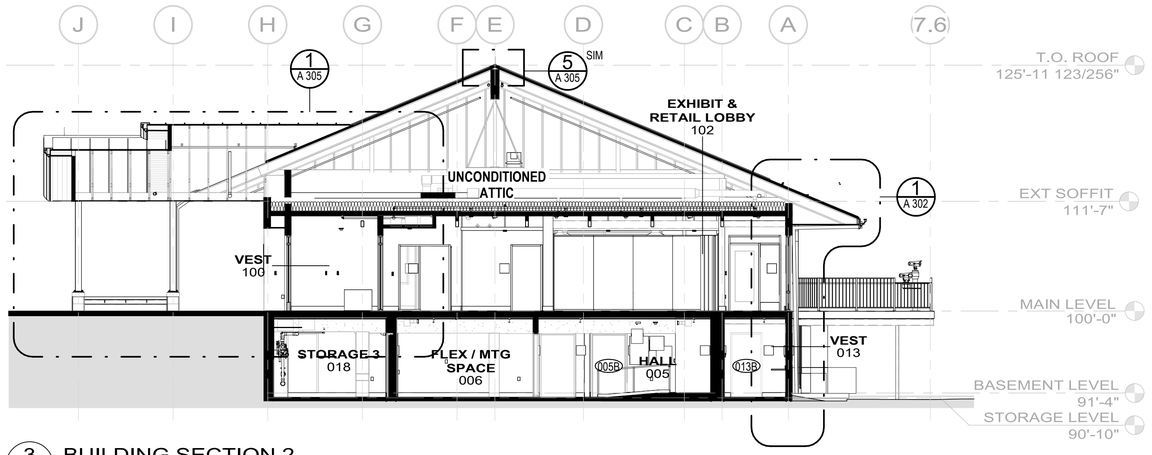
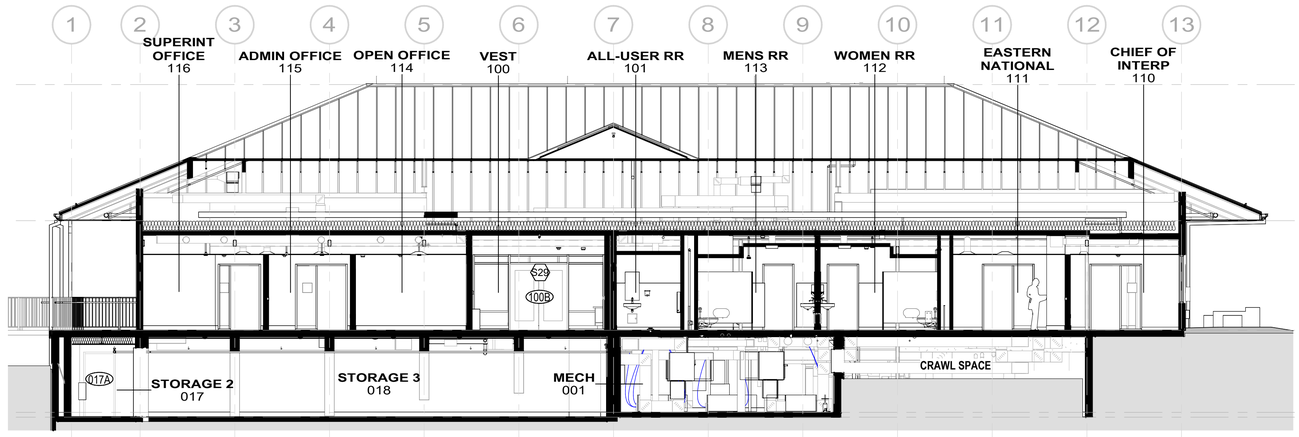
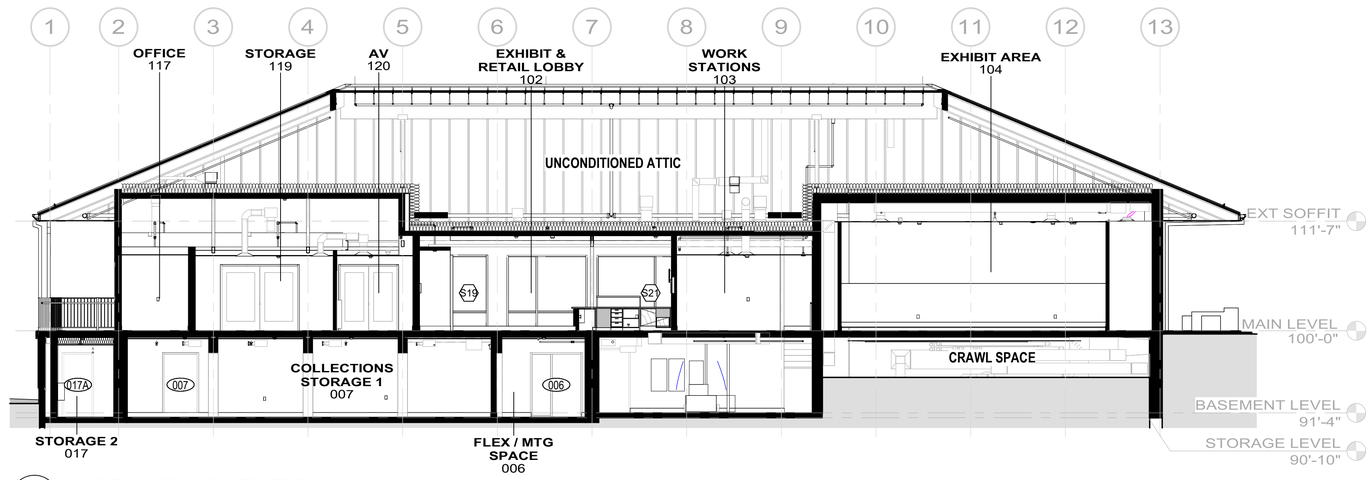


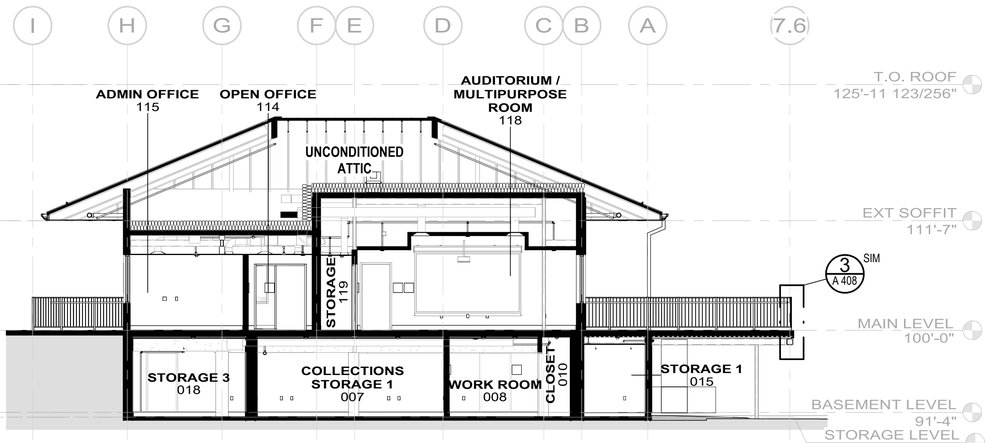

3 A 301 BUILDING SECTION 2
SCALE (A)




2 A 301 N/S BUILDING SECTION 1
SCALE (A)




1 A 301 N/S BUILDING SECTION 2
SCALE (A)




4 A 301 E/W BUILDING SECTION 3
SCALE (A)

## GENERAL NOTES

1. BUILDING SECTIONS INTENDED TO SHOW GENERAL LAYOUT AND RELATIONSHIP BETWEEN SPACES RE: FLOOR PLANS, RCP, INTERIOR ELEVATIONS, AND WALL SECTIONS.
2. RE: CIVIL, STRUC, MECH, PLUMB, ELEC, AND FIRE PROTECTION FOR MORE INFO.
3. FF: 100'-0" = NAVD88 657.3' +/- AT EXT DOORS.
4. PROVIDE ADEQUATE BLOCKING, BACKING OR STRUC SUPPORT AS REQ'D FOR PROPER INSTALLATION OF ALL MOUNTED ASSEMBLIES, INCLUDING ALL ATTACHED EQUIP, ACCESSORIES (CONTRACTOR FURNISHED ITEMS), PLUMB FIXTURES, MILLWORK, AND CASEWORK.

SCALE: (A) 8 0 8 16
SCALE OF FEET

LICENSED PROFESSIONAL ARCHITECT ELIZABETH HALLAS 07416 IOWA 5/29/26

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|---|---|---|---|---|
| A/E FIRM: | DESIGNED: | SUB SHEET NO. | TITLE OF SHEET | DRAWING NO. |
| PRIME: OTAK, INC. REDMOND, WA | LB/CM | A 301 | BUILDING SECTIONS | 394 194248 |
| SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | CADD LB/MM | | EFFIGY MOUNDS NATIONAL MONUMENT | PMIS/PKG NO. 318539 |
| | TECH REVIEW: ACN / EMH | | | SHEET 35 OF 142 |
| | DATE: 05/29/2026 | | | |

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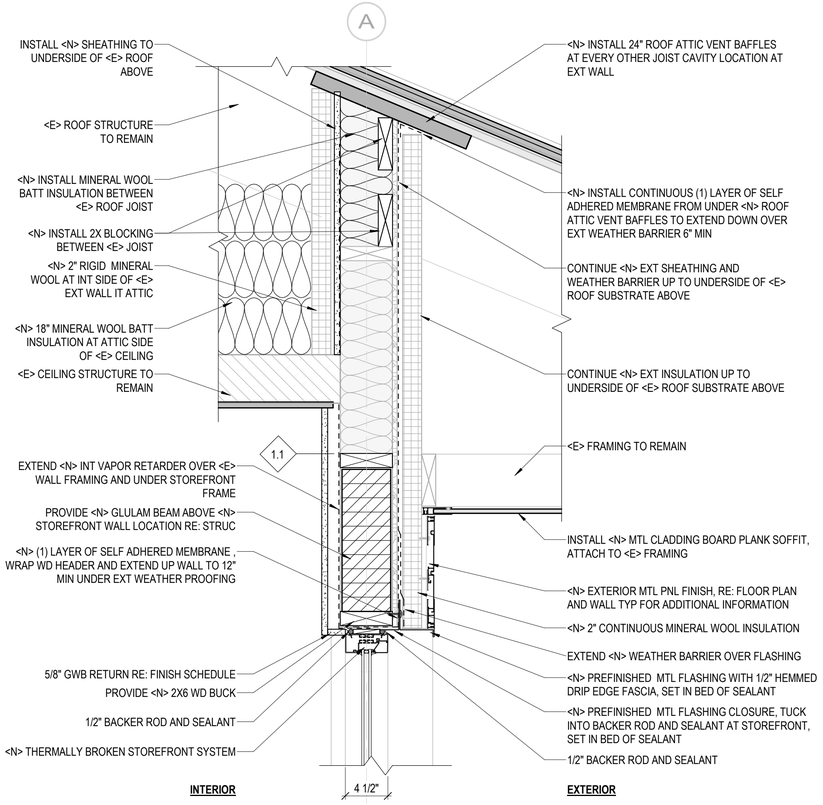
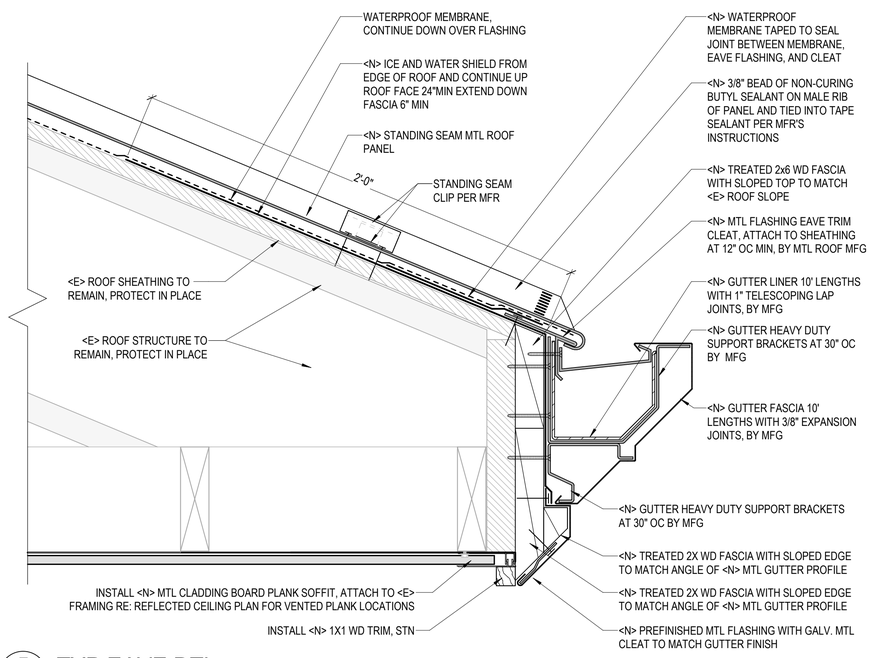



4 A 302 STOREFRONT HEAD DTL AT EAST LOBBY
SCALE B




5 A 302 TYP EAVE DTL
SCALE B

# OPTION NOTES

**OPTION A: HVAC & FIRE SUPPRESSION**

**OPTION B EXTERIOR ENVELOPE REHABILITATION**
REMOVE EXT SIDING AND ROOFING, REFRAME <N> OPENINGS, INSTALL <N> ROOF, REHABILITATE <E> BRICK, INSTALL <N> WEATHER BARRIER, INSULATION, AL CLADDING, MTL PANELS, EXT WINDOWS AND DOORS.

**OPTION C INTERIOR ACCESSIBILITY COMPLIANCE, PLUMBING, & ELECTRICAL**
PERFORM HAZARDOUS MATERIAL ABATEMENT. REMOVE INT FINISHES AND DEMO SELECT INT WALLS FOR INT RECONFIGURATION, INSTALL <N> WALLS AT RECONFIGURED RESTROOMS AND OFFICE SPACES.

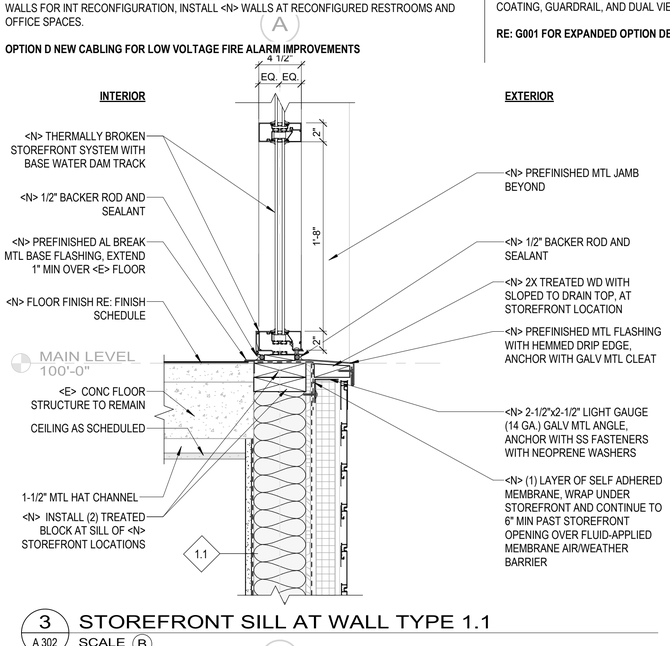
**OPTION D NEW CABLING FOR LOW VOLTAGE FIRE ALARM IMPROVEMENTS**

**OPTION E LIGHTNING PROTECTION**

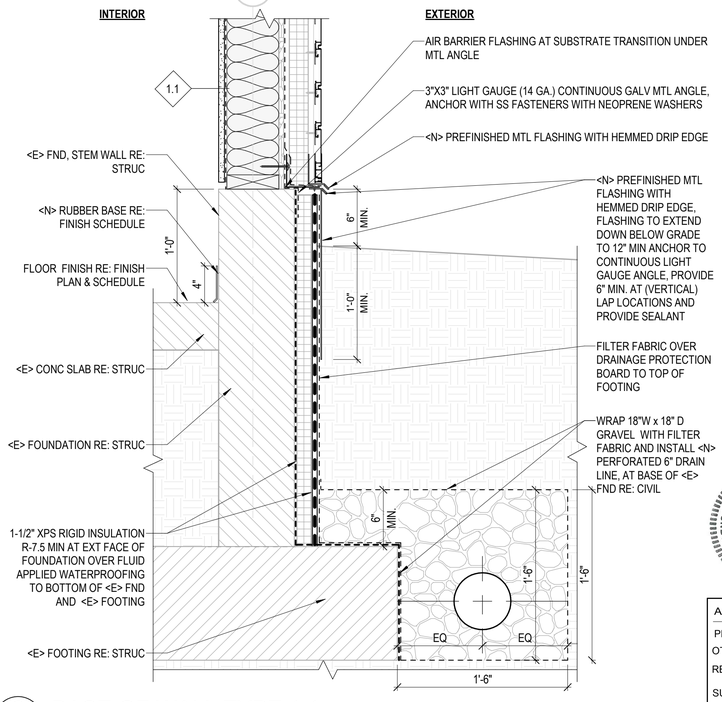
**OPTION F MEETING ROOM/AUDITORIUM AV & EXHIBIT AREA RENOVATION**
<N> OPERABLE PARTITION AND STL BEAM TO SUPPORT PARTITION, WINDOW SHADES, AV, ASSISTED LISTENING SYSTEM, AND AUDITORIUM FURNISHINGS.

**OPTION G EXPAND DECK AT NE CORNER**
SALVAGE EXISTING GUARDRAIL AND SELECTIVE DEMO OF EXISTING DECK. INSTALL NEW STEEL COLUMNS, BEAMS, METAL DECK, CIP CONCRETE, ADDITIONAL TRAFFIC COATING, GUARDRAIL, AND DUAL VIEWER.

**RE: G001 FOR EXPANDED OPTION DESCRIPTIONS**



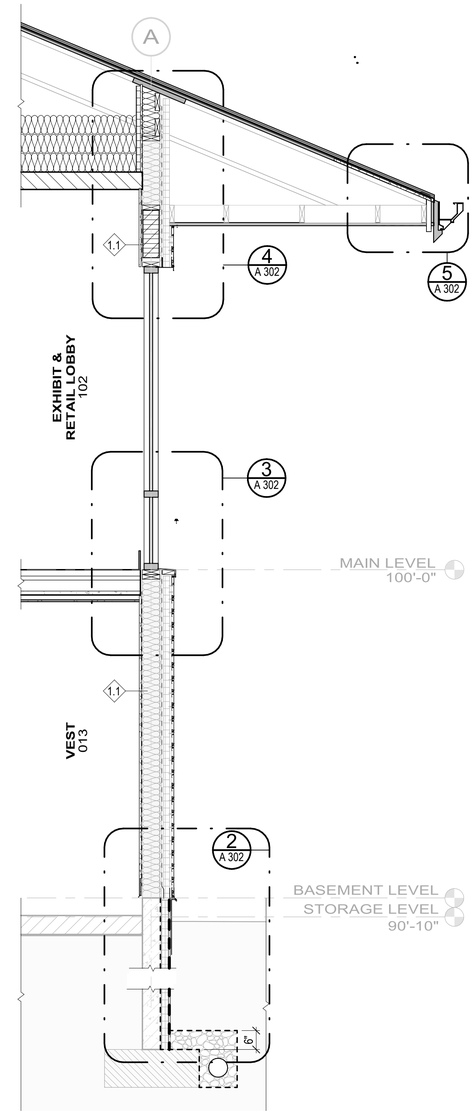

3 A 302 STOREFRONT SILL AT WALL TYPE 1.1
SCALE B




2 A 302 BASE OF WALL TYPE 1.1
SCALE B

# GENERAL NOTES

1. WALL SECTIONS INTENDED TO SHOW GENERAL LAYOUT AND RELATIONSHIP BETWEEN SPACES RE: FLOOR PLANS, RCP, INTERIOR ELEVATIONS, AND WALL SECTIONS.
2. REFER TO STOREFRONT AND DOOR SCHEDULE FOR DIMS AND MORE INFORMATION.
3. REFER TO INDEX FOR SYMBOL INFORMATION.
4. MECH AND ELEC EQUIPMENT IS SHOWN FOR REFERENCE ONLY. SEE MECH AND ELEC SHEETS FOR TYPE AND LOCATIONS.




1 A 302 WALL SECTION AT NORTH EXT WALL
SCALE A

SCALE: B
12 6 0 12
SCALE OF INCHES

SCALE: A
2 0 2 4
SCALE OF FEET

LICENSED PROFESSIONAL ARCHITECT
ELIZABETH HALLAS
07418
IOWA

| | | | | |
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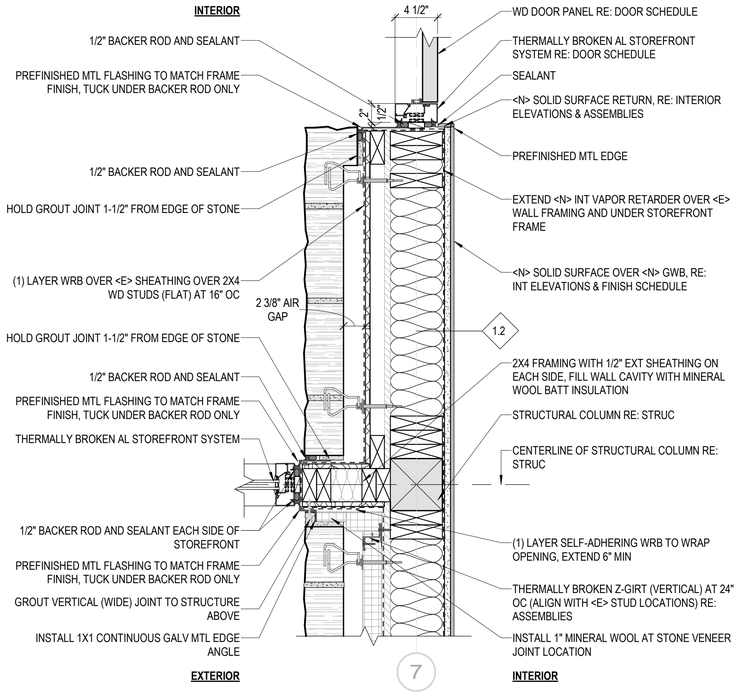
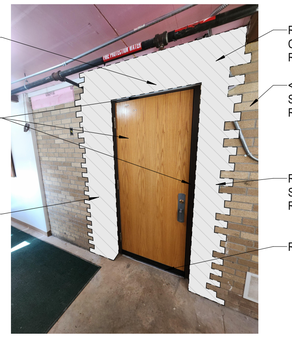
| A/E FIRM: ANDERSON HALLAS ARCHITECTS, GOLDEN, CO | DESIGNED: LB/CM | SUB SHEET NO. | TITLE OF SHEET: WALL SECTIONS | DRAWING NO. 394 194248 |
| PRIME: OTAK, INC. REDMOND, WA | CADD: LB/MM | A 302 | EFFIGY MOUNDS NATIONAL MONUMENT | PMIS/PKG NO. 318539 |
| SUBCONTRACTOR | TECH REVIEW: ACN / EMH | | | SHEET 36 OF 142 |
| | DATE: 05/29/2026 | | | |



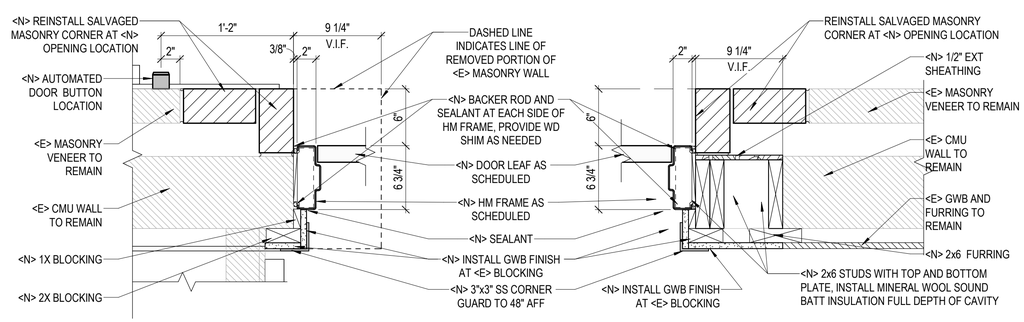

## 6 A 303 JAMB OF ENTRY STOREFRONT PLAN DTL

SCALE B




## 5 A 303 PHOTO OF DOOR 005A AT BASEMENT - DEMO

N.T.S.




## 4 A 303 LT AND RT JAMB OF ENLARGED OPENING PLAN DTL AT DOOR NO. 005A

SCALE B

# OPTION NOTES

**OPTION A: HVAC & FIRE SUPPRESSION**

**OPTION B EXTERIOR ENVELOPE REHABILITATION**
REMOVE EXT SIDING AND ROOFING, REFRAME <N> OPENINGS, INSTALL <N> ROOF, REHABILITATE <E> BRICK, INSTALL <N> WEATHER BARRIER, INSULATION, AL CLADDING, MTL PANELS, EXT WINDOWS AND DOORS.

**OPTION C INTERIOR ACCESSIBILITY COMPLIANCE, PLUMBING, & ELECTRICAL**
PERFORM HAZARDOUS MATERIAL ABATEMENT. REMOVE INT FINISHES AND DEMO SELECT INT WALLS FOR INT RECONFIGURATION, INSTALL <N> WALLS AT RECONFIGURED RESTROOMS AND OFFICE SPACES.

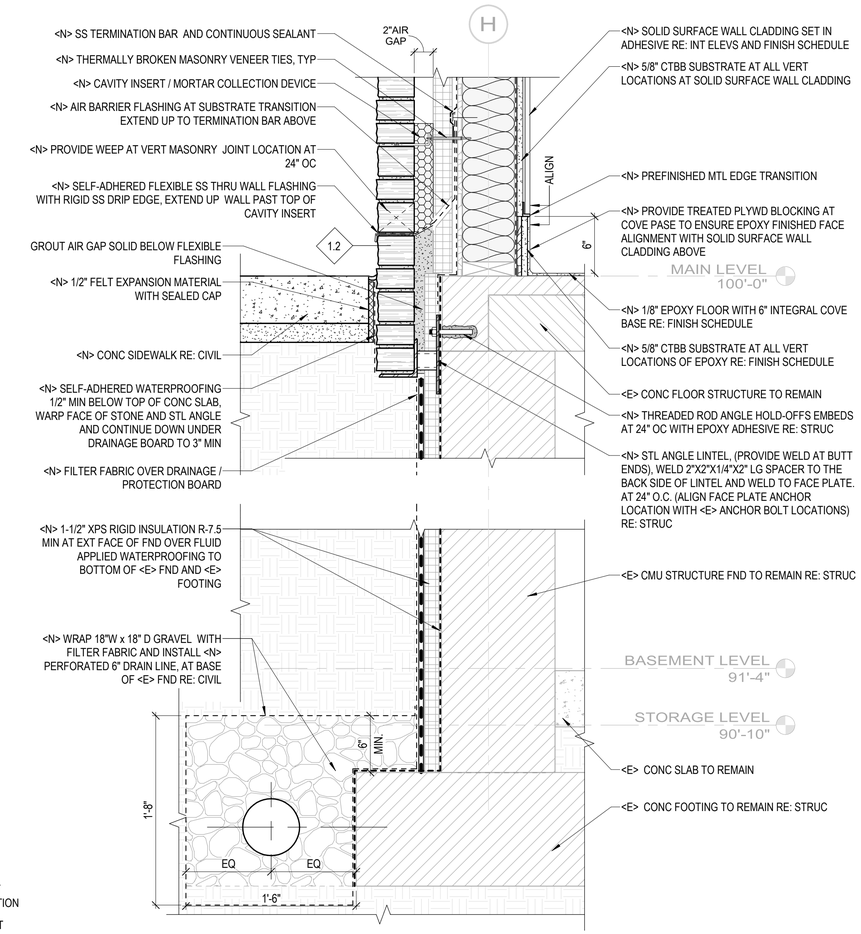
**OPTION D NEW CABLING FOR LOW VOLTAGE FIRE ALARM IMPROVEMENTS**

**OPTION E LIGHTNING PROTECTION**

**OPTION F MEETING ROOM/AUDITORIUM AV & EXHIBIT AREA RENOVATION**
<N> OPERABLE PARTITION AND STL BEAM TO SUPPORT PARTITION, WINDOW SHADES, AV, ASSISTED LISTENING SYSTEM, AND AUDITORIUM FURNISHINGS.

**OPTION G EXPAND DECK AT NE CORNER**
SALVAGE EXISTING GUARDRAIL AND SELECTIVE DEMO OF EXISTING DECK. INSTALL NEW STEEL COLUMNS, BEAMS, METAL DECK, CIP CONCRETE, ADDITIONAL TRAFFIC COATING, GUARDRAIL, AND DUAL VIEWER.

**RE: G001 FOR EXPANDED OPTION DESCRIPTIONS**

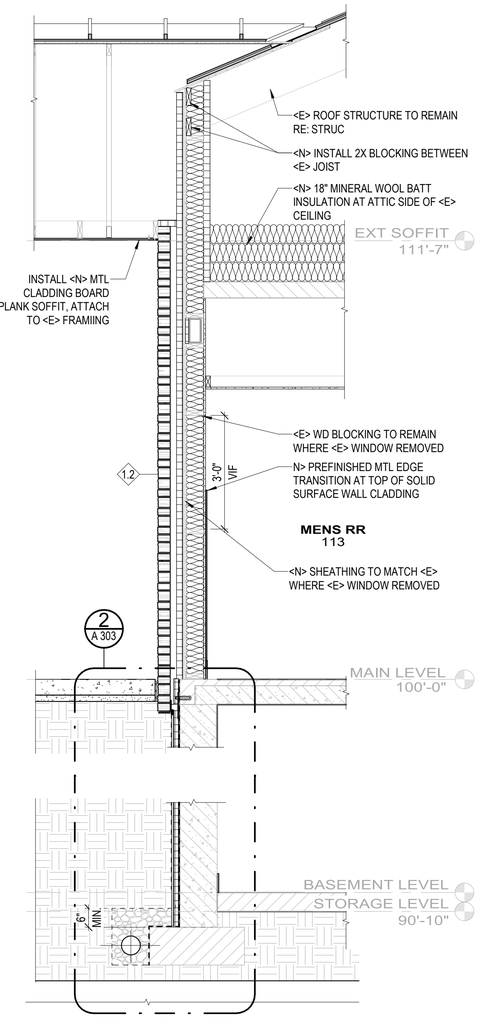



## 2 A 303 SECTION AT WALL BASE WITH STONE VENEER

SCALE B

# GENERAL NOTES

1. WALL SECTIONS INTENDED TO SHOW GENERAL LAYOUT AND RELATIONSHIP BETWEEN SPACES RE: FLOOR PLANS, RCP, INTERIOR ELEVATIONS, AND WALL SECTIONS.
2. REFER TO STOREFRONT AND DOOR SCHEDULE FOR DIMS AND MORE INFORMATION.
3. REFER TO INDEX FOR SYMBOL INFORMATION.
4. MECH AND ELEC EQUIPMENT IS SHOWN FOR REFERENCE ONLY. SEE MECH AND ELEC SHEETS FOR TYPE AND LOCATIONS.




## 1 A 303 WALL SECTION AT STONE VENEER

SCALE A

SCALE: B 12 6 0 12 SCALE OF INCHES

SCALE: A 2 0 2 4 SCALE OF FEET

LICENSED PROFESSIONAL ARCHITECT — ELIZABETH HALLAS 07416 — IOWA

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA; SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM; CADD: LB/MM; TECH REVIEW: ACN / EMH; DATE: 05/29/2026 | SUB SHEET NO. A 303 | TITLE OF SHEET: WALL SECTIONS — EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248; PMIS/PKG NO. 318539; SHEET 37 OF 142 |

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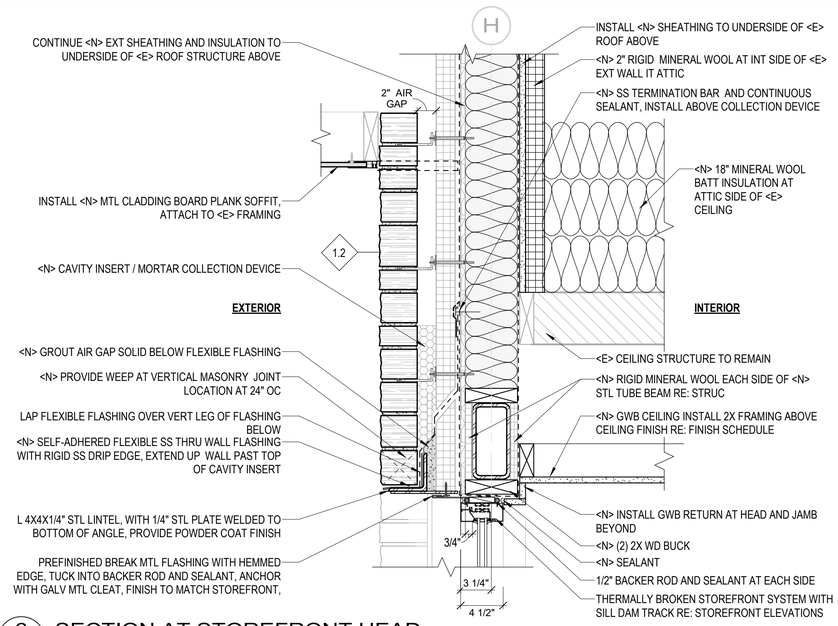
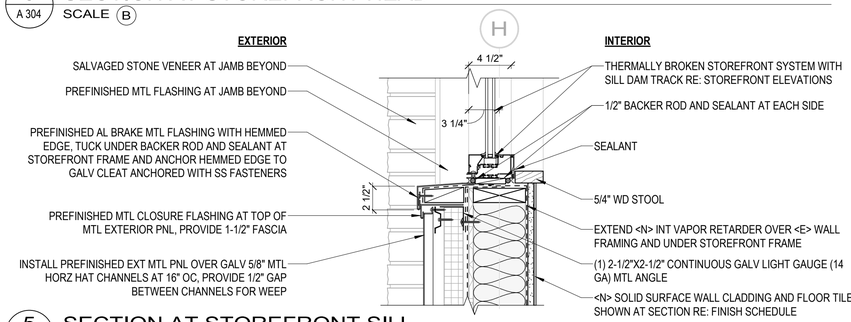
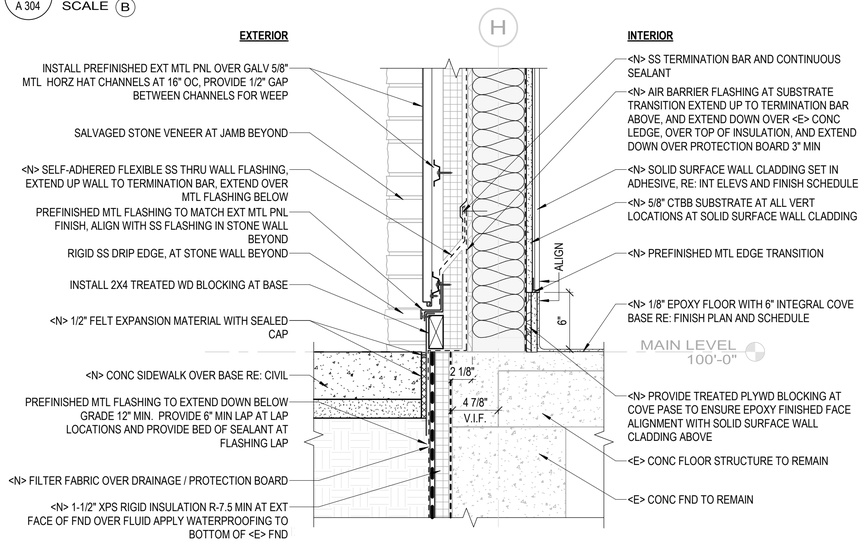



6 / A 304 SECTION AT STOREFRONT HEAD
SCALE B




5 / A 304 SECTION AT STOREFRONT SILL
SCALE B




4 / A 304 SECTION AT WALL BASE AT MTL PANEL
SCALE B

# OPTION NOTES

**OPTION A: HVAC & FIRE SUPPRESSION**

**OPTION B EXTERIOR ENVELOPE REHABILITATION**
REMOVE EXT SIDING AND ROOFING, REFRAME <N> OPENINGS, INSTALL <N> ROOF, REHABILITATE <E> BRICK, INSTALL <N> WEATHER BARRIER, INSULATION, AL CLADDING, MTL PANELS, EXT WINDOWS AND DOORS.

**OPTION C INTERIOR ACCESSIBILITY COMPLIANCE, PLUMBING, & ELECTRICAL**
PERFORM HAZARDOUS MATERIAL ABATEMENT. REMOVE INT FINISHES AND DEMO SELECT INT WALLS FOR INT RECONFIGURATION, INSTALL <N> WALLS AT RECONFIGURED RESTROOMS AND OFFICE SPACES.

**OPTION D NEW CABLING FOR LOW VOLTAGE FIRE ALARM IMPROVEMENTS**

**OPTION E LIGHTNING PROTECTION**

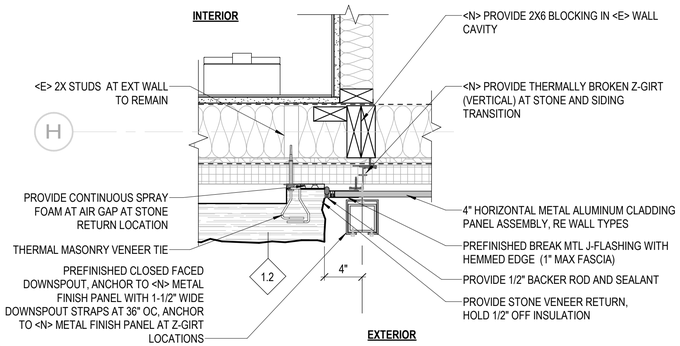
**OPTION F MEETING ROOM/AUDITORIUM AV & EXHIBIT AREA RENOVATION**
<N> OPERABLE PARTITION AND STL BEAM TO SUPPORT PARTITION, WINDOW SHADES, AV, ASSISTED LISTENING SYSTEM, AND AUDITORIUM FURNISHINGS.

**OPTION G EXPAND DECK AT NE CORNER**
SALVAGE EXISTING GUARDRAIL AND SELECTIVE DEMO OF EXISTING DECK. INSTALL NEW STEEL COLUMNS, BEAMS, METAL DECK, CIP CONCRETE, ADDITIONAL TRAFFIC COATING, GUARDRAIL, AND DUAL VIEWER.

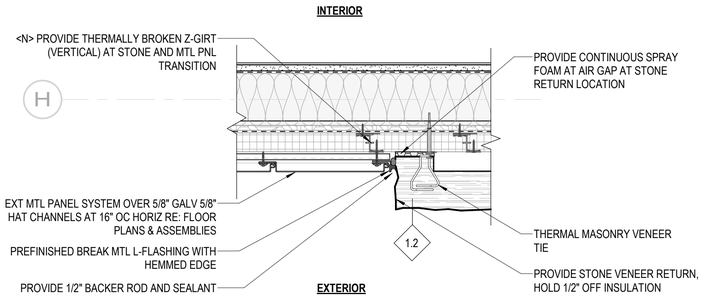
**RE: G001 FOR EXPANDED OPTION DESCRIPTIONS**

# GENERAL NOTES

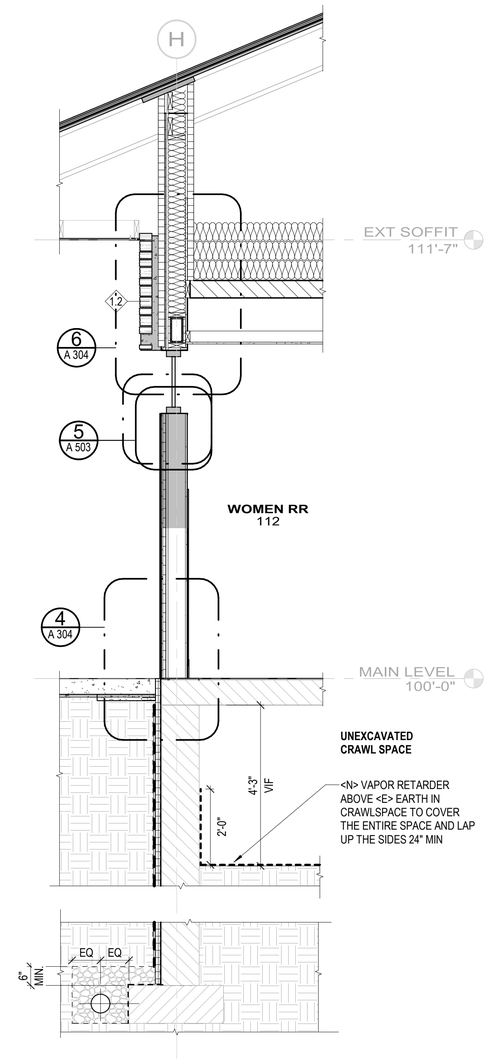
1. WALL SECTIONS INTENDED TO SHOW GENERAL LAYOUT AND RELATIONSHIP BETWEEN SPACES RE: FLOOR PLANS, RCP, INTERIOR ELEVATIONS, AND WALL SECTIONS.
2. REFER TO STOREFRONT AND DOOR SCHEDULE FOR DIMS AND MORE INFORMATION.
3. REFER TO INDEX FOR SYMBOL INFORMATION.
4. MECH AND ELEC EQUIPMENT IS SHOWN FOR REFERENCE ONLY. SEE MECH AND ELEC SHEETS FOR TYPE AND LOCATIONS.




3 / A 304 PLAN DETAIL AT STONE & PLANK SIDING TRANSITION
SCALE A




2 / A 304 PLAN DETAIL AT STONE & MTL PANEL TRANSITION
SCALE A




1 / A 304 WALL SECTION AT STOREFRONT
SCALE B

SCALE: B 12 6 0 12 SCALE OF INCHES
SCALE: A 2 0 2 4 SCALE OF FEET

LICENSED PROFESSIONAL ARCHITECT ELIZABETH HALLAS 07418 IOWA

| | | | | |
|---|---|---|---|---|
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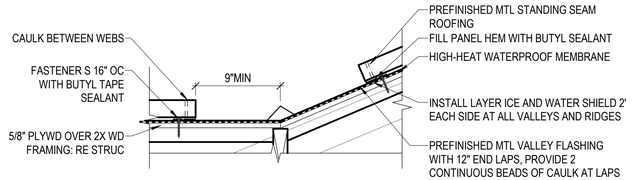
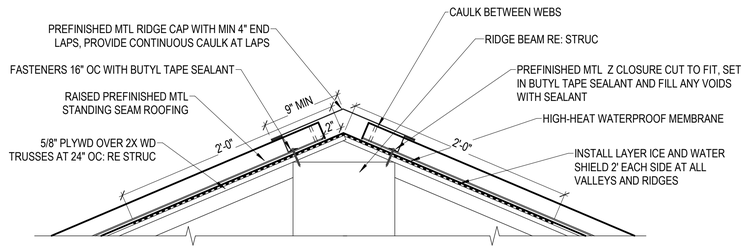
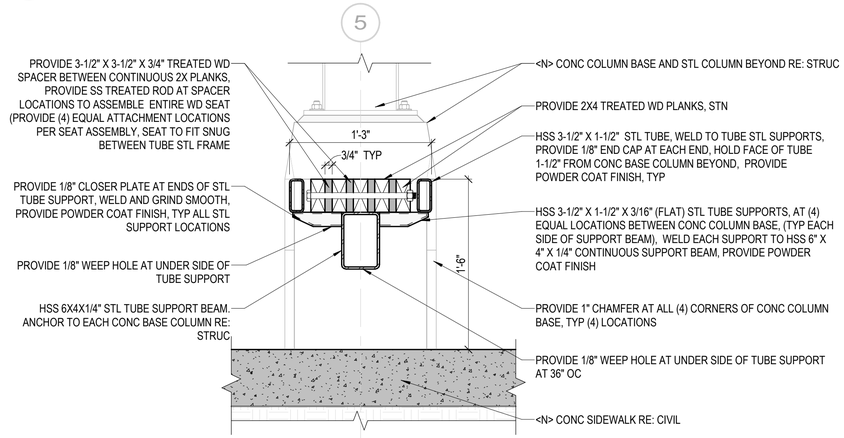


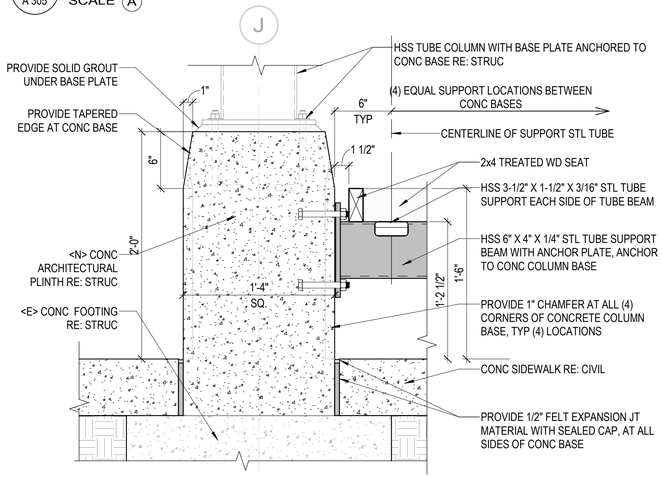

## 6 SECTION AT ROOF VALLEY
A 305 SCALE A



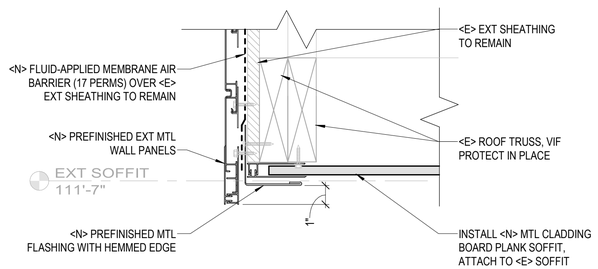

## 5 SECTION AT ROOF RIDGE
A 305 SCALE A




## 4 SECTION AT BENCH
A 305 SCALE A




## 3 SECTION AT COLUMN BASE
A 305 SCALE A




## 2 SECTION AT SOFFIT EDGE
A 305 SCALE B

# OPTION NOTES

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**OPTION B EXTERIOR ENVELOPE REHABILITATION**
REMOVE EXT SIDING AND ROOFING, REFRAME <N> OPENINGS, INSTALL <N> ROOF, REHABILITATE <E> BRICK, INSTALL <N> WEATHER BARRIER, INSULATION, AL CLADDING, MTL PANELS, EXT WINDOWS AND DOORS.

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**OPTION D NEW CABLING FOR LOW VOLTAGE FIRE ALARM IMPROVEMENTS**

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**RE: G001 FOR EXPANDED OPTION DESCRIPTIONS**

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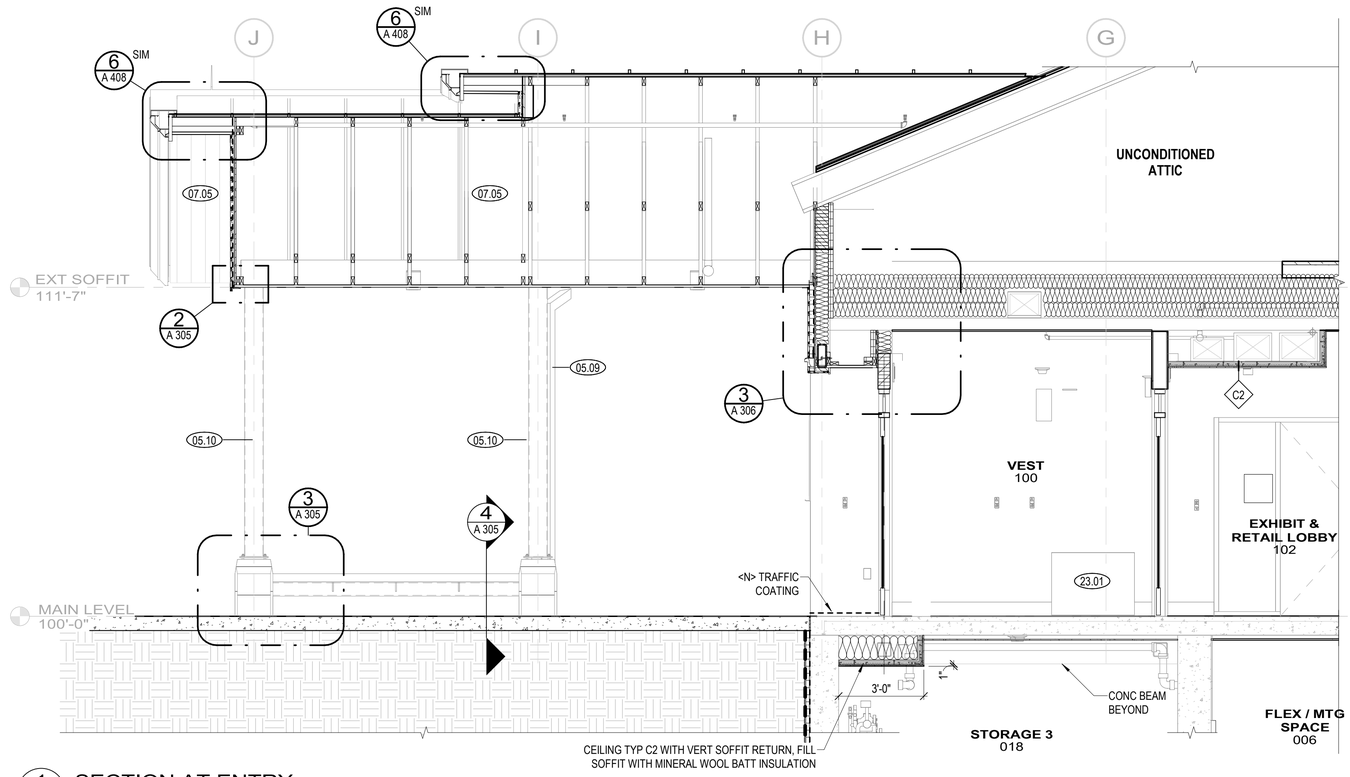
# KEYNOTES

| | |
|---|---|
| 05.09 | OPTION B: <N> PREFINISHED MTL 4"x5" DOWNSPOUT RE: BUILDING ELEVATIONS |
| 05.10 | OPTION B: <N> HSS COLUMN IN LOCATION OF REMOVED <E> WD COLUMN RE: STRUC |
| 07.05 | OPTION B: INSTALL <N> MTL PANEL SOFFIT WITH ORIENTATION WITH LENGTH ALONG GABLE |
| 23.01 | OPTION A: <N> MECHANICAL EQUIPMENT RE: MECH |




## 1 SECTION AT ENTRY
A 305 SCALE B

SCALE: C 6 3 0 6 SCALE OF INCHES
SCALE: B 2 0 3 6 SCALE OF FEET
SCALE: A 12 6 0 12 SCALE OF INCHES

LICENSED PROFESSIONAL ARCHITECT ELIZABETH HALLAS 07418 IOWA

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM DRAWN: LB/MM TECH REVIEW: ACN / EMH DATE: 05/29/2026 | SUB SHEET NO. A 305 | TITLE OF SHEET: WALL SECTIONS — EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 39 OF 142 |

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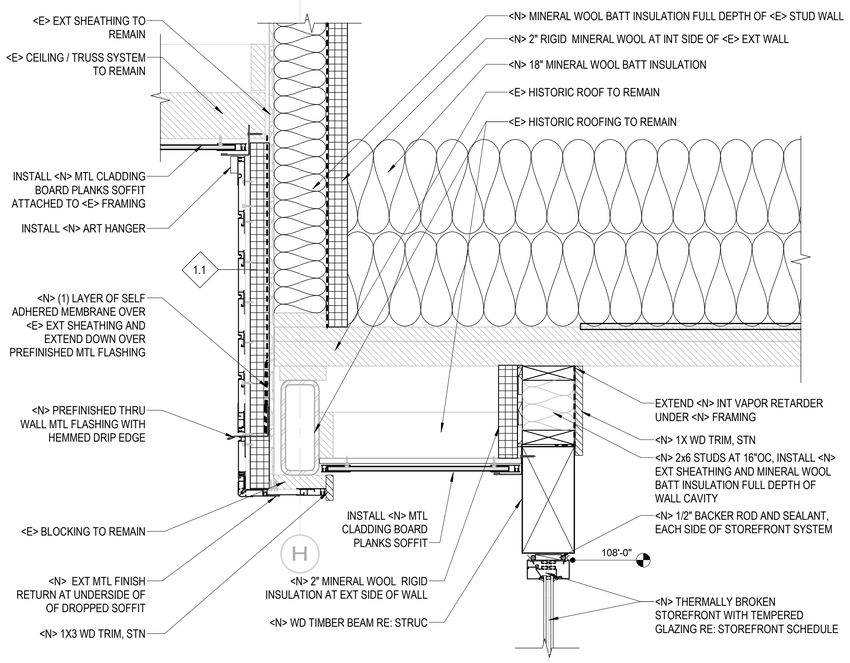


3 / A 306 SECTION AT ENTRY HEADER
SCALE A

# OPTION NOTES

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**OPTION B EXTERIOR ENVELOPE REHABILITATION**
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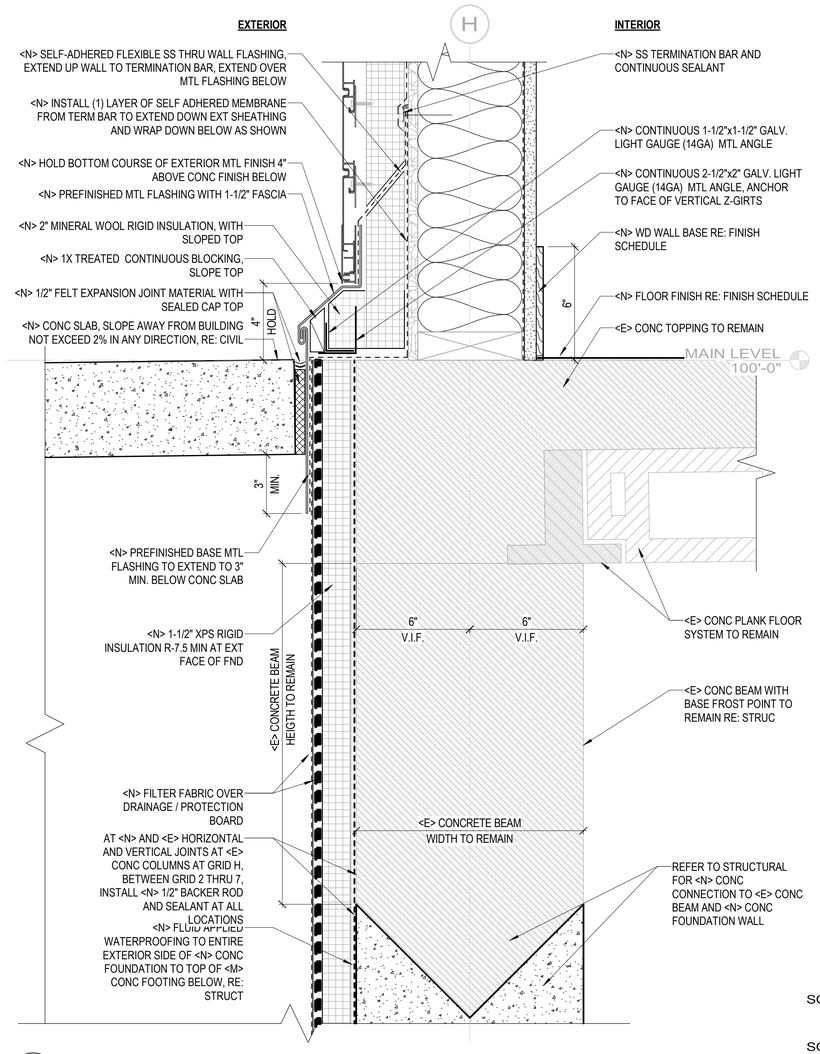
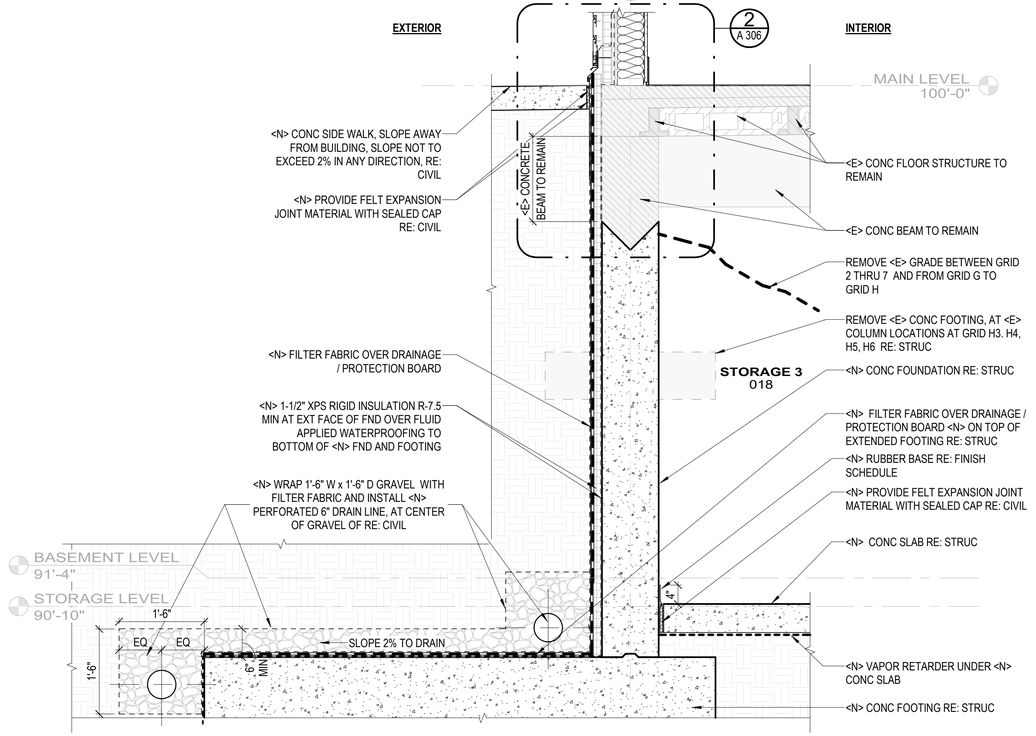



2 / A 306 SECTION AT EXT FDN WALL AT GRID H BETWEEN GRID 2 & 7
SCALE B




1 / A 306 WALL SECTION AT NEW CONC SLAB
SCALE A

SCALE: C 12 6 0 12 SCALE OF INCHES
SCALE: B 6 3 0 6 SCALE OF INCHES
SCALE: A 1 0 1 2 3 SCALE OF FEET

LICENSED PROFESSIONAL ARCHITECT — ELIZABETH HALLAS — 07418 — IOWA

| | | | | |
|---|---|---|---|---|
| A/E FIRM: OTAK, INC. REDMOND, WA<br>PRIME:<br>SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM<br>CADD LB/MM<br>TECH REVIEW: ACN / EMH<br>DATE: 05/29/2026 | SUB SHEET NO. A 306 | TITLE OF SHEET: WALL SECTIONS AT GRID H<br>EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248<br>PMIS/PKG NO. 318539<br>SHEET 40 OF 142 |

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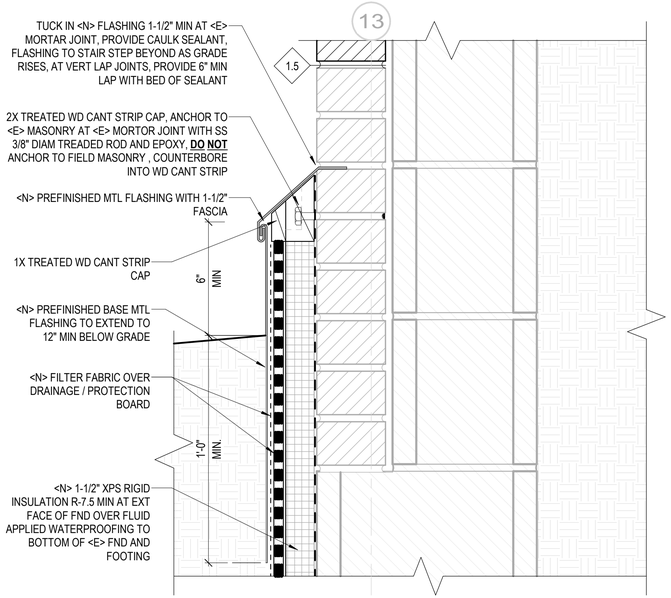
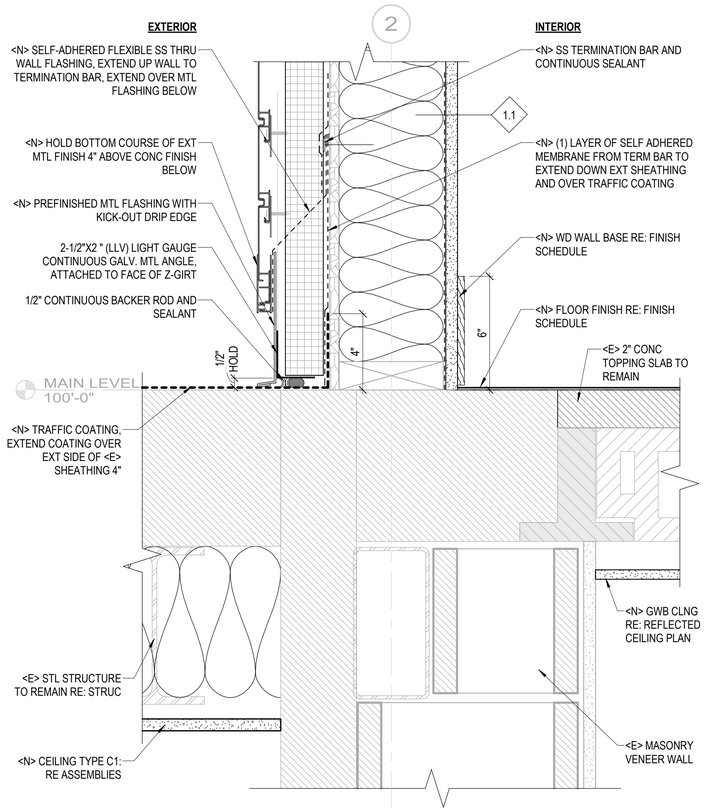
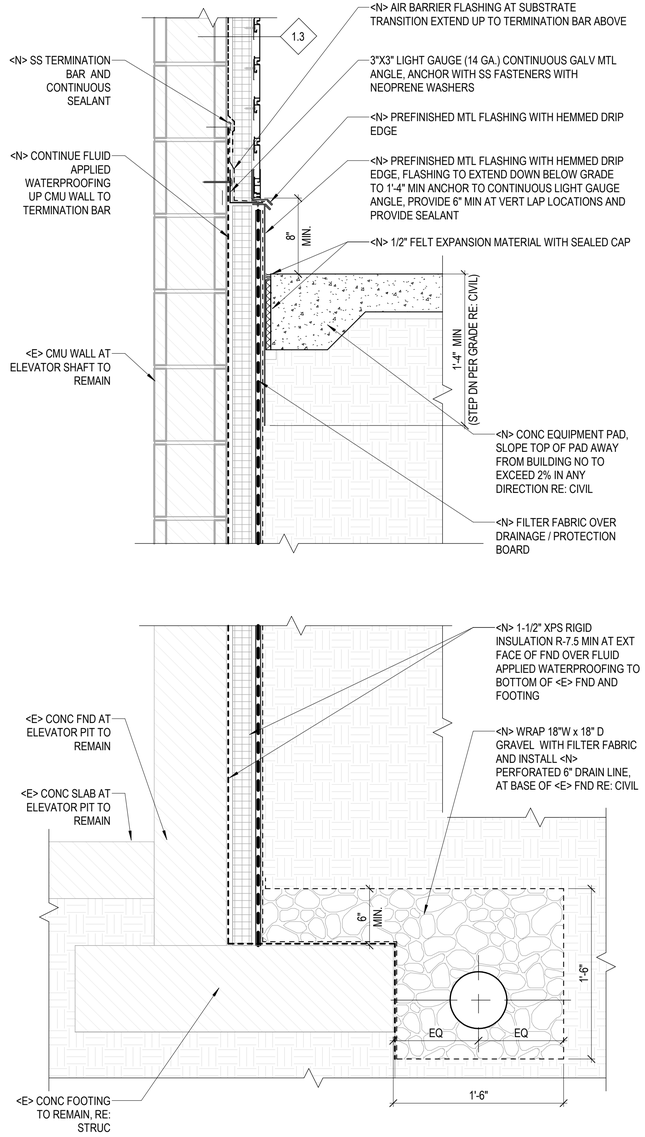



4 A 307 SECTION DETAIL AT MASONRY WALL BASE
SCALE C




3 A 307 SECTION AT BASE OF EXT WALL AT DECK
SCALE C




2 A 307 WALL SECTION AT ELEVATOR SHAFT WALL
SCALE B

## OPTION NOTES

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**OPTION B EXTERIOR ENVELOPE REHABILITATION**
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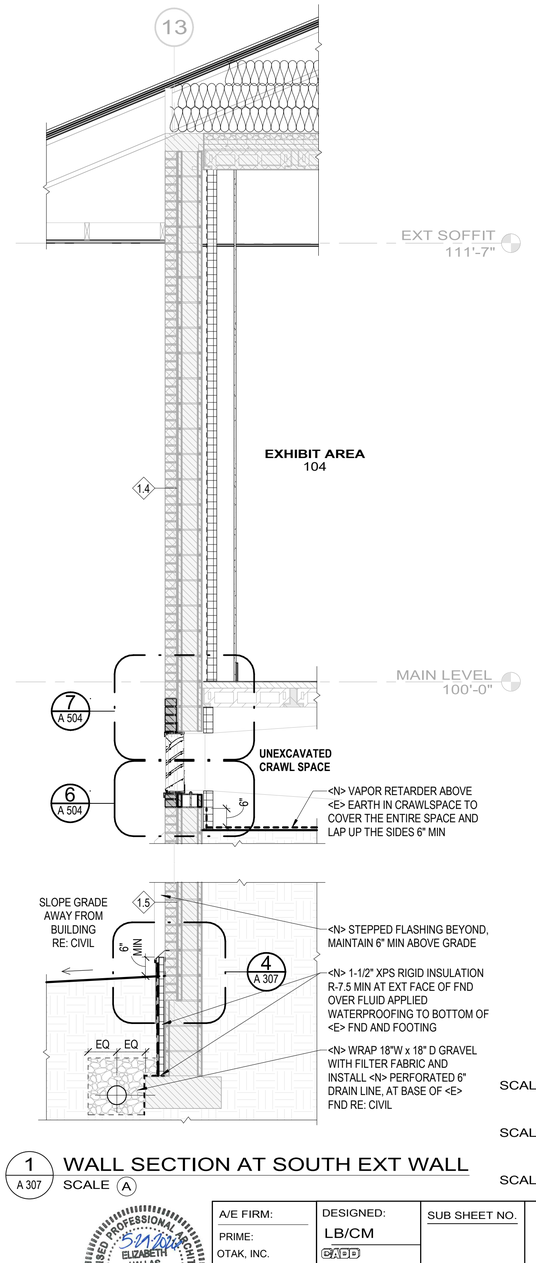
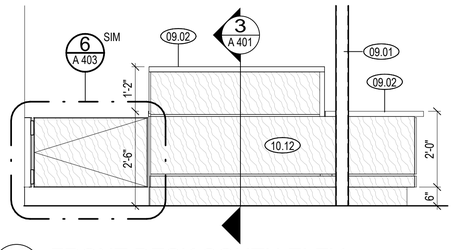



1 A 307 WALL SECTION AT SOUTH EXT WALL
SCALE A

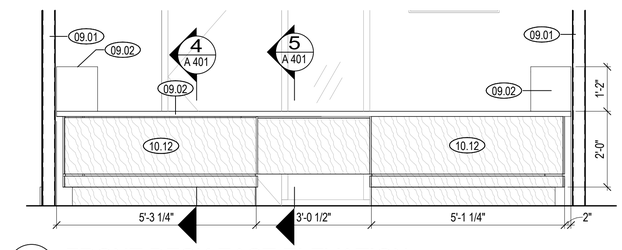
SCALE: C 6 3 0 6 SCALE OF INCHES
SCALE: B 12 6 0 12 SCALE OF INCHES
SCALE: A 2 0 2 4 SCALE OF FEET

LICENSED PROFESSIONAL ARCHITECT ELIZABETH HALLAS 07418 IOWA 5/12/2026

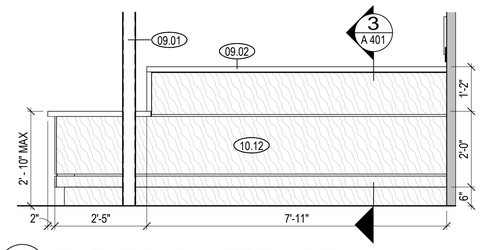
| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM CADD LB/MM TECH REVIEW: ACN / EMH DATE: 05/29/2026 | SUB SHEET NO. A 307 | TITLE OF SHEET WALL SECTIONS EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 41 OF 142 |



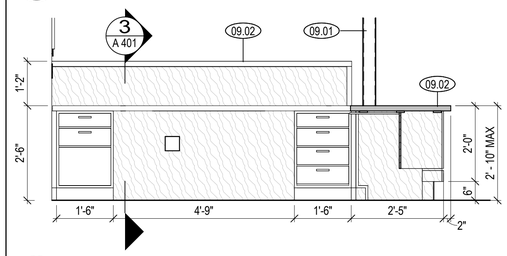

12 A 401 FRONT DESK SOUTH ELEV
SCALE A



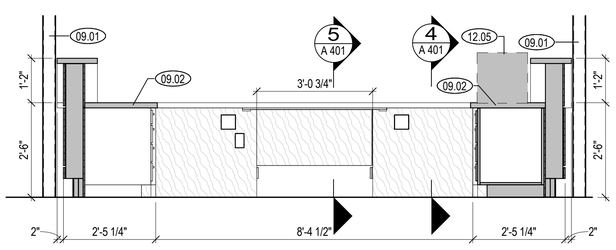

11 A 401 FRONT DESK EAST ELEVATION
SCALE A



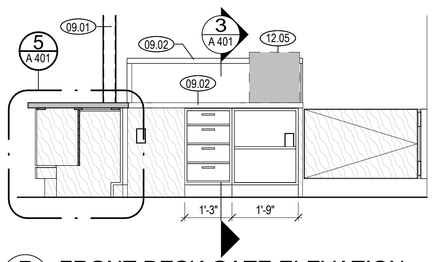

10 A 401 FRONT DESK NORTH ELEV
SCALE A




9 A 401 FRONT DESK WORKSPACE
SCALE A




8 A 401 FRONT DESK WEST ELEVATION
SCALE A



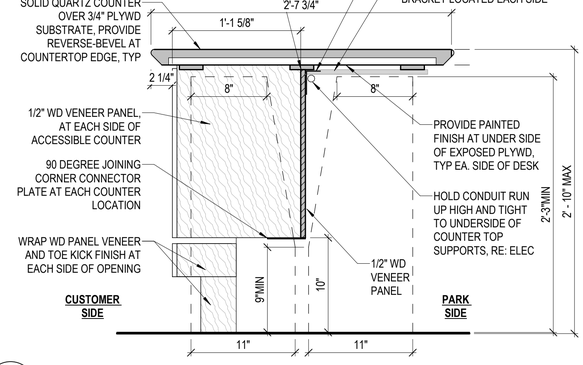

7 A 401 FRONT DESK GATE ELEVATION
SCALE A

## GENERAL NOTES

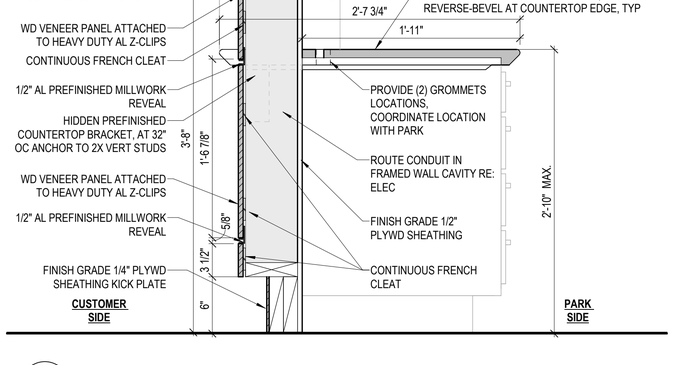
1. RE: CIVIL, LAND, STRUC, MECH, PLUMB, ELEC, AND FIRE PROTECTION FOR MORE INFO.
2. FF: MAIN LEVEL 100'-0" = NAVD88 657.3' +/- AT EXT DOORS.
3. ALL PLAN DIMS LOCATING EXT ASSEMBLIES ARE TO GRID LINE OR FACE OF FINISH, UNO. <E> WALLS WITHOUT WALL TAGS SHALL REMAIN.
4. VERIFY EQUIP ROUGH-IN DIMS WITH MFR FOR ALL EQUIP INCLUDING <E>, REUSED, OR FURNISHED BY CO.
5. PROVIDE ADEQUATE BLOCKING, BACKING OR STRUC SUPPORT AS REQ'D FOR PROPER INSTALLATION OF ALL MOUNTED ASSEMBLIES, INCLUDING ALL ATTACHED EQUIP, ACCESSORIES (CONTRACTOR FURNISHED ITEMS), PLUMB FIXTURES, MILLWORK, AND CASEWORK.
6. FLOOR DRAIN LOCATIONS SHOWN ARE APPROXIMATE. VIF EXACT LOCATIONS PER PLUMB WITH STRUC COORDINATION. SLOPE CONC TO DRAIN AS INDICATED.
7. ALL CONDUIT & PIPING TO BE ROUTED WITHIN WALLS OR FURRING WALLS UON. CONTRACTOR TO COORDINATE IN FIELD ALL EXPOSED CONDUIT AND PIPING ROUTING WITH CO.
8. DOORS AND STOREFRONTS OCCUPY <E> FRAMED OPENINGS UON. VIF DIMS IN FIELD. ALIGN ALL <N> ROUGH OPENINGS WITH <E> MASONRY JOINTS UON: RE:STOREFRONT & DOOR SCHEDULE FOR DIMS AND MORE INFO.
9. PATCH AND REPAIR ALL FINISHES IMPACTED BY SYSTEMS AND DEMO WORK RE: FINISH SCHEDULE UON. MATCH ADJ IN MATERIAL, COLOR, TEXTURE AND FINISH.
10. FURNITURE SHOWN IS FOR REFERENCE ONLY AND IS NOT INCLUDED IN THE CONTRACT UON.

## KEYNOTES

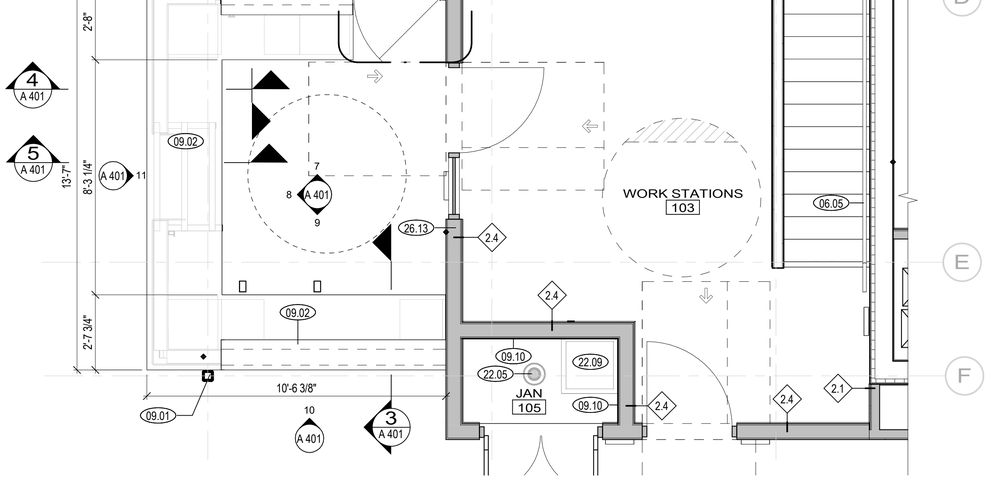
| | |
|---|---|
| 06.05 | OPTION C: <N> 1-1/2" DIAM WD HANDRAIL AT 36" AFF ATTACHED TO <E> CMU WALL WITH WALL-MOUNT PREFINISHED MTL BRACKETS; PROVIDE 12" MIN HANDRAIL EXTENSIONS PAST NOSING AT TOP AND BOTTOM OF STAIR |
| 09.01 | OPTION C: PT <E> STL COLUMN RE: FINISH SCHEDULE |
| 09.02 | OPTION C: <N> QUARTZ COUNTERTOP ON HEAVY DUTY HIDDEN MTL WALL BRACKETS |
| 09.10 | OPTION C: INSTALL <N> FRP FROM FLOOR TO CLNG EACH SIDE OF MOP SINK RE: FINISH SCHEDULE |
| 10.12 | OPTION C: <N> WD VENEER PANEL ATTACHED TO HEAVY DUTY Z-CLIPS |
| 12.05 | OPTION C: EASTERN NATIONAL REGISTER SPACE |
| 22.05 | OPTION C: <N> FLOOR DRAIN, GRIND CONC TOPPING SLAB TO SLOPE FLOOR (2% MIN) 2' FROM DRAIN IN ALL DIRECTIONS, MAINTAIN CONSISTENT FLOOR FINISH RE: PLUMB |
| 22.09 | OPTION C: <N> ABAAS COMPLIANT MOP SINK RE: PLUMB |
| 26.13 | OPTION C: <N> DIGITAL DISPLAY RE: ELEC & TECH |




5 A 401 SECTION AT MODESTY PANEL
SCALE B

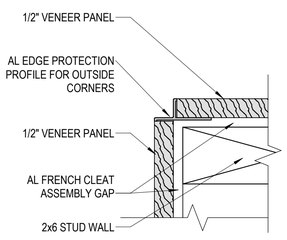



3 A 401 SECTION AT HIGHER DESK
SCALE B

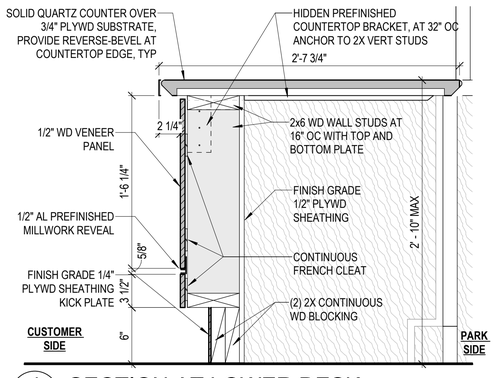



1 A 401 FRONT DESK AREA
SCALE A

NOTE:
REFER TO A403 FOR ADDITIONAL FRONT DESK DETAILS




6 A 401 TYP PLAN DTL AT CORNER
SCALE D




4 A 401 SECTION AT LOWER DESK
SCALE B

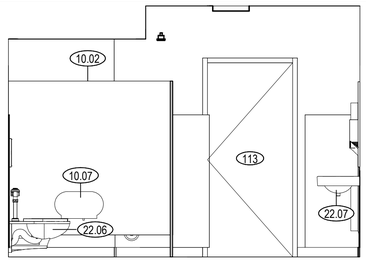
2 A 401 FRONT DESK ISOMETRIC
N.T.S.

SCALE: D 2 0 2 4 SCALE OF INCH
SCALE: C 6 3 0 6 SCALE OF INCHES
SCALE: B 12 6 0 12 SCALE OF INCHES
SCALE: A 2 0 3 6 SCALE OF FEET

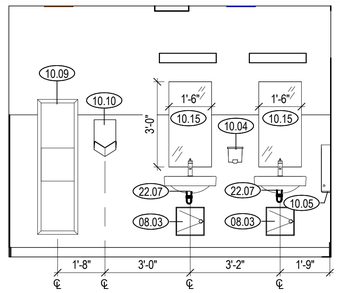
LICENSED PROFESSIONAL ARCHITECT ELIZABETH HALLAS 07418 IOWA

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM DRAWN: LB/MM TECH REVIEW: ACN / EMH DATE: 05/29/2026 | SUB SHEET NO. A 401 | TITLE OF SHEET ENLARGED - FRONT DESK EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 42 OF 142 |

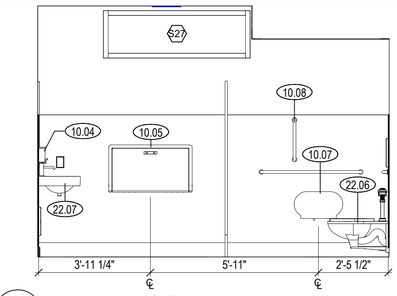
7/7/2026 1:20:31 PM Autodesk Docs://2025-110 NPS EFMO Visitor Center DD-CD/2024250 NPS EFMO Visitor Center - v24_BIM360.rvt



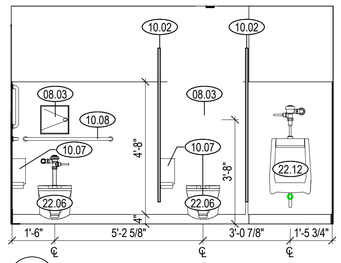

13 A 402 MEN RR NORTH
SCALE C



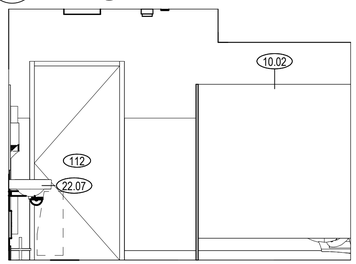

12 A 402 MEN RR EAST
SCALE A



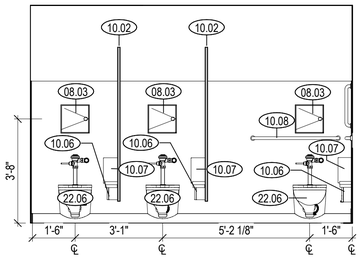

11 A 402 MEN RR SOUTH
SCALE A



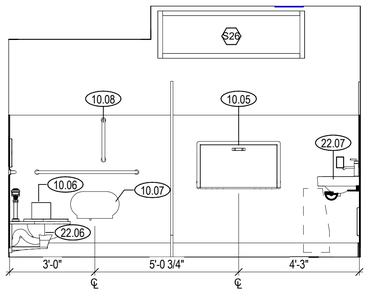

10 A 402 MEN RR WEST
SCALE A



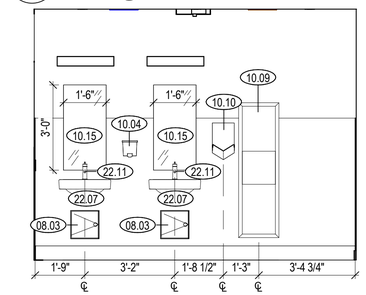

9 A 402 WOMEN'S RR NORTH
SCALE A



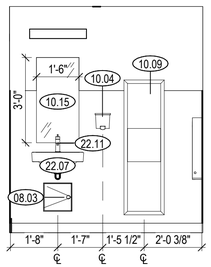

8 A 402 WOMEN'S RR EAST
SCALE A



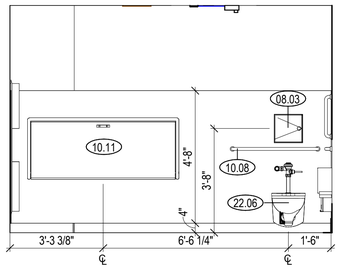

7 A 402 WOMEN'S RR SOUTH
SCALE A



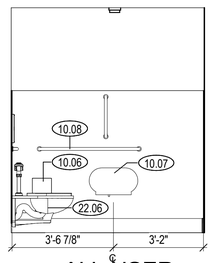

6 A 402 WOMEN'S RR WEST
SCALE A



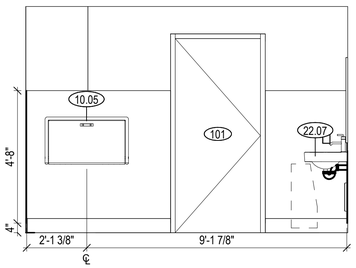

5 A 402 ALL-USER RR NORTH
SCALE A




4 A 402 ALL-USER RR EAST
SCALE A

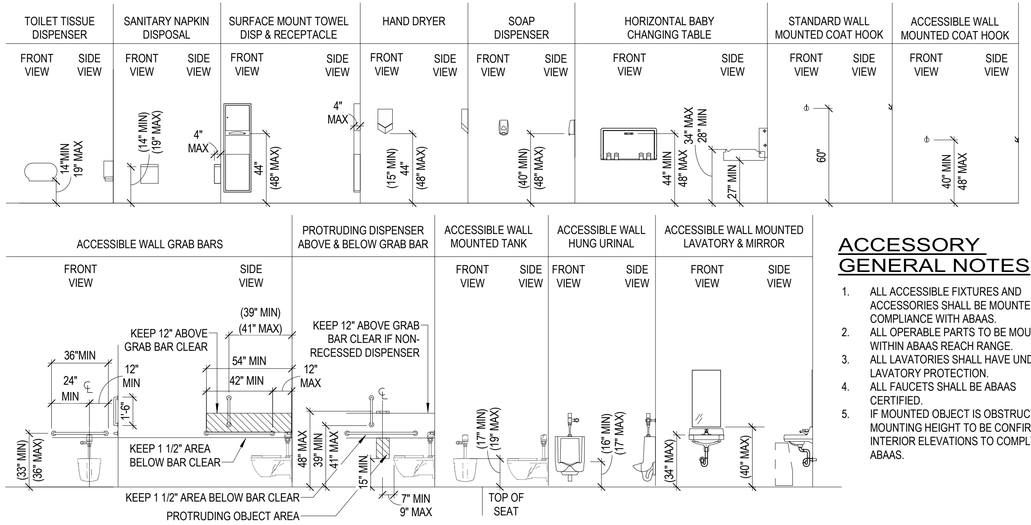



3 A 402 ALL-USER RR SOUTH
SCALE A




2 A 402 ALL-USER RR WEST
SCALE A

## MOUNTING HEIGHTS




## ACCESSORY GENERAL NOTES

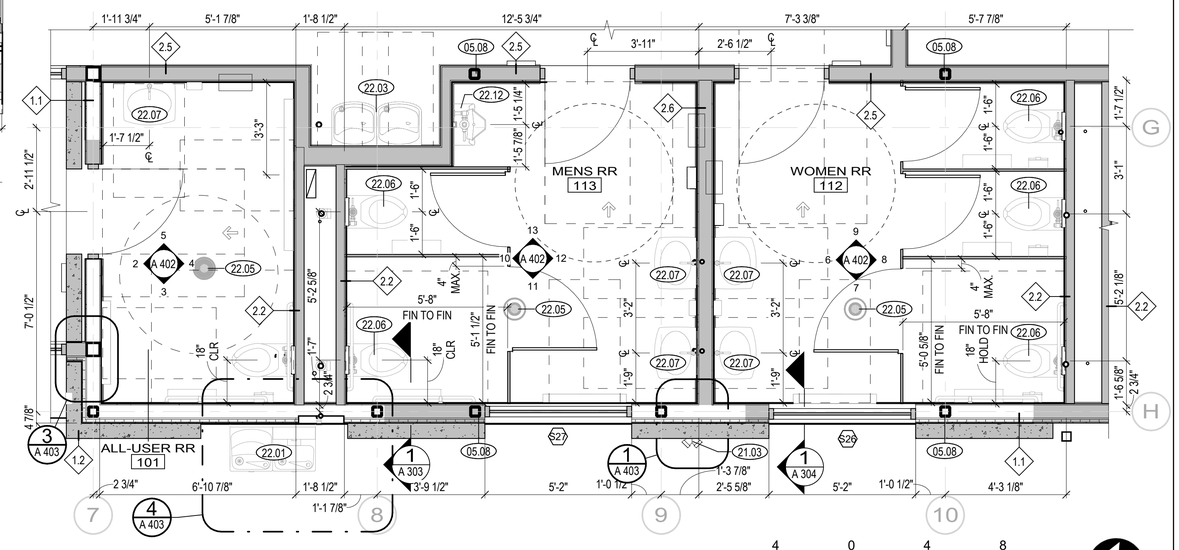
1. ALL ACCESSIBLE FIXTURES AND ACCESSORIES SHALL BE MOUNTED IN COMPLIANCE WITH ABAAS.
2. ALL OPERABLE PARTS TO BE MOUNTED WITHIN ABAAS REACH RANGE.
3. ALL LAVATORIES SHALL HAVE UNDER LAVATORY PROTECTION.
4. ALL FAUCETS SHALL BE ABAAS CERTIFIED.
5. IF MOUNTED OBJECT IS OBSTRUCTED, MOUNTING HEIGHT TO BE CONFIRMED IN INTERIOR ELEVATIONS TO COMPLY WITH ABAAS.

## GENERAL NOTES

1. RE: CIVIL, LAND, STRUC, MECH, PLUMB, ELEC, AND FIRE PROTECTION FOR MORE INFO.
2. FF: MAIN LEVEL 100'-0" = NAVD88 657.3' +/- AT EXT DOORS.
3. ALL PLAN DIMS LOCATING EXT ASSEMBLIES ARE TO GRID LINE OR FACE OF FINISH, UNO. <E> WALLS WITHOUT WALL TAGS SHALL REMAIN.
4. VERIFY EQUIP ROUGH-IN DIMS WITH MFR FOR ALL EQUIP INCLUDING <E>, REUSED, OR FURNISHED BY CO.
5. PROVIDE ADEQUATE BLOCKING, BACKING OR STRUC SUPPORT AS REQ'D FOR PROPER INSTALLATION OF ALL MOUNTED ASSEMBLIES, INCLUDING ALL ATTACHED EQUIP, ACCESSORIES (CONTRACTOR FURNISHED ITEMS), PLUMB FIXTURES, MILLWORK, AND CASEWORK.
6. FLOOR DRAIN LOCATIONS SHOWN ARE APPROXIMATE. VIF EXACT LOCATIONS PER PLUMB WITH STRUC COORDINATION. SLOPE CONC TO DRAIN AS INDICATED.
7. ALL CONDUIT & PIPING TO BE ROUTED WITHIN WALLS OR FURRING WALLS UON. CONTRACTOR TO COORDINATE IN FIELD ALL EXPOSED CONDUIT AND PIPING ROUTING WITH CO.
8. DOORS AND STOREFRONTS OCCUPY <E> FRAMED OPENINGS UON. VIF DIMS IN FIELD. ALIGN ALL <N> ROUGH OPENINGS WITH <E> MASONRY JOINTS UON: RE:STOREFRONT & DOOR SCHEDULE FOR DIMS AND MORE INFO.
9. PATCH AND REPAIR ALL FINISHES IMPACTED BY SYSTEMS AND DEMO WORK RE: FINISH SCHEDULE UON. MATCH ADJ IN MATERIAL, COLOR, TEXTURE AND FINISH.
10. FURNITURE SHOWN IS FOR REFERENCE ONLY AND IS NOT INCLUDED IN THE CONTRACT UON.

## KEYNOTES

| | |
|---|---|
| 05.08 | OPTION C: <N> HSS COLUMN, PROVIDE RIGID INSULATION AT EXT SIDE OF COLUMN BETWEEN SHEATHING AND STL RE: STRUC |
| 08.03 | OPTION C: <N> 12"X12" PREFINISHED MTL PLUMB ACCESS PANEL RE: PLUMB |
| 10.02 | OPTION C: <N> FLOOR MOUNTED SOLID PLASTIC PHENOLIC TOILET PARTITION, MOUNT IN COMPLIANCE WITH ABAAS RE: MOUNTING HEIGHTS |
| 10.04 | OPTION C: <N> WALL MOUNTED FOAM SOAP DISPENSER, MOUNT IN COMPLIANCE WITH ABAAS RE: MOUNTING HEIGHTS |
| 10.05 | OPTION C: <N> WALL MOUNTED BABY CHANGING STATION, MOUNT IN COMPLIANCE WITH ABAAS RE: MOUNTING HEIGHTS |
| 10.06 | OPTION C: <N> SANITARY NAPKIN DISPOSAL, MOUNT IN COMPLIANCE WITH ABAAS RE: MOUNTING HEIGHTS |
| 10.07 | OPTION C: <N> JUMBO ROLL TOILET PAPER DISPENSER, MOUNT IN COMPLIANCE WITH ABAAS RE: MOUNTING HEIGHTS |
| 10.08 | OPTION C: <N> ACCESSIBLE MTL GRAB BARS, MOUNT IN COMPLIANCE WITH ABAAS RE: MOUNTING HEIGHTS |
| 10.09 | OPTION C: <N> PAPER TOWEL DISPENSER MOUNTED IN A RECESSED SS FRAME WITH AN INTEGRAL TRASH RECEPTACLE, MOUNT IN COMPLIANCE WITH ABAAS RE: MOUNTING HEIGHTS |
| 10.10 | OPTION C: <N> AUTOMATIC HAND DRYER, MOUNT IN COMPLIANCE WITH ABAAS RE: MOUNTING HEIGHTS |
| 10.11 | OPTION C: <N> WALL MOUNTED SUPPLEMENTARY CHANGING STATION, MOUNT IN COMPLIANCE WITH ABAAS RE: MOUNTING HEIGHTS |
| 10.15 | OPTION C: <N> WALL MOUNTED MIRROR, MOUNT IN COMPLIANCE WITH ABAAS RE: MOUNTING HEIGHTS |
| 21.03 | OPTION A: <N> DOUBLE-HEAD FIRE DEPARTMENT CONNECTION WHERE <E> REMOVED RE: FIRE PROTECTION |
| 22.01 | OPTION C: <N> EXT WALL MOUNTED WATER FOUNTAIN WITH BOTTLE FILLER RE: PLUMB |
| 22.03 | OPTION C: <N> INT DRINKING FOUNTAINS WITH BOTTLE FILLER RE: PLUMB |
| 22.05 | OPTION C: <N> FLOOR DRAIN, GRIND CONC TOPPING SLAB TO SLOPE FLOOR (2% MIN) 2' FROM DRAIN IN ALL DIRECTIONS, MAINTAIN CONSISTENT FLOOR FINISH RE: PLUMB |
| 22.06 | OPTION C: <N> ABAAS COMPLIANT WALL HUNG WATER CLOSET RE: PLUMB |
| 22.07 | OPTION C: <N> ABAAS COMPLIANT WALL MOUNTED LAVATORY RE: PLUMB |
| 22.11 | OPTION C: <N> ABAAS COMPLIANT FAUCET RE: PLUMB |
| 22.12 | OPTION C: <N> ABAAS COMPLIANT WALL MOUNTED URINAL RE: PLUMB |




1 A 402 ENLARGED RESTROOM PLAN
SCALE A

SCALE: B 4 0 4 8 SCALE OF FEET

SCALE: A 2 0 3 6 SCALE OF FEET

ELIZABETH HALLAS 07418 LICENSED PROFESSIONAL ARCHITECT IOWA

| | | | | | |
|---|---|---|---|---|---|
| A/E FIRM: | DESIGNED: | SUB SHEET NO. | TITLE OF SHEET | DRAWING NO. | |
| PRIME: OTAK, INC. REDMOND, WA | LB/CM | A 402 | ENLARGED - PUBLIC RESTROOMS | 394 194248 | |
| SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DRAWN: LB/MM | | EFFIGY MOUNDS NATIONAL MONUMENT | PMIS/PKG NO. 318539 | |
| | TECH REVIEW: ACN / EMH | | | SHEET 43 OF 142 | |
| | DATE: 05/29/2026 | | | | |

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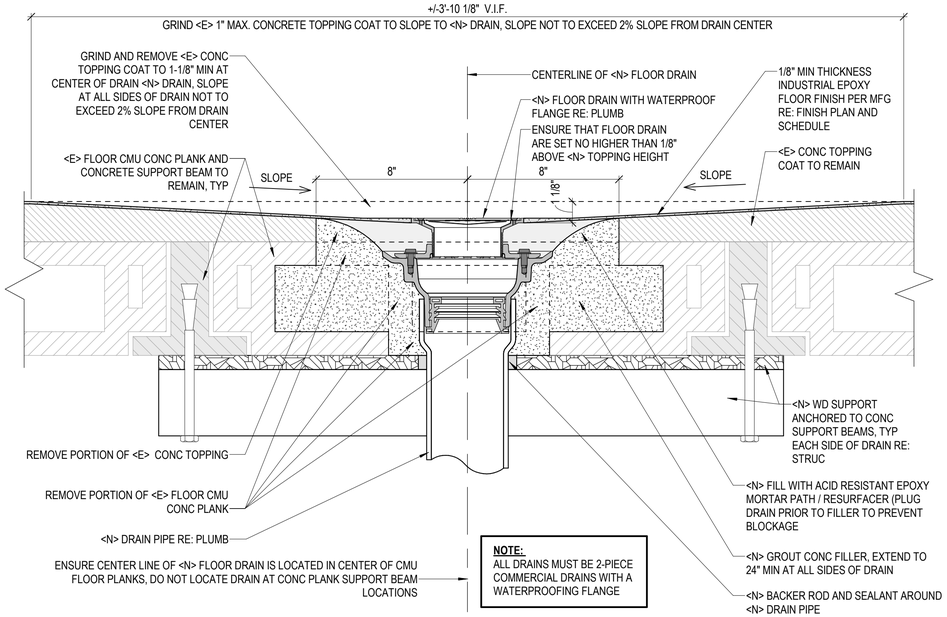
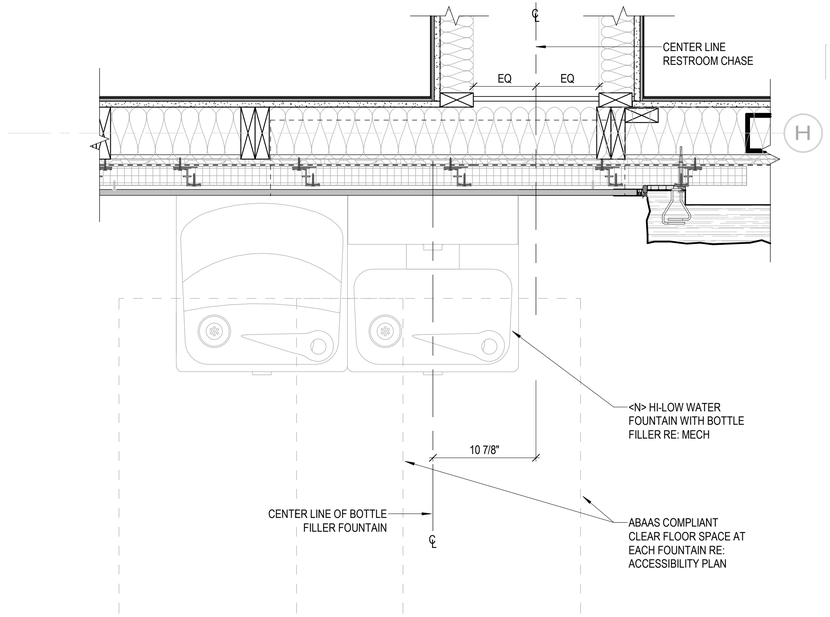



## 8 TYP FLOOR DRAIN SECTION DETAIL
A 403 SCALE B

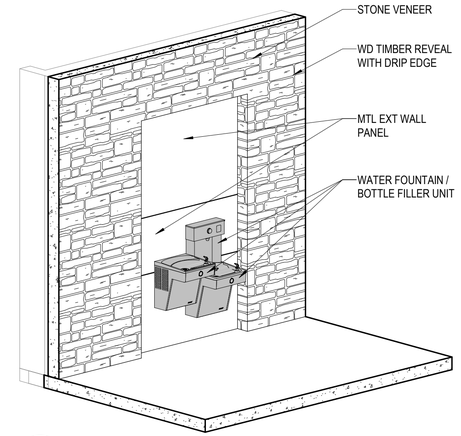



## 4 ENLARGED PLAN AT EXTERIOR WATER FOUNTAIN
A 403 SCALE A

WATER FOUNTAIN IS OPTION C WITH PLUMBING SCOPE.

# GENERAL NOTES

1. RE: CIVIL, LAND, STRUC, MECH, PLUMB, ELEC, AND FIRE PROTECTION FOR MORE INFO.
2. FF: MAIN LEVEL 100'-0" = NAVD88 657.3' +/- AT EXT DOORS.
3. ALL PLAN DIMS LOCATING EXT ASSEMBLIES ARE TO GRID LINE OR FACE OF FINISH, UNO. <E> WALLS WITHOUT WALL TAGS SHALL REMAIN.
4. VERIFY EQUIP ROUGH-IN DIMS WITH MFR FOR ALL EQUIP INCLUDING <E>, REUSED, OR FURNISHED BY CO.
5. PROVIDE ADEQUATE BLOCKING, BACKING OR STRUC SUPPORT AS REQ'D FOR PROPER INSTALLATION OF ALL MOUNTED ASSEMBLIES, INCLUDING ALL ATTACHED EQUIP, ACCESSORIES (CONTRACTOR FURNISHED ITEMS), PLUMB FIXTURES, MILLWORK, AND CASEWORK.
6. FLOOR DRAIN LOCATIONS SHOWN ARE APPROXIMATE. VIF EXACT LOCATIONS PER PLUMB WITH STRUC COORDINATION. SLOPE CONC TO DRAIN AS INDICATED.
7. ALL CONDUIT & PIPING TO BE ROUTED WITHIN WALLS OR FURRING WALLS UON. CONTRACTOR TO COORDINATE IN FIELD ALL EXPOSED CONDUIT AND PIPING ROUTING WITH CO.
8. DOORS AND STOREFRONTS OCCUPY <E> FRAMED OPENINGS UON. VIF DIMS IN FIELD. ALIGN ALL <N> ROUGH OPENINGS WITH <E> MASONRY JOINTS UON: RE:STOREFRONT & DOOR SCHEDULE FOR DIMS AND MORE INFO.
9. PATCH AND REPAIR ALL FINISHES IMPACTED BY SYSTEMS AND DEMO WORK RE: FINISH SCHEDULE UON. MATCH ADJ IN MATERIAL, COLOR, TEXTURE AND FINISH.
10. FURNITURE SHOWN IS FOR REFERENCE ONLY AND IS NOT INCLUDED IN THE CONTRACT UON.



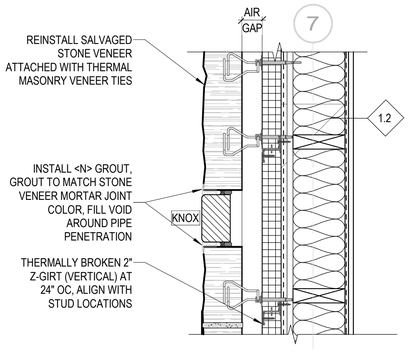

## 2 ISO AT EXT WATER FOUNTAIN
A 403 N.T.S.

WATER FOUNTAIN IS OPTION C WITH PLUMBING SCOPE.

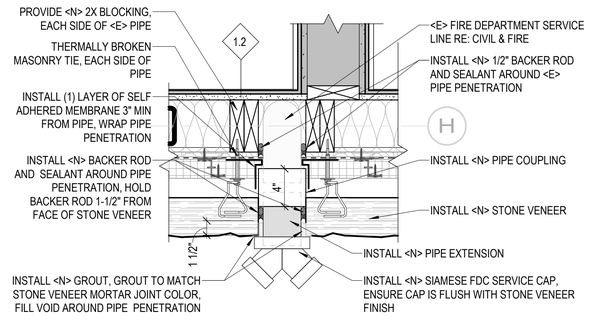



## 6 FRONT DESK ACCESS GATE ELEVATION
A 403 SCALE A

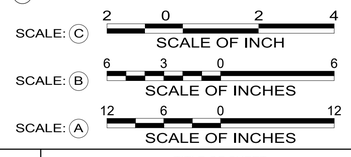


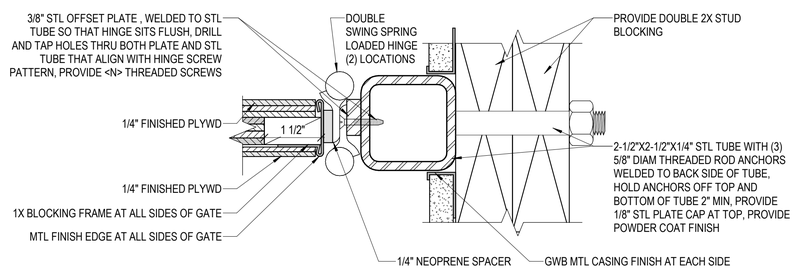




## 3 KNOX BOX PLAN DETAIL
A 403 SCALE A

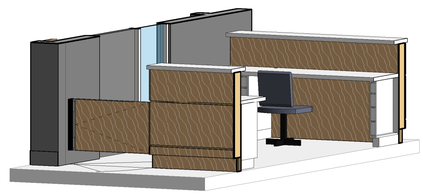



## 1 ENLARGED PLAN DETAIL AT FDC SERVICE
A 403 SCALE A




## 7 PLAN DETAIL AT FRONT DESK ACCESS GATE
A 403 SCALE A



## 5 ISO AT FRONT DESK ACCESS GATE
A 403 N.T.S.

LICENSED PROFESSIONAL ARCHITECT
ELIZABETH HALLAS
07418
IOWA

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM CADD LB/MM TECH REVIEW: ACN / EMH DATE: 05/29/2026 | SUB SHEET NO. A 403 | TITLE OF SHEET: ENLARGED - DRINKING FOUNTAIN & GATE — EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 44 OF 142 |

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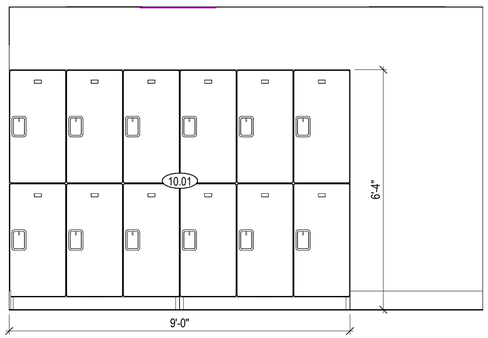
# GENERAL NOTES

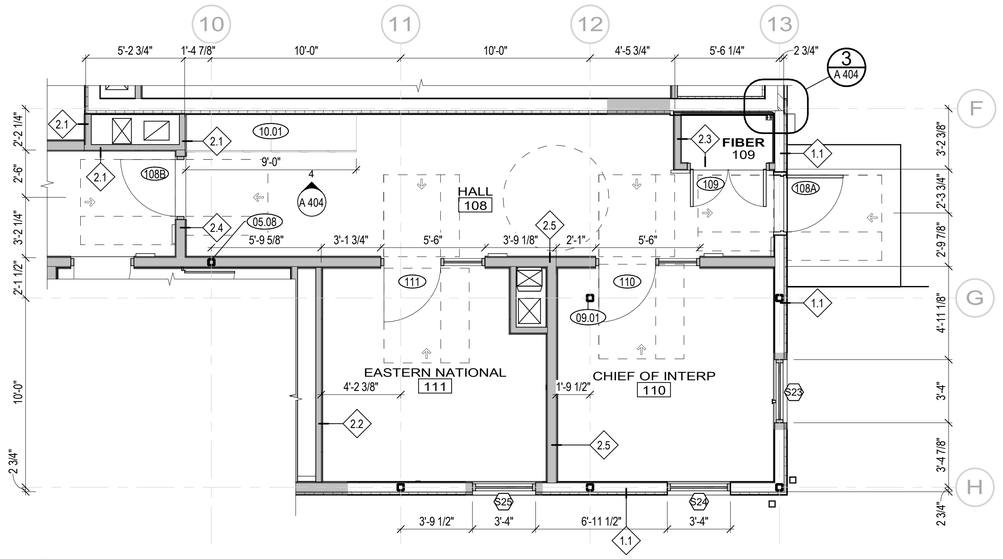
1. RE: CIVIL, LAND, STRUC, MECH, PLUMB, ELEC, AND FIRE PROTECTION FOR MORE INFO.
2. FF: MAIN LEVEL 100'-0" = NAVD88 657.3' +/- AT EXT DOORS.
3. ALL PLAN DIMS LOCATING EXT ASSEMBLIES ARE TO GRID LINE OR FACE OF FINISH, UNO. <E> WALLS WITHOUT WALL TAGS SHALL REMAIN.
4. VERIFY EQUIP ROUGH-IN DIMS WITH MFR FOR ALL EQUIP INCLUDING <E>, REUSED, OR FURNISHED BY CO.
5. PROVIDE ADEQUATE BLOCKING, BACKING OR STRUC SUPPORT AS REQ'D FOR PROPER INSTALLATION OF ALL MOUNTED ASSEMBLIES, INCLUDING ALL ATTACHED EQUIP, ACCESSORIES (CONTRACTOR FURNISHED ITEMS), PLUMB FIXTURES, MILLWORK, AND CASEWORK.
6. FLOOR DRAIN LOCATIONS SHOWN ARE APPROXIMATE. VIF EXACT LOCATIONS PER PLUMB WITH STRUC COORDINATION. SLOPE CONC TO DRAIN AS INDICATED.
7. ALL CONDUIT & PIPING TO BE ROUTED WITHIN WALLS OR FURRING WALLS UON. CONTRACTOR TO COORDINATE IN FIELD ALL EXPOSED CONDUIT AND PIPING ROUTING WITH CO.
8. DOORS AND STOREFRONTS OCCUPY <E> FRAMED OPENINGS UON. VIF DIMS IN FIELD. ALIGN ALL <N> ROUGH OPENINGS WITH <E> MASONRY JOINTS UON: RE:STOREFRONT & DOOR SCHEDULE FOR DIMS AND MORE INFO.
9. PATCH AND REPAIR ALL FINISHES IMPACTED BY SYSTEMS AND DEMO WORK RE: FINISH SCHEDULE UON. MATCH ADJ IN MATERIAL, COLOR, TEXTURE, AND FINISH.
10. FURNITURE SHOWN IS FOR REFERENCE ONLY AND IS NOT INCLUDED IN THE CONTRACT UON.. ALL SHELVING TO BE FURNITURE SOLUTION AND SHOWN FOR REFERENCE ONLY.

# KEYNOTES

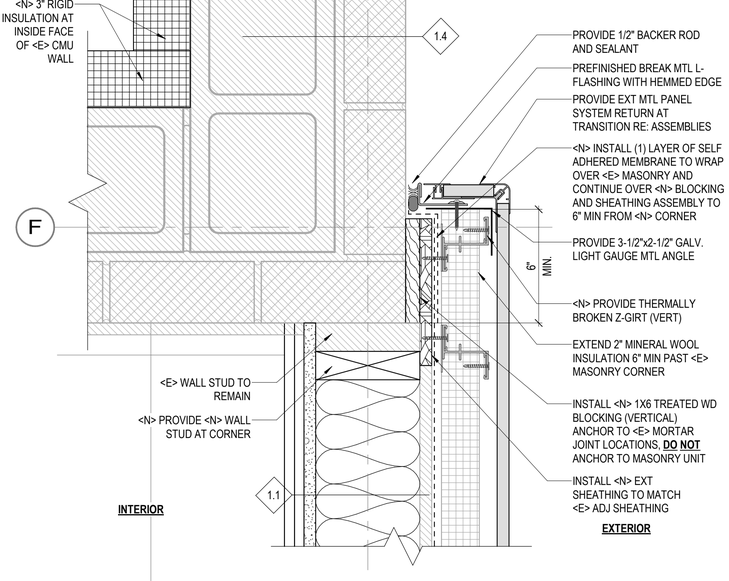
| | |
|---|---|
| 05.08 | OPTION C: <N> HSS COLUMN, PROVIDE RIGID INSULATION AT EXT SIDE OF COLUMN BETWEEN SHEATHING AND STL RE: STRUC |
| 09.01 | OPTION C: PT <E> STL COLUMN RE: FINISH SCHEDULE |
| 10.01 | OPTION C: <N> TWO TIERED EXTRA WIDE WD LOCKERS, LOWER LOCKERS SHALL BE PROVIDED WITH AN ADJUSTABLE SHELF LESS THAN 15" AFF IN COMPLIANCE WITH ABAAS. |
| 23.01 | OPTION A: <N> MECHANICAL EQUIPMENT RE: MECH |
| 26.07 | OPTION C: <N> ABAAS DOOR PUSH BUTTON RE: ELEC |
| 27.02 | OPTION C: <N> AV RACK RE: TECH |



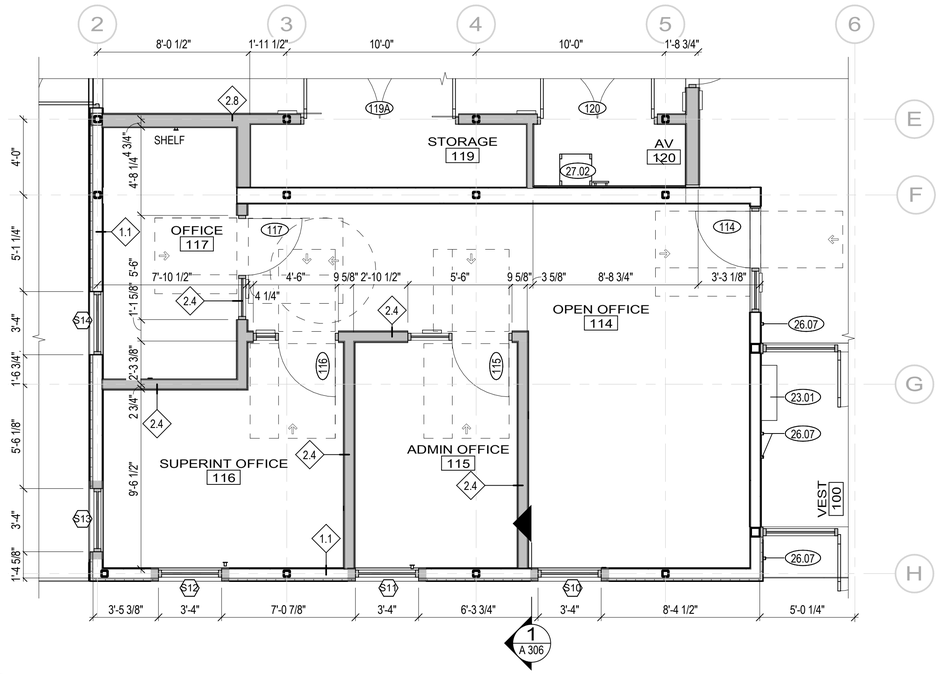

4 / A 404 INTERIOR ELEVATION - HALL LOCKERS
SCALE (C)




2 / A 404 ENLARGED - OFFICE AREA 2
SCALE (A)




3 / A 404 PLAN DTL AT MATERIAL TRANSITION AT EXTERIOR WALL
SCALE (B)




1 / A 404 ENLARGED - OFFICE AREA 1
SCALE (A)

SCALE: (C) 2 0 2 4 SCALE OF FEET
SCALE: (B) 6 3 0 6 SCALE OF INCHES
SCALE: (A) 4 0 4 8 SCALE OF FEET

LICENSED PROFESSIONAL ARCHITECT ELIZABETH HALLAS 07418 IOWA

| | | | |
|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA<br>SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM<br>DRAWN: LB/MM<br>TECH REVIEW: ACN / EMH<br>DATE: 05/29/2026 | SUB SHEET NO. A 404 | TITLE OF SHEET: ENLARGED - OFFICE AREA<br>EFFIGY MOUNDS NATIONAL MONUMENT |

DRAWING NO. 394 194248
PMIS/PKG NO. 318539
SHEET 45 OF 142

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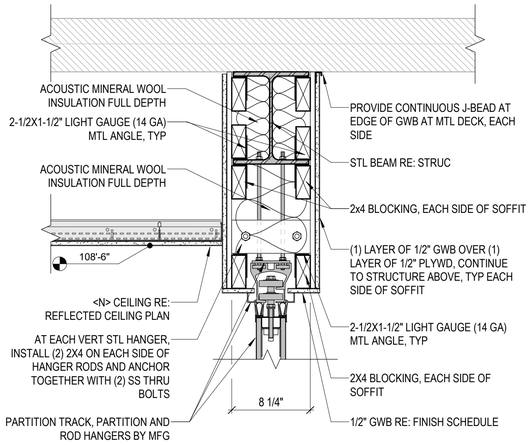
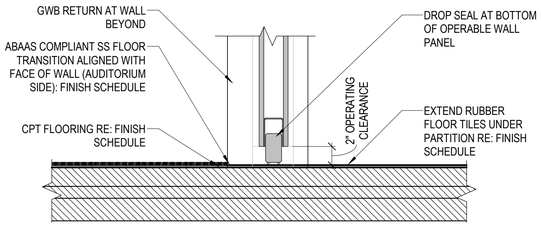
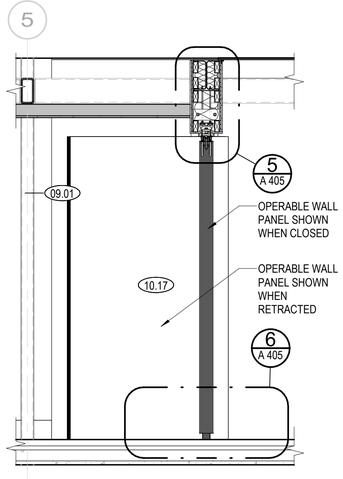


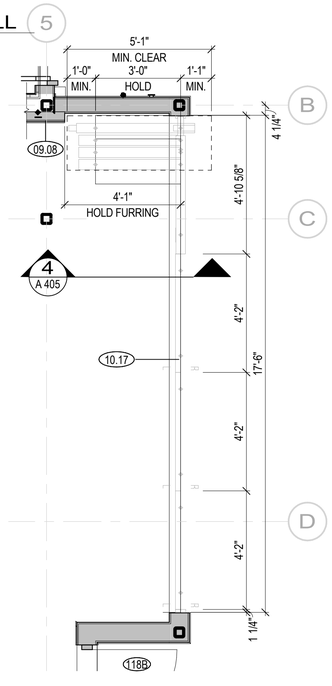

5 A 405 SECTION AT OPERABLE PARTITION HEAD
SCALE C



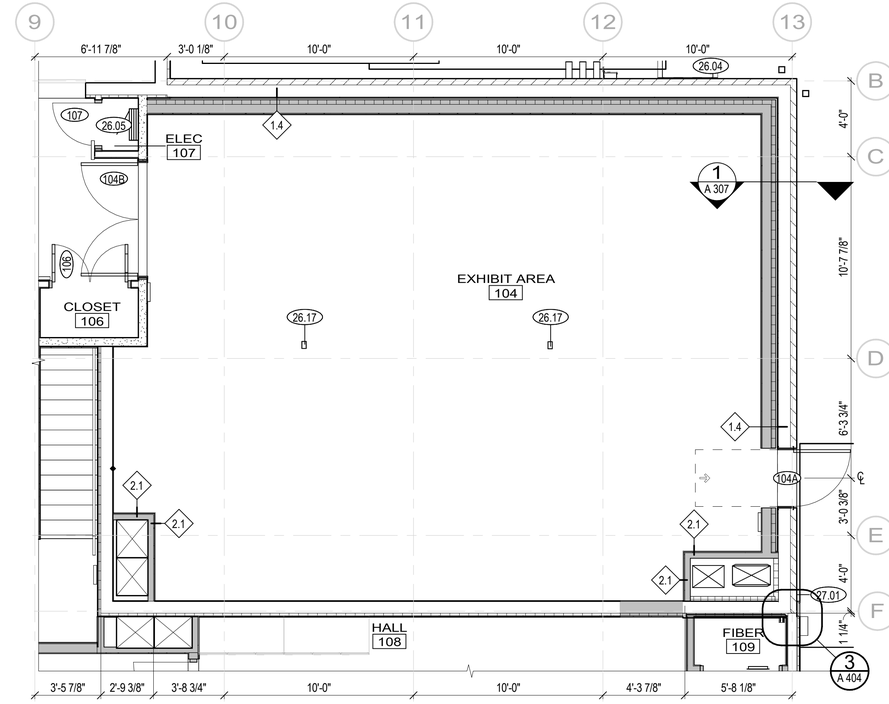

6 A 405 SECTION AT OPERABLE PARTITION SILL
SCALE B



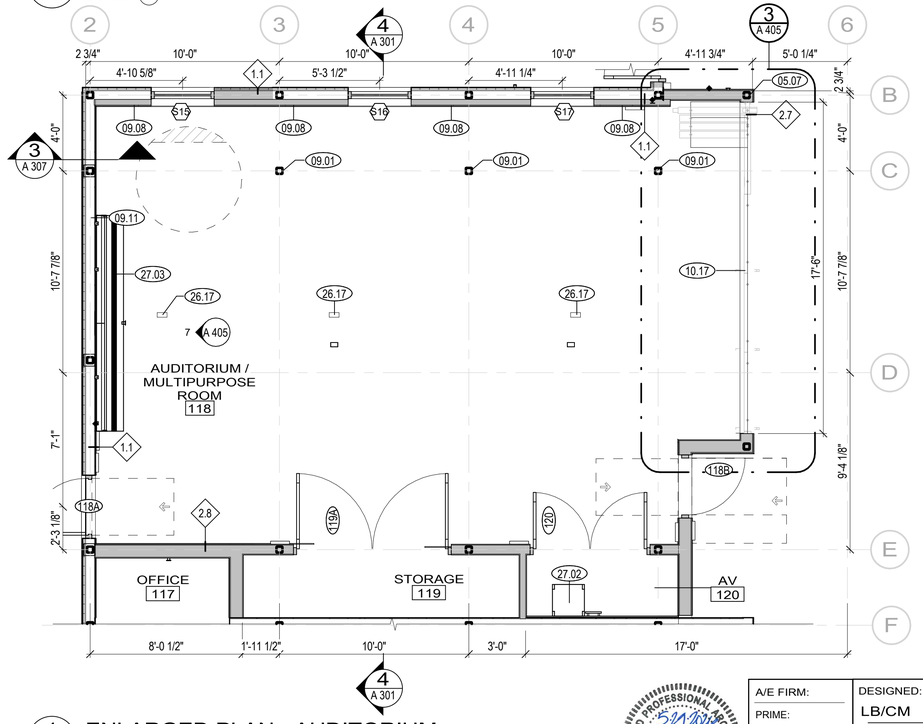

4 A 405 SECTION AT OPERABLE WALL
SCALE B




3 A 405 ENLARGED PLAN - OPERABLE WALL
SCALE B




2 A 405 EXHIBIT AREA ENLARGED PLAN
SCALE A




1 A 405 ENLARGED PLAN - AUDITORIUM
SCALE A




7 A 405 AUDITORIUM NORTH ELEVATION
SCALE A

# GENERAL NOTES

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10. FURNITURE SHOWN IS FOR REFERENCE ONLY AND IS NOT INCLUDED IN THE CONTRACT UON.. ALL SHELVING TO BE FURNITURE SOLUTION AND SHOWN FOR REFERENCE ONLY.

# KEYNOTES

| | |
|---|---|
| 05.07 | OPTION C: <N> HSS COLUMN TO SUPPORT NEW HSS BEAM FOR FOLDING PARTITION |
| 09.01 | OPTION C: PT <E> STL COLUMN RE: FINISH SCHEDULE |
| 09.08 | INSTALL <N> INT VAPOR BARRIER AT INSIDE FACE OF <E> WALL FRAMING RE: ASSEMBLIES, INSTALL <N> 2X4 WD FRAMING AT 16" OC (ALIGN WITH <E> FRAMING, EXTEND WD FURRING AND 5/8" GWB TO 6" ABOVE CLNG RE: FINISH SCHEDULE |
| 09.11 | OPTION C: INSTALL <N> 5/8" GWB FINISH FROM FLOOR TO 6" ABOVE CLNG RE: FINISH SCHEDULE |
| 10.17 | OPTION F: <N> FOLDING WALL PARTITION |
| 26.04 | OPTION C: <N> ELEC EQUIP RE: ELEC |
| 26.05 | OPTION C: <E> ELEC PANEL TO REMAIN, PROTECT IN PLACE RE: ELEC |
| 26.17 | OPTION C: <N> FLOOR BOX, HEIGHT DIFFERENCE BETWEEN BOX AND ADJ FLOOR SHALL NOT EXCEED 1/4", FLOOR BOX AND INSTALLATION SHALL BE ABAAS COMPLIANT RE: ELEC |
| 27.01 | OPTION C: <E> TELECOM/TELEPHONE BOX, PROTECT IN PLACE RE: ELEC |
| 27.02 | OPTION C: <N> AV RACK RE: TECH |
| 27.03 | OPTION F: <N> WALL MOUNTED PROJECTION SCREEN RE: TECH |

SCALE: C 12 6 0 12 SCALE OF INCHES
SCALE: B 2 0 2 4 SCALE OF FEET
SCALE: A 4 0 4 8 SCALE OF FEET

LICENSED PROFESSIONAL ARCHITECT ELIZABETH HALLAS 07418 IOWA

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|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM DRAWN: LB/MM TECH REVIEW: ACN / EMH DATE: 05/29/2026 | SUB SHEET NO. A 405 | TITLE OF SHEET: ENLARGED - AUDITORIUM & EXHIBITS — EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 46 OF 142 |

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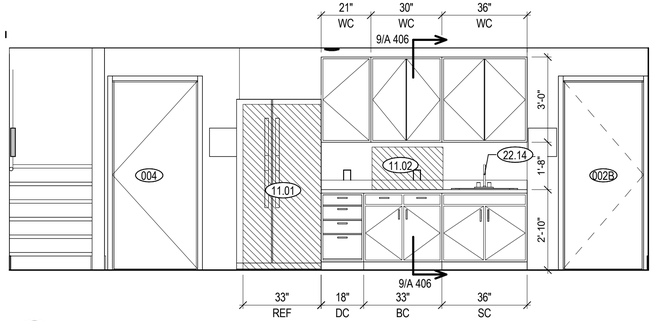
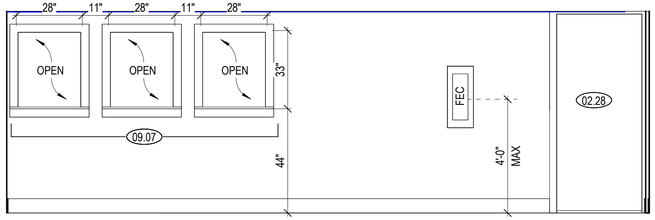
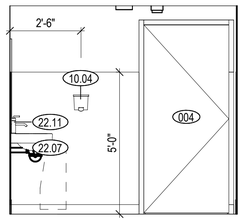


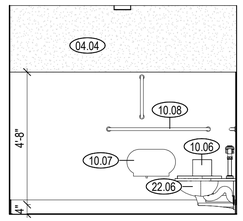

2 A 406 BREAK ROOM SOUTH
SCALE A



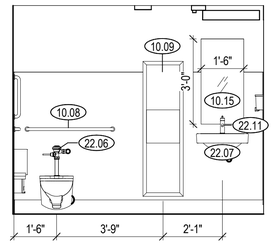

1 A 406 BREAK ROOM NORTH
SCALE A



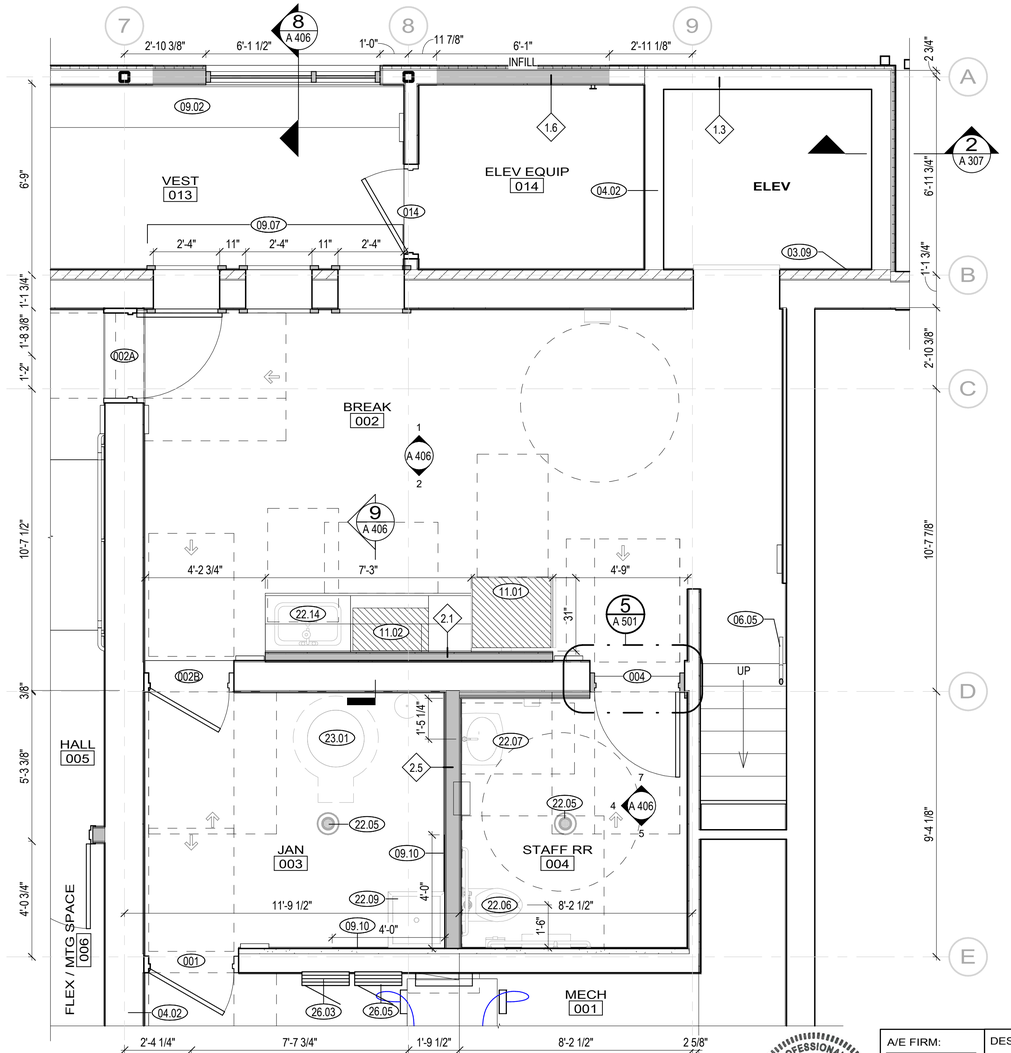

7 A 406 STAFF RR NORTH
SCALE A



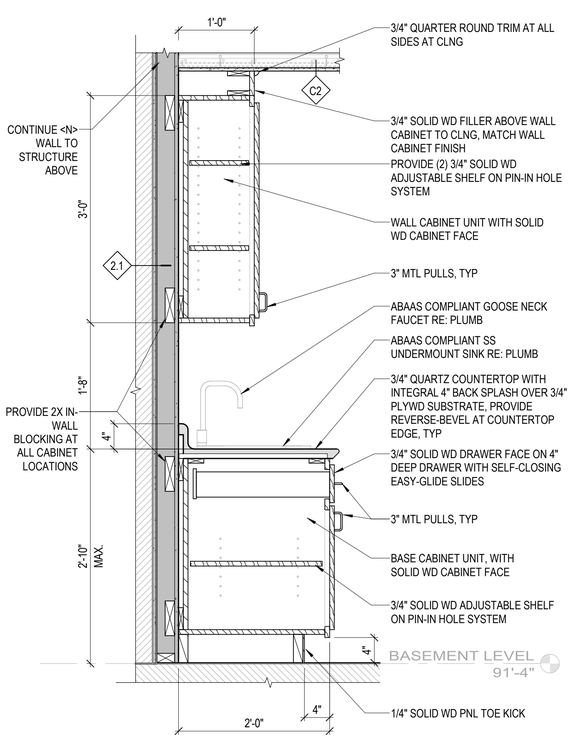

5 A 406 STAFF RR SOUTH
SCALE A




4 A 406 STAFF RR WEST
SCALE A




3 A 406 ENLARGED STAFF AREA PLAN
SCALE A




9 A 406 WALL AND BASE CABINET TYP DTL
SCALE C

# GENERAL NOTES

1. RE: CIVIL, LAND, STRUC, MECH, PLUMB, ELEC, AND FIRE PROTECTION FOR MORE INFO.
2. FF: MAIN LEVEL 100'-0" = NAVD88 657.3' +/- AT EXT DOORS.
3. ALL PLAN DIMS LOCATING EXT ASSEMBLIES ARE TO GRID LINE OR FACE OF FINISH, UNO. <E> WALLS WITHOUT WALL TAGS SHALL REMAIN.
4. VERIFY EQUIP ROUGH-IN DIMS WITH MFR FOR ALL EQUIP INCLUDING <E>, REUSED, OR FURNISHED BY CO.
5. PROVIDE ADEQUATE BLOCKING, BACKING OR STRUC SUPPORT AS REQ'D FOR PROPER INSTALLATION OF ALL MOUNTED ASSEMBLIES, INCLUDING ALL ATTACHED EQUIP, ACCESSORIES (CONTRACTOR FURNISHED ITEMS), PLUMB FIXTURES, MILLWORK, AND CASEWORK.
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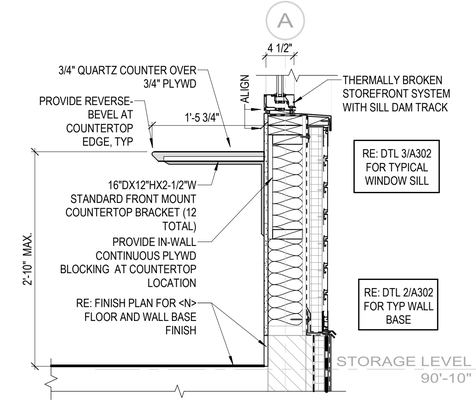
8. DOORS AND STOREFRONTS OCCUPY <E> FRAMED OPENINGS UON. VIF DIMS IN FIELD. ALIGN ALL <N> ROUGH OPENINGS WITH <E> MASONRY JOINTS UON: RE:STOREFRONT & DOOR SCHEDULE FOR DIMS AND MORE INFO.
9. PATCH AND REPAIR ALL FINISHES IMPACTED BY SYSTEMS AND DEMO WORK RE: FINISH SCHEDULE UON. MATCH ADJ IN MATERIAL, COLOR, TEXTURE AND FINISH.
10. FURNITURE SHOWN IS FOR REFERENCE ONLY AND IS NOT INCLUDED IN THE CONTRACT UON.

# KEYNOTES

| | |
|---|---|
| 02.28 | <E> ELEVATOR DOOR TO REMAIN, PROTECT IN PLACE |
| 03.09 | PATCH CONC AT BASE OF EAST WALL OF ELEVATOR SHAFT |
| 04.02 | OPTION A: INFILL <E> OPENING (11X28 APPROXIMATE SIZE, VIF) IN CMU. SEAL, AND FINISH TO MATCH ADJ |
| 04.04 | OPTION C: <E> CMU WALL, PROVIDE BLOCK FILLER AND PNT RE: FINISH SCHEDULE |
| 06.05 | OPTION C: <N> 1-1/2" DIAM WD HANDRAIL AT 36" AFF ATTACHED TO <E> CMU WALL WITH WALL-MOUNT PREFINISHED MTL BRACKETS; PROVIDE 12" MIN HANDRAIL EXTENSIONS PAST NOSING AT TOP AND BOTTOM OF STAIR |
| 09.02 | OPTION C: <N> QUARTZ COUNTERTOP ON HEAVY DUTY HIDDEN MTL WALL BRACKETS |
| 09.07 | OPTION C: AT BOTH SIDES OF WALL PROVIDE <N> 1X4 WD TRIM AT HEAD AND JAMBS OF CASED OPENING, STN. BELOW <N> 1X SILLS PROVIDE 1X4 APRON, STN. AFTER DEMO'ING FINISH ON WALL CONFIRM WITH CO ON THIS DIRECTION BEFORE PROCEEDING WITH WORK. |
| 09.10 | OPTION C: INSTALL <N> FRP FROM FLOOR TO CLNG EACH SIDE OF MOP SINK RE: FINISH SCHEDULE |
| 10.04 | OPTION C: <N> WALL MOUNTED FOAM SOAP DISPENSER, MOUNT IN COMPLIANCE WITH ABAAS RE: MOUNTING HEIGHTS |
| 10.06 | OPTION C: <N> SANITARY NAPKIN DISPOSAL, MOUNT IN COMPLIANCE WITH ABAAS RE: MOUNTING HEIGHTS |
| 10.07 | OPTION C: <N> JUMBO ROLL TOILET PAPER DISPENSER, MOUNT IN COMPLIANCE WITH ABAAS RE: MOUNTING HEIGHTS |
| 10.08 | OPTION C: <N> ACCESSIBLE MTL GRAB BARS, MOUNT IN COMPLIANCE WITH ABAAS RE: MOUNTING HEIGHTS |
| 10.09 | OPTION C: <N> PAPER TOWEL DISPENSER MOUNTED IN A RECESSED SS FRAME WITH AN INTEGRAL TRASH RECEPTACLE, MOUNT IN COMPLIANCE WITH ABAAS RE: MOUNTING HEIGHTS |
| 10.15 | OPTION C: <N> WALL MOUNTED MIRROR, MOUNT IN COMPLIANCE WITH ABAAS RE: MOUNTING HEIGHTS |
| 11.01 | OPTION C: <N> REFRIGERATOR WITH INTEGRAL ICE MAKER AND WATER DISPENSER. |
| 11.02 | MICROWAVE, PARK PROVIDED, NIC |
| 22.05 | OPTION C: <N> FLOOR DRAIN, GRIND CONC TOPPING SLAB TO SLOPE FLOOR (2% MIN) 2' FROM DRAIN IN ALL DIRECTIONS, MAINTAIN CONSISTENT FLOOR FINISH RE: PLUMB |
| 22.06 | OPTION C: <N> ABAAS COMPLIANT WALL HUNG WATER CLOSET RE: PLUMB |
| 22.07 | OPTION C: <N> ABAAS COMPLIANT WALL MOUNTED LAVATORY RE: PLUMB |
| 22.09 | OPTION C: <N> ABAAS COMPLIANT MOP SINK RE: PLUMB |
| 22.11 | OPTION C: <N> ABAAS COMPLIANT FAUCET RE: PLUMB |
| 22.14 | OPTION C: <N> ABAAS COMPLIANT SS UNDERMOUNT SINK RE: PLUMB |
| 23.01 | OPTION A: <N> MECHANICAL EQUIPMENT RE: MECH |
| 26.03 | OPTION C: <N> ELEC PANEL RE: ELEC |
| 26.05 | OPTION C: <E> ELEC PANEL TO REMAIN, PROTECT IN PLACE RE: ELEC |




8 A 406 SECTION AT COUNTER TOP
SCALE B

SCALE: C 1 0 1 2 SCALE OF FEET

SCALE: B 12 6 0 12 SCALE OF INCHES

SCALE: A 2 0 3 6 SCALE OF FEET

ELIZABETH HALLAS 07416 LICENSED PROFESSIONAL ENGINEER IOWA

| | | | | | |
|---|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM DRAWN: LB/MM TECH REVIEW: ACN / EMH DATE: 05/29/2026 | SUB SHEET NO. A 406 | TITLE OF SHEET ENLARGED - BASEMENT STAFF AREA EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 47 OF 142 | |

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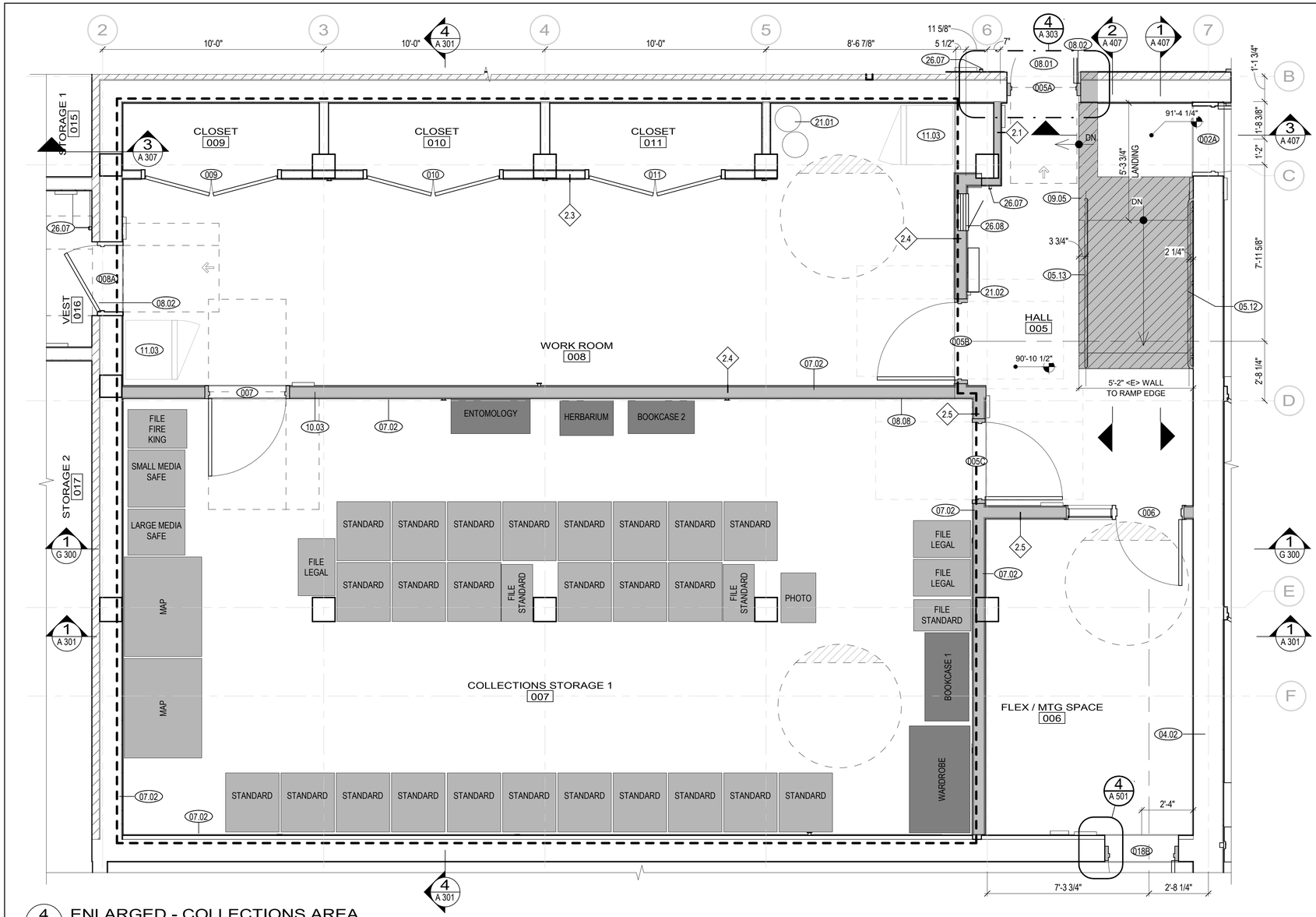
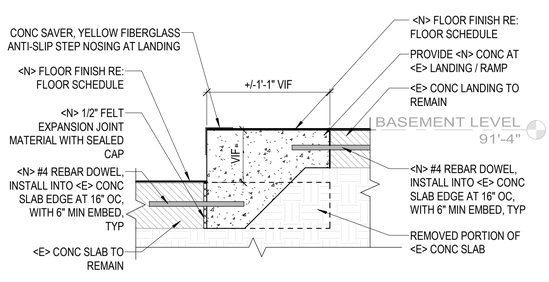
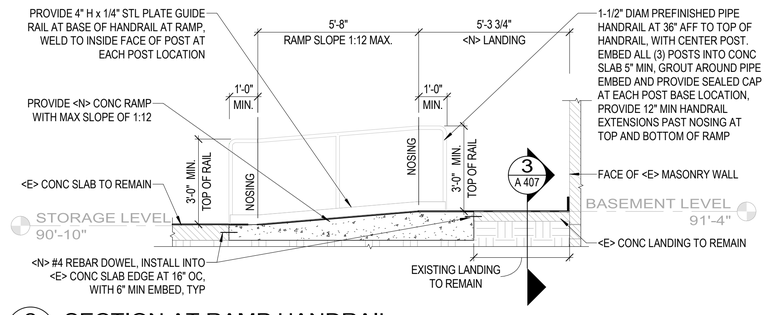


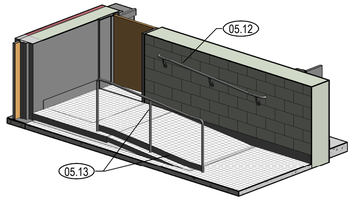

## 4 A 407 ENLARGED - COLLECTIONS AREA
SCALE (A)




## 3 A 407 SECTION AT RAMP LANDING
SCALE (C)




## 2 A 407 SECTION AT RAMP HANDRAIL
SCALE (B)

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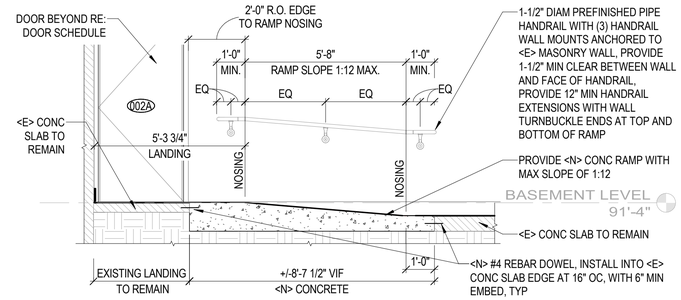
# KEYNOTES

| | |
|---|---|
| 04.02 | OPTION A: INFILL <E> OPENING (11X28 APPROXIMATE SIZE, VIF) IN CMU. SEAL, AND FINISH TO MATCH ADJ |
| 05.12 | OPTION C: <N> 1-1/2" DIA. PRE-FINISHED PIPE HANDRAIL AT 36" TO TOP OF RAIL, INSTALL (3) HANDRAIL WALL MOUNTS TO <E> MASONRY WALL, PROVIDE 1-1/2" MIN CLEAR BETWEEN WALL AND FACE OF HANDRAIL, PROVIDE 12" MIN HANDRAIL EXTENTIONS FROM RAMP NOSING AT TOP AND BOTTOM OF RAMP. RE: 1/A407 |
| 05.13 | OPTION C: 1-1/2" DIAM PRE-FINISHED PIPE HANDRAIL AT 36" TO TOP OF HANDRAIL, WITH CENTER POST. EMBED ALL (3) POST INTO CONC SLAB 5" MIN, GROUT AROUND PIPE EMBED AND PROVIDE SEALED CAP AT EACH POST BASE, PROVIDE 12" MIN HANDRAIL EXTENSIONS PAST NOSING AT TOP AND BOTTOM OF RAMP, AT HANDRAIL BASE PROVIDE 4"H x 1/4" STL PLATE GUIDE RAIL, WELD TO INSIDE FACE OF POST AT EACH LOCATION RE: 2/A407 |
| 07.02 | OPTION C: <N> 1 LAYER 5/8" TYPE X FIRE-RESISTANT GWB FINISH, SEAL ALL PENETRATIONS |
| 08.01 | OPTION C: ADJUST DOOR OPENING TO ACCOMMODATE <N> RAMP WIDTH |
| 08.02 | OPTION C: ADJUST DOOR FRAME TO ALIGN WITH INT FACE OF WALL |
| 08.08 | OPTION C: <N> ABAAS DOOR PUSH BUTTON |
| 09.05 | OPTION C: <N> SAFETY ANTI-SLIP STEP EDGE (BLACK AND YELLOW) AT STEP LOCATION AT LANDING |
| 10.03 | OPTION C: <N> RECESSED KEY STORAGE PANEL BOX, MOUNT IN COMPLIANCE WITH ABAAS RE: MOUNTING HEIGHTS |
| 11.03 | OPTION C: RELOCATED <E> COLLECTIONS FREEZER, NIC |
| 21.01 | OPTION A: <E> CLEAN AGENT TANK TO REMAIN, PROTECT IN PLACE RE: FIRE PROTECTION |
| 21.02 | OPTION A: RELOCATED <E> AGENT RELEASE CONTROL PANEL RE: FIRE PROTECTION |
| 26.07 | OPTION C: <N> ABAAS DOOR PUSH BUTTON RE: ELEC |
| 26.08 | OPTION C: REINSTALL <E> ELEC PANEL RE: ELEC |

# COLLECTIONS STORAGE LEGEND

- SHORT STORAGE FURNITURE (NIC) SHOWN FOR REFERENCE ONLY
- TALL STORAGE FURNITURE (NIC) SHOWN FOR REFERENCE ONLY
- INSTALL PORTION OF <N> CONC RAMP AND ADJ FLOOR SLAB
- INDICATE WALLS TO BE SEALED TO THE UNDERSIDE OF ABOVE STRUCTURE FOR PROPER OPERATION OF CLEAN AGENT SYSTEM




## 1 A 407 SECTION AT RAMP AT WALL
SCALE (B)

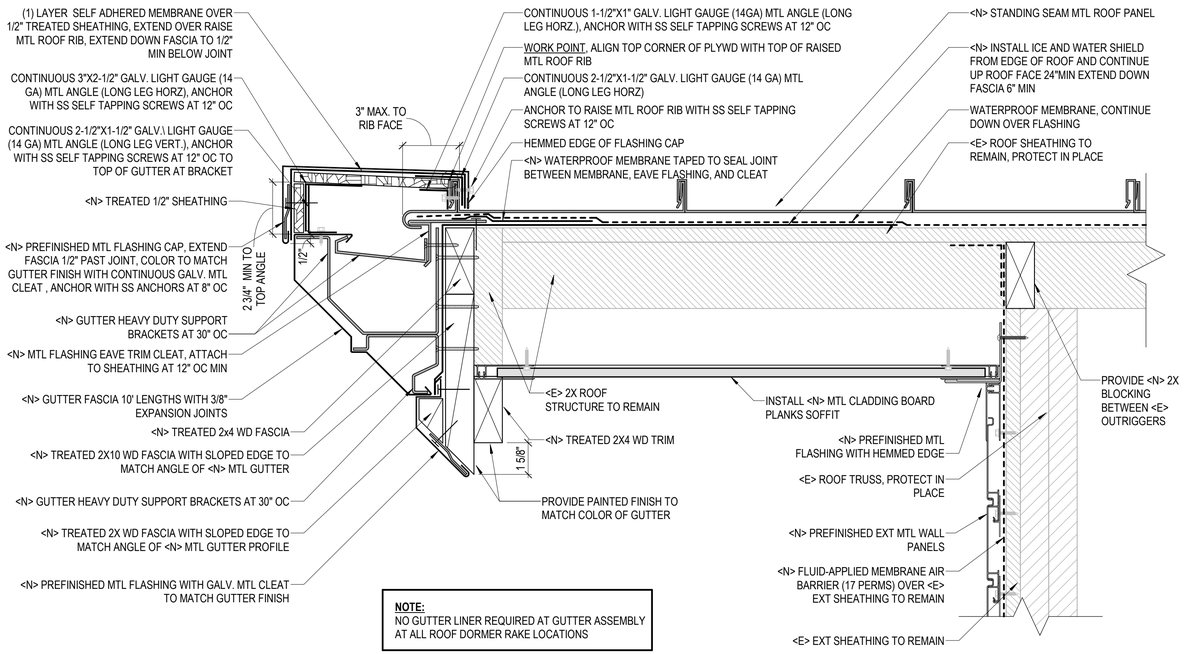
SCALE: (C) 12 6 0 12 SCALE OF INCHES

SCALE: (B) 2 0 3 6 SCALE OF FEET

SCALE: (A) 8 0 8 16 SCALE OF FEET

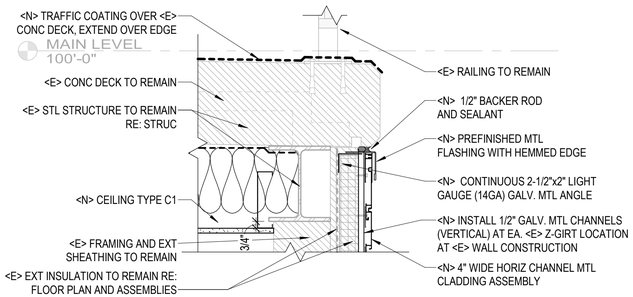
LICENSED PROFESSIONAL ARCHITECT ELIZABETH HALLAS 07418 IOWA

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM CADD LB/MM TECH REVIEW: ACN / EMH DATE: 05/29/2026 | SUB SHEET NO. A 407 | TITLE OF SHEET ENLARGED - COLLECTIONS STORAGE EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 48 OF 142 |

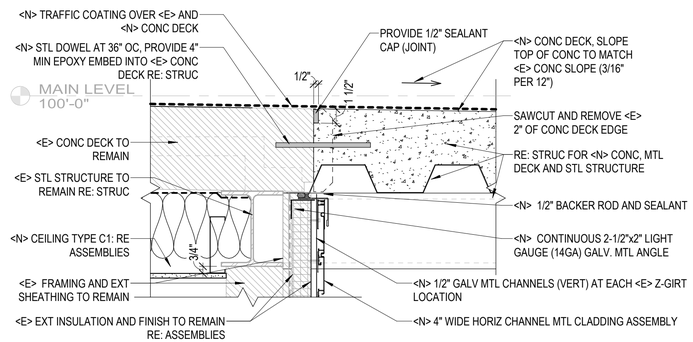



**NOTE:**
NO GUTTER LINER REQUIRED AT GUTTER ASSEMBLY AT ALL ROOF DORMER RAKE LOCATIONS

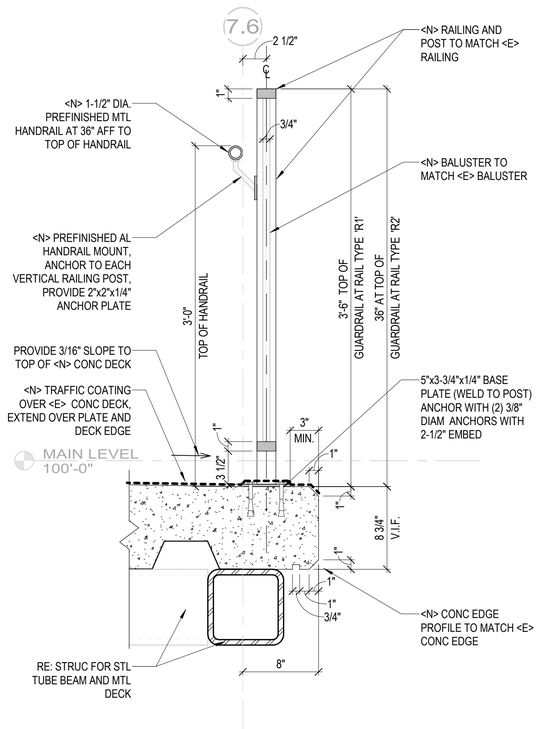
## 6 A 408 SECTION DETAIL AT RAKE OF ROOF
SCALE (C)




## 5 A 408 SECTION AT GUARDRAIL /CONC EDGE
SCALE (B)



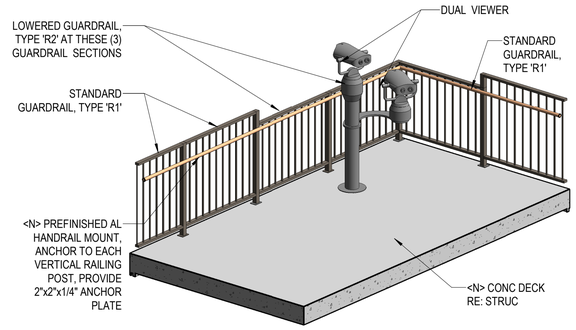

## 4 A 408 OPTION G SECTION AT EXISTING CONCRETE DECK EDGE
SCALE (B)




## 3 A 408 SECTION DETAIL AT GUARDRAIL - CONCRETE DECK EDGE
SCALE (B)

# GENERAL NOTES

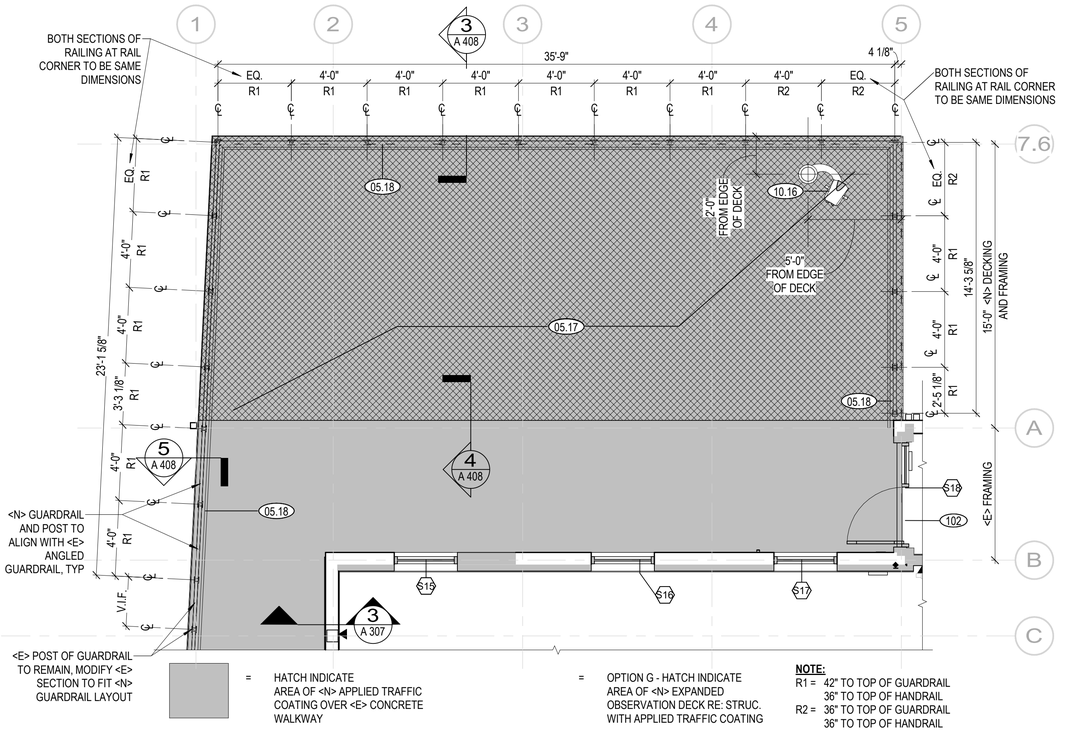
1. RE: CIVIL, LAND, STRUC, MECH, PLUMB, ELEC, AND FIRE PROTECTION FOR MORE INFO.
2. FF: MAIN LEVEL 100'-0" = NAVD88 657.3' +/- AT EXT DOORS.
3. ALL PLAN DIMS LOCATING EXT ASSEMBLIES ARE TO GRID LINE OR FACE OF FINISH, UNO. <E> WALLS WITHOUT WALL TAGS SHALL REMAIN.
4. VERIFY EQUIP ROUGH-IN DIMS WITH MFR FOR ALL EQUIP INCLUDING <E>, REUSED, OR FURNISHED BY CO.
5. PROVIDE ADEQUATE BLOCKING, BACKING OR STRUC SUPPORT AS REQ'D FOR PROPER INSTALLATION OF ALL MOUNTED ASSEMBLIES, INCLUDING ALL ATTACHED EQUIP, ACCESSORIES (CONTRACTOR FURNISHED ITEMS), PLUMB FIXTURES, MILLWORK, AND CASEWORK.
6. FLOOR DRAIN LOCATIONS SHOWN ARE APPROXIMATE. VIF EXACT LOCATIONS PER PLUMB WITH STRUC COORDINATION. SLOPE CONC TO DRAIN AS INDICATED.
7. ALL CONDUIT & PIPING TO BE ROUTED WITHIN WALLS OR FURRING WALLS UON. CONTRACTOR TO COORDINATE IN FIELD ALL EXPOSED CONDUIT AND PIPING ROUTING WITH CO.
8. DOORS AND STOREFRONTS OCCUPY <E> FRAMED OPENINGS UON. VIF DIMS IN FIELD. ALIGN ALL <N> ROUGH OPENINGS WITH <E> MASONRY JOINTS UON: RE:STOREFRONT & DOOR SCHEDULE FOR DIMS AND MORE INFO.
9. PATCH AND REPAIR ALL FINISHES IMPACTED BY SYSTEMS AND DEMO WORK RE: FINISH SCHEDULE UON. MATCH ADJ IN MATERIAL, COLOR, TEXTURE, AND FINISH.
10. FURNITURE SHOWN IS FOR REFERENCE ONLY AND IS NOT INCLUDED IN THE CONTRACT UON.. ALL SHELVING TO BE FURNITURE SOLUTION AND SHOWN FOR REFERENCE ONLY.




## 2 A 408 ISO-1 AT DUAL VIEWER
N.T.S.

# KEYNOTES

| | |
|---|---|
| 05.17 | OPTION G: EXPAND DECK AT NORTHEAST CORNER. ADDITION OF CONC OVER MTL DECK, STRUCTURAL STL, <N> MTL RAILINGS. |
| 05.18 | OPTION G: <N> VERT PICKET STYLE MTL RAILING TO MATCH <E> (PT) WITH MTL HANDRAIL AT 36" AFF (STN) OPTION G AT EAST DECK EXTENSION ONLY |
| 10.16 | OPTION G: <N> ACCESSIBLE PREFINISHED MTL DUAL VIEWING MACHINE |




**NOTE:**
R1 = 42" TO TOP OF GUARDRAIL
36" TO TOP OF HANDRAIL
R2 = 36" TO TOP OF GUARDRAIL
36" TO TOP OF HANDRAIL

## 1 A 408 OPTION G - EXPANDED OBSERVATION DECK
SCALE (A)

SCALE: (C) 6 3 0 6 SCALE OF INCHES
SCALE: (B) 12 6 0 12 SCALE OF INCHES
SCALE: (A) 4 0 4 8 SCALE OF FEET

LICENSED PROFESSIONAL ARCHITECT ELIZABETH HALLAS 07418 IOWA

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA<br>SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM<br>DRAWN: LB/MM<br>TECH REVIEW: ACN / EMH<br>DATE: 05/29/2026 | SUB SHEET NO. A 408 | TITLE OF SHEET: ENLARGED - OBSERVATION DECK<br>EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248<br>PMIS/PKG NO. 318539<br>SHEET 49 OF 142 |

7/7/2026 1:21:00 PM Autodesk Docs://2025-110 NPS EFMO Visitor Center DD-CD/2024250 NPS EFMO Visitor Center - v24_BIM360.rvt

# DOOR SCHEDULE

| NUMBER | N/E | PANEL TYPE | FRAME TYPE | SIZE THICKNESS | SIZE WIDTH | SIZE HEIGHT | DETAILS HEAD | DETAILS JAMB | DETAILS THRESH | HARDWARE SET NO. | COMMENTS |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 001 | <N> | A | <E> | 1 3/4" | 3'-0" | 6'-8" | ** | ** | ** | 07 | PROVIDE <N> 1HR FIRE RATED DOOR - <E> RATED FRAME TO REMAIN |
| 002A | <E> | - | - | 1 3/4" | 3'-0" | 6'-8" | ** | ** | ** | 11 | |
| 002B | <N> | A | <E> | 1 3/4" | 2'-10" | 6'-8" | ** | ** | ** | 07 | PROVIDE <N> WD DOOR IN <E> FRAME |
| 004 | <N> | A | A | 1 3/4" | 3'-0" | 6'-8" | ** | 5/A501 | ** | 13 | |
| 005A | <N> | C | A | 1 3/4" | 3'-0" | 6'-8" | ** | ** | ** | 16 | ADJUST DOOR OPENING TO ACCOMODATE <N> RAMP WIDTH |
| 005B | <N> | C | A | 1 3/4" | 3'-6" | 6'-8" | ** | ** | ** | 4B | SEALED DOOR BOTTOM FOR CLEAN AGENT |
| 005C | <N> | A | A | 1 3/4" | 3'-6" | 6'-8" | ** | ** | ** | 4A | PROVIDE <N> 45-MIN FIRE RATED DOOR AND FRAME, SEALED DOOR BOTTOM FOR CLEAN AGENT |
| 006 | <N> | A | B | 1 3/4" | 3'-0" | 6'-8" | ** | ** | ** | 20 | |
| 007 | <N> | C | A | 1 3/4" | 3'-6" | 6'-8" | ** | ** | ** | 04 | PROVIDE <N> 45-MIN FIRE RATED DOOR, GLAZING AND FRAME |
| 008A | <E> | - | - | 1 3/4" | 3'-0" | 6'-8" | ** | ** | ** | 12 | SEALED DOOR BOTTOM FOR CLEAN AGENT |
| 009 | <E> | - | - | 1 3/4" | 6'-0" | 6'-8" | ** | ** | ** | 15 | |
| 010 | <E> | - | - | 1 3/4" | 6'-0" | 6'-8" | ** | ** | ** | 15 | |
| 011 | <E> | - | - | 1 3/4" | 6'-0" | 6'-8" | ** | ** | ** | 15 | |
| 013A | <N> | F | STOREFRONT | 1 3/4" | 3'-0" | 6'-10" | ** | ** | ** | A01 | |
| 013B | <E> | - | - | 1 3/4" | 3'-0" | 6'-8" | ** | ** | ** | 06 | |
| 014 | <N> | - | - | 1 3/4" | 3'-0" | 6'-8" | ** | ** | ** | 06 | PROVIDE <N> 1-HR FIRE RATED DOOR PANEL AND FRAME. IF <E> DOOR AND FRAME MEETS ELEVATOR CODE WITHOUT REPLACEMENT, RETAIN <E> |
| 015 | <E> | - | - | 1 3/4" | 3'-0" | 6'-8" | ** | ** | ** | 06 | |
| 016 | <E> | - | - | 1 3/4" | 3'-0" | 6'-8" | ** | ** | ** | 01 | SEAL SIDES OF MTL DOOR JAMB AT BTTM 12" EACH SIDE. |
| 017A | <E> | - | - | 1 3/4" | 3'-0" | 6'-8" | ** | ** | ** | 06 | |
| 018A | <N> | A | A | 1 3/4" | 3'-0" | 6'-8" | ** | ** | ** | 06 | |
| 018B | <N> | A | A | 1 3/4" | 3'-0" | 6'-8" | 3/A501 | 4/A501 | ** | 06 | |
| 100A | <N> | E | STOREFRONT | 1 3/4" | 6'-0" | 7'-0" | ** | ** | ** | A02 | |
| 100B | <N> | E | STOREFRONT | 1 3/4" | 6'-0" | 7'-0" | ** | ** | ** | A05 | |
| 101 | <N> | A | STOREFRONT | 1 3/4" | 3'-0" | 6'-10" | ** | ** | ** | 14 | |
| 102 | <N> | F | STOREFRONT | 1 3/4" | 3'-0" | 7'-0" | ** | ** | ** | A03A | |
| 103A | <N> | A | STOREFRONT | 1 3/4" | 3'-0" | 6'-10" | ** | ** | ** | 05 | |
| 103B | <N> | C | STOREFRONT | 1 3/4" | 3'-0" | 6'-10" | ** | ** | ** | 5A | |
| 104A | <N> | G | A | 1 3/4" | 3'-0" | 6'-8" | ** | ** | ** | 03 | |
| 104B | <N> | B | A | 1 3/4" | 6'-0" | 6'-10" | ** | ** | ** | 17 | PROVIDE <N> 45-MIN FIRE RATED DOOR, GLAZING AND FRAME, UV FILM ON GLAZING |
| 105 | <N> | D | STOREFRONT | 1 3/4" | 4'-0" | 6'-10" | ** | ** | ** | 09 | |
| 106 | <N> | D | STOREFRONT | 1 3/4" | 4'-0" | 6'-10" | ** | ** | ** | 09 | |
| 107 | <N> | A | STOREFRONT | 1 3/4" | 2'-3" | 6'-10" | ** | ** | ** | 10 | |
| 108A | <E> | - | - | 1 3/4" | 3'-0" | 6'-8" | ** | ** | ** | 02 | |
| 108B | <N> | C | STOREFRONT | 1 3/4" | 3'-0" | 6'-10" | ** | ** | ** | 5A | |
| 109 | <N> | D | STOREFRONT | 1 3/4" | 4'-0" | 6'-10" | ** | ** | ** | 09 | |
| 110 | <N> | A | STOREFRONT | 1 3/4" | 3'-0" | 6'-10" | ** | ** | ** | 20 | |
| 111 | <N> | A | STOREFRONT | 1 3/4" | 3'-0" | 6'-10" | ** | ** | ** | 20 | |
| 112 | <N> | A | STOREFRONT | 1 3/4" | 3'-0" | 6'-10" | ** | ** | ** | 18 | |
| 113 | <N> | A | STOREFRONT | 1 3/4" | 3'-0" | 6'-10" | ** | ** | ** | 18 | |
| 114 | <N> | A | STOREFRONT | 1 3/4" | 3'-0" | 6'-10" | ** | ** | ** | 5A | |
| 115 | <N> | A | STOREFRONT | 1 3/4" | 3'-0" | 6'-10" | ** | ** | ** | 20 | |
| 116 | <N> | A | STOREFRONT | 1 3/4" | 3'-0" | 6'-10" | ** | ** | ** | 20 | |
| 117 | <N> | A | STOREFRONT | 1 3/4" | 3'-0" | 6'-10" | ** | ** | ** | 20 | |
| 118A | <N> | G | A | 1 3/4" | 3'-0" | 7'-0" | ** | ** | ** | 03 | |
| 118B | <N> | A | STOREFRONT | 1 3/4" | 3'-0" | 6'-10" | ** | ** | ** | 21 | |
| 119A | <N> | D | STOREFRONT | 1 3/4" | 8'-0" | 6'-10" | ** | ** | ** | 08 | |
| 120 | <N> | D | STOREFRONT | 1 3/4" | 6'-0" | 6'-10" | ** | ** | ** | 08 | |

** = REFERENCE STOREFRONT SCHEDULE FOR ALL DTLS RELATED TO DOORS

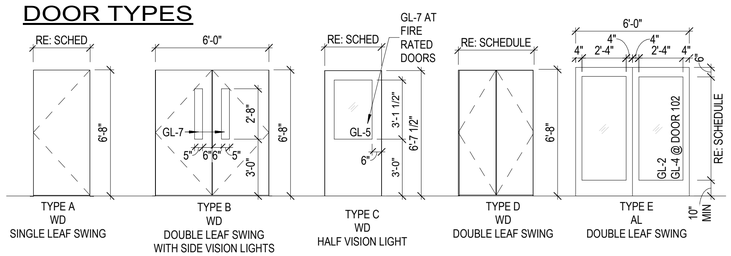
# GLAZING SCHEDULE

| TYPE | LOCATION | CRITERIA | | BASIS OF DESIGN |
|---|---|---|---|---|
| GL-1 | SOUTH & WEST ELEVATIONS | ENERGY PERFORMANCE | 1" INSULATED GLAZING UNITS | |
| | | COATINGS | LOW-E COATING (#2 SURFACE) | SNX 51/23 ON #2 AIR FILLED, SHGC 0.23 |
| | | GLAZING STRENGTH | FLOAT GLASS | |
| | | APPEARANCE | CLEAR W/ BIRD FRIENDLY ETCHING | BIRD 1ST ETCH DX 22 ON #1 |
| GL-2 | SOUTH & WEST ELEVATIONS & TEMPERED | ENERGY PERFORMANCE | 1" INSULATED GLAZING UNITS | |
| | | COATINGS | LOW-E COATING (#2 SURFACE) | SNX 51/23 ON #2 AIR FILLED, SHGC 0.23 |
| | | GLAZING STRENGTH | TEMPERED GLASS | |
| | | APPEARANCE | CLEAR W/ BIRD FRIENDLY ETCHING | BIRD 1ST ETCH DX 22 ON #1 |
| GL-3 | NORTH & EAST ELEVATIONS | ENERGY PERFORMANCE | 1" INSULATED GLAZING UNITS | AIR FILLED |
| | | COATINGS | NONE | |
| | | GLAZING STRENGTH | FLOAT GLASS | |
| | | APPEARANCE | CLEAR W/ BIRD FRIENDLY ETCHING | BIRD 1ST ETCH DX 22 ON #1 |
| GL-4 | NORTH & EAST ELEVATIONS & TEMPERED | ENERGY PERFORMANCE | 1" INSULATED GLAZING UNITS | AIR FILLED |
| | | COATINGS | NONE | |
| | | GLAZING STRENGTH | TEMPERED GLASS | |
| | | APPEARANCE | CLEAR W/ BIRD FRIENDLY ETCHING | BIRD 1ST ETCH DX 22 ON #1 |
| GL-5 | INTERIOR LOCATIONS | ENERGY PERFORMANCE | NOT INSULATED | |
| | | COATINGS | NONE | |
| | | GLAZING STRENGTH | SINGLE PANE TEMPERED GLASS | |
| | | APPEARANCE | CLEAR | |
| GL-6 | WEST ELEVATION | ENERGY PERFORMANCE | 1" INSULATED GLAZING UNITS | |
| | | COATINGS | 50% FROST ETCHING (#2 SURFACE) | |
| | | GLAZING STRENGTH | TEMPERED GLASS | |
| | | APPEARANCE | FROSTED | |
| GL-7 | INTERIOR FIRE RATED DOOR LOCATIONS | FIRE PERFORMANCE | 45 MINUTE RATED | |
| | | COATINGS | NONE | |
| | | GLAZING STRENGTH | SINGLE PANE TEMPERED GLASS | |
| | | APPEARANCE | CLEAR | |

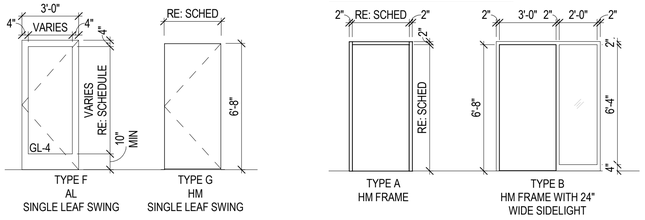
# GENERAL NOTES

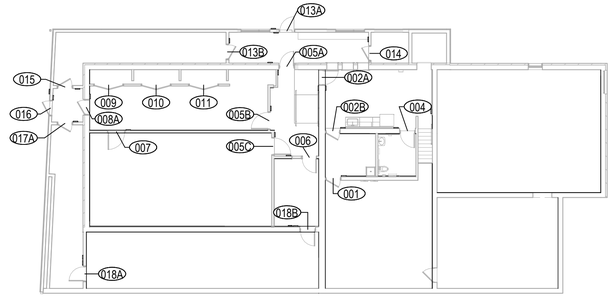
1. SAFETY GLAZING IS TO BE UTILIZED IN THE FOLLOWING LOCATIONS AS REQUIRED BY CODE:
   A. WITHIN ANY DOOR
   B. ADJ TO ANY DOOR WHERE THE NEAREST VERT EDGE OF THE GLAZING IS WITHIN A 24" ARC OF EITHER VERT EDGE OF THE DOOR IN A CLOSED POSITION AND WHERE THE BOTTOM EXPOSED EDGE OF THE GLAZING IS LESS THAN 60" ABOVE THE WALKING SURFACE
   C. FIXED OR OPERABLE PANELS THAT MEET ALL THE FOLLOWING CONDITIONS:
      A. THE EXPOSED AREA OF THE INDIVIDUAL PANE IS GREATER THAN 9 SF
      B. THE BOTTOM EDGE OF THE GLAZING IS LESS THAN 18" AFF
      C. THE TOP EDGE OF THE GLAZING IS GREATER THAN 36" AFF
      D. ONE OR MORE WALKING SURFACES ARE WITHIN 36", MEASURED HORIZ AND IN A STRAIGHT LINE, OF THE PLANE OF GLAZING
2. REFER TO DOOR TYPES, STOREFRONT TYPES, GLAZING SCHEDULE & SPECS FOR GLAZING REQUIREMENTS.
3. INSTALL ABAAS COMPLIANT DOOR HARDWARE AT ALL DOORS. REFER TO SPECS FOR DOOR HARDWARE SETS.
   DOOR HANDLES, PULLS, LATCHES, LOCKS AND OTHER OPERATING, DEVICES ON DOORS REQUIRED TO BE ACCESSIBLE BY IBC 2024 CHAPTER 11 SHALL NOT REQUIRE TIGHT GRASPING, TIGHT PINCHING OR TWISTING OF THE WRIST TO OPERATE, PER 2024 IBC 1008.1.9.1.
4. DOORS AND STOREFRONTS OCCUPY <E> FRAMED OPENINGS UON. VIF DIMS IN FIELD. ALIGN ALL <N> ROUGH OPENINGS WITH <E> MASONRY JOINTS UON: RE:STOREFRONT & DOOR SCHEDULE FOR DIMS AND MORE INFO.
5. PATCH AND REPAIR ALL FINISHES IMPACTED BY SYSTEMS AND DEMO WORK RE: FINISH SCHEDULE UON. MATCH ADJ IN MATERIAL, COLOR, TEXTURE AND FINISH.
6. OPTIONS:
   A. OPTION B - EXTERIOR DOORS, STOREFRONTS, AND WINDOWS
   B. OPTION C - INTERIOR DOORS AND WINDOWS

# DOOR TYPES

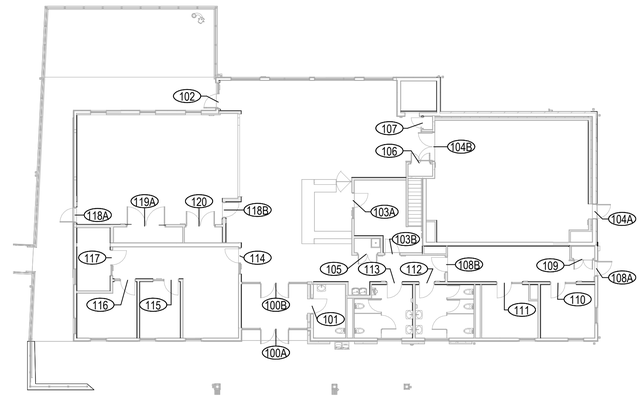



# FRAME TYPES



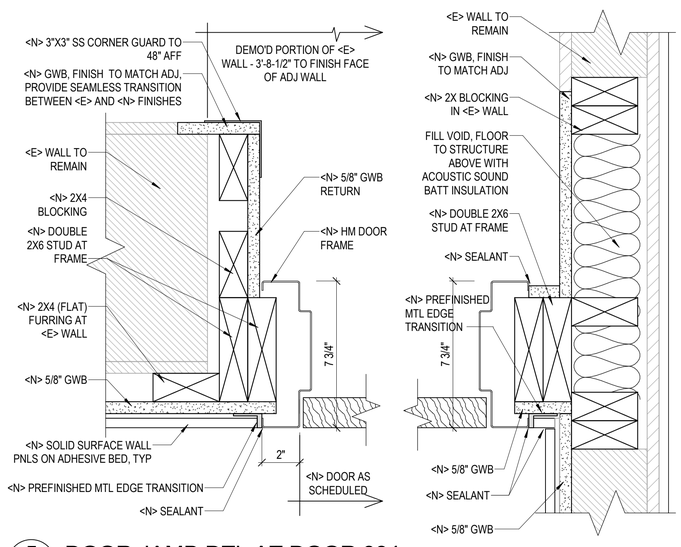




1 / A 501 DOOR KEY PLAN - BASEMENT LEVEL
SCALE (A)

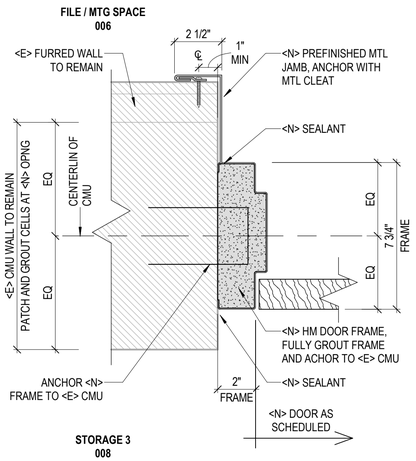



2 / A 501 DOOR KEY PLAN - MAIN LEVEL
SCALE (A)

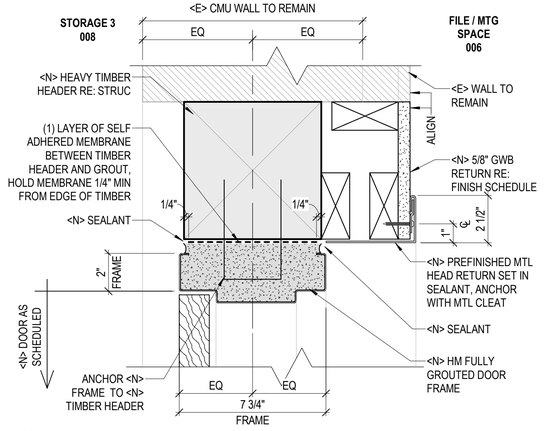
SCALE: (B) 6 3 0 6 SCALE OF INCHES
SCALE: (A) 16 0 16 32 SCALE OF FEET




5 / A 501 DOOR JAMB DTL AT DOOR 004
SCALE (B)




4 / A 501 DOOR JAMB DTL AT DOOR 018B
SCALE (B)




3 / A 501 DOOR HEAD DETAIL AT DOOR 018B
SCALE (B)

| | | | |
|---|---|---|---|
| A/E FIRM: | DESIGNED: LB/CM | SUB SHEET NO. | TITLE OF SHEET: DOOR SCHEDULE |
| PRIME: OTAK, INC. REDMOND, WA | CADD LB/MM | A 501 | EFFIGY MOUNDS NATIONAL MONUMENT |
| SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | TECH REVIEW: ACN / EMH | | DRAWING NO. 394 194248 |
| | DATE: 05/29/2026 | | PMIS/PKG NO. 318539 |
| | | | SHEET 50 OF 142 |

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7/7/2026 1:21:04 PM

# STOREFRONT SCHEDULE

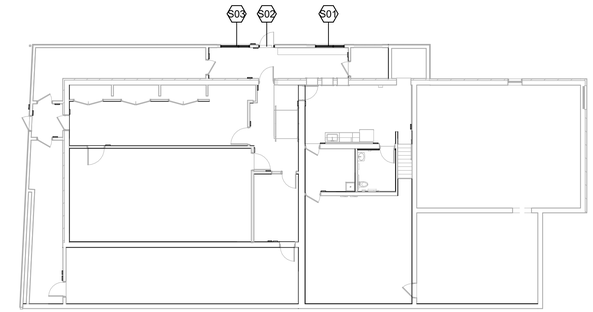
| MARK | TYPE | WIDTH | HEIGHT | DETAILS | | | THERMAL/INTERIOR | COMMENTS |
|---|---|---|---|---|---|---|---|---|
| | | | | SILL | JAMB | HEAD | | |
| S01 | J | 6'-1 1/2" | 4'-0" | 5/A503 | - | 4/A302 SIM. | THERMAL | PROVIDE <N> MANUAL SINGLE ROLLER SHADE |
| S02 | I | 3'-4" | 7'-0" | 5/A503 | - | 4/A302 SIM. | THERMAL | |
| S03 | J | 6'-1 1/2" | 4'-0" | 5/A503 | - | 4/A302 SIM. | THERMAL | PROVIDE <N> MANUAL SINGLE ROLLER SHADE |
| S10 | C | 3'-4" | 5'-2" | 5/A503 | - | 5/A504 | THERMAL | PROVIDE <N> MANUAL SINGLE ROLLER SHADE, PROVIDE OPENING CONTROL DEVICE TO LIMIT OPERABILITY TO 4 IN |
| S11 | C | 3'-4" | 5'-2" | 5/A503 | - | 5/A504 | THERMAL | PROVIDE <N> MANUAL SINGLE ROLLER SHADE, PROVIDE OPENING CONTROL DEVICE TO LIMIT OPERABILITY TO 4 IN |
| S12 | C | 3'-4" | 5'-2" | 5/A503 | - | 5/A504 | THERMAL | PROVIDE <N> MANUAL SINGLE ROLLER SHADE, PROVIDE OPENING CONTROL DEVICE TO LIMIT OPERABILITY TO 4 IN |
| S13 | C | 3'-4" | 5'-2" | 5/A503 | - | 5/A504 | THERMAL | PROVIDE <N> MANUAL SINGLE ROLLER SHADE, PROVIDE OPENING CONTROL DEVICE TO LIMIT OPERABILITY TO 4 IN |
| S14 | C | 3'-4" | 5'-2" | 5/A503 | - | 5/A504 | THERMAL | PROVIDE <N> MANUAL SINGLE ROLLER SHADE, PROVIDE OPENING CONTROL DEVICE TO LIMIT OPERABILITY TO 4 IN |
| S15 | C.1 | 3'-4" | 5'-2" | 2/A504 | - | 3/A504 | THERMAL | OPTION C: PROVIDE <N> MANUAL DUAL ROLLER SHADE |
| S16 | C.1 | 3'-4" | 5'-2" | 2/A504 | - | 3/A504 | THERMAL | OPTION C: PROVIDE <N> MANUAL DUAL ROLLER SHADE |
| S17 | C.1 | 3'-4" | 5'-2" | 2/A504 | - | 3/A504 | THERMAL | OPTION C: PROVIDE <N> MANUAL DUAL ROLLER SHADE |
| S18 | D | 5'-6" | 8'-0" | 3, 4/A503 | 8, 9, A503 | 1/A504 | THERMAL | |
| S19 | E | 6'-1 1/2" | 8'-0" | 3/A302 | - | 4/A302 | THERMAL | PROVIDE <N> MANUAL SINGLE ROLLER SHADE |
| S20 | G | 8'-6" | 8'-0" | 3/A302 | - | 4/A302 | THERMAL | PROVIDE <N> MANUAL SINGLE ROLLER SHADE |
| S21 | F | 8'-6" | 8'-0" | 3/A302 | - | 4/A302 | THERMAL | PROVIDE <N> MANUAL SINGLE ROLLER SHADE |
| S22 | E | 6'-1 1/2" | 8'-0" | 3/A302 | - | 4/A302 | THERMAL | PROVIDE <N> MANUAL SINGLE ROLLER SHADE |
| S23 | C | 3'-4" | 5'-2" | 5/A503 | | 4/A302 SIM. | THERMAL | PROVIDE <N> MANUAL SINGLE ROLLER SHADE, PROVIDE OPENING CONTROL DEVICE TO LIMIT OPERABILITY TO 4 IN |
| S24 | C | 3'-4" | 5'-2" | 5/A503 | | 4/A302 SIM. | THERMAL | PROVIDE <N> MANUAL SINGLE ROLLER SHADE, PROVIDE OPENING CONTROL DEVICE TO LIMIT OPERABILITY TO 4 IN |
| S25 | C | 3'-4" | 5'-2" | 5/A503 | | 4/A302 SIM. | THERMAL | PROVIDE <N> MANUAL SINGLE ROLLER SHADE, PROVIDE OPENING CONTROL DEVICE TO LIMIT OPERABILITY TO 4 IN |
| S26 | H | 5'-2" | 1'-8" | 3/A304 | - | 4/A304 | THERMAL | |
| S27 | H | 5'-2" | 1'-8" | 3/A304 | - | 4/A304 | THERMAL | |
| S28 | A | 13'-10 3/4" | 8'-0" | 3, 4/A503 | 6/A303 | 2/A307 | THERMAL | |
| S29 | B | 13'-10 3/4" | 8'-0" | 1, 2,/A503 | - | - | THERMAL | |
| S30 | K | 5'-6" | 7'-0" | | 6, 7 /A503 | | INTERIOR | |
| S31 | K | 5'-6" | 7'-0" | 1/A503 | 6, 7 /A503 | | INTERIOR | |
| S32 | L | 4'-6" | 7'-0" | 1/A503 | 6, 7 /A503 | | INTERIOR | |
| S33 | K | 5'-6" | 7'-0" | 1/A503 | 6, 7 /A503 | | INTERIOR | |
| S34 | K | 5'-6" | 7'-0" | 1/A503 | 6, 7 /A503 | | INTERIOR | |
| S35 | K | 5'-6" | 7'-0" | 1/A503 | 6, 7 /A503 | | INTERIOR | |
| S36 | I | 3'-4" | 7'-0" | - | 7/A503 | - | INTERIOR | |
| S37 | I | 3'-4" | 7'-0" | - | 7/A503 | - | INTERIOR | |
| S38 | I | 3'-4" | 7'-0" | - | 7/A503 | - | INTERIOR | |
| S39 | I | 3'-4" | 7'-0" | - | 7/A503 | - | INTERIOR | |
| S40 | I | 4'-4" | 7'-0" | - | 7/A503 | - | INTERIOR | |
| S41 | I | 3'-4" | 7'-0" | - | 7/A503 | - | INTERIOR | |
| S42 | I | 6'-4" | 7'-0" | - | 7/A503 | - | INTERIOR | |
| S43 | I | 8'-4" | 7'-0" | - | 7/A503 | - | INTERIOR | |
| S45 | I | 4'-4" | 7'-0" | - | 7/A503 | - | INTERIOR | |
| S46 | I | 4'-4" | 7'-0" | - | 7/A503 | - | INTERIOR | |
| S48 | I | 2'-7" | 7'-0" | - | 7/A503 | - | INTERIOR | |
| S49 | K | 5'-6" | 8'-0" | 1/A503 | 6, 7 /A503 | - | INTERIOR | |
| S50 | I | 3'-4" | 8'-0" | - | 7/A503 | - | INTERIOR | |

# GLAZING SCHEDULE

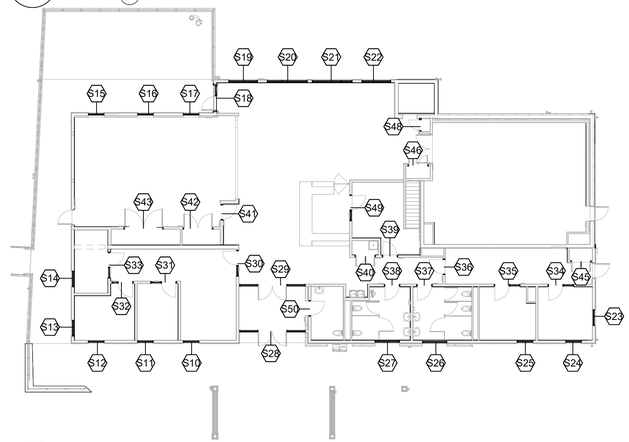
| TYPE | LOCATION | CRITERIA | | BASIS OF DESIGN |
|---|---|---|---|---|
| GL-1 | SOUTH & WEST ELEVATIONS | ENERGY PERFORMANCE | 1" INSULATED GLAZING UNITS | |
| | | COATINGS | LOW-E COATING (#2 SURFACE) | SNX 51/23 ON #2 AIR FILLED, SHGC 0.23 |
| | | GLAZING STRENGTH | FLOAT GLASS | |
| | | APPEARANCE | CLEAR W/ BIRD FRIENDLY ETCHING | BIRD 1ST ETCH DX 22 ON #1 |
| GL-2 | SOUTH & WEST ELEVATIONS & TEMPERED | ENERGY PERFORMANCE | 1" INSULATED GLAZING UNITS | |
| | | COATINGS | LOW-E COATING (#2 SURFACE) | SNX 51/23 ON #2 AIR FILLED, SHGC 0.23 |
| | | GLAZING STRENGTH | TEMPERED GLASS | |
| | | APPEARANCE | CLEAR W/ BIRD FRIENDLY ETCHING | BIRD 1ST ETCH DX 22 ON #1 |
| GL-3 | NORTH & EAST ELEVATIONS | ENERGY PERFORMANCE | 1" INSULATED GLAZING UNITS | AIR FILLED |
| | | COATINGS | NONE | |
| | | GLAZING STRENGTH | FLOAT GLASS | |
| | | APPEARANCE | CLEAR W/ BIRD FRIENDLY ETCHING | BIRD 1ST ETCH DX 22 ON #1 |
| GL-4 | NORTH & EAST ELEVATIONS & TEMPERED | ENERGY PERFORMANCE | 1" INSULATED GLAZING UNITS | AIR FILLED |
| | | COATINGS | NONE | |
| | | GLAZING STRENGTH | TEMPERED GLASS | |
| | | APPEARANCE | CLEAR W/ BIRD FRIENDLY ETCHING | BIRD 1ST ETCH DX 22 ON #1 |
| GL-5 | INTERIOR LOCATIONS | ENERGY PERFORMANCE | NOT INSULATED | |
| | | COATINGS | NONE | |
| | | GLAZING STRENGTH | SINGLE PANE TEMPERED GLASS | |
| | | APPEARANCE | CLEAR | |
| GL-6 | WEST ELEVATION | ENERGY PERFORMANCE | 1" INSULATED GLAZING UNITS | |
| | | COATINGS | 50% FROST ETCHING (#2 SURFACE) | |
| | | GLAZING STRENGTH | TEMPERED GLASS | |
| | | APPEARANCE | FROSTED | |
| GL-7 | INTERIOR FIRE RATED DOOR LOCATIONS | FIRE PERFORMANCE | 45 MINUTE RATED | |
| | | COATINGS | NONE | |
| | | GLAZING STRENGTH | SINGLE PANE TEMPERED GLASS | |
| | | APPEARANCE | CLEAR | |

# GENERAL NOTES

1. SAFETY GLAZING IS TO BE UTILIZED IN THE FOLLOWING LOCATIONS AS REQUIRED BY CODE:
   - A. WITHIN ANY DOOR
   - B. ADJ TO ANY DOOR WHERE THE NEAREST VERT EDGE OF THE GLAZING IS WITHIN A 24" ARC OF EITHER VERT EDGE OF THE DOOR IN A CLOSED POSITION AND WHERE THE BOTTOM EXPOSED EDGE OF THE GLAZING IS LESS THAN 60" ABOVE THE WALKING SURFACE
   - C. FIXED OR OPERABLE PANELS THAT MEET ALL THE FOLLOWING CONDITIONS:
     - A. THE EXPOSED AREA OF THE INDIVIDUAL PANE IS GREATER THAN 9 SF
     - B. THE BOTTOM EDGE OF THE GLAZING IS LESS THAN 18" AFF
     - C. THE TOP EDGE OF THE GLAZING IS GREATER THAN 36" AFF
     - D. ONE OR MORE WALKING SURFACES ARE WITHIN 36", MEASURED HORIZ AND IN A STRAIGHT LINE, OF THE PLANE OF GLAZING
2. REFER TO DOOR TYPES, STOREFRONT TYPES, GLAZING SCHEDULE & SPECS FOR GLAZING REQUIREMENTS.
3. INSTALL ABAAS COMPLIANT DOOR HARDWARE AT ALL DOORS. REFER TO SPECS FOR DOOR HARDWARE SETS.
   DOOR HANDLES, PULLS, LATCHES, LOCKS AND OTHER OPERATING, DEVICES ON DOORS REQUIRED TO BE ACCESSIBLE BY IBC 2024 CHAPTER 11 SHALL NOT REQUIRE TIGHT GRASPING, TIGHT PINCHING OR TWISTING OF THE WRIST TO OPERATE, PER 2024 IBC 1008.1.9.1.
4. DOORS AND STOREFRONTS OCCUPY <E> FRAMED OPENINGS UON. VIF DIMS IN FIELD. ALIGN ALL <N> ROUGH OPENINGS WITH <E> MASONRY JOINTS UON: RE:STOREFRONT & DOOR SCHEDULE FOR DIMS AND MORE INFO.
5. PATCH AND REPAIR ALL FINISHES IMPACTED BY SYSTEMS AND DEMO WORK RE: FINISH SCHEDULE UON. MATCH ADJ IN MATERIAL, COLOR, TEXTURE AND FINISH.
6. OPTIONS:
   - A. OPTION B - EXTERIOR DOORS, STOREFRONTS, AND WINDOWS
   - B. OPTION C - INTERIOR DOORS AND WINDOWS



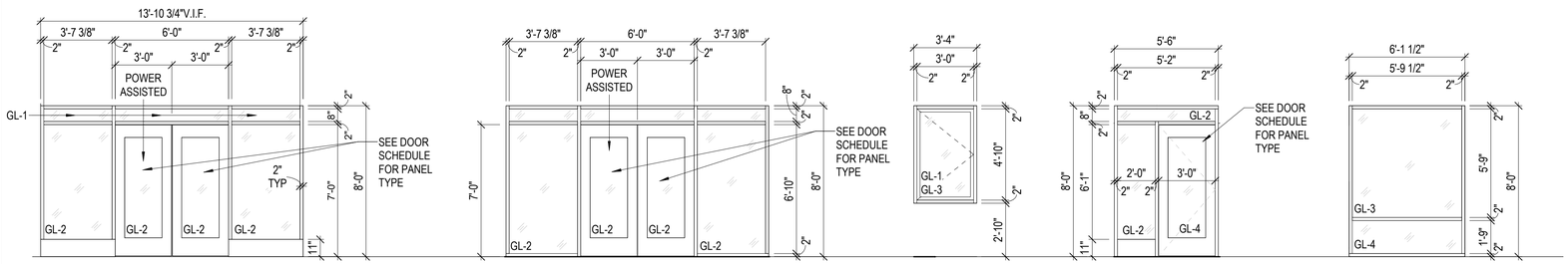

2 A 502 STOREFRONT KEY PLAN - BASEMENT LEVEL
SCALE B

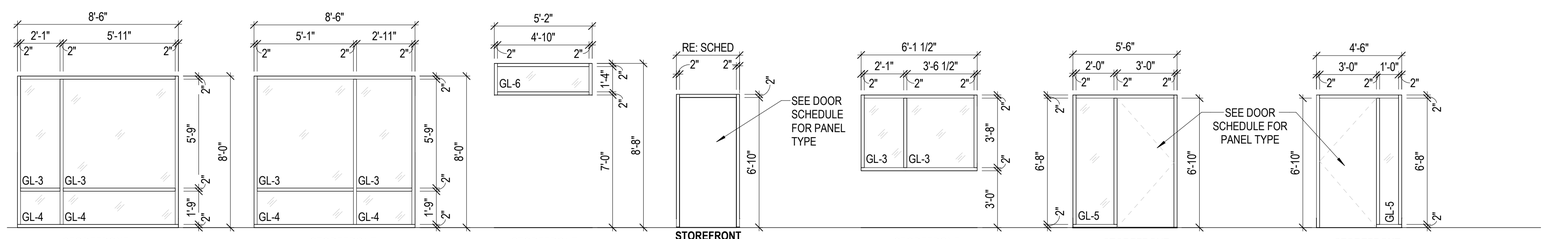



1 A 502 STOREFRONT KEY PLAN - MAIN LEVEL
SCALE B

## STOREFRONT TYPES

ALL STOREFRONT SHOWN BELOW ARE EXTERIOR STOREFRONT ELEVATIONS:




SCALE: B 16 0 16 32 SCALE OF FEET

SCALE: A 8 0 8 16 SCALE OF FEET

LICENSED PROFESSIONAL ARCHITECT ELIZABETH HALLAS 07418 IOWA

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM CADD LB/MM TECH REVIEW: ACN / EMH DATE: 05/29/2026 | SUB SHEET NO. A 502 | TITLE OF SHEET STOREFRONT SCHEDULE EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 51 OF 142 |

7/7/2026 1:21:08 PM Autodesk Docs://2025-110 NPS EFMO Visitor Center DD-CD/2024250 NPS EFMO Visitor Center - v24_BIM360.rvt

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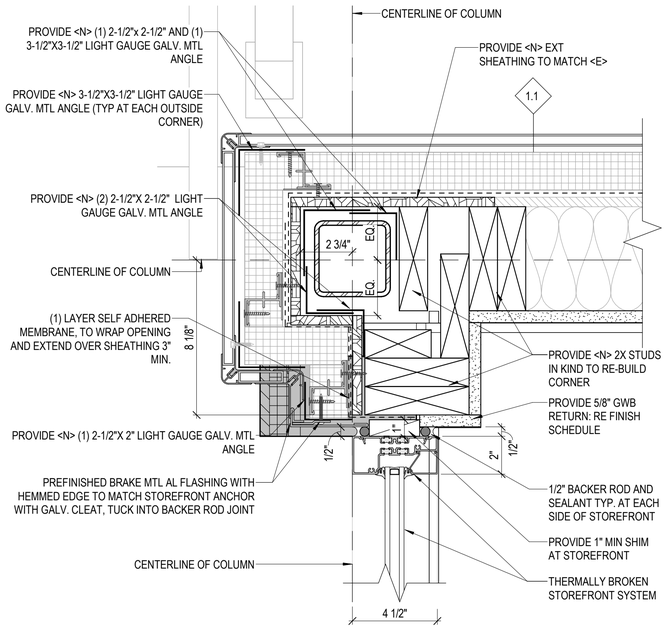
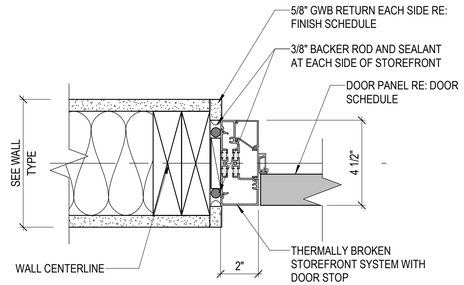
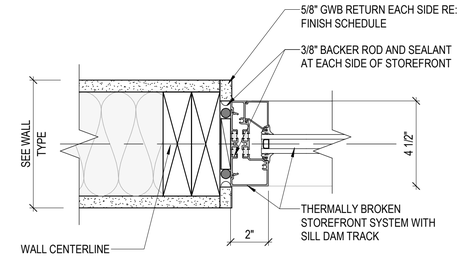


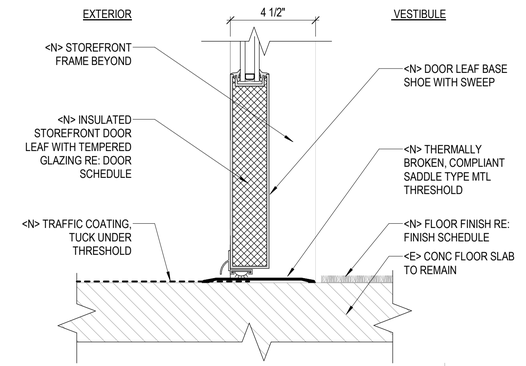

9 A 503 PLAN DETAIL AT STOREFRONT JAMB
SCALE A



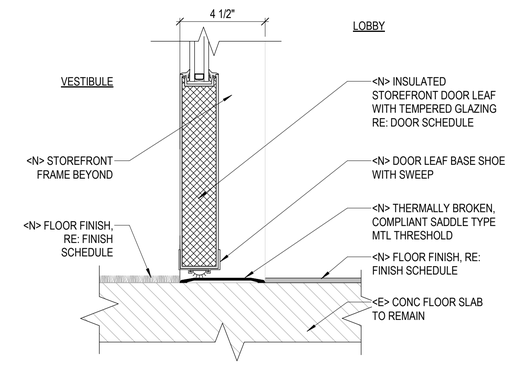

7 A 503 TYP SECTION DETAIL AT INT STOREFRONT - JAMB DOOR PNL
SCALE A



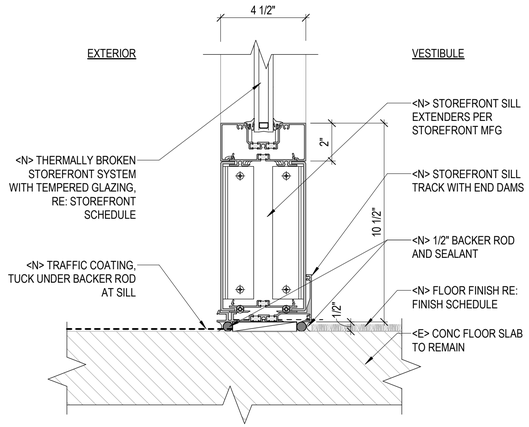

6 A 503 TYP SECTION DETAIL AT INT STOREFRONT - JAMB SIDELIGHT
SCALE A



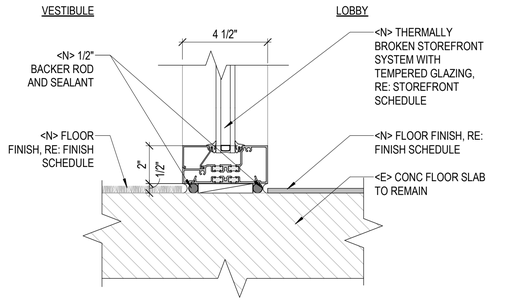

4 A 503 SECTION AT EXT VESTIBULE THRESHOLD
SCALE A



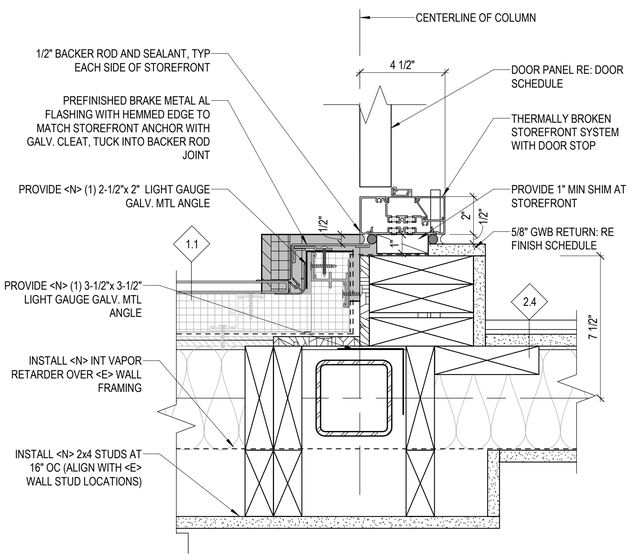

2 A 503 SECTION AT INT VESTIBULE THRESHOLD
SCALE A



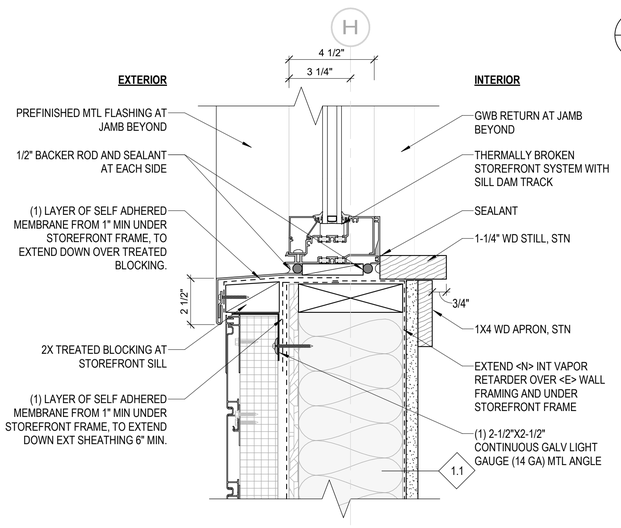

3 A 503 SECTION AT EXT VESTIBULE STOREFRONT - SILL
SCALE A




1 A 503 SECTION AT INT VESTIBULE STOREFRONT - SILL
SCALE A



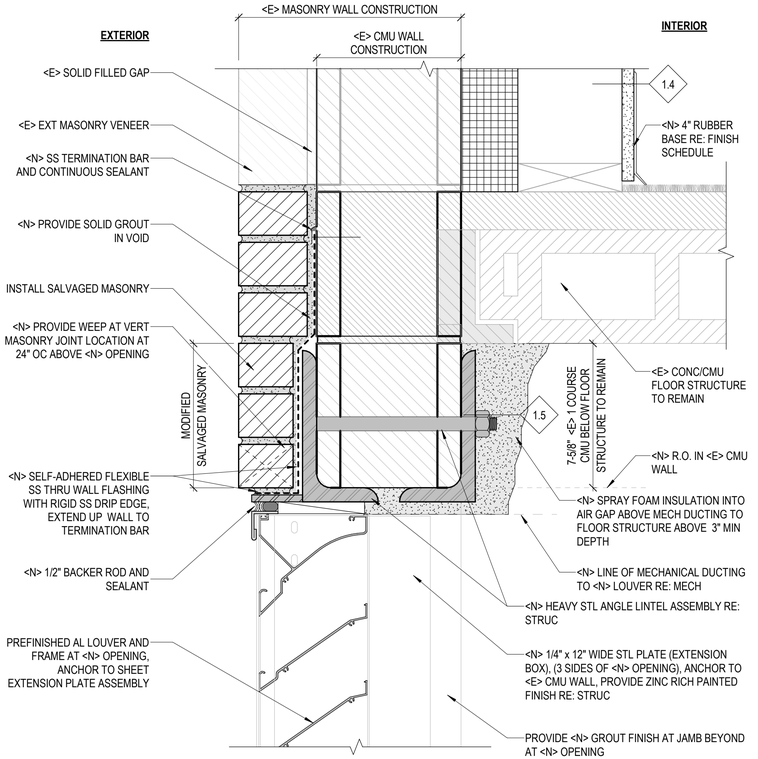

8 A 503 PLAN DETAIL AT STOREFRONT JAMB
SCALE A



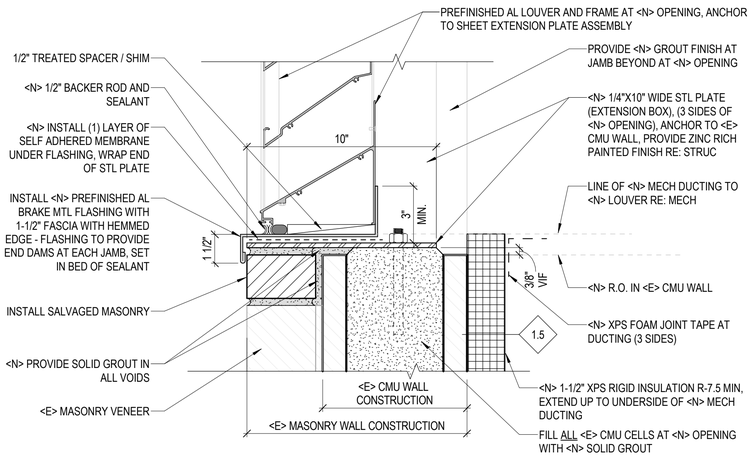

5 A 503 SECTION AT STOREFRONT SILL AT EXT MTL CLADDING WALL
SCALE A

SCALE: A 6 3 0 6
SCALE OF INCHES

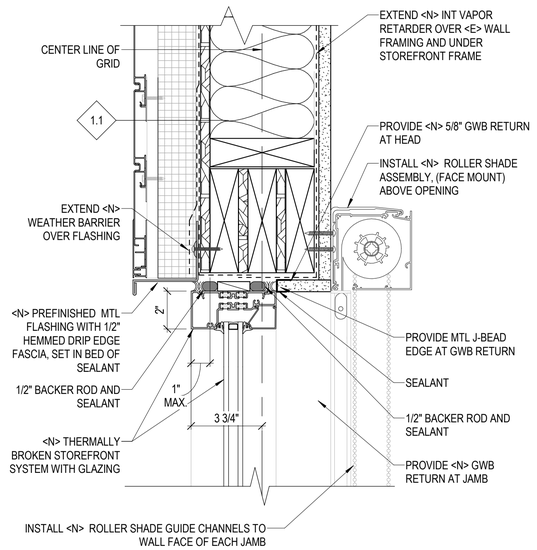
| | | | |
|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA<br>SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM<br>DRAWN: LB/MM<br>TECH REVIEW: ACN / EMH<br>DATE: 05/29/2026 | SUB SHEET NO. A 503 | TITLE OF SHEET: STOREFRONT DETAILS<br>EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248<br>PMIS/PKG NO. 318539<br>SHEET 52 OF 142 |



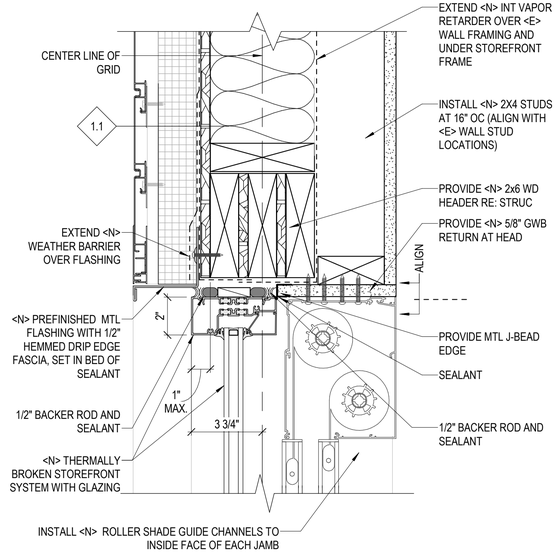

7 A 504 SECTION AT MECHANICAL LOUVER - HEAD
SCALE A



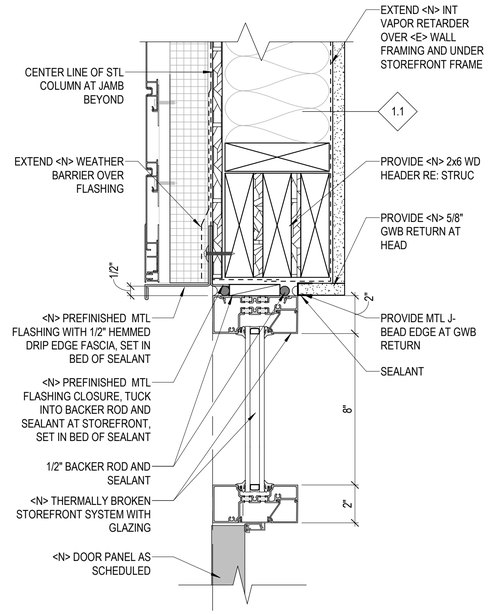

6 A 504 SECTION DETAIL AT MECHANICAL LOUVER - SILL
SCALE A



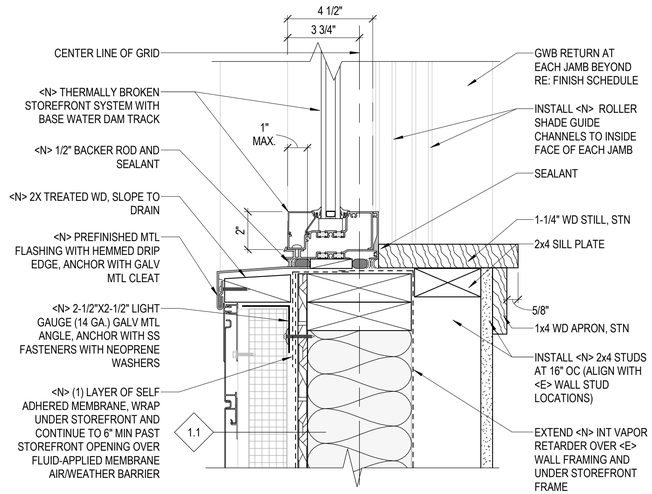

4 A 504 SECTION AT STOREFRONT HEAD AT OFFICE
SCALE A




3 A 504 SECTION AT STOREFRONT HEAD AT AUDITORIUM
SCALE A




1 A 504 SECTION AT STOREFRONT HEAD
SCALE A




2 A 504 SECTION AT STOREFRONT SILL AT AUDITORIUM
SCALE A

SCALE: A 6 3 0 6
SCALE OF INCHES

LICENSED PROFESSIONAL ARCHITECT
ELIZABETH HALLAS
07418
IOWA
5/29/2026

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | DESIGNED: LB/CM CADD LB/MM TECH REVIEW: ACN / EMH DATE: 05/29/2026 | SUB SHEET NO. A 504 | TITLE OF SHEET STOREFRONT & MECH LOUVER DETAILS EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 53 OF 142 |

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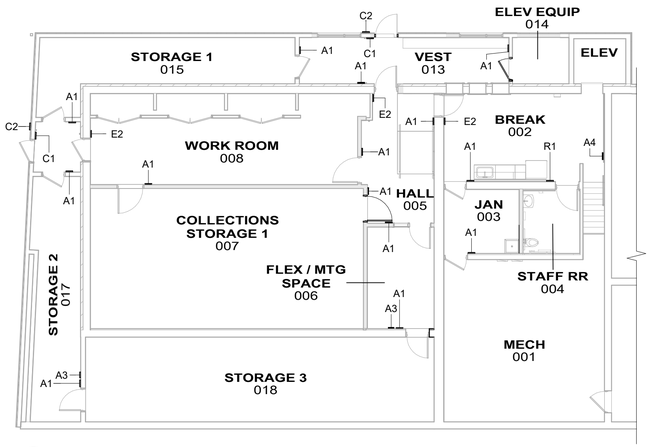
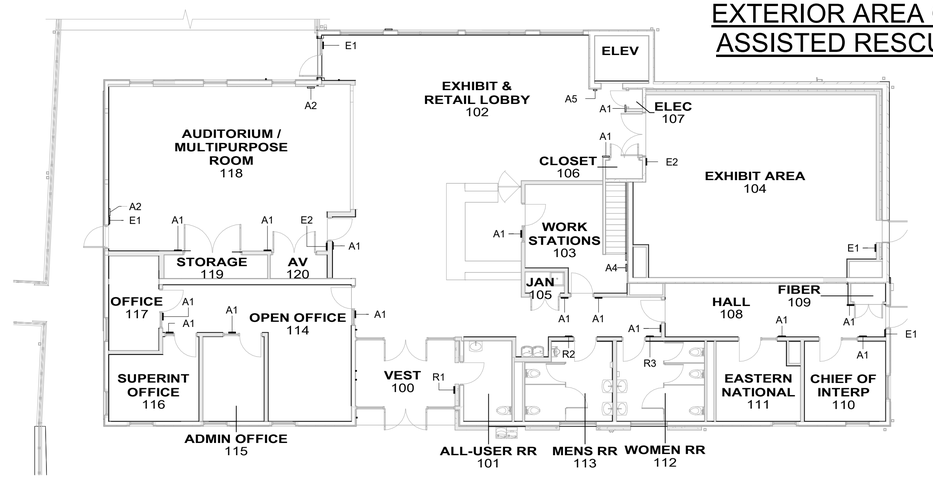
| SIGNAGE SCHEDULE | | |
|---|---|---|
| LEVEL | TEXT FOR SIGN | SIGNAGE TYPE |
| MAIN LEVEL | AUDITORIUM/MULTI-PURPOSE ROOM 118 | A1 - ROOM NAME |
| MAIN LEVEL | AV CLOSET 120 | A1 - ROOM NAME |
| MAIN LEVEL | STORAGE CLOSET 119 | A1 - ROOM NAME |
| MAIN LEVEL | WORK STATIONS 103 | A1 - ROOM NAME |
| MAIN LEVEL | WORK STATIONS 103 | A1 - ROOM NAME |
| MAIN LEVEL | JANITORIAL CLOSET 105 | A1 - ROOM NAME |
| MAIN LEVEL | EMPLOYEE AREA 108 | A1 - ROOM NAME |
| MAIN LEVEL | EASTERN NATIONAL 111 | A1 - ROOM NAME |
| MAIN LEVEL | CHIEF OF INTERPRETATION 110 | A1 - ROOM NAME |
| MAIN LEVEL | FIBER CLOSET 109 | A1 - ROOM NAME |
| MAIN LEVEL | OFFICE 117 | A1 - ROOM NAME |
| MAIN LEVEL | SUPERINTENDANT OFFICE 116 | A1 - ROOM NAME |
| MAIN LEVEL | OFFICE 115 | A1 - ROOM NAME |
| MAIN LEVEL | EMPLOYEE AREA 114 | A1 - ROOM NAME |
| MAIN LEVEL | * | E1 - EXIT |
| MAIN LEVEL | * | E1 - EXIT |
| MAIN LEVEL | * | R1 - RESTROOM - ALL GENDER |
| MAIN LEVEL | * | R2 - MEN'S RESTROOM |
| MAIN LEVEL | * | R3 - WOMEN'S RESTROOM |
| MAIN LEVEL | * | E1 - EXIT |
| MAIN LEVEL | * | E1 - EXIT |
| MAIN LEVEL | * | E2 - EXIT ACCESS |
| MAIN LEVEL | * | E2 - EXIT ACCESS |
| MAIN LEVEL | * | A5 - ELEVATOR |
| MAIN LEVEL | * | A4 - STAIRS |
| MAIN LEVEL | CLOSET 106 | A1 - ROOM NAME |
| MAIN LEVEL | ELECTRICAL 107 | A1 - ROOM NAME |
| MAIN LEVEL | * | A2 - ASSISTED LISTENING DEVICE |
| MAIN LEVEL | * | A2 - ASSISTED LISTENING DEVICE |
| BASEMENT LEVEL | WORKROOM 008 | A1 - ROOM NAME |
| BASEMENT LEVEL | COLLECTIONS STORAGE 007 | A1 - ROOM NAME |
| BASEMENT LEVEL | FLEX/MEETING 006 | A1 - ROOM NAME |
| BASEMENT LEVEL | STORAGE 018 | A1 - ROOM NAME |
| BASEMENT LEVEL | STORAGE 018 | A1 - ROOM NAME |
| BASEMENT LEVEL | * | C1 - EXTERIOR AREA FOR ASSISTED RESCUE |
| BASEMENT LEVEL | * | E2 - EXIT ACCESS |
| BASEMENT LEVEL | BREAK ROOM 002 | A1 - ROOM NAME |
| BASEMENT LEVEL | * | E2 - EXIT ACCESS |
| BASEMENT LEVEL | * | R1 - RESTROOM - ALL GENDER |
| BASEMENT LEVEL | JANITORIAL 003 | A1 - ROOM NAME |
| BASEMENT LEVEL | MECHANICAL 001 | A1 - ROOM NAME |
| BASEMENT LEVEL | STORAGE 015 | A1 - ROOM NAME |
| BASEMENT LEVEL | ELEVATOR EQUIPMENT 014 | A1 - ROOM NAME |
| BASEMENT LEVEL | COLLECTION STORAGE 007 | A1 - ROOM NAME |
| BASEMENT LEVEL | * | E2 - EXIT ACCESS |
| BASEMENT LEVEL | STORAGE 2 017 | A1 - ROOM NAME |
| BASEMENT LEVEL | STORAGE 015 | A1 - ROOM NAME |
| BASEMENT LEVEL | * | C1 - EXTERIOR AREA FOR ASSISTED RESCUE |
| BASEMENT LEVEL | * | C2 - TWO-WAY COMMUNICATION |
| BASEMENT LEVEL | * | C2 - TWO-WAY COMMUNICATION |
| BASEMENT LEVEL | * | A4 - STAIRS |
| BASEMENT LEVEL | HALL 005 | A1 - ROOM NAME |
| BASEMENT LEVEL | * | A3 - WATCH YOUR HEAD |
| BASEMENT LEVEL | * | A3 - WATCH YOUR HEAD |

* = REFER TO DETAIL ON THIS SHEET




1 / A 505 SIGNAGE KEY PLAN - BASEMENT LEVEL

3/32" = 1'-0"



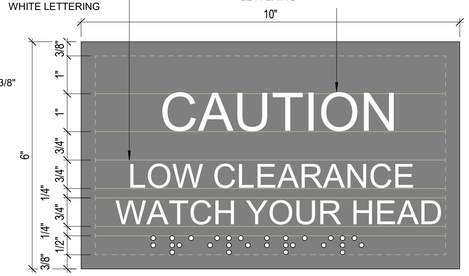

2 / A 505 SIGNAGE KEY PLAN - MAIN LEVEL

3/32" = 1'-0"

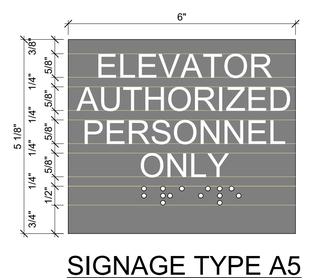
## SIGNAGE GENERAL NOTES

1. ALL SIGNAGE MATERIAL (FIELD) SHALL BE DARK GREY (L-010-50) DUET MATERIAL
2. ALL LETTERING SHALL BE ENGLISH (FONT IS HELVETICA MEDIUM) 1/32" RAISED LETTERS 5/8" HIGH CENTERED ON SIGN WHITE LETTERING UON.
3. ALL SIGNAGE TO BE 1/8" PHOTOPOLYMER MOUNTED TO 1/8" THICK ACRYLIC BACKER.
4. ALL BRAILLE TO BE GRADE II BRAILE CENTERED ON SIGN.
5. REFER TO CIVIL FOR EXTERIOR SIGNAGE.
6. PER 2024 IBC SECTION 10.13.1, EXIT SIGNAGE EXCEPTIONS:
   A. EXIT SIGNS ARE NOT REQUIRED IN ROOMS OR AREAS THAT REQUIRE ONLY ONE EXIT.
   B. MAIN EXTERIOR EXIT DOORS THAT ARE OBVIOUSLY & CLEARLY IDENTIFIABLE AS EXITS NEED NOT HAVE EXIT SIGNS WHERE APPROVED BY THE BUILDING OFFICIAL.
7. SIGNAGE FOR EXTERIOR AREA OF ASSISTED RESCUE SHALL COMPLY WITH 2024 IBC SECTION 1009.11 REQUIREMENTS.
8. ALL SIGNS THAT ARE A1, R1, R2, & R3 SHALL BE LOCATED ON THE EXTERIOR SIDE OF THE ROOM UNLESS NOTED OTHERWISE. WHEN POSSIBLE, LOCATE SIGNS ON THE WALL ON THE STRIKE SIDE OF THE DOOR TO THAT ROOM.
9. ALL SIGNS THAT ARE TYPE C1, E1, & E2 SHALL BE LOCATED ON THE INTERIOR SIDE OF THE DOOR WHEN FACING THE EXTERIOR OF THAT SPACE OR THE BUILDING.
10. ALL SIGNS SHALL INCLUDE ADHESIVE FOAM MOUNTING STRIPS ON THE BACK TO HANG SIGN ON WALL.
11. SIGNS CONTAINING RAISED CHARACHERS AND BRAILLE SHALL BE LOCATED SO THAT A CLEAR FLOOR AREA 18"X18" MINIMUM, CENTERED ON THE RAISED CAHRACTERS IS PROVIDED BEYOND THE ARC OF ANY DOOR SWING BETWEEN THE CLOSED POSITION & 45 DEGREE OPEN POSITION PER A117.1 SECTION 703.
12. RAISED CHARACTERS & BRAILLE SHALL BE 48" MINIMUM & 60" MAXIMUM ABOVE THE FLOOR, MEASURED TO THE BASELINE OF THE BRAILLE CELLS & TO THE BASELINE OF THE HIGHEST RAISED CHARACTER PER ABAAS & A117.1 SECTIONS 703.3.10 & 703.4.5.
13. ALL WORK THIS SHEET ARE OPTION C.

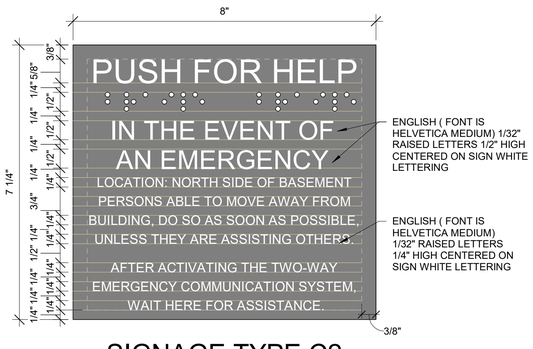
## SIGNAGE TYPE C1 EXTERIOR AREA OF ASSISTED RESCUE



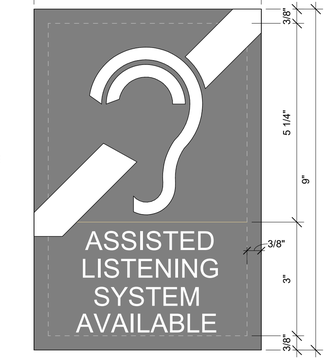

## SIGNAGE TYPE A3 LOW CLEARANCE



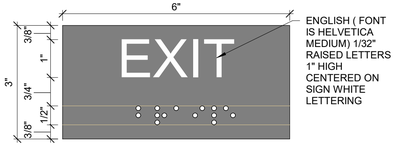

## SIGNAGE TYPE A5 ELEVATOR



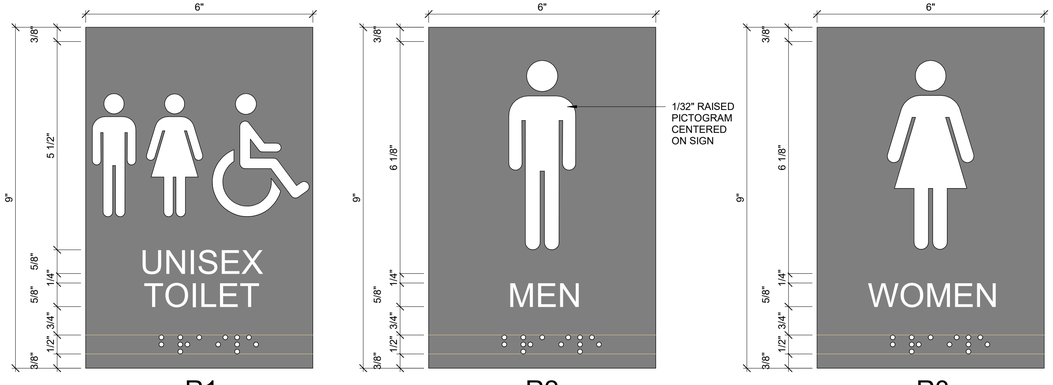

## SIGNAGE TYPE C2 TWO WAY COMMUNICATIONS




## SIGNAGE TYPE A2 ASSISTED LISTENING SYSTEM

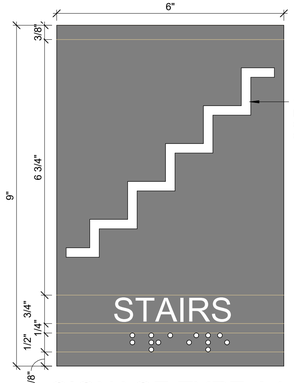



## SIGNAGE TYPE E1 - EXIT

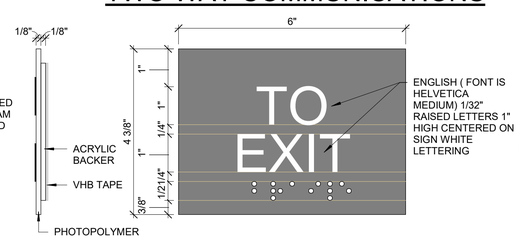



R1 R2 R3

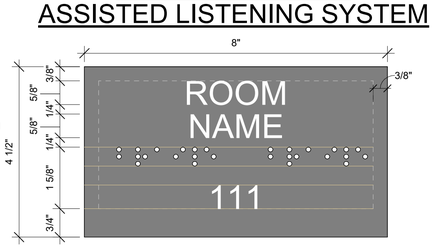
## SIGNAGE TYPE R1, R2 & R3 - RESTROOMS




## SIGNAGE TYPE A4 STAIRS




## SIGNAGE TYPE E2 - TO EXIT




## SIGNAGE TYPE A1 - ROOM NAMES

| | | | |
|---|---|---|---|
| A/E FIRM: | DESIGNED: | SUB SHEET NO. | TITLE OF SHEET |
| PRIME: OTAK, INC. REDMOND, WA | LB/CM | A 505 | SIGNAGE SCHEDULE & DETAILS |
| SUBCONTRACTOR ANDERSON HALLAS ARCHITECTS GOLDEN, CO | CADD LB/MM | | EFFIGY MOUNDS NATIONAL MONUMENT |
| | TECH REVIEW: ACN / EMH | | |
| | DATE: 05/29/2026 | | |

LICENSED PROFESSIONAL ARCHITECT — ELIZABETH HALLAS — 07418 — IOWA

| | |
|---|---|
| DRAWING NO. | 394 194248 |
| PMIS/PKG NO. | 318539 |
| SHEET | 54 OF 142 |

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11/18/2025 4:19:22 PM Autodesk Docs://2025-110 NPS EFMO Visitor Center DD-CD/33278AE - NPS EFMO Visitor Center - Structure - v24.rvt

GENERAL NOTES:

1) THE DESIGN AND CONSTRUCTION OF THIS PROJECT IS GOVERNED BY THE 2024 INTERNATIONAL BUILDING CODE (IBC), THE 2024 INTERNATIONAL EXISTING BUILDING CODE (IEBC), RP 10-22, AND NPS BUILDING REQUIREMENTS.
2) THE STRUCTURAL DRAWINGS SHALL BE UTILIZED IN CONJUNCTION WITH OTHER DESIGN CONSULTANT'S DRAWINGS (ARCHITECTURAL, MEP, CIVIL, ETC.). IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO COORDINATE THE REQUIREMENTS OF THE DRAWINGS INTO THEIR SHOP DRAWINGS AND CONSTRUCTION.
3) THE STRUCTURAL DRAWINGS ARE INTENDED TO SHOW THE GENERAL CHARACTER AND EXTENT OF THE PROJECT; NOT INTENDED TO SHOW ALL DETAILS OF THE WORK. USE DETAILS MARKED "TYPICAL" WHEREVER THEY APPLY.
4) THE STRUCTURAL DRAWINGS SHALL BE UTILIZED IN CONJUNCTION WITH THE PROJECT SPECIFICATIONS. IN THE EVENT OF A CONFLICT BETWEEN THE TWO, THE GENERAL STRUCTURAL NOTES SHALL SUPERSEDE THE PROJECT SPECIFICATIONS. ANY DISCREPANCY SHALL BE BROUGHT TO THE ATTENTION OF THE CONTRACTING OFFICER (CO) PRIOR TO THE COMMENCEMENT OF WORK.
5) CONSTRUCTION SEQUENCE AND METHODS:
   A. THE STRUCTURAL DRAWINGS ARE INTENDED FOR THE STRUCTURE TO ACT AS A WHOLE ONCE CONSTRUCTION IS COMPLETE. IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO ENSURE SAFETY STABILITY (I.E. TEMPORARY SHORING AND BRACING IF REQUIRED) DURING CONSTRUCTION AS A RESULT OF CONSTRUCTION METHODS AND SEQUENCES.
   B. THE CONTRACTOR SHALL TAKE INTO ACCOUNT COLD WEATHER CONSTRUCTION AND THE EFFECTS OF THERMAL MOVEMENT DURING THE CONSTRUCTION SCHEDULE.
   C. NON-CANTILEVERED OR RESTRAINED RETAINING WALLS SHALL NOT BE BACKFILLED UNTIL THE WALL HAS BEEN TIED INTO THE LOWER AND UPPER SLAB SUPPORTS UNLESS ADEQUATE ENGINEERED BRACING HAS BEEN PROVIDED.
6) SUBMITTALS:
   A. GRAVITY SUPPORT, SEISMIC BRACING, AND RESTRAINT TO THE SUPERSTRUCTURE OF ANY MEP EQUIPMENT, MACHINERY, AND ASSOCIATED PIPING SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR. CONNECTIONS NOT IN COMPLIANCE WITH THE SMACNA (SHEET METAL AND AIR CONDITIONING CONTRACTOR'S NATIONAL ASSOCIATION) OR THE MEP DESIGN DRAWINGS, SHALL BEAR THE SEAL OF A REGISTERED ENGINEER IN THE STATE OF IOWA AND SHALL BE SUBMITTED ALONG WITH CALCULATIONS TO THE CONTRACTING OFFICER FOR APPROVAL PRIOR TO FABRICATION.
   B. SUBMITTAL TABLE:

SUBMITTAL TABLE

| ITEM | SUBMITTAL (1) (3) | DEFERRED SUBMITTAL (2) (3) | COMMENTS |
|---|---|---|---|
| CONCRETE MIX DESIGNS | X | | |
| CONCRETE REINFORCEMENT (INCLUDING MILL TEST REPORTS) | X | | |
| MEP EQUIPMENT GRAVITY SUPPORT, ANCHORAGE AND SEISMIC BRACING | | X | |
| GLU-LAMINATED MEMBERS | X | | |
| STRUCTURAL STEEL (INCLUDING MILL TEST REPORTS) | X | | |
| EMBEDDED STEEL ITEMS | X | | |
| LADDERS AND RAILINGS | | X | |
| ATTACHMENT OF EXTERIOR CLADDING TO PRIMARY STRUCTURE | | X | |
| FENCING | | X | |
| HELICAL ANCHORS (OR MICROPILES) | | X | |

SUBMITTAL TABLE NOTES:
(1) SHOP DRAWINGS SHALL BE SUBMITTED TO THE CONTRACTING OFFICER (CO) PRIOR TO FABRICATION AND CONSTRUCTION OF STRUCTURAL ITEMS. IF THE SHOP DRAWINGS DIFFER FROM OR ADD TO THE DESIGN OF THE STRUCTURAL DRAWINGS, THEY SHALL BEAR THE SEAL AND SIGNATURE OF A STRUCTURAL ENGINEER REGISTERED IN THE STATE OF IOWA. ANY CHANGES TO THE STRUCTURAL DRAWINGS SHALL BE SUBMITTED TO THE CONTRACTING OFFICER AND ARE SUBJECT TO REVIEW AND ACCEPTANCE OF THE CONTRACTING OFFICER.
(2) DESIGN DRAWINGS, SHOP DRAWINGS, AND CALCULATIONS FOR THE DESIGN AND FABRICATION OF ITEMS THAT ARE DESIGNED BY OTHERS SHALL BEAR THE SEAL AND SIGNATURE OF A STRUCTURAL ENGINEER REGISTERED IN THE STATE OF IOWA, AND SHALL BE SUBMITTED TO THE CO PRIOR TO FABRICATION. CALCULATIONS AND BIDDER DESIGN DRAWINGS SHALL INCLUDE THE DESIGN, CONNECTION TO THE STRUCTURE, AND ACCOUNTING OF ANY LOCALIZED EFFECTS THE CONNECTIONS OR SYSTEMS MAY INDUCE ON THE STRUCTURE. DESIGN SHALL BE BASED ON THE REQUIREMENTS NOTED UNDER "DESIGN CRITERIA."
(3) THE USE OF REPRODUCTIONS OR PHOTO COPIES OF THE CONTRACT DRAWINGS SHALL NOT BE PERMITTED. WHEN CAD OR REVIT FILES ARE PROVIDED TO THE CONTRACTORS, IT IS THE RESPONSIBILITY OF THE DETAILERS TO REMOVE ALL INFORMATION NOT DIRECTLY RELEVANT TO THE CREATION OF THE PLACING DRAWINGS AS WELL AS ALL REFERENCES TO THE OUTSIDE SOURCE FILES.

7) DESIGN CRITERIA:
   A. CODE: 2024 IBC, 2024 IEBC, 2022 ASCE 7, AND RP10-22
   B. LOADS AND DESIGN CRITERIA: THE FOLLOWING LIVE LOADS AND CRITERIA WERE USED IN ADDITION TO THE DEAD LOAD OF THE STRUCTURE.

LIVE LOADS:
ROOF.....20 PSF
FLOOR (SLAB ON GRADE).....100 PSF

SNOW LOADS:
SNOW IMPORTANCE FACTOR.....Is = 1.0
GROUND SNOW LOAD.....Pg = 47.0 PSF
THERMAL FACTOR.....Ct = 1.2
EXPOSURE FACTOR.....Ce = 1.0
ROOF SLOPE FACTOR.....Cs = AS DETERMINED BASED ON ROOF SLOPE
MINIMUM UNIFORM SNOW LOAD.....20 PSF x Is = 20 PSF
RAIN OR SNOW SURCHARGE.....5 PSF (FOR ROOF SLOPES < 1:12) ADDED TO MINIMUM UNIFORM SNOW LOAD)
SNOW DRIFTS.....PER THE 2024 IBC

SOIL CRITERIA: (PER CODE MINIMUMS)
ALLOWABLE SOIL BEARING VALUES.....1,500 PSF (w/ 1/3 INCREASE FOR SEISMIC/WIND LOADS)
PASSIVE.....250 PSF
FRICTION COEFFICIENT.....0.25
LATERAL BEARING PRESSURE...150 PSF

RAIN LOADS:
RAIN.....3.4 IN/HR

LATERAL CRITERIA:
WIND.....BASIC WIND SPEED, Vult = 108 MPH
RISK CATEGORY II
WIND EXPOSURE C
Kzt = 1.0; Kd = 0.85; Ke = 1.0; Kz = 0.94
INTERNAL PRESSURE COEFFICIENT, Gcpi = +/- 0.18
COMPONENTS AND CLADDING: MAX (ULTIMATE) DESIGN PRESSURE = -81 PSF

GENERAL NOTES, DESIGN CRITERIA, CONTINUED:

SEISMIC.....Ie = 1.0; RISK CATEGORY II
SITE CLASS D
Sds = 0.08g
Ss = 0.077; S1 = 0.052; Sd1 = 0.076
SEISMIC DESIGN CATEGORY B
LIGHT-FRAMED WOOD SHEATHED WALLS (EXTERIOR SHELL)
R = 6.5 Wo = 2.0 Cd = 4.0
ORDINARY REINFORCED MASONRY (CMU) (INTERIOR)
R = 2.0 Wo = 2.5 Cd = 1.75

8) THE CONTRACTOR SHALL BE FAMILIAR WITH THE RECOMMENDATIONS OF THE PROJECT'S GEOTECHNICAL REPORT REFERENCED IN SECTION 7 "DESIGN CRITERIA" PRIOR TO START OF CONSTRUCTION. SUBGRADE PREPARATION, DRAINAGE, BACKFILL, EXCAVATION, SHORING, ETC. SHALL BE CONSTRUCTED AND INSTALLED PER THE GEOTECHNICAL MINIMUMS.

CAST-IN-PLACE CONCRETE
1) CONCRETE WORK SHALL CONFORM TO ACI 318-19 AND THE 2024 INTERNATIONAL BUILDING CODE. THE MINIMUM 28 DAY CONCRETE STRENGTHS SHALL BE AS FOLLOWS:
f'c = 4,000 PSI.....ALL USES UNLESS NOTED OTHERWISE
1 INCH MAXIMUM AGGREGATE
0.48 MAXIMUM WATER/CEMENT RATIO
2) CONCRETE MIX DESIGNS ALONG WITH TEST DATA AS REQUIRED, SHALL BE SUBMITTED BY THE CONTRACTOR AN ADEQUATE AMOUNT OF TIME PRIOR TO CONCRETE POURS. ALL HORIZONTAL EXPOSED SURFACES SHALL HAVE MIX DESIGNS SUBMITTED WITH AN AIR ENTRAINMENT ADMIXTURE INCLUDED PROVIDING BETWEEN 4% AND 6% AIR ENTRAINMENT.
3) SPECIFIED CONCRETE STRENGTHS SHALL BE VERIFIED BY STANDARD 28-DAY CYLINDER TESTS PER ASTM C39.
4) A 20% MAXIMUM OF THE CEMENT CONTENT MAY BE SUBSTITUTED WITH FLYASH CONFORMING TO ASTM C618 TYPE F OR C, OR SLAG CEMENT CONFORMING TO ASTM C989, GRADE 100 OR 120. HIGHER PERCENTAGES MAY BE UTILIZED WITH ACCEPTANCE AND APPROVAL BY THE CONTRACTING OFFICER. ANY CONCRETE MIX UTILIZING FLYASH OR SLAG SHALL BE VERIFIED WITH TEST DATA.
5) ADDITIONAL WATER SHALL NOT BE ADDED TO THE CONCRETE MIX AT THE JOBSITE UNLESS THE BATCH PLANT TICKET INDICATES WATER WAS WITHHELD AT THE BATCH PLANT FOR LATER INCLUSION AT THE JOBSITE. THE SPECIAL INSPECTOR SHALL APPROVE AND MONITOR ANY ADDITION OF WATER AT THE JOBSITE. WATER REDUCING ADMIXTURES CONFORMING TO ASTM C494 MAY BE UTILIZED IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS.
6) IF CONCRETE IS TO BE POURED AGAINST AN EXISTING CONCRETE SURFACE, THE EXISTING SURFACE SHALL BE CLEANED AND ROUGHENED TO A MINIMUM 1/4" AMPLITUDE.
7) SLEEVES, OPENINGS, CONDUIT, AND OTHER EMBEDDED ITEMS NOT SHOWN ON THE STRUCTURAL DRAWINGS SHALL BE APPROVED BY THE CONTRACTING OFFICER BEFORE POURING. CONDUITS EMBEDDED IN SLABS SHALL NOT BE LARGER IN OUTSIDE DIMENSION THAN ONE THIRD OF THE THICKNESS OF THE SLAB AND 2" (WHICHEVER IS SMALLER) AND SHALL NOT BE SPACED CLOSER THAN THREE DIAMETERS ON CENTER OR 6" (WHICHEVER IS GREATER).
8) PROVIDE 3/4" CHAMFERS ON ALL EXPOSED CONCRETE EDGES UNLESS NOTED OTHERWISE.
9) CONCRETE COLUMNS/PILASTERS WHICH ARE TIED TO WALLS SHALL BE POURED INTEGRALLY AS ONE POUR.

10) REINFORCING STEEL:
A. REINFORCING STEEL SHALL BE DETAILED, FABRICATED AND INSTALLED ACCORDING TO THE "MANUAL OF STANDARD PRACTICE OF REINFORCED CONCRETE CONSTRUCTION" BY THE CONCRETE REINFORCING INSTITUTE (CRSI).
B. REINFORCING STEEL SHALL CONFORM TO ASTM A615, GRADE 60 OR WELDABLE ASTM A706, GRADE 60. SMOOTH BARS OR WELDED WIRE FABRIC SHALL CONFORM TO ASTM A185.
C. ALL LAP SPLICES OF REINFORCEMENT SHALL CONFORM TO CLASS B LAPS AS SHOWN ON THE LAP SPLICE SCHEDULE UNLESS NOTED OTHERWISE.

REINFORCING SPLICE LENGTHS (CLASS B) GRADE 60 f'c = 4.0 ksi

| BAR SIZE | #3 | #4 | #5 | #6 | #7 | #8 |
|---|---|---|---|---|---|---|
| UNCOATED | 2'-0" | 2'-6" | 3'-6" | 4'-0" | 5'-0" | 5'-6" |

D. DEVELOPMENT LENGTH NOTED IN THESE DRAWINGS SHALL BE EQUAL TO 80% OF THE LENGTH NOTED IN THE LAP SPLICE SCHEDULE.
E. ALL 90° AND 180° HOOKS SHOWN ON THE DRAWINGS SHALL BE STANDARD HOOKS AS DEFINED BY ACI 318-19 AND CRSI, UNLESS NOTED OTHERWISE. STIRRUP/TIE HOOKS AND SEISMIC HOOKS NOTED ON THE DRAWINGS SHALL BE 135° HOOKS AS DEFINED BY ACI 318-19 AND CRSI.
F. CONSECUTIVE CROSSTIES (WITH 90° HOOK ON ONE END AND 135° HOOK ON OTHER END) WITHIN COLUMNS OR BEAMS ENGAGING THE SAME LONGITUDINAL BARS SHALL BE ALTERNATED END-FOR-END ALONG THE LONGITUDINAL REINFORCEMENT.
G. UNLESS NOTED OTHERWISE, PLACE (2) #5 BARS ON ALL SIDES OF SLAB OPENINGS EXTENDED 24" BEYOND OPENING. PROVIDE (1) OR (2) 3-6" LONG DIAGONAL #5 BARS AT EACH CORNER OF THE OPENING MATCHING THE LAYERS OF REINFORCING.
H. UNLESS OTHERWISE NOTED ON PLANS, CONCRETE COVER SHALL BE:

| USE | COVER |
|---|---|
| SLAB BARS | 1" TOP COVER |
| WALL BARS: | |
| INTERIOR FACES | 3/4" |
| EXPOSED TO EARTH OR WEATHER | 1-1/2" (#5 AND SMALLER)<br>2" (#6 AND LARGER) |
| FOOTING BARS | 3" BOTTOM COVER |
| ALL OTHER LOCATIONS | 1-1/2" |

11) CONCRETE WALLS:
A. PROVIDE MINIMUM WALL REINFORCING SHOWN BELOW UNLESS NOTED OTHERWISE:

| WALL THICKNESS | REINFORCING |
|---|---|
| 6" | #4 BARS AT 12" o.c. EA. WAY AT CL OF WALL |
| 8" | #5 BARS AT 12" o.c. EA. WAY AT CL OF WALL |
| 10" | #4 BARS AT 12" o.c. EA. WAY, EA. FACE OF WALL |
| 12" | #5 BARS AT 12" o.c. EA. WAY, EA. FACE OF WALL |

B. HOOKED DOWELS FROM FOUNDATIONS SHALL BE PROVIDED TO MATCH VERTICAL WALL REINFORCING.
C. UNLESS NOTED OTHERWISE, PROVIDE HOOKED DOWELS MATCHING SLAB REINFORCING FROM WALLS TO SLABS OR HOOK SLAB REINFORCEMENT INTO WALLS. HOOK SHALL BE PLACED WITH 2" COVER AT FAR FACE OF WALL FROM SLAB.
D. UNLESS NOTED OTHERWISE, PLACE (2) #5 BARS ON ALL SIDES OF WALL OPENINGS EXTENDED 30" BEYOND OPENING. PROVIDE (1) OR (2) 3-6" LONG DIAGONAL #5 BARS AT EACH CORNER OF THE OPENING MATCHING THE LAYERS OF REINFORCING.
E. VERTICAL AND HORIZONTAL BARS SHALL BE PLACED/TERMINTED WITH 2" COVER AT ENDS AND TOPS/BOTTOMS OF WALLS.

GENERAL NOTES, CONCRETE AND REINFORCING STEEL, CONCRETE WALLS, CONTINUED:

12) ADDITIONAL CONCRETE ITEMS:
A. PERMANENTLY EXPOSED EMBEDDED PLATES AND ANGLES SHALL BE HOT-DIPPED, GALVANIZED AFTER FABRICATION, UNLESS OTHERWISE NOTED. NO LOADS OR WELDS SHALL BE PLACED ON EMBEDDED PLATES OR ANGLES FOR A MINIMUM OF 7 DAYS AFTER PAINTING.
B. HYDROPHILIC WATERSTOP SHALL BE PLACED WHERE NOTED ON DRAWINGS/SPECIFICATIONS AND AT BELOW GRADE CONCRETE CONSTRUCTION JOINTS WHERE EXTERIOR/INTERIOR SPACE OCCUR ON EACH SIDE OF JOINT OR SLAB ON GRADE OCCURS ON ONE SIDE OF JOINTS. REFERENCE WATERSTOP SPECIFICATIONS. VERIFY ALL WATERSTOP LOCATIONS AND REQUIREMENTS WITH CONTRACTING OFFICER PRIOR TO CONCRETE POUR. PLACE WATERSTOP ALONG CONCRETE CONSTRUCTION JOINT TO ALLOW FOR 2" CONCRETE COVER AND 3/4" CLEAR FROM ANY REINFORCING, TYPICAL.
C. ALL NON-SHRINK GROUT SHALL BE NON-METALLIC AND HAVE A MINIMUM 6,000 PSI COMPRESSIVE STRENGTH AT 28-DAYS.
D. WEDGE ANCHORS, SCREW ANCHORS, AND EPOXY ANCHORS SHALL BE AS INDICATED OR AN APPROVED EQUAL SUBMITTED WITH ICC (OR IAPMO) REPORTS TO THE CONTRACTING OFFICER FOR REVIEW. ALL ANCHORS SHALL BE APPROVED FOR CRACKED CONCRETE. CONTRACTOR SHALL TAKE MEASURES TO ENSURE THAT ALL POST INSTALLED ANCHORS SHALL NOT DAMAGE OR CUT ANY EXISTING REINFORCEMENT. IF NOT INDICATED USE HILTI KB-TZ2, HUS-EZ, OR HIT-HY200 V3 EPOXY
E. A VAPOR/MOISTURE BARRIER SHALL BE PLACED BETWEEN ALL CONCRETE POURS AND WOOD.
F. CHLORIDES SHALL NOT BE PRESENT IN ANY CONCRETE ADDITIVES UTILIZED.

13) POST-INSTALLED CONCRETE ANCHORS
A. WHERE THE AUTHORITY HAVING JURISDICTION OVER THIS PROJECT REQUIRES ADHERENCE TO ACI 318-19 SECTION 17.8.2.2, MANUFACTURER'S FIELD REPRESENTATIVE SHALL PROVIDE INSTALLATION TRAINING FOR ALL PRODUCTS TO BE USED PRIOR TO THE COMMENCEMENT OF WORK. ONLY TRAINED INSTALLERS SHALL PERFORM POST-INSTALLED ANCHOR INSTALLATION. A RECORD OF TRAINING SHALL BE KEPT ON SITE AND BE MADE AVAILABLE TO THE CONTRACTING OFFICER AS REQUIRED.
B. ADHESIVE ANCHORS SHALL BE INSTALLED IN CONCRETE HAVING A MINIMUM AGE OF 21 DAYS AT THE TIME OF ANCHOR INSTALLATION IN ACCORDANCE WITH ACI 318-19 SECTION 17.4.5.2.
C. WHERE THE AUTHORITY HAVING JURISDICTION OVER THIS PROJECT REQUIRES ADHERENCE TO ACI 318-19 SECTION 17.8.2.2, INSTALLATION OF ADHESIVE ANCHORS IN HORIZONTAL TO VERTICALLY OVERHEAD ORIENTATION SHALL BE DONE BY A CERTIFIED ADHESIVE ANCHOR INSTALL (AAI) AS CERTIFIED THROUGH ACI AND IN ACCORDANCE WITH ACI 318-19. PROOF OF CURRENT CERTIFICATION SHALL BE SUBMITTED TO THE CONTRACTING OFFICER FOR APPROVAL PRIOR TO COMMENCEMENT OF INSTALLATION.

STRUCTURAL STEEL
1) STEEL DESIGN, FABRICATION, AND ERECTION SHALL CONFORM WITH "AISC SPECIFICATION FOR THE DESIGN, FABRICATION, AND ERECTION OF STRUCTURAL STEEL BUILDINGS" AND THE "CODE OF STANDARD PRACTICE FOR STEEL BUILDINGS AND BRIDGES." CONSTRUCTION SHALL CONFORM TO CHAPTER 22 OF THE IBC. ALL STRUCTURAL STEEL SHALL BE PAINTED, U.N.O.
2) THE GRADE AND SPECIFICATION OF THE STEEL MEMBERS SHALL BE AS FOLLOWS:

| | | |
|---|---|---|
| WIDE FLANGE SHAPES | ASTM A992, GR. 50 | |
| CHANNELS, PLATES & ANGLES | ASTM A36 | FY=36KSI |
| HSS | ASTM A500 GRADE C | FY=50KSI |
| WELDING ELECTRODES | E70XX UNLESS OTHERWISE NOTED | |
| PIPE | ASTM A53 | FY=35KSI |
| BOLTS | ASTM A307 | |
| HIGH-STRENGTH BOLTS | ASTM A325 AND A490 | |
| THREADED RODS | ASTM F1554 GR 36 U.N.O. | |
| NUTS | ASTM A563 B, HEX | |
| WASHERS | ASTM F436 | |

3) WELDING SHALL CONFORM TO AWS D1.1 AND VISUALLY CONFORM TO AWS SECTION 6 AND TABLE 6.1. WELDERS SHALL BE QUALIFIED FOR THE SPECIFIC PREQUALIFIED JOINTS REQUIRED BY THE DESIGN AND CERTIFIED IN ACCORDANCE WITH AWS REQUIREMENTS. WELDING SHALL BE DONE IN ACCORDANCE WITH APPROPRIATE WELD PROCEDURE SPECIFICATIONS (WPSS); WELDERS SHALL BE FAMILIAR WITH THE APPLICABLE WSPSS. WELDING SHALL BE DONE WITH AWS PREQUALIFIED WELDING PROCESSES UNLESS OTHERWISE APPROVED. WELDER QUALIFICATIONS AND WPSS SHALL BE MAINTAINED AT THE SITE OF THE WORK AND SHALL BE READILY AVAILABLE FOR INSPECTION UPON REQUEST, BOTH IN THE SHOP AND IN THE FIELD.
4) HIGH-STRENGTH BOLTS SHALL BE SLIP-CRITICAL (SC) UNLESS NOTED OTHERWIES. ALL FAYING SURFACES FOR SC JOINTS SHALL BE PREPARED AS REQUIRED FOR CLASS A OR BETTER.
5) BOLTS UTILIZED IN SEISMIC RESISTING ELEMENTS SHALL BE FULLY TENSIONED.
6) ALL BOLTED AND THREADED ROD CONNECTIONS SHALL UTILIZE WASHERS UNLESS NOTED OTHERWISE.
7) DISCONTINUITIES CREATED BY ERRORS OR BY FABRICATION OR ERECTION OPERATIONS, SUCH AS TACK WELDS, ERECTION AIDS, AIR-ARC GOUGING, AND FLAME CUTTING, SHALL BE REPAIRED AS REQUIRED BY THE CONTRACTING OFFICER.
8) WELDS SHALL BE A MINIMUM OF 3/16" IN SIZE UNLESS NOTED OTHERWISE.
9) ALL FLARE BEVEL OR FLARE-V WELDS NOTED ON THE DRAWINGS SHALL BE CONSTRUCTED WITH FULL EFFECTIVE THROAT ACHIEVABLE.
10) ALL WELDS SPECIFIED ON THE DRAWINGS AS FIELD WELDS MAY BE SHOP WELDED AT CONTRACTOR'S OPTION IF STEEL ERECTION CAN STILL BE ACHIEVED.
11) ALL ANCHOR BOLTS SHALL BE HEADED AND BE ASTM F1554 GR 36 UNLESS NOTED OTHERWISE.
12) ALL CONNECTIONS SHALL HAVE BOLT HOLE DIAMETER SIZE QUAL TO BOLT DIAMETER + 1/16", UNLESS NOTED OTHERWISE.
13) ALL HSS MEMBERS TO HAVE 1/4" CAP PLATE WELDED WITH 1/8" EFFECTIVE THROAT PARTIAL PEN UNLESS NOTED OTHERWISE.

SAWN LUMBER
1) ALL WOOD FRAMING MEMBERS INCLUDING, BUT NOT LIMITED TO, WALL STUDS AND JOISTS ARE INTENDED TO ACT AS A SYSTEM AS DETAILED IN THE STRUCTURAL DRAWINGS AND ONCE CONSTRUCTION IS COMPLETE. IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO ENSURE SAFETY AND STABILITY OF WOOD FRAMING SYSTEMS (I.E. TEMPORARY BRACING IF REQUIRED) DURING CONSTRUCTION AS A RESULT OF CONSTRUCTION METHODS AND SEQUENCES. REFERENCE ARCHITECTURAL DRAWINGS FOR ALL SAWN LUMBER FINISH REQUIREMENTS.
2) ALL SAWN LUMBER SHALL CONFORM TO THE WESTERN WOOD PRODUCTS ASSOCIATION OR THE WEST COAST LUMBER INSPECTION BUREAU GRADING RULES. LUMBER SHALL BE OF THE SPECIES AND GRADE SHOWN BELOW:

| MEMBER | GRADE |
|---|---|
| 2x AND 4x FRAMING..... | DOUGLAS FIR-LARCH NO. 2 |
| 5x AND GREATER BEAMS..... | DOUGLAS FIR-LARCH NO. 1 |
| POSTS/COLUMNS..... | DOUGLAS FIR-LARCH NO. 1 |

3) WOOD STUDS IN BEARING WALLS SHALL BE OF THE SIZE, GRADE, AND SPACING NOTED BELOW UNLESS NOTED OTHERWISE ON THE DRAWINGS.ALL STUD BEARING WALLS REQUIRE SHEATHING ON A MINIMUM OF ONE SIDE OF THE WALL.

GENERAL NOTES, SAWN LUMBER, CONTINUED:

4) WHERE ARCHITECTURAL FINISHES ARE APPLIED TO FLOORS ABOVE PRIOR TO THE MINIMUM SHEATHING REQUIREMENTS BEING INSTALLED ON THE STUD BEARING WALLS, IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO VERIFY THE ADEQUACY OF THE UNSHEATHED BEARING WALL AND TO PROVIDE BRACING AS REQUIRED.

| STUD MEMBER AND SPACING | GRADE |
|---|---|
| REF PLANS (2x6 @ 16" O.C.) | DF No. 2 |

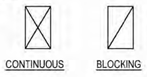
5) STORAGE OF ALL LUMBER AND TIMBER ON SITE SHALL BE KEPT OFF THE GROUND, UNDER COVER, AND PROTECTED FROM DAMAGE.
6) ALL DIMENSIONAL LUMBER AND TIMBER SHALL BE CERTIFIED BY THE SUPPLIER IN WRITING TO BE KILN DRIED.
7) STRUCTURE SHALL NOT BE ENCLOSED UNLESS LUMBER MOISTURE CONTENT HAS BEEN VERIFIED TO BE AT OR BELOW 15%. ANY SIGNS OF MOLD SHALL BE REMOVED AND TREATED IN ACCORDANCE WITH THE PROJECT SPECIFICATIONS OR INDUSTRY STANDARDS.
8) ALL LUMBER IN CONTACT WITH THE GROUND OR CONCRETE SHALL BE PRESSURE TREATED. CONTRACTOR MAY SUBMIT FOR APPROVAL, A MOISTURE BARRIER IN LIEU OF THE PRESSURE TREATED WOOD.
9) FASTENERS (SCREWS, NAILS, BOLTS, ETC.) ATTACHING TIMBER CONNECTORS (JOIST HANGERS, POST CAPS AND BASES, ETC.) TO PT WOOD SHALL HAVE SIMILAR CORROSION RESISTANCE PROPERTIES ( MATCHING PROTECTIVE TREATMENTS) AS THE PROTECTED CONNECTOR. FASTENERS ATTACHING SAWN TIMBER MEMBERS OR SHEATHING TO PT WOOD ARE TO BE CORROSION RESISTANT; NAILS AND LAG BOLTS SHALL BE EITHER HOT-DIP GALVANIZED OR STAINLESS STEEL. VERIFY THE SUITABILITY OF THE FASTENER PROTECTION/COATING WITH THE WOOD TREATMENT CHEMICAL MANUFACTURER/SUPPLIER.
10) ALL PLATES AND LEDGERS SHALL BE FASTENED WITH A MINIMUM OF (3) ANCHORS PER PIECE. INTERIOR NON-BEARING WALLS SHALL BE FASTENED WITH 1/2"Ø GALVANIZED ANCHOR BOLTS AT 6'-0" O.C. WITH (1) BOLT LOCATED 1'-0" FROM THE END OF EACH SILL PLATE.
11) ALL METAL HARDWARE AND FRAMING ACCESSORIES SHALL BE MANUFACTURED BY SIMPSON STRONG-TIE COMPANY. ALL ITEMS SHALL BE INSTALLED PER MANUFACTURER'S REQUIREMENTS. ALL NAIL HOLES SHALL BE FILLED WITH THE RECOMMENDED FASTENER (MAXIMUM NAILING).
12) ALL WALLS SHALL HAVE DOUBLE TOP PLATES AND SHALL BE SPLICED PER THE TYPICAL TOP PLATE SPLICE DETAIL UNLESS NOTED OTHERWISE. TOP PLATES AT WALL INTERSECTIONS SHALL BE LAPPED AND NAILED WITH (3) 16d NAILS.
13) HOLES FOR BOLTS SHALL BE DRILLED WITH A BIT OF THE SAME NOMINAL DIAMETER AS THE BOLT +1/16". LEAD HOLES FOR LAG SCREWS SHALL BE BORED PER NDS 11.1.3.
14) ALL BOLTS, CARRIAGE BOLTS, LAG SCREWS, EXPANSION BOLTS, AND EPOXY BOLTS SHALL BE INSTALLED WITH STANDARD CUT WASHERS UNDER THE BOLT HEADS AND NUTS THAT BEAR DIRECTLY ON THE WOOD. ALL NUTS SHALL BE TIGHTENED AT THE TIME OF INSTALLATION AND RE-TIGHTENED IF NECESSARY DUE TO WOOD SHRINKAGE, PRIOR TO CLOSE-IN OR AT THE COMPLETION OF THE PROJECT. BOLTS AND LAG SCREWS SHALL CONFORM TO ANSI/ASME STANDARD B18.2.1. WOOD SCREWS SHALL CONFORM TO B18.6.1. ALL BOLTS SHALL CONFORM TO ASTM A307 UNLESS NOTED OTHERWISE. THE MINIMUM STRENGTHS FOR LAG SCREWS AND WOOD SCREWS SHALL BE AS FOLLOWS:

| WOOD SCREW DIAMETER - INCHES | MIN. BENDING YIELD STRENGTH (PSI) |
|---|---|
| 0.138 (#6) | 100,000 |
| 0.151 (#7) | 90,000 |
| 0.164 (#8) | 90,000 |
| 0.177 (#9) | 90,000 |
| 0.190 (#10) | 80,000 |
| 0.216 (#12) | 80,000 |
| 0.246 (#14) | 70,000 |

| LAG SCREW DIAMETER - INCHES | MIN. BENDING YIELD STRENGTH (PSI) |
|---|---|
| 1/4" | 70,000 |
| 5/16" | 60,000 |
| 3/8" AND GREATER | 45,000 |

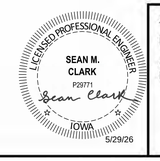
15) CUTTING AND NOTCHING OF SAWN LUMBER JOISTS, SAWN LUMBER RAFTERS (DOES NOT INCLUDE TRUSSES) AND STUDS/POSTS SHALL BE IN CONFORMANCE WITH THE FOLLOWING CRITERIA:
A. JOISTS NOTCHES AT THE ENDS OF JOISTS SHALL NOT EXCEED 1/5 THE JOIST DEPTH. HOLES BORED IN JOISTS SHALL BE WITHIN 2-1/2" OF THE TOP OR BOTTOM OF THE JOIST, AND THE DIAMETER OF ANY SUCH HOLE SHALL NOT EXCEED 1/4 THE DEPTH OF THE JOIST. NOTCHES IN THE TOP OR BOTTOM OF THE JOISTS SHALL NOT EXCEED 1/6 THE DEPTH AND SHALL NOT BE LOCATED IN THE MIDDLE THIRD OF THE SPAN.
B. RAFTERS NOTCHES AT THE ENDS OF RAFTERS OR CEILING JOISTS SHALL NOT EXCEED 1/5 THE DEPTH. NOTCHES IN THE TOP OR BOTTOM OF SHALL NOT EXCEED 1/6 THE DEPTH AND SHALL NOT BE LOCATED IN THE MIDDLE THIRD OF THE SPAN. HOLES BORED IN THE MEMBER SHALL BE WITHIN 2-1/2" OF THE TOP OR BOTTOM, AND THE DIAMETER OF ANY SUCH HOLE SHALL NOT EXCEED 1/4 THE DEPTH OF THE MEMBER.
C. STUDS ALL BORED HOLES IN STUDS SHALL BE CENTERED IN THE WIDTH OF THE STUD. A MAXIMUM 2-1/4" DIAMETER NEATLY BORED HOLE MAY BE PLACED WITH NO ADDITIONAL REINFORCEMENT REQUIRED (MINIMUM 2x6 STUDS).

16) WOOD SYMBOLS:




17) ALL NAILS FOR STRUCTURAL WORK SHALL BE COMMON WIRE NAILS UNLESS NOTED OR DETAILED OTHERWISE MEETING ASTM F1667. HOLES SHALL BE PRE-DRILLED WHERE NECESSARY TO PREVENT SPLITTING. NAILS SHALL HAVE THE MINIMUM PROPERTIES SPECIFIED BELOW:

| NAIL TYPE | SHANK DIA. - INCHES | MIN. PENETRATION -INCHES | MIN. BENDING YIELD STRENGTH (PSI) |
|---|---|---|---|
| 6d | 0.113 | 1.13 | 100,000 |
| 8d | 0.131 | 1.31 | 100,000 |
| 10d | 0.148 | 1.48 | 90,000 |
| 12d | 0.148 | 1.48 | 90,000 |
| 16d | 0.162 | 1.62 | 90,000 |
| 20d | 0.192 | 1.92 | 80,000 |

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| A/E FIRM | |
| PRIME: OTAK INC. REDMOND, WA | |
| DESIGNED: | KEM |
| CADD | OC |
| TECH REVIEW: | SMC |
| DATE: | 5/29/2026 |
| SUB SHEET NO. | S001 |
| TITLE OF SHEET | GENERAL NOTES |
| | EFFIGY MOUNDS NATIONAL MONUMENT |
| DRAWING NO. | 394 194248 |
| PMIS/PKG NO. | 318539 |
| SHEET | 55 OF 142 |

GENERAL NOTES (CONTINUED...):

GLUED LAMINATED MEMBERS:

1. GLUED LAMINATED MEMBERS SHALL CONFORM TO THE REQUIREMENTS OF ONE OF THE FOLLOWING STANDARDS AND PUBLICATIONS:

   A. AMERICAN NATIONAL STANDARD FOR STRUCTURAL GLUED LAMINATED TIMBER.
   B. ANSI/AITC STANDARD A190.1 AND ASTM D 3737
   C. ANY CODE-APPROVED STANDARD OR PUBLICATION. APPROVAL MUST BE OBTAINED FROM CONTRACTING OFFICER

2. THE MINIMUM GLUE LAMINATED MEMBER GRADES SHALL BE AS FOLLOWS UNLESS NOTED OTHERWISE:

| MEMBER | GRADE |
|---|---|
| SIMPLE SPAN | 24F-V4 |
| CONTINUOUS/CANTILEVER | 24F-V8 |
| COLUMN/POST | DF-L1D |

3. APPEARANCE SHALL BE FRAMING INDUSTRIAL FOR HIDDEN MEMBERS AND ARCHITECTURAL FOR EXPOSED MEMBERS UNLESS NOTED OTHERWISE. REF. ARCHITECTURAL DRAWINGS FOR ADDITIONAL REQUIREMENTS.
4. ALL STRUCTURAL GLUED LAMINATED MEMBERS EXPOSED TO WEATHER AND NOT IN CONTACT WITH SOIL SHALL BE PRESSURE TREATED WITH PERMAPOST HI-CLEAR II OR APPROVED EQUAL. REF. ARCH.
5. ALL BEAMS SHALL HAVE NO CAMBER UNLESS OTHERWISE NOTED ON THE DRAWINGS
6. NO NOTCHING OR BORING OF HOLES IN BEAMS IS ALLOWED WITHOUT APPROVAL BY CONTRACTING OFFICER
7. GLUE SHALL BE WET-USE EXTERIOR WATERPROOF GLUE.
8. WHERE HANGERS ARE REQUIRED BUT NOT SPECIFICALLY SIZED, SIMPSON GLT HANGERS SHALL BE USED. ALL ITEMS SHALL BE INSTALLED PER THE MANUFACTURERS INSTALLATION REQUIREMENTS
9. GLUED LAMINATED WOOD SYMBOLS:

**GLUED LAMINATED BEAM**

**GLUED LAMINATED BEAM ELEVATION**

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LICENSED PROFESSIONAL ENGINEER
SEAN M. CLARK
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5/29/26

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**OPTIONS**

(A) OPTION A: HVAC SYSTEMS INCLUDING EQUIPMENT, PIPING, DUCTWORK, AND CONTROLS. FIRE SUPPRESSION INCLUDING WET AND DRY SPRINKLER SYSTEMS INCLUDING ENTRY EQUIPMENT, FIRE ALARM SYSTEM, AND REFURBISHMENT OF CLEAN AGENT SYSTEM. INCLUDES ALSO SEALING COLLECTIONS SPACE FOR CLEAN AGENT SYSTEM FUNCTION.

(B) OPTION B: EXTERIOR FINISH AND INSULATION REPLACEMENT FROM EXTERIOR FACE OF EXTERIOR SHEATHING OUT AND NEW ROOF. INCLUDES REFRAMING AND REPLACING WINDOWS AND DOORS, REPLACEMENT OF EXTERIOR COLUMNS, AND FINALIZING SITE WORK.

(C) OPTION C: INTERIOR LAYOUT RECONFIGURATION, INCLUDING RELOCATING WALLS, PLUMBING, AND ELECTRICAL. ALL INTERIOR FINISH AND INSULATION REPLACEMENT FROM INTERIOR FACE OF EXTERIOR SIDING IN. INCLUDES CASEWORK, LIGHT FIXTURES, MONITORS, AND TOILET ACCESSORIES.

(D) OPTION D: CABLING FOR LOW VOLTAGE FIRE ALARM IMPROVEMENTS.

(E) OPTION E: INSTALLATION OF NEW LIGHTNING PROTECTION SYSTEM.

(F) OPTION F: MAINTAIN (E) ASSISTIVE LISTENING SYSTEM. PROVIDE ALL (N) MEETINGROOM/AUDITORIUM AV EQUIP. INSTALL (N) FOLDING DOOR WITH STRUC SUPPORTS AND (N) WINDOW TREATMENTS

(G) OPTION G: DECK ADDITION WITH DUAL VIEWER, STRUC STL, (N) RAILINGS, AND (E) RETAINING WALL REALIGNMENT.

**SHEET NOTES**

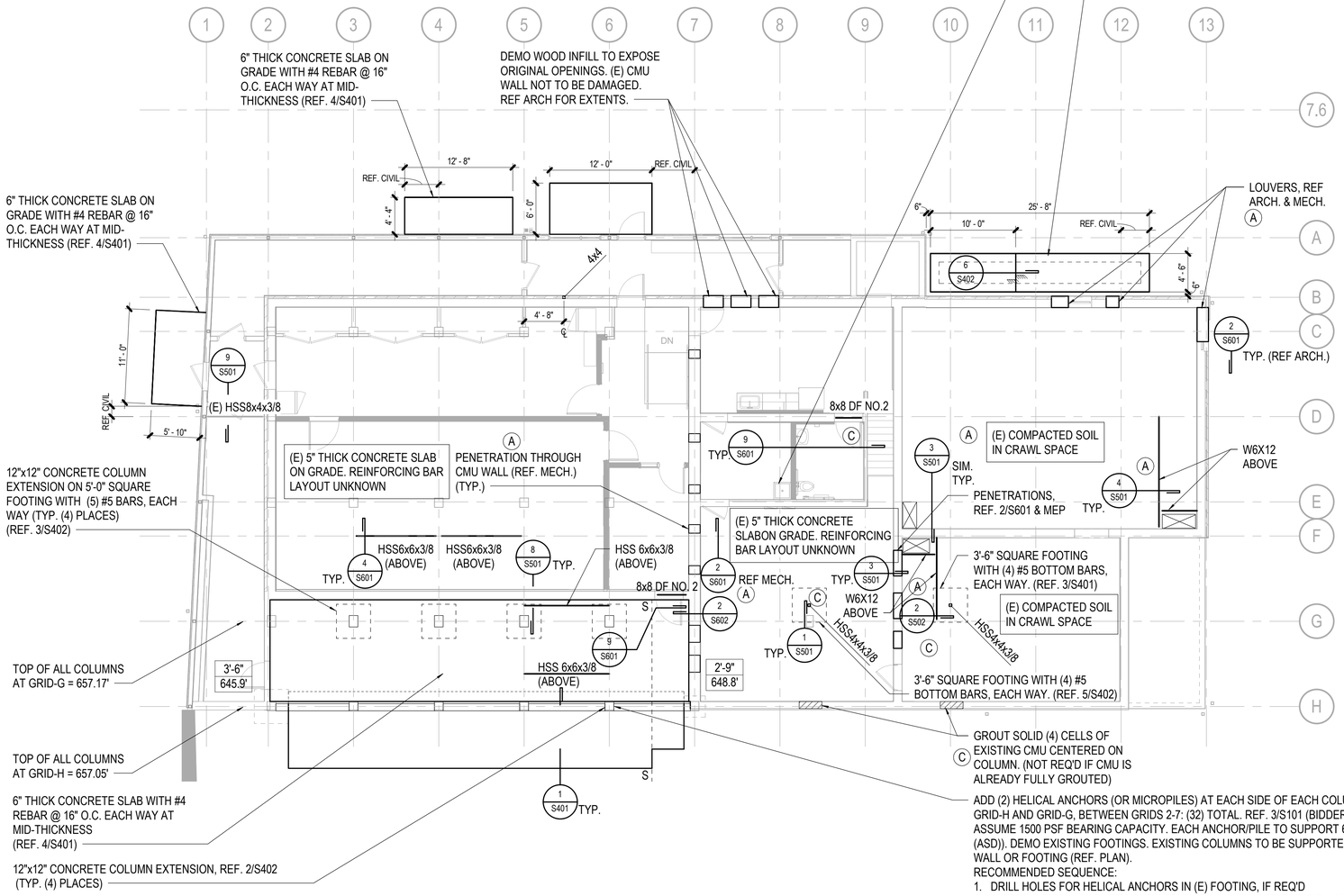
1) CONSTRUCTION AND DESIGN SHALL CONFORM TO THE IBC 2024, IEBC 2024, ASCE 7-22, ASCE 41-23 AND RP 10-22.

2) VERIFY ALL DIMENSIONS WITH CONTRACTING OFFICER AND IN THE FIELD PRIOR TO CONSTRUCTION. NOTIFY CONTRACTING OFFICER OF ANY DISCREPANCIES.

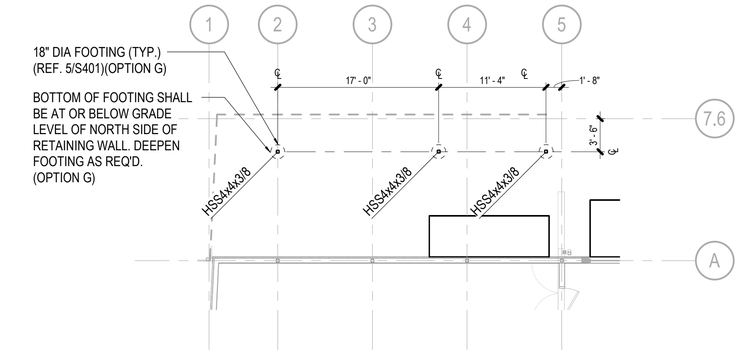
3) ALL FOOTINGS SHALL BE CENTERED BELOW COLUMNS OR WALLS UNLESS NOTED OTHERWISE.

4) X'-X" INDICATES SQUARE FOOTING LENGTH AS INDICATED ON THE 1963 AS-BUILT DRAWINGS.
XXX.X' INDICATES TOP OF FOOTING ELEVATION IN FEET AS INDICATED ON 1963 AS-BUILT DRAWINGS.

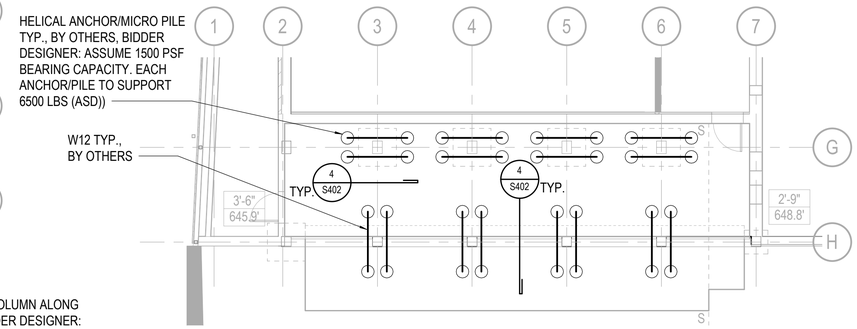
5) S INDICATES STEP IN FOOTING (REF. 6/S401)
S




1 S101 STRUCTURAL BASEMENT LEVEL FLOOR PLAN
SCALE: A




2 S101 BASEMENT FLOOR PLAN - OPTION G
SCALE: A




3 S101 RECOMMENDED SHORING PLAN
SCALE: A

SCALE: A 8 0 8 16
SCALE OF FEET

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**OPTIONS**

(A) OPTION A: HVAC SYSTEMS INCLUDING EQUIPMENT, PIPING, DUCTWORK, AND CONTROLS. FIRE SUPPRESSION INCLUDING WET AND DRY SPRINKLER SYSTEMS INCLUDING ENTRY EQUIPMENT, FIRE ALARM SYSTEM, AND REFURBISHMENT OF CLEAN AGENT SYSTEM. INCLUDES ALSO SEALING COLLECTIONS SPACE FOR CLEAN AGENT SYSTEM FUNCTION.

(B) OPTION B: EXTERIOR FINISH AND INSULATION REPLACEMENT FROM EXTERIOR FACE OF EXTERIOR SHEATHING OUT AND NEW ROOF. INCLUDES REFRAMING AND REPLACING WINDOWS AND DOORS, REPLACEMENT OF EXTERIOR COLUMNS, AND FINALIZING SITE WORK.

(C) OPTION C: INTERIOR LAYOUT RECONFIGURATION, INCLUDING RELOCATING WALLS, PLUMBING, AND ELECTRICAL. ALL INTERIOR FINISH AND INSULATION REPLACEMENT FROM INTERIOR FACE OF EXTERIOR SIDING IN. INCLUDES CASEWORK, LIGHT FIXTURES, MONITORS, AND TOILET ACCESSORIES.

(D) OPTION D: CABLING FOR LOW VOLTAGE FIRE ALARM IMPROVEMENTS.

(E) OPTION E: INSTALLATION OF NEW LIGHTNING PROTECTION SYSTEM.

(F) OPTION F: MAINTAIN (E) ASSISTIVE LISTENING SYSTEM. PROVIDE ALL (N) MEETINGROOM/AUDITORIUM AV EQUIP. INSTALL (N) FOLDING DOOR WITH STRUC SUPPORTS AND (N) WINDOW TREATMENTS

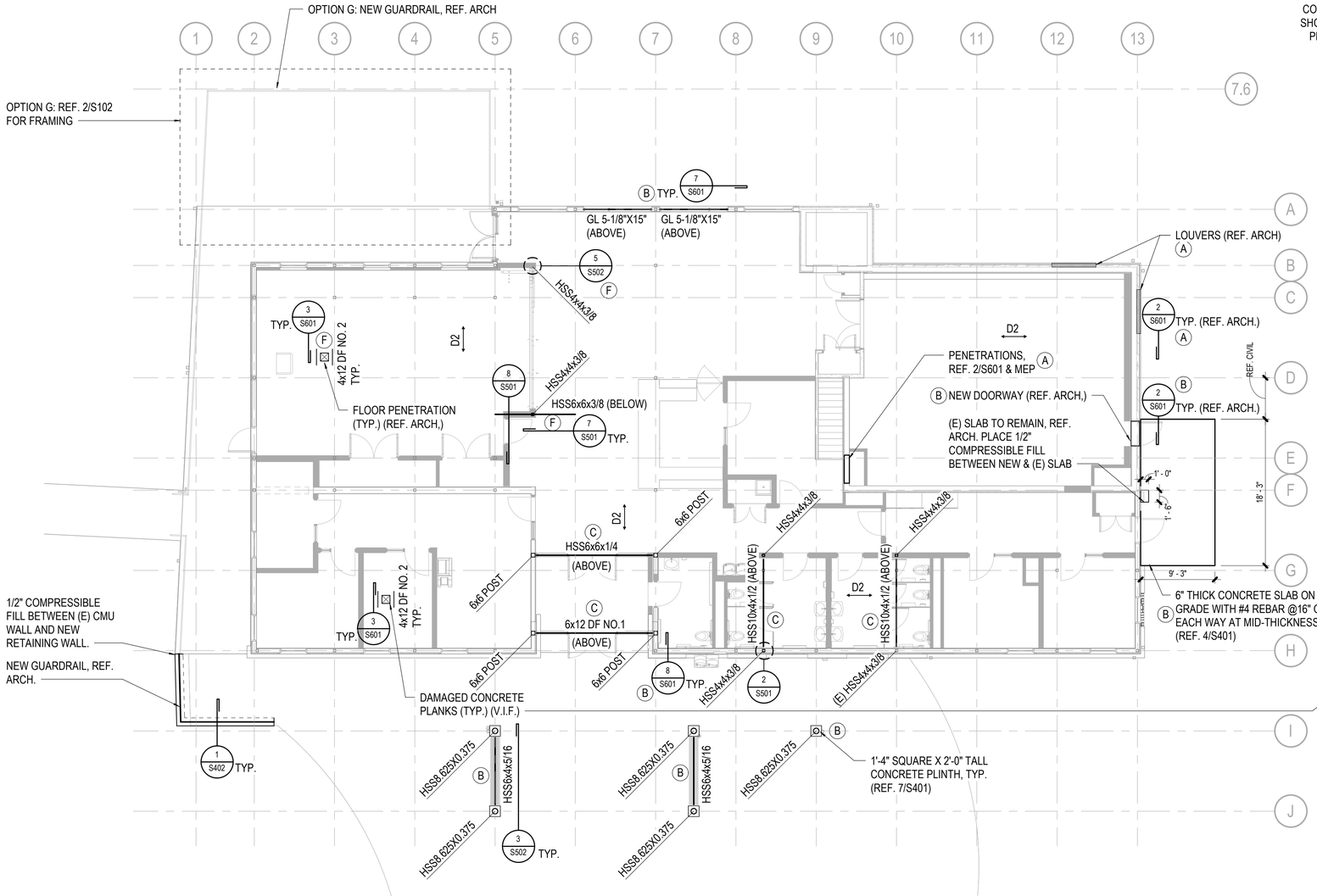
(G) OPTION G: DECK ADDITION WITH DUAL VIEWER, STRUC STL, (N) RAILINGS, AND (E) RETAINING WALL REALIGNMENT.

**SHEET NOTES**

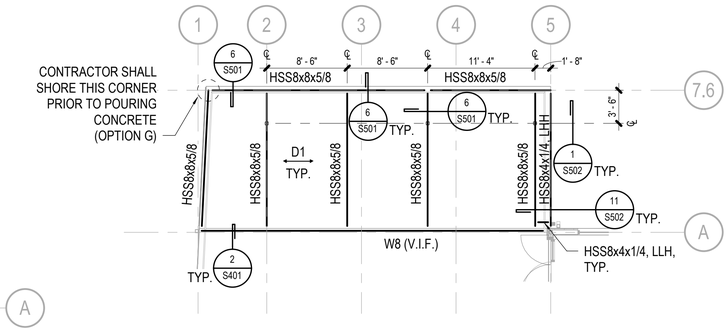
1) VERIFY ALL DIMENSIONS WITH CONTRACTING OFFICER AND IN THE FIELD PRIOR TO CONSTRUCTION. NOTIFY CONTRACTING OFFICER OF ANY DISCREPANCIES.

2) D1 INDICATES SPAN DIRECTION OF 18 GA. 3" W-DECK FILLED W/ 3" NORMAL WEIGHT CONCRETE TOPPING WITH #4 @ 16" O.C. EACH WAY. PUDDLE WELD AT EACH LOW FLUTE AND BUTTON PUNCH SIDE LAPS AT 12" O.C.

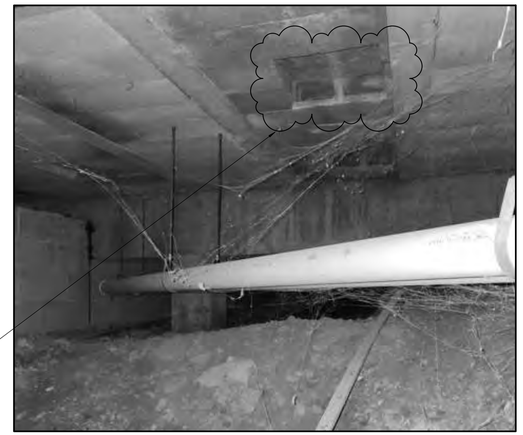
3) D2 (E) DOX PLANK FLOOR SYSTEM ASSUMED TO CONSIST OF HORIZONTAL CONCRETE MASONRY UNITS, (3) CELLS WIDE, SPANNING BETWEEN CONCRETE BEAMS WITHIN THE DEPTH OF THE UNITS SPACED AT 30" O.C. OVERLAIN WITH 2" CONCRETE TOPPING. CMU BLOCKS MAY HAVE INTEGRAL TABS THAT ATTACH OVER THE BEAMS. DOX PLANK CALLED OUT AS 6" ON S102 AND 8" ON S103.
(E) F&A FLOOR SYSTEM IS BRITTLE.




1 S102 STRUCTURAL MAIN LEVEL FLOOR PLAN
SCALE: A




2 S102 FIRST FLOOR PLAN - OPTION G
SCALE: A



SCALE: A 8 0 8 16 SCALE OF FEET

| | | | |
|---|---|---|---|
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**OPTIONS**

(A) OPTION A: HVAC SYSTEMS INCLUDING EQUIPMENT, PIPING, DUCTWORK, AND CONTROLS. FIRE SUPPRESSION INCLUDING WET AND DRY SPRINKLER SYSTEMS INCLUDING ENTRY EQUIPMENT, FIRE ALARM SYSTEM, AND REFURBISHMENT OF CLEAN AGENT SYSTEM. INCLUDES ALSO SEALING COLLECTIONS SPACE FOR CLEAN AGENT SYSTEM FUNCTION.

(B) OPTION B: EXTERIOR FINISH AND INSULATION REPLACEMENT FROM EXTERIOR FACE OF EXTERIOR SHEATHING OUT AND NEW ROOF. INCLUDES REFRAMING AND REPLACING WINDOWS AND DOORS, REPLACEMENT OF EXTERIOR COLUMNS, AND FINALIZING SITE WORK.

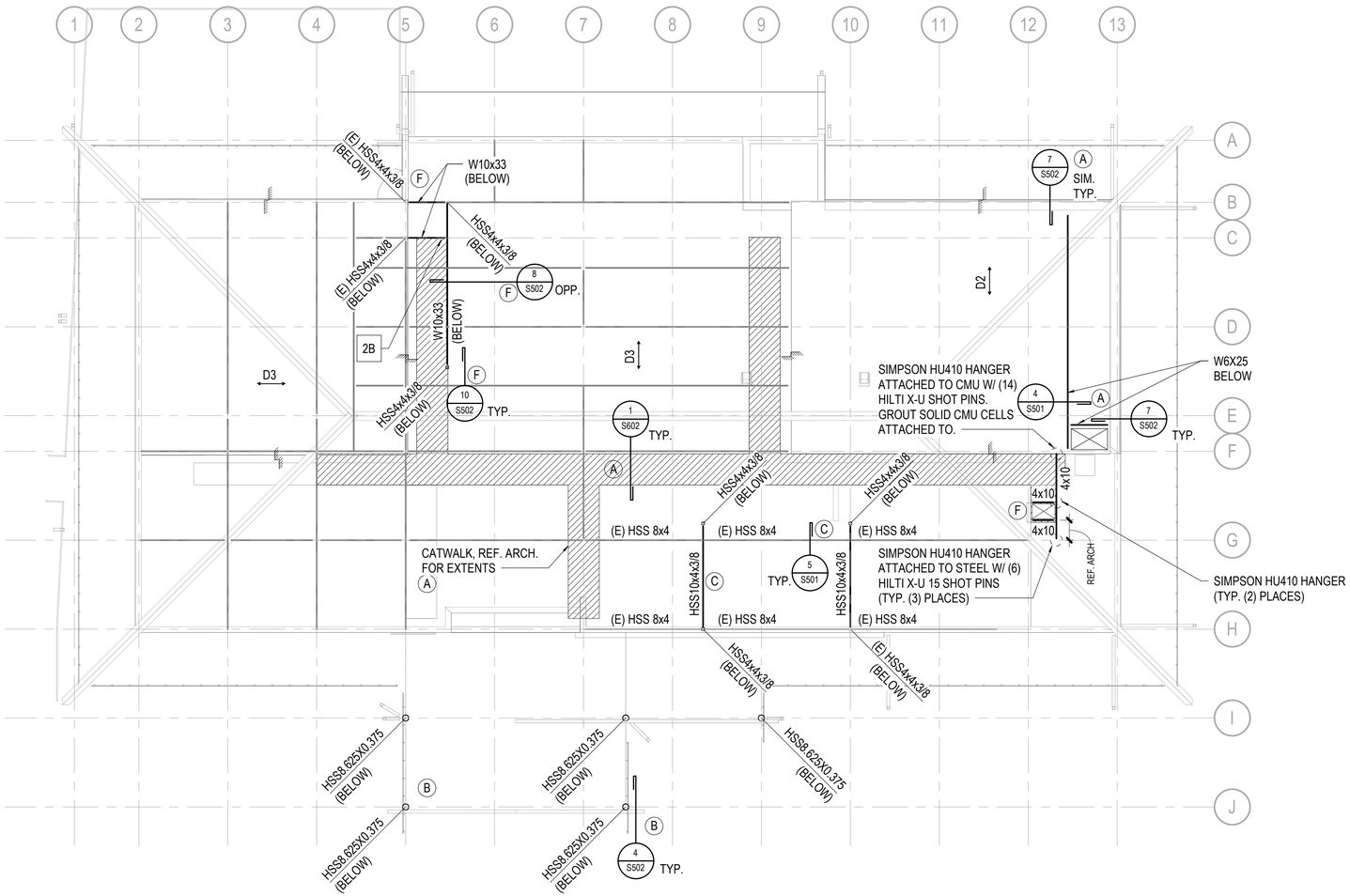
(C) OPTION C: INTERIOR LAYOUT RECONFIGURATION, INCLUDING RELOCATING WALLS, PLUMBING, AND ELECTRICAL. ALL INTERIOR FINISH AND INSULATION REPLACEMENT FROM INTERIOR FACE OF EXTERIOR SIDING IN. INCLUDES CASEWORK, LIGHT FIXTURES, MONITORS, AND TOILET ACCESSORIES.

(D) OPTION D: CABLING FOR LOW VOLTAGE FIRE ALARM IMPROVEMENTS.

(E) OPTION E: INSTALLATION OF NEW LIGHTNING PROTECTION SYSTEM.

(F) OPTION F: MAINTAIN (E) ASSISTIVE LISTENING SYSTEM. PROVIDE ALL (N) MEETINGROOM/AUDITORIUM AV EQUIP. INSTALL (N) FOLDING DOOR WITH STRUC SUPPORTS AND (N) WINDOW TREATMENTS

(G) OPTION G: DECK ADDITION WITH DUAL VIEWER, STRUC STL, (N) RAILINGS, AND (E) RETAINING WALL REALIGNMENT.




1 S103 LOW ROOF PLAN
SCALE: (A)

**SHEET NOTES**

1) VERIFY ALL DIMENSIONS WITH CONTRACTING OFFICER AND IN THE FIELD PRIOR TO CONSTRUCTION. NOTIFY CONTRACTING OFFICER OF ANY DISCREPANCIES.

2) D2 (E) DOX PLANK FLOOR SYSTEM ASSUMED TO CONSIST OF HORIZONTAL CONCRETE MASONRY UNITS, (3) CELLS WIDE, SPANNING BETWEEN CONCRETE BEAMS WITHIN THE DEPTH OF THE UNITS SPACED AT 30" O.C. OVERLAIN WITH 2" CONCRETE TOPPING. CMU BLOCKS MAY HAVE INTEGRAL TABS THAT ATTACH OVER THE BEAMS. DOX PLANK CALLED OUT AS 6" ON S102 AND 8" ON S103.
(E) F&A FLOOR SYSTEM IS BRITTLE.

3) D3 (E) WOOD ROOF SYSTEM CALLED OUT AS: 3" T&G PLANKS

4) INDICATES STEP IN ELEVATION

5) XX INDICATES CONNECTION REQUIRED, REF. 9/S502

SCALE: (A) 8 0 8 16
SCALE OF FEET

LICENSED PROFESSIONAL ENGINEER
SEAN M. CLARK
P28771
IOWA
5/29/26

| | | | |
|---|---|---|---|
| A/E FIRM<br>PRIME:<br>OTAK INC.<br>REDMOND, WA | DESIGNED: KEM<br>CADD: OC<br>TECH REVIEW: SMC<br>DATE: 03/03/2026 | SUB SHEET NO.<br>S103 | TITLE OF SHEET<br>STRUCTURAL ATTIC PLAN<br>EFFIGY MOUNDS NATIONAL MONUMENT |

DRAWING NO. 394 194248
PMIS/PKG NO. 318539
SHEET 59 OF 142

5/22/2026 9:02:10 AM Autodesk Docs://2025-110 NPS EFMO Visitor Center DD-CD/33278AE - NPS EFMO Visitor Center - Structure - v24.rvt

**OPTIONS**

(A) OPTION A: HVAC SYSTEMS INCLUDING EQUIPMENT, PIPING, DUCTWORK, AND CONTROLS. FIRE SUPPRESSION INCLUDING WET AND DRY SPRINKLER SYSTEMS INCLUDING ENTRY EQUIPMENT, FIRE ALARM SYSTEM, AND REFURBISHMENT OF CLEAN AGENT SYSTEM. INCLUDES ALSO SEALING COLLECTIONS SPACE FOR CLEAN AGENT SYSTEM FUNCTION.

(B) OPTION B: EXTERIOR FINISH AND INSULATION REPLACEMENT FROM EXTERIOR FACE OF EXTERIOR SHEATHING OUT AND NEW ROOF. INCLUDES REFRAMING AND REPLACING WINDOWS AND DOORS, REPLACEMENT OF EXTERIOR COLUMNS, AND FINALIZING SITE WORK.

(C) OPTION C: INTERIOR LAYOUT RECONFIGURATION, INCLUDING RELOCATING WALLS, PLUMBING, AND ELECTRICAL. ALL INTERIOR FINISH AND INSULATION REPLACEMENT FROM INTERIOR FACE OF EXTERIOR SIDING IN. INCLUDES CASEWORK, LIGHT FIXTURES, MONITORS, AND TOILET ACCESSORIES.

(D) OPTION D: CABLING FOR LOW VOLTAGE FIRE ALARM IMPROVEMENTS.

(E) OPTION E: INSTALLATION OF NEW LIGHTNING PROTECTION SYSTEM.

(F) OPTION F: MAINTAIN (E) ASSISTIVE LISTENING SYSTEM. PROVIDE ALL (N) MEETINGROOM/AUDITORIUM AV EQUIP. INSTALL (N) FOLDING DOOR WITH STRUC SUPPORTS AND (N) WINDOW TREATMENTS

(G) OPTION G: DECK ADDITION WITH DUAL VIEWER, STRUC STL, (N) RAILINGS, AND (E) RETAINING WALL REALIGNMENT.

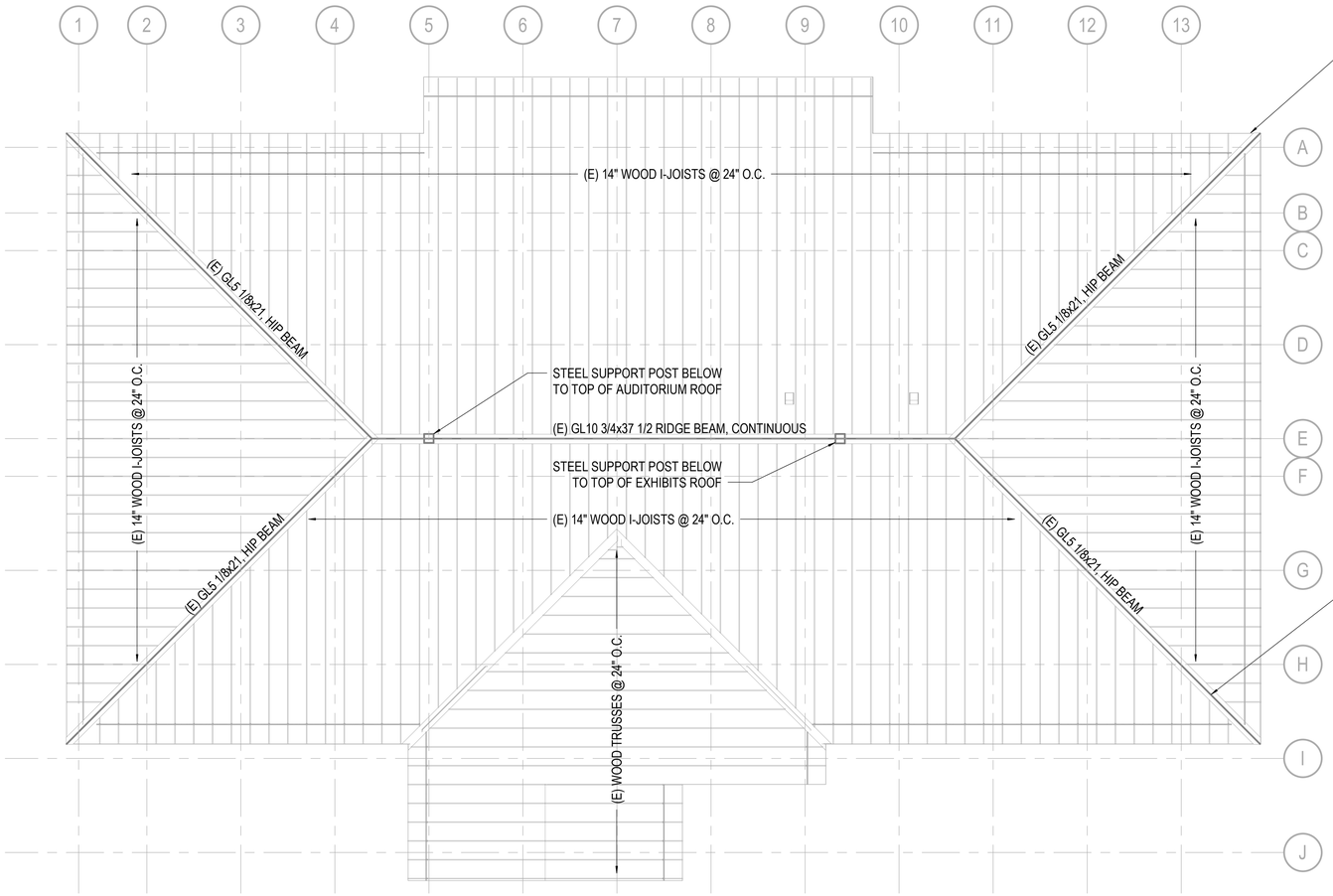
**SHEET NOTES**

1) VERIFY ALL DIMENSIONS WITH CONTRACTING OFFICER AND IN THE FIELD PRIOR TO CONSTRUCTION. NOTIFY CONTRACTING OFFICER OF ANY DISCREPANCIES.

2) (E) INDICATES EXISTING

3) D4 ⟷ INDICATES SPAN DIRECTION OF (E) ROOF DIAPHRAGM CALLED OUT AS: 3/4" THICK PLYWOOD. NAILING PATTERN UNKNOWN.

4) REFERENCE DETAILS 10/S601 & 11/S601 FOR TYPICAL ROOF PENETRATION DETAILS. REFERENCE ARCH AND MEP DRAWINGS FOR LOCATIONS. (A)




REPAIR OF WATER DAMAGED RAFTERS AT FASCIA:
1) INVESTIGATE EAVES TO DETERMINE AREA SIZE AND LOCATION OF DAMAGE.
2) ASSUME 20% OF ROOF EAVE AREA REQUIRES REPAIR (600 SQUARE FEET)

REPAIR PLAN:
FOR MINIMAL WATER DAMAGE (MAXIMUM OF 18-LINEAL-INCHES): SISTER 2x MEMBER (HEIGHT TO MATCH JOIST, 14" TYPICAL, 10" AT CORNERS) TO SIDE OF (E) I-JOIST AND NAIL TO TOP AND BOTTOM FLANGE WITH 10d COMMON NAILS @ 4" O.C.(REF. 5/S601) EXTEND 2x MEMBER TWICE THE LENGTH OF THE WATER DAMAGE (ASSUME 2-FT LENGTHS UNTIL INVESTIGATION IS COMPLETE)

REPAIR OF ALL RAFTER CONNECTIONS TO HIP/VALLEY BEAMS:
ASSUME STEEL BEARING ANGLE UNDERNEATH EACH RAFTER AND A SIDE CLIP (COLD-FORM BENT PLATE, MINIMUM 16GA, MINIMUM SIZE = 8" SQ.) AT EACH RAFTER (REF. 6/S601).
BASE ASSUMPTION OF (120) CONNECTIONS TO BE REPAIRED, ACCOUNT FOR CONTINGENCY.

1 S104 STRUCTURAL ROOF PLAN
SCALE: (A)

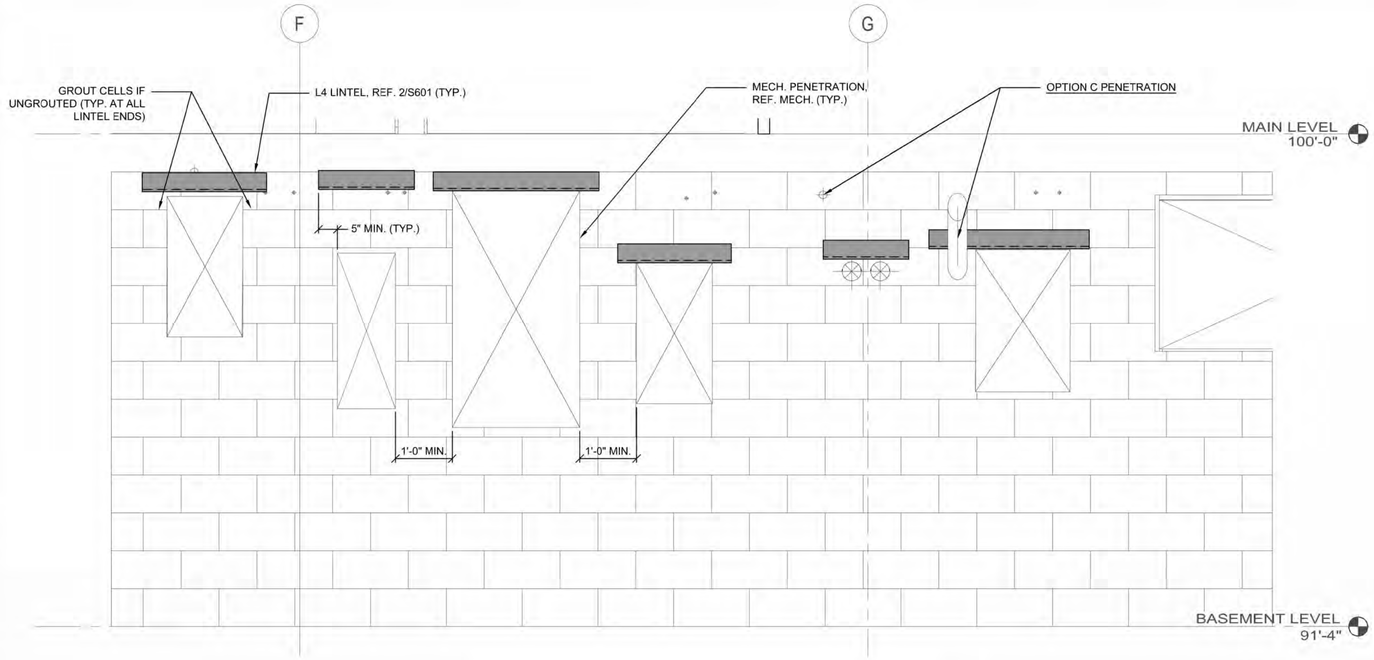
SCALE: (A) 8 0 8 16
SCALE OF FEET

LICENSED PROFESSIONAL ENGINEER
SEAN M. CLARK
P29771
IOWA
5/29/26

| | | | |
|---|---|---|---|
| A/E FIRM<br>PRIME:<br>OTAK INC.<br>REDMOND, WA | DESIGNED: KEM<br>CADD OC<br>TECH REVIEW: SMC<br>DATE: 03/03/2026 | SUB SHEET NO.<br>S104 | TITLE OF SHEET<br>STRUCTURAL ROOF PLAN<br>EFFIGY MOUNDS NATIONAL MONUMENT |

DRAWING NO. 394 194248
PMIS/PKG NO. 318539
SHEET 60 OF 142

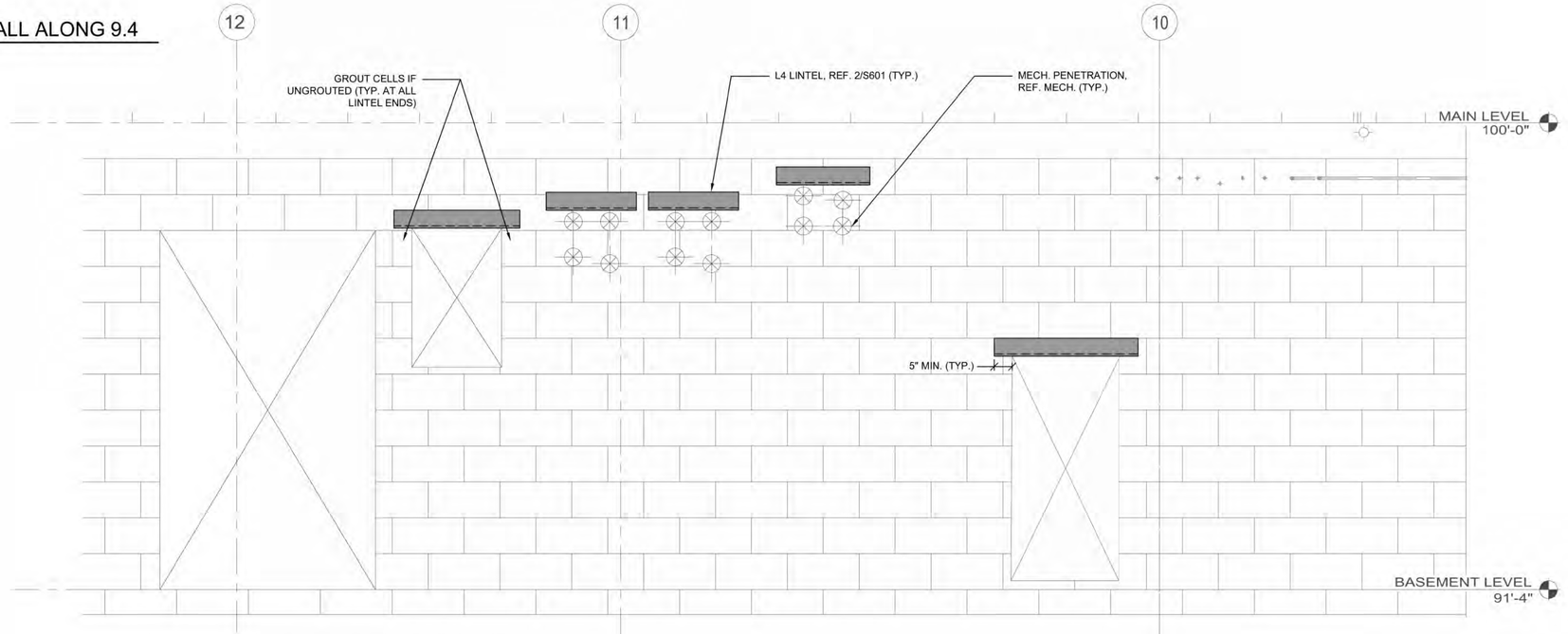
5/22/2026 9:02:11 AM Autodesk Docs://2025-110 NPS EFMO Visitor Center DD-CD/33278AE - NPS EFMO Visitor Center - Structure - v24.rvt




1 S300 MECHANICAL/CRAWLSPACE WALL ALONG 9.4

3/4" = 1'-0"

NOTES:
1) 1" ANNULAR SPACE REQUIRED AROUND EACH MECHANICAL PENETRATION NOT SHOWN FOR CLARITY.
2) ALL PENETRATIONS FOR OPTION A UNLESS NOTED OTHERWISE.




2 S300 CRAWLSPACE WALL ALONG F

3/4" = 1'-0"

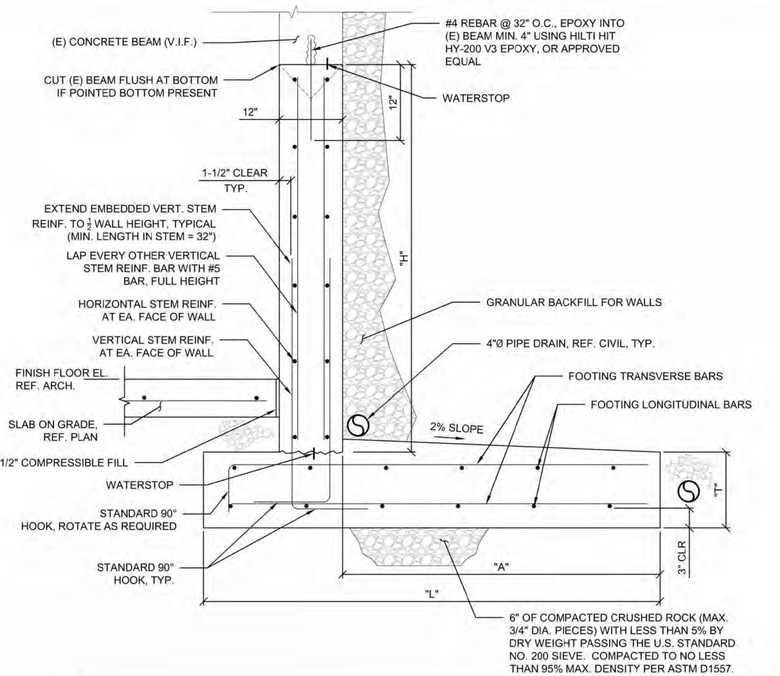
NOTES:
1) 1" ANNULAR SPACE REQUIRED AROUND EACH MECHANICAL PENETRATION NOT SHOWN FOR CLARITY.
2) ALL PENETRATIONS FOR OPTION A UNLESS NOTED OTHERWISE.

0 1.5 3
SCALE IN FEET

LICENSED PROFESSIONAL ENGINEER
SEAN M. CLARK
P29771
IOWA
5/29/26

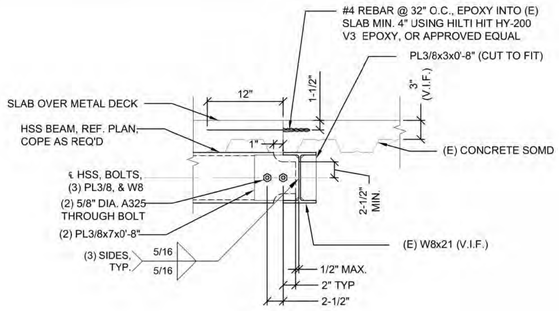
| | | | | |
|---|---|---|---|---|
| A/E FIRM<br>PRIME:<br>OTAK INC.<br>REDMOND, WA | DESIGNED: KEM<br>CADD: OC<br>TECH REVIEW: SMC<br>DATE: 5/29/2026 | SUB SHEET NO.<br>S300 | TITLE OF SHEET<br>STRUCTURAL SECTIONS<br>EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248<br>PMIS/PKG NO. 318539<br>SHEET 61 OF 142 |

11/18/2025 4:19:22 PM Autodesk Docs://2025-110 NPS EFMO Visitor Center DD-CD/33278AE - NPS EFMO Visitor Center - Structure - v24.rvt

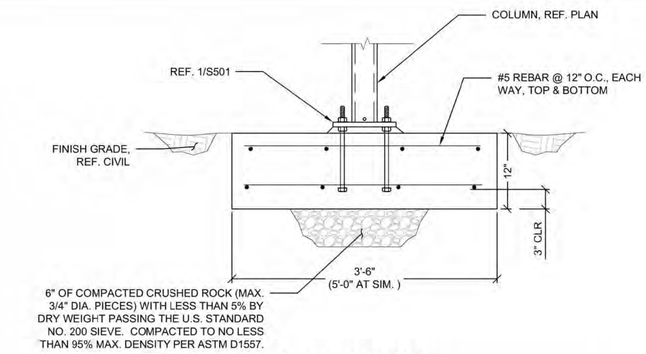



| RETAINING WALL SCHEDULE | | | | | | | |
|---|---|---|---|---|---|---|---|
| H (MAX.) | L | A | T | STEM REINF. VERT. | STEM REINF. HORIZ. | TRANSVERSE FTG. REINF. | LONGITUDINAL FTG. REINF. |
| 11'-2" | 9'-0" | 7'-0" | 1'-2" | #5 @ 3" O.C. | #5 @ 12" O.C. | #5 @ 3" O.C. | #5 @ 12" O.C. |
| 8'-0" | 6'-6" | 4'-6" | 1'-2" | #5 @ 9" O.C. | #5 @ 12" O.C. | #5 @ 9" O.C. | #5 @ 12" O.C. |

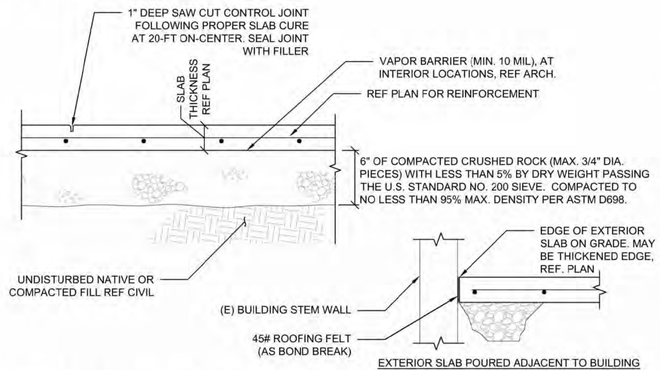
1 S401 — RETAINING WALL SCHEDULE
1" = 1'-0"
BASE OPTION



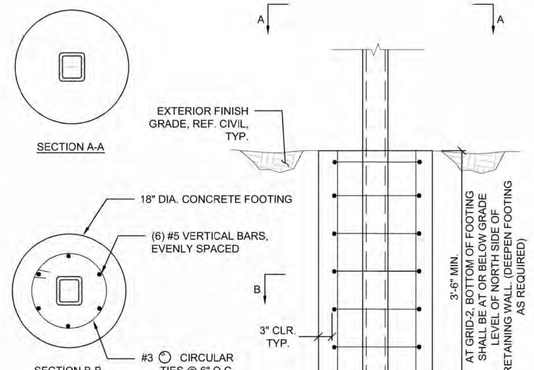

2 S401 — ATTACH NEW SOMD TO (E) STRUCTURE
1" = 1'-0"
OPTION G



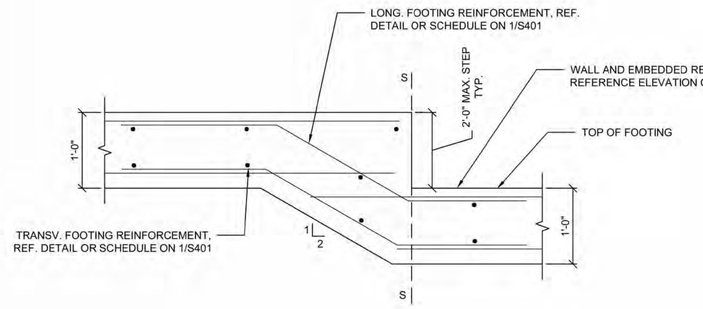

3 S401 — NEW SPREAD FOOTING
1" = 1'-0"
OPTION C



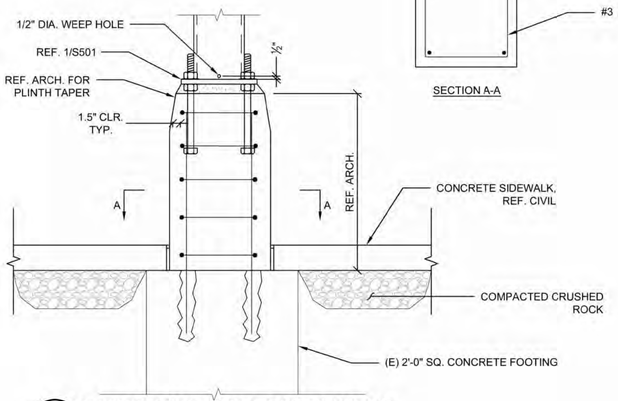

4 S401 — CONCRETE SLAB ON GRADE
1" = 1'-0"
BASE OPTION



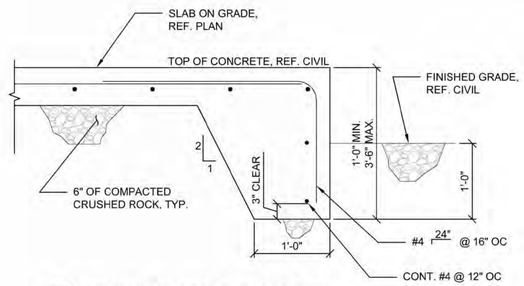

5 S401 — NEW COLUMN AND FLAGPOLE FOOTING
1" = 1'-0"
OPTION G




6 S401 — STEPPED CONT. FOOTING
1" = 1'-0"
BASE OPTION




7 S401 — ENTRYWAY COLUMN PLINTHS
1" = 1'-0"
OPTION B




8 S401 — THICKENED SLAB EDGE
1" = 1'-0"
BASE OPTION

0 1'-0" 2'-0"
SCALE IN FEET

LICENSED PROFESSIONAL ENGINEER
SEAN M. CLARK
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IOWA
5/29/26

| | | | |
|---|---|---|---|
| A/E FIRM<br>PRIME:<br>OTAK INC.<br>REDMOND, WA | DESIGNED: KEM<br>CADD OC<br>TECH REVIEW: SMC<br>DATE: 5/29/2026 | SUB SHEET NO.<br>S401 | TITLE OF SHEET<br>CONCRETE DETAILS<br>EFFIGY MOUNDS NATIONAL MONUMENT |

DRAWING NO. 394 194248
PMIS/PKG NO. 318539
SHEET 62 OF 142

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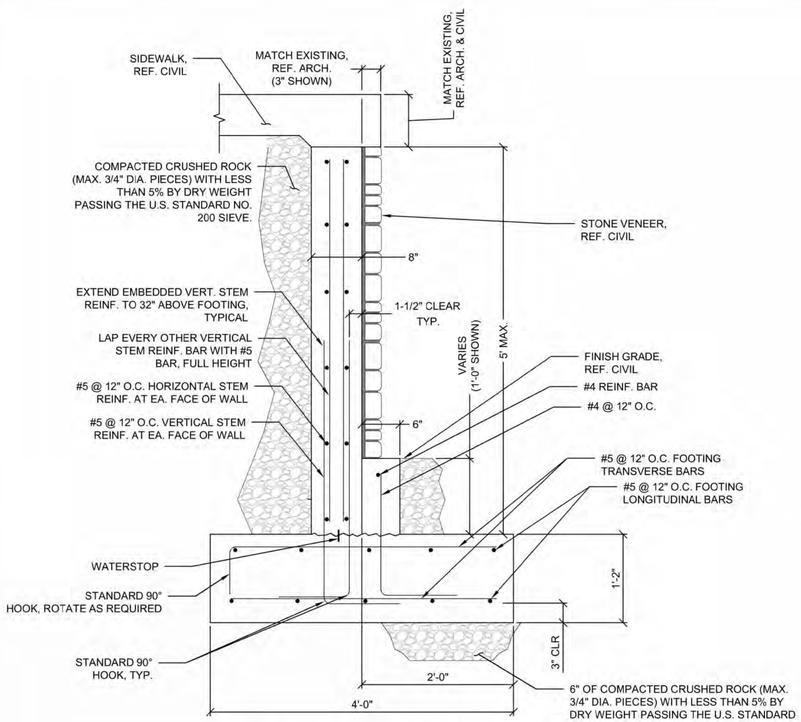
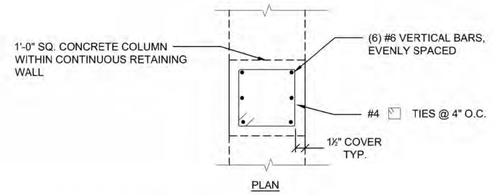
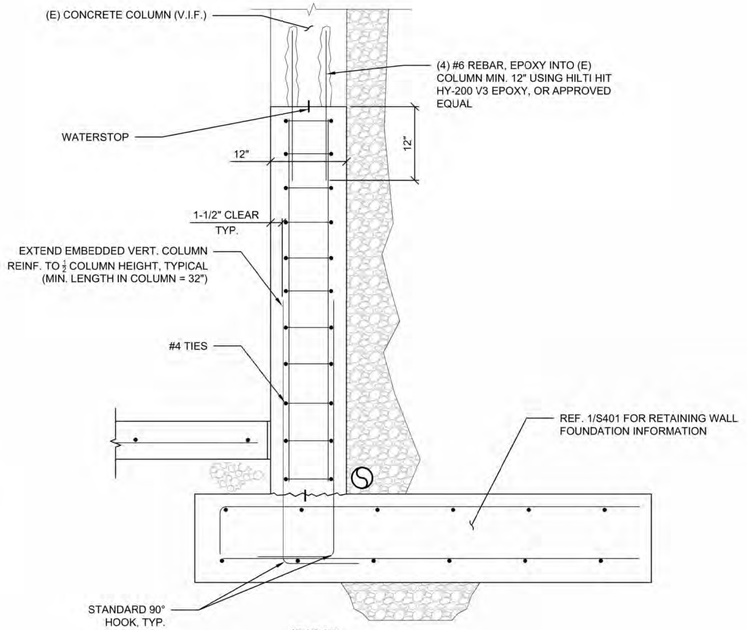



1 S402 PATIO RETAINING WALL
1" = 1'-0"
BASE OPTION

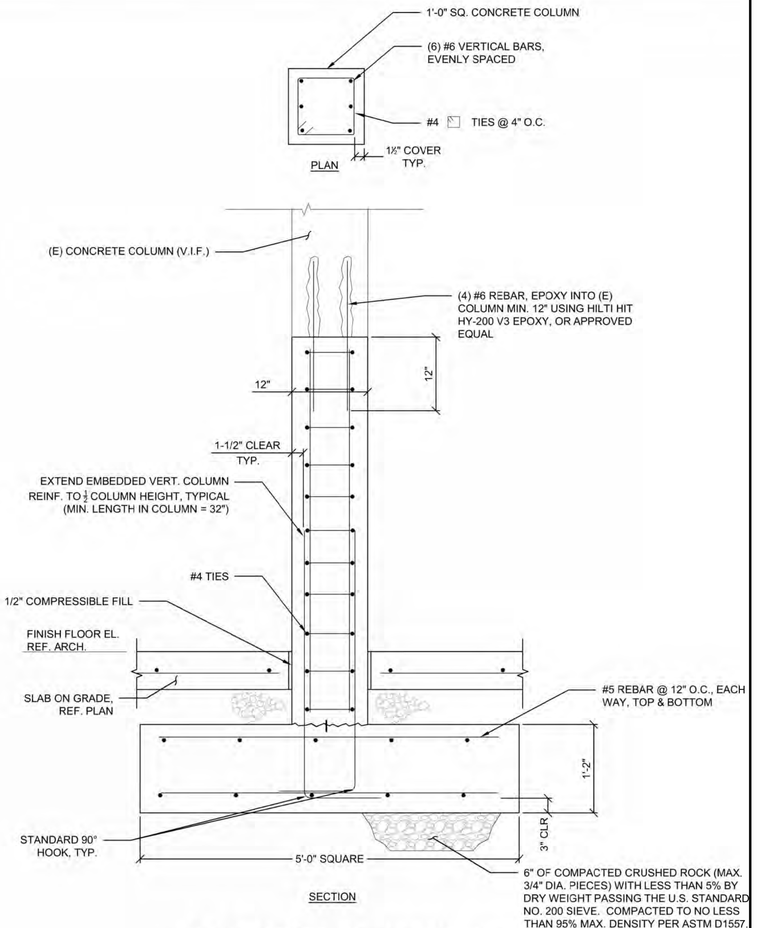



PLAN




SECTION

2 S402 COLUMN EXTENSION - GRID H
1" = 1'-0"
BASE OPTION

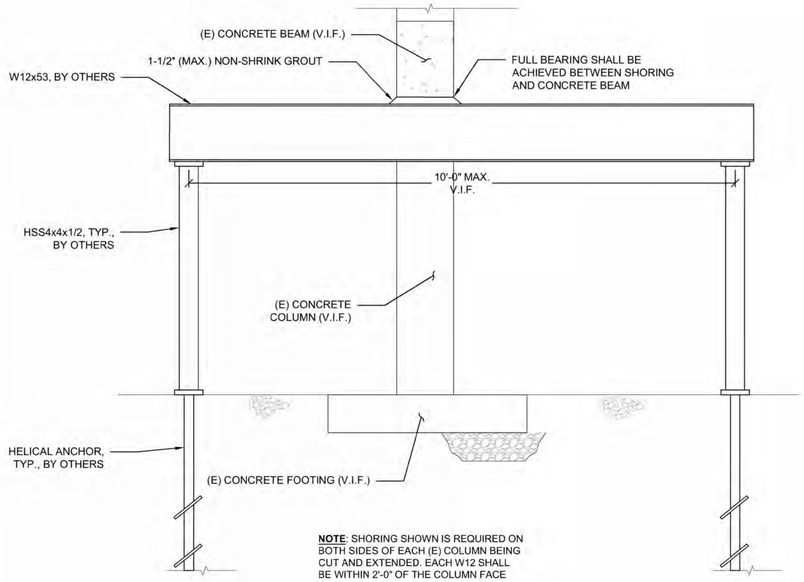



PLAN



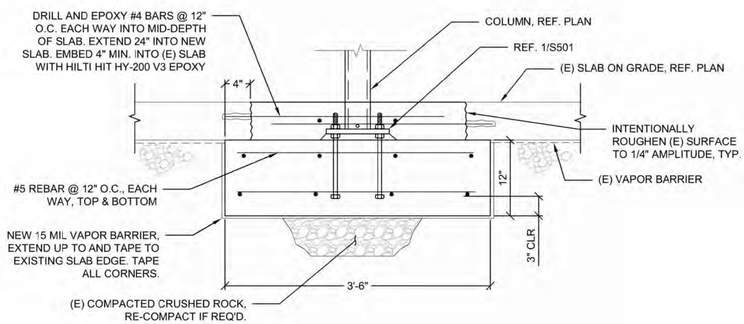

SECTION

3 S402 COLUMN EXTENSION - GRID G
1" = 1'-0"
BASE OPTION

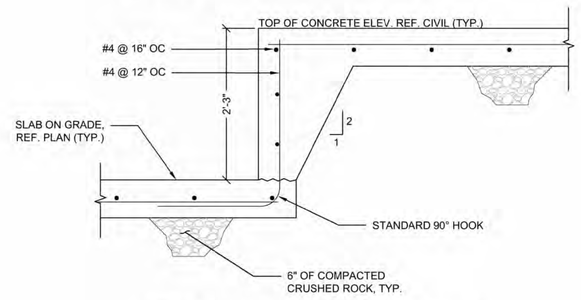



**NOTE**: SHORING SHOWN IS REQUIRED ON BOTH SIDES OF EACH (E) COLUMN BEING CUT AND EXTENDED. EACH W12 SHALL BE WITHIN 2'-0" OF THE COLUMN FACE

4 S402 RECOMMENDED SHORING
3/4" = 1'-0"
BASE OPTION




5 S402 FOOTING PEN. THRU SLAB
1" = 1'-0"
OPTION C




6 S402 STEPPED SLAB ON GRADE
1" = 1'-0"
BASE OPTION

0 1'-0" 2'-0"
SCALE IN FEET

SEAN M. CLARK
P2771
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IOWA
5/29/26

| | | | |
|---|---|---|---|
| A/E FIRM<br>PRIME:<br>OTAK INC.<br>REDMOND, WA | DESIGNED: KEM<br>CADD: OC<br>TECH REVIEW: SMC<br>DATE: 5/29/2026 | SUB SHEET NO.<br>S402 | TITLE OF SHEET<br>CONCRETE DETAILS 2<br>EFFIGY MOUNDS NATIONAL MONUMENT |

DRAWING NO. 394 194248
PMIS/PKG NO. 318539
SHEET 63 OF 142

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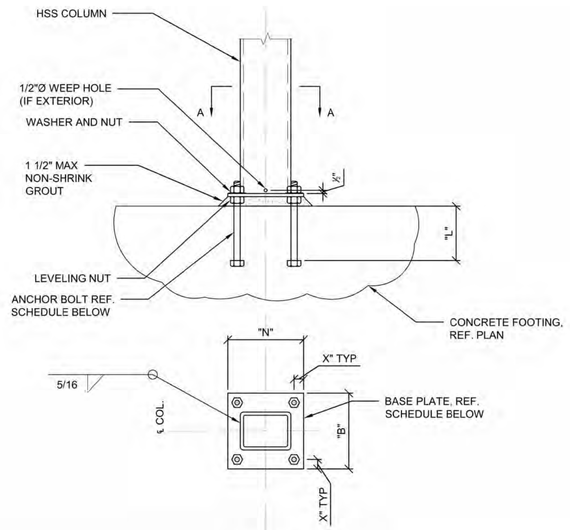


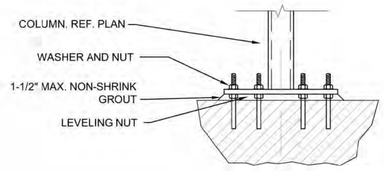

SECTION A-A

| TYP. GRAVITY COLUMN BASE PLATE SCHEDULE | | | | | |
|---|---|---|---|---|---|
| COL. SIZE | N(IN.) | B(IN.) | T(IN.) | X(IN.) | ANCHOR BOLTS (No.) DIA. xL |
| HSS4x4 | 10 | 10 | 3/4 | 1 1/2 | (4) 5/8"Ø x 9" |
| HSS8.625 | 12 | 12 | 3/4 | 1 3/4 | (4) 5/8"Ø x 9" |

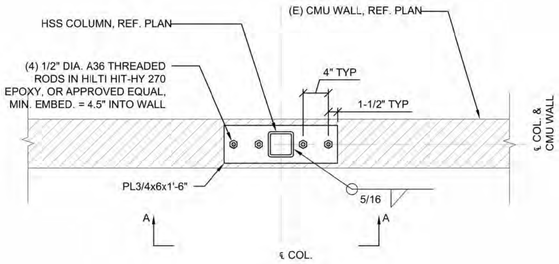
NOTES:
1. ANCHOR BOLTS SHALL BE HEADED GALVANIZED, ASTM F1554 GR 36 U.N.O.
2. ANCHOR BOLT HOLE DIAMETERS ($d_h$) SHALL BE AS FOLLOWS:
   $d_b \le 1"$ : $dh = d_b + 5/16"$
   $1" < d_b \le 2"$ : $d_h = d_b + 1/2"$
   $d_b > 2"$ : $d_h = d_b + 1"$
3. PROVIDE MILL FINISH AT BOTTOM OF COLUMN AND TOP OF BASE PLATE, TYP.

## 1 S501 HSS BASE PLATE DETAIL & SCHEDULE
1" = 1'-0"

OPTION C



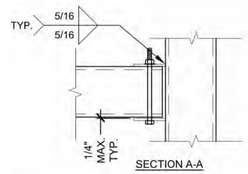

SECTION A-A



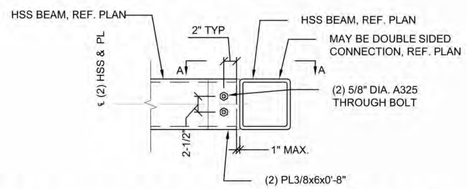

NOTES:
1. REF. 1/S501 FOR DETAILS NOT SHOWN
2. IF (E) WALL IS NOT FULLY GROUTED, GROUT SOLID (4) CELLS OF WALL, CENTERED ON COLUMN.

## 2 S501 HSS BASE PLATE ON (E) CMU WALL
1" = 1'-0"

OPTION C

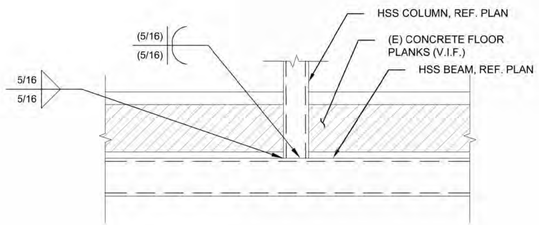



SECTION A-A



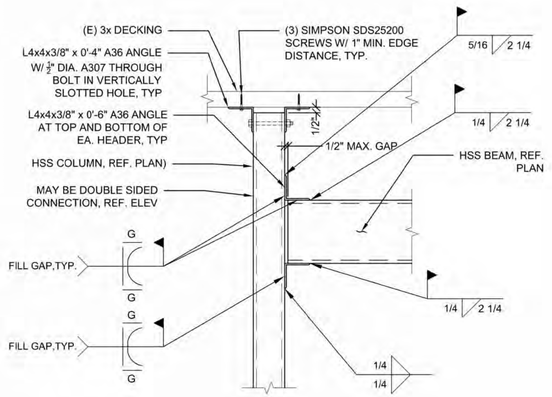

## 6 S501 HSS BEAM CONNECTION TO HSS BEAM
1" = 1'-0"

OPTION G



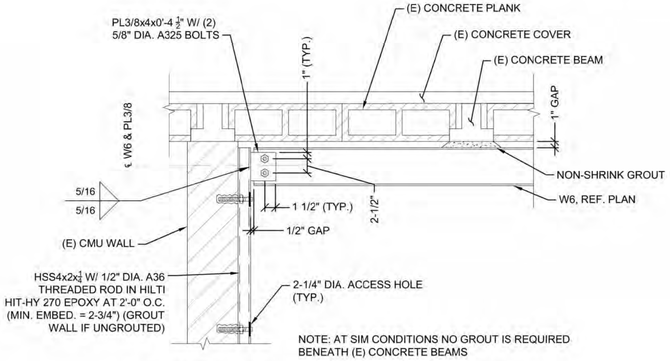

## 7 S501 HSS COLUMN ATTACHMENT TO HSS BEAM
1" = 1'-0"

OPTION F




## 5 S501 HSS BEAM CONNECTION TO HSS COLUMN
1" = 1'-0"

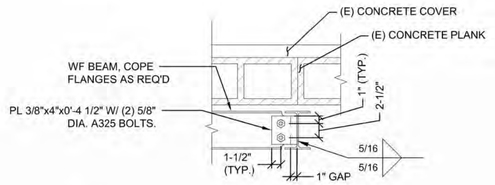
OPTION C/F




NOTE: AT SIM CONDITIONS NO GROUT IS REQUIRED BENEATH (E) CONCRETE BEAMS

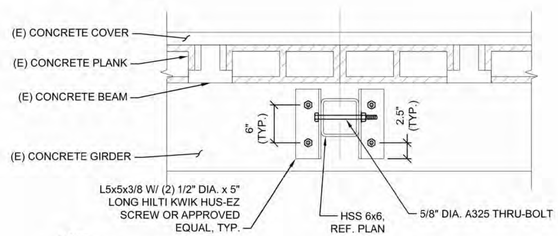
## 3 S501 W6 CONNECTION TO CMU WALL
1" = 1'-0"

OPTION A




## 4 S501 W6 BEAM TO BEAM CONNECTION
1" = 1'-0"

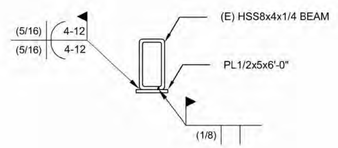
OPTION A




NOTE:
DO NOT DAMAGE (E) REBAR DURING DRILLING (MAY REQUIRE GPR)

## 8 S501 HSS BEAM CONNECTION TO CONC GIRDER
1" = 1'-0"

BASE OPTION




## 9 S501 BURST HSS BEAM REPAIR
1" = 1'-0"

BASE OPTION

0 1'-0" 2'-0"
SCALE IN FEET

LICENSED PROFESSIONAL ENGINEER
SEAN M. CLARK
P29771
IOWA
5/29/26

| | | | |
|---|---|---|---|
| A/E FIRM<br>PRIME:<br>OTAK INC.<br>REDMOND, WA | DESIGNED: KEM<br>CADD: OC<br>TECH REVIEW: SMC<br>DATE: 5/29/2026 | SUB SHEET NO.<br>S501 | TITLE OF SHEET<br>STEEL DETAILS<br>EFFIGY MOUNDS NATIONAL MONUMENT |

DRAWING NO. 394 194248
PMIS/PKG NO. 318539
SHEET 64 OF 142

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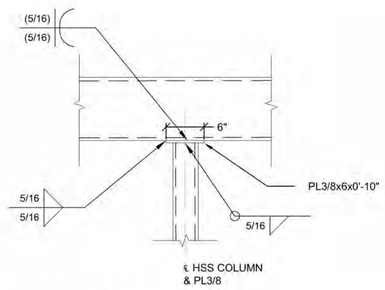
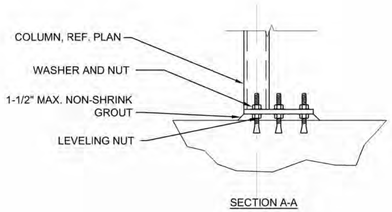
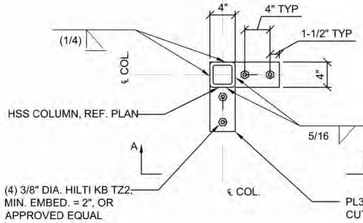
11/18/2025 4:19:22 PM Autodesk Docs://2025-110 NPS EFMO Visitor Center DD-CD/33278AE - NPS EFMO Visitor Center - Structure - v24.rvt




## 1 S502 HSS BEAM TO HSS COLUMN
1" = 1'-0"

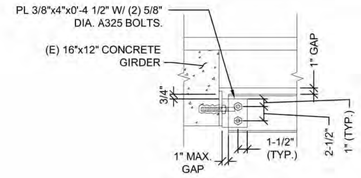
OPTION G

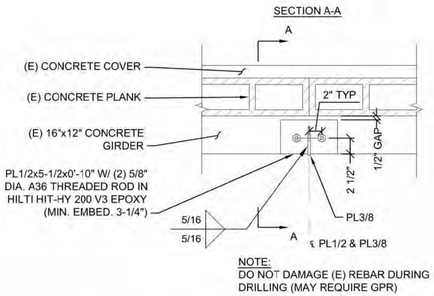



## 5 S502 HSS BASE PLATE IN CORNER
1" = 1'-0"

OPTION F

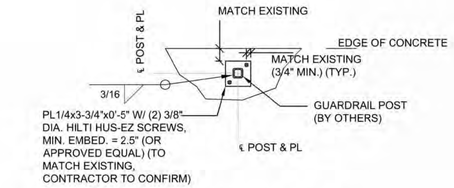



NOTE:
DO NOT DAMAGE (E) REBAR DURING
DRILLING (MAY REQUIRE GPR)

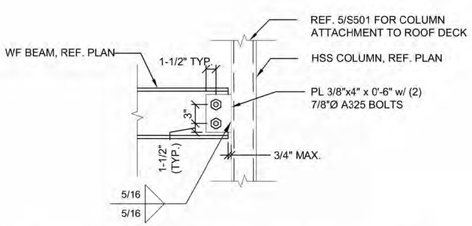
## 2 S502 W6 ATTACHMENT TO CONC. GIRDER
1" = 1'-0"

OPTION A



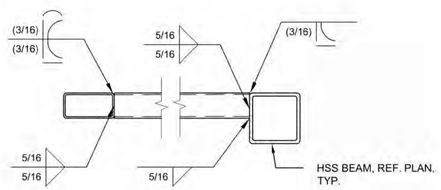

## 6 S502 GUARDRAIL ATTACHMENT
1" = 1'-0"

BASE OPTION



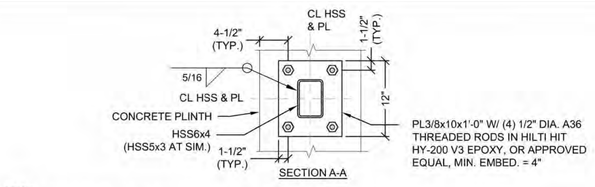

## 10 S502 TYP. WF BEAM TO HSS COLUMN CONN.
1" = 1'-0"

OPTION F

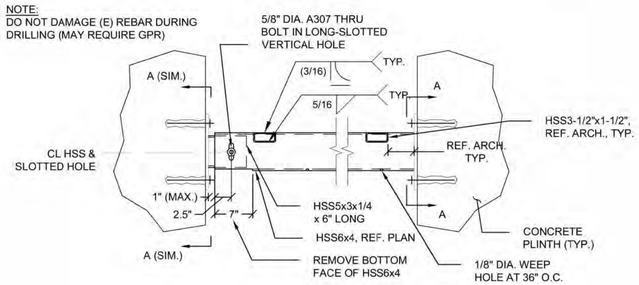



## 11 S502 HORIZONTAL HSS BEAM CONN.
1" = 1'-0"

OPTION G

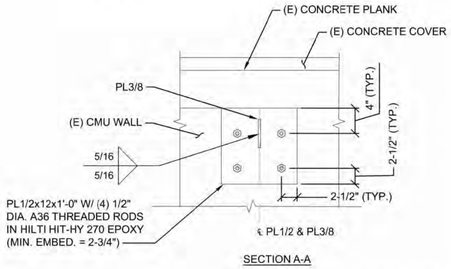



NOTE:
DO NOT DAMAGE (E) REBAR DURING
DRILLING (MAY REQUIRE GPR)

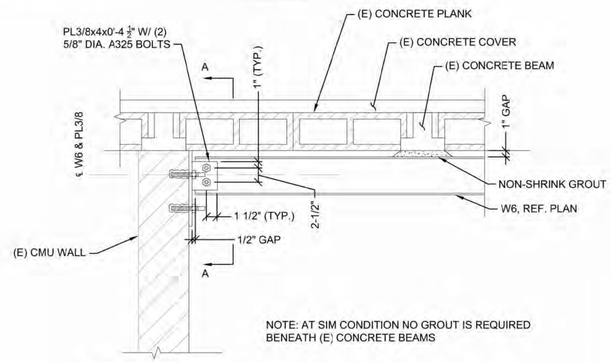



## 3 S502 HSS BENCH CONNECTIONS
1" = 1'-0"

OPTION B



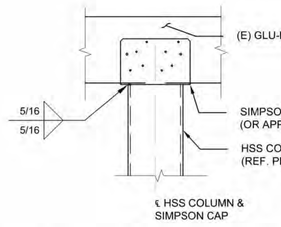

SECTION A-A




NOTE: AT SIM CONDITION NO GROUT IS REQUIRED
BENEATH (E) CONCRETE BEAMS

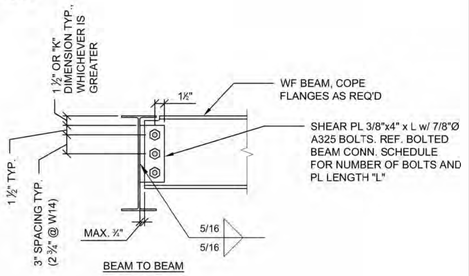
## 7 S502 W6 ATTACHMENT AT ATTIC LID
1" = 1'-0"

OPTION A

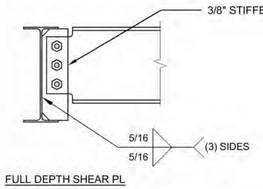



## 4 S502 COLUMN CAP CONNECTION
1" = 1'-0"

OPTION B




BEAM TO BEAM




FULL DEPTH SHEAR PL

SEE PLAN FOR LOCATIONS OF FULL DEPTH SHEAR PLATE INDICATED *

## 8 S502 TYP. BOLTED WF BEAM CONNECTION
1" = 1'-0"

OPTION F

| TYP. BOLTED BEAM SCHEDULE | | | | |
|---|---|---|---|---|
| CONN. SYMBOL | SHEAR PL LENGTH "L" | NUMBER OF 7/8" DIA. BOLTS | WIDE FLANGE BEAM (SEE NOTE 1) | |
| 2B | 6" | 2 | W8, W10 | STANDARD HOLE |

NOTES:
1) BOLT HOLES SHALL BE STANDARD HOLES (BOLT DIA. + 1/16")
2) CONNECTION SHALL BE USED FOR DEPTH OF WIDE FLANGE BEAM LISTED UNLESS NOTED OTHERWISE ON PLAN

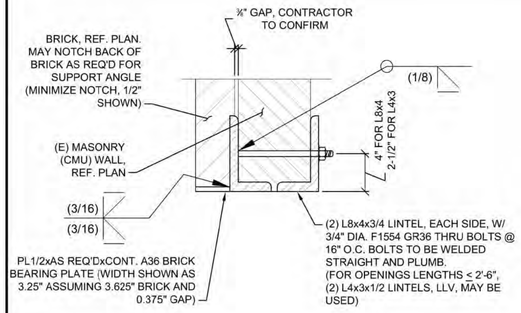
## 9 S502 TYP. BOLTED WF BEAM CONN. SCHEDULE
1" = 1'-0"

OPTION F

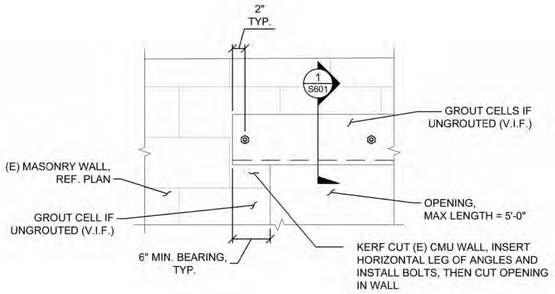
0 1'-0" 2'-0"
SCALE IN FEET

LICENSED PROFESSIONAL ENGINEER
SEAN M. CLARK
P29771
IOWA
5/29/26

| | | | | |
|---|---|---|---|---|
| A/E FIRM<br>PRIME:<br>OTAK INC.<br>REDMOND, WA | DESIGNED: KEM<br>CADD: OC<br>TECH REVIEW: SMC<br>DATE: 5/29/2026 | SUB SHEET NO.<br>S502 | TITLE OF SHEET<br>STEEL DETAILS 2<br>EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248<br>PMIS/PKG NO. 318539<br>SHEET 65 OF 142 |

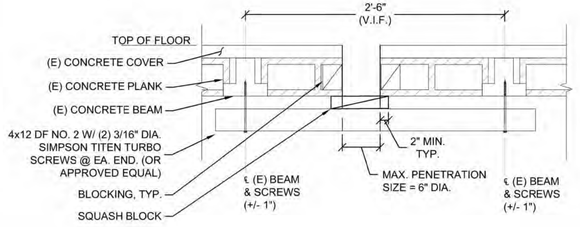



1 S601 MASONRY LINTEL
1-1/2" = 1'-0"
OPTION A/B

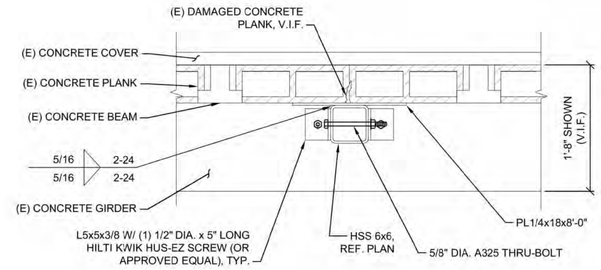



**NOTE**: TEMPORARY SHORING BY CONTRACTOR REQUIRED AT TIME OF INSTALL.

2 S601 MASONRY LINTEL BEARING
1" = 1'-0"
OPTION A/B

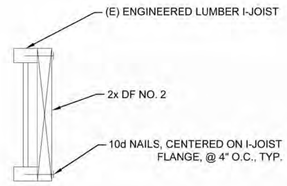



3 S601 WOOD INFILL AT VERT. PENETRATIONS
1" = 1'-0"
BASE OPTION

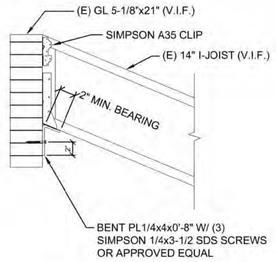



**NOTE**:
1) IF (E) PLANKS ARE NOT LEVEL, USE 1" MAX NON-SHRINK GROUT BETWEEN PL1/4 AND THE (E) PLANKS TO ACHIEVE FULL BEARING.
2) DO NOT DAMAGE (E) REBAR DURING DRILLING (MAY REQUIRE GPR)

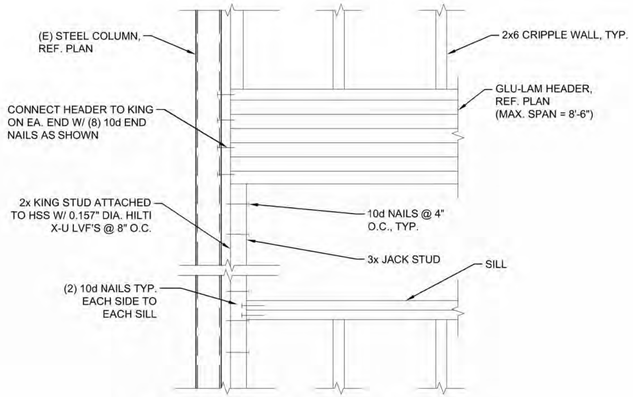
4 S601 CRACKED CONCRETE PLANK REPAIR
1" = 1'-0"
BASE OPTION



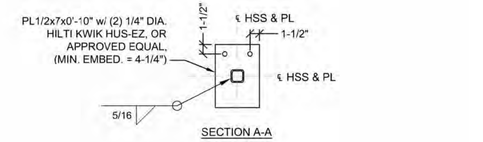

5 S601 JOIST STRENGTHENING
1-1/2" = 1'-0"
BASE OPTION



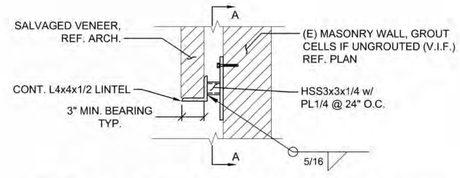

6 S601 I-JOIST CONNECTION STRENGTHENING
1-1/2" = 1'-0"
BASE OPTION



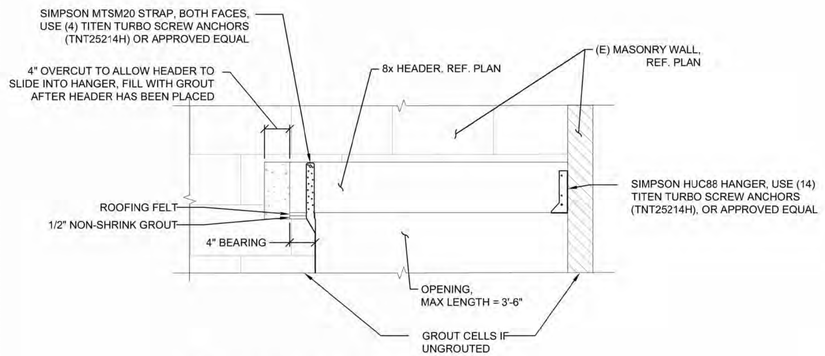

7 S601 WINDOW HEADER SUPPORT
1" = 1'-0"
OPTION B



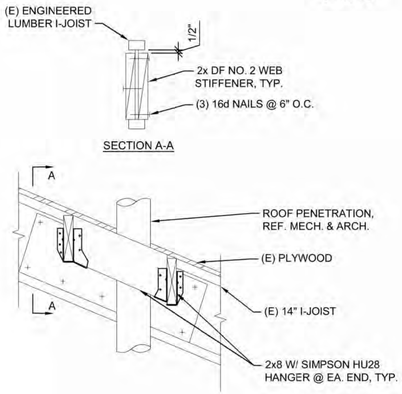

SECTION A-A




8 S601 STEEL LEDGER FOR VENEER
1" = 1'-0"
OPTION B



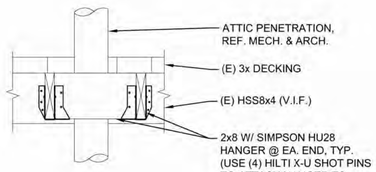

9 S601 WOOD HEADER ATTACHMENT
1" = 1'-0"
BASE OPTION



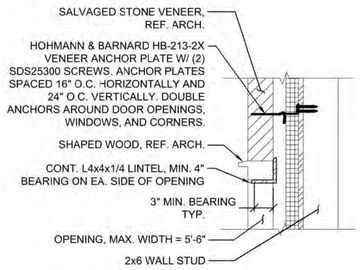

SECTION A-A

**NOTE:** AT SIM. CONDITION (REMOVED PENETRATIONS) SHEATH OVER REMOVED PENETRATION AND ATTACH TO 2x8s, WHERE OCCURS.

10 S601 TYP. ROOF PENETRATION FRAMING
1" = 1'-0"
OPTION A




11 S601 TYP. ATTIC PENETRATION FRAMING
1" = 1'-0"
OPTION A




12 S601 TYP. LOOSE LINTEL AT SALVAGED STONE
1" = 1'-0"
OPTION B

0 1'-0" 2'-0"
SCALE IN FEET

LICENSED PROFESSIONAL ENGINEER
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5/29/26

| | | | | |
|---|---|---|---|---|
| A/E FIRM<br>PRIME:<br>OTAK INC.<br>REDMOND, WA | DESIGNED: KEM<br>CADD: OC<br>TECH REVIEW: SMC<br>DATE: 5/29/2026 | SUB SHEET NO.<br>S601 | TITLE OF SHEET<br>FRAMING DETAILS<br>EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248<br>PMIS/PKG NO. 318539<br>SHEET 66 OF 142 |

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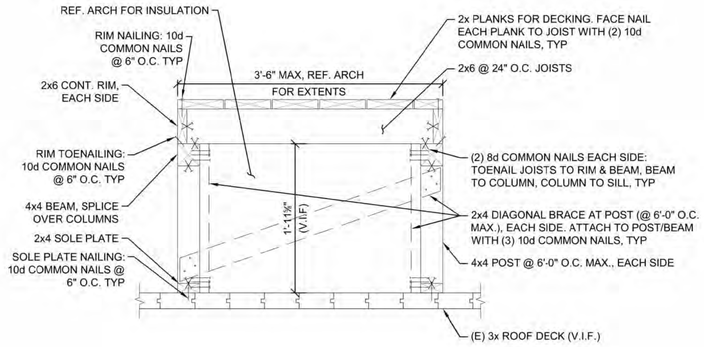
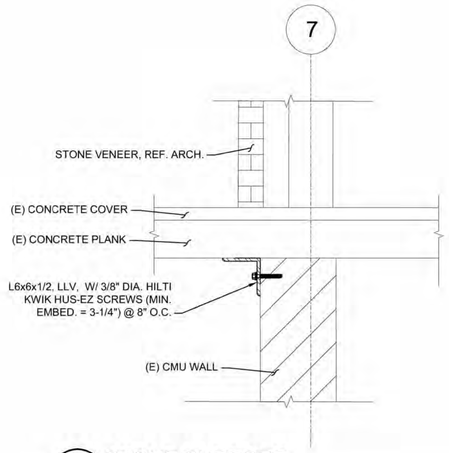



1 / S602 ATTIC CATWALK
1" = 1'-0"
OPTION A




2 / S602 STONE SUPPORT
1" = 1'-0"
OPTION B

0 1'-0" 2'-0"
SCALE IN FEET

LICENSED PROFESSIONAL ENGINEER
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IOWA
5/29/26

| | | | |
|---|---|---|---|
| A/E FIRM<br>PRIME:<br>OTAK INC.<br>REDMOND, WA | DESIGNED: KEM<br>CADD: OC<br>TECH REVIEW: SMC<br>DATE: 5/29/2026 | SUB SHEET NO.<br>S602 | TITLE OF SHEET<br>FRAMING DETAILS 2<br>EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248<br>PMIS/PKG NO. 318539<br>SHEET 67 OF 142 |

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2/27/2026 4:54:25 PM Autodesk Docs://2025-110 NPS EFMO Visitor Center DD-CD/NPS EFMO Visitor Center_MP R24.rvt

## MECHANICAL NOTES

1. GENERAL

1. ALL WORK SHALL BE IN ACCORDANCE WITH SMACNA STANDARDS AND SPECIFICATIONS, AND LOCAL AUTHORITY HAVING JURISDICTION.
2. THESE DRAWINGS ARE DIAGRAMMATIC IN NATURE AND NOT INTENDED TO SHOW ALL TRANSITIONS, OFFSETS, ETC. CONTRACTOR SHALL FIELD VERIFY EXISTING CONDITIONS AND PROVIDE ALL NECESSARY FITTINGS TO COMPLETE THE INTENT OF THE DRAWINGS. ANY DISCREPANCIES BETWEEN DRAWINGS AND FIELD CONDITIONS SHALL BE REPORTED TO THE CONTRACTING OFFICER FOR RESOLUTION. CONTRACTOR MAY LOCATE MECHANICAL EQUIPMENT DIFFERENTLY THAN SHOWN ON DRAWINGS DUE TO CONFLICTS, AS LONG AS FUNCTION AND/OR APPEARANCE ARE NOT AFFECTED.
3. COORDINATE SPACE REQUIREMENTS, SUPPORTS, AND INSTALLATION OF MECHANICAL WORK, WHICH ARE INDICATED DIAGRAMMATICALLY ON THE DRAWINGS. FOLLOW ROUTING SHOWN FOR PIPES AND DUCTS AS CLOSELY AS PRACTICABLE; PLACE RUNS PARALLEL WITH LINES OF BUILDING. UTILIZE SPACES EFFICIENTLY TO MAXIMIZE ACCESSIBILITY FOR OTHER INSTALLATIONS, FOR MAINTENANCE, AND FOR REPAIRS.
4. COMPLY WITH MANUFACTURER'S INSTRUCTIONS INCLUDING EACH STEP IN SEQUENCE. SHOULD MANUFACTURERS' INSTRUCTIONS CONFLICT WITH THE DRAWINGS REQUEST CLARIFICATION FROM THE CONTRACTING OFFICER BEFORE PROCEEDING.
5. DUCT SIZES ARE INSIDE DIMENSION.
6. CONTRACTOR SHALL REVIEW THESE DOCUMENTS CAREFULLY. CONTRACTOR SHALL CONTACT THE CONTRACTING OFFICER, FOR RESOLUTION OF ANY DISCREPANCIES, OMISSIONS, OR CLARIFICATIONS, BEFORE BID DATE. IN THE EVENT THAT AN INTERPRETATION OF BID DOCUMENTS IS NECESSARY AFTER THE BID DATE, THE DECISION OF THE CONTRACTING OFFICER SHALL BE FINAL AND BINDING.
7. PRODUCT DELIVERY, STORAGE, AND HANDLING: PROVIDE EQUIPMENT AND PERSONNEL TO HANDLE PRODUCTS BY METHODS TO PREVENT DAMAGE. PROMPTLY INSPECT SHIPMENTS TO ENSURE THAT PRODUCTS ARE UNDAMAGED. STORE AND PROTECT PRODUCTS IN ACCORDANCE WITH MANUFACTURERS' INSTRUCTIONS.
8. ALL REMOVED EQUIPMENT SHALL BE DISPOSED OF PER THEIR DIRECTION AND ALL APPLICABLE LOCAL, STATE, AND FEDERAL GUIDELINES.
9. THE CONTRACTOR IS RESPONSIBLE FOR THE COSTS OF ALL CHANGE ORDERS, WHICH THE NATIONAL PARK SERVICE (NPS) AND CONTRACTING OFFICER HAVE NOT APPROVED IN WRITING PRIOR TO THE EXECUTION OF THE ASSOCIATED WORK.
10. IN THE CASE OF A CONFLICT, UNLESS OTHERWISE NOTED, KEYNOTES ON MECHANICAL PLANS SHALL SUPERCEDE ANY GENERAL NOTES ON THE PLANS.
11. CONTRACTOR SHALL COORDINATE WORK WITH OTHER TRADES AND NOTIFY THE CONTRACTING OFFICER IF ANY CONFLICTS OCCUR.
12. THERMOSTAT LOCATIONS TO BE COORDINATED WITH THE BUILDING MANAGEMENT FOR NPS, VERIFIED BY THE CONTRACTING OFFICER. THERMOSTAT HEIGHTS SHALL BE 48" AFF TO MATCH LIGHTSWITCH HEIGHTS AND INSTALLED TO MEET ICC A117.1. CONTRACTOR TO CALIBRATE ALL THERMOSTATS SHOWN ON THIS PLAN.
13. PROVIDE VOLUME DAMPERS AT ALL DIFFUSER TAKEOFFS.
14. ALL TAKEOFFS, RUNOUTS, AND FLEX DUCTWORK TO DIFFUSERS SHALL BE THE SAME SIZE AS DIFFUSER INLET UNLESS OTHERWISE NOTED.
15. THERMOSTAT CONTROL LINES SHALL BE SUPPORTED AT INTERVALS NOT TO EXCEED 4'-0".
16. PROVIDE EXTERNAL INSULATION ON ALL NEW RIGID ROUND DUCTWORK.
17. PROVIDE ACCESS DOORS IN DUCTWORK AS REQUIRED FOR ACCESS TO FIRE DAMPERS, FIRE/SMOKE DAMPERS, OR ANY OTHER MECHANICAL EQUIPMENT REQUIRING MAINTENANCE OF SERVICE.
18. ALL PIPE AND DUCT PENETRATIONS THOUGH RATED WALLS SHALL BE SEALED PER 2025 IBC.

CONTROLS INTENT

ERV UNITS (TYPICAL OF 1)

OCCUPIED MODE:
UNIT SHALL BE PROVIDED WITH A TIME CLOCK. DURING OCCUPIED HOURS, THE ERV SHALL RUN CONTINUOUSLY.

LOUVERS (TYPICAL OF L-1,2)

WHEN THE ERV IS COMMANDED ON, THE ASSOCIATE INTAKE AND EXHAUST LOUVER SHALL OPEN.
WHEN THE ERVS IS COMMANDED OFF, THE ASSOCIATED INTAKE AND EXHAUST LOUVER SHALL CLOSE.

RADON DETECTION, TESTING, AND MITIGATION DESIGN NOTES:

- ALL WORK SHALL BE COMPLETED IN ACCORDANCE WITH ALL APPLICABLE ANSI & AARST STANDARDS AND SPECIFICATIONS, AND LOCAL AUTHORITY HAVING JURISDICTION.
- ALL DETECTION AND TESTING SHALL BE PERFORMED BY QUALIFIED RADON MITIGATION PROFESSIONAL WITH ACTIVE STATE LICENSURE AND ACTIVE CERFTIFICATION THROUGH THE NATIONAL RADON PROFICIENCY PROGRAM AND/OR NATIONAL RADON SAFETY BOARD.
- ALL DESIGN, INCLUDING BUT NOT LIMITED TO; FAN(S) SELECTION, FAN(S) LOCATION, DUCT SIZING, DUCT ROUTING, DISCHARGE AIR LOCATION, DISCHARGE AIR SURROUNDING CLEARANCE, MATERIAL SELECTIONS, SYSTEM COMMISIONING, MAINTENENCE AND OR TESTING ACCESS REQUIREMENTS, ANY APPLICABLE OWNER TRAINING, AND ANY OTHER COMPONENTS REQUIRED FOR AN EFFECTIVE AND EFFICIENT RADON MITIGATION SYSTEM SHALL BE PERFORMED BY QUALIFIED RADON MITIGATION PROFESSIONAL WITH ACTIVE STATE LICENSURE AND ACTIVE CERFTIFICATION THROUGH THE NATIONAL RADON PROFICIENCY PROGRAM AND/OR NATIONAL RADON SAFETY BOARD.
- RESULTS FROM TESTING AND THE MAXIMUM ALLOWABLE CONCENTRATION OF RADON SET IN PLACE BY APPLICABLE AUTHORITIES HAVING JURISDICTION SHALL BE USED TO DETERMINE IF A RADON MITIGATION SYSTEM SHALL BE REQUIRED.
- IF MITIGATION IS REQUIRED, QUALIFIED RADON MITIGATION PROFESSIONAL SHALL DETERMINE WHICH MITIGATION STRATEGY WILL BE MOST EFFECTIVE TO MAINTAIN LEVELS OF RADON BELOW MAXIMUM ALLOWABLE CONCENTRATION IN ALL APPLICABLE AREAS OF THE BUILDING.
- TESTING AFTER SYSTEM INSTALLATION SHALL BE PERFORMED TO CONFIRM LEVELS OF RADON ARE BELOW MAXIMUM ALLOWABLE CONCENTRATION IN ALL APPLICABLE AREAS OF THE BUILDING.
- IF SYSTEM DOES NOT MEET OR EXCEED THE MAXIMUM ALLOWABLE RADON CONCENTRATION, THE CONTRACTOR IS RESPONSIBLE FOR ACTIONS REQUIRED TO DETERMINE REASON(S) OF FAILURE AND PROVIDE THE REDESIGN AND/OR REINSTALLATION OF SYSTEM.
- CONTRACTOR TO ACKNOWLEDGE AND ACCEPT FULL RESPONSIBILTY THAT THE RESULTS OF ALL TESTING, DESIGN, AND INSTALATION PROVIDE ACCEPTABLE CONCENTRATIONS OF RADON IN THE BUILDING AND THE CONTRACT IS NOT COMPLETE UNTIL ALL TESTING REQUIREMENTS ARE MET.
- THESE DRAWINGS ARE TO BE REFERENCED FOR BUILDING LAYOUT, ANTICIPATED BUILDING AREAS EXPOSED TO SOILS, CHASE(S) AND OR SHAFT(S) COMMUNICATING BETWEEN LEVELS, AND COORDINATION FOR POTENTIAL FUTURE INTERIOR OR EXTERIOR DUCT ROUTING ONLY. NO ASSUMPTIONS ARE TO BE USED FROM THESE DRAWINGS IN THE DESIGN OF A RADON MITIGATION SYSTEM, CONTRACTOR TO CONFIRM ALL EXISTING CONDITIONS IN THE FIELD.
- PROVIDE PRICING FOR MOST LIKELY AND HIGHEST PERFORMING RADON MITIGATION SYSTEM.

## MECHANICAL LEGEND

PIPING SERVICES

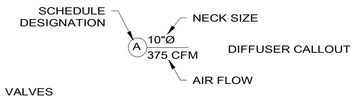
- LPG — PROPANE
- CD — CONDENSATE DRAIN
- RS — REFRIGERANT SUCTION
- RL — REFRIGERANT LIQUID

PIPING SYSTEM COMPONENTS

- ELBOW DOWN
- ELBOW UP
- TEE DOWN
- TEE UP
- CAP
- UNION
- FLEXIBLE CONNECTION
- DRAIN VALVE
- STRAINER
- (FS) FLOW SENSOR
- (S) TEMPERATURE & HUMIDISTAT SENSOR
- AUTOMATIC AIR CENT
- GAUGE
- THERMOMETER
- MANUAL AIR VENT
- P&T TAP
- IN LINE PUMP
- (M) METER
- (T) THERMOSTAT
- (P) PRESSURE SENSOR
- $CO_2$ CARBON DIOXIDE SENSOR
- CO CARBON MONOXIDE SENSOR

HVAC SYSTEM COMPONENTS

- 18X12 RECTANGULAR DUCT
- 10"Ø ROUND DUCT
- MANUAL DAMPER
- M MOTORIZED DAMPER
- BD BACKDRAFT DAMPER
- RECT. SUPPLY DUCT
- RECT. RETURN DUCT
- ECCENTRIC REDUCER
- DN DUCT SLOPE
- SUPPLY DIFFUSER/RETURN GRILLE
- SUPPLY DIFFUSER W/ FLEX CONNECTION
- SLOT DIFFUSER
- FIRE/SMOKE DAMPER (F/S)
  SMOKE DAMPER (SD)




VALVES

- GATE VALVE
- BALL VALVE
- PLUG VALVE
- CHECK VALVE
- PRESSURE REDUCING VALVE
- SAFETY RELIEF VALVE
- BACK FLOW PREVENTER
- BALANCING VALVE
- BUTTERFLY VALVE

DRAWING NOTATION

- NEW TO EXISTING
- NEW WORK
- EXISTING ITEMS
- EXISTING ITEMS TO BE DEMOLISHED

## MECHANICAL ABBREVIATIONS

| | |
|---|---|
| AC | AIR CONDITIONER |
| ADJ | ADJUSTABLE |
| AFF | ABOVE FINISHED FLOOR |
| AHJ | AUTHORITY HAVING JURISDICTION |
| AHU | AIR HANDLING UNIT |
| BAS | BUILDING AUTOMATION SYSTEM |
| BTU | BRITISH THERMAL UNIT |
| BLDG | BUILDING |
| BTUH | BRITISH THERMAL UNIT PER HOUR |
| CA | COMPRESSED AIR |
| CD | CONDENSATE DRAIN |
| CFH | CUBIC FEET PER HOUR |
| CFM | CUBIC FEET PER MINUTE |
| CU | CONDENSING UNIT |
| CUH | CABINET UNIT HEATER |
| (D) | DEMOLISH |
| DIA | DIAMETER |
| DN | DOWN |
| DX | DIRECT EXPANSION |
| (E) | EXISTING |
| EA | EXHAUST AIR |
| FFA | FROM FLOOR ABOVE |
| FFB | FROM FLOOR BELOW |
| FD | FIRE DAMPER |
| FPM | FEET PER MINUTE |
| F/S | COMBINATION FIRE+SMOKE DAMPER |
| GPM | GALLONS PER MINUTE |
| HWR | HEATING WATER RETURN |
| HWS | HEATING WATER SUPPLY |
| I.D. | INSIDE DIAMETER |
| LVR | LOUVER |
| MAX | MAXIMUM |
| MIN | MINIMUM |
| (N) | NEW |
| NIC | NOT IN CONTRACT |
| N.O. | NORMALLY OPEN |
| N.C. | NORMALLY CLOSED |
| OA | OUTSIDE AIR |
| O.C. | ON CENTER |
| O.D. | OUTSIDE DIAMETER |
| PSIA | POUNDS PER SQUARE INCH (ABSOLUTE) |
| PSIG | POUNDS PER SQUARE INCH (GAUGE) |
| (R) | RELOCATED |
| RA | RETURN AIR |
| RE:__ | REFERENCE ____ |
| REQ | REQUIRED/REQUIREMENT |
| RFA | RELIEF AIR |
| RL | REFRIGERANT LIQUID |
| RS | REFRIGERANT SUCTION |
| SA | SUPPLY AIR |
| SF | SQUARE FEET |
| SL | SEA LEVEL |
| S.S. | STAINLESS STEEL |
| STD | STANDARD |
| TA | TRANSFER AIR |
| TYP | TYPICAL |
| UNO | UNLESS NOTED OTHERWISE |
| VA | VENTILATION AIR |

## MECHANICAL SHEET LIST



| NUMBER | SHEET NAME |
|---|---|
| M000 | MECHANICAL LEGENDS AND NOTES |
| MD100 | MECHANICAL DEMO PLAN - BASEMENT |
| MD101 | MECHANICAL DEMO PLAN - MAIN LEVEL |
| M100 | MECHANICAL PLAN - BASEMENT |
| M101 | MECHANICAL PLAN - MAIN LEVEL |
| M102 | MECHANICAL PLAN - ATTIC |
| M120 | MECHANICAL PIPING PLAN - BASEMENT |
| M121 | MECHANICAL PIPING PLAN - MAIN LEVEL |
| M300 | MECHANICAL SECTIONS |
| M400 | MECHANICAL ENLARGED PLAN - SUPPLY |
| M401 | MECHANICAL ENLARGED PLAN - RETURN |
| M500 | MECHANICAL DETAILS |
| M600 | MECHANICAL SCHEDULES |
| M800 | MECHANICAL CALCS |



**ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.**

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR 360 ENGINEERING GOLDEN, CO | DESIGNED: AF/BG/MD DRAWN: AF/MD TECH REVIEW: BG DATE: 5/29/2026 | SUB SHEET NO. M000 | TITLE OF SHEET: MECHANICAL LEGENDS AND NOTES — EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248; PMIS/PKG NO. 318539; SHEET 68 OF 142 |

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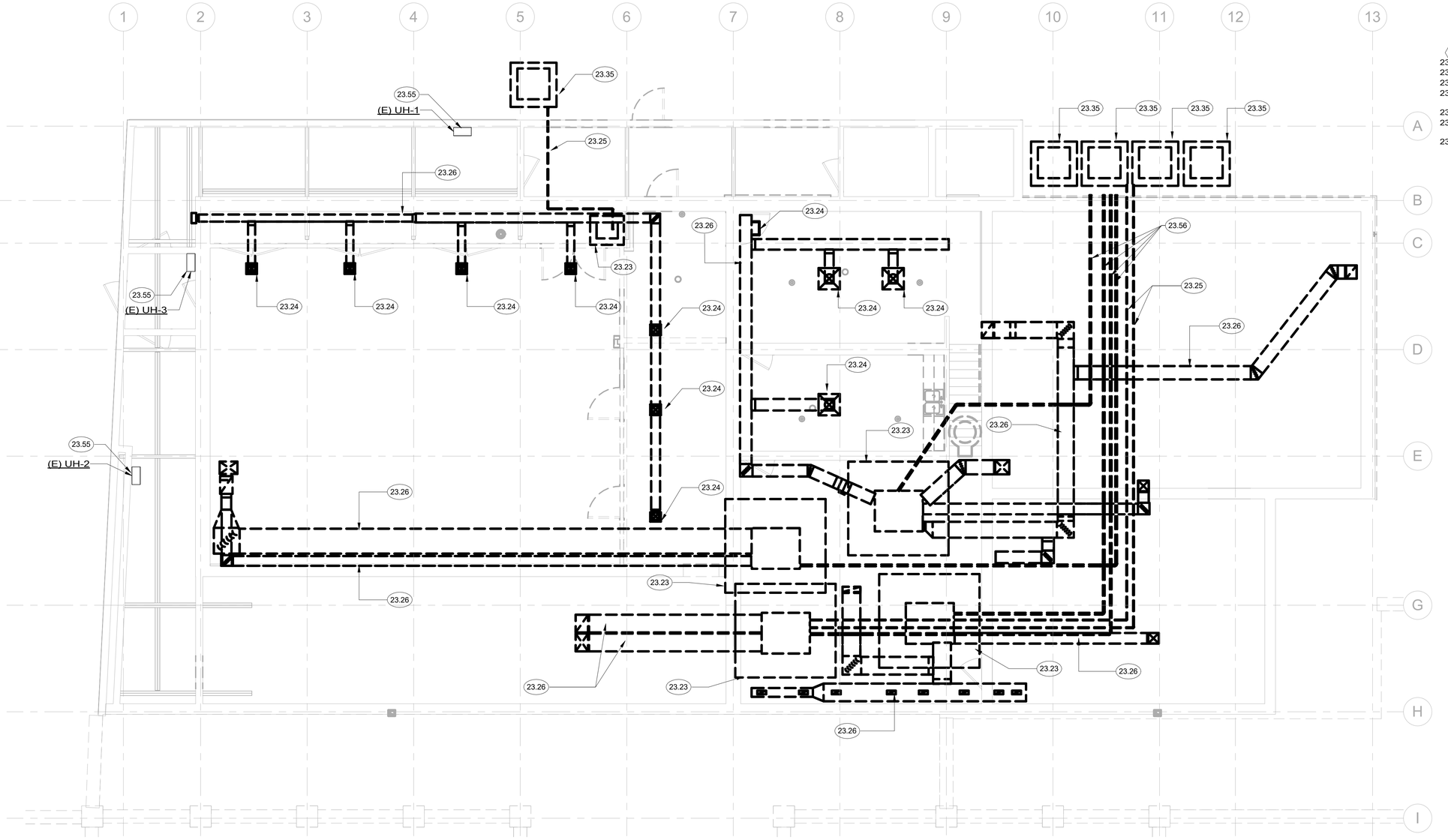
GENERAL NOTES:

1. EXISTING CONDITIONS ARE BASED ON PROVIDED AS-BUILT DRAWINGS. ACTUAL CONDITIONS MAY DIFFER FROM THESE DRAWINGS.
2. DEMO ALL EXISTING MECHANICAL EQUIPMENT, DUCTS AND MECHANICAL PIPING.
3. ALL EQUIPMENT, DUCTWORK, REFRIDGERANT PIPING, ETC. TO BE PROPERLY REMOVED AND DISPOSED/RECYCLED. RECOVER AND DISPOSE OF REFRIDGERANT PER CODE.




KEYNOTES:

| # | |
|---|---|
| 23.23 | DEMOLISH EXISTING PROPANE FIRED, SPLIT-SYSTEM FURNACE. |
| 23.24 | DEMOLISH EXISTING GRILLE/DIFFUSER AT THIS LOCATION. |
| 23.25 | DEMOLISH REFRIGERANT LIQUID AND SUCTION PIPING AS SHOWN. |
| 23.26 | DEMOLISH SUPPLY & RETURN DUCTS AS SHOWN. IF DUCTWORK GOES THROUGH FLOOR, PATCH FLOOR. |
| 23.35 | DEMOLISH EXISTING OUTDOOR SPLIT-SYSTEM FURNACE. |
| 23.55 | EXISTING UNIT HEATER TO REMAIN. REMOVE DURING DEMOLITION AND REINSTALL DURING CONSTRUCTION. |
| 23.56 | DEMOLISH EXISTING EXHAUST PVC & COMBUSTION AIR PIPES. |

**ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.**

1 MD100 MECHANICAL DEMO PLAN - BASEMENT
SCALE B

SCALE B 8 0 4 8
SCALE OF FEET

DENISE M. DIHLE P26897 5/29/26 PROFESSIONAL ENGINEER LICENSED IOWA

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|---|---|---|---|---|
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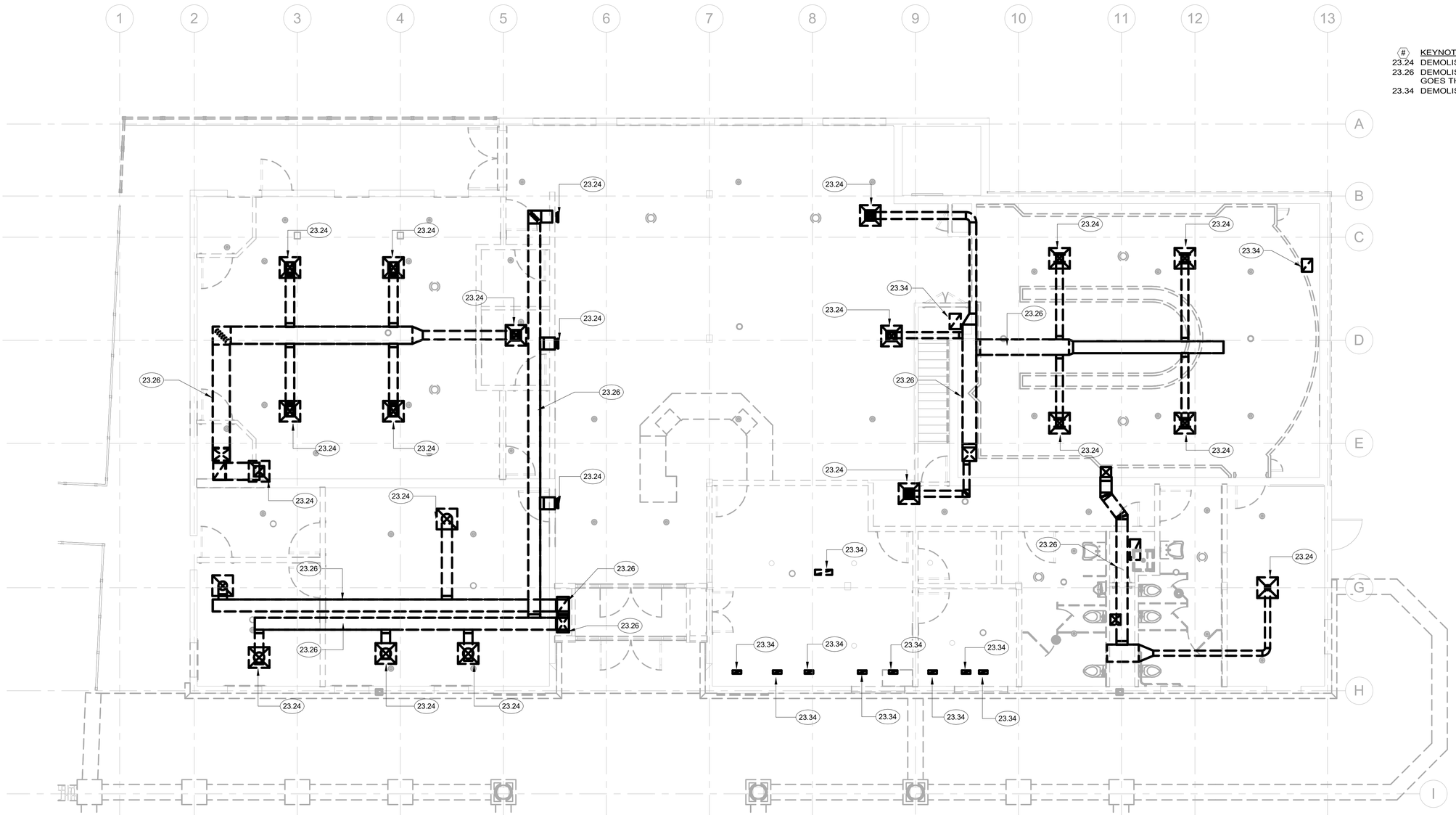
GENERAL NOTES:

1. EXISTING CONDITIONS ARE BASED ON PROVIDED AS-BUILT DRAWINGS. ACTUAL CONDITIONS MAY DIFFER FROM THESE DRAWINGS.
2. DEMO ALL EXISTING MECHANICAL EQUIPMENT, DUCTS AND MECHANICAL PIPING.

(#) KEYNOTES:

23.24 DEMOLISH EXISTING GRILLE/DIFFUSER AT THIS LOCATION.
23.26 DEMOLISH SUPPLY & RETURN DUCTS AS SHOWN. IF DUCTWORK GOES THROUGH FLOOR, PATCH FLOOR.
23.34 DEMOLISH FLOOR GRILLES AND PATCH FLOOR.




ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.

1 MECHANICAL PLAN DEMO - MAIN LEVEL
MD101 SCALE B

SCALE B 8 0 4 8
SCALE OF FEET

DENISE M. DIKLE P26897 5/29/26 PROFESSIONAL ENGINEER LICENSED IOWA

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|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR 360 ENGINEERING GOLDEN, CO | DESIGNED: AF/BG/MD CADD AF/MD TECH REVIEW: BG DATE: 5/29/2026 | SUB SHEET NO. MD101 | TITLE OF SHEET: MECHANICAL DEMO PLAN - MAIN LEVEL<br>EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248<br>PMIS/PKG NO. 318539<br>SHEET 70 OF 142 |

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GENERAL NOTES:

1. STACK DUCTS AND PIPES THROUGH FOUNDATION WALLS.
2. THERMOSTAT HEIGHTS SHALL BE 48" AFF TO MATCH LIGHTSWITCH HEIGHTS AND INSTALLED TO MEET ICC A117.1.
3. REFER TO M-000 FOR RADON MITIGATION DESIGN-BUILD REQUIREMENTS.

(#) KEYNOTES:

23.27 INSTALL NEW AIR HANDLING UNIT WITHIN MECHANICAL ROOM. ROUTE 4" COMBUSTION AIR AND 4" FLUE VENT THROUGH CRAWLSPACE TO CONCENTRIC VENT KITS AT BUILDING EXTERIOR WALL AS SHOWN. SEE DETAIL 5 ON M500 FOR MORE DETAILS. STACK SUPPLY AND RETURN DUCTS AT FOUNDATION WALL PENETRATIONS. STACK SUPPLY AND RETURN DUCTS AT FOUNDATION WALL PENETRATIONS.

23.29 INSTALL NEW OUTSIDE AIR LOUVER WITHIN EXTERIOR WALL.

23.30 INSTALL NEW RELIEF AIR LOUVER WITHIN EXTERIOR WALL.

23.31 INSTALL NEW CABINET UNIT HEATER RECESSED WITHIN CEILING.

23.36 ELEVATOR AND ELEVATOR MACHINE ROOM EQUIPMENT TO BE SERVICED AND INSPECTED. ANY DEFICIENCIES OR ISSUES TO BE REPORTED TO CONTRACTING OFFICER.

23.37 PIPES SHALL BE SPACED EXACTLY 4" APART PER MANUFACTURE'S RECOMMENDATION. SEE DETAIL 5 ON M500 FOR MORE DETAILS.

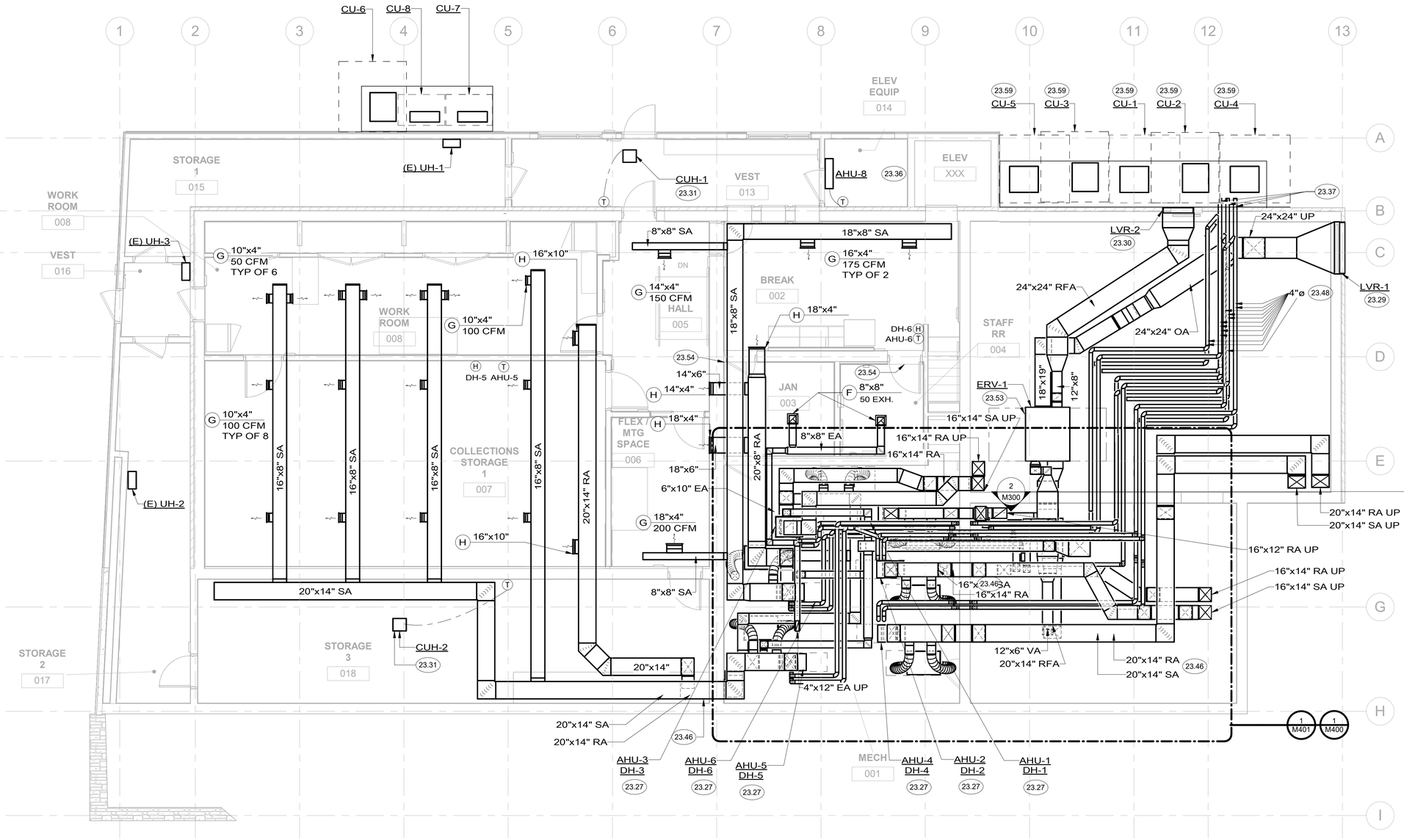
23.46 RETURN AIR DUCT STACKED UNDERNEATH SUPPLY AIR DUCT.

23.48 PIPES STACKED FOR CLARITY. EACH ARROWHEAD REPRESENTS A PIPE. QTY: OF 12

23.53 INSTALL ERV-1 IN CRAWLSPACE. MAINTAIN MANUFACTURER'S CLEARANCES.

23.54 1/2" UNDERCUT DOOR.

23.59 CONDENSING UNITS FRONT SHALL BE STAGGERED BY 90 DEGREES TO REDUCE TOTAL FOOTPRINT AND STILL BE INSTALLED PER MANUFACTURE REQUIREMENTS.




1 M100 MECHANICAL PLAN - BASEMENT
SCALE B

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.

SCALE B 8 0 4 8
SCALE OF FEET

DENISE M. DIHLE P28807 5/29/26 IOWA LICENSED PROFESSIONAL ENGINEER

| | | | | | |
|---|---|---|---|---|---|
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GENERAL NOTES:

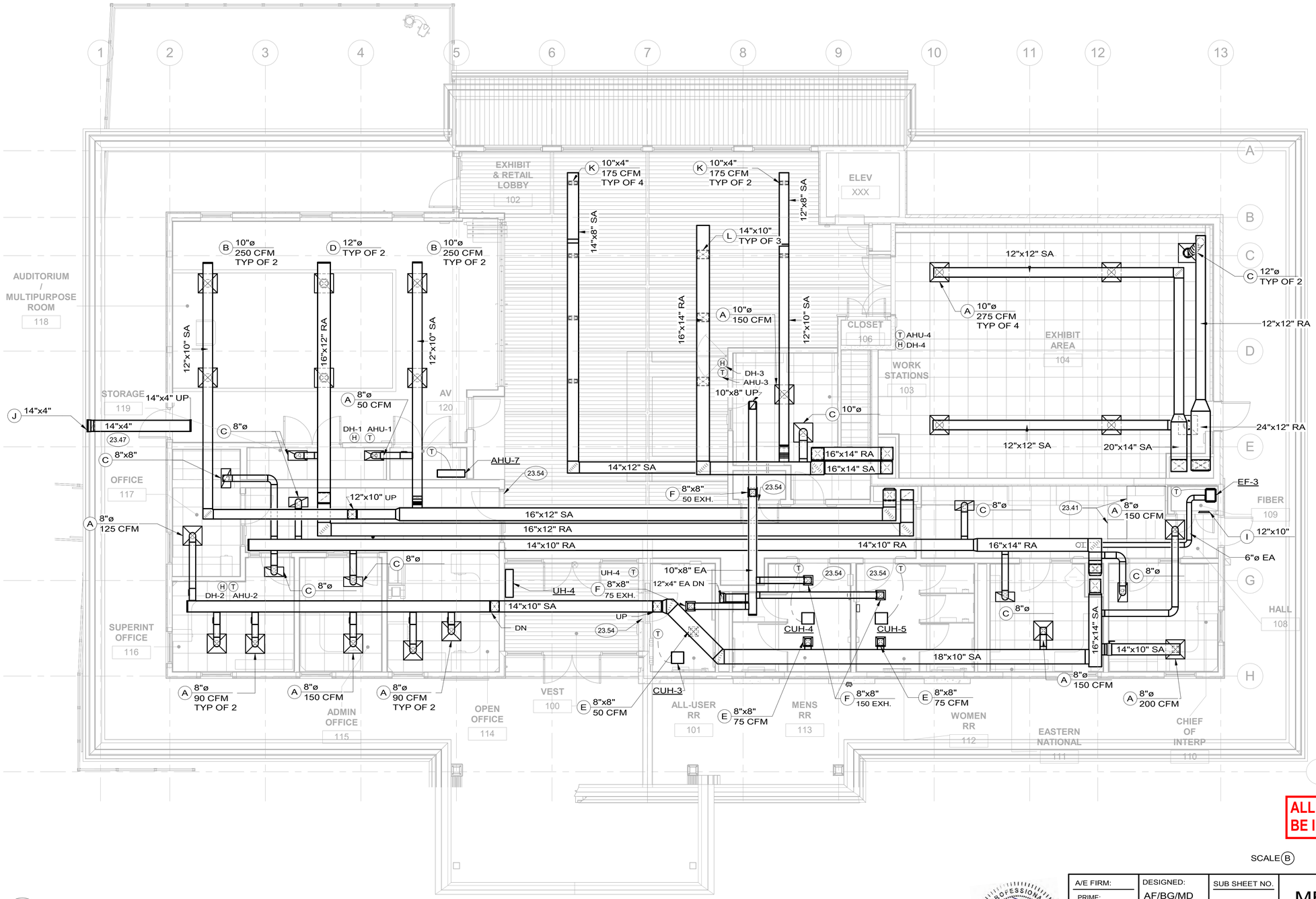
1. ALL DUCTWORK SHOWN ON M-101 IS TO BE ABOVE CEILING AND IN ATTIC SPACE. DUCTWORK IS SHOWN ON THIS PLAN FOR CLARITY.

(#) KEYNOTES:

23.41 ATTIC ACCESS PANEL. RE. ARCHITECTURAL DRAWINGS FOR MORE DETAILS.

23.47 EXHAUST DUCT TO RUN PARALLEL TO ROOF AND IN-BETWEEN TRUSSES. DUCT TO TERMINATE TO EXHAUST GRILLE IN THE BOTTOM OF THE DUCT. REFER TO M100 FOR CONTINUATION.

23.54 1/2" UNDERCUT DOOR.




ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.

SCALE B 8 0 4 8 SCALE OF FEET

1 M101 MECHANICAL PLAN - MAIN LEVEL
SCALE B

| | | | | |
|---|---|---|---|---|
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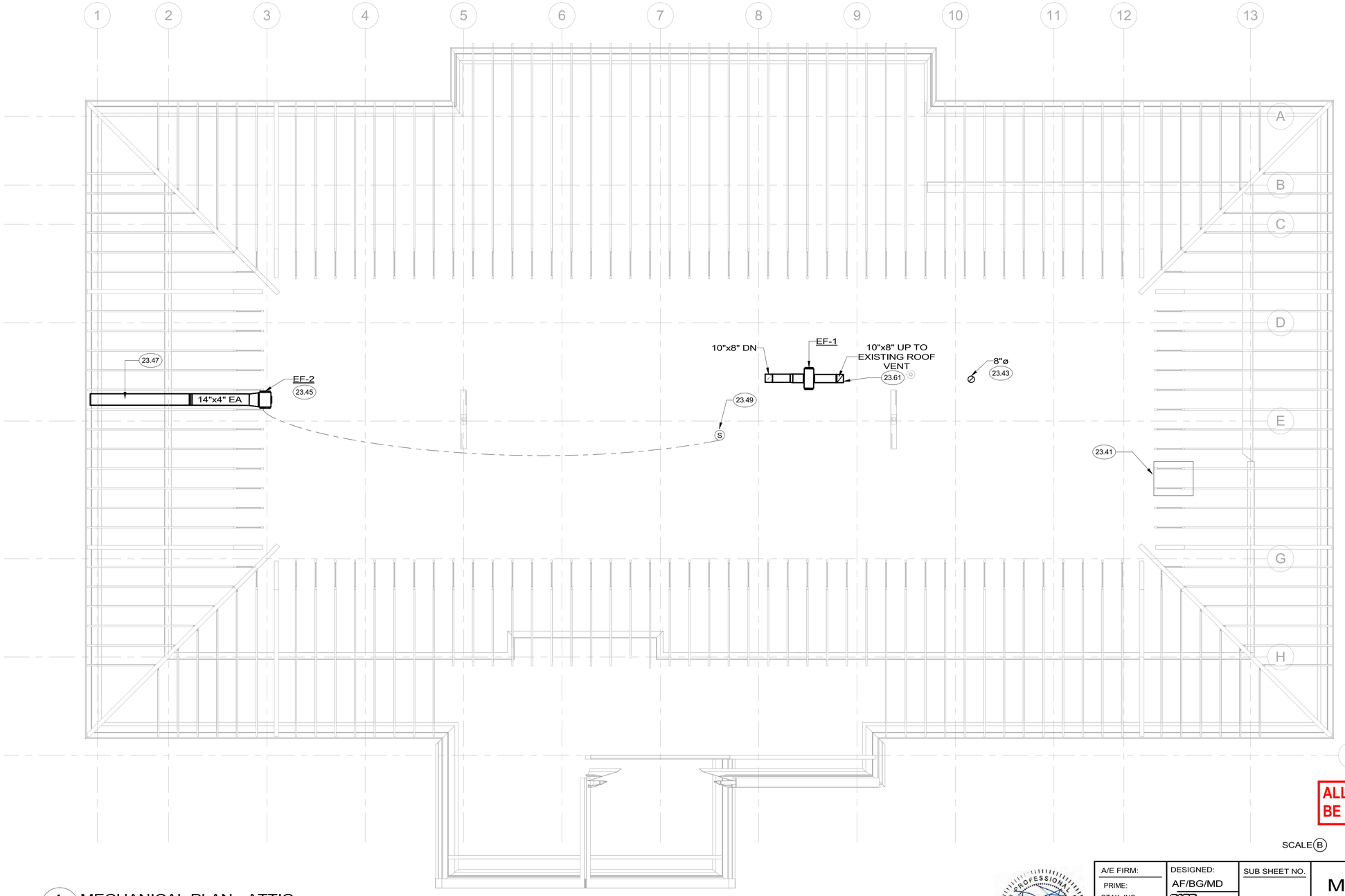


1 M102 MECHANICAL PLAN - ATTIC
SCALE B

KEYNOTES:

| # | |
|---|---|
| 23.41 | ATTIC ACCESS PANEL. RE. ARCHITECTURAL DRAWINGS FOR MORE DETAILS. |
| 23.43 | TO EXISTING ROOF VENT FOR ATTIC VENTILATION. |
| 23.45 | ATTIC VENTILATION FAN SUSPENDED WITHIN ATTIC, CONTROL VIA HUMIDITY AND TEMPERATURE SENSING THERMOSTAT. ACTIVATE FAN WHEN HUMIDITY RISES ABOVE 60% OR ATTIC TEMPERATURE IS ABOVE 100 DEGREES. |
| 23.47 | EXHAUST DUCT TO RUN PARALLEL TO ROOF AND IN-BETWEEN TRUSSES. DUCT TO TERMINATE TO EXHAUST GRILLE IN THE BOTTOM OF THE DUCT. REFER TO M100 FOR CONTINUATION. |
| 23.49 | COMBO TEMPERATURE & HUMIDISTAT SENSOR TO BE SUSPENDED IN ATTIC. |
| 23.61 | EXISTING ROOF VENT TO BE CLEANED. RE. ARCHITECTURAL DRAWINGS FOR MORE DETAIL. |

**ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.**

SCALE B
8 0 4 8
SCALE OF FEET

DENISE M. DIHLE P26897 5/29/26 PROFESSIONAL ENGINEER IOWA

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|---|---|---|---|
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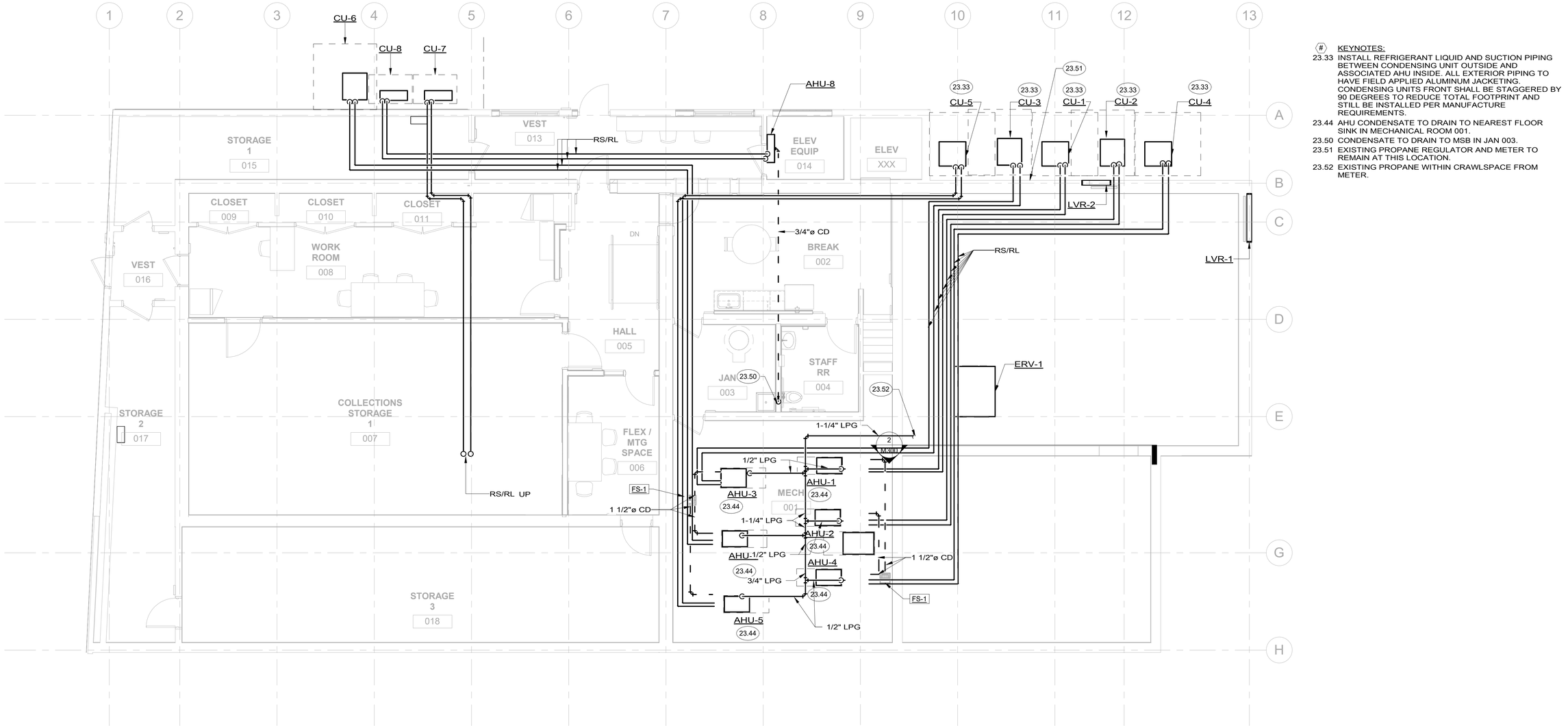


(#) KEYNOTES:

23.33 INSTALL REFRIGERANT LIQUID AND SUCTION PIPING BETWEEN CONDENSING UNIT OUTSIDE AND ASSOCIATED AHU INSIDE. ALL EXTERIOR PIPING TO HAVE FIELD APPLIED ALUMINUM JACKETING. CONDENSING UNITS FRONT SHALL BE STAGGERED BY 90 DEGREES TO REDUCE TOTAL FOOTPRINT AND STILL BE INSTALLED PER MANUFACTURE REQUIREMENTS.

23.44 AHU CONDENSATE TO DRAIN TO NEAREST FLOOR SINK IN MECHANICAL ROOM 001.

23.50 CONDENSATE TO DRAIN TO MSB IN JAN 003.

23.51 EXISTING PROPANE REGULATOR AND METER TO REMAIN AT THIS LOCATION.

23.52 EXISTING PROPANE WITHIN CRAWLSPACE FROM METER.

1 / M120 MECHANICAL PIPING PLAN - BASEMENT
SCALE B

**ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.**

SCALE B 8 0 4 8
SCALE OF FEET

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DENISE M. DIKLE P26897 5/29/26 IOWA LICENSED PROFESSIONAL ENGINEER

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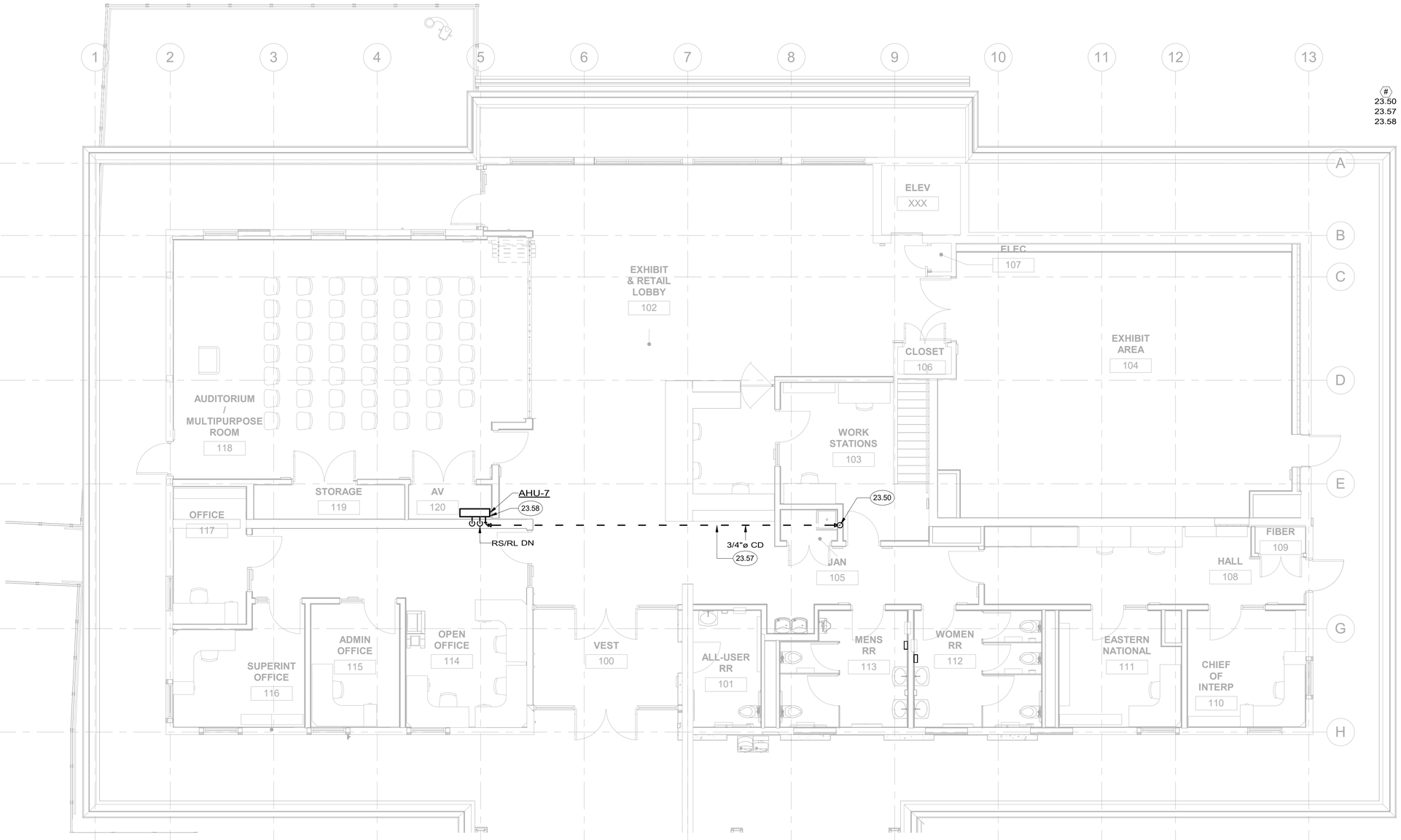



(#) KEYNOTES:

23.50 CONDENSATE TO DRAIN TO MSB IN JAN 003.

23.57 CONDENSATE TO ROUTE ABOVE CEILING.

23.58 INSTALL REFRIGERANT LIQUID AND SUCTION PIPING BETWEEN CONDENSING UNIT OUTSIDE AND ASSOCIATED AHU INSIDE. ALL EXTERIOR PIPING TO HAVE FIELD APPLIED ALUMINUM JACKETING.

**ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.**

1 / M121 MECHANICAL PIPING PLAN - MAIN LEVEL

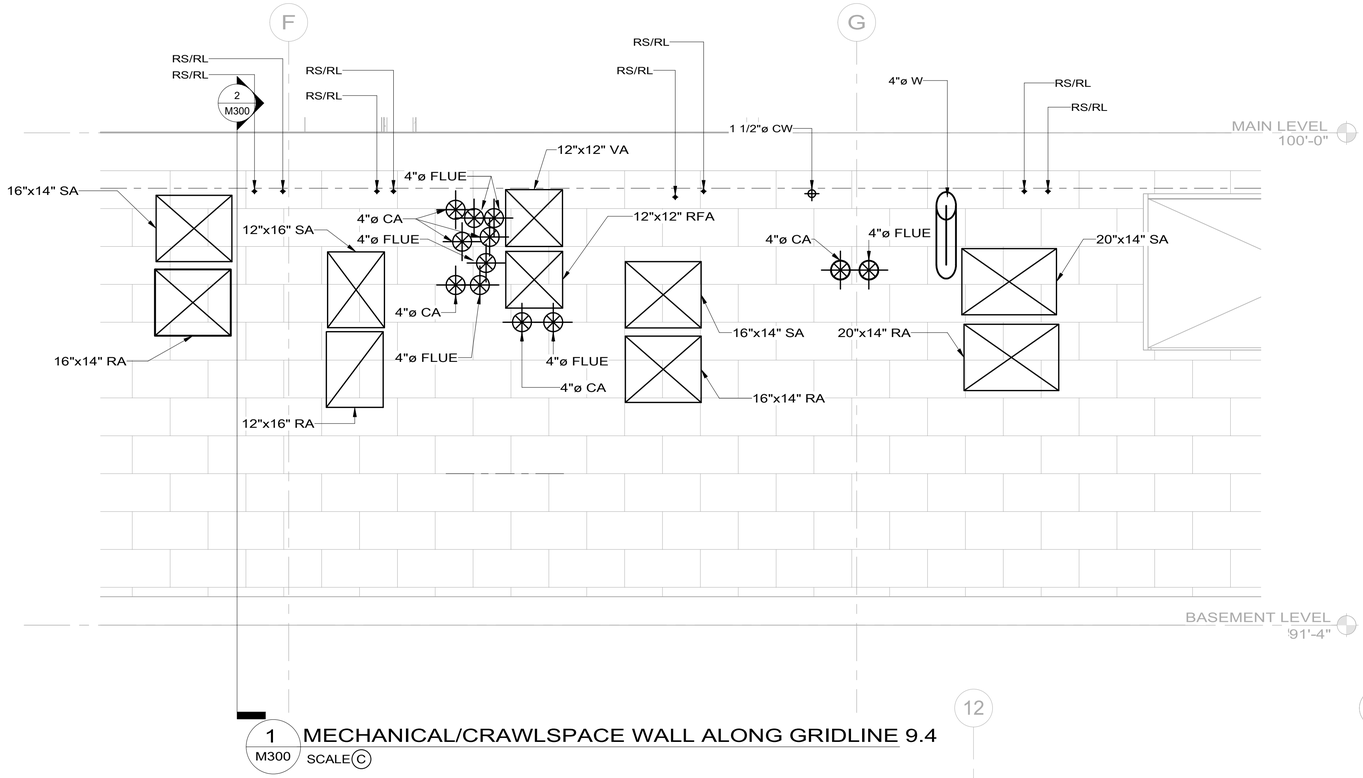
SCALE (B)

SCALE (B) 8 0 4 8

SCALE OF FEET

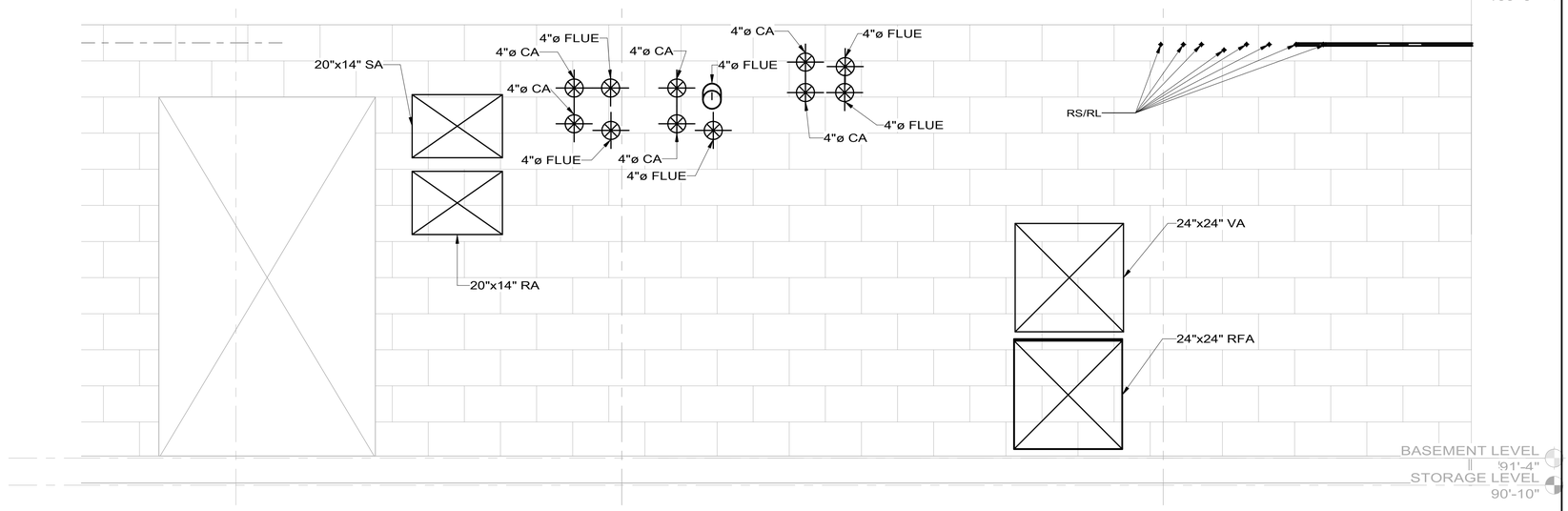
LICENSED PROFESSIONAL ENGINEER
DENISE M. DIHLE
P26897
5/29/26
IOWA

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA<br>SUBCONTRACTOR 360 ENGINEERING GOLDEN, CO | DESIGNED: AF/BG/MD<br>CADD AF/MD<br>TECH REVIEW: BG<br>DATE: 5/29/2026 | SUB SHEET NO. M121 | TITLE OF SHEET: MECHANICAL PIPING PLAN - MAIN LEVEL<br>EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248<br>PMIS/PKG NO. 318539<br>SHEET 75 OF 142 |




1 M300 MECHANICAL/CRAWLSPACE WALL ALONG GRIDLINE 9.4

SCALE C




2 M300 CRAWLSPACE WALL ALONG GRIDLINE F

SCALE C

SCALE C 1.5 0 1.5 3

SCALE OF FEET

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.

PROFESSIONAL ENGINEER — DENISE M. DIHLE P26897 — IOWA — 5/29/26

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA<br>SUBCONTRACTOR 360 ENGINEERING GOLDEN, CO | DESIGNED: AF/BG/MD<br>CADD AF/MD<br>TECH REVIEW: BG<br>DATE: 5/29/2026 | SUB SHEET NO. M300 | TITLE OF SHEET MECHANICAL SECTIONS<br>EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248<br>PMIS/PKG NO. 318539<br>SHEET 76 OF 142 |

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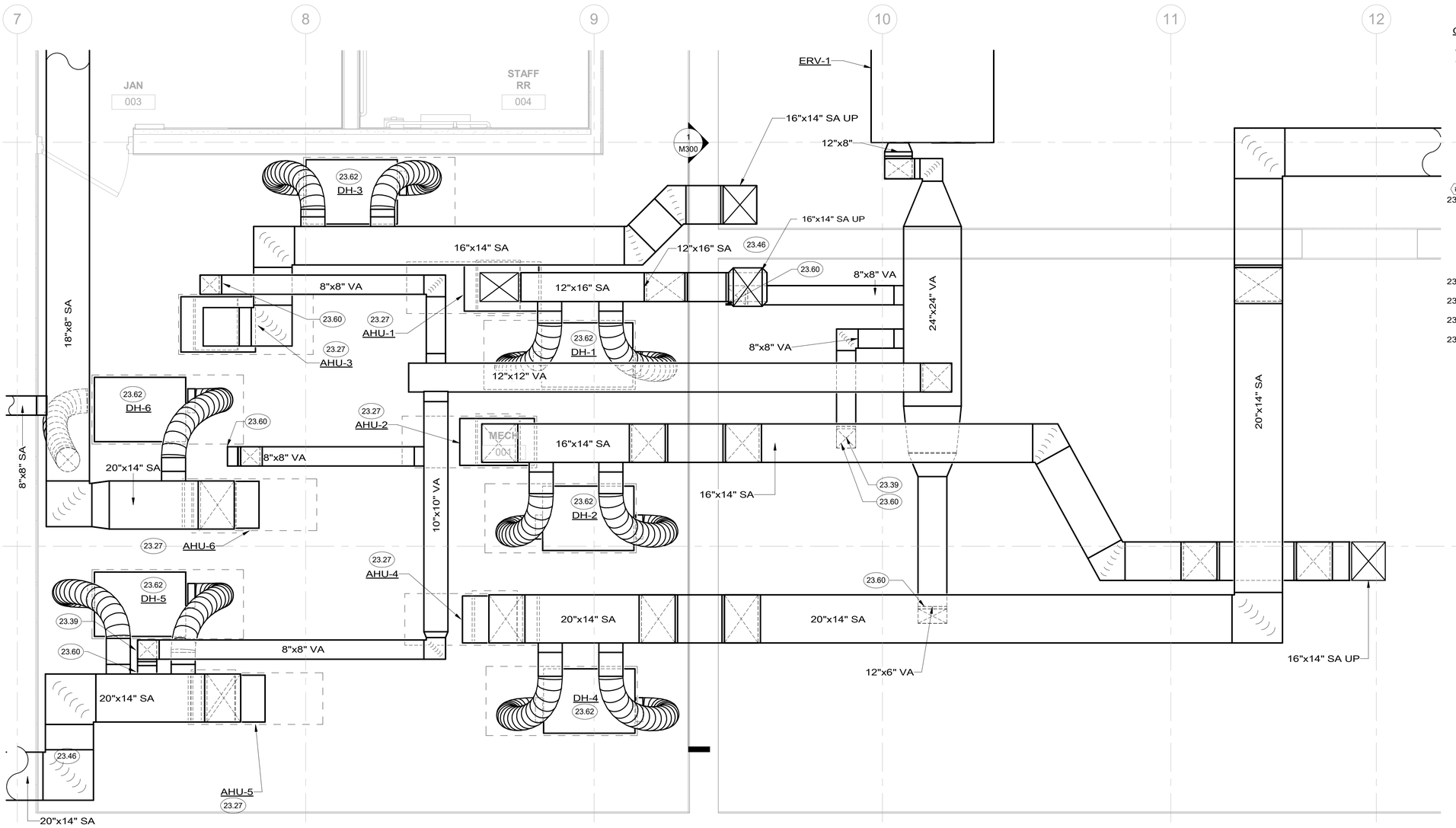


GENERAL NOTES:

1. STACK ALL DUCTS THROUGH LOAD BEARING WALLS TO LIMIT WALL PRENTRATIONS. SEE DETAIL 5 ON M500 FOR MORE DETAILS ON AHU DUCTWORK.

(#) KEYNOTES:

23.27 INSTALL NEW AIR HANDLING UNIT WITHIN MECHANICAL ROOM. ROUTE 4" COMBUSTION AIR AND 4" FLUE VENT THROUGH CRAWLSPACE TO CONCENTRIC VENT KITS AT BUILDING EXTERIOR WALL AS SHOWN. SEE DETAIL 5 ON M500 FOR MORE DETAILS. STACK SUPPLY AND RETURN DUCTS AT FOUNDATION WALL PENETRATIONS. STACK SUPPLY AND RETURN DUCTS AT FOUNDATION WALL PENETRATIONS.

23.39 INSTALL VERTICAL VOLUME DAMPER ON DUCT BEFORE MAIN.

23.46 RETURN AIR DUCT STACKED UNDERNEATH SUPPLY AIR DUCT.

23.60 VENTILATION TO CONNECT INTO RETURN DUCT, SEE DETAIL 4 ON M-500 FOR MORE DETAILS.

23.62 INSTALL NEW DEHUMIDIFIER ON GROUND NEXT TO ASSOCIATED AIR HANDLING UNIT. INSTALL WITH 10" SUPPLY AND RETURN CONNECTIONS BETWEEN DEHUMIDIFIER AND AHU SUPPLY DUCT MAIN.

1 M400 MECHANICAL PLAN - BASEMENT - ENLARGED MECHANICAL ROOM - SUPPLY AND VENTILATION

SCALE D

**ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.**

SCALE D 2 0 2 4 SCALE OF FEET

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR 360 ENGINEERING GOLDEN, CO | DESIGNED: AF/BG/MD CADD AF/MD TECH REVIEW: BG DATE: 5/29/2026 | SUB SHEET NO. M400 | TITLE OF SHEET MECHANICAL ENLARGED PLAN - SUPPLY EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 77 OF 142 |

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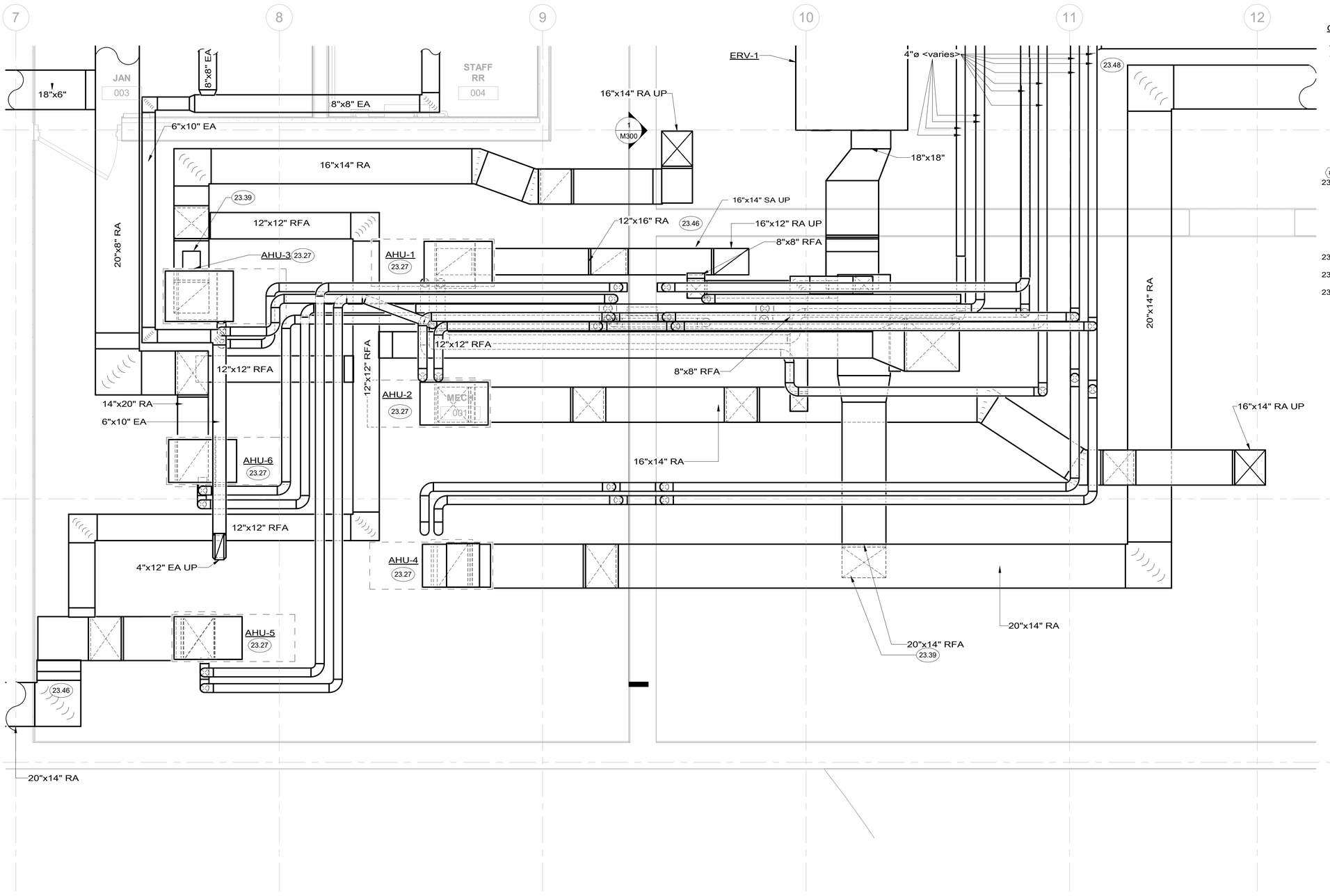


GENERAL NOTES:

1. STACK ALL DUCTS THROUGH LOAD BEARING WALLS TO LIMIT WALL PRENTRATIONS. SEE DETAIL 5 ON M500 FOR MORE DETAILS ON AHU DUCTWORK.

(#) KEYNOTES:

23.27 INSTALL NEW AIR HANDLING UNIT WITHIN MECHANICAL ROOM. ROUTE 4" COMBUSTION AIR AND 4" FLUE VENT THROUGH CRAWLSPACE TO CONCENTRIC VENT KITS AT BUILDING EXTERIOR WALL AS SHOWN. SEE DETAIL 5 ON M500 FOR MORE DETAILS. STACK SUPPLY AND RETURN DUCTS AT FOUNDATION WALL PENETRATIONS. STACK SUPPLY AND RETURN DUCTS AT FOUNDATION WALL PENETRATIONS.

23.39 INSTALL VERTICAL VOLUME DAMPER ON DUCT BEFORE MAIN.

23.46 RETURN AIR DUCT STACKED UNDERNEATH SUPPLY AIR DUCT.

23.48 PIPES STACKED FOR CLARITY. EACH ARROWHEAD REPRESENTS A PIPE. QTY: OF 12

1 M401 MECHANICAL PLAN - BASEMENT - ENLARGED MECHANICAL ROOM - RETURN AND EXHAUST

SCALE D

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.

SCALE D 2 0 2 4

SCALE OF FEET

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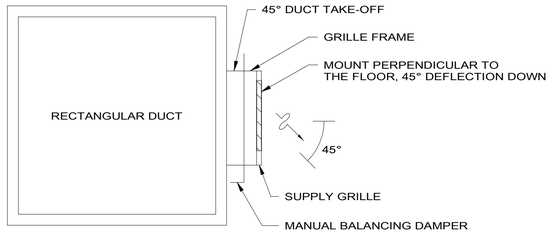
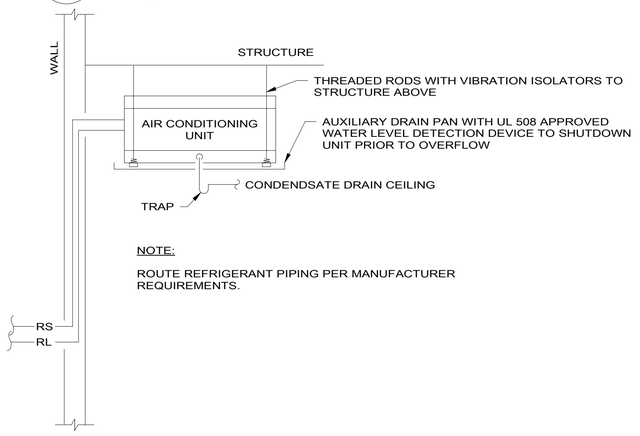
LICENSED PROFESSIONAL ENGINEER
DENISE M. DIHLE
P26897
5/29/26
IOWA

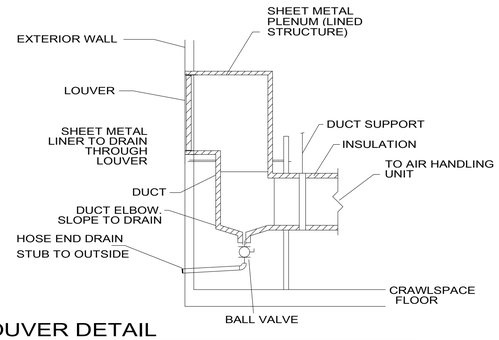
| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR 360 ENGINEERING GOLDEN, CO | DESIGNED: AF/BG/MD CADD AF/MD TECH REVIEW: BG DATE: 5/29/2026 | SUB SHEET NO. M401 | TITLE OF SHEET MECHANICAL ENLARGED PLAN - RETURN EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 78 OF 142 |



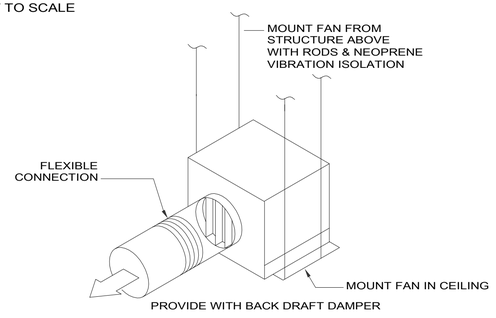

## 8 M500 RECTANGULAR DUCT GRILLE CONNECTION DETAIL
SCALE A



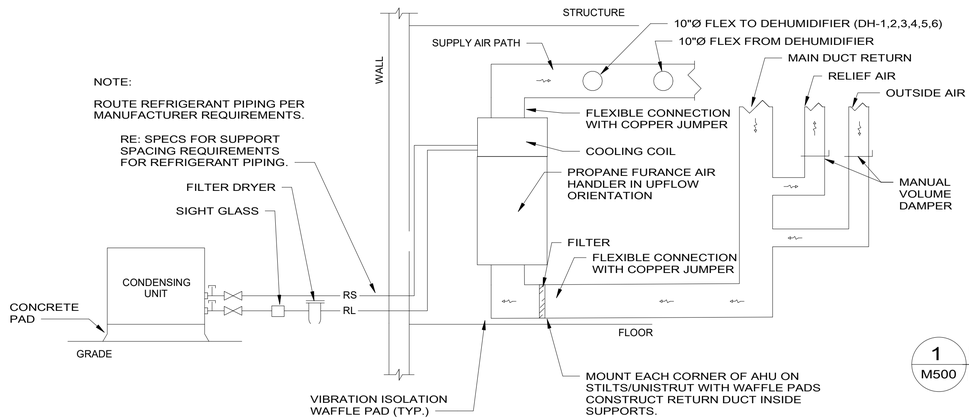

## 9 M500 SPLIT SYSTEM DETAIL
NOT TO SCALE



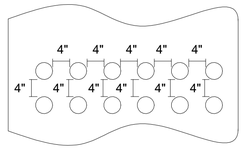

## 10 M500 LOUVER DETAIL
NOT TO SCALE



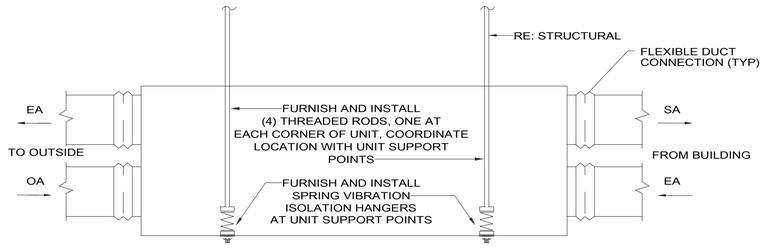

## 11 M500 CEILING MOUNTED EXHAUST FAN DETAIL
NOT TO SCALE




## 4 M500 SPLIT SYSTEM DETAIL
NOT TO SCALE

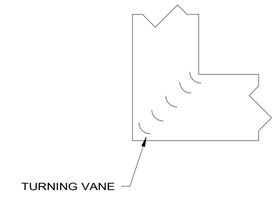



## 5 M500 CONCENTRIC VENT PENERTRATION SPACING DETAIL
NOT TO SCALE



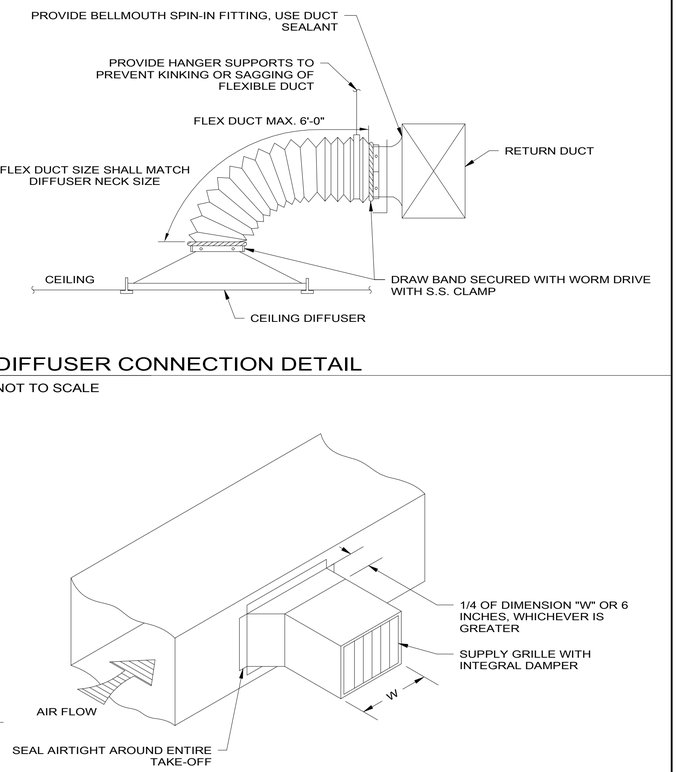

NOTES:
PROVIDE MINIMUM OF 36" CLEARANCE ON CONTROL PANEL SIDE OF UNIT TO ACCESS FILTERS, MOTORS, ETC.

## 6 M500 ERV UNIT DETAIL
NOT TO SCALE




NOTE: VANES ARE SUPPORTED BY SEPERATE FRAME BOLTED OR RIVETED TO ELBOW.

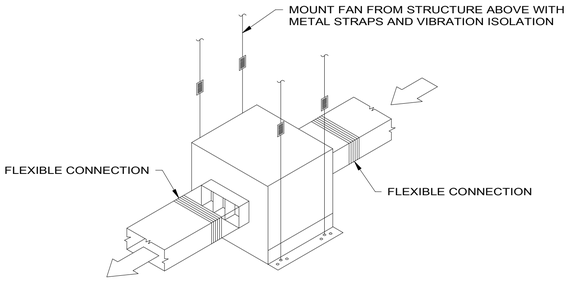
## 7 M500 TURNING VANE DETAIL
NOT TO SCALE




## 1 M500 DIFFUSER CONNECTION DETAIL
NOT TO SCALE




## 2 M500 RECTANGULAR DUCT MOUNTED GRILLE TAKE-OFF DETAIL
NOT TO SCALE




## 3 M500 INLINE FAN DETAIL
NOT TO SCALE

**ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.**

DENISE M. DIHLE P26897 5/29/26 PROFESSIONAL ENGINEER IOWA

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA; SUBCONTRACTOR 360 ENGINEERING GOLDEN, CO | DESIGNED: AF/BG/MD; CADD: AF/MD; TECH REVIEW: BG; DATE: 5/29/2026 | SUB SHEET NO. M500 | TITLE OF SHEET: MECHANICAL DETAILS; EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248; PMIS/PKG NO. 318539; SHEET 79 OF 142 |

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## ENERGY RECOVERY VENTILATION UNIT SCHEDULE

| GENERAL TAG | MANUFACTURER | MODEL | LOCATION | AIRFLOWS SUPPLY [CFM] | EXHAUST [CFM] | ESP [IN. W.C.] | SPEED [RPM] | POWER [W/CFM] | SIZE [HP] | ERV SUMMER PERFORMANCE OA DB [°F] | OA WB [°F] | RA DB [°F] | RA WB [°F] | SA DB [°F] | SA WB [°F] | ERV WINTER PERFORMANCE OA DB [°F] | OA WB [°F] | RA DB [°F] | RA WB [°F] | SA DB [°F] | SA WB [°F] | ELECTRICAL VOLTAGE [V] | PHASE | FREQ. [Hz] | MCA [A] | MOCP [A] | PHYSICAL LENGTH [IN] | WIDTH [IN] | HEIGHT [IN] | WEIGHT [LB] | NOTES |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ERV-1 | GREENHECK | ERV-20-15L | CRAWLSPACE | 1105 | 725 | 0.5 | 1725 | .498 | 3/4 | 91.0 | 75.0 | 75.0 | 59.5 | 80.9 | 67.8 | -10.0 | -11.1 | 72.0 | 55.7 | 38.5 | 34.4 | 230 | 1 | 60 | 14.2 | 20 | 62 | 51 | 34.4 | 577 | 1,2,3,4 |

NOTES
1. PROVIDE WITH MERV 13 FILTER.
2. PROVIDE WITH STANDALONE UNIT CONTROL. INSTALL CONTROL INTERFACE WITHIN MECHANICAL ROOM, COORDINATE OCCUPANCY SCHEDULE WITH BUILDING MANAGER CONTRACTING OFFICER.
3. DISCONNECT BY DIV 26.
4. PROVIDE WITH ELECTRIC PRE-HEATER FROST CONTROLS. ENABLE PRE-HEATER WHEN EXHAUST AIRFLOW TEMPERATURE IS BELOW 36°F.

## GRILLES, REGISTERS, AND DIFFUSERS SCHEDULE

| GENERAL TAG | MANUFACTURER | MODEL | SERVICE | MATERIAL | FACE SIZE | NOTES |
|---|---|---|---|---|---|---|
| A | PRICE | SCD | SUPPLY | ALUMINUM | 24"x24" | 1,5 |
| B | PRICE | SCD | SUPPLY | ALUMINUM | 24"x24" | 4,5 |
| C | PRICE | SCD | RETURN | ALUMINUM | REFER TO PLANS | 1 |
| D | PRICE | SCD | RETURN | ALUMINUM | REFER TO PLANS | 4 |
| E | PRICE | SCD | SUPPLY | ALUMINUM | 8"x8" | 4,5 |
| F | PRICE | SCD | EXHAUST | ALUMINUM | 12"x12" | 1 |
| G | PRICE | 510L | SUPPLY | ALUMINUM | REFER TO PLANS | 2,3 |
| H | PRICE | 510Z | RETURN | ALUMINUM | REFER TO PLANS | 3 |
| I | PRICE | STG | TRANSFER | ALUMINUM | 12"x10" | |
| J | PRICE | STG | EXHAUST | STEEL | 14"x4" | 7 |
| K | PRICE | 510L | SUPPLY | ALUMINUM | 10"x4" | 2,6 |
| L | PRICE | 510Z | RETURN | ALUMINUM | 14"x10" | 6 |

NOTES
1. PROVIDE WITH LAY IN CEILING MOUNTING FRAME.
2. OPPOSED BLADE DAMPER IN FACE OF GRILLE.
3. MOUNT ON SIDE OF DUCT WITH DUCT FRAME.
4. PROVIDE WITH GYPSUM BOARD MOUNTING FRAME FOR GYPSUM BOARD HARD LID CEILING/WALL - SEE PLAN.
5. PROVIDE WITH VC DAMPER.
6. PROVIDE WITH MOUNTING FRAME FOR WOODEN SLATS. INSTALL FRAME PARALLEL TO WOODEN SLATS.
7. PROVIDE GRILLE WITH COATING FOR EXTERIOR USE.

## DEHUMIDIFIER SCHEDULE

| GENERAL TAG | MANUFACTURER | MODEL | DESCRIPTION | SERVES | WEIGHT [LBS] | NOTES |
|---|---|---|---|---|---|---|
| DH-1 | SANTA FE | ULTRA V205 | DUCTED | AHU-1 | 143 | 1,2,3,4,5 |
| DH-2 | SANTA FE | ULTRA V205 | DUCTED | AHU-2 | 143 | 1,2,3,4,5 |
| DH-3 | SANTA FE | ULTRA V205 | DUCTED | AHU-3 | 143 | 1,2,3,4,5 |
| DH-4 | SANTA FE | ULTRA V205 | DUCTED | AHU-4 | 143 | 1,2,3,4,5 |
| DH-5 | SANTA FE | ULTRA V205 | DUCTED | AHU-5 | 143 | 1,2,3,4,5 |
| DH-6 | SANTA FE | ULTRA V205 | DUCTED | AHU-6 | 143 | 1,2,3,4,5 |

NOTES:
1. 6.3 PINTS/kWh MOISTURE REMOVAL EFFICIENCY.
2. 10" DUCT CONNECTIONS, INSTALL 10" FLEX DUCT INLET AND OUTLET TO AHU SUPPLY DUCT. MAX LENGTH 6'.
3. 1,115 WATTS, 120V, 60Hz, 10' CORD LENGTH FOR PLUG-IN RECEPTACLE.
4. 3/4" CONDENSATE DRAIN. INSTALL WITH TRAP AND ROUTE TO MECH ROOM FLOOR DRAIN PER MFG REQUIREMENTS.
5. CONTROLLED LOCALLY BY HUMIDISTAT IN SPACE TO ENABLE WHEN RH IS ABOVE 55% AND DISABLE WHEN RH IS BELOW 40% (ADJ.)

## ELECTRIC HEATER SCHEDULE

| GENERAL TAG | MANUFACTURER | MODEL | LOCATION | TYPE/MOUNTING | ELECTRICAL POWER [kW] | VOLTAGE [V] | PHASE | FREQ. [Hz] | FLA [A] | PHYSICAL LENGTH [IN] | WIDTH [IN] | HEIGHT [IN] | WEIGHT [LB] | NOTES |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CUH-1 | INDEECO | 931IF-F4004-N | VEST - 013 | CEILING RECESSED | 2 | 240 | 1 | 60 | 9.8 | 19 | 15 | 3-3/4 | 25 | 1,2,3,4 |
| CUH-2 | INDEECO | 931IF-F4007-N | STORAGE - 018 | CEILING RECESSED | 3 | 240 | 1 | 60 | 16.7 | 19 | 15 | 3-3/4 | 25 | 1,2,3,4 |
| CUH-3 | INDEECO | 931IF-F1500-N | RR - 101 | CEILING RECESSED | 1.5 | 120 | 1 | 60 | 7.5 | 19 | 15 | 3-3/4 | 25 | 1,2,3,4 |
| CUH-4 | INDEECO | 931IF-F1500-N | RR - 113 | CEILING RECESSED | 1.5 | 120 | 1 | 60 | 7.5 | 19 | 15 | 3-3/4 | 25 | 1,2,3,4 |
| CUH-5 | INDEECO | 931IF-F1500-N | RR - 112 | CEILING RECESSED | 1.5 | 120 | 1 | 60 | 7.5 | 19 | 15 | 3-3/4 | 25 | 1,2,3,4 |
| UH-4 | INDEECO | 922IC-35 | VEST - 100 | WALL MOUNTED | 2 | 240 | 1 | 60 | 9 | 35 | 9-3/4 | 26-3/8 | 25 | 1,2,3,5 |

NOTES
1. PROVIDE WITH FIELD INSTALLED LINE VOLTAGE, WALL-MOUNTED THERMOSTAT WITH COVER.
2. PROVIDE WITH INTEGRAL ELECTRICAL DISCONNECT.
3. PROVIDE WITH MANUFACTURER'S MOUNTING BRACKET AND HARDWARE.
4. PROVIDE WITH KIT FOR FULLY-RECESSED MOUNTING.
5. PROVIDE WITH KIT FOR WALL MOUNTING.

## FAN SCHEDULE

| GENERAL TAG | MANUFACTURER | MODEL | LOCATION | TYPE | PERFORMANCE (@ 700FT) AIRFLOW [CFM] | ESP [IN. W.C.] | SPEED [RPM] | SIZE [HP] | SOUND [SONES] | ELECTRICAL VOLTAGE [V] | PHASE | MCA [A] | MOCP [A] | FLA [A] | PHYSICAL LENGTH [IN.] | WIDTH [IN.] | HEIGHT [IN.] | WEIGHT [LB] | NOTES |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| EF-1 | GREENHECK | CSP-A700-VG | ATTIC | INLINE | 525 | 0.5 | 1164 | 0.14 | 1.6 | 115 | 1 | 5.1 | 15 | 4.1 | 12 | 24 | 12 | 39 | 1,2,3,4 |
| EF-2 | GREENHECK | CSP-A510-VG | ATTIC | INLINE | 200 | 0.5 | 1110 | 0.06 | 0.7 | 115 | 1 | 3.1 | 15 | 2.5 | 14 | 18 | 15 | 36 | 2,3,4,6 |
| EF-3 | GREENHECK | SP-B150 | 109 - FIBER | CEILING | 125 | 0.37 | 1050 | 0.07 | 3.5 | 115 | 1 | 2.2 | 15 | 1.8 | 12 | 14 | 7 | 10 | 2,3,5 |

NOTES
EXHAUST FAN
1. PROVIDE WITH TIMECLOCK, COORDINATE SCHEDULE WITH BUILDING CONTRACTING OFFICER.
2. PROVIDE WITH BACKDRAFT DAMPER.
3. UNIT WEIGHT DOES NOT INCLUDE ACCESSORIES.
4. PROVIDE SPEED CONTROL SWITCH FOR BALANCING ONLY.
5. CONTROLED BY THERMOSTAT WITH COVER.
6. CONTROLLED BY COMBO THERMOSTAT AND HUMIDSTAT WITH COVER.

## SPLIT SYSTEMS SCHEDULE - OUTDOOR UNIT

| GENERAL TAG | MANUFACTURER | MODEL | INDOOR UNIT | REFRIGERANT | DX COOLING NOMINAL COOLING [BTU/H] | COOLING COIL MODEL NUMBER [°F] | ELECTRICAL VOLTAGE [V] | PHASE | FREQ. [Hz] | MCA [A] | MOCP [A] | RLA [A] | SEER | PHYSICAL LENGTH [IN] | WIDTH [IN] | HEIGHT [IN] | WEIGHT [LB] | NOTES |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CU-1 | TRANE | 5TTR5036N1000A | AHU-1 | R-454B | 36,000 | 4TXCC007DS3HCB | 230 | 1 | 60 | 24 | 40 | 18.5 | 16 | 33 | 30 | 33 | 252 | 3,4,5,6,7,8 |
| CU-2 | TRANE | 5TTR5036N1000A | AHU-2 | R-454B | 36,000 | 4TXCC007DS3HCB | 230 | 1 | 60 | 24 | 40 | 18.5 | 16 | 33 | 30 | 33 | 252 | 3,4,5,6,7,8 |
| CU-3 | TRANE | 5TTR5036N1000A | AHU-3 | R-454B | 36,000 | 4TXCC007DS3HCB | 230 | 1 | 60 | 24 | 40 | 18.5 | 16 | 33 | 30 | 33 | 252 | 3,4,5,6,7,8 |
| CU-4 | TRANE | 5TTR5036N1000A | AHU-4 | R-454B | 36,000 | 4TXCC007DS3HCB | 230 | 1 | 60 | 24 | 40 | 18.5 | 16 | 33 | 30 | 33 | 252 | 3,4,5,6,7,8 |
| CU-5 | TRANE | 5TTR5024N1000A | AHU-5 | R-454B | 24,000 | 4TXCC007DS3HCB | 230 | 1 | 60 | 24 | 40 | 18.5 | 16 | 33 | 30 | 33 | 252 | 1,3,4,5,6,7,8 |
| CU-6 | TRANE | 5TTR5024N1000A | AHU-6 | R-454B | 24,000 | 4TXCC007DS3HCB | 230 | 1 | 60 | 24 | 40 | 18.5 | 16 | 33 | 30 | 33 | 252 | 3,4,5,6,7,8 |
| CU-7 | TRANE / MITSUBISHI | PUY-AK12NL | AHU-7 | R-454B | 12,000 | - | 230 | 1 | 60 | 16 | 20 | - | 21.1 | 12 | 34 | 25 | 252 | 1,2,3,4,5,6,7 |
| CU-8 | TRANE / MITSUBISHI | PUY-AK12NL | AHU-8 | R-454B | 12,000 | - | 230 | 1 | 60 | 16 | 20 | - | 21.1 | 12 | 34 | 25 | 252 | 1,2,3,4,5,6,7 |

NOTES
1. PROVIDE WITH LOW AMBIENT KIT.
2. PROVIDE WITH HAIL GUARDS.
3. PROVIDE WITH CONCRETE HOUSEKEEPING PAD. RE. STRUCUTURAL FOR MORE DETAILS. PROVIDE WITH LEVELING BASE MOUNTS FOR A SLOPED CONCRETE PAD.
4. PROVIDE WITH SINGLE POWER POINT CONNECTION.
5. DISCONNECT BY DIVISION 26.
6. ALL REFRIGERANT PIPING, INCLUDING VALVES, FITTINGS, AND CONNECTIONS, TO BE PROVIDED, SIZED, AND ROUTED PER MANUFACTURER'S RECOMMENDATIONS.
7. PROVIDE ALUMINUM JACKETING FOR ALL EXTERIOR REFRIGERANT PIPING, SEE SPECIFICATIONS FOR ALUMINUM JACKETING REQUIREMENTS.
8. PROVIDE WITH TRANE WEATHERGUARD ACCESSORY (BAYTWGR0003) OR SIMILAR.

## SPLIT SYSTEMS SCHEDULE - INDOOR UNIT

| GENERAL TAG | MANUFACTURER | MODEL | LOCATION | REFRIGERANT | OUTDOOR UNIT | SUPPLY FAN AIRFLOW [CFM] | VENTILATION AIR [CFM] | ESP [IN. W.C.] | DX COOLING NOMINAL COOLING [BTU/H] | OPERATING RANGE [°F] | PROPANE HEATING NOMINAL HEATING [BTU/H] | ELECTRICAL VOLTAGE [V] | PHASE | FREQ. [Hz] | MCA [A] | MOCP [A] | PHYSICAL LENGTH [IN] | WIDTH [IN] | HEIGHT [IN] | WEIGHT [LB] | NOTES |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| AHU-1 | TRANE | S9V2B060U4 | MECH-001 | R-454B | CU-1 | 1050 | 230 | 1.0 | 36,000 | - | 60,000 | 120 | 1 | 60 | 10.9 | 15 | 38.75 | 17.5 | 24 | 119 | 1,3,4,5,6,8,9 |
| AHU-2 | TRANE | S9V2B080U4 | MECH-001 | R-454B | CU-2 | 1335 | 135 | 1.0 | 36,000 | - | 80,000 | 120 | 1 | 60 | 10.9 | 15 | 38.75 | 17.5 | 24 | 119 | 1,3,4,5,6,8,9 |
| AHU-3 | TRANE | S9V2B080U4 | MECH-001 | R-454B | CU-3 | 1200 | 220 | .75 | 36,000 | - | 80,000 | 120 | 1 | 60 | 10.9 | 15 | 38.75 | 17.5 | 24 | 119 | 1,3,4,5,6,8,9 |
| AHU-4 | TRANE | S9V2B060U4 | MECH-001 | R-454B | CU-4 | 1100 | 225 | .75 | 36,000 | - | 60,000 | 120 | 1 | 60 | 10.9 | 15 | 38.75 | 17.5 | 24 | 119 | 1,3,4,5,6,8,9 |
| AHU-5 | TRANE | S9V2B080U4 | MECH-001 | R-454B | CU-5 | 850 | 125 | .75 | 24,000 | - | 60,000 | 120 | 1 | 60 | 10.9 | 15 | 38.75 | 17.5 | 24 | 119 | 1,3,4,5,6,8,9 |
| AHU-6 | TRANE | S9V2B060U4 | MECH-001 | R-454B | CU-6 | 700 | 170 | .75 | 24,000 | - | 60,000 | 120 | 1 | 60 | 10.9 | 15 | 38.75 | 17.5 | 24 | 119 | 1,3,4,5,6,8,9 |
| AHU-7 | TRANE / MITSUBISHI | PKA-AL12NL | AV CLOSET | R-454B | CU-7 | 250 | NONE | N/A | 12,000 | -40 | - | OUTDOOR | OUTDOOR | OUTDOOR | - | - | 9.35 | 35.35 | 11.78 | 28 | 2,4,5,6,7 |
| AHU-8 | TRANE / MITSUBISHI | PKA-AL12NL | ELEVATOR MECH | R-454B | CU-8 | 250 | NONE | N/A | 12,000 | -40 | - | OUTDOOR | OUTDOOR | OUTDOOR | - | - | 9.35 | 35.35 | 11.78 | 28 | 2,4,5,6,7 |

NOTES
1. CONDENSATE SHALL GRAVITY DRAIN TO MECHANICAL ROOM FLOOR DRAIN.
2. POWERED BY OUTDOOR UNIT.
3. INDOOR UNIT AND OUTDOOR UNIT POWERED SEPARATELY.
4. DISCONNECT BY DIVISION 26.
5. PROVIDE WITH 7-DAY PROGRAMMABLE THERMOSTAT WITH COVER.
6. ALL REFRIGERANT PIPING, INCLUDING VALVES, FITTINGS, AND CONNECTIONS, TO BE PROVIDED, SIZED, AND ROUTED PER MANUFACTURER'S RECOMMENDATIONS.
7. PROVIDE WITH INTERNAL BLUE DIAMOND CONDENSATE PUMP (X86-011).
8. PROVIDE WITH FURANCE CONDENSATE NEUTRALIZER.
9. PROVIDE WITH MERV 13 FILTER.

## LOUVER SCHEDULE

| GENERAL TAG | MANUFACTURER | MODEL | FINISH | SERVICE | PERFORMANCE AIRFLOW [CFM] | FREE AREA [SF] | FACE VELOCITY [FPM] | PRESSURE DROP [IN W.C.] | PHYSICAL LENGTH [IN] | WIDTH [IN] | HEIGHT [IN] | WEIGHT [LB] | NOTES |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| LVR-1 | GREENHECK | ESD-635-60x18 | ALUMINUM | INTAKE | 1225 | 2.9 | 424 | 0.03 | 6 | 60 | 18 | 29 | 1,2,4,5 |
| LVR-2 | GREENHECK | ESD-635-36x24 | ALUMINUM | RELIEF | 700 | 2.8 | 386 | 0.02 | 6 | 36 | 24 | 23 | 1,2,3,4 |

NOTES
1. PROVIDE WITH STAINLESS STEEL BIRDSCREEN.
2. PROVIDE WITH 2 COATS OF KYNAR PAINT. COORDINATE FINAL COLOR SELECTION WITH CONTRACTING OFFICER PRIOR TO ORDERING.
3. PROVIDE WITH BACKDRAFT DAMPER UPSTREAM OF LOUVER MOUNTED IN DUCT, GREENHECK MODEL EM-30, MATCH LOUVER SIZE. ENSURE THAT THE DAMPER ACTUATOR IS INSTALLED EXTERIOR OF THE AIR STREAM.
4. PROVIDE WITH EXTENDED SILL, FLANGED FRAME, AND WELDED CONSTRUCTION.
5. PROVIDE WITH CONTROL DAMPER AND NORMALLY OPEN 2-POS ACTUATOR AND END SWITCH, GREENHECK MODEL VCD-23, MATCH LOUVER SIZE. ENSURE THAT THE DAMPER ACTUATOR IS INSTALLED EXTERIOR OF THE AIR STREAM.

**ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.**

PROFESSIONAL ENGINEER LICENSED IOWA DENISE M. DIHLE P26997 5/29/26

| | |
|---|---|
| A/E FIRM: | |
| PRIME: | OTAK, INC. |
| | REDMOND, WA |
| SUBCONTRACTOR | 360 ENGINEERING |
| | GOLDEN, CO |
| DESIGNED: | AF/BG/MD |
| CADD | AF/MD |
| TECH REVIEW: | BG |
| DATE: | 5/29/2026 |
| SUB SHEET NO. | M600 |
| TITLE OF SHEET | MECHANICAL SCHEDULES |
| | EFFIGY MOUNDS NATIONAL MONUMENT |
| DRAWING NO. | 394 / 194248 |
| PMIS/PKG NO. | 318539 |
| SHEET | 80 OF 142 |

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## ASHRAE 15 REFRIGERANT CALCULATIONS

| System Information | | | | | Space Information (See space volume table) | | Effective Dispersal Volume Charge (Maximum allowable refrigerant charge) | | | | | NOTES |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| System Name | System Type | Refrigerant | Refrigerant Type | Refrigerant Charge (lbs) | Space Tag(s) | Total Volume (ft3) | EDVC - RCL (lbs) | EDVC - RCL ADJ. FOR ALTITUDE (lbs) | RCL COMPLIANT? | EDVC - LFL (lbs) | LFL COMPLIANT? | |
| AHU-1 | Split | R-454B | A2L | 3.5 | A | 4152 | 12.9 | 12.7 | YES | 91.3 | YES | 1,2,3,4 |
| AHU-2 | Split | R-454B | A2L | 3.5 | A | 4152 | 12.9 | 12.7 | YES | 91.3 | YES | 1,2,3,4 |
| AHU-3 | Split | R-454B | A2L | 3.5 | A | 4152 | 12.9 | 12.7 | YES | 91.3 | YES | 1,2,3,4 |
| AHU-4 | Split | R-454B | A2L | 3.5 | A | 4152 | 12.9 | 12.7 | YES | 91.3 | YES | 1,2,3,4 |
| AHU-5 | Split | R-454B | A2L | 3.5 | A | 4152 | 12.9 | 12.7 | YES | 91.3 | YES | 1,2,3,4 |
| AHU-6 | Split | R-454B | A2L | 3.75 | A | 4152 | 12.9 | 12.7 | YES | 91.3 | YES | 1,2,3,4 |
| AHU-7 | Split | R-454B | A2L | 4.5 | B,D | 2094 | 6.5 | 6.4 | YES | 46.1 | YES | 1,2,3,4 |
| AHU-8 | Split | R-454B | A2L | 4.5 | C,D | 1842 | 5.7 | 5.6 | YES | 40.5 | YES | 1,2,3,4 |

NOTES:

1. SPACE VOLUME INCLUDES SPACES ADJACENT TO ROOM, CONNECTED BY DUCTS AS SHOWN ON PLAN.
2. SPACE VOLUME INCLUDES OVERHEAD PLENUM SPACE, CONNECTED TO ROOM M001 BY DUCTS AS SHOWN ON PLAN.
3 CALCULATIONS ARE BASED ON ASHRAE 15-2024 AND COMPLY WITH THE 2024 INTERNATIONAL MECHANICAL CODE, SECTION 1101.1.1.
4 CALCULATIONS ARE ADJUSTED FOR ALTITUDE EQUAL TO 700 FEET ABOVE SEA LEVEL.

## GAS CALCULATIONS (LINE SIZING PER NFPA 54 2021 TABLE 12.22)

**NEW GAS LINE CONNECTED LOAD**

| EQUIPMENT | INPUT EACH AT SEA LEVEL (MBH) |
|---|---|
| (D) HVAC-1 | -115 |
| (D) HVAC-2 | -115 |
| (D) HVAC-3 | -90 |
| (D) HVAC-5 | -100 |
| (N) AHU-1 | 80 |
| (N) AHU-2 | 80 |
| (N) AHU-3 | 80 |
| (N) AHU-4 | 60 |
| (N) AHU-5 | 60 |
| (N) AHU-6 | 60 |
| NET LOAD CHANGE (MBH) | 0 |
| TOTAL EQUIVALENT LENGTH (FT) | 85 |
| SYSTEM DESIGN PRESSURE ("WC) | 14 |
| SYSTEM DESIGN PRESSURE DROP ("WC) | 0.5 |

**NO CHANGE IN OVERALL GAS INPUT, EXISTING GAS MAIN SIZE IS SUFFICIENT.**

### AHU-1

| System Name and Number | Condition Analyzed (impacts Ez) | Occupancy Category | Zone Floor Area Az (sq ft) | Are you using default value for zone population? | Zone Population Pz people | | People Outdoor Air Rate Rp (cfm per person) | Area Outdoor Air Rate Ra (cfm per sq ft) | Breathing Zone Outdoor Airflow Vbz (cfm) | Zone Air Distribution Effectiveness Ez | Zone Outdoor Airflow Voz (cfm) | Outdoor air intake flow provided (measured or design) (cfm) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Default | Manual | | | Rp Pz + Ra Az | | Vbz / Ez | |
| AHU-1 | Cooling | Auditorium seating area | 900 | No | | 25 | 5 | 0.06 | 179 | 0.8 | 223.8 | 230 |

### AHU-2

| System Name and Number | Condition Analyzed (impacts Ez) | Occupancy Category | Zone Floor Area Az (sq ft) | Are you using default value for zone population? | Zone Population Pz people | | People Outdoor Air Rate Rp (cfm per person) | Area Outdoor Air Rate Ra (cfm per sq ft) | Breathing Zone Outdoor Airflow Vbz (cfm) | Zone Air Distribution Effectiveness Ez | Zone Outdoor Airflow Voz (cfm) | Outdoor air intake flow provided (measured or design) (cfm) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Default | Manual | | | Rp Pz + Ra Az | | Vbz / Ez | |
| Offices | Cooling | Office space | 1215 | Yes | 6.075 | | 5 | 0.06 | 103.275 | 0.8 | 129.1 | 135 |

### AHU-3

| System Name and Number | Condition Analyzed (impacts Ez) | Occupancy Category | Zone Floor Area Az (sq ft) | Are you using default value for zone population? | Zone Population Pz people | | People Outdoor Air Rate Rp (cfm per person) | Area Outdoor Air Rate Ra (cfm per sq ft) | Breathing Zone Outdoor Airflow Vbz (cfm) | Zone Air Distribution Effectiveness Ez | Zone Outdoor Airflow Voz (cfm) | Outdoor air intake flow provided (measured or design) (cfm) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Default | Manual | | | Rp Pz + Ra Az | | Vbz / Ez | |
| Exhibit & Lobby | Cooling | Lobbies | 1545 | No | | 16 | 5 | 0.06 | 172.7 | 0.8 | 215.9 | 220 |

### AHU-4

| System Name and Number | Condition Analyzed (impacts Ez) | Occupancy Category | Zone Floor Area Az (sq ft) | Are you using default value for zone population? | Zone Population Pz people | | People Outdoor Air Rate Rp (cfm per person) | Area Outdoor Air Rate Ra (cfm per sq ft) | Breathing Zone Outdoor Airflow Vbz (cfm) | Zone Air Distribution Effectiveness Ez | Zone Outdoor Airflow Voz (cfm) | Outdoor air intake flow provided (measured or design) (cfm) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Default | Manual | | | Rp Pz + Ra Az | | Vbz / Ez | |
| AHU-4 | Cooling | Museums/galleries | 950 | No | | 16 | 7.5 | 0.06 | 177 | 0.8 | 221.3 | 225 |

### AHU-5

| System Name and Number | Condition Analyzed (impacts Ez) | Occupancy Category | Zone Floor Area Az (sq ft) | Are you using default value for zone population? | Zone Population Pz people | | People Outdoor Air Rate Rp (cfm per person) | Area Outdoor Air Rate Ra (cfm per sq ft) | Breathing Zone Outdoor Airflow Vbz (cfm) | Zone Air Distribution Effectiveness Ez | Zone Outdoor Airflow Voz (cfm) | Outdoor air intake flow provided (measured or design) (cfm) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Default | Manual | | | Rp Pz + Ra Az | | Vbz / Ez | |
| AHU-5 | Cooling | Office space | 1137 | Yes | 5.685 | | 5 | 0.06 | 96.645 | 0.8 | 120.8 | 125 |

### AHU-6

| System Name and Number | Condition Analyzed (impacts Ez) | Occupancy Category | Zone Floor Area Az (sq ft) | Are you using default value for zone population? | Zone Population Pz people | | People Outdoor Air Rate Rp (cfm per person) | Area Outdoor Air Rate Ra (cfm per sq ft) | Breathing Zone Outdoor Airflow Vbz (cfm) | Zone Air Distribution Effectiveness Ez | Zone Outdoor Airflow Voz (cfm) | Outdoor air intake flow provided (measured or design) (cfm) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Default | Manual | | | Rp Pz + Ra Az | | Vbz / Ez | |
| AHU-6 | Cooling | Breakrooms | 585 | No | | 12 | 5 | 0.12 | 130.2 | 0.8 | 162.8 | 170 |

**ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.**

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR 360 ENGINEERING GOLDEN, CO | DESIGNED: AF/BG/MD CADD AF/MD TECH REVIEW: BG DATE: 5/29/2026 | SUB SHEET NO. M800 | TITLE OF SHEET MECHANICAL CALCS EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 81 OF 142 |

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## PLUMBING NOTES

1. GENERAL
   1. THESE DRAWINGS ARE DIAGRAMMATIC IN NATURE AND NOT INTENDED TO SHOW ALL TRANSITIONS, OFFSETS, ETC. CONTRACTOR SHALL FIELD VERIFY EXISTING CONDITIONS AND PROVIDE ALL NECESSARY FITTINGS TO COMPLETE THE INTENT OF THE DRAWINGS. ANY DISCREPANCIES BETWEEN DRAWINGS AND FIELD CONDITIONS SHALL BE REPORTED TO THE CONTRACTING OFFICER FOR RESOLUTION.
   2. CONTRACTOR SHALL COORDINATE WORK WITH OTHER TRADES AND NOTIFY THE CONTRACTING OFFICER IF ANY CONFLICTS OCCUR.
   3. CONTRACTOR SHALL REVIEW THESE DOCUMENTS CAREFULLY. CONTRACTOR SHALL CONTACT THE THE CONTRACTING OFFICER FOR RESOLUTION OF ANY DISCREPANCIES, OMISSIONS, OR CLARIFICATIONS, BEFORE BID DATE. IN THE EVENT THAT AN INTERPRETATION OF BID DOCUMENTS IS NECESSARY AFTER THE BID DATE, THE DECISION OF THE CONTRACTING OFFICER SHALL BE FINAL AND BINDING.
   4. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE COSTS OF ALL CHANGE ORDERS, WHICH THE CONTRACTING OFFICER HAS NOT APPROVED IN WRITING PRIOR TO THE EXECUTION OF THE ASSOCIATED WORK.
   5. IN THE CASE OF A CONFLICT, UNLESS OTHERWISE NOTED, KEYNOTES ON PLUMBING PLANS SHALL SUPERCEDE ANY GENERAL NOTES ON THE PLANS.
2. EXECUTION
   1. ALL PLUMBING WORK SHALL COMPLY WITH LOCAL CODES AND ORDINANCES.
   2. PITCH WASTE LINES NOT LESS THAN 1/4" PER FOOT. (UNLESS NOTED OTHERWISE).
   3. RUN ALL PIPING ON WARM SIDE OF BUILDING INSULATION. PIPE INSULATION IS NOT CONSIDERED FREEZE PROTECTION.
   4. PROVIDE DIELECTRIC UNIONS AT CONNECTIONS BETWEEN DISSIMILAR METALS, I.E., IRON VALVES AND COPPER TUBING.
   5. PROVIDE PIPE HANGERS OF THE SAME MATERIAL AS THE PIPING SYSTEM OR USE COATED HANGERS.
   6. SET FLOOR DRAINS SO THAT TOP WILL BE SLIGHTLY LOWER THAN SURROUNDING FLOOR.
   7. PROVIDE BALL VALVES AND UNIONS ON ALL LINES TO EQUIPMENT FOR ISOLATION AND REMOVAL.
   8. PLUMBING CONTRACTOR SHALL CHECK OPERATION AND CONDITION OF ALL EXISTING PLUMBING EQUIPMENT WITHIN THE CONFINES OF THIS SPACE, AND PREPARE A WRITTEN LIST OF DEFICIENCIES IN EQUIPMENT OPERATION OR CONDITION. LIST SHALL BE SUBMITTED TO THE CONTRACTING OFFICER WITHIN 2 WEEKS OF CONTRACT AWARD DATE.
   9. THE CONTRACTOR SHALL X-RAY THE FLOOR AND OBTAIN THE CONTRACTING OFFICER APPROVAL BEFORE CUTTING ANY FLOOR PENETRATIONS.
   10. WHERE REMOVAL OR RELOCATION OF EXISTING FIXTURES OCCURS, CONTRACTOR SHALL REMOVE ALL WASTE AND VENT BRANCH LINES NOT REQUIRED FOR NEW, OR REMAINING FIXTURE LOCATIONS.
   11. ALL PIPE PENETRATIONS THROUGH RATED WALLS SHALL BE SEALED PER 2024 IBC.
   12. PROVIDE ADHESIVE, MULTICOLOR PIPE LABELS FOR ALL SYSTEMS. PROVIDE EVERY 50 FEET, AND EVERY 25 FEET IN CONGESTED AREAS.

## PLUMBING LEGEND

### PIPING SERVICES

- CA — COMPRESSED AIR
- G — NATURAL GAS
- LPG — PROPANE
- COLD WATER (CW)
- HOT WATER (HW)
- HOT WATER RECIRCULATION (HWC)
- VENT (V)
- WASTE ABOVE (W)
- WASTE BELOW (W)
- CONDENSATE ABOVE (CD)

### DRAWING NOTATION

- NEW TO EXISTING
- NEW WORK
- EXISTING ITEMS
- EXISTING ITEMS TO BE DEMOLISHED

### VALVES

- GATE VALVE
- BALL VALVE
- PLUG VALVE
- CHECK VALVE
- PRESSURE REDUCING VALVE
- SAFETY RELIEF VALVE
- BACK FLOW PREVENTER
- BALANCING VALVE
- BUTTERFLY VALVE

### PIPING SYSTEM COMPONENTS

- ELBOW DOWN
- ELBOW UP
- TEE DOWN
- TEE UP
- CAP
- UNION
- FLEXIBLE CONNECTION
- FLOOR DRAIN
- FLOOR SINK
- DRAIN VALVE
- VENTURI TYPE BALANCING VALVE
- STRAINER
- FS — FLOW SENSOR
- S — TEMPERATURE SENSOR
- GRADE CLEAN OUT
- FLOOR CLEAN OUT
- WALL CLEAN OUT
- AUTOMATIC AIR CENT
- GAUGE
- THERMOMETER
- MANUAL AIR VENT
- P&T TAP
- IN LINE PUMP
- M — METER
- T — THERMOSTAT
- P — PRESSURE SENSOR

## PLUMBING ABBREVIATIONS

| | |
|---|---|
| ADJ | ADJUSTABLE |
| AFF | ABOVE FINISHED FLOOR |
| AHJ | AUTHORITY HAVING JURISDICTION |
| BLDG | BUILDING |
| CA | COMPRESSED AIR |
| CD | CONDENSATE DRAIN |
| CFH | CUBIC FEET PER HOUR |
| CI | CAST IRON |
| CW | COLD WATER |
| CWV | COMBINATION WASTE AND VENT |
| Cx | COMMISSIONING |
| (D) | DEMOLISH |
| DIA | DIAMETER |
| DFU | DRAINAGE FIXTURE UNIT |
| (E) | EXISTING |
| FD | FLOOR DRAIN |
| FFA | FROM FLOOR ABOVE |
| FFB | FROM FLOOR BELOW |
| FPM | FEET PER MINUTE |
| GPM | GALLONS PER MINUTE |
| GW | GREASE WASTE |
| GPM | GALLON PER MINUTE |
| HW | HOT WATER |
| HWC | HOT WATER CIRCULATION |
| HT | HEAT TRACE |
| ID | INDIRECT DRAIN |
| I.D. | INSIDE DIAMETER |
| LPG | LIQUEFIED PETROLEUM GAS (PROPANE) |
| MAX | MAXIMUM |
| MIN | MINIMUM |
| (N) | NEW |
| NIC | NOT IN CONTRACT |
| O.C. | ON CENTER |
| O.D. | OUTSIDE DIAMETER |
| PSIA | POUNDS PER SQUARE INCH (ABSOLUTE) |
| PSIG | POUNDS PER SQUARE INCH (GAUGE) |
| (R) | RELOCATED |
| RE:__ | REFERENCE ____ |
| REQ | REQUIRED/REQUIREMENT |
| SF | SQUARE FEET |
| STD | STANDARD |
| TFA | TO FLOOR ABOVE |
| TFB | TO FLOOR BELOW |
| TYP | TYPICAL |
| UNO | UNLESS NOTED OTHERWISE |
| V | VENT |
| VTR | VENT THRU ROOF |
| W | WASTE |
| W.C. | WATER COLUMN |
| WSFU | WATER SUPPLY FIXTURE UNIT |

## PLUMBING SHEET LIST



| NUMBER | SHEET NAME |
|---|---|
| P000 | PLUMBING LEGENDS AND NOTES |
| PD100 | PLUMBING DEMO PLAN - BASEMENT |
| PD101 | PLUMBING DEMO PLAN - MAIN LEVEL |
| P110 | PLUMBING DOMESTIC PLAN - BASEMENT |
| P111 | PLUMBING DOMESTIC PLAN - MAIN LEVEL |
| P120 | PLUMBING WASTE AND VENT PLAN - BASEMENT |
| P121 | PLUMBING WASTE AND VENT PLAN - MAIN LEVEL |
| P500 | PLUMBING DETAILS |
| P600 | PLUMBING SCHEDULES |
| P800 | PLUMBING CALCS |
| P900 | PLUMBING WASTE AND VENT ISOMETRICS |



**ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C.**

LICENSED PROFESSIONAL ENGINEER — DENISE M. DIHLE P26897 — 5/29/26 — IOWA

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA<br>SUBCONTRACTOR 360 ENGINEERING GOLDEN, CO | DESIGNED: AF/BG/MD<br>CADD: AF/MD<br>TECH REVIEW: BG<br>DATE: 5/29/2026 | SUB SHEET NO. P000 | TITLE OF SHEET: PLUMBING LEGENDS AND NOTES<br>EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248<br>PMIS/PKG NO. 318539<br>SHEET 82 OF 142 |

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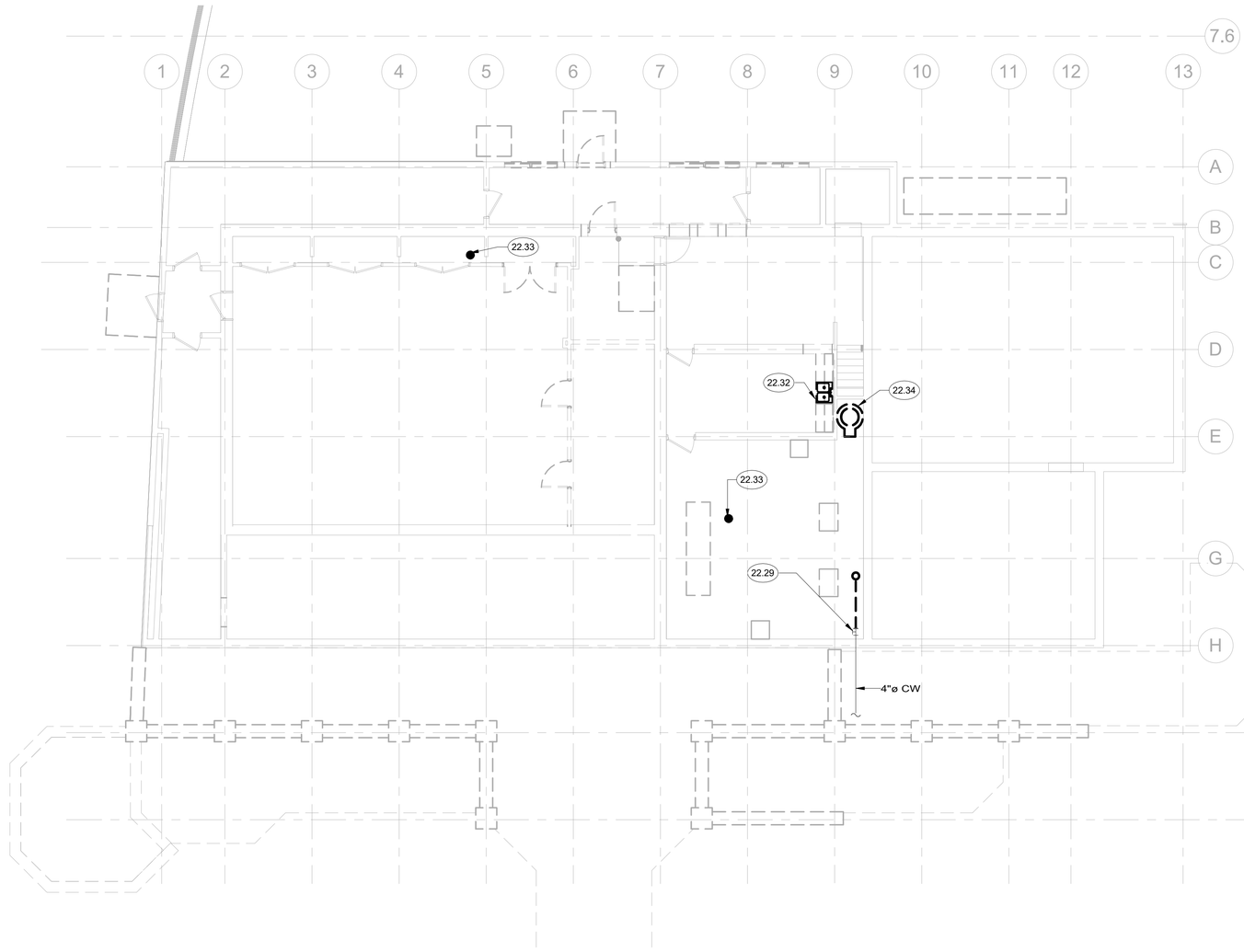
GENERAL NOTES:

1. EXISTING CONDITIONS ARE BASED ON PROVIDED AS-BUILT DRAWINGS. ACTUAL CONDITIONS MAY DIFFER FROM THESE DRAWINGS.
2. DEMO ALL EXISTING PLUMBING, HW, CW, WASTE, AND VENT PIPING ABOVE GRADE. EXISTING PROPANE PIPING TO REMAIN IN CRAWLSPACE UP TO MECHANICAL ROOM FOR RECONNECTION.

(#) KEYNOTES:

22.29 DEMOLISH EXISTING PLUMBING UP TO TEE FOR FIRE PROTECTION SYSTEM. CAP AND PREPARE FOR CONNECTION TO NEW.
22.32 DEMOLISH DOUBLE COMP. SINK AND CONNECTED PIPING.
22.33 DEMOLISH EXISTING FLOOR DRAIN.
22.34 DEMOLISH EXISTING WATER HEATER.




1 PD100 BASEMENT PLUMBING PLAN - DEMO
SCALE A

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C.

SCALE A 8 0 8 16
SCALE OF FEET

DENISE M. DIHLE P26997 5/29/26 PROFESSIONAL ENGINEER LICENSED IOWA

| | | | | |
|---|---|---|---|---|
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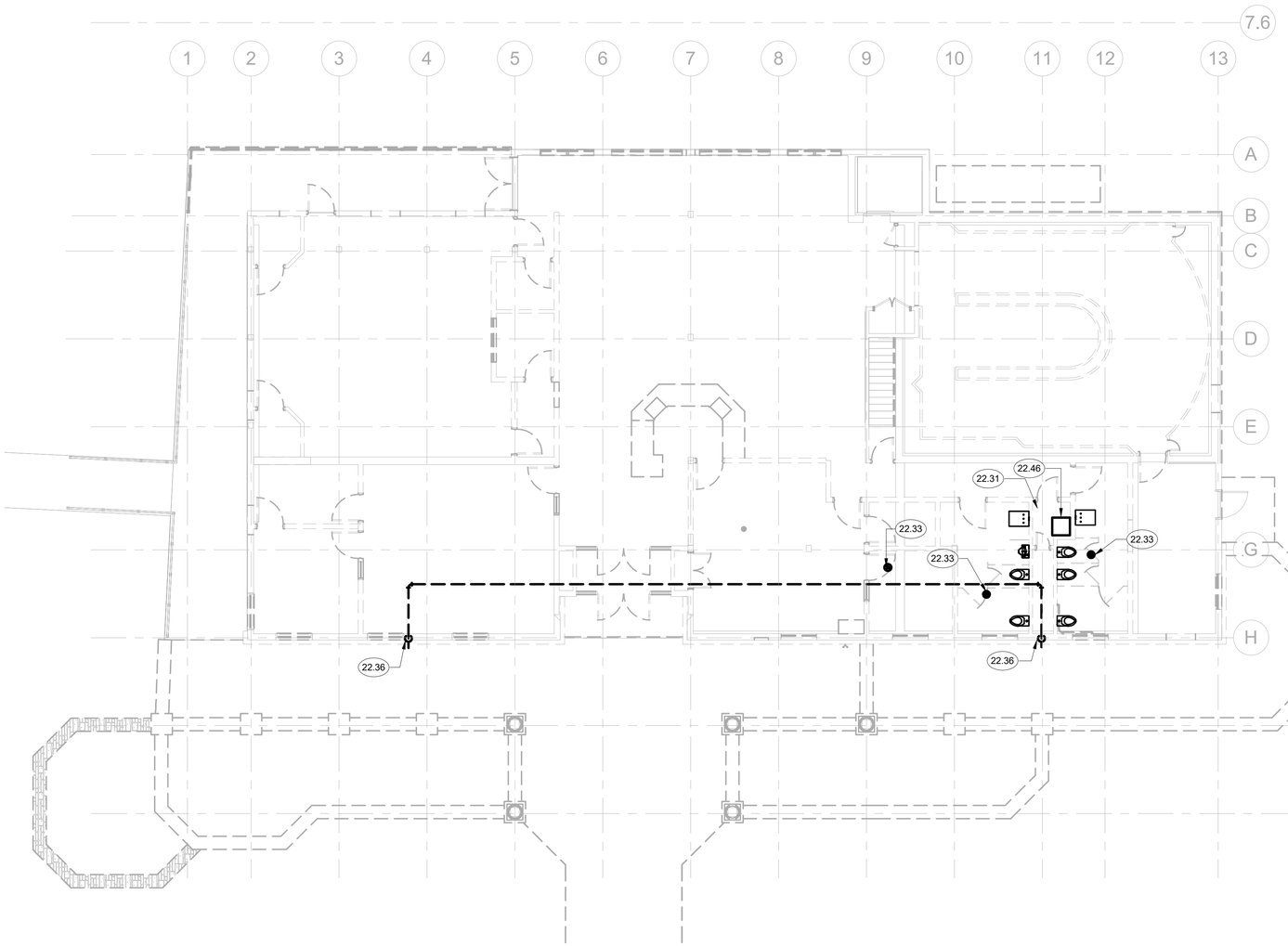
GENERAL NOTES:

1. EXISTING CONDITIONS ARE BASED ON PROVIDED AS-BUILT DRAWINGS. ACTUAL CONDITIONS MAY DIFFER FROM THESE DRAWINGS.
2. DEMO ALL EXISTING PLUMBING, HW, CW, WASTE, AND VENT PIPING ABOVE GRADE.

(#) KEYNOTES:

22.31 DEMOLISH EXISTING RESTROOM FIXTURES AND ALL ASSOCIATED PIPING.
22.33 DEMOLISH EXISTING FLOOR DRAIN.
22.36 DEMOLISH EXISTING WALL HYDRANTS. REPAIR EXTERIOR WALL TO MATCH EXISTING AND PREPARE FOR NEW WALL HYDRANT..
22.46 DEMOLISH EXISTING MOB SINK BASIN.




1 PD101 MAIN LEVEL PLUMBING PLAN - DEMO
SCALE A

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C.

SCALE A 8 0 8 16
SCALE OF FEET

DENISE M. DIHLE P26997 5/29/26 LICENSED PROFESSIONAL ENGINEER IOWA

| | | | | | |
|---|---|---|---|---|---|
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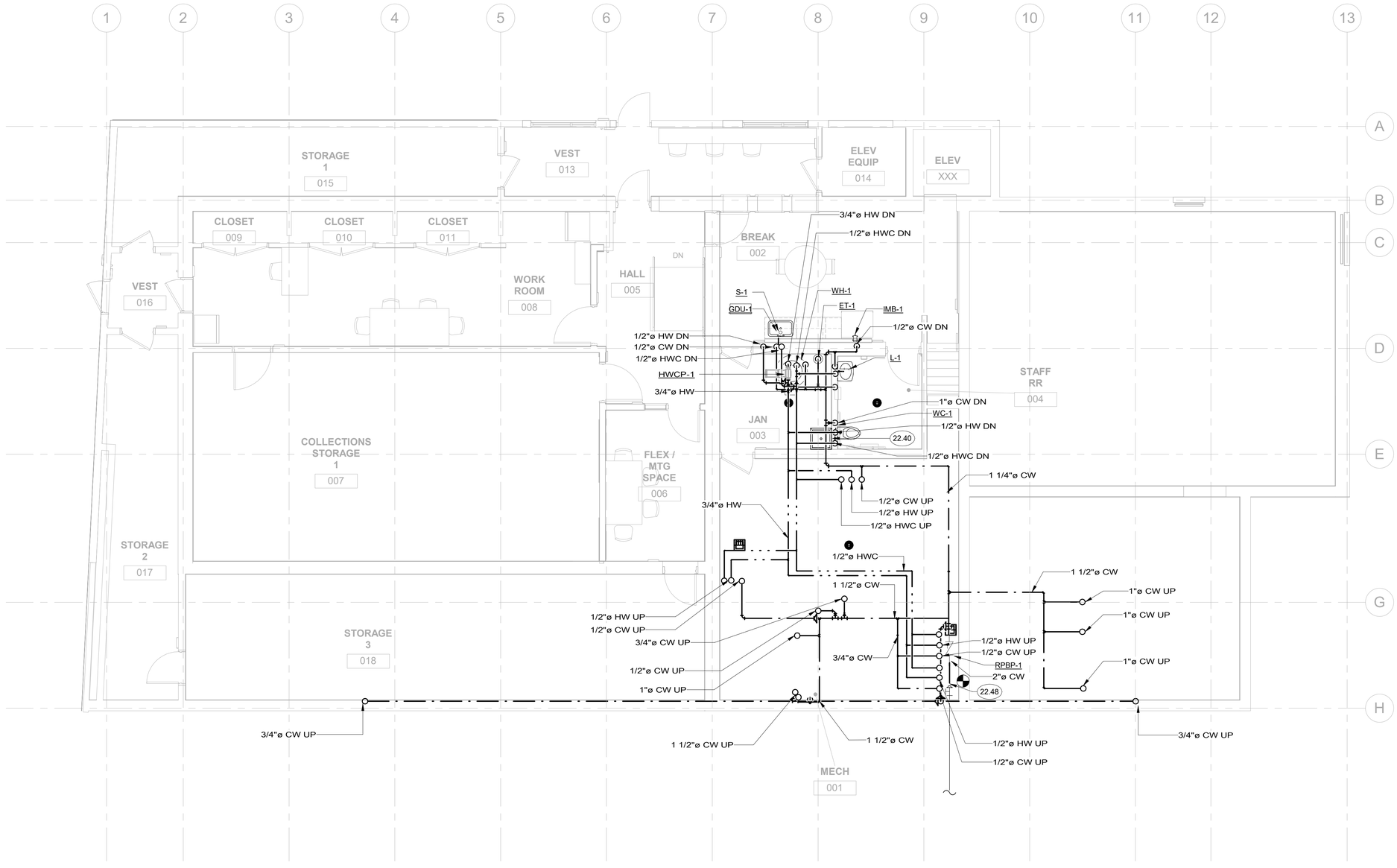


(#) KEYNOTES:

22.40 INSTALL THERMAL MIXING VALVE ON HWC. SET TO 115°F.

22.48 CONNECT NEW 2" DOMESTIC COLD WATER TO EXISTING WATER ENTRY.

1 P110 BASEMENT PLUMBING PLAN - DOMESTIC

SCALE (B)

**ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C.**

SCALE (B) 4 0 4 8

SCALE OF FEET

LICENSED PROFESSIONAL ENGINEER DENISE M. DIHLE P26897 5/21/26 IOWA

| | | | | |
|---|---|---|---|---|
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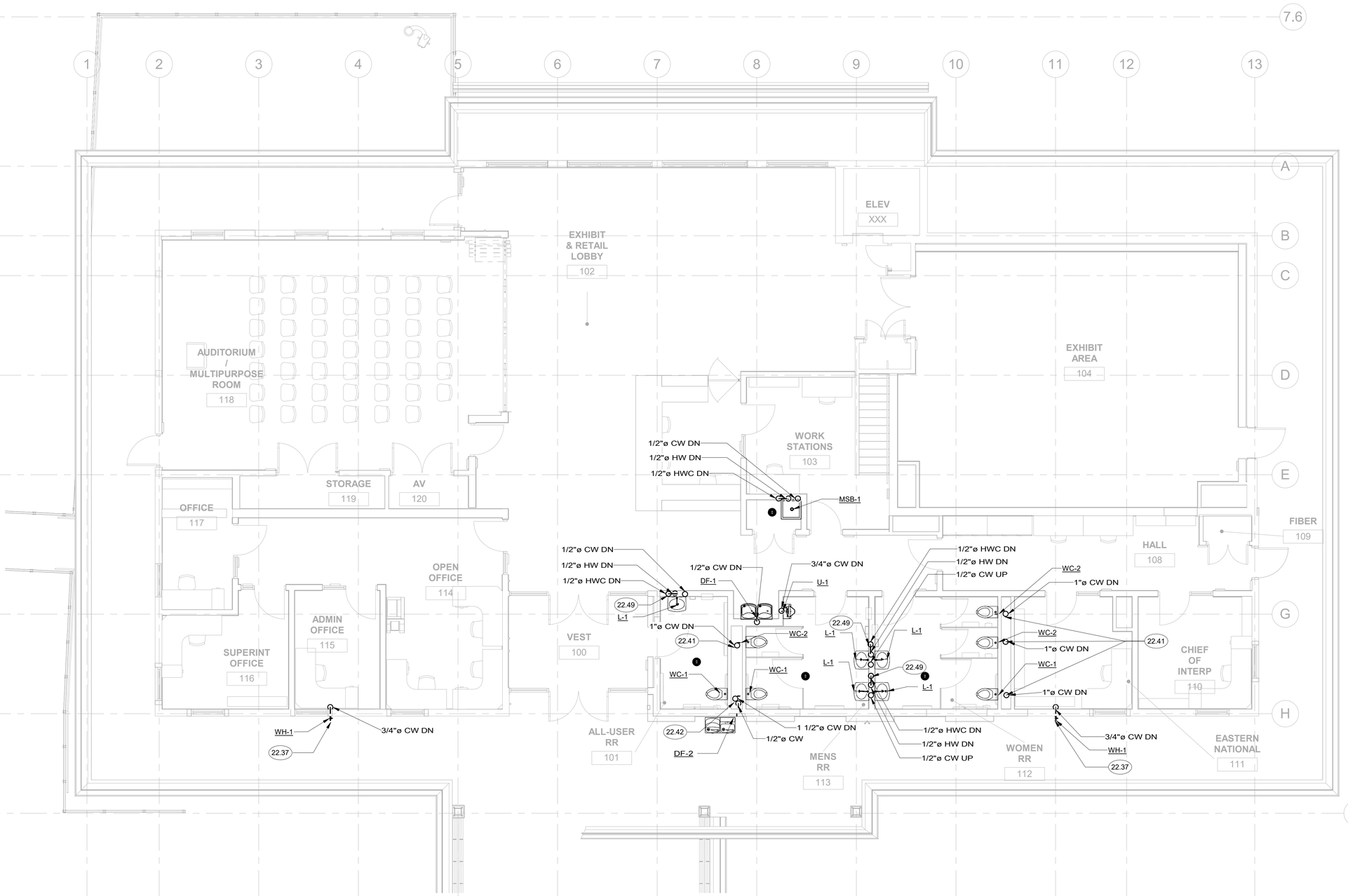


1 P111 MAIN LEVEL PLUMBING PLAN - DOMESTIC
SCALE B

(#) KEYNOTES:
22.37 PROVIDE AND INSTALL NEW WALL HYDRANT.
22.41 INSTALL A SIOUX CHIEF SIZE "A" WATER HAMMER.
22.42 INSTALL A SIOUX CHIEF SIZE "B" WATER HAMMER.
22.49 INSTALL 1/2" BALANCING VALVE ON VERTICAL HWC. BALANCE TO 1 GPM.

**ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C.**

SCALE B
4 0 4 8
SCALE OF FEET

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR 360 ENGINEERING GOLDEN, CO | DESIGNED: AF/BG/MD CADD AF/MD TECH REVIEW: BG DATE: 5/29/2026 | SUB SHEET NO. P111 | TITLE OF SHEET: PLUMBING DOMESTIC PLAN - MAIN LEVEL<br>EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248<br>PMIS/PKG NO. 318539<br>SHEET 86 OF 142 |

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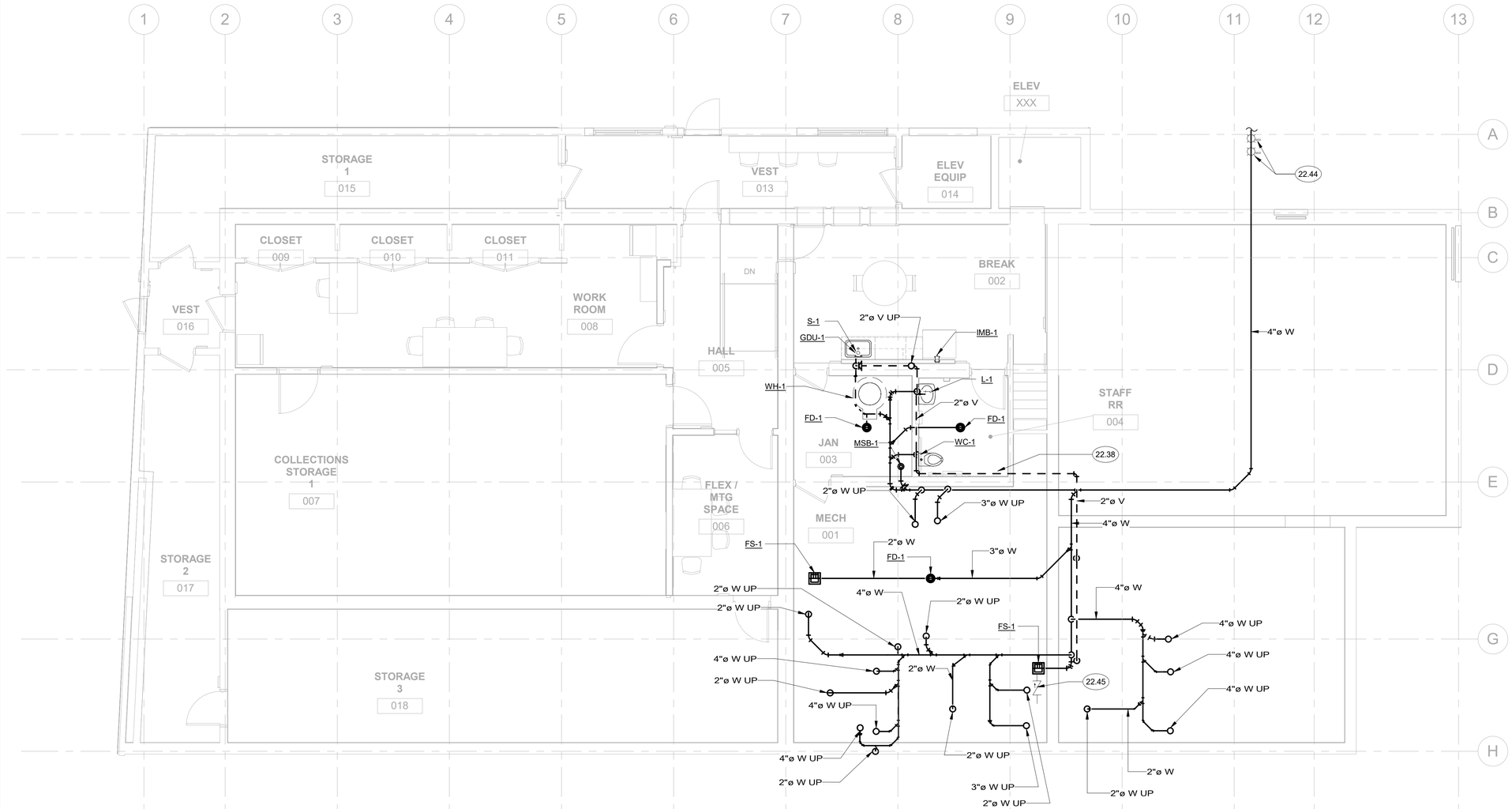


(#) KEYNOTES:
22.38 VENT PIPING IN MECHANICAL ROOM SHALL BE WALL-MOUNTED.
22.44 INSTALL A DOUBLE GRADE CLEAN OUT ON EXISTING WASTE LINE.
22.45 INSTALL 4" DRAIN FROM BACKFLOW PREVENTER TO FLOOR SINK.

**ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C.**

SCALE (B) 4 0 4 8
SCALE OF FEET

1 P120 BASEMENT PLUMBING PLAN - WASTE & VENT
SCALE (B)

DENISE M. DIHLE P26897 5/21/26 LICENSED PROFESSIONAL ENGINEER IOWA

| | | | | |
|---|---|---|---|---|
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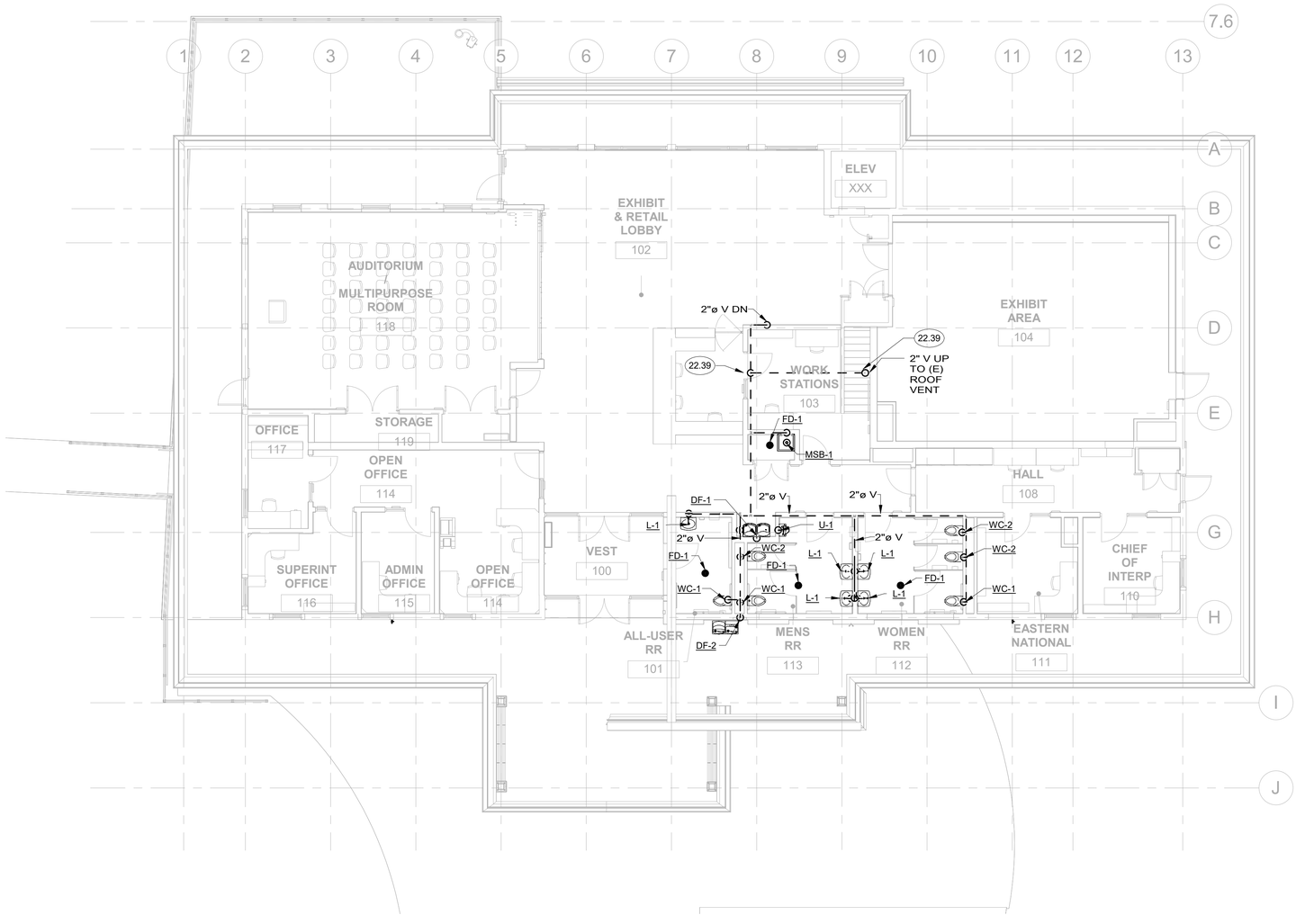



1 P121 MAIN LEVEL PLUMBING PLAN - WASTE & VENT

SCALE A

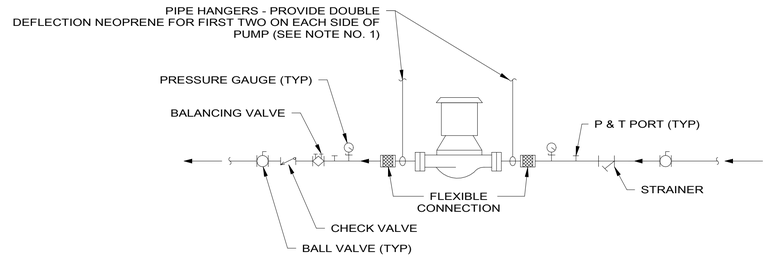
(#) KEYNOTES:
22.23 INSTALL NEW 3" VTR. TRANSITION TO 2" VENT BELOW ROOF LINE.
22.39 VENT TO RISE ABOVE CEILING INTO ATTIC.

**ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C.**

SCALE A 8 0 8 16
SCALE OF FEET

DENISE M. DIHLE P26897 5/29/26 LICENSED PROFESSIONAL ENGINEER IOWA

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR 360 ENGINEERING GOLDEN, CO | DESIGNED: AF/BG/MD DRAWN: AF/MD TECH REVIEW: BG DATE: 5/29/2026 | SUB SHEET NO. P121 | TITLE OF SHEET: PLUMBING WASTE AND VENT PLAN - MAIN LEVEL<br>EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248<br>PMIS/PKG NO. 318539<br>SHEET 88 OF 142 |

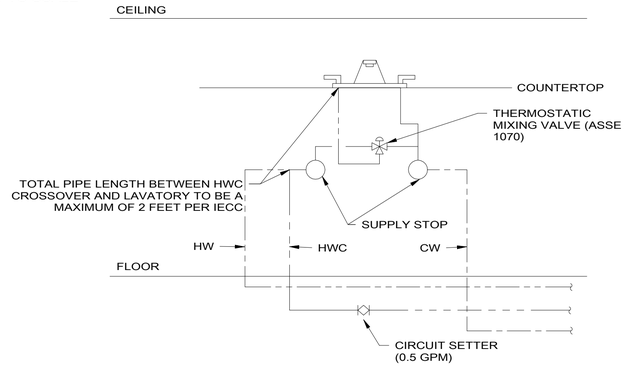



NOTES:

1. SUPPORT PUMP FROM PIPING ONLY. DO NOT SUPPORT PUMP FROM MOTOR.

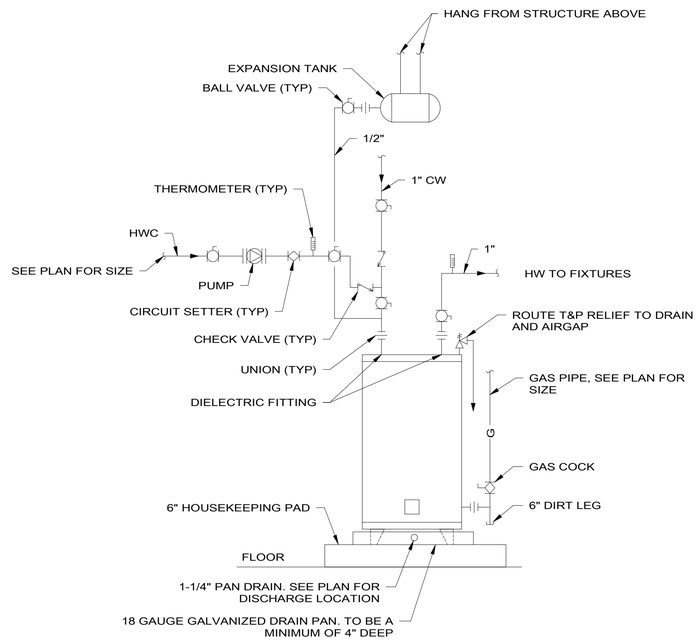
## 4 P500 HORIZONTAL IN-LINE PUMP DETAIL

NOT TO SCALE



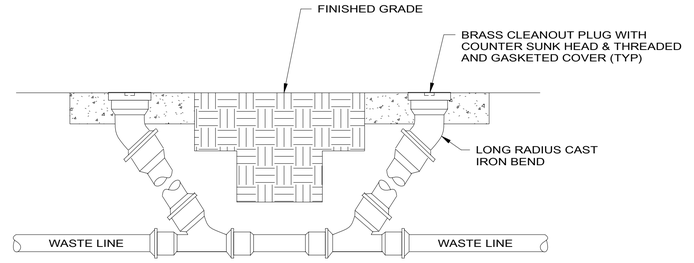

## 5 P500 LAVATORY HOT WATER RECIRCULATION

NOT TO SCALE



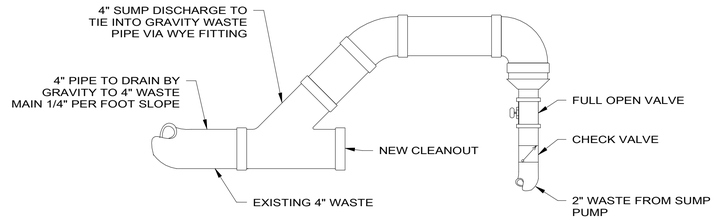

## 6 P500 WATER HEATER DETAIL

NOT TO SCALE



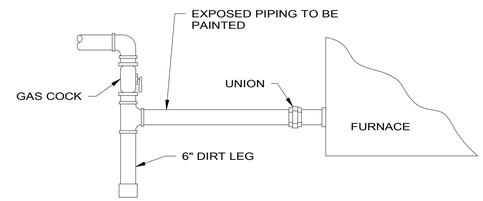

## 1 P500 2-WAY GRADE CLEANOUT DETAIL

NOT TO SCALE




## 2 P500 SUMP DISCHARGE CONNECTION TO WASTE PIPE DETAIL

NOT TO SCALE




## 3 P500 GAS EQUIPMENT CONNECTION DETAIL

NOT TO SCALE

**ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C.**

DENISE M. DIHLE P26897 5/29/26 IOWA PROFESSIONAL ENGINEER LICENSED

| | | | | |
|---|---|---|---|---|
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## PUMP SCHEDULE

| GENERAL TAG | MANUFACTURER | MODEL | SYSTEM | PERFORMANCE FLOW [GPM] | PERFORMANCE HEAD [FT] | ELECTRICAL POWER [HP] | VOLTAGE [V] | PHASE | FREQ. [Hz] | SPEED [RPM] | NOTES |
|---|---|---|---|---|---|---|---|---|---|---|---|
| HWCP-1 | BELL & GOSSETT | PL-30 | HW | 3 | 18 | 1/12 | - | - | - | - | 1,2 |

NOTES
1. LEAD FREE BRONZE
2. PROVIDE WITH AQUASTAT AND TIMECLOCK TO RUN DURING OCCUPIED HOURS.

## PLUMBING FIXTURE SCHEDULE

| FIXTURE TAG | DESCRIPTION | MANUFACTURER | MODEL | FINISH | ACCESSORIES MANUFACTURER | ACCESSORIES MODEL | ACCESSORIES FINISH | INFO GPM/GPF | INFO ELECTRICAL | NOTES |
|---|---|---|---|---|---|---|---|---|---|---|
| DF-1 | DRINKING FOUNTAIN (ADA) HI/LOW | ELKAY | EDFP217C | CHROME | - | - | - | - | - | 1,2 |
| DF-2 | OUTDOOR DRINKING FOUNTAIN HI/LOW W/ WATER BOTTLE FILLER | ELKAY | LVRCTL8WSK | CHROME | - | - | - | - | - | 1,2,23 |
| FD-1 | FLOOR DRAIN | JAY R SMITH | 2005-Y-A | NICKEL BRONZE | - | - | - | - | - | 3,4,7 |
| FS-1 | FLOOR SINK - 8" ROUND, 6" DEEP | JAY R SMITH | 3413-Y | WHITE, ACID RESISTANT | - | - | - | - | - | 3,4,5,7 |
| GDU-1 | GARBAGE DISPOSAL UNIT | INSINKERATOR | BADGER 5 | - | - | - | - | - | 120V/60/1 - 1/2 HP | - |
| IMB-1 | ICE MAKER BOX | SIOUX CHIEF | 696-G1010MF | WHITE | - | - | - | - | - | 6 |
| L-1 | LAVATORY - UNDERCOUNTER MOUNTED (ADA) | AMERICAN STANDARD | 0356.421 | WHITE | SLOAN | SF-2150 | CHROME | 0.5 | - | 1,8,9,10,11,22 |
| MSB-1 | MOP SERVICE BASIN | FIAT | MSB2424 | WHITE | CHICAGO | 445-897SRXKCP | CHROME | 2.2 | - | 12,13 |
| RPBP-1 | REDUCED PRESSURE BACKFLOW PREVENTER | APOLLO | RPLF4A 2" | - | - | - | - | - | - | 14 |
| S-1 | SINK - SINGLE COMPARTMENT (ADA) | JUST | SL-ADA-2125-A-GR | STAINLESS STEEL (18 GA) | DELTA | 9191-PR-DST | CHROME | 1.8 | - | 8,9,15,22 |
| U-1 | URINAL (ADA) | AMERICAN STANDARD | 6590 | WHITE | SLOAN | 8186 | CHROME | 0.125 | - | 1,11,16 |
| WC-1 | WATER CLOSET- WALL MOUNTED (ADA) | AMERICAN STANDARD | 3351 | WHITE | SLOAN BEMIS | 8111 1955CT | CHROME WHITE | 1.28 | - | 1,11,16,17 |
| WC-2 | WATER CLOSET- WALL MOUNTED | AMERICAN STANDARD | 3351 | WHITE | SLOAN BEMIS | 8111 1955CT | CHROME WHITE | 1.28 | - | 1,11,16,17 |
| WH-1 | WALL HYDRANT | WOODFORD | B67 3/4" | CHROME | - | - | - | - | - | 18,19,20,21 |

NOTES
1. PROVIDE WITH CONCEALED FLOOR MOUNTED CARRIER
2. 17 GA. P-TRAP, LOOSE KEY ANGLE STOPS, STAINLESS STEEL BRAIDED SUPPLIES. PROVIDE WITH 3-PACK OF FILTERS FOR STOCK.
3. PROVIDE WITH TRAP GUARD.
4. MOUNT FLUSH WITH FLOOR.
5. 1/2 GRATE, DOME STRAINER.
6. WITH "AA" WATER HAMMER ARRESTERS.
7. BODY/CONNECTION MATERIAL TO MATCH WASTE PIPING.
8. PROVIDE LEONARD #270-LF MIXING VALVE UNDER FIXTURE. ASSE 1070 RATED.
9. PROVIDE WITH TRUEBRO #103 E-Z P-TRAP AND SUPPLIES INSULATION KIT.
10. BATTERY POWERED FAUCET.
11. SEE ARCHITECTURAL PLANS FOR MOUNTING HEIGHT.
12. PAIL HOOK, WALL BRACKET, THREAD END, VACUUM BREAKER, INTEGRAL CHECKS & SHUTOFF STOPS.
13. PROVIDE WITH HOSE AND HOSE BRACKET, MOP HANGER, SS WALL GUARD.
14. PROVIDE WITH STRAINER.
15. 6" DEEP BOWL, REAR DRAIN LOCATION.
16. PROVIDE WITH AUTOMATIC BATTERY FLUSH VALVE.
17. PROVIDE WITH FLUSH HANDLE ON OPEN/ACCESSIBLE SIDE OF WATER CLOSET.

NOTES
18. COORDINATE LENGTH WITH WALL.
19. VACUUM BREAKER.
20. ASSE 1052 RATED.
21. WITH ENCLOSED WALL MOUNTED BOX.
22. 17 GA. P-TRAP, LOOSE KEY ANGLE STOPS, STAINLESS STEEL BRAIDED SUPPLIES.
23. PROVIDE WITH CANE DETECTION APRON, 98324C OR EQUIVALENT, TO BOTTOM OF HIGH FOUNTAIN.

## PLUMBING FIXTURE CONNECTION SCHEDULE

| GENERAL TAG | CW [IN.] | HW [IN.] | HWC [IN.] | WASTE [IN.] | VENT [IN.] | NOTES |
|---|---|---|---|---|---|---|
| DF-1 | 1/2" | - | - | 2" | 1 1/2" | - |
| DF-2 | 1/2" | - | - | 2" | 1 1/2" | - |
| EWH-1 | 3/4" | 3/4" | - | - | - | - |
| FD-1 | - | - | - | 2" | 2" | - |
| FS-1 | - | - | - | 4" | 2" | - |
| GDU-1 | - | - | - | - | - | - |
| IMB-1 | 1/2" | - | - | - | - | - |
| L-1 | 1/2" | 1/2" | 1/2" | 2" | 1 1/2" | 1 |
| MSB-1 | 1/2" | 1/2" | - | 3" | 2" | - |
| RPBP-1 | SEE PLAN | - | - | DRAIN TO FS-1 | - | - |
| S-1 | 1/2" | 1/2" | 1/2" | 2" | 1 1/2" | - |
| U-1 | 3/4" | - | - | 2" | 1 1/2" | - |
| WC-1 | 1" | - | - | 4" | 2" | - |
| WC-2 | 1" | - | - | 4" | 2" | - |
| WH-1 | 3/4" | - | - | - | - | - |

NOTES
1. CONNECT HOT WATER RECIRUCLATION TO HOT WATER LINE PER 2024 IECC TABLE C404.5.1.

## ELECTRIC WATER HEATER SCHEDULE

| GENERAL TAG | MANUFACTURER | MODEL | HEATING POWER [kW] | HEATING RECOVERY AT 90°F RISE | HEATING CAPACITY [GAL] | PHYSICAL HEIGHT [IN] | PHYSICAL DIAMETER [IN.] | PHYSICAL WEIGHT [LB] | ELECTRICAL VOLTAGE [V] | ELECTRICAL PHASE | NOTES |
|---|---|---|---|---|---|---|---|---|---|---|---|
| EWH-1 | AO SMITH | DEN-40 | 6 | 55 GPH | 50 | 59 | 20.5 | 125 | 120 | 1 | 1,2 |

NOTES
1. PROVIDE WITH EXPANSION TANK (ET-1).
2. PROVIDE WITH ANODE ROD.

## PLUMBING MISCELLANEOUS SCHEDULE

| GENERAL TAG | DESCRIPTION | MANUFACTURER | MODEL | NOTES |
|---|---|---|---|---|
| ET-1 | EXPANSION TANK | WATTS | PLT-12 | 1 |

NOTES
1. 4.5 GALLON ACCEPTANCE VOLUME.

**ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C.**

| | | | | | |
|---|---|---|---|---|---|
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## Sanitary Drainage Fixture Unit Calculation

**(DFU Units per IPC Table 709.1 / Waste Size and Slope per IPC Table 710.1(1))**

| Fixture | Number of Instances | DFU Units | Total Load |
|---|---|---|---|
| DRINKING FOUNTAIN | 2 | 0.5 | 1 |
| FLOOR DRAIN | 7 | 2 | 14 |
| FLOOR SINK | 1 | 2.0 | 2 |
| KITCHEN SINK WITH WASTE DISPOSER, DISHWASHER OR BOTH | 1 | 2 | 2 |
| LAVATORY | 6 | 1 | 6 |
| SERVICE SINK / MOP SERVICE BASIN | 4 | 2 | 8 |
| URINAL | 1 | 4 | 4 |
| WATER CLOSET - PUBLIC (1.6 GPF) | 7 | 4 | 28 |
| | | Total | 65 |

A TOTAL DFU OF 65 REQUIRES A 4" LINE AT A 1/4" PER FOOT SLOPE

## Water Supply Fixture Unit Calculation

**(Calculations Based on IPC Table E103.3(2))**

| Fixture | Number of Instances | Cold (WSFU) | Hot (WSFU) | Total Load (WSFU) | Total Combined Load (WSFU) |
|---|---|---|---|---|---|
| WALL HYDRANT | 2 | 5 | – | 5 | 10 |
| MOP SINK BASIN | 4 | 3 | 3 | 4 | 16 |
| DRINKING FOUNTAIN/ICE MAKER BOX | 3 | 0.25 | 0 | 0.25 | 0.75 |
| KITCHEN SINK (PRIVATE) | 1 | 1 | 1 | 1.4 | 1.4 |
| LAVATORY (PUBLIC) | 6 | 1.5 | 1.5 | 2 | 12 |
| URINAL (3/4" VALVE) | 1 | 5 | 0 | 5 | 5 |
| WATER CLOSET (VALVE/PUBLIC) | 7 | 10 | 0 | 10 | 70 |
| TOTAL | | | | | 115.2 |

A WSFU OF 115.15 IS EQUAL TO 71.7 GPM REQUIRING A 2" LINE TO LIMIT VELOCITY TO UNDER 10 FT/SECOND

**ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C.**

LICENSED PROFESSIONAL ENGINEER
DENISE M. DIHLE
P26897
5/24/24
IOWA

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR 360 ENGINEERING GOLDEN, CO | DESIGNED: AF/BG/MD CADD AF/MD TECH REVIEW: BG DATE: 5/29/2026 | SUB SHEET NO. P800 | TITLE OF SHEET PLUMBING CALCS EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 91 OF 142 |

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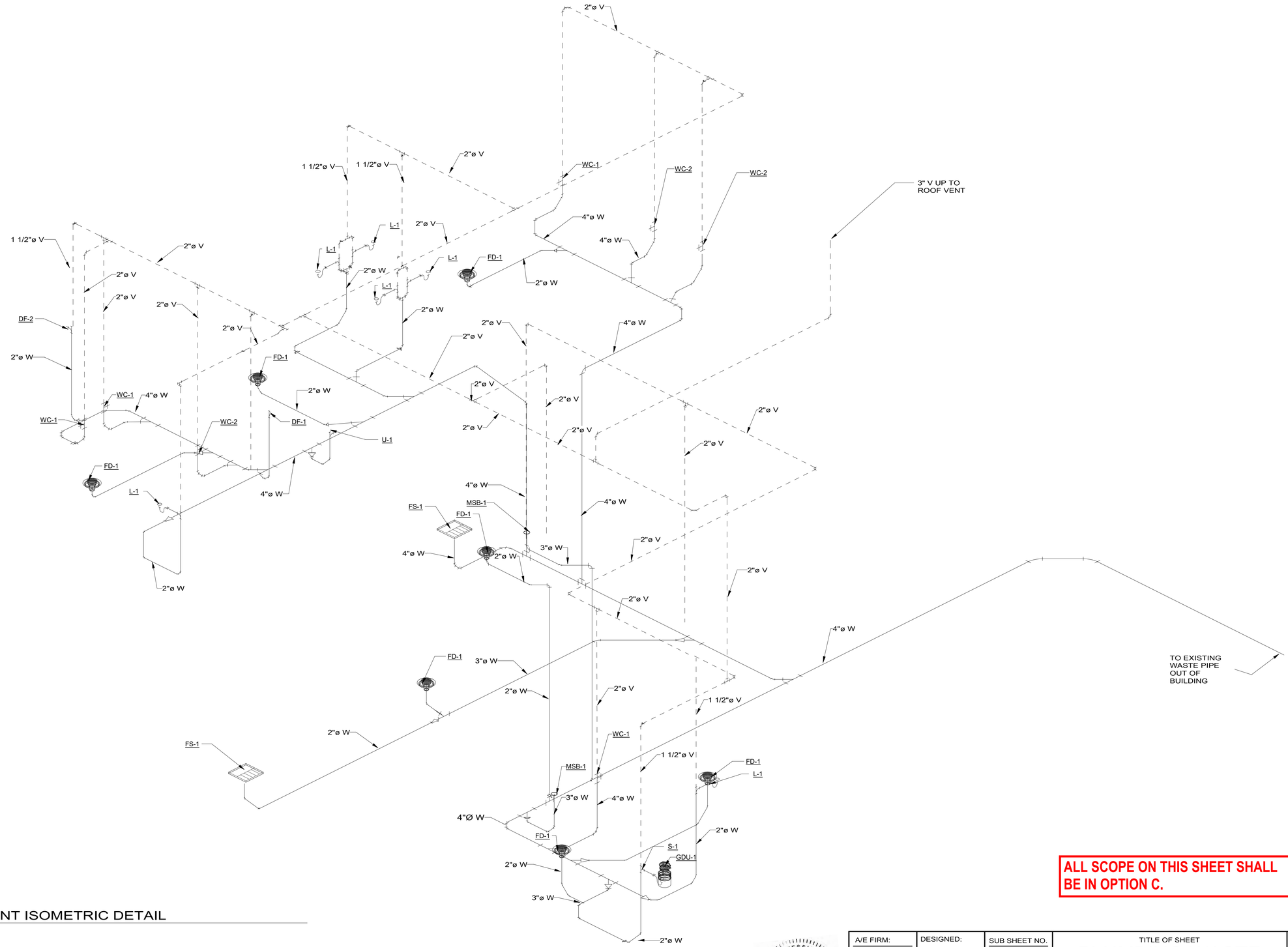



1 P900 WASTE AND VENT ISOMETRIC DETAIL
SCALE A

**ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C.**

DENISE M. DIHLE P26897 5/29/26 PROFESSIONAL ENGINEER LICENSED IOWA

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA<br>SUBCONTRACTOR 360 ENGINEERING GOLDEN, CO | DESIGNED: AF/BG/MD<br>CADD AF/MD<br>TECH REVIEW: BG<br>DATE: 5/29/2026 | SUB SHEET NO. P900 | TITLE OF SHEET<br>PLUMBING WASTE AND VENT ISOMETRICS<br>EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248<br>PMIS/PKG NO. 318539<br>SHEET 92 OF 142 |

# FIRE PROTECTION SYMBOL LIST

NOTE: This is a standard symbol list and not all items listed may be used.

## Abbreviations

| | |
|---|---|
| (A) | ABANDON IN PLACE |
| (E) | EXISTING |
| (F) | FUTURE |
| (N) | NEW |
| (R) | RELOCATE/RELOCATED LOCATION |
| (X) | DEMOLITION |
| AFF | ABOVE FINISHED FLOOR |
| AS | AUTOMATIC SPRINKLER |
| BOB | BOTTOM OF BEAM |
| BOD | BOTTOM OF DECK |
| BOP | BOTTOM OF PIPE |
| BOR | BOTTOM OF RISER |
| BV | BUTTERFLY VALVE |
| C | CENTER LINE |
| CV | CHECK VALVE |
| DDCV | DOUBLE DETECTOR CHECK VALVE ASSEMBLY |
| DN | DROP NIPPLE |
| EC | EXTENDED COVERAGE |
| EL | ELEVATION |
| F | FAHRENHEIT |
| FDC | FIRE DEPARTMENT CONNECTION |
| FF | FINISHED FLOOR |
| FFL | FLOOR FLANGE |
| FHC | FIRE HOSE CABINET |
| FHS | FIRE HOSE STATION |
| FS | FLOW SWITCH |
| FT | FEET |
| G | GRADE |
| GPM | GALLONS PER MINUTE |
| GV | GATE VALVE |
| H | HANGER |
| HSW | HORIZONTAL SIDE WALL |
| HV | HOSE VALVE |
| ID | INSIDE DIAMETER |
| IN | INCHES |
| MAX | MAXIMUM |
| MIN | MINIMUM |
| N&C | NIPPLE AND CAP |
| NIC | NOT IN CONTRACT |
| NO | NUMBER |
| NTS | NOT TO SCALE |
| OBJ | OPEN BAR JOIST |
| OD | OUTSIDE DIAMETER |
| OS & Y | OUTSIDE SCREW & YOKE |
| PIV | POST INDICATOR VALVE |
| PRV | PRESSURE REDUCING VALVE |
| PS | PRESSURE SWITCH |
| RM | ROOF MANIFOLD |
| RN | RISER NIPPLE |
| SB | SWAY BRACE |
| SF | SQUARE FEET |
| SOV | SHUT OFF VALVE |
| SP | STANDPIPE |
| SSP | STANDARD SPRAY PENDENT |
| SSU | STANDARD SPRAY UPRIGHT |
| TOB | TOP OF BEAM |
| TOP | TOP OF PIPE |
| TOR | TOP OF RISER |
| TOS | TOP OF STEEL |
| TS | TAMPER SWITCH |
| TYP | TYPICAL |
| UNO | UNLESS NOTED OTHERWISE |

## Control

- BELL/GONG
- CONTROL PANEL
- FLOW DETECTION SWITCH
- PRESSURE GAUGE
- TAMPER DETECTOR

## Miscellaneous

- AUXILIARY DRAIN
- CONTINUATION OF PIPE
- NEW PIPE
- POINT OF CONNECTION
- RISER
- SCREWED CAP

## Valves

- DRY PIPE VALVE, PLAN VIEW
- PREACTION VALVE

# FIRE PROTECTION DEMOLITION NOTES

A. DEMOLISH THE EXISTING WET PIPE FIRE SPRINKLER SYSTEM IN ITS ENTIRETY, INCLUDING ALL PIPING, FITTINGS, VALVES, FLOW AND TAMPER SWITCHES, AND OTHER ASSOCIATED APPURTENANCES. DEMOLISH SYSTEM BACK TO EXISTING TEE WITHIN MECHANICAL ROOM CONNECTED TO THE INCOMING COMBINED FIRE/DOMESTIC LINE.

B. PROVIDE PROTECTION FOR ALL SURFACES, COLLECTIONS, OR OTHER BUILDING CONTENTS DURING DEMOLITION. SEE DIVISION 01 FOR ADDITIONAL INFORMATION.

C. PROVIDE FIRE WATCH WHEN THE BUILDING FIRE PROTECTION SYSTEM IS NOT OPERABLE.

D. RELOCATE EXISTING MAIN DRAIN FROM FRONT ENTRY TO WEST SIDE OF THE BUILDING ON THE BASEMENT LEVEL.

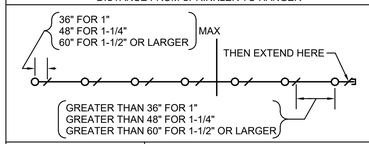
E. DEMOLISH THE EXISTING CLEAN AGENT FIRE EXTINGUISHING SYSTEM PIPING, FITTINGS, AND NOZZLES AS NECESSARY TO ACCOMMODATE PROTECTION FOR THE NEW REMODELED SPACE. SEE DIVISION 28 FOR ADDITIONAL CLEAN AGENT FIRE EXTINGUISHING DEMOLITION SCOPE.

**MAXIMUM DISTANCE BETWEEN HANGERS (FT-IN)**

| STEEL PIPE | NOMINAL PIPE SIZE (IN) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 1 | 1-1/4 | 1-1/2 | 2 | 2-1/2 | 3 | 4 | 6 |
| | 12-0 | 12-0 | 15-0 | 15-0 | 15-0 | 15-0 | 15-0 | 15-0 |

FOR STEEL PIPE, THE MAXIMUM LENGTH FOR UNSUPPORTED ARMOVER SHALL BE 2'-0". WHEN PRESSURE EXCEEDS 100 PSI, UNSUPPORTED LENGTH SHALL BE 1'-0".

DISTANCE FROM SPRINKLER TO HANGER




| HANGER ROD SIZE | | |
|---|---|---|
| PIPE SIZE | DIA. OF ROD | NOTE: TYPICAL TICK MARKS AS SHOWN ON PIPING PLAN MAY NOT REFLECT ACTUAL FIELD INSTALLATION. FINAL HANGER INSTALLATION SHALL BE IN ACCORDANCE WITH THE REQUIREMENTS OF NFPA 13 CHAPTER 9. |
| UP TO AND INCLUDING 4" | 3/8" | |
| 6" AND 8" | 1/2" | |

# GENERAL FIRE PROTECTION NOTES

A. PROVIDE A NEW WET PIPE SPRINKLER SYSTEM FOR THE TEMPERED AREAS WITHIN THE BUILDING, EXCLUDING THE COLLECTIONS AND WORKROOM AREA.

B. PROVIDE A DRY PIPE SPRINKLER SYSTEM FOR AREAS WITHIN THE BUILDING SUBJECT TO TEMPERATURES BELOW 40 DEGREES FAHRENHEIT.

C. PROVIDE A DOUBLE-INTERLOCK PREACTION SPRINKLER SYSTEM FOR THE COLLECTIONS AND WORKROOM AREA.

D. PROVIDE MODIFICATION/EXTENSION OF THE EXISTING CLEAN AGENT FIRE EXTINGUISHING SYSTEM TO ACCOMMODATE THE REMODEL. DESIGN IS TO BE IN ACCORDANCE WITH NFPA 2001, STANDARD ON CLEAN AGENT FIRE EXTINGUISHING SYSTEMS (2022 EDITION).

E. DESIGN AND INSTALL SYSTEMS IN ACCORDANCE WITH NFPA 13, STANDARD FOR THE INSTALLATION OF SPRINKLER SYSTEMS, 2022 EDITION, AND NFPA 72, NATIONAL FIRE ALARM AND SIGNALING CODE, 2022 EDITION.

F. PROVIDE COMPLETE SHOP DRAWINGS AS REQUIRED BY NFPA 13.

G. PROVIDE SCHEDULE 40, BLACK STEEL, PIPING FOR ABOVEGROUND FIRE SPRINKLER SYSTEM PIPING.

H. PITCH DRY AND PREACTION SPRINKLER SYSTEM TO DRAIN PIPING IN ACCORDANCE WITH NFPA 13. WHERE PIPING CANNOT BE PITCHED, PROVIDE LOW POINT DRAINS.

I. PROVIDE THREADED, CUT-GROOVED, OR MECHANICAL FITTINGS.

J. IN EXPOSED LOCATIONS, PAINT FIRE SPRINKLER PIPING AND FITTINGS TO MATCH ADJACENT FINISHES.

K. PROVIDE NITROGEN GENERATOR SYSTEM WITH ASSOCIATED CONTROL AND MONITORING SYSTEM, COMPRESSOR, PURGE VALVES AND/OR VENTS, AND OTHER APPURTENANCES AS REQUIRED FOR COMPLETE SUPERVISORY GAS FOR THE DRY AND PREACTION SPRINKLER SYSTEMS. NITROGEN GENERATOR SYSTEM TO ACHIEVE MINIMUM 98% CONCENTRATION PURITY.

L. PROVIDE RELAY OUTPUTS FROM NITROGEN SYSTEM CONTROLLER MONITORED BY THE FIRE ALARM SYSTEM AS REQUIRED BY CODE AND PER MANUFACTURER REQUIREMENTS.

M. PROVIDE QUICK-RESPONSE, CONCEALED SPRINKLERS IN ALL FINISHED AREAS. PROVIDE DRY TYPE SPRINKLERS FOR ALL PENDENT SPRINKLERS CONNECTED TO THE DRY AND PREACTION SPRINKLER SYSTEMS.

N. PROVIDE NEW BACKFLOW PREVENTER WITHIN THE EXISTING MECHANICAL ROOM FOR THE NEW INCOMING 6" FIRE LINE.

O. FIRE PROTECTION SYSTEM TO BE DESIGNED BY SOMEONE MEETING OR EXCEEDING ON MORE OF THE FOLLOWING:

   a. A NICET III TECHNICIAN FOR THE TYPE OF SYSTEM DESIGNED.
   b. A FIRE PROTECTION ENGINEER.
P. c. AS APPROVED BY THE NPS AHJ.

A. FIRE PROTECTION SYSTEMS SHALL BE INSTALLED BY SOMEONE MEETING THE FOLLOWING:

   a. A NICET II TECHNICIAN FOR THE TYPE OF SYSTEM INSTALLED.
   b. STATE OR MUNICIPAL CERTIFIED/LICENSED SPRINKLER CONTRACTOR.
   c. AS APPROVED BY THE NPS AHJ.

# SPRINKLER HEAD LEGEND

| SYMBOL | DESCRIPTION | K-FACTOR | ORIFICE SIZE | RESPONSE | TEMP | COVERAGE | FINISH | ESCHUTCHEON | COUNT | COMMENTS |
|---|---|---|---|---|---|---|---|---|---|---|
| | DRY BARREL SIDEWALL SPRINKLER | 5.6 | 1/2" | Quick | 155 | Standard | CHROME | RECESSED | 3 | |
| | HORIZONTAL SIDEWALL SPRINKLER | 5.6 | 1/2" | Quick | 155 | Standard | CHROME | RECESSED | 1 | |
| | UPRIGHT SPRINKLER | 5.6 | 1/2" | Quick | 155 | Standard | BRASS | NONE | 48 | |
| | SINGLE-SIDED ATTIC SPRINKLER | 5.6 | 1/2" | Quick | 200 | Standard | BRASS | NONE | 22 | |
| | DOUBLE-SIDED ATTIC SPRINKLER | 8 | 3/4" | Quick | 200 | Standard | BRASS | NONE | 9 | |
| | HIP-JOINT ATTIC SPRINKLER | 5.6 | 1/2" | Quick | 200 | Standard | BRASS | NONE | 28 | |
| | CONCEALED PENDENT SPRINKLER | 5.6 | 1/2" | Quick | 155 | Standard | WHITE POLY | COVERPLATE | 61 | COORDINATE ALL FINISHES WITH ARCHITECT PRIOR TO CONSTRUCTION. |

TOTAL SPRINKLERS: 172

# SHEET INDEX


FP001 SYMBOL LIST AND GENERAL NOTES - FIRE PROTECTION

FP101 SITE PLAN - FIRE PROTECTION

FP113 DEMOLITION REFLECTED CEILING PLANS - FIRE PROTECTION

FP114 REFLECTED CEILING PLANS - FIRE PROTECTION
FP115 ATTIC PLAN - FIRE PROTECTION

FP500 SECTIONS - FIRE PROTECTION

FP600 DETAILS - FIRE PROTECTION
FP601 DETAILS - FIRE PROTECTION


ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.

LICENSED PROFESSIONAL ENGINEER
KENTON AIKENS
P30101
IOWA
Exp: 12/31/26

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA<br>SUBCONTRACTOR: INTERFACE ENGINEERING INC. PORTLAND, OR | DESIGNED: ETB<br>CADD: KRA<br>TECH REVIEW: KJA<br>DATE: 05/29/2026 | SUB SHEET NO. FP001 | TITLE OF SHEET: SYMBOLS LIST AND GENERAL NOTES - FIRE PROTECTION<br>EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248<br>PMIS/PKG NO. 318539<br>SHEET 93 OF 142 |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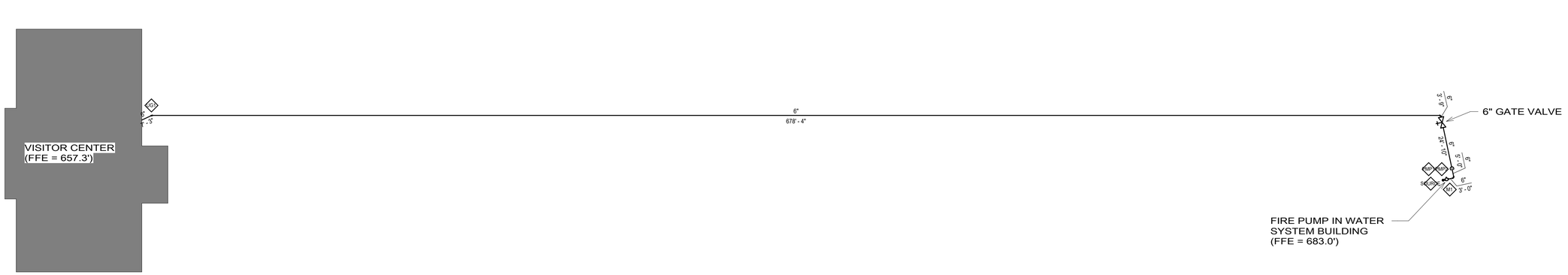
## GENERAL NOTES

A. SITE PLAN IS PROVIDED FOR INFORMATION PURPOSES ONLY. REFER TO CIVIL FOR ALL SITE RELATED WORK.




1 FP101 SITE PLAN - FIRE PROTECTION
SCALE A

SCALE: A 8 0 8 16
SCALE OF FEET

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.

LICENSED PROFESSIONAL ENGINEER
KENTON J. AIKEN
P30101
IOWA
Exp: 12/31/26

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA<br>SUBCONTRACTOR: INTERFACE ENGINEERING INC. PORTLAND, OR | DESIGNED: ETB<br>CADD: KRA<br>TECH REVIEW: KJA<br>DATE: 05/29/2026 | SUB SHEET NO. FP101 | TITLE OF SHEET: SITE PLAN - FIRE PROTECTION<br>EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248<br>PMIS/PKG NO. 318539<br>SHEET 94 OF 142 |

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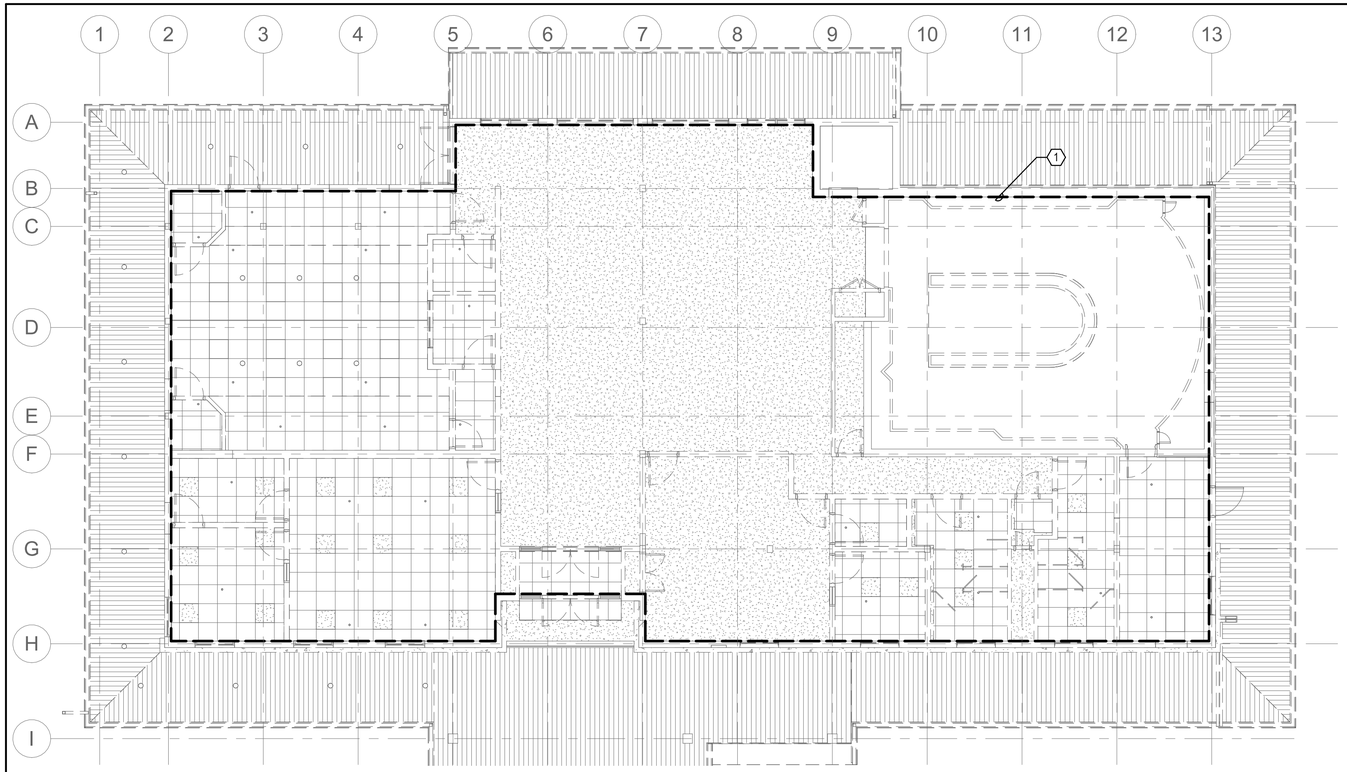
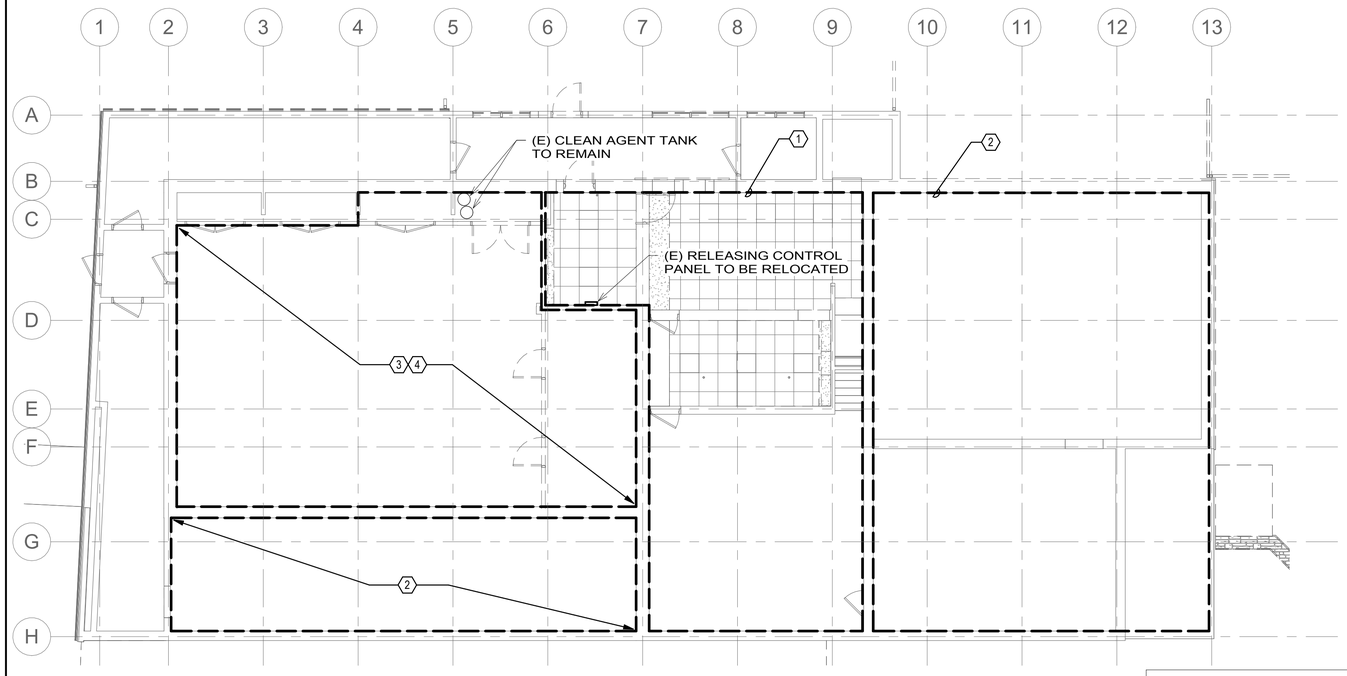



2 FP113 FIRST FLOOR DEMO FIRE PROTECTION PLAN - OVERALL
SCALE A




1 FP113 BASEMENT LEVEL DEMO FIRE PROTECTION PLAN - OVERALL
SCALE A

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.

## SHEET KEYNOTES

1. DEMOLISH EXISTING SPRINKLER SYSTEM IN ITS ENTIRETY.
2. NO EXISTING SPRINKLERS, PIPING, FITTINGS, ETC. IN THIS AREA.
3. DEMOLISH EXISTING CLEAN AGENT SYSTEM PIPING, FITTINGS, AND NOZZLES AS REQUIRED TO ACCOMMODATE THE NEW SPACE CONFIGURATION. SEE DIVISION 28 FOR ADDITIONAL WORK FOR THE CLEAN AGENT SYSTEM.

SCALE: A
8 0 8 16
SCALE OF FEET

LICENSED PROFESSIONAL ENGINEER
KENTON J. AIKENS
P30101
IOWA
Exp: 12/31/26

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA<br>SUBCONTRACTOR: INTERFACE ENGINEERING INC. PORTLAND, OR | DESIGNED: ETB<br>CADD KRA<br>TECH REVIEW: KJA<br>DATE: 05/29/2026 | SUB SHEET NO. FP113 | TITLE OF SHEET: DEMOLITION REFLECTED CEILING PLANS - FIRE PROTECTION<br>EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248<br>PMIS/PKG NO. 318539<br>SHEET 95 OF 142 |

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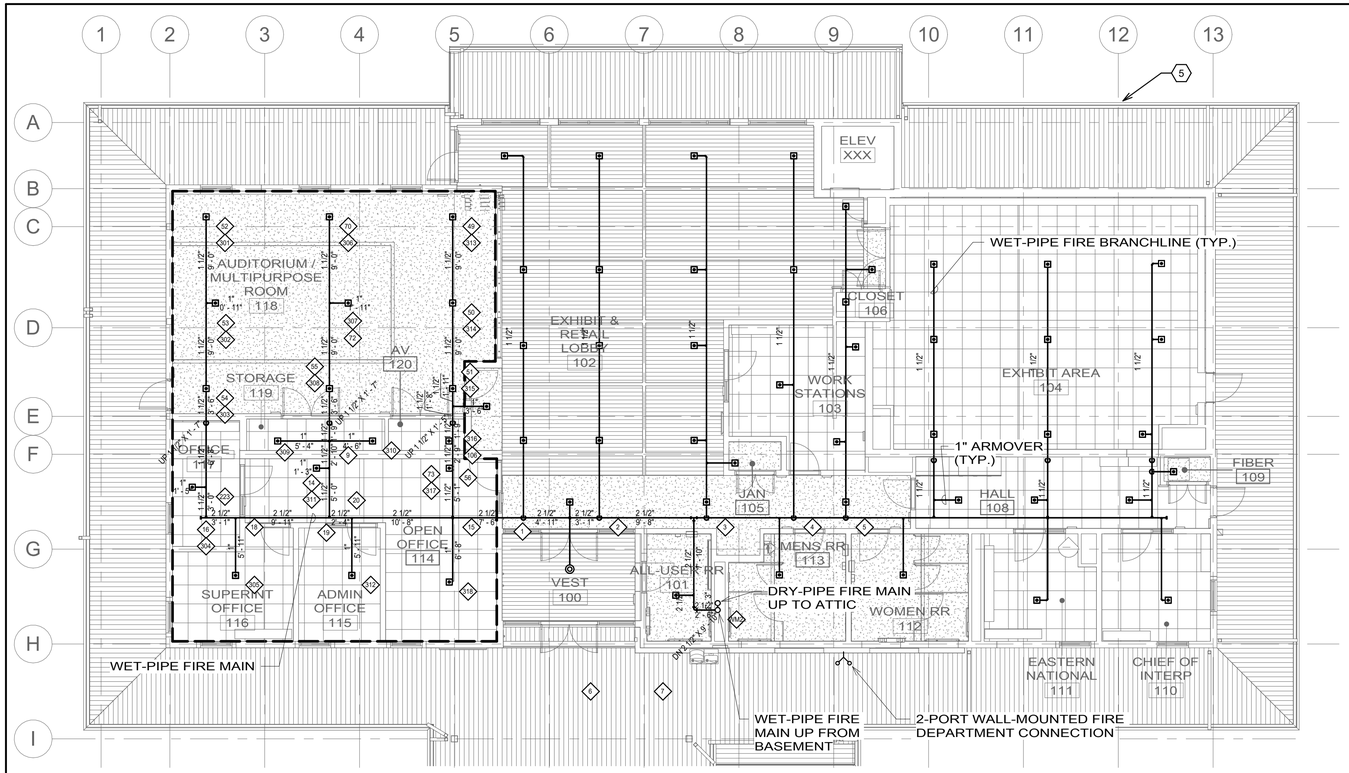
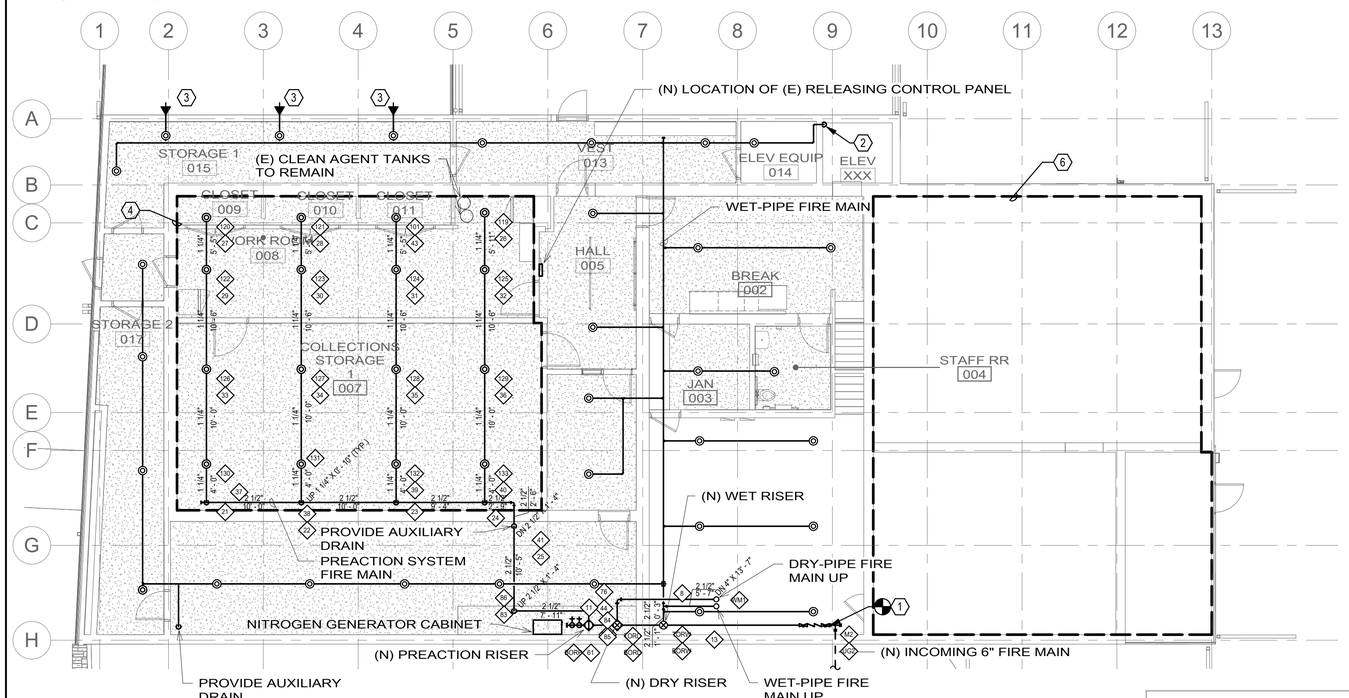



2 FP114 FIRST FLOOR FIRE PROTECTION PLAN - OVERALL
SCALE A

| Hydraulic Information | |
|---|---|
| Remote Area Name | LEVEL 1 |
| Remote Area Size | 1585 SF |
| Occupancy Classification | Light Hazard |
| Density | 0.10 GPM/SF |
| K-Factor | 5.6 |
| Flowing Heads | 18 @ 9.00 |
| Head Coverage | 168 MAX. SF |
| Flow at Base of Riser | 365.27 GPM |
| Pressure at Base of Riser | 71.21 PSI |
| Flow at Source | 465.27 GPM |
| Pressure at Source | 67.64 PSI |
| Total Hose Streams | 100.00 GPM |
| Safety Margin | +12.78 (18.9%) |
| Comments: | |
| NFPA 13 - 2022 | |

## SHEET KEYNOTES

1. PROVIDE 6" BACKFLOW PREVENTER BEFORE INCOMING LINE CONNECTS TO SPRINKLER SYSTEM RISERS.
2. DOWN TO FEED SIDEWALL SPRINKLER IN ELEVATOR PIT. PROVIDE 1" SHUTOFF VALVE, ZERO DELAY FLOWSWITCH, AND TEE WITH CAP TO FACILITATE DRAINAGE.
3. DRY SIDEWALL SPRINKLER ONLY TO BE PROVIDED IF DECK IS EXPANDED PER BID OPTION D.
4. MODIFY EXISTING CLEAN AGENT FIRE EXTINGUISHING SYSTEM PIPING, NOZZLES, AND TANK FILL BASED ON NEW HYDRAULIC CALCULATIONS TO ACCOMMODATE THE NEW SPACE CONFIGURATION. PROVIDE SEALING OF ROOM TO ACHIEVE THE REQUIRED AGENT CONCENTRATION IN ACCORDANCE WITH NFPA 2001 AND MANUFACTURER REQUIREMENTS. PROVIDE DOOR FAN TEST TO CONTRACING OFFICER AS REQUESTED.
5. SPRINKLER PROTECTION NOT REQUIRED IN ACCORDANCE WITH NFPA 13 (2022 EDITION) SECTION 9.2.3.2.
6. SPRINKLER PROTECTION NOT REQUIRED IN ACCORDANCE WITH NFPA 13 (2022 EDITION) SECTION 9.2.2.




1 FP114 BASEMENT LEVEL FIRE PROTECTION PLAN - OVERALL
SCALE A

| Hydraulic Information | |
|---|---|
| Remote Area Name | REMOTE AREA 1 - COLLECTIONS STORAGE |
| Remote Area Size | 1262 SF |
| Occupancy Classification | Ordinary Hazard Group 1 |
| Density | 0.15 GPM/SF |
| K-Factor | 5.6 |
| Flowing Heads | 16 @ 12.13 |
| Head Coverage | 130 MAX. SF |
| Flow at Base of Riser | 340.97 GPM |
| Pressure at Base of Riser | 59.71 PSI |
| Flow at Source | 590.97 GPM |
| Pressure at Source | -24.50 PSI |
| Total Hose Streams | 250.00 GPM |
| Safety Margin | +25.31 (45.6%) |
| Comments: | ENTIRE PREACTION SYSTEM IS FLOWING AND HYDRAULICALLY CALCULATED. DRY CAPACITY = 26.3 GAL |
| NFPA 13 - 2022 | |

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.

SCALE: A 8 0 8 16
SCALE OF FEET

KENTON J. AIKENS P30101 IOWA LICENSED PROFESSIONAL ENGINEER
Exp: 12/31/26

| | | | |
|---|---|---|---|
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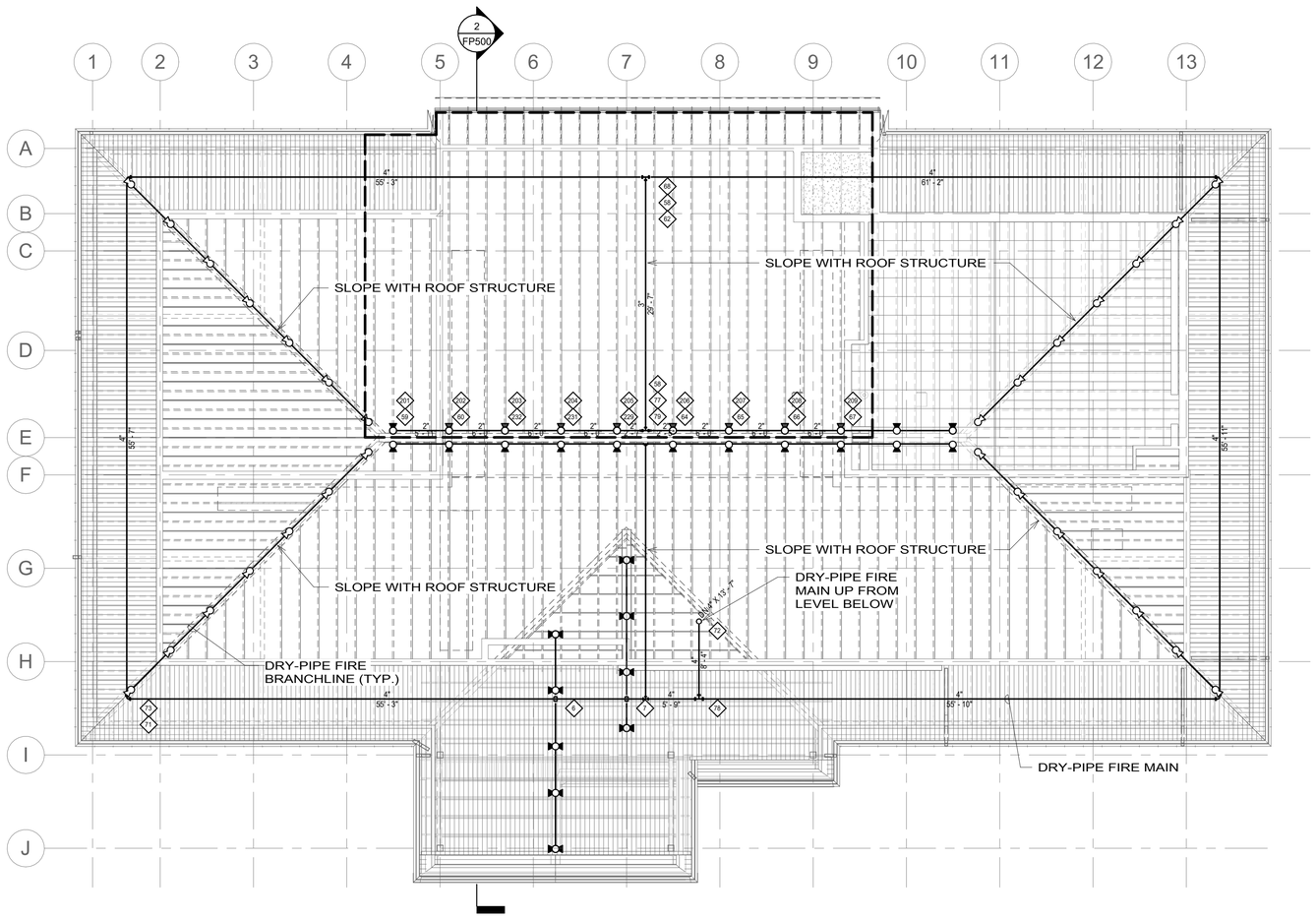


1 FP115 ATTIC PLAN - FIRE PROTECTION
SCALE A

| Hydraulic Information | |
|---|---|
| Remote Area Name | REMOTE AREA - ATTIC |
| Remote Area Size | 2160 SF |
| Occupancy Classification | Light Hazard |
| Density | 0.10 GPM/SF |
| K-Factor | 5.6 |
| Flowing Heads | 9 @ 36.00 |
| Head Coverage | 240 SF MAX SF |
| Flow at Base of Riser | 330.88 GPM |
| Pressure at Base of Riser | 72.81 PSI |
| Flow at Source | 430.88 GPM |
| Pressure at Source | 68.19 PSI |
| Total Hose Streams | 100.00 GPM |
| Safety Margin | +12.97 (19.0%) |
| Comments: NFPA 13 - 2022 | DRY SYSTEM CAPACITY = 332.8 GAL NINE SPRINKLERS FLOWING PER MANUFACTURER'S CALCULATION REQUIREMENTS |

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.

SCALE: A 8 0 8 16
SCALE OF FEET


LICENSED PROFESSIONAL ENGINEER
KENTON J. AIKENS
P30101
IOWA
Exp: 12/31/26

A/E FIRM:
PRIME:
OTAK, INC.
REDMOND, WA
SUBCONTRACTOR:
INTERFACE ENGINEERING INC.
PORTLAND, OR


DESIGNED: ETB
CADD: KRA
TECH REVIEW: KJA
DATE: 05/29/2026

SUB SHEET NO. FP115

TITLE OF SHEET
ATTIC PLAN - FIRE PROTECTION

EFFIGY MOUNDS NATIONAL MONUMENT

DRAWING NO. 394 194248
PMIS/PKG NO. 318539
SHEET 97 OF 142


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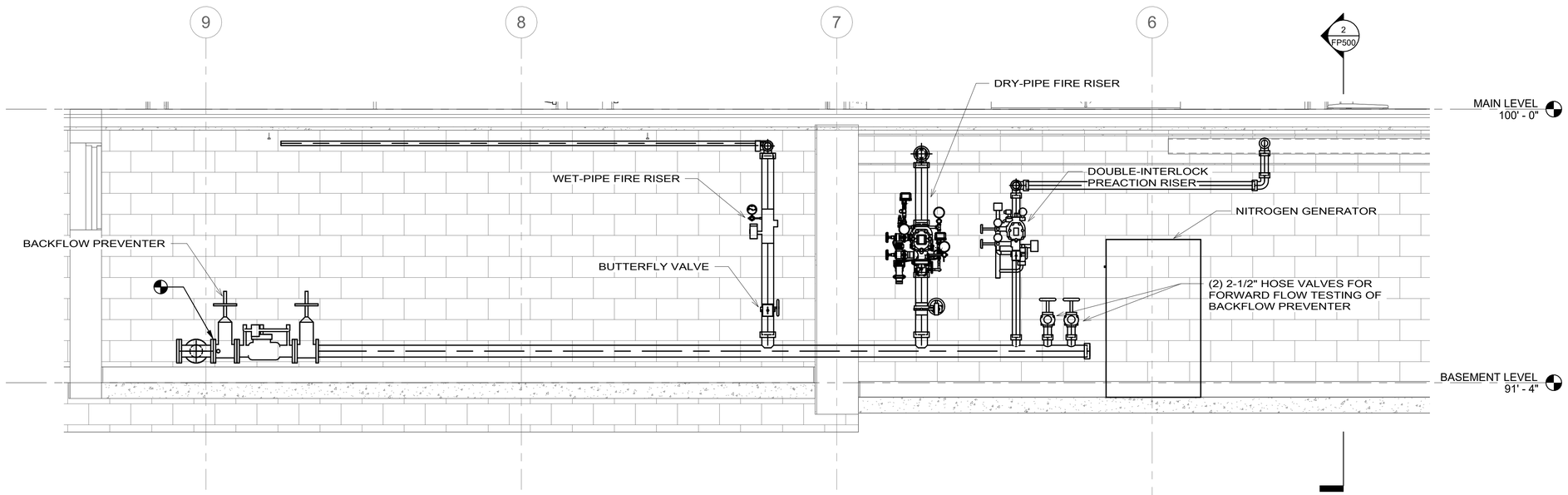
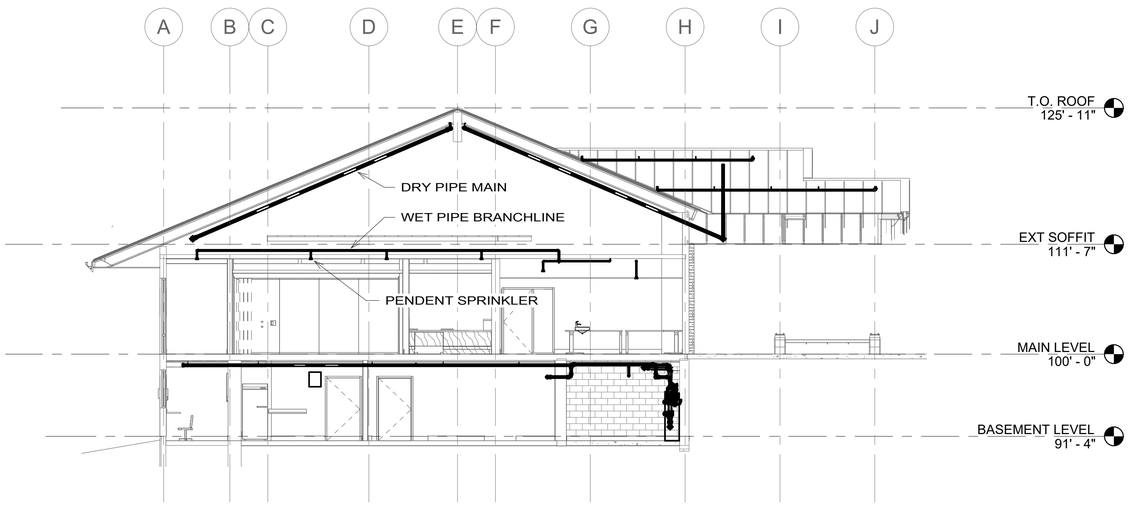



1 FP500 MANIFOLD SECTION - FIRE PROTECTION

SCALE (D)




2 FP500 BUILDING SECTION - FIRE PROTECTION

SCALE (A)

SCALE: (D) 2 0 2 4 SCALE OF FEET

SCALE: (A) 8 0 8 16 SCALE OF FEET

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.

LICENSED PROFESSIONAL ENGINEER
KENTON J. AIKEN
P30101
IOWA
Exp: 12/31/26

| | | | | | |
|---|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA<br>SUBCONTRACTOR: INTERFACE ENGINEERING INC. PORTLAND, OR | DESIGNED: ETB<br>CADD KRA<br>TECH REVIEW: KJA<br>DATE: 05/29/2026 | SUB SHEET NO. FP500 | TITLE OF SHEET SECTIONS - FIRE PROTECTION<br>EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248<br>PMIS/PKG NO. 318539<br>SHEET 98 OF 142 |

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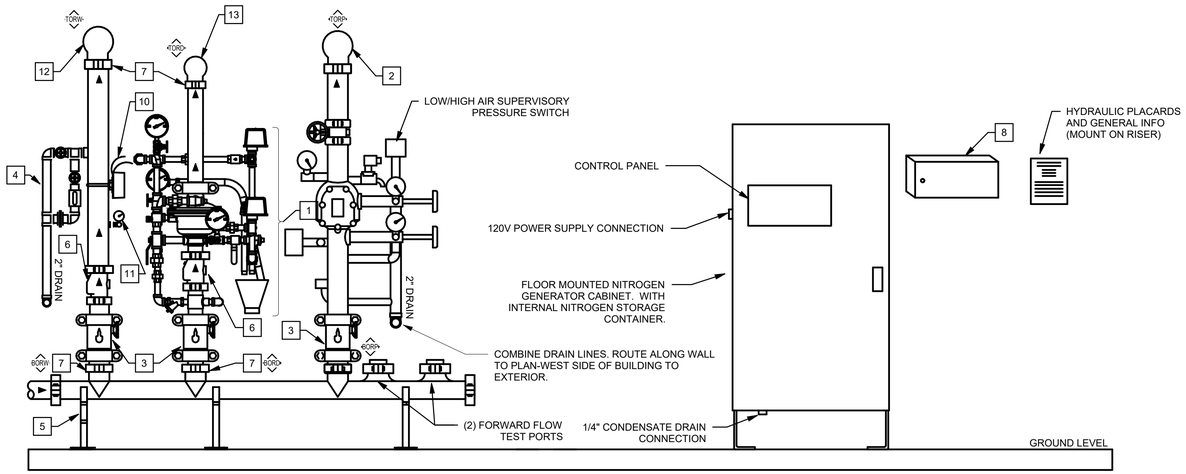
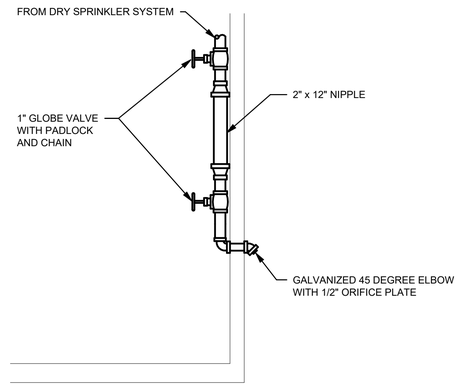



## 1 RISER MANIFOLD - FIRE PROTECTION

NO SCALE

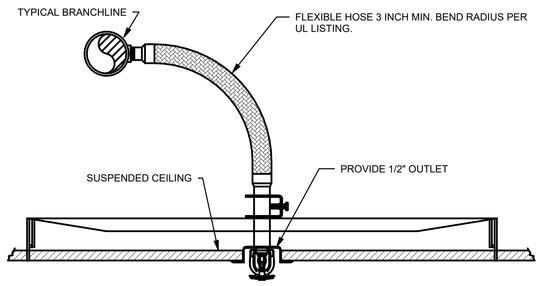
### DETAIL KEYNOTES

1 DRY PIPE VALVE WITH TRIM (COORDINATE CONNECTION OF AIR SUPERVISORY SWITCH AND WATERFLOW PRESSURE SWITCH).
2 TO PREACTION SYSTEM
3 BUTTERFLY VALVE (COORDINATE CONNECTION WITH FIRE ALARM CONTRACTOR)
4 SECTIONAL TEST AND DRAIN W/ PRESSUR RELIEF VAVLE. ROUTE TO EXTERIOR SAFE LOCATION.
5 FLOOR STAND.
6 CHECK VALVE.
7 GROOVED COUPLING.
8 SPARE SPRINKLER HEAD BOX.
9 ELECTRIC BELL. PROVIDE 120V POWER (COORDINATE POWER SUPPLY WITH ELECTRICAL).
10 WATERFLOW SWITCH (COORDINATE CONNECTION TO FIRE ALARM PANEL AND ELECTRIC BELL).
11 300 PSI PRESSURE GAUGE.
12 TO WET-PIPE SYSTEM.
13 TO DRY-PIPE SYSTEM.



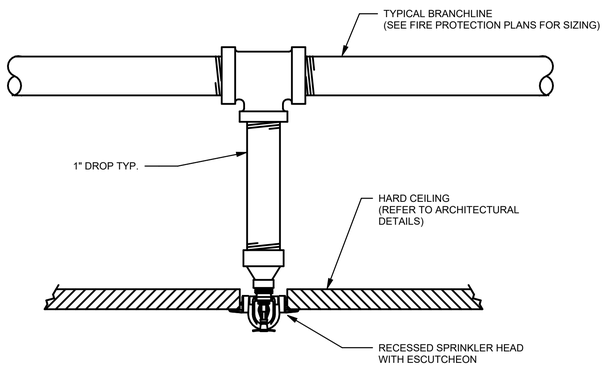

## 5 DRY SYSTEM - 1 INCH INSPECTOR'S TEST

NO SCALE



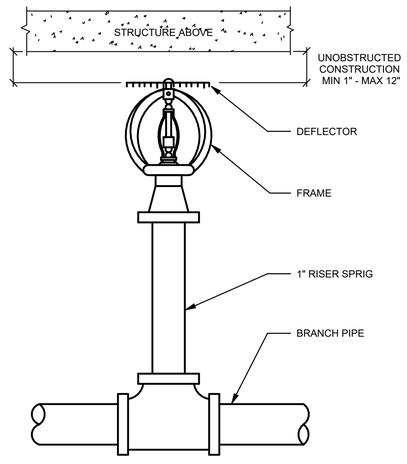

## 2 SPRINKLER ON FLEX DROP

NO SCALE



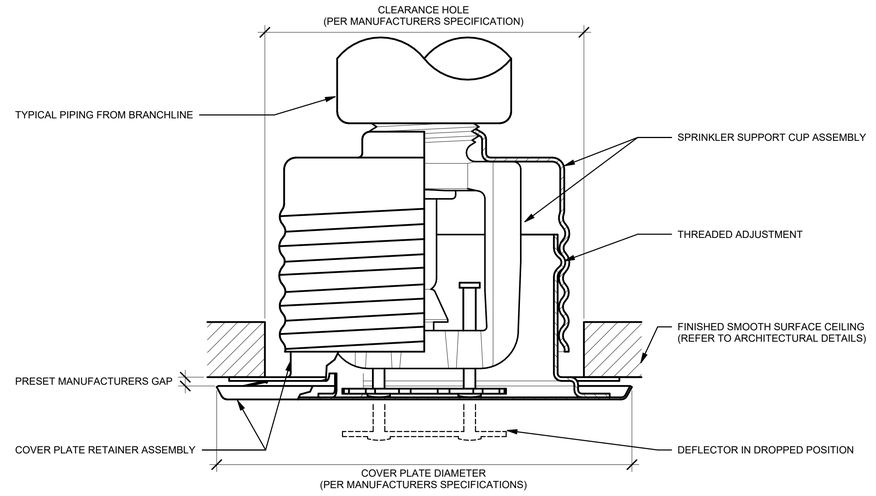

## 3 PENDENT IN HARD LID CEILING

NO SCALE




## 4 UPRIGHT HEAD ON SPRIG UP

NO SCALE




## 6 CONCEALED FLAT PLATE SPRINKLER

NO SCALE

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.

LICENSED PROFESSIONAL ENGINEER — KENTON J. AIKENS — P30101 — IOWA — Exp: 12/31/26

| | | | | | |
|---|---|---|---|---|---|
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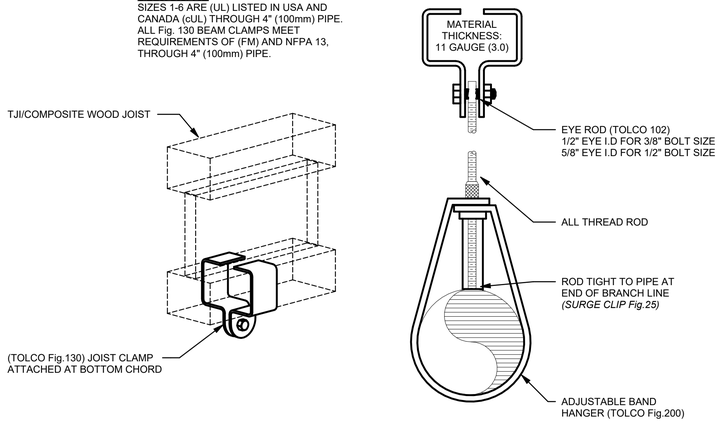
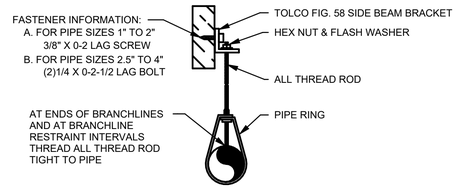
**APPROVALS:**
SIZES 1-6 ARE (UL) LISTED IN USA AND CANADA (cUL) THROUGH 4" (100mm) PIPE. ALL Fig. 130 BEAM CLAMPS MEET REQUIREMENTS OF (FM) AND NFPA 13, THROUGH 4" (100mm) PIPE.



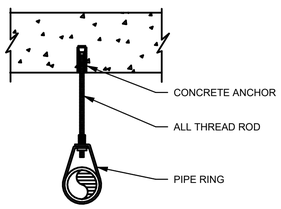

## 1 COMPOSITE WOOD JOIST CLAMP

NO SCALE



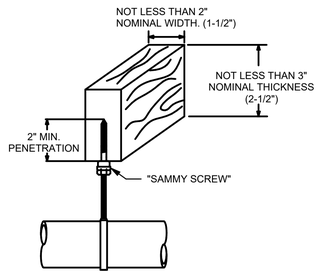

## 4 SIDE BEAM BRACKET HANGER

NO SCALE




## 2 DRILLED CONCRETE ANCHOR HANGER

NO SCALE




## 3 WOOD JOIST "SAMMY" HANGER

NO SCALE

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.

| | | | | | |
|---|---|---|---|---|---|
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# FIRE ALARM SYMBOL LIST

NOTE: This is a standard symbol list and not all items listed may be used.

## Abbreviations

| | |
|---|---|
| (E) | EXISTING |
| (R) | REUSE/REINSTALL/RELOCATE |
| AHJ | AUTHORITY HAVING JURISDICTION |
| AUX | AUXILIARY |
| AWG | AMERICAN WIRE GAUGE |
| BAS | BUILDING AUTOMATION SYSTEM |
| C | CONDUIT |
| CLG | CEILING |
| CO | CARBON MONOXIDE |
| COORD | COORDINATE |
| DIA | DIAMETER |
| DWG | DRAWING |
| EA | EACH |
| EMT | ELECTRICAL METALLIC TUBING |
| FA | FIRE ALARM |
| FAA | FIRE ALARM ANNUNCIATOR |
| FACP | FIRE ALARM CONTROL PANEL |
| FF | FINISHED FLOOR |
| IN | INCH, INCHES |
| LED | LIGHT EMITTING DIODE |
| LV | LOW VOLTAGE |
| mA | MILLIAMPERE |
| MIN | MINIMUM |
| MISC | MISCHELLANEOUS |
| N/A | NOT APPLICABLE |
| NAC | NOTIFICATION APPLINACE CIRCUIT |
| NTS | NOT TO SCALE |
| PG | PROTECTIVE GUARD |
| PNL | PANEL |
| PVC | PANEL |
| PWR | POLYVINYL CHLORIDE |
| QTY | POWER |
| REF | QUANTITY |
| RFI | REFERENCE |
| SHT | REQUEST FOR INFORMATION |
| SIM | SHEET |
| SLC | SIMILAR |
| STD | SIGNALING LINE CIRCUIT |
| TBD | STANDARD |
| TYP | TO BE DETERMINED |
| UG | TYPICAL |
| UL | UNDERGROUND |
| V | UNDERWRITERS LABORATORIES |
| WG | VOLT, VOLTS |
| WP | WIREGUARD |
| | WEATHERPROOF |

## Fire Alarm

| | |
|---|---|
| FACP | COMBINATION FIRE ALARM/PRE ACTION CONTROL PANEL |
| NAC | NOTIFICATION APPLIANCE CIRCUIT PANEL |
| FAA | FIRE ALARM ANNUNCIATOR PANEL |
| FAC | FIRE ALARM COMMUNICATOR, CELLULAR |
| S | SMOKE DETECTOR |
| S CO | SMOKE/CARBON MONOXIDE DETECTOR |
| CO | CARBON MONOXIDE DETECTOR |
| SOV | SOLENOID VALVE |
| DOC | DOCUMENT BOX |
| H +F | HEAT DETECTOR (+F = FIXED, HIGH TEMP) |
| F | PULL STATION |
| WF | SPRINKLER FLOW SWITCH INPUT MODULE |
| VS | SPRINKLER TAMPER SWITCH INPUT MODULE |
| PS | SPRINKLER PRESSURE SWITCH INPUT MODULE |
| # | HORN/STROBE COMBINATION (# INDICATES MINIMUM CANDELA RATING, C = CEILING MOUNTED) |
| # | STROBE, CEILING MOUNTED (# INDICATES MINIMUM CANDELA RATING) |
| AIM | ADDRESSABLE INPUT MODULE |
| AOM | ADDRESSABLE OUTPUT MODULE |
| REL | RELEASING MODULE |
| SS | SURGE SUPPRESSION DEVICE |

## General

| | |
|---|---|
| X / X | DETAIL NUMBER AND SHEET LOCATION |
| X | KEYED NOTE |

## Clean Agent - Fire Suppression

| | |
|---|---|
| FSCP | FIRE SUPPRESSION CONTROL PANEL - CLEAN AGENT |
| ASD | AIR SAMPLING TYPE SMOKE DETECTOR |
| S AS | AIR SAMPLING POINT |
| S CA | SMOKE DETECTOR |
| R | MANUAL RELEASE STATION |
| A | ABORT STATION |
| # CA | HORN/STROBE COMBINATION (# INDICATES MINIMUM CANDELA RATING, C = CEILING MOUNTED) |
| # CA | STROBE, WALL MOUNTED (# INDICATES MINIMUM CANDELA RATING) |
| MM CA | MONITOR MODULE |

## GENERAL FIRE ALARM AND CLEAN AGENT SYSTEMS DEMO NOTES

A. DEMOLISH EXISTING FIRE ALARM SYSTEM IN ITS ENTIRETY FOR THE VISITOR CENTER.

B. RETAIN EXISTING CLEAN AGENT SYSTEM COMPONENTS EXCEPT IN AREAS WHICH ARE SLATED FOR DEMOLITION OR REMODEL. REMOVE DEVICES IN THOSE AREAS AND RETAIN FOR REUSE. DEMOLISH ALL UNUSED CONDUIT AND WIRING.

C. PROVIDE PROTECTION FOR ALL SURFACES, COLLECTIONS OR OTHER BUILDING CONTENTS DURING DEMOLITION.

D. PROVIDE FIRE WATCH WHEN THE BUILDING FIRE DETECTION SYSTEM IS NOT OPERABLE.

## CLEAN AGENT FIRE EXTINGUISHING CONTROL SYSTEM SCOPE NOTES

A. RETAIN EXISTING CLEAN AGENT SYSTEM CONTROL PANEL AND DEVICES FOR REUSE.

B. RELOCATE THE CLEAN AGENT CONTROL PANEL, MAINTENANCE BYPASS SWITCH AND OTHER COMPONENTS TO NEW LOCATION. EXTEND WIRING AND CONDUIT OR PROVIDE NEW WIRING AND CONDUIT AS NEEDED FOR NEW LOCATION. PROVIDE NEW SURGE PROTECTION FOR CLEAN AGENT CONTROL PANEL AC POWER CIRCUIT.

C. PROVIDE ADDITIONAL COMPONENTS INCLUDING DISCHARGE HORNS AND STROBES, SMOKE DETECTORS, MANUAL RELEASE STATIONS, SIGNS, AND OTHER APPURTENANCES TO ACCOMMODATE THE REMODEL.

D. PROVIDE AIR SAMPLING SMOKE DETECTION IN THE COLLECTIONS, WORK ROOM AREA AND CLOSETS, AND CONNECT TO CLEAN AGENT CONTROL PANEL.

E. REWIRE EXISTING CLEAN AGENT SYSTEM SO THAT THE CROSS ZONE CIRCUITS ARE CONFIGURED WITH THE AIR SAMPLING SMOKE DETECTOR ACTING AS ONE DETECTOR INPUT AND THE SPOT SMOKE DETECTORS ACTING AS THE CROSS ZONE SECOND DETECTOR INPUT FOR AGENT RELEASE.

F. REFERENCE DIVISION 21 FOR ADDITIONAL CLEAN AGENT SYSTEM SCOPE.

## FIRE ALARM SCOPE NOTES

A. PROVIDE NEW ADDRESSABLE FIRE ALARM SYSTEM THROUGHOUT THE VISITOR CENTER. PROVIDE FIRE ALARM NOTIFICATION, DETECTION, MONITORING, AND CONTROLS FOR FIRE SAFETY CONTROL FUNCTIONS.

- a. PROVIDE AUDIBLE NOTIFICATION THROUGHOUT THE BUILDING.
- b. PROVIDE VISUAL NOTIFICATION IN ALL PUBLIC AND COMMON USE SPACES.
- c. PROVIDE MONITORING OF THE EXISTING CLEAN AGENT RELEASING CONTROL PANEL.
- d. PROVIDE RELEASING CONTROL FOR PREACTION SPRINKLER SYSTEM.
- e. PROVIDE SMOKE DETECTION CONNECTED TO THE FIRE ALARM SYSTEM. LOCATE THROUGHOUT THE BUILDING WITH THE EXCEPTION OF THE RESTROOMS, COLLECTIONS STORAGE, WORKROOM AND CLOSETS, AND SPACES NO SUITABLE FOR SMOKE DETECTION. IN SPACES NOT SUITABLE FOR SMOKE DETECTION, THE FIRE SPRINKLERS WILL ACT AS THE FIRE ALARM SYSTEM INITIATING DEVICES.
- f. PROVIDE MONITORING OF THE FIRE PUMP AND ASSOCIATED VALVES AND FLOW SWITCHES LOCATED IN THE PUMP HOUSE.
- g. PROVIDE MONITORING OF VALVES, PRESSURE SWITCHES, AND FLOW SWITCHES IN THE VISITOR CENTER.

B. DESIGN AND INSTALL SYSTEM IN ACCORDANCE WITH APPLICABLE CODES.

C. CONCEAL WIRING AND CONDUIT IN ALL FINISHED SPACES. NO EXPOSED WIRING ALLOWED.

## APPLICABLE CODES

A. 2024 INTERNATIONAL BUILDING CODE (IBC)
B. 2024 INTERNATIONAL FIRE CODE (IFC)
C. ARCHITECTURAL BARRIERS ACT ACCESSIBILITY STANDARDS (ABAAS)
D. 2023 NATIONAL ELECTRICAL CODE (NEC)
E. 2023 NATIONAL MECHANICAL CODE (NMC)
F. 2022 NATIONAL FIRE ALARM AND SIGNALING CODE (NFPA 72)
G. MPS STRUCTURAL FIRE MANAGEMENT REFERENCE MANUAL 58 (RM58)

## SHEET INDEX


| | |
|---|---|
| FA001 | SYMBOL LIST AND GENERAL NOTES - FIRE ALARM |
| FA101 | SITE PLAN - FIRE ALARM |
| FA113 | DEMOLITION FLOOR PLANS - FIRE ALARM |
| FA114 | FLOOR PLANS - FIRE ALARM |
| FA600 | DETAILS - FIRE ALARM |
| FA601 | DETAILS - FIRE ALARM |


ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.

LICENSED PROFESSIONAL ENGINEER
KENTON J. AIKENS
P30101
IOWA
Exp: 12/31/26

| | | | | |
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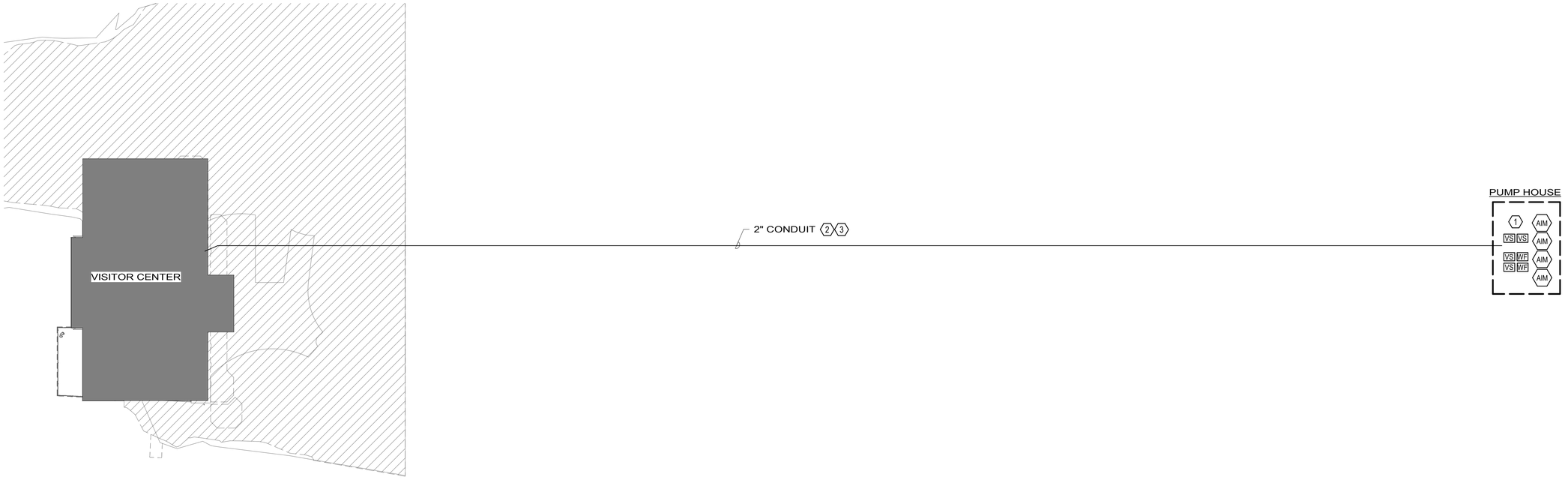
## SHEET KEYNOTES

1. PROVIDE MONITORING OF THE FIRE PUMP CONTROLLER IN THE PUMP HOUSE BUILDING AND ALL ASSOCIATED VALVES AND FLOW SWITCHES. COORDINATE WITH FIRE PROTECTION FOR EXACT QUANTITIES, LOCATIONS, AND CONNECTIONS TO VALVES AND CONTROLLER.
2. UTILIZE SAME TRENCH AS FIRE SPRINKLER MAIN.
3. CONDUIT PROVIDED IN BASE SCOPE.




1 FA101 SITE PLAN - FIRE ALARM
SCALE A

SCALE: A 8 0 8 16
SCALE OF FEET

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION D UNLESS NOTED OTHERWISE

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KENTON J. AIKENS
P30101
IOWA
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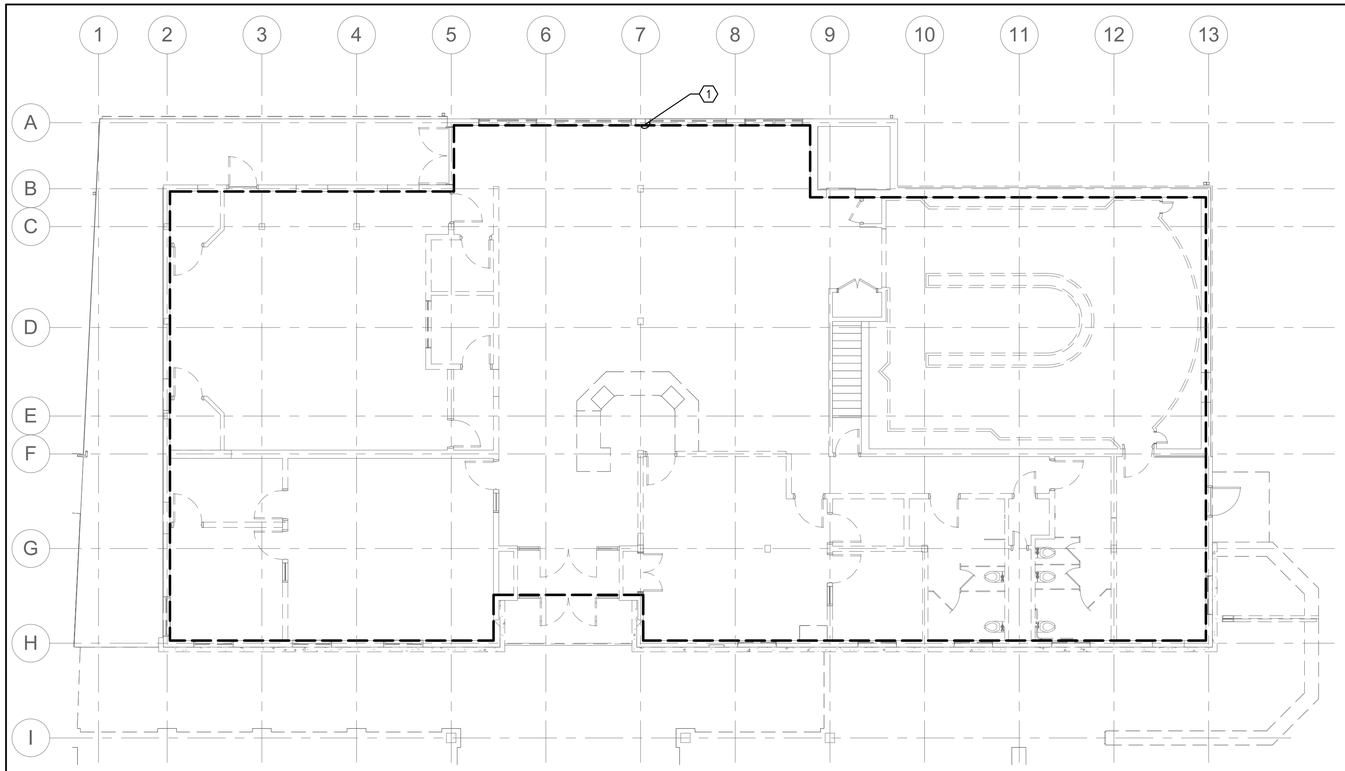
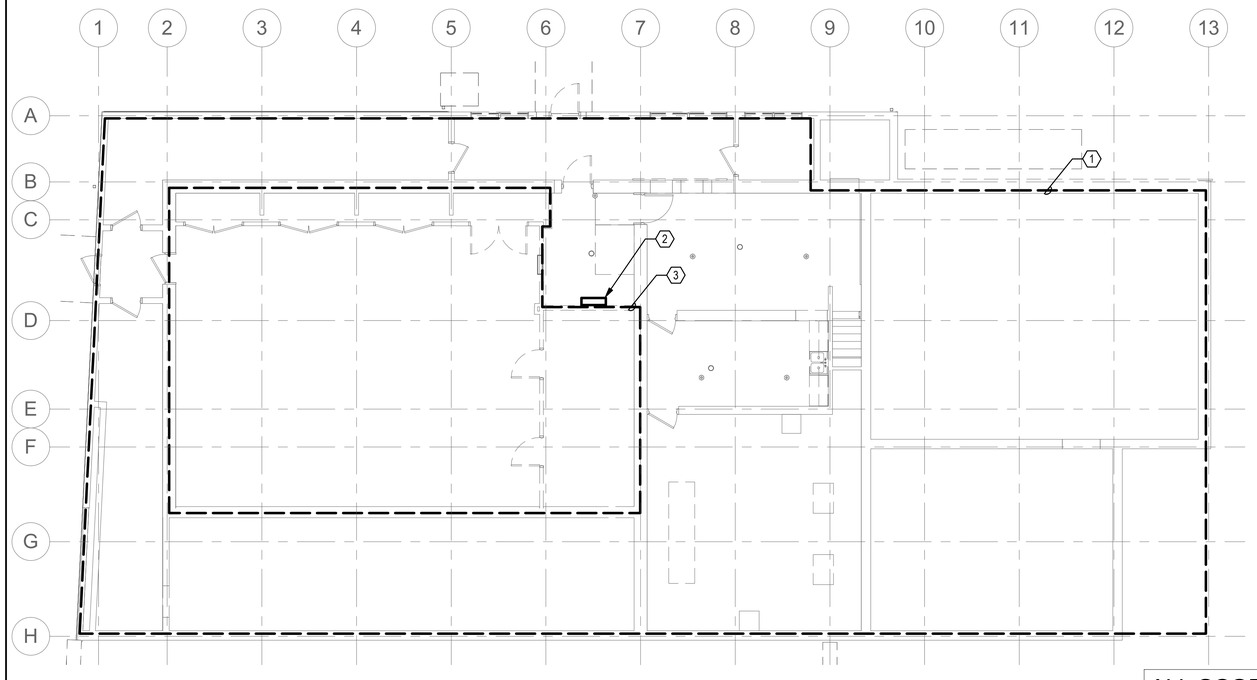



2 FA113 FIRST FLOOR DEMO FIRE ALARM PLAN - OVERALL
SCALE A




1 FA113 BASEMENT LEVEL DEMO FIRE ALARM PLAN - OVERALL
SCALE A

## SHEET KEYNOTES

1. DEMOLISH EXISTING FIRE ALARM SYSTEM IN ITS ENTIRETY.
2. REMOVE EXISTING CLEAN AGENT PANEL, BYPASS SWITCH, AND OTHER CLEAN AGENT SYSTEM COMPONENTS FROM WALL SCHEDULED FOR DEMOLITION. RETAIN EQUIPMENT FOR RELOCATION.
3. REMOVE AND RELOCATE CLEAN AGENT DEVICES, APPLIANCES, CONDUIT, ELECTRICAL BOXES, WIRING, ETC., AS NEEDED TO ACCOMMODATE THE RECONFIGURATION OF THE PROTECTED SPACE. DO NOT LEAVE ANY UNUSED CONDUIT OR WIRING.

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.

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KENTON J. KIKENS
P30101
IOWA
Exp. 12/31/26

SCALE: A 8 0 8 16
SCALE OF FEET

| | | | | |
|---|---|---|---|---|
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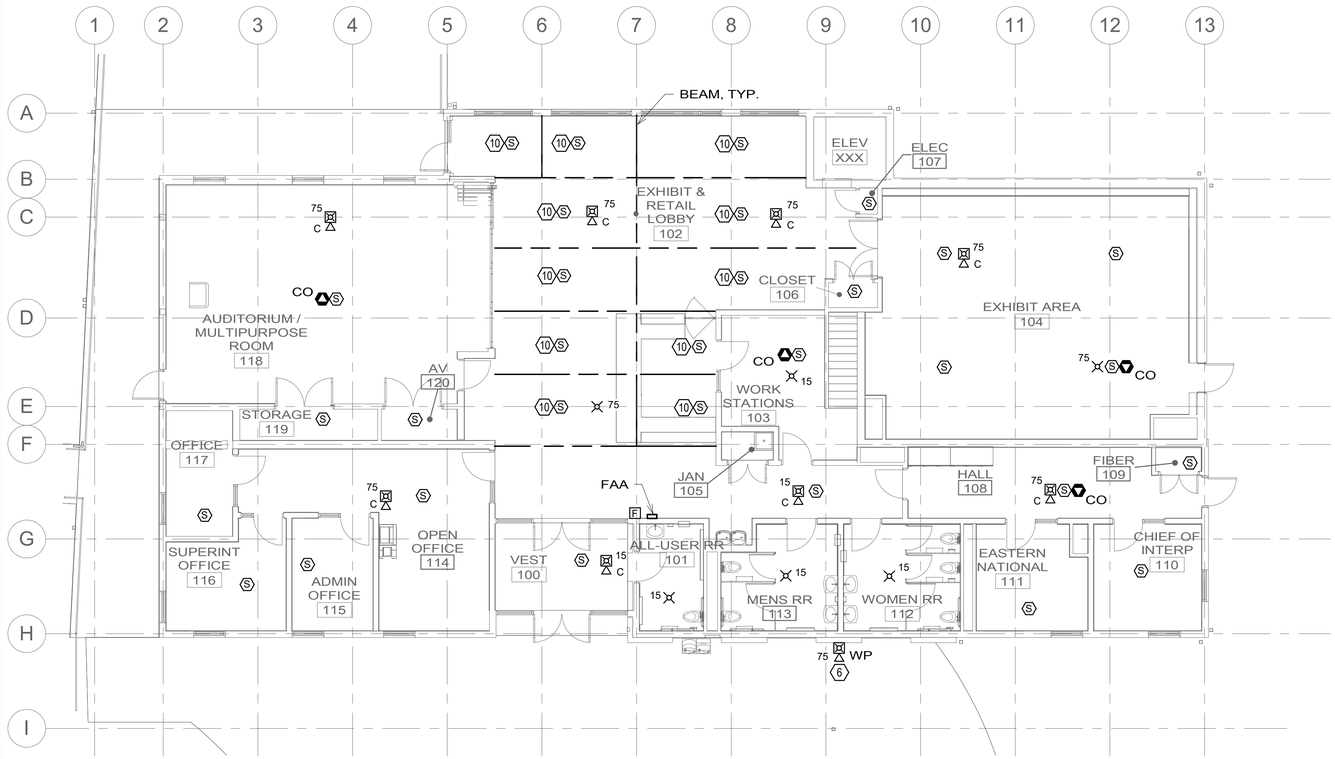
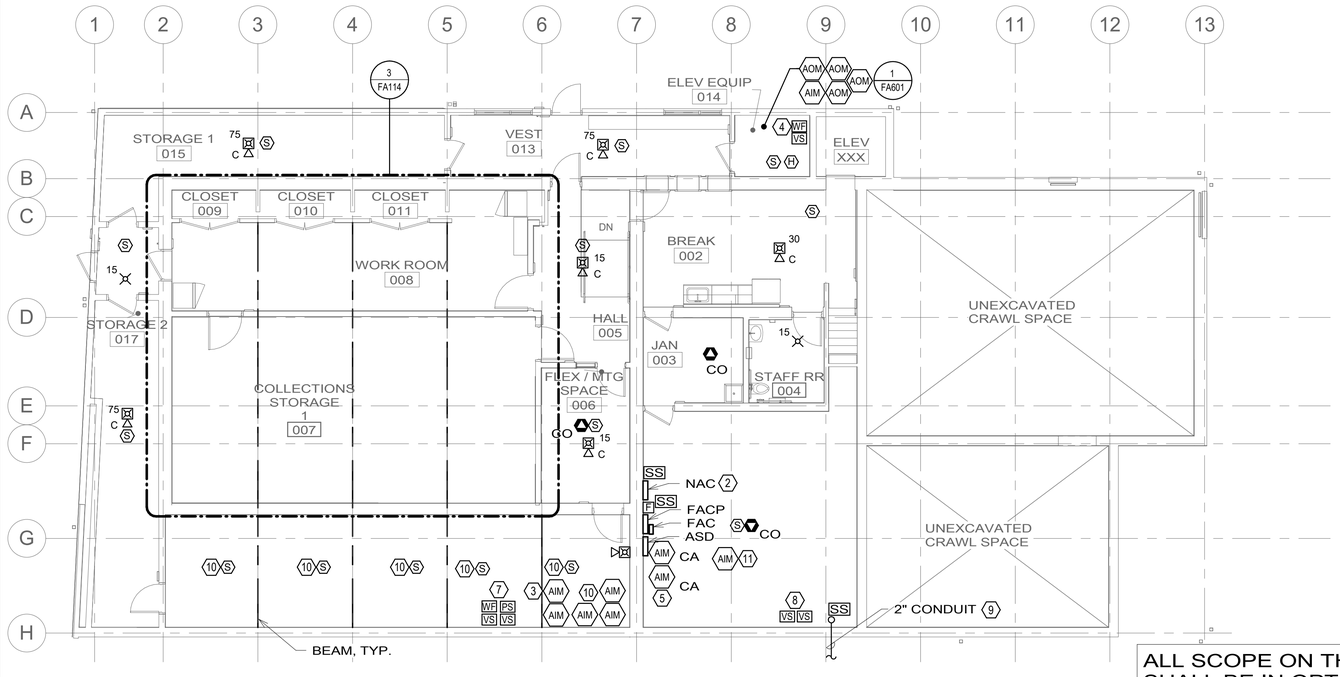


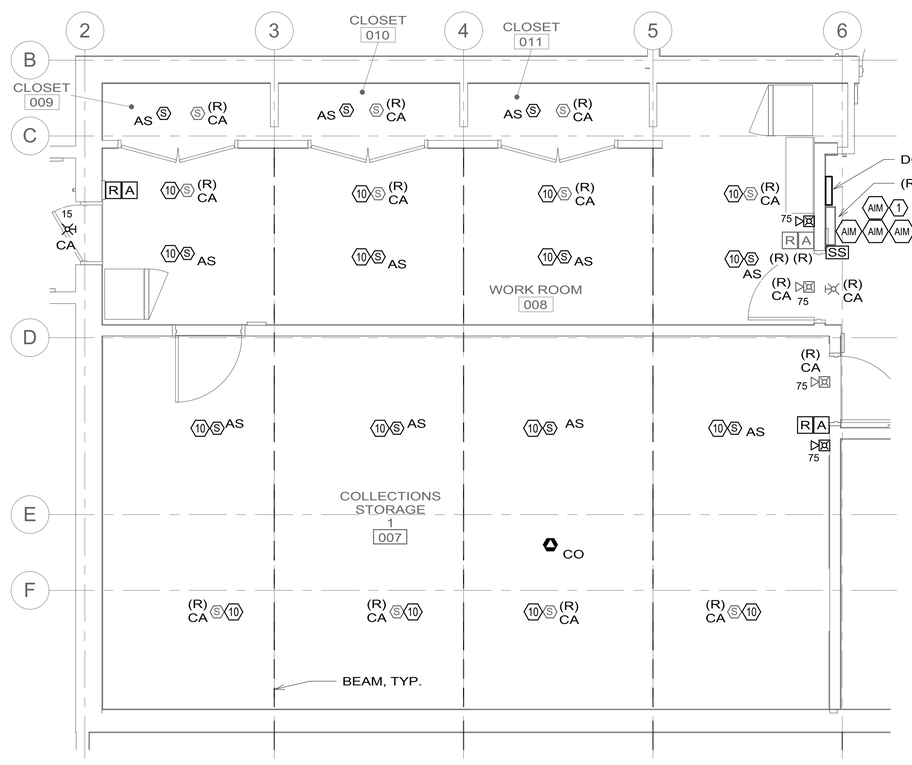

2 FA114 FIRST FLOOR FIRE ALARM PLAN - OVERALL
SCALE A




1 FA114 BASEMENT LEVEL FIRE ALARM PLAN - OVERALL
SCALE A

## SHEET KEYNOTES

1. PROVIDE ADDRESSABLE INPUT MODULES TO MONITOR CLEAN AGENT RELEASING CONTROL PANEL FOR ALARM, TROUBLE, SUPERVISORY, AND DISCHARGE CONDITIONS.
2. PROVIDE QUANTITY OF FIRE ALARM POWER SUPPLIES REQUIRED FOR A COMPLETE AND OPERABLE FIRE ALARM SYSTEM.
3. PROVIDE ADDRESSABLE INPUT MODULES TO MONITOR THE NITROGEN GENERATOR SYSTEM FOR TROUBLE AND SUPERVISORY CONDITIONS.
4. PROVIDE MONITORING OF FLOW SWITCH FOR ELEVATOR PIT SPRINKLER. FIRE ALARM TO PERFORM PRIMARY RECALL FUNCTION UPON ACTIVATION OF FLOW SWITCH.
5. PROVIDE MONITOR MODULE FOR FSCP TO MONITOR AIR SAMPLING SMOKE DETECTOR FOR ALARM CONDITION.
6. PROVIDE HORN/STROBE TO SOUND UPON FIRE SPRINKLER WATER FLOW. HORN TO SILENCE ONLY UPON CEASE OF WATER FLOW. INSTALL ABOVE FDC.
7. PROVIDE ADDRESSABLE INPUT MODULES TO MONITOR THE FIRE SPRINKLER WET RISER AND DRY RISER. COORDINATE WITH FIRE SPRINKLER DESIGN FOR EXACT QUANTITY AND LOCATION OF CONNECTIONS.
7. PROVIDE ADDRESSABLE INPUT MODULES TO MONITOR THE FIRE SPRINKLER WET RISER AND DRY RISER. COORDINATE WITH FIRE SPRINKLER DESIGN FOR EXACT QUANTITY AND LOCATION OF CONNECTIONS.
8. PROVIDE ADDRESSABLE INPUT MODULES TO MONITOR THE FIRE SPRINKLER BACKFLOW PREVENTER. COORDINATE WITH FIRE SPRINKLER DESIGN FOR EXACT QUANTITY AND LOCATION OF CONNECTIONS.
9. 2" CONDUIT TO THE FIRE PUMP HOUSE BUILDING PROVIDED IN BASE SCOPE. SEE FA101 FOR MONITORING OF THE FIRE PUMP.
10. INSTALL DETECTOR WITHIN BEAM POCKET.
11. PROVIDE ADDRESSBLE INPUT MODULE FOR FACP TO MONITOR ASD FOR LOW LEVEL ALERT. LOW LEVEL ALERT TO INITIATE A SUPERVISORY ALARM ON THE FIRE ALARM SYSTEM.




3 BASEMENT LEVEL ENLARGED FIRE ALARM PLAN
0' 2' 4' 8'
1/4" = 1'-0"

SCALE: A
8 0 8 16
SCALE OF FEET

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A UNLESS NOTED OTHERWISE

LICENSED PROFESSIONAL ENGINEER KENTON J. AIKENS 30101 IOWA Exp: 12/31/26

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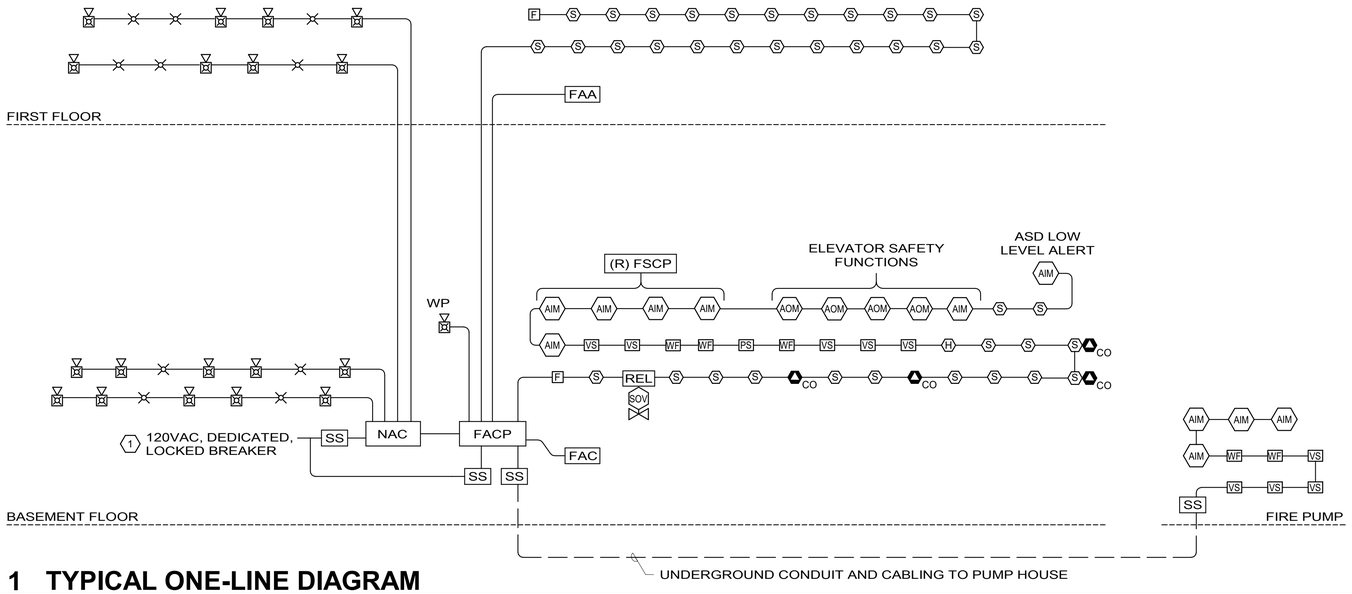
## GENERAL DETAIL NOTES

A. ONE-LINE DIAGRAM IS DIAGRAMMATIC AND NOT ALL DEVICES, APPLIANCES, OR EQUIPMENT IS SHOWN.

## DETAIL KEYNOTES

1. 120VAC PRIMARY POWER SOURCE SHALL BE A MECHANICALLY PROTECTED BRANCH CIRCUIT. THE CIRCUIT DISCONNECTING MEANS SHALL HAVE A RED MARKING AND IDENTIFIED AS "FIRE ALARM CIRCUIT".




## 1 TYPICAL ONE-LINE DIAGRAM

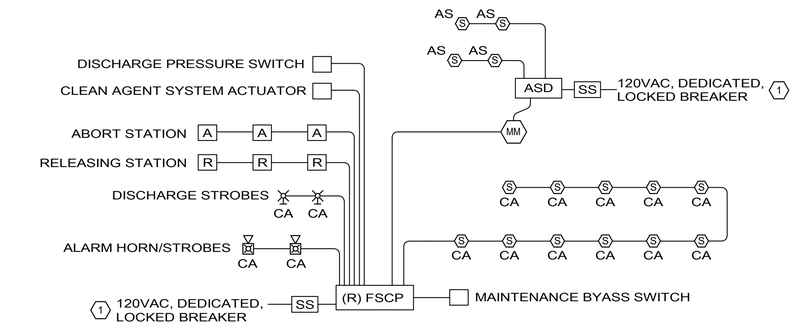
NO SCALE

## GENERAL DETAIL NOTES

A. ONE-LINE DIAGRAM IS DIAGRAMMATIC AND NOT ALL DEVICES, APPLIANCES, OR EQUIPMENT IS SHOWN.

## DETAIL KEYNOTES

1. 120VAC PRIMARY POWER SOURCE SHALL BE A MECHANICALLY PROTECTED BRANCH CIRCUIT. THE CIRCUIT DISCONNECTING MEANS SHALL HAVE A RED MARKING AND IDENTIFIED AS "FIRE ALARM CIRCUIT".




NOTE: FINAL CONFIGURATION OF CLEAN AGENT SYSTEM DEPICTED.

## 2 TYPICAL CLEAN AGENT ONE-LINE DIAGRAM

NO SCALE

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.

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## DETAIL KEYNOTES

F1 MOUNT 135° FIXED TEMPERATURE HEAT DETECTOR TO WITHIN 24 INCHES OF EACH SPRINKLER HEAD IN ELEVATOR MACHINE ROOM.

F2 MOUNT SMOKE DETECTOR IN ELEVATOR LOBBY ON CEILING TO WITHIN 21 FEET OF CENTER OF ELEVATOR DOOR.

F3 PROVIDE ADDRESSABLE FIRE ALARM CONTROL RELAY AND WIRING TO ELEVATOR CONTROLLER/DRIVE FOR ELEVATOR PRIMARY RECALL.

F4 PROVIDE ADDRESSABLE FIRE ALARM CONTROL RELAY AND WIRING TO ELEVATOR CONTROLLER FOR ELEVATOR ALTERNATE RECALL.

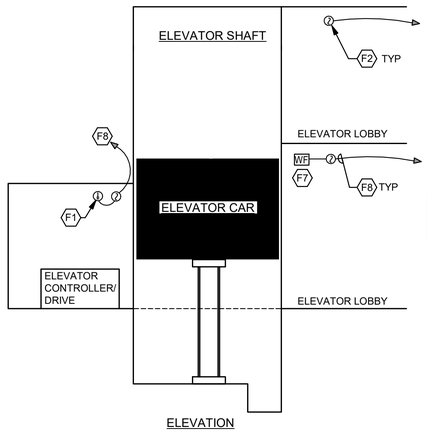
F5 PROVIDE ADDRESSABLE FIRE ALARM CONTROL RELAY AND WIRING TO ELEVATOR CONTROLLER TO ACTIVATE FIREMAN'S HAT LIGHT IN ELEVATOR CAB UPON INITIATION OF SMOKE DETECTION IN MACHINE ROOM OR WATERFLOW IN PIT.

F6 PROVIDE ADDRESSABLE FIRE ALARM CONTROL RELAY AND WIRING TO SHUNT TRIP CONTROL OF ELEVATOR POWER.

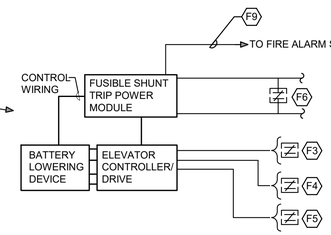
F7 PROVIDE ADDRESSABLE INPUT MODULE TO MONITOR FLOW SWITCH MONITORING ELEVATOR PIT SPRINKLER. FLOW SWITCH TO PERFORM PRIMARY RECALL.

F8 CONNECT TO FIRE ALARM SYSTEMS SIGNALING LINE CIRCUIT.

F9 CONNECT TO FIRE ALARM SYSTEM FOR MONITORING OF CONTROL POWER.



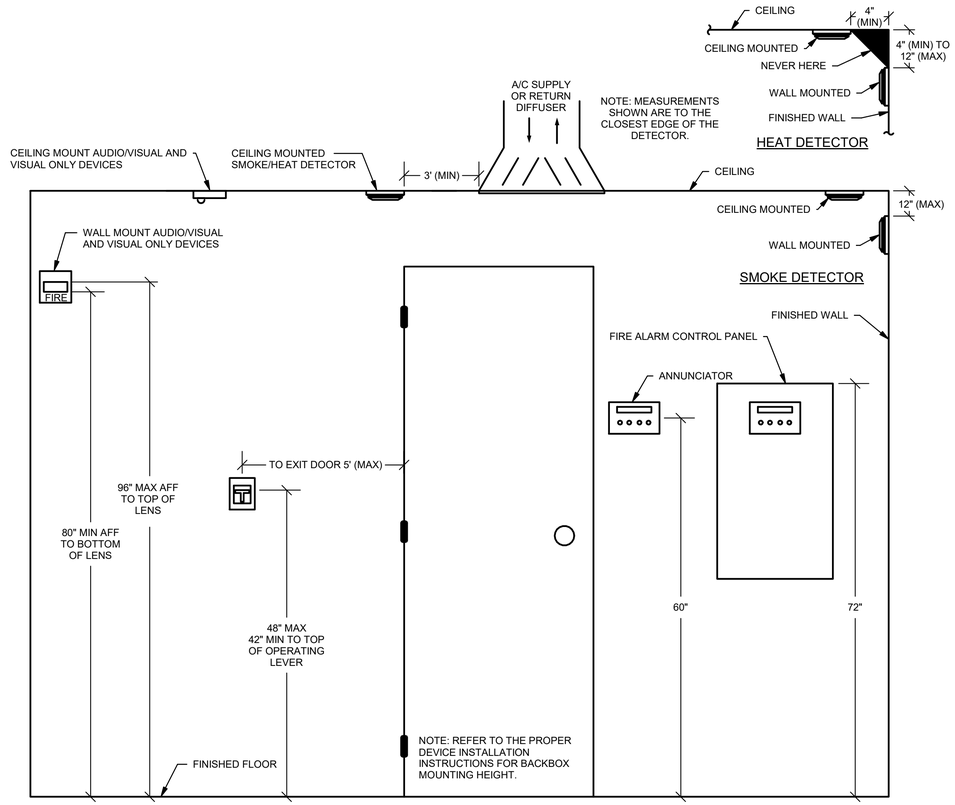

ELEVATION




WIRING SCHEMATIC

## 1 HYDRAULIC TYPE ELEVATOR FIRE ALARM

NO SCALE




## 2 TYPICAL MOUNTING HEIGHTS

NO SCALE

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION A.

LICENSED PROFESSIONAL ENGINEER
KENTON J. KIKENS
#30101
IOWA
Exp. 12/31/26

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# ELECTRICAL GENERAL NOTES

1. THE CONTRACTOR SHALL PROVIDE ALL LABOR AND MATERIAL NECESSARY FOR A COMPLETE AND FUNCTIONING ELECTRICAL SYSTEM.
2. MATERIALS AND INSTALLATION SHALL COMPLY WITH CODES, LAWS AND ORDINANCES OF FEDERAL, STATE AND LOCAL GOVERNING BODIES HAVING JURISDICTION. CODES FOR THIS SCOPE OF WORK INCLUDE THE 2024 IBC AND IEBC, 2024 IECC, 2023 NEC, 2024 IFC, AND 2016 ASHRAE 90.1.
3. MATERIALS AND EQUIPMENT SHALL BE LISTED AND/OR LABELED BY U.L., ETL, CSA OR ANOTHER RECOGNIZED TESTING LAB.
4. ALL WORK REQUIRED FOR THE INSTALLATION AS SHOWN ON DRAWINGS INCLUDING LABOR, EQUIPMENT AND MATERIALS SHALL BE IN STRICT COMPLIANCE WITH THE BUILDING STANDARDS, EXCEPT AS NOTED OTHERWISE.
5. THE CONTRACTOR SHALL SECURE AND PAY FOR ALL PERMITS, GOVERNMENTAL FEES, TAXES AND LICENSES NECESSARY FOR THE PROPER EXECUTION AND COMPLETION OF THE ELECTRICAL WORK.
6. THE CONTRACTOR SHALL PREPARE AND SUBMIT TO GOVERNMENTAL AGENCIES AND UTILITY COMPANIES SHOP DRAWINGS, WHICH ARE REQUIRED BY THESE AGENCIES, FOR THEIR APPROVAL.
7. THE CONTRACTOR SHALL NOTIFY THE CONTRACTING OFFICER OF ANY MATERIALS OR APPARATUS BELIEVED TO BE INADEQUATE, UNSUITABLE, IN VIOLATION OF LAWS, ORDINANCES, RULES OR REGULATIONS OF AUTHORITIES HAVING JURISDICTION.
8. FOR ALL JOBS THAT INCLUDE DEMOLITION WORK BY THE ELECTRICAL CONTRACTOR, DURING AND AFTER DEMOLITION, EC SHALL MAINTAIN CIRCUIT CONTINUITY TO ALL EXISTING DEVICES THAT ARE TO REMAIN. EC SHALL REMOVE, RELOCATE, AND/OR REWORK ANY CONDUIT AND WIRING TO FACILITATE THE NEW CONSTRUCTION SCOPE OF WORK.
9. THE CONTRACTOR SHALL CAREFULLY EXAMINE THE CONTRACT DOCUMENTS AND THOROUGHLY BECOME FAMILIAR WITH THE BUILDING STANDARDS AND LOCAL CONDITIONS RELATING TO THE WORK. FAILURE TO DO SO WILL NOT RELIEVE THE CONTRACTOR OF THE OBLIGATIONS OF THE CONTRACT.
10. ALL MATERIALS, AND EQUIPMENT SHALL BE ERECTED, INSTALLED, CONNECTED, CLEANED, ADJUSTED, TESTED, CONDITIONED, AND PLACED IN SERVICE IN ACCORDANCE WITH THE MANUFACTURER'S DIRECTIONS AND RECOMMENDATIONS.
11. ALL CUTTING, DRILLING AND PATCHING OF MASONRY, STEEL OR IRON WORK BELONGING TO THE BUILDING MUST BE DONE BY THIS CONTRACTOR IN ORDER THAT THEIR WORK MAY BE PROPERLY INSTALLED, BUT UNDER NO CONDITIONS MAY STRUCTURAL WORK BE CUT, EXCEPT AT THE DIRECTION OF THE CONTRACTING OFFICER OR THEIR REPRESENTATIVE.
12. ALL MATERIAL, EQUIPMENT, WIRING DEVICES, ETC. SHALL BE NEW, UNLESS SPECIFICALLY INDICATED AS EXISTING TO BE REUSED.
13. CONTRACTOR SHALL OBTAIN AND VERIFY EXACT UTILITY COMPANY DRAWINGS AND REQUIREMENTS. ELECTRICAL CONTRACTOR IS TO SUBMIT A COMPLETE CONSTRUCTION DRAWING SET TO THE ELECTRICAL UTILITY COMPANY WITHIN 10 DAYS OF AWARD OF CONTRACT. COORDINATE TIMELINE OF THE REVIEW, APPROVAL, ALL ASSOCIATED DOWN TIME, CONSTRUCTION SCHEDULING, DELIVERY, AND INSTALLATION OF THE UTILITY TRANSFORMER. NOTIFY THE CONTRACTING OFFICER OF SCHEDULING CONFLICTS.
14. THE CONTRACTOR SHALL BE RESPONSIBLE FOR ALL COORDINATION WITH, APPROVAL FROM, AND ASSOCIATED FEES REQUIRED BY THE POWER UTILITY COMPANY (ALLAMAKEE-CLAYTON ELECTRIC COOPERATIVE, JASON TROENDLE 1.563.605.6018, JTROENDLE@ACREC.COOP). CONTRACTOR SHALL CARRY THE FOLLOWING ALLOWANCE TO COVER THE COSTS OF THE CONNECTION TO THE UTILITY PROVIDER. ALLOWANCE IS TO BE $75,000.00.
15. ALL NEW CIRCUIT BREAKERS FOR NEW OR EXISTING PANELBOARDS SHALL MATCH EXISTING BUILDING PANELBOARD MANUFACTURER AND BREAKER TYPE. THE CONTRACTOR SHALL PROVIDE NEW TYPE WRITTEN PANEL DIRECTORIES FOR ALL NEW PANELS AND EXISTING PANELS WHICH HAVE CHANGED. PANELBOARD SHALL BE MARKED WHERE THE SOURCE OF POWER SUPPLY ORIGINATES AND THEIR LISTED AMPERE RATING.
16. DO NOT SHARE NEUTRAL CONDUCTORS FOR MULTIWIRE BRANCH CIRCUITS. WHERE SHARED NEUTRAL CONDUCTORS ARE REQUIRED (SUCH AS POWERED FURNITURE SYSTEMS), HANDLE TIES SHALL BE PROVIDED ON THE CIRCUIT BREAKERS, WITH SHARED NEUTRALS, SUCH THAT IT WILL SIMULTANEOUSLY DISCONNECT ALL UNGROUNDED CONDUCTORS. ALL HANDLE TIES ARE REQUIRED TO BE INDICATED ON THE PANELBOARD SHOP DRAWINGS.
17. SHOULD ACTUAL FIELD CONDITIONS REQUIRE CIRCUIT DESIGNATIONS INDICATED ON PLANS TO VARY, ELECTRICAL CONTRACTOR SHALL PROVIDE EQUIPMENT LABELS AND CIRUIT DIRECTORIES TO MATCH INSTALLED CONDITIONS IN COMPLIANCE WITH NEC 408.4. MODIFIED CIRCUIT DESCRIPTION AND NUMBER USED SHALL BE INDICATED ON THE "AS-BUILT" DRAWINGS PROVIDED TO THE CONTRACTING OFFICER.
19. ALL SERVICE EQUIPMENT SHALL BE LEGIBLY MARKED IN THE FIELD BY THE ELECTRICAL CONTRACTOR WITH THE MAXIMUM AVAILABLE FAULT CURRENT AS INDICATED WITHIN THESE DOCUMENTS. THE FIELD MARKING(S) SHALL COMPLY WITH ELECTRICAL SPECIFICATIONS FOR READABILITY AND DURABILITY.
20. EACH SWITCH, LIGHT, RECEPTACLE AND OTHER MISCELLANEOUS DEVICE SHALL BE PROVIDED WITH A GALVANIZED OR PRESSED STEEL OUTLET BOX OF THE KNOCKOUT TYPE, OF NOT LESS THAN NO. 14 U.S. GAUGE STEEL. CONDUITS SHALL BE FASTENED WITH LOCKNUTS AND BUSHINGS AND ALL UNUSED KNOCKOUTS MUST BE LEFT SEALED. THERE MUST BE SUFFICIENT ROOM FOR WIRES AND BUSHINGS AND DEEP BOXES SHALL BE INSTALLED WHERE REQUIRED. BOXES SHALL BE SECURELY AND ADEQUATELY SUPPORTED.
21. IN EXPOSED AND SUSPENDED CEILING APPLICATIONS, ROUTE CONDUIT AS CLOSE TO STRUCTURAL SLAB OR DECK AS POSSIBLE, AND SUPPORT CONDUIT AND JUNCTION BOXES DIRECTLY FROM THE STRUCTURAL SLAB, DECK, OR FRAMING PROVIDED FOR THAT PURPOSE. LIGHTING BRANCH CIRCUIT CONDUITS SHALL NOT BE CLIPPED TO THE CEILING SYSTEM UNLESS IT HAS BEEN SPECIFICALLY DESIGNED FOR THAT PURPOSE.
22. ALL EXPOSED CONDUIT SHALL BE CONCEALED TO THE GREATEST EXTENT POSSIBLE, AND SHALL BE INSTALLED PARALLEL AND CLOSE TO STRUCTURAL MEMBERS. GENERAL CONTRACTOR SHALL PAINT CONDUIT TO MATCH ADJACENT FINISHES.
23. WHERE FLOOR FITTINGS REQUIRE PENETRATION OF THE FLOOR SLAB, THEY SHALL BE STANDARD DEVICE LISTED BY UL FOR THE PURPOSE AND HAVE A UL FIRE RATING EQUAL TO THE FLOOR RATING. FLOOR SERVICE BOXES SHALL BE MODULAR, ADJUSTABLE FLUSH TYPE, DUAL SERVICE UNITS SUITABLE FOR WIRING METHOD USED. COMPARTMENT BARRIERS SHALL SEPARATE POWER FROM LOW VOLTAGE CABLING. PROVIDE RECTANGULAR SERVICE PLATE WITH SATIN FINISH.
24. LUMINAIRES SHOWN AS SHADED SHALL BE PROVIDED WITH EMERGENCY BATTERY BACKUP POWER. EMERGENCY LUMINAIRES SHALL SENSE UNSWITCHED POWER TO THE SPACE AND OPERATE AUTOMATICALLY UPON LOSS OF NORMAL POWER. ALL SHADED LUMINAIRES WITH LED SOURCES SHALL BE PROVIDED WITH 90 MINS OF BACKUP POWER VIA INTEGRAL BATTERY OR EXTERNAL MINI/MICRO INVERTERS AS SPECIFIED IN THE LUMINAIRE SCHEDULE. ALL EMERGENCY LUMINAIRES WITH INTEGRAL BATTERY, EMERGENCY LIGHTING UNITS, AND INVERTERS SHALL HAVE INTEGRAL OR REMOTE TEST SWITCHES WITH VISIBLE INDICATOR LIGHTS, AS INDICATED IN THE FIXTURE SCHEDULE. THE EMERGENCY BATTERY OR INVERTER SHALL BE CONNECTED TO THE UNSWITCHED LEG OF THE LIGHTING CIRCUIT INDICATED TO DETECT THE LOSS OF NORMAL POWER.
25. UNLESS OTHERWISE NOTED, LUMINAIRES DESIGNATED AS NIGHT LIGHT (NL) SHALL BE CONNECTED AHEAD OF LOCAL SWITCHING AND REMAIN ON 24 HOURS A DAY.
26. ALL DIMMED LIGHTING CIRCUITS ARE TO RECEIVE DEDICATED NEUTRALS. DO NOT SHARE NEUTRALS ON DIMMED LIGHTING CIRCUITS.
27. PROVIDE CONTRACTING OFFICER WITH A COMPLETE LISTING OF ALL LAMPS UTILIZED ON THE PROJECT INCLUDING MANUFACTURER AND CATALOG INFORMATION. PROVIDE A SUGGESTED SOURCE, INCLUDING CONTACT NAME AND PHONE NUMBER, FOR REORDERING.
28. THE CONTRACTOR SHALL VERIFY THE CEILING TYPE BEFORE ORDERING LIGHTING FIXTURES.
29. ROUGH-IN FOR MECHANICAL EQUIPMENT SHALL ONLY OCCUR AFTER MECHANICAL EQUIPMENT SUBMITTALS ARE THOROUGHLY REVIEWED FOR CHANGES. NOTIFY CONTRACTING OFFICER OF ANY DISCREPANCIES.
30. EC SHALL COORDINATE ELECTRIC WATER COOLER RECEPTACLE PLACEMENT SUCH THAT THE RECEPTACLE IS ACCESSIBLE WITHIN THE WATER COOLER SHROUD, YET CONCEALED BY THE SHROUD PER NEC 422.33(A). PROVIDE 5mA GFCI CIRCUIT BREAKER IN ELECTRICAL PANEL PER NEC SECTION 422.
31. THE POWER AND CONTROL REQUIREMENTS FOR ALL EQUIPMENT CONNECTIONS SHALL BE CONFIRMED WITH APPROVED SHOP DRAWINGS PRIOR TO ELECTRICAL ROUGH-IN. FINAL POWER REQUIREMENTS, DIMENSIONED ROUGH-IN LOCATIONS, LOW VOLTAGE SYSTEM CONNECTIONS, ETC. SHALL BE CONFIRMED AND MODIFIED AS REQUIRED.
32. ALL DEVICES IN OR ABOVE COUNTERS SHALL HAVE LOCATIONS AND MOUNTING HEIGHTS CONFIRMED WITH ARCHITECTURAL ELEVATIONS PRIOR TO ROUGH-IN. ANY ADJUSTMENTS TO MOUNTING HEIGHTS REQUIRED BY LACK OF COORDINATION WILL BE AT THE CONTRACTOR'S EXPENSE.
33. G.C. SHALL INCLUDE IN THEIR COST THE REMOVAL OF ALL EXISTING ELECTRICAL DEVICES, CONDUITS, FIXTURES AND EQUIPMENT. TURN EQUIPMENT OVER TO CONTRACTING OFFICER AS INDICATED OR RECYCLE/DISCARD ALL EQUIPMENT AS REQUIRED. E.C. SHALL BE RESPONSIBLE FOR DISCONNECTING PRIMARY SERVICE AND TEMPORARY POWER.

ALL INFORMATION SHOWN ON THIS SHEET IS COMMON TO ALL OPTIONS.

LICENSED PROFESSIONAL ENGINEER
05/21/2026
JEFF MULLIKIN
P23384
ARIZONA

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| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR AE DESIGN DENVER, CO | DESIGNED: JR/JM/TW CADD: AE DESIGN TECH REVIEW: MRS/ESN DATE: 5/29/2026 | SUB SHEET NO. E000 | TITLE OF SHEET: ELECTRICAL COVER SHEET — EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 107 OF 142 | |

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## ABBREVIATIONS

| | |
|---|---|
| A | AMPERE(S) |
| AC | ABOVE COUNTER |
| AFF | ABOVE FINISHED FLOOR |
| AFG | ABOVE FINISHED GRADE |
| AHJ | AUTHORITY HAVING JURISDICTION |
| AIC | AMPERES INTERRUPTING CAPACITY |
| ATS | AUTOMATIC TRANSFER SWITCH |
| BFF | BELOW FINISHED FLOOR |
| BOF | BOTTOM OF FIXTURE |
| C | CONDUIT |
| CATV | CABLE TELEVISION |
| CB | CIRCUIT BREAKER |
| CCT | CORRELATED COLOR TEMPERATURE |
| CLG | CEILING |
| CO | CONTRACTING OFFICER |
| CT | CURRENT TRANSFORMER |
| DED | DEDICATED CIRCUIT |
| DISC | DISCONNECT |
| DW | DISHWASHER |
| DWG(S) | DRAWING(S) |
| (E) | EXISTING TO REMAIN |
| EC | ELECTRICAL CONTRACTOR |
| EF | EXHAUST FAN |
| (EL) | EXISTING LOCATION |
| (ER) | EXISTING TO BE RELOCATED |
| EM | EMERGENCY |
| EPO | EMERGENCY POWER OFF |
| EWC | ELECTRIC WATER COOLER |
| (R) | EXISTING TO BE REMOVED |
| F | FUSE |
| FLA | FULL LOAD AMPS |
| GD | GARBAGE DISPOSAL |
| G | GROUND |
| GC | GENERAL CONTRACTOR |
| GFI | GROUND FAULT CIRCUIT INTERRUPTER |
| HP | HORSEPOWER |
| IDF | INTERMEDIATE DISTRIBUTION FACILITY |
| IG | ISOLATED GROUND |
| ISC | SHORT CIRCUIT CURRENT |
| KVA | KILOVOLT AMPERE(S) |
| KW | KILOWATT(S) |
| LTG | LIGHTING |
| MCA | MINIMUM CIRCUIT AMPERE(S) |
| MCB | MAIN CIRCUIT BREAKER |
| MDP | MAIN DISTRIBUTION CENTER |
| MDF | MAIN DISTRIBUTION FACILITY |
| MLO | MAIN LUGS ONLY |
| MTS | MANUAL TRANSFER SWITCH |
| MW | MICROWAVE |
| NC | NORMALLY CLOSED |
| NL | NIGHT LIGHT - SEE GENERAL NOTES |
| NO | NORMALLY OPEN |
| OAE | OR APPROVED EQUAL |
| OFH | OVERALL FIXTURE HEIGHT |
| OH | OVERHEAD |
| P | POLE |
| PART | PARTIAL CIRCUIT |
| PH | PHASE |
| PNL | PANEL |
| RCPT | RECEPTACLE |
| REF | REFRIGERATOR |
| RFD | RECESSED FIXTURE DEPTH |
| (RL) | RELOCATED LOCATION |
| SPD | SURGE PROTECTION DEVICE |
| SM | SURFACE-MOUNTED |
| UC | UNDER COUNTER/CABINET |
| UG | UNDERGROUND |
| UON | UNLESS OTHERWISE NOTED |
| V | VOLT(S) |
| W | WATT(S) OR WIRE |
| WFD | WALL FIXTURE DEPTH |
| WG | WIRE GUARD |
| WP | WEATHERPROOF |
| XFMR | TRANSFORMER |

## WIRING DEVICES

- DUPLEX RECEPTACLE
- FOUR PLEX RECEPTACLE
- SINGLE RECEPTACLE
- COMBO RECEPTACLE/SWITCH
- SWITCHED DUPLEX RECEPTACLE
- SWITCHED FOUR PLEX RECEPTACLE
- EMERGENCY POWERED DUPLEX RECEPTACLE
- SPECIAL PURPOSE RECEPTACLE
- FLOOR MOUNTED SPECIAL PURPOSE RECEPTACLE
- FLOOR MOUNTED RECEPTACLE DUPLEX/QUAD
- CEILING MOUNTED RECEPTACLE DUPLEX/QUAD (CLG)
- SURFACE RACEWAY (SR)
- CLOCK RECEPTACLE
- JUNCTION BOX
- WALL MOUNTED J-BOX
- FLOOR MOUNTED JUNCTION BOX
- MOLDED CASE CIRCUIT BREAKER IN ENCLOSURE
- NON-FUSED DISCONNECT SWITCH
- FUSED DISCONNECT SWITCH
- MAGNETIC CONTROLLER (STARTER)
- COMBINATION STARTER/DISCONNECT SWITCH
- MOTOR (HP)
- RELAY (R)
- TIME CLOCK (TC)
- THERMAL OVERLOAD SWITCH ($^{TO}$)
- SINGLE POLE SWITCH, LINE VOLTAGE ($)
- 3-WAY SWITCH, LINE VOLTAGE ($^{3}$)
- 4-WAY SWITCH, LINE VOLTAGE ($^{4}$)
- KEY OPERATED SWITCH ($^{K}$)
- DIMMER SWITCH, LINE VOLTAGE ($^{D}$)
- RECESSED DOOR SWITCH ($^{DOOR}$)
- LIGHTING CONTROL DEVICE, REFER TO DETAILS FOR CONTROL INTENT (X)

## DISTRIBUTION AND RACEWAY

- MDC — MAIN DISTRIBUTION CENTER (MDC)
- SURFACE MTD PANELBOARD
- RECESSED PANELBOARD
- T — TRANSFORMER
- BRANCH CIRCUIT HOMERUN
- CONDUIT CONCEALED IN FLOOR OR UNDERGROUND
- CONDUIT EXPOSED OR CONCEALED IN WALL OR CEILING
- RACEWAY UP
- RACEWAY DOWN
- CAPPED CONDUIT
- CURRENT TRANSFORMER
- CIRCUIT BREAKER SWITCH
- FUSED SWITCH
- GROUNDING ELECTRODE CONDUCTOR
- M — METER
- GFP — GROUND FAULT PROTECTION

## FIRE ALARM

- FACP — FIRE ALARM CONTROL PANEL
- FAA — FIRE ALARM ANNUNCIATOR/GRAPHIC MAP
- FA-RPS — FIRE ALARM REMOTE POWER SUPPLY
- C — CONTROL MODULE
- M — MONITOR MODULE
- MANUAL PULLDOWN STATION
- S — WALL MOUNTED ADA STROBE
- ADA HORN OR SPEAKER WITH STROBE
- MINI HORN / STROBE
- MH — ELECTROMAGNETIC DOOR HOLD OPEN
- FS — SPRINKLER FLOW SWITCH
- TS — SPRINKLER TAMPER SWITCH
- T — THERMAL DETECTOR
- S — PHOTOELECTRIC SMOKE DETECTOR
- CO — CARBON MONOXIDE DETECTOR
- S/CO — COMBINATION SMOKE/CARBON MONOXIDE DETECTOR
- DUCT SMOKE DETECTOR, SUPPLY OR RETURN
- TS — REMOTE INDICATING LIGHT (TEST SWITCH)
- D — 120V. MOTORIZED SMOKE DAMPER
- RA — RESCUE ASSISTANCE PHONE
- F — FIRE FIGHTERS PHONE JACK

## LIGHTING FIXTURES

- A / 1 a — LUMINAIRE TYPE, REFERENCING LUMINAIRE SCHEDULE, TYPICAL ALL FIXTURES. SUBSCRIPT, IF SHOWN, REFERENCES WALL SWITCH OR RELAY/ZONE CONTROL
- WALL MOUNTED LUMINAIRE
- SURFACE OR PENDANT MOUNTED LUMINAIRE
- RECESSED LUMINAIRE
- RECESSED DOWNLIGHT LUMINAIRE
- SURFACE CEILING LUMINAIRE
- PENDANT LUMINAIRE
- ARROW INDICATES DIRECTIONAL LUMINAIRE
- MONOPOINT LUMINAIRE
- SURFACE OR PENDANT TRACK LUMINAIRE REFER TO FIXTURE SCHEDULE FOR HEAD QTY.
- LED TAPE LUMINAIRE
- FESTOON LIGHTING
- RECESSED MULTI-HEAD LUMINAIRE
- FLOOR OR TABLE LAMP
- EXIT LUMINAIRE - SHADED INDICATES FACE / DIRECTIONAL ARROWS AS SHOWN
- BATTERY PACK EMERGENCY LUMINAIRE
- HATCH INDICATES EMERGENCY LUMINAIRE
- K — PORCELAIN KEYLESS LAMP HOLDER
- STEP LIGHT TYPE LUMINAIRE
- IN-GRADE UPLIGHT
- BOLLARD LUMINAIRE
- PEDESTRIAN POLE OR POST TOP LUMINAIRE
- EXTERIOR AREA LIGHT

## SYMBOLS

- X — POOL EQUIPMENT SCHEDULE NOTATION
- X — KITCHEN EQUIPMENT SCHEDULE NOTATION
- X / X — MECHANICAL EQUIPMENT SCHEDULE NOTATION
- X — DETAIL NOTE
- X — DELTA REVISION NOTE
- X — ELECTRICAL WIRE SIZE
- T1 — LIGHTING CONTROLS SEQUENCE OF OPERATION

## LIGHTNING PROTECTION

- AIR TERMINAL
- GROUND ROD

ALL INFORMATION SHOWN ON THIS SHEET IS COMMON TO ALL OPTIONS.

LICENSED PROFESSIONAL ENGINEER — 05/21/2026 — JEFF MULLIKIN — P25384 — POWER

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| | | | | SHEET 108 OF 142 | |

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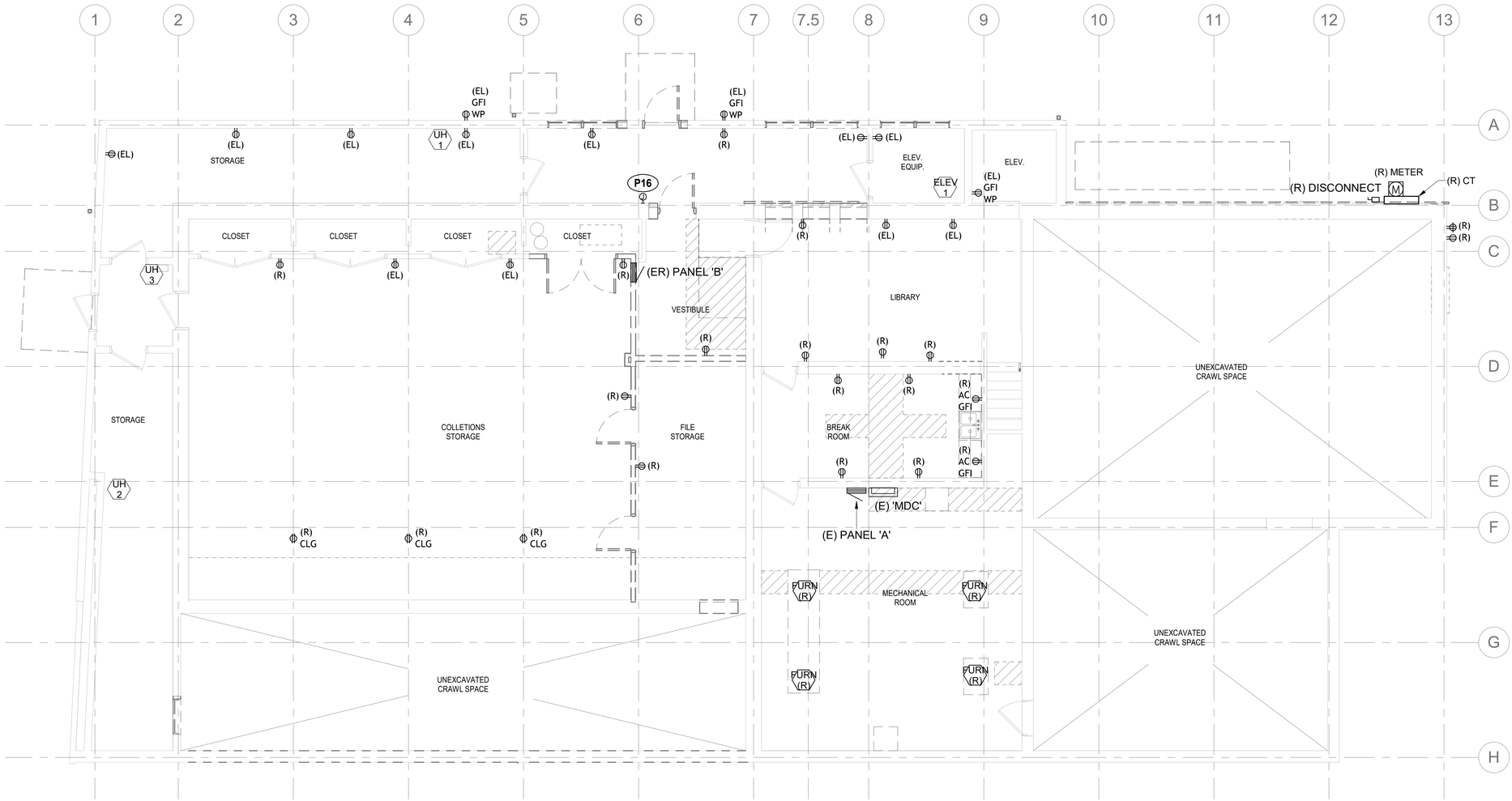


1 ED100 BASEMENT ELECTRICAL DEMOLITION PLAN

SCALE A

## ELECTRICAL DEMOLITION REQUIREMENTS

| | |
|---|---|
| 1. | THE BASIS OF THESE DRAWINGS WERE SITE OBSERVATIONS, ORIGINAL BUILDING DRAWINGS AND VARIOUS OTHER SOURCES. EVERY ATTEMPT HAS BEEN MADE TO DOCUMENT THE ACTUAL CONDITIONS. HOWEVER, THE CONTRACTOR SHALL CAREFULLY EXAMINE THE CONTRACT DOCUMENTS AND THOROUGHLY BECOME FAMILIAR WITH THE BUILDING/NPS STANDARDS RELATING TO THE WORK. FAILURE TO DO SO WILL NOT RELIEVE THE CONTRACTOR OF THE OBLIGATIONS OF THE CONTRACT. |
| 2. | THE ELECTRICAL CONTRACTOR SHALL NOTIFY THE CONTRACTING OFFICER OF ANY MATERIALS OR APPARATUS BELIEVED TO BE INADEQUATE, UNSUITABLE, IN VIOLATION OF LAWS, ORDINANCES, RULES OR REGULATIONS OF AUTHORITIES HAVING JURISDICTION. |
| 3. | THE ELECTRICAL CONTRACTOR SHALL INCLUDE IN THEIR COST THE REMOVAL OF ALL EXISTING ELECTRICAL DEVICES, CONDUITS, FIXTURES AND EQUIPMENT. DISCARD ALL EQUIPMENT AS REQUIRED, UNLESS NOTED OTHERWISE. |
| 4. | EXISTING DEVICES NOT NOTED AS EXISTING (E) OR INDICATED ON PLANS SHALL REMAIN, AS THEY PRESENTLY EXIST. FOR DEMOLISHED DEVICES IN WALLS WHICH ARE TO REMAIN, THE ELECTRICAL CONTRACTOR SHALL INSTALL A BLANK FACE PLATE ON THE BACKBOX. |
| 5. | THE DEMOLITION OF SOME DEVICES MAY INTERRUPT POWER TO DEVICES DOWN STREAM. THE ELECTRICAL CONTRACTOR IS RESPONSIBLE FOR REWORKING THESE CIRCUITS TO MAINTAIN POWER TO DOWN STREAM DEVICES WHICH WILL REMAIN. |
| 6. | ALL UNENERGIZED/DEMOLISHED CIRCUITRY SHALL HAVE CONDUCTORS REMOVED FROM CONDUIT AND CONDUIT SHALL BE MARKED "EMPTY" WITH INDELIBLE MARKER. |

## POWER GENERAL NOTES

| | |
|---|---|
| A. | DEVICES NOTED WITH (EL) SHALL MAINTAIN EXISTING LOCATION IN ORDER TO LIMIT IMPACT TO EXISTING CONDITIONS. DEVICE AND FACEPLATE SHALL BE REMOVED AND REPLACED WITH NEW. BACKBOX, IF PRESENT AND CAPABLE, AND CONDUIT SHALL BE MAINTAINED FOR REUSE. IF BACKBOX IS NOT PRESENT OR CAPABLE, A NEW BOX SHALL BE PROVIDED. NEW BRANCH CIRCUIT FEEDERS SHALL BE PROVIDED. REFER TO NEW POWER PLANS. |

## KEYNOTE LEGEND

| KEY VALUE | KEYNOTE TEXT |
|---|---|
| P16 | EC SHALL DETERMINE IF EXISTING EXPOSED CONDUIT AND JUNCTION BOX ARE TO REMAIN TO BE REUSED. IF J-BOX AND CONDUIT ARE TO BE REUSED, EC SHALL RE-ROUTE CONDUIT AND RELOCATE J-BOX TO BE ABOVE CEILING TO HIDE IT FROM VIEW. |

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.

SCALE: A 4 0 4 8 12 SCALE OF FEET

LICENSED PROFESSIONAL ENGINEER 05/21/2026 JEFF MULLIKIN P25384 ARMY

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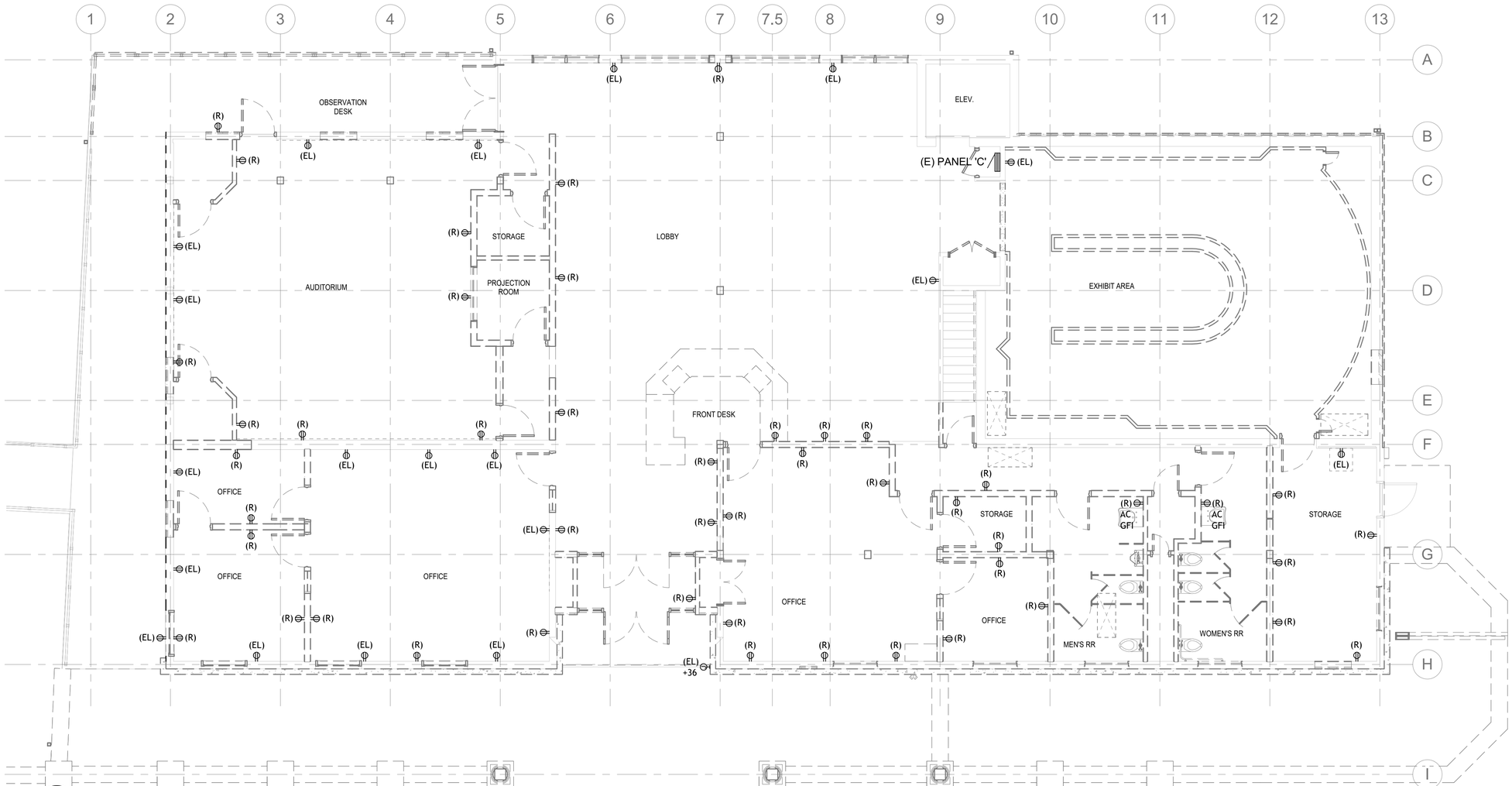


1 ED101 FIRST FLOOR ELECTRICAL DEMOLITION PLAN
SCALE A

## ELECTRICAL DEMOLITION REQUIREMENTS

| | |
|---|---|
| 1. | THE BASIS OF THESE DRAWINGS WERE SITE OBSERVATIONS, ORIGINAL BUILDING DRAWINGS AND VARIOUS OTHER SOURCES. EVERY ATTEMPT HAS BEEN MADE TO DOCUMENT THE ACTUAL CONDITIONS. HOWEVER, THE CONTRACTOR SHALL CAREFULLY EXAMINE THE CONTRACT DOCUMENTS AND THOROUGHLY BECOME FAMILIAR WITH THE BUILDING/NPS STANDARDS RELATING TO THE WORK. FAILURE TO DO SO WILL NOT RELIEVE THE CONTRACTOR OF THE OBLIGATIONS OF THE CONTRACT. |
| 2. | THE ELECTRICAL CONTRACTOR SHALL NOTIFY THE CONTRACTING OFFICER OF ANY MATERIALS OR APPARATUS BELIEVED TO BE INADEQUATE, UNSUITABLE, IN VIOLATION OF LAWS, ORDINANCES, RULES OR REGULATIONS OF AUTHORITIES HAVING JURISDICTION. |
| 3. | THE ELECTRICAL CONTRACTOR SHALL INCLUDE IN THEIR COST THE REMOVAL OF ALL EXISTING ELECTRICAL DEVICES, CONDUITS, FIXTURES AND EQUIPMENT. DISCARD ALL EQUIPMENT AS REQUIRED, UNLESS NOTED OTHERWISE. |
| 4. | EXISTING DEVICES NOT NOTED AS EXISTING (E) OR INDICATED ON PLANS SHALL REMAIN, AS THEY PRESENTLY EXIST. FOR DEMOLISHED DEVICES IN WALLS WHICH ARE TO REMAIN, THE ELECTRICAL CONTRACTOR SHALL INSTALL A BLANK FACE PLATE ON THE BACKBOX. |
| 5. | THE DEMOLITION OF SOME DEVICES MAY INTERRUPT POWER TO DEVICES DOWN STREAM. THE ELECTRICAL CONTRACTOR IS RESPONSIBLE FOR REWORKING THESE CIRCUITS TO MAINTAIN POWER TO DOWN STREAM DEVICES WHICH WILL REMAIN. |
| 6. | ALL UNENERGIZED/DEMOLISHED CIRCUITRY SHALL HAVE CONDUCTORS REMOVED FROM CONDUIT AND CONDUIT SHALL BE MARKED "EMPTY" WITH INDELIBLE MARKER. |

## POWER GENERAL NOTES

| | |
|---|---|
| A. | DEVICES NOTED WITH (EL) SHALL MAINTAIN EXISTING LOCATION IN ORDER TO LIMIT IMPACT TO EXISTING CONDITIONS. DEVICE AND FACEPLATE SHALL BE REMOVED AND REPLACED WITH NEW. BACKBOX, IF PRESENT AND CAPABLE, AND CONDUIT SHALL BE MAINTAINED FOR REUSE. IF BACKBOX IS NOT PRESENT OR CAPABLE, A NEW BOX SHALL BE PROVIDED. NEW BRANCH CIRCUIT FEEDERS SHALL BE PROVIDED. REFER TO NEW POWER PLANS. |

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.

SCALE: A 4 0 4 8 12 SCALE OF FEET

LICENSED PROFESSIONAL ENGINEER 05/21/2026 JEFF MULLIKIN P25384 IOWA

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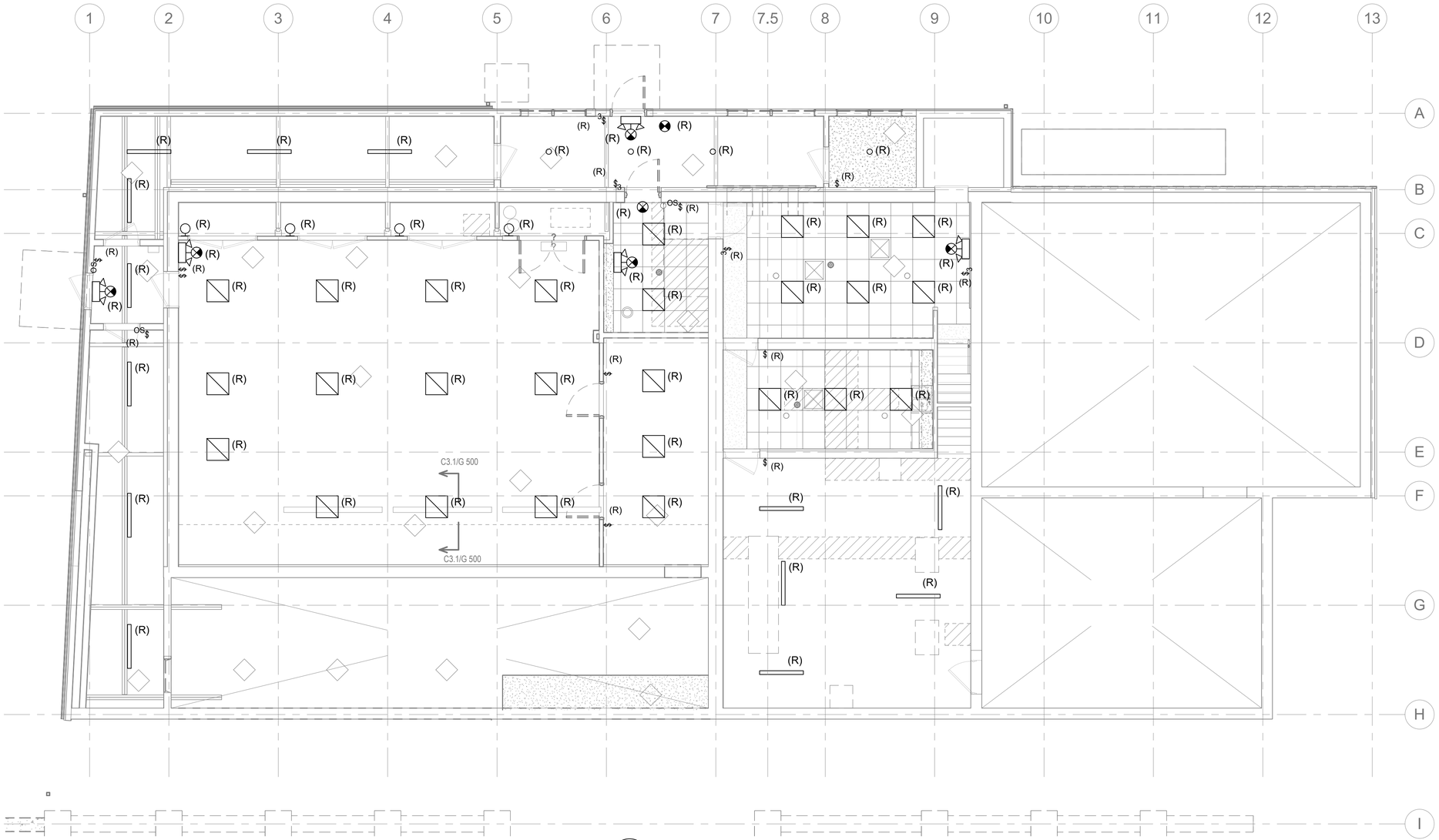


1 ED200 BASEMENT DEMOLITION LIGHTING PLAN

SCALE A

## ELECTRICAL DEMOLITION REQUIREMENTS

| | |
|---|---|
| 1. | THE BASIS OF THESE DRAWINGS WERE SITE OBSERVATIONS, ORIGINAL BUILDING DRAWINGS AND VARIOUS OTHER SOURCES. EVERY ATTEMPT HAS BEEN MADE TO DOCUMENT THE ACTUAL CONDITIONS. HOWEVER, THE CONTRACTOR SHALL CAREFULLY EXAMINE THE CONTRACT DOCUMENTS AND THOROUGHLY BECOME FAMILIAR WITH THE BUILDING/NPS STANDARDS RELATING TO THE WORK. FAILURE TO DO SO WILL NOT RELIEVE THE CONTRACTOR OF THE OBLIGATIONS OF THE CONTRACT. |
| 2. | THE ELECTRICAL CONTRACTOR SHALL NOTIFY THE CONTRACTING OFFICER OF ANY MATERIALS OR APPARATUS BELIEVED TO BE INADEQUATE, UNSUITABLE, IN VIOLATION OF LAWS, ORDINANCES, RULES OR REGULATIONS OF AUTHORITIES HAVING JURISDICTION. |
| 3. | THE ELECTRICAL CONTRACTOR SHALL INCLUDE IN THEIR COST THE REMOVAL OF ALL EXISTING ELECTRICAL DEVICES, CONDUITS, FIXTURES AND EQUIPMENT. DISCARD ALL EQUIPMENT AS REQUIRED, UNLESS NOTED OTHERWISE. |
| 4. | EXISTING DEVICES NOT NOTED AS EXISTING (E) OR INDICATED ON PLANS SHALL REMAIN, AS THEY PRESENTLY EXIST. FOR DEMOLISHED DEVICES IN WALLS WHICH ARE TO REMAIN, THE ELECTRICAL CONTRACTOR SHALL INSTALL A BLANK FACE PLATE ON THE BACKBOX. |
| 5. | THE DEMOLITION OF SOME DEVICES MAY INTERRUPT POWER TO DEVICES DOWN STREAM. THE ELECTRICAL CONTRACTOR IS RESPONSIBLE FOR REWORKING THESE CIRCUITS TO MAINTAIN POWER TO DOWN STREAM DEVICES WHICH WILL REMAIN. |
| 6. | ALL UNENERGIZED/DEMOLISHED CIRCUITRY SHALL HAVE CONDUCTORS REMOVED FROM CONDUIT AND CONDUIT SHALL BE MARKED "EMPTY" WITH INDELIBLE MARKER. |

## LIGHTING DEMOLITION NOTES

| | |
|---|---|
| A. | ALL EXISTING LIGHTING FIXTURES AND DEVICES SHALL BE REMOVED INCLUDING ASSOCIATED CONTROLS, CONDUIT, AND WIRING BACK TO SOURCE UNLESS NOTED OTHERWISE. |

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.

SCALE: A 4 0 4 8 12 SCALE OF FEET

LICENSED PROFESSIONAL ENGINEER 05/21/2026 JEFF MULLIKIN P25384 COLORADO

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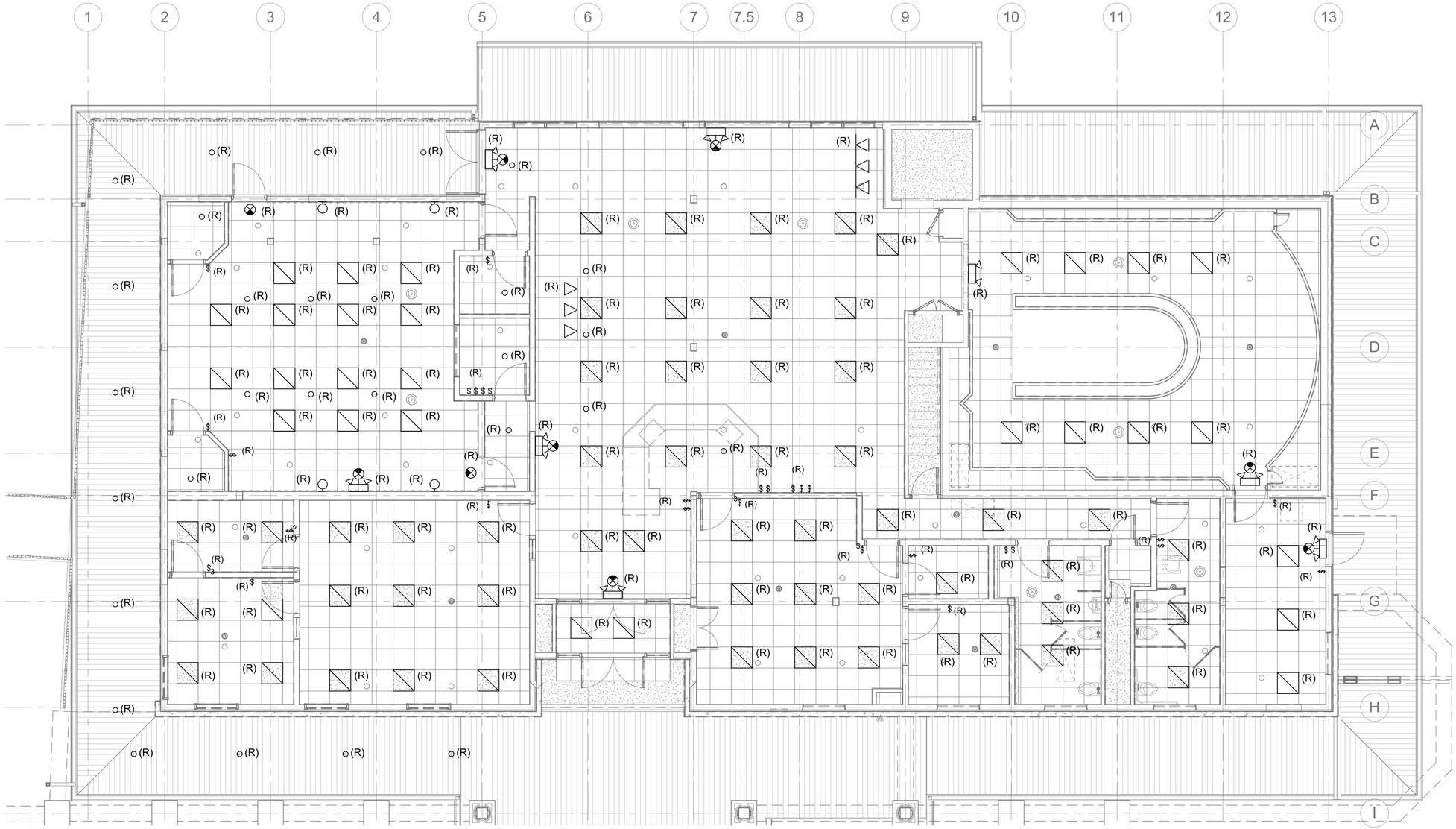


## ELECTRICAL DEMOLITION REQUIREMENTS

| | |
|---|---|
| 1. | THE BASIS OF THESE DRAWINGS WERE SITE OBSERVATIONS, ORIGINAL BUILDING DRAWINGS AND VARIOUS OTHER SOURCES. EVERY ATTEMPT HAS BEEN MADE TO DOCUMENT THE ACTUAL CONDITIONS. HOWEVER, THE CONTRACTOR SHALL CAREFULLY EXAMINE THE CONTRACT DOCUMENTS AND THOROUGHLY BECOME FAMILIAR WITH THE BUILDING/NPS STANDARDS RELATING TO THE WORK. FAILURE TO DO SO WILL NOT RELIEVE THE CONTRACTOR OF THE OBLIGATIONS OF THE CONTRACT. |
| 2. | THE ELECTRICAL CONTRACTOR SHALL NOTIFY THE CONTRACTING OFFICER OF ANY MATERIALS OR APPARATUS BELIEVED TO BE INADEQUATE, UNSUITABLE, IN VIOLATION OF LAWS, ORDINANCES, RULES OR REGULATIONS OF AUTHORITIES HAVING JURISDICTION. |
| 3. | THE ELECTRICAL CONTRACTOR SHALL INCLUDE IN THEIR COST THE REMOVAL OF ALL EXISTING ELECTRICAL DEVICES, CONDUITS, FIXTURES AND EQUIPMENT. DISCARD ALL EQUIPMENT AS REQUIRED, UNLESS NOTED OTHERWISE. |
| 4. | EXISTING DEVICES NOT NOTED AS EXISTING (E) OR INDICATED ON PLANS SHALL REMAIN, AS THEY PRESENTLY EXIST. FOR DEMOLISHED DEVICES IN WALLS WHICH ARE TO REMAIN, THE ELECTRICAL CONTRACTOR SHALL INSTALL A BLANK FACE PLATE ON THE BACKBOX. |
| 5. | THE DEMOLITION OF SOME DEVICES MAY INTERRUPT POWER TO DEVICES DOWN STREAM. THE ELECTRICAL CONTRACTOR IS RESPONSIBLE FOR REWORKING THESE CIRCUITS TO MAINTAIN POWER TO DOWN STREAM DEVICES WHICH WILL REMAIN. |
| 6. | ALL UNENERGIZED/DEMOLISHED CIRCUITRY SHALL HAVE CONDUCTORS REMOVED FROM CONDUIT AND CONDUIT SHALL BE MARKED "EMPTY" WITH INDELIBLE MARKER. |

1 ED201 FIRST FLOOR DEMOLITION LIGHTING PLAN

SCALE A

## LIGHTING DEMOLITION NOTES

| | |
|---|---|
| A. | ALL EXISTING LIGHTING FIXTURES AND DEVICES SHALL BE REMOVED INCLUDING ASSOCIATED CONTROLS, CONDUIT, AND WIRING BACK TO SOURCE UNLESS NOTED OTHERWISE. |

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.

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LICENSED PROFESSIONAL ENGINEER 05/21/2026 JEFF MULLIKIN P25384 ROYAL

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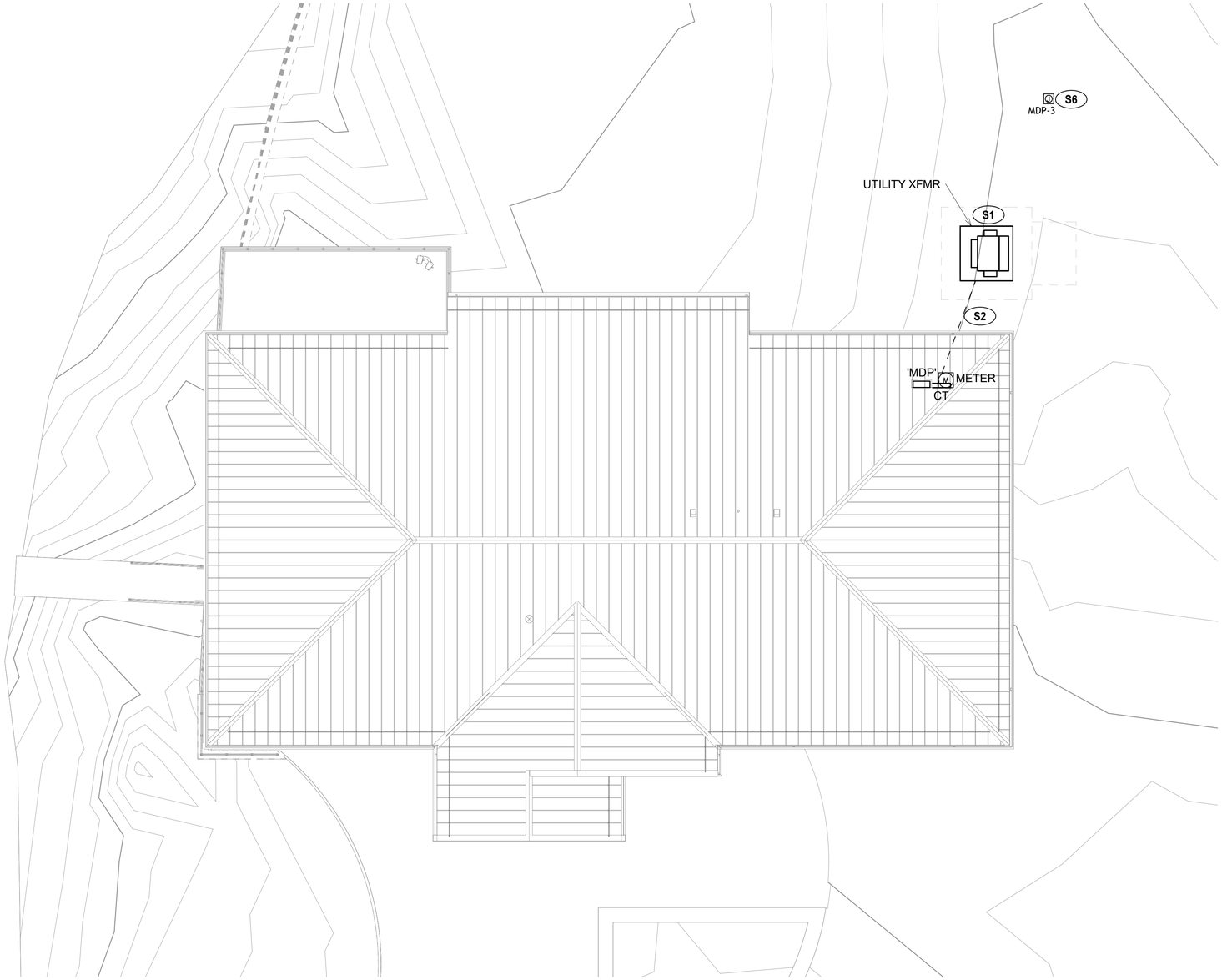
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR AE DESIGN DENVER, CO | DESIGNED: JR/JM/TW CADD AE DESIGN TECH REVIEW: MRS/ESN DATE: 5/29/2026 | SUB SHEET NO. ED201 | TITLE OF SHEET FIRST FLOOR DEMOLITION LIGHTING PLAN EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 112 OF 142 | |




1 E010 ELECTRICAL SITE PLAN - OVERALL
SCALE A

| KEYNOTE LEGEND | |
|---|---|
| KEY VALUE | KEYNOTE TEXT |
| S1 | EXISTING UTILITY TRANSFORMER AT THIS LOCATION SHALL BE REPLACED WITH NEW. ALLAMAKEE CLAYTON ELECTRIC COOPERATIVE (ACEC) SHALL PROVIDE NEW UTILITY TRANSFORMER ON EXISTING PAD. EXISTING UTILITY PRIMARY CONDUIT AND CONDUCTORS ARE EXISTING TO REMAIN. THE EC SHALL PERFORM ALL TRENCHING AND BACKFILLING ON THE SECONDARY SIDE OF THE TRANSFORMER. ACEC SHALL MAKE ALL CONNECTIONS OF PRIMARY AND SECONDARY CABLING AT THE TRANSFORMER LANDINGS. ACEC SHALL FURNISH, INSTALL, AND CONNECT THE METER IN THE EC PROVIDED METER HOUSING. REFER TO ELECTRICAL ONE-LINE DIAGRAM FOR MORE INFORMATION. |
| S2 | APPROXIMATE ROUTING OF NEW UNDERGROUND SECONDARY ELECTRICAL CONDUIT FROM UTILITY TRANSFORMER TO CT CABINET. REFER TO ELECTRICAL ONE-LINE DIAGRAM FOR MORE INFORMATION. |
| S6 | EXISTING LOCATION OF LIFT STATION. EC SHALL DISCONNECT AND PRESERVE CONDUCTORS AND CONDUITS DURING DEMOLITION FOR REUSE AND REINSTALLATION TO NEW MDP. |

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.

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LICENSED PROFESSIONAL ENGINEER
05/21/2026
JEFF MULLIKIN
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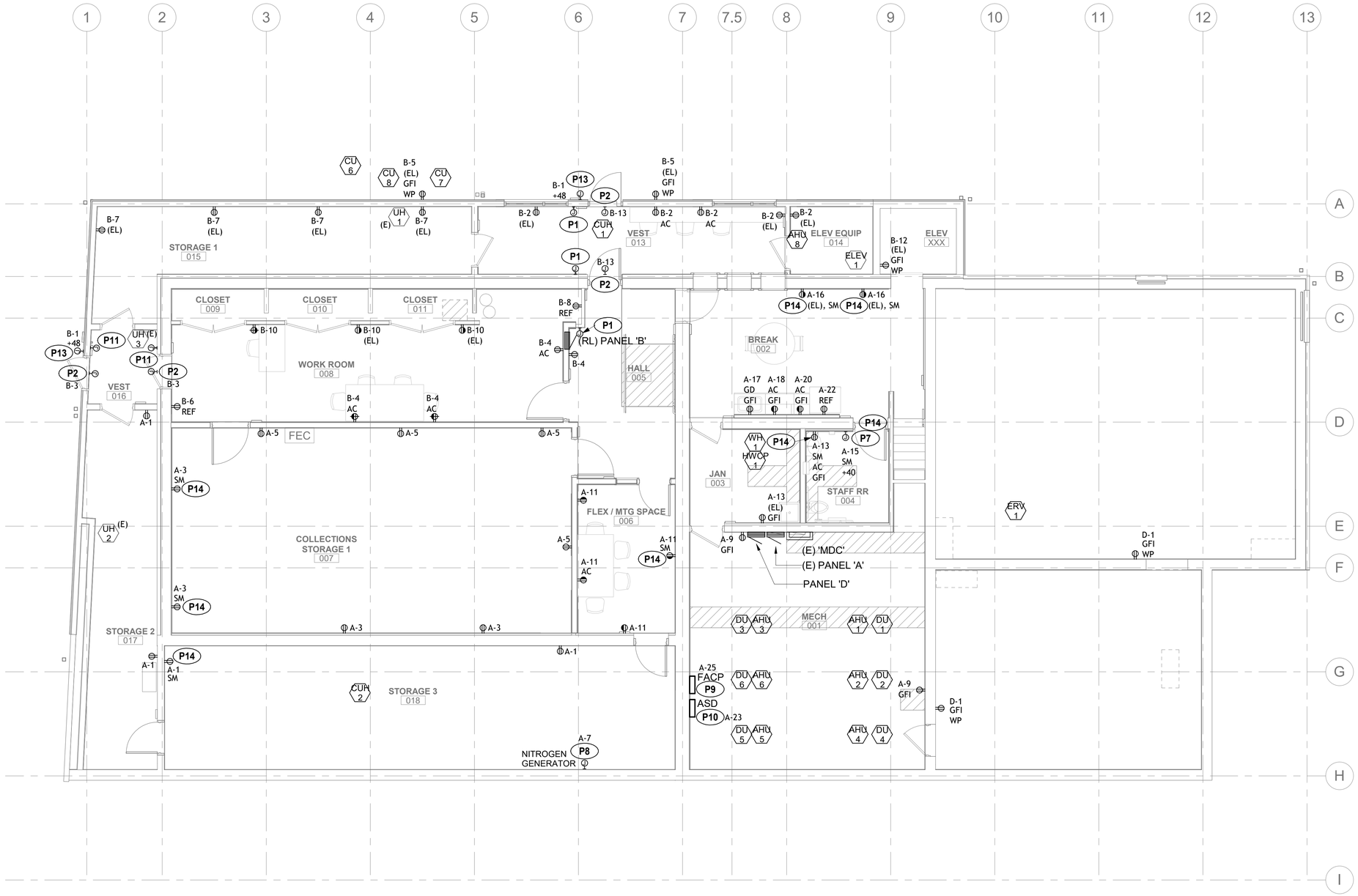


1 E100 BASEMENT ELECTRICAL PLAN

SCALE A

## POWER GENERAL NOTES

A. DEVICES NOTED WITH (EL) SHALL MAINTAIN EXISTING LOCATION IN ORDER TO LIMIT IMPACT TO EXISTING CONDITIONS. DEVICE AND FACEPLATE SHALL BE REMOVED AND REPLACED WITH NEW. BACKBOX, IF PRESENT AND CAPABLE, AND CONDUIT SHALL BE MAINTAINED FOR REUSE. IF BACKBOX IS NOT PRESENT OR CAPABLE, A NEW BOX SHALL BE PROVIDED. NEW BRANCH CIRCUIT FEEDERS SHALL BE PROVIDED.

B. ALL CONDUITS ROUTED BELOW FIRST LEVEL FLOOR DECK SHALL HAVE ANCHORS ATTACHED TO THE CONCRETE BEAMS ONLY. DO NOT ANCHOR TO THE CMU FLOOR PLANKS UNDER THE CONCRETE FLOOR TOPING. REFER TO ASSEMBLIES DETAIL C3.1/G500 FOR MORE INFORMATION.

## KEYNOTE LEGEND

| KEY VALUE | KEYNOTE TEXT |
|---|---|
| P1 | WALL MOUNTED JUNCTION BOX WITH 3/4" CONDUIT FOR INSTALLATION OF MOTORIZED ASSIST PUSH PAD DEVICE. COORDINATE EXACT LOCATION AND CONNECTION REQUIREMENTS WITH APPROVED SHOP DRAWINGS AND CO PRIOR TO ROUGH-IN. |
| P2 | JUNCTION BOX FOR 120V POWER CONNECTION TO MOTORIZED AUTOMATED ACCESS DOOR OPERATOR. EC SHALL MAKE ALL NECESSARY INTERCONNECTIONS BETWEEN JUNCTION BOX, OPERATOR MOTOR, AND PUSH PAD LOCATIONS PER MANUFACTURER'S REQUIREMENTS. COORDINATE EXACT LOCATIONS WITH APPROVED EQUIPMENT SHOP DRAWINGS. REFER TO KEYNOTE #1 FOR LOCATION OF ACTUATOR PUSH PAD. |
| P7 | WALL MOUNTED JUNCTION BOX FOR 120V, 20A HARDWIRED POWER CONNECTION TO HAND DRYER. EC SHALL CONFIRM EXACT FINAL LOCATION AND MOUNTING HEIGHT PRIOR TO ROUGH-IN. |
| P8 | EC SHALL PROVIDE 120V, 20A DEDICATED POWER CONNECTION TO NITROGEN GENERATOR. CONFIRM EXACT LOCATION AND CONNECTION REQUIREMENTS WITH FIRE PROTECTION CONTRACTOR PRIOR TO INSTALLATION. |
| P9 | EC SHALL PROVIDE 120V, 20A DEDICATED POWER CONNECTION TO FIRE ALARM CONTROL PANEL (FACP). CONFIRM EXACT LOCATION AND CONNECTION REQUIREMENTS WITH FIRE PROTECTION CONTRACTOR PRIOR TO INSTALLATION. |
| P10 | EC SHALL PROVIDE 120V, 20A DEDICATED POWER CONNECTION TO AIR SAMPLING DETECTION SYSTEM (ASD). CONFIRM EXACT LOCATION AND CONNECTION REQUIREMENTS WITH FIRE PROTECTION CONTRACTOR PRIOR TO INSTALLATION. |
| P11 | WALL MOUNTED JUNCTION BOX WITH 3/4" CONDUIT FOR INSTALLATION OF MOTORIZED ASSIST PUSH PAD DEVICE. PROVIDE RETROFIT CONDITION CLOSERS AT THIS LOCATION. COORDINATE EXACT LOCATION AND CONNECTION REQUIREMENTS WITH APPROVED SHOP DRAWINGS AND CO PRIOR TO ROUGH-IN. |
| P13 | PROVIDE WEATHERPROOF, TWO-WAY COMMUNICATIONS, AREA OF REFUGE, CALL STATION AT EXTERIOR EGRESS DOOR. DEVICE SHALL BE SURFACE MOUNTED AT +48" TO CENTER OF UNIT. THE TWO-WAY COMMUNICATION SYSTEM SHALL INCLUDE BOTH AUDIBLE AND VISIBLE SIGNALS. ALL CABLING BETWEEN COMMAND UNIT AND EACH CALL STATION AND POWER SUPPLY SHALL BE ROUTED WITHIN 3/4" CONDUIT. INCLUDE COMPLETE AND OPERATING SYSTEM, PROGRAMMING, SIGNAGE, DIRECTIONS, AND INSTRUCTIONS. REFER TO SHEET E101 FOR LOCATION OF COMMAND UNIT. ALL DEVICE LOCATIONS SHALL BE APPROVED BY THE LOCAL FIRE DEPARTMENT PRIOR TO INSTALLATION. |
| P14 | PROVIDE WIREMOLD SERIES 2400 (OR EQUAL) SINGLE CHANNEL RACEWAY ROUTED UP TO ABOVE CEILING. VERIFY FINAL RACEWAY AND BOX FINISH WITH CONTRACTING OFFICER. |

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.

SCALE: A 4 0 4 8 12
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LICENSED PROFESSIONAL ENGINEER
05/21/2026
JEFF MULLIKIN
P25384
ARIZONA

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| PRIME: | OTAK, INC. |
| | REDMOND, WA |
| SUBCONTRACTOR | AE DESIGN |
| | DENVER, CO |
| DESIGNED: | JR/JM/TW |
| CADD | AE DESIGN |
| TECH REVIEW: | MRS/ESN |
| DATE: | 5/29/2026 |
| SUB SHEET NO. | E100 |
| TITLE OF SHEET | BASEMENT ELECTRICAL PLAN |
| | EFFIGY MOUNDS NATIONAL MONUMENT |
| DRAWING NO. | 394 194248 |
| PMIS/PKG NO. | 318539 |
| SHEET | 114 OF 142 |

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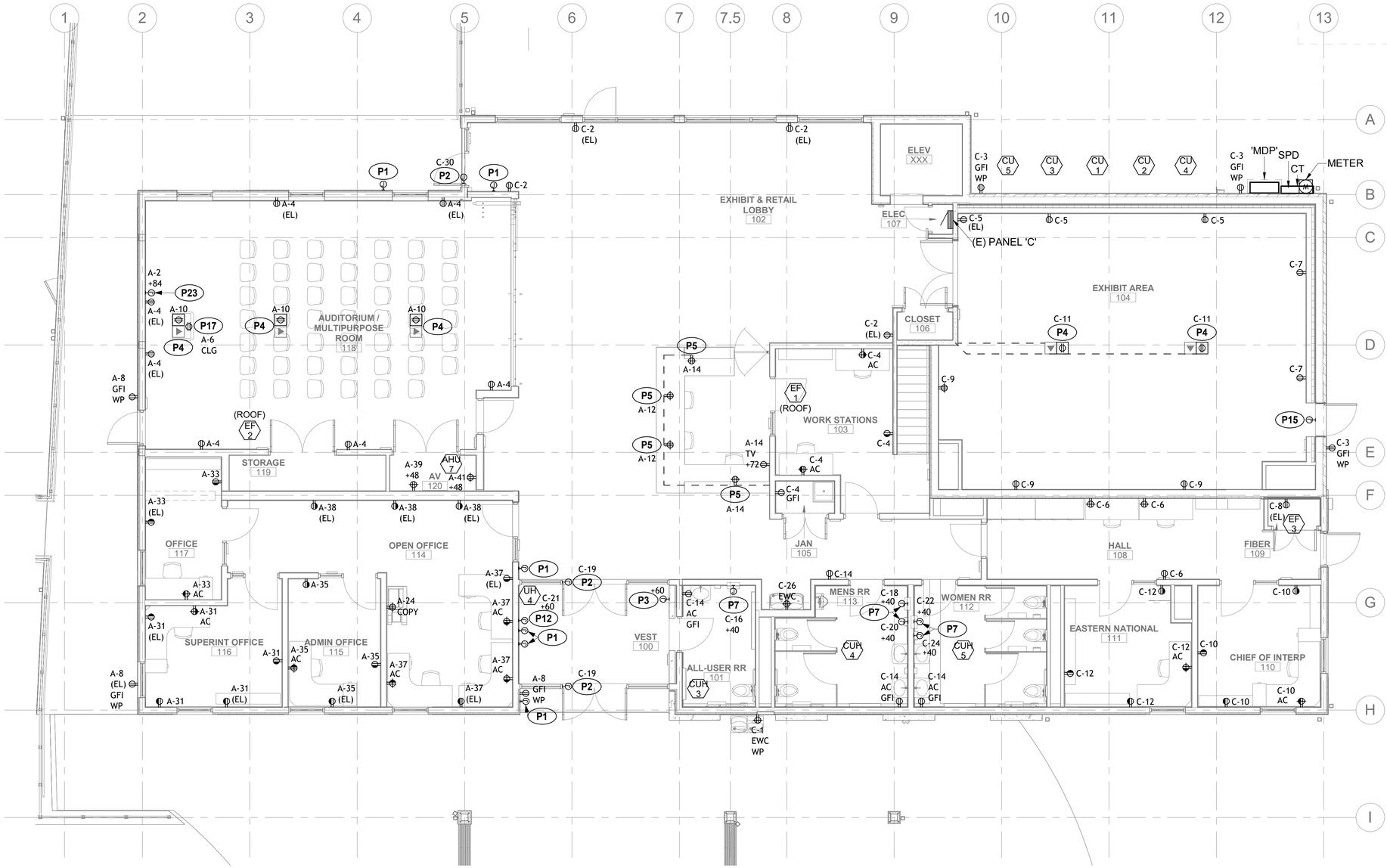


1 E101 FIRST FLOOR ELECTRICAL PLAN

SCALE A

## POWER GENERAL NOTES

| | |
|---|---|
| A. | DEVICES NOTED WITH (EL) SHALL MAINTAIN EXISTING LOCATION IN ORDER TO LIMIT IMPACT TO EXISTING CONDITIONS. DEVICE AND FACEPLATE SHALL BE REMOVED AND REPLACED WITH NEW. BACKBOX, IF PRESENT AND CAPABLE, AND CONDUIT SHALL BE MAINTAINED FOR REUSE. IF BACKBOX IS NOT PRESENT OR CAPABLE, A NEW BOX SHALL BE PROVIDED. NEW BRANCH CIRCUIT FEEDERS SHALL BE PROVIDED. |
| B. | ALL CONDUITS ROUTED BELOW FIRST LEVEL FLOOR DECK SHALL HAVE ANCHORS ATTACHED TO THE CONCRETE BEAMS ONLY. DO NOT ANCHOR TO THE CMU FLOOR PLANKS UNDER THE CONCRETE FLOOR TOPING. REFER TO ASSEMBLIES DETAIL C3.1/G500 FOR MORE INFORMATION. |

## KEYNOTE LEGEND

| KEY VALUE | KEYNOTE TEXT |
|---|---|
| P1 | WALL MOUNTED JUNCTION BOX WITH 3/4" CONDUIT FOR INSTALLATION OF MOTORIZED ASSIST PUSH PAD DEVICE. COORDINATE EXACT LOCATION AND CONNECTION REQUIREMENTS WITH APPROVED SHOP DRAWINGS AND CO PRIOR TO ROUGH-IN. |
| P2 | JUNCTION BOX FOR 120V POWER CONNECTION TO MOTORIZED AUTOMATED ACCESS DOOR OPERATOR. EC SHALL MAKE ALL NECESSARY INTERCONNECTIONS BETWEEN JUNCTION BOX, OPERATOR MOTOR, AND PUSH PAD LOCATIONS PER MANUFACTURER'S REQUIREMENTS. COORDINATE EXACT LOCATIONS WITH APPROVED EQUIPMENT SHOP DRAWINGS. REFER TO KEYNOTE #1 FOR LOCATION OF ACTUATOR PUSH PAD. |
| P3 | WALL MOUNTED JUNCTION BOX FOR 120V, 20A DEDICATED POWER CONNECTION TO FIRE ALARM PANEL. COORDINATE EXACT LOCATION AND REQUIREMENTS WITH APPROVED FIRE ALARM SHOP DRAWINGS PRIOR TO ROUGH-IN. |
| P4 | EC SHALL PROVIDE DUAL-SERVICE POKE-THRU FLOORBOX FOR FUTURE EXHIBIT INSTALLATION POWER REQUIREMENTS. PROVIDE POWER DEVICE AS INDICATED ON PLAN WITH BLANK OUTLET FOR FUTURE AV/DATA CONNECTIONS. ROUTE 3/4" CONDUIT FOR POWER AND 1-1/2" CONDUIT FOR LOW-VOLTAGE CABLING TO NEAREST ADJACENT WALL AND EXTEND TO ABOVE ACCESSIBLE CEILING. FLOORBOX DEVICE TO BE LEGRAND #4ATP2RSB-NA, OR APPROVED EQUAL. VERIFY EXACT LOCATION AND FINISH WITH CONTRACTING OFFICER PRIOR TO ROUGH-IN. |
| P5 | EC SHALL ROUTE CONDUIT FROM WALL, THROUGH MILLWORK TO DESK RECEPTACLES. COORDINATE FINAL LOCATIONS OF RECEPTACLES AND CONDUIT WITH CONTRACTING OFFICER PRIOR TO ROUGH-IN. |
| P7 | WALL MOUNTED JUNCTION BOX FOR 120V, 20A HARDWIRED POWER CONNECTION TO HAND DRYER. EC SHALL CONFIRM EXACT FINAL LOCATION AND MOUNTING HEIGHT PRIOR TO ROUGH-IN. |
| P12 | PROVIDE TWO WAY COMMUNICATIONS AREA OF REFUGE COMMAND UNIT RECESSED WITHIN WALL AND MOUNTED AT +60" TO CENTER OF UNIT. PROVIDE REMOTE POWER SUPPLY ABOVE ACCESSIBLE CEILING IN OPEN OFFICE 114. ALL CABLING BETWEEN COMMAND UNIT AND EACH CALL STATION AND POWER SUPPLY SHALL BE ROUTED WITHIN ¾" CONDUIT. PROVIDE TIMED AUTOMATIC TELEPHONE DIAL-OUT CAPABILITY TO A MONITORING LOCATION OR 911. INCLUDE COMPLETE AND OPERATING SYSTEM, PROGRAMMING, SIGNAGE, DIRECTIONS, AND INSTRUCTIONS. REFER TO SHEET E100 FOR LOCATIONS OF CALL STATIONS AT EACH EXTERIOR EGRESS DOOR. ALL DEVICE LOCATIONS SHALL BE APPROVED BY THE LOCAL FIRE DEPARTMENT PRIOR TO INSTALLATION. |
| P15 | EC SHALL PROVIDE 120V POWER CONNECTION TO DOOR HARDWARE. COORDINATE EXACT LOCATION AND CONNECTION REQUIREMENTS WITH APPROVED SHOP DRAWINGS AND CO PRIOR TO ROUGH-IN. |
| P17 | EC SHALL INSTALL CEILING MOUNTED RECEPTACLE WITHIN 'PB1' PLENUM BOX AS SHOWN IN DETAIL #6/T501. REFER TO SHEET T201 FOR LOCATION. |

## KEYNOTE LEGEND

| KEY VALUE | KEYNOTE TEXT |
|---|---|
| P23 | **OPTION F:** WALL MOUNTED JUNCTION BOX FOR 120V, 20A DEDICATED POWER CONNECTION TO MOTORIZED PROJECTOR SCREEN. EC SHALL CONFIRM EXACT FINAL LOCATION AND MOUNTING HEIGHT PRIOR TO ROUGH-IN. |

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.

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05/21/2026 JEFF MULLIKIN P25384 KANSAS LICENSED PROFESSIONAL ENGINEER

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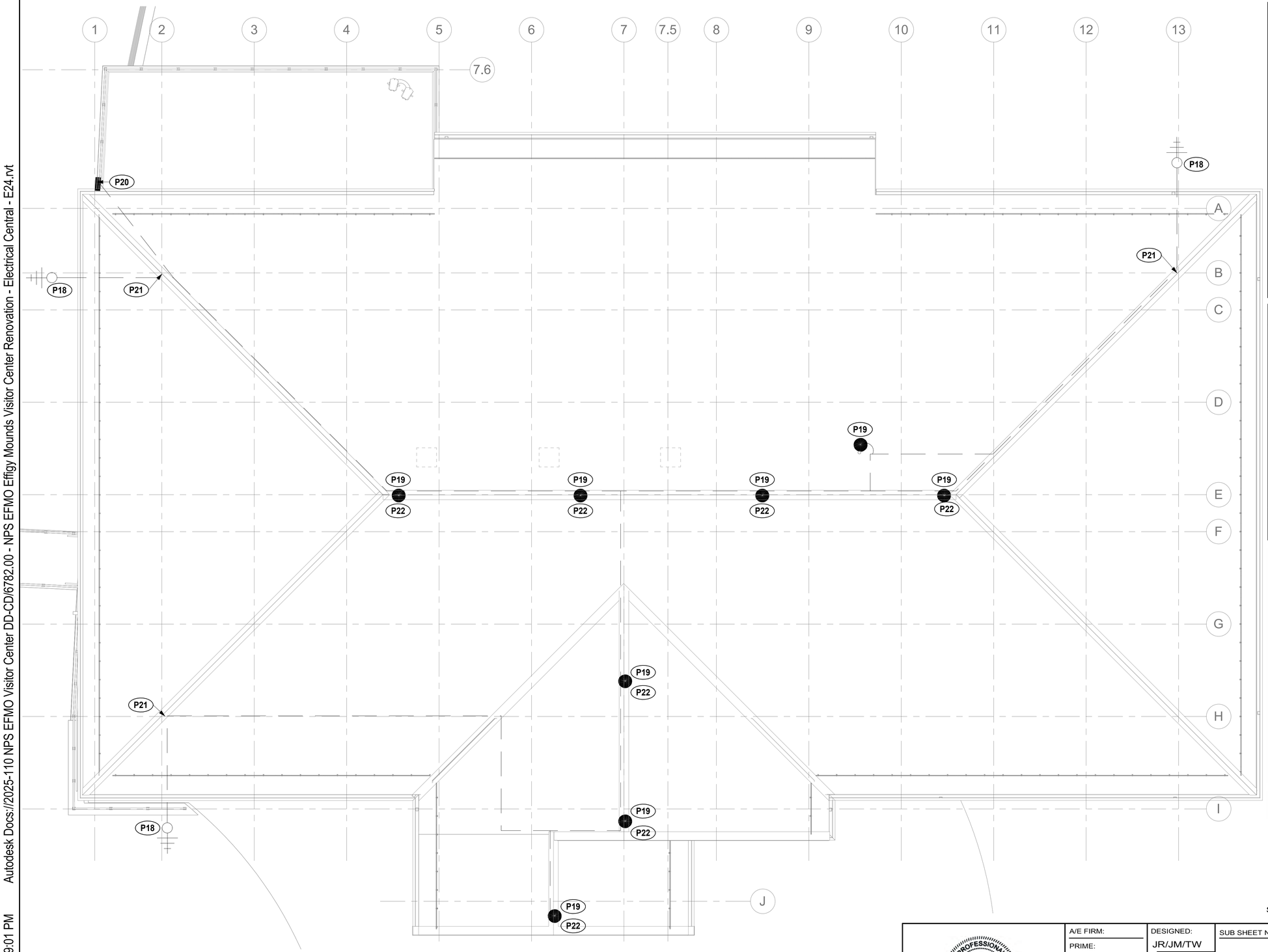
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR AE DESIGN DENVER, CO | DESIGNED: JR/JM/TW DRAWN: AE DESIGN TECH REVIEW: MRS/ESN DATE: 5/29/2026 | SUB SHEET NO. E101 | TITLE OF SHEET FIRST FLOOR ELECTRICAL PLAN EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 115 OF 142 | |




1 E102 ELECTRICAL LIGHTNING PROTECTION PLAN
SCALE A

## GENERAL NOTES

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|---|---|
| A. | VERIFY ALL AIR TERMINAL, DOWNLEAD, AND GROUND ROD LOCATIONS WITH APPROVED SHOP DRAWINGS AND EXISTING FIELD CONDITIONS PRIOR TO INSTALLATION. |
| B. | ALL CONDUCTIVE SERVICES (ELECTRICAL, COMMUNICATIONS, ETC.) AND BURIED METALLIC PIPING (WATER, GAS, FIRE, SANITARY, STORM, ETC.) WHICH ENTERS THE PROTECTED BUILDING SHALL BE BONDED TO THE LIGHTNING PROTECTION SYSTEM WITHIN 12 FT (3.6M) OF GRADE, PER NFPA 780 4.13. GAS TO BE BONDED ON CUSTOMER SIDE OF METER. |
| C. | THE LIGHTNING PROTECTION SYSTEM REQUIRES A SURGE PROTECTION DEVICE TO BE INSTALLED ON EACH POWER SERVICE ENTRANCE. THESE SURGE PROTECTION DEVICES MUST BE A TYPE 1 OR TYPE 2 SURGE PROTECTION DEVICE THAT IS LISTED TO COMPLY WITH UL 1449, 4TH EDITION AND HAVE A NOMINAL DISCHARGE CURRENT OF 20KA PER PHASE, PER NFPA 780 4.18. |
| D. | SURGE PROTECTION DEVICES SHALL BE PROVIDED FOR ALL COMMUNICATION SYSTEMS, INCLUDING BUT NOT LIMITED TO CATV, ALARM, DATA, AND ANTENNA SYSTEMS AT FACILITY ENTRANCES, PER NFPA 780 4.18. |
| E. | REFER TO SHEET E403 FOR ADDITIONAL LIGHTNING PROTECTION INSTALLATION DETAILS. |

## KEYNOTE LEGEND

| KEY VALUE | KEYNOTE TEXT |
|---|---|
| P18 | LIGHTNING SYSTEM IS REQUIRED TO BE COMMONLY BONDED WITH FULL SIZED MAIN CONDUCTOR TO WATER, FIRE SUPPRESSION, AND ELECTRICAL GROUND. PVC PROVIDED BY ELECTRICAL CONTRACTOR. |
| P19 | DUE TO DISSIMILAR METALS, ALUMINIUM AIR TERMINALS MUST BE USED OVER THE STANDING SEAM ROOF. |
| P20 | BOND STRUCTURAL BEAM BELOW SLAB TO METAL HANDRAIL IN LIEU OF AIR TERMINALS. BUILDING STEEL SHALL BE BONDED TO THE LIGHTNING PROTECTION SYSTEM AT THE DOWNLEAD WITHIN EXTERIOR WALL. EC SHALL TEST CONTINUITY BETWEEN THE GUARDRAIL AND DOWNLEADS. BONDING CONDUCTORS SHALL BE SIZED PER NFPA 780. |
| P21 | DOWNLEADS ARE TO ROUTE DOWN THROUGH EXTERIOR WALL AT THIS LOCATION. RE: #4/E403. |
| P22 | PROVIDE THRU ROOF BASE AND AIR TERMINAL SUCH THAT CONDUCTOR BETWEEN AIR TERMINALS IS HIDDEN FROM VIEW, ROUTED WITHIN RAFTER SPACE. |

## LIGHTNING PROTECTION SYSTEM NOTES

ELECTRICAL CONTRACTOR SHALL PROVIDE 'OPTION B' PRICING TO PROVIDE AND INSTALL A LIGHTNING PROTECTION SYSTEM WITH AN UNDERWRITER'S LABORATORIES LISTED MASTER C LABEL. REFER TO PERFORMANCE SPECIFICATION SECTION 26 4113 FOR SYSTEM REQUIREMENTS. COORDINATE ALL AIR TERMINAL LOCATIONS ON ROOF WITH CONTRACTING OFFICER WHILE DEVELOPING SUBMITTALS. COORDINATE ALL DOWN CONDUCTOR LOCATIONS WITH CONTRACTING OFFICER AND MANUFACTURER DURING SUBMITTALS. SUBMITTALS REQUIRE FULL ROOF PLANS WITH ALL AIR TERMINALS (AND ALL OTHER EQUIPMENT) TO BE INDICATED FOR REVIEW PRIOR TO SUBMITTAL APPROVAL. SYSTEM SHALL BE INSTALLED IN A NEAT AND WORKMANLIKE MANNER. CONDUCTORS SHALL BE RUN PARALLEL TO BUILDING SURFACES AND CONCEALED FROM VIEW TO THE GREATEST EXTENT POSSIBLE, WHERE VISIBLE, PROVIDE STRAIGHT RUNS, MINIMIZE LOOSE TURNS, AND PROVIDE SUPPORTS AT NO MORE THAN 3' INTERVALS. MAINTAIN A MINIMUM OF 8" BEND RADIUS ON ALL TURNS TO COMPLY WITH NFPA REQUIREMENTS.

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION E.

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LICENSED PROFESSIONAL ENGINEER
05/21/2026
JEFF MULLIKIN
P25384
WASHINGTON

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| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR AE DESIGN DENVER, CO | DESIGNED: JR/JM/TW CADD: AE DESIGN TECH REVIEW: MRS/ESN DATE: 5/29/2026 | SUB SHEET NO. E102 | TITLE OF SHEET ELECTRICAL OPTION LIGHTNING PROTECTION PLAN EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 116 OF 142 | |

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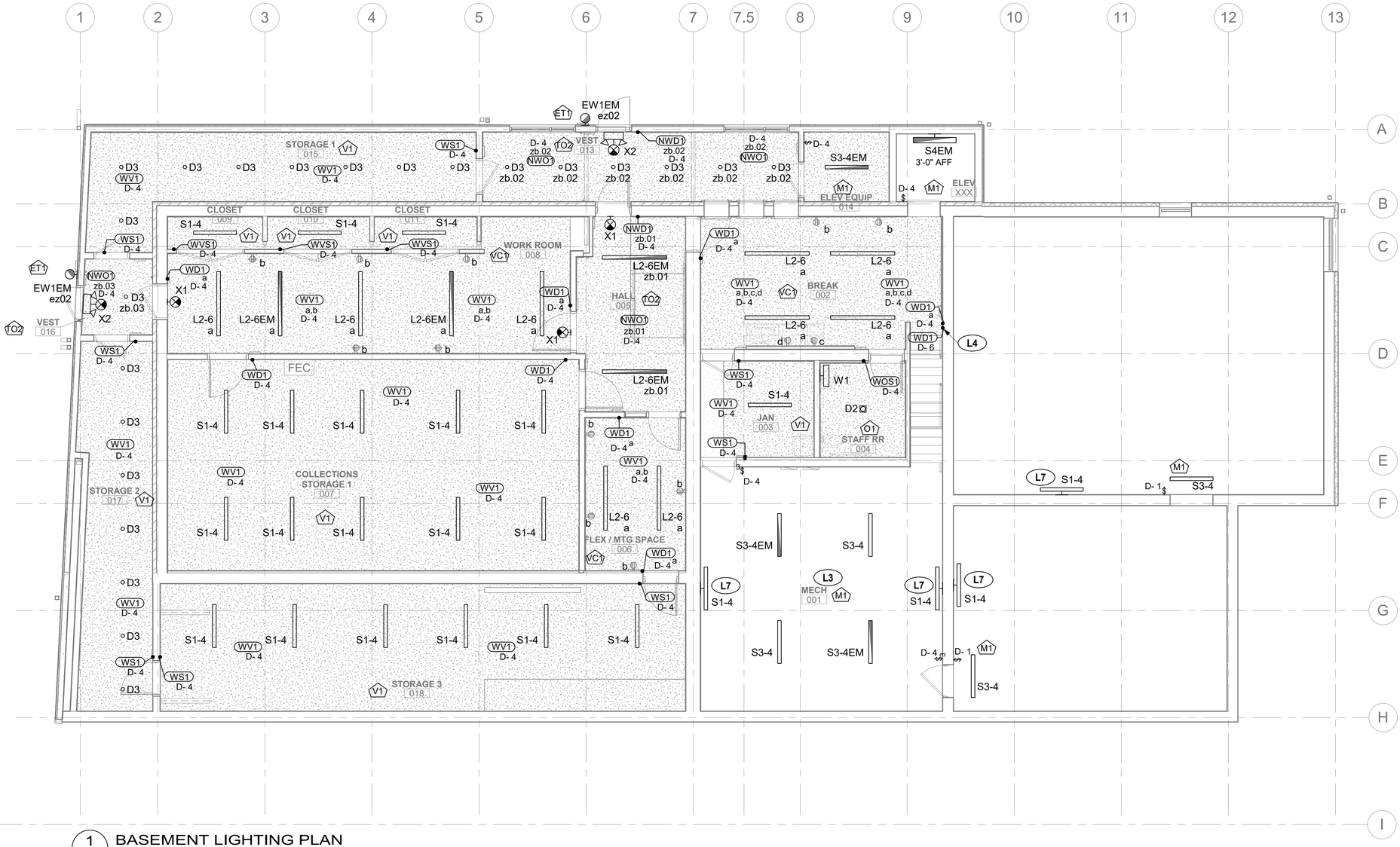


1 E200 BASEMENT LIGHTING PLAN
SCALE A

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.

## LIGHTING GENERAL NOTES

A. DAYLIGHT ZONE LIGHTING POWER DENSITY IS SHOWN ON PLAN FOR REFERENCE ONLY. PHOTOCELLS FOR DAYLIGHT HARVESTING SHALL ONLY BE PROVIDED IF SHOWN ON PLAN.

B. CONTROLLED RECEPTACLES WITH CONTROL DESIGNATION SHALL BE CONTROLLED AS INDICATED, VIA EITHER LOCAL ROOM/LOAD CONTROLLER CONNECTED TO LIGHTING CONTROL SENSORS, OR VIA TIMECLOCK BASED SYSTEM WITH LOCAL OVERRIDE SWITCH AND SCHEDULING CAPABILITIES TO AUTOMATICALLY TURN OFF AS REQUIRED BY IECC 2021. REFER TO LIGHTING CONTROL SCHEDULES, AND CONTROLLED RECEPTACLE AND ROOM/LOAD CONTROLLER DETAILS FOR ADDITIONAL INFORMATION.

C. ALL CONDUITS ROUTED BELOW FIRST LEVEL FLOOR DECK SHALL HAVE ANCHORS ATTACHED TO THE CONCRETE BEAMS ONLY. DO NOT ANCHOR TO THE CMU FLOOR PLANKS UNDER THE CONCRETE FLOOR TOPING. REFER TO ASSEMBLIES DETAIL C3.1/G500 FOR MORE INFORMATION.

## KEYNOTE LEGEND

| KEY VALUE | KEYNOTE TEXT |
|---|---|
| L3 | EC SHALL COORDINATE INSTALLATION OF LIGHTING FIXTURES WITH MECHANICAL DUCTWORK IN THIS SPACE. |
| L4 | SWITCH SHALL BE CIRCUITED TO CONTROL TROFFER FIXTURES LOCATED ABOVE THE STAIRS AT THIS LOCATION. REFER TO FIRST FLOOR LIGHTING PLAN ON SHEET E201 FOR FIXTURE LOCATIONS AND ZONING. |
| L7 | S1-4 FIXTURE SHALL BE SURFACE MOUNTED TO THE WALL AT THIS LOCATION. |

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.

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LICENSED PROFESSIONAL ENGINEER 05/21/2026 JEFF MULLIKIN P25384 ROHNERT PARK CA

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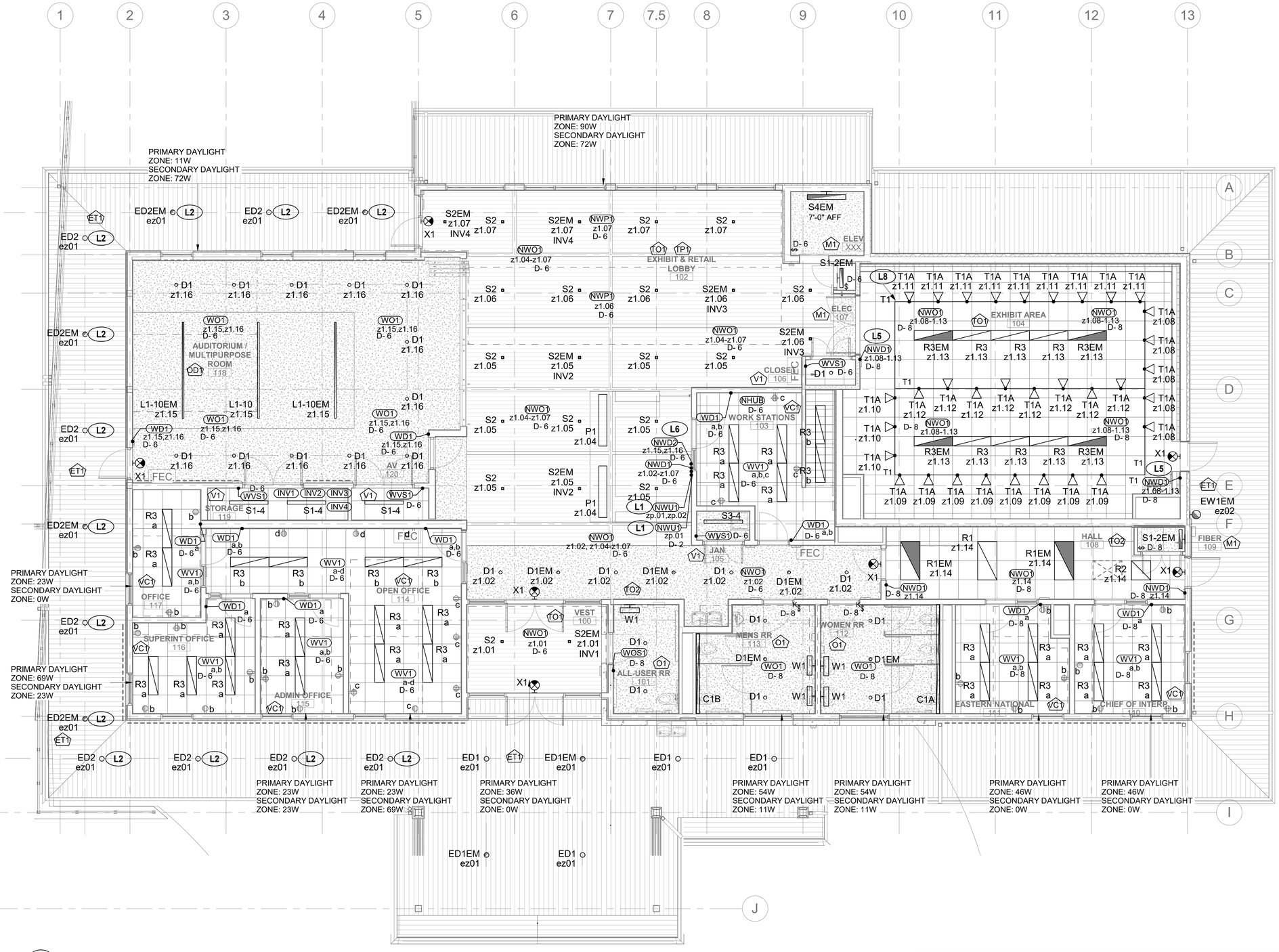
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR AE DESIGN DENVER, CO | DESIGNED: JR/JM/TW CADD AE DESIGN TECH REVIEW: MRS/ESN DATE: 5/29/2026 | SUB SHEET NO. E200 | TITLE OF SHEET BASEMENT LIGHTING PLAN EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 117 OF 142 |

5/21/2026 1:49:05 PM Autodesk Docs://2025-110 NPS EFMO Visitor Center DD-CD/6782.00 - NPS EFMO Effigy Mounds Visitor Center Renovation - Electrical Central - E24.rvt




1 E201 FIRST FLOOR LIGHTING PLAN
SCALE A

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.

## LIGHTING GENERAL NOTES

A. DAYLIGHT ZONE LIGHTING POWER DENSITY IS SHOWN ON PLAN FOR REFERENCE ONLY. PHOTOCELLS FOR DAYLIGHT HARVESTING SHALL ONLY BE PROVIDED IF SHOWN ON PLAN.

B. CONTROLLED RECEPTACLES WITH CONTROL DESIGNATION SHALL BE CONTROLLED AS INDICATED, VIA EITHER LOCAL ROOM/LOAD CONTROLLER CONNECTED TO LIGHTING CONTROL SENSORS, OR VIA TIMECLOCK BASED SYSTEM WITH LOCAL OVERRIDE SWITCH AND SCHEDULING CAPABILITIES TO AUTOMATICALLY TURN OFF AS REQUIRED BY IECC 2021. REFER TO LIGHTING CONTROL SCHEDULES, AND CONTROLLED RECEPTACLE AND ROOM/LOAD CONTROLLER DETAILS FOR ADDITIONAL INFORMATION.

C. ALL CONDUITS ROUTED BELOW FIRST LEVEL FLOOR DECK SHALL HAVE ANCHORS ATTACHED TO THE CONCRETE BEAMS ONLY. DO NOT ANCHOR TO THE CMU FLOOR PLANKS UNDER THE CONCRETE FLOOR TOPING. REFER TO ASSEMBLIES DETAIL C3.1/G500 FOR MORE INFORMATION.

## KEYNOTE LEGEND

| KEY VALUE | KEYNOTE TEXT |
|---|---|
| L1 | NWU1 DEVICE SHALL BE A WIRELESS TIMER SWITCH TO CONTROL EXTERIOR PARKING LOT BOLLARD LIGHTS. ONE NWU1 DEVICE SHALL CONTROL ALL BOLLARDS, AND THE OTHER SHALL CONTROL EAST BOLLARDS ONLY. BOTH NWU1 DEVICES SHALL BE INSTALLED IN A SINGLE FACEPLATE. REFER TO PARKING LOT DRAWINGS (PMIS 224159) FOR BOLLARD ZONING. |
| L2 | EC SHALL INSTALL RETROFIT ED2/ED2EM FIXTURE AS A ONE FOR ONE REPLACEMENT OF EXISTING DOWNLIGHT AT THIS LOCATION. |
| L5 | PROGRAMMING OF LIGHTS WITHIN THE EXHIBIT AREA SHALL INCLUDE A PROGRAMMING MASK TO PREVENT OCCUPANTS FROM TURNING OFF LIGHTS DURING OPEN HOURS. |
| L6 | NWD2 DEVICE AT THIS LOCATION PROVIDED TO CONTROL LIGHTS WITHIN AUDITORIUM/MULTIPURPOSE ROOM 118. DEVICE SHALL BE INSTALLED ADJACENT TO AV CONTROL DEVICE FOR AUDITORIUM/MULTIPURPOSE ROOM 118. |
| L8 | **OPTION F:** ALL EXHIBIT AREA TRACK LIGHTING |

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.

SCALE: A 4 0 4 8 12 SCALE OF FEET

| | | | | | |
|---|---|---|---|---|---|
| LICENSED PROFESSIONAL ENGINEER 05/21/2026 JEFF MULLIKIN P25384 | A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR AE DESIGN DENVER, CO | DESIGNED: JR/JM/TW CADD AE DESIGN TECH REVIEW: MRS/ESN DATE: 5/29/2026 | SUB SHEET NO. E201 | TITLE OF SHEET FIRST FLOOR LIGHTING PLAN EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 118 OF 142 |

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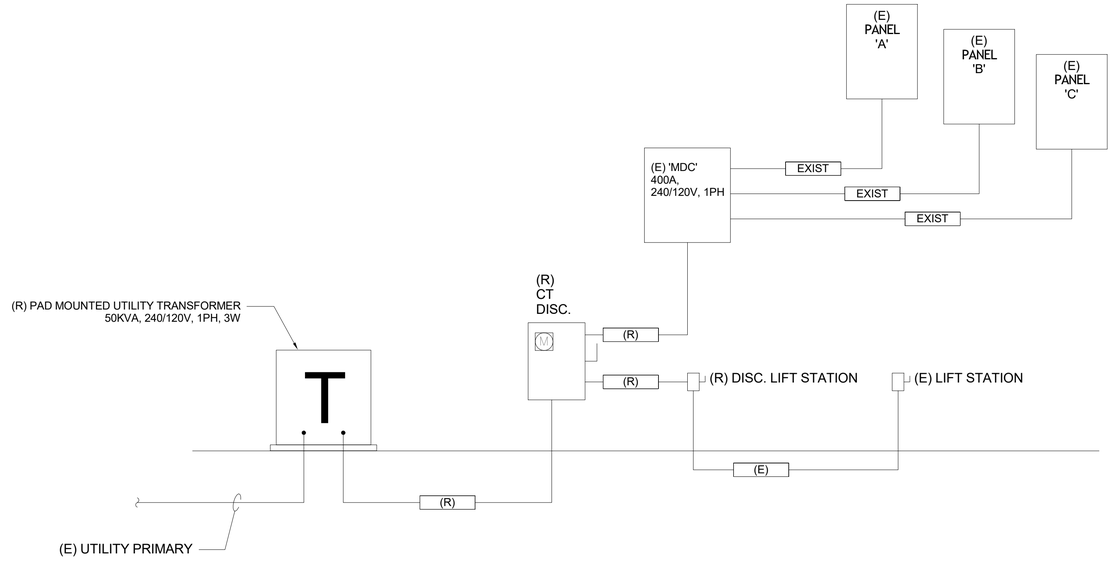
## FEEDER SCHEDULE

| KEY/ AMPS | FEEDER CONDUIT AND CONDUCTORS | KEY/ AMPS | FEEDER CONDUIT AND CONDUCTORS |
|---|---|---|---|
| COPPER SINGLE PHASE | | | |
| SERVICE ENTRANCE FEEDERS | | SDS XFMR FEEDERS (NOTE 1) | |
| 100N | 4#1, 1-1/2"C | 30S | 4#10, 1#8G, 3/4"C |
| 200N | 4#3/0, 2"C | 50S | 3#6, 1#8G, 1-1/4"C |
| 400N | 2[3#3/0, 2"C] | 100S | 3#1, 1#6G, 1-1/2"C |
| 600N | 2[3#350, 3"C] | 150S | 3#1/0, 1#6G, 2"C |
| 800N | 2[3#500, 3-1/2"C] | 250S | 3#250, 1#2G, 3"C |
| 1000N | 3[3#400, 3-1/2"C] | 400S | 2[3#3/0, 1#2G, 2-1/2"C] |
| 1200N | 4[3#350, 3"C] | 500S | 2[3#250, 1#1/0G, 3"C] |
| 1600N | 5[3#400, 3-1/2"C] | 800S | 2[3#500, 1#2/0G, 3"C] |
| 2000N | 6[3#400, 3-1/2"C] | 1000S | 3[3#400, 1#3/0G, 3-1/2"C] |
| 2500N | 7[3#500, 3-1/2"C] | 1600S | 5[3#400, 1#3/0G, 3-1/2"C] |
| 3000N | 8[3#500, 3-1/2"C] | 2500S | 7[3#500, 1#3/0G, 3-1/2"C] |
| 3500N | 10[3#500, 3-1/2"C] | -- | -- |
| 4000N | 11[3#500, 3-1/2"C] | -- | -- |
| EQUIPMENT FEEDERS | | | |
| 20NG | 3#12, #12G, 3/4"C | 20G | 2#12, #12G, 3/4"C |
| 30NG | 3#10, 1#10G, 3/4"C | 30G | 2#10, 1#10G, 3/4"C |
| 40NG | 3#8, 1#10G, 1"C | 40G | 2#8, 1#10G, 1"C |
| 50NG | 3#6, 1#10G, 1-1/4"C | 50G | 2#6, 1#10G, 1"C |
| 60NG | 3#4, 1#10G, 1-1/4"C | 60G | 2#4, 1#10G, 1"C |
| 70NG | 3#4, 1#8G, 1-1/4"C | 70G | 2#4, 1#8G, 1-1/4"C |
| 80NG | 3#3, 1#8G, 1-1/4"C | 80G | 2#3, 1#8G, 1-1/4"C |
| 90NG | 3#2, 1#8G, 1-1/2"C | 90G | 2#2, 1#8G, 1-1/4"C |
| 100NG | 3#1, 1#8G, 1-1/2"C | 100G | 2#1, 1#8G, 1-1/2"C |
| 110NG | 3#1, 1#6G, 2"C | 110G | 2#1, 1#6G, 1-1/2"C |
| 125NG | 3#1/0, 1#6G, 2"C | 125G | 2#1/0, 1#6G, 1-1/2"C |
| 150NG | 3#1/0, 1#6G, 2"C | 150G | 2#1/0, 1#6G, 1-1/2"C |
| 175NG | 3#2/0, 1#6G, 2"C | 175G | 2#2/0, 1#6G, 2"C |
| 200NG | 3#3/0, 1#6G, 2-1/2"C | 200G | 2#3/0, 1#6G, 2"C |
| 225NG | 3#4/0, 1#4G, 2-1/2"C | 225G | 2#4/0, 1#4G, 2"C |
| 250NG | 3#250, 1#4G, 3"C | 250G | 2#250, 1#4G, 2-1/2"C |
| 300NG | 3#350, 1#4G, 3"C | 300G | 2#350, 1#4G, 2-1/2"C |
| 350NG | 3#500, 1#3G, 3-1/2"C | 350G | 2#500, 1#3G, 3"C |
| 400NG | 2[3#3/0, 1#3G, 2-1/2"C] | 400G | 2[2#3/0, 1#3G, 2"C] |
| 450NG | 2[3#4/0, 1#2G, 2-1/2"C] | 450G | 2[2#4/0, 1#2G, 2"C] |
| 500NG | 2[3#250, 1#2G, 3"C] | 500G | 2[2#250, 1#2G, 2-1/2"C] |
| 600NG | 2[3#350, 1#1G, 3"C] | 600G | 2[2#350, 1#1G, 2-1/2"C] |
| 700NG | 2[3#500, 1#1/0G, 3-1/2"C] | 700G | 2[2#500, 1#1/0G, 3"C] |
| 800NG | 2[3#500, 1#1/0G, 3-1/2"C] | 800G | 2[2#500, 1#1/0G, 3"C] |
| 1000NG | 3[3#400, 1#2/0G, 3-1/2"C] | 1000G | 3[2#400, 1#2/0G, 3"C] |
| 1200NG | 4[3#350, 1#3/0G, 3"C] | 1200G | 4[2#350, 1#3/0G, 3"C] |
| 1600NG | 5[3#400, 1#4/0G, 3-1/2"C] | 1600G | 5[2#400, 1#4/0G, 3"C] |
| 2000NG | 6[3#400, 1#250G, 3-1/2"C] | 2000G | 6[2#400, 1#250G, 3"C] |
| 2500NG | 7[3#500, 1#350G, 3-1/2"C] | 2500G | 7[2#500, 1#350G, 3"C] |
| 3000NG | 8[3#500, 1#400G, 3-1/2"C] | 3000G | 8[2#500, 1#400G, 3"C] |
| 3500NG | 10[3#500, 1#500G, 3-1/2"C] | 3500G | 10[2#500, 1#500G, 3"C] |
| 4000NG | 11[3#500, 1#500G, 3-1/2"C] | 4000G | 11[2#500, 1#500G, 3"C] |

| GROUNDING CONDUCTORS | | ABBREVIATIONS | |
|---|---|---|---|
| G8 | 1#8, 3/4" C | MECH | SEE MECH SCHEDULE |
| G6 | 1#6, 3/4" C | XFMR | SEE XFMR SCHEDULE |
| G4 | 1#4, 3/4" C | | |
| G2 | 1#2, 3/4" C | | |
| G10 | 1#1/0, 3/4" C | | |
| G20 | 1#2/0, 3/4" C | | |
| G30 | 1#3/0, 3/4" C | | |

NOTES:

1. FEEDER FOR SECONDARY OF SEPARATELY DERIVED SYSTEM (SDS). GROUND SIZE PER NEC TABLE INCLUDED IN ARTICLE 250.66.
2. ALL CONDUCTORS ARE SINGLE CONDUCTOR COPPER THWN UNLESS NOTED OTHERWISE. AMPACITY BASED ON THE NEC TABLE INCLUDED IN ARTICLE 310.
3. ALL CONDUITS ARE EMT UNLESS NOTED OTHERWISE, FILL RATIOS BASED ON NEC ANNEX C TABLE C.1.

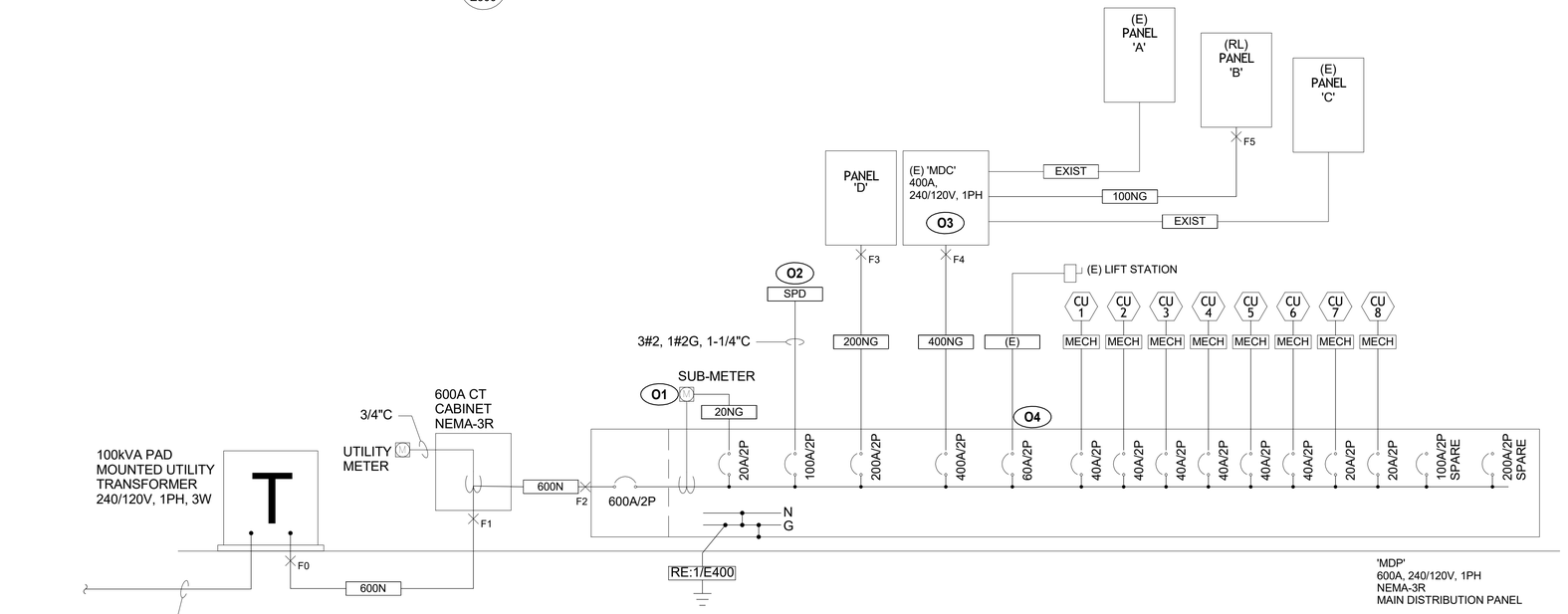



2 E300 DEMOLITION ELECTRICAL ONE-LINE DIAGRAM

## KEYNOTE LEGEND

| KEY VALUE | KEYNOTE TEXT |
|---|---|
| O1 | CONTRACTOR SHALL PROVIDE SUB-METERING IN COMPLIANCE WITH EPACT 2005 REQUIREMENTS. METER SHALL RECORD KWH ENERGY USAGE DATA AND BE CAPABLE OF WIRELESS AMR AND TCP/IP COMMUNICATION PROTOCOL. AS A BASIS OF DESIGN, PROVIDE LEVITON VERIFEYE SERIES 1000 1K240-08W (OR APPROVED EQUAL) RATED FOR OUTDOOR USE. PROVIDE AND INSTALL CTS ON INCOMING SERVICE FEEDER PER MANUFACTURER'S RECOMMENDATIONS. |
| O2 | CONTRACTOR SHALL PROVIDE UL LISTED EXTERNAL SURGE PROTECTION DEVICE (SPD) WITH 100KA PER MODE (200KA PER PHASE) MINIMUM FOR ELECTRICAL EQUIPMENT INDICATED. AS BASIS OF DESIGN, SPD SHALL BE CURRENT TECHNOLOGY, CURRENTGUARD PLUS #CGP-100-120/240-2G (OR APPROVED EQUAL). VERIFY FINAL BREAKER AND WIRE SIZE PER SELECTED MANUFACTURER'S RECOMMENDATION. |
| O3 | EC SHALL ISOLATE ALL NEUTRAL AND GROUND WIRES WITHIN PANEL TO RESPECTIVE BUSSING. NEUTRAL-GROUND BOND WILL BE MOVED TO THE MDP. |
| O4 | EC SHALL INTERCEPT EXISTING FEEDER/CONDUIT AND EXTEND INTO NEW 'MDP'. |

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.




1 E300 ELECTRICAL ONE-LINE DIAGRAM

| | | | | | |
|---|---|---|---|---|---|
| LICENSED PROFESSIONAL ENGINEER 05/21/2026 JEFF MULLIKIN P25384 ROYAL | A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR AE DESIGN DENVER, CO | DESIGNED: JR/JM/TW CADD AE DESIGN TECH REVIEW: MRS/ESN DATE: 5/29/2026 | SUB SHEET NO. E300 | TITLE OF SHEET ELECTRICAL ONE LINE DIAGRAM EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 119 OF 142 |

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.

# GENERAL GROUNDING NOTES

A. ALL CABLES TO BE TERMINATED ONTO BUS BAR WITH TWO HOLE COMPRESSION LUGS AND ATTACHED TO BUS BAR WITH TAB COMPRESSION BELLEVILLE WASHERS AND TORK BOLT ASSEMBLY.

B. ALL GROUND CONNECTORS SHALL BE STRANDED.

C. ALL BUS BARS SHALL BE ATTACHED TO SURFACE WITH NON-CONDUCTIVE STAND-OFFS.

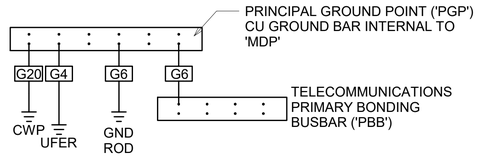
D. GROUND BUS BAR AND GROUNDING SYSTEM SHALL BE U.L. LISTED AND COMPLY WITH MANUFACTURERS INSTALLATION INSTRUCTIONS.

# GROUNDING ELECTRODE SYSTEMS NOTES

1. METAL UNDERGROUND WATER PIPE - MAKE CONNECTION TO METAL UNDERGROUND WATER PIPE INDIRECT CONTRACT WITH THE EARTH FOR 10FT AND ELECTRICALLY CONTINUOUS TO THE POINTS OF CONNECTION TO THE GROUNDING ELECTRODE CONDUCTOR AND BONDING CONDUCTORS. CONNECTION POINT TO BE AT A MAXIMUM OF 5FT OF THE POINT OF ENTRANCE ON THE INTERIOR OF THE BUILDING.

2. UFER GROUND (CONCRETE-ENCASED ELECTRODE) - AN ELECTRODE ENCASED BY AT LEAST 2" OF CONCRETE, LOCATED WITHIN AND NEAR THE BOTTOM OF A CONCRETE FOUNDATION OR FOOTING THAT IS IN DIRECT CONTACT WITH EARTH, CONSISTING OF AT LEAST 20' OF ONE OR MORE BARE OR ZINC GALVANIZED OR OTHER ELECTRICALLY CONDUCTIVE COATED STEEL REINFORCING BARS OR RODS OF NOT LESS THAN 1/2" IN DIAMETER, OR CONSISTING OF AT LEAST 20FT OF BARE COPPER CONDUCTOR NOT SMALLER THAN NO. 4 AWG. REINFORCING BARS SHALL BE PERMITTED TO BE BONDED TOGETHER BY THE USUAL STEEL TIE WIRES OR OTHER EFFECTIVE MEANS.

3. GROUND ROD - ROD IS TO BE 8FT IN LENGTH AND SHALL BE MADE OF IRON OR STEEL AT LEAST 5/8" DIAMETER. INSTALLATION METHODS FOR GROUND ROD SHALL BE IN COMPLIANCE WITH THE NEC SUCH THAT AT LEAST 8FT OF LENGTH IS IN CONTACT WITH THE EARTH.




1 E400 GROUNDING DETAIL & NOTES

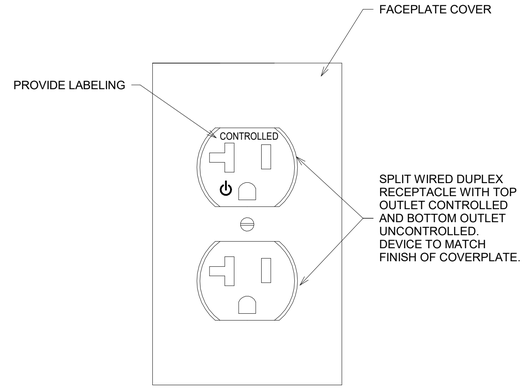
# LOAD SUMMARY: MDP

| LOAD DESCRIPTION | LOAD | NOTES |
|---|---|---|
| PANEL 'D' | 33423 VA | |
| MDC | 67830 VA | |
| (E) LIFT STATION | 9600 VA | |
| CU-1 | 3848 VA | |
| CU-2 | 3848 VA | |
| CU-3 | 3848 VA | |
| CU-4 | 3848 VA | |
| CU-5 | 3848 VA | |
| CU-6 | 3848 VA | |
| CU-7 | 3528 VA | |
| CU-8 | 3528 VA | |

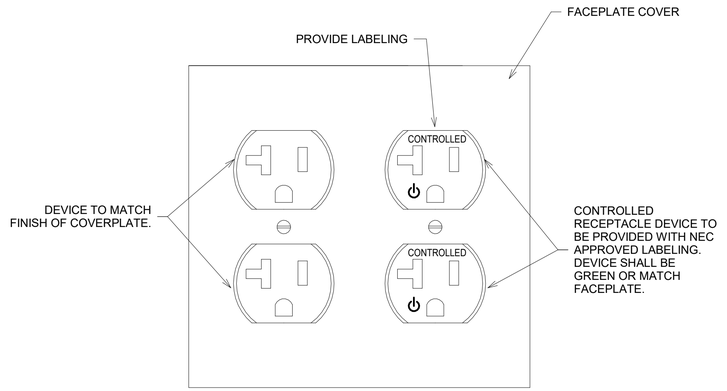
| | |
|---|---|
| **TOTAL NEW LOAD** | 141 kVA |
| **@ 240/120V, 3PH** | 587 A |
| **TOTAL EST. DEMAND** | 137 kVA |
| **@240/120V, 3PH** | 572 A |

**THEREFORE THE 600A SERVICE IS JUSTIFIED.**

**Notes:**
AVAILABLE SPARE CAPACITY INDICATED IS BASED UPON TOTAL CONNECTED LOAD. DEMAND LOAD UNDER NORMAL OPERATING CONDITIONS IS ANTICIPATED AT APPROXIMATELY 50% OVERALL CAPACITY. THE SERVICE SIZE IS LIMITED TO THE INDICATED 600A BASED ON COORDINATION DISCUSSIONS WITH THE UTILITY COMPANY, AND APPROVED BY THE NATIONAL PARK SERVICE AS PART OF THE VALUE METHODOLOGY DISCUSSIONS REGARDING MAINTAINING THE EXISTING UTILITY PRIMARILY LINES TO THE SITE, AS DOCUMENTED IN THE SCHEMATIC DESIGN PREFERRED ALTERNATIVE DOCUMENTATION.




2 E400 RECEPTACLE - CONTROLLED SPLIT WIRE STRAIGHT BLADE




3 E400 RECEPTACLE - CONTROLLED SPLIT WIRE STRAIGHT BLADE QUAD

## FAULT CURRENT CALCULATION SCHEDULE

| POINT | LOCATION DESCRIPTION | LENGTH (L) (ft) | VOLTAGE (E L-L) | VOLTAGE (E L-N) | PHASE | WIRE SIZE | CONDUCTOR MATERIAL | CONDUCTOR TYPE | CONDUIT MATERIAL | VOLTAGE CLASS | C VALUE | # OF PARALLEL RUNS | Isc AVAILABLE UPSTREAM | Isc AT EQUIP (I 3PH) OR (I L-L) | POINT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| F0 | UTILITY XFMR | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | 29,600 | F0 |
| F1 | CT CABINET | 25 | 240 | 120 | 1 | 350 | COPPER | THREE SINGLE CONDUCTORS | STEEL | 600V | 19703 | 2 | 29,600 | 25,595 | F1 |
| F2 | 'MDP' | 5 | 240 | 120 | 1 | 350 | COPPER | THREE SINGLE CONDUCTORS | STEEL | 600V | 19703 | 2 | 25,595 | 24,920 | F2 |
| F3 | PANEL 'D' | 85 | 240 | 120 | 1 | 3X | COPPER | THREE SINGLE CONDUCTORS | STEEL | 600V | 12843 | 1 | 24,920 | 10,495 | F3 |
| F4 | 'MDC' | 85 | 240 | 120 | 1 | 3X | COPPER | THREE SINGLE CONDUCTORS | STEEL | 600V | 12843 | 2 | 24,920 | 14,770 | F4 |
| F5 | PANEL 'B' | 50 | 240 | 120 | 1 | 1 | COPPER | THREE SINGLE CONDUCTORS | STEEL | 600V | 7292 | 1 | 14,770 | 8,010 | F5 |

NOTES:
1. ALL CALCULATIONS WERE DONE USING BUSSMAN "POINT-TO-POINT" METHOD.
2. REFER TO PLANS FOR ASSUMED UTILITY TRANSFORMER SIZE UTILIZED FOR CALCULATIONS.
3. TRANSFORMER IMPEDANCES USED IN THE CALCULATIONS WERE TAKEN FROM EATON'S PUBLISHED IMPEDANCES FOR DOE 2016 DRY-TYPE TRANSFORMERS.
4. CONDUCTOR LENGTHS INDICATED IN THIS SCHEDULE ARE FOR THE PURPOSES OF FAULT CURRENT CALCULATIONS ONLY. THESE LENGTHS ASSUME WORST CASE SHORTEST DISTANCE CONDITIONS AND SHOULD NOT BE UTILIZED BY THE ELECTRICAL CONTRACTOR FOR PRICING PURPOSES. THE ELECTRICAL CONTRACTOR SHALL BE RESPONSIBLE FOR ESTIMATING AND MEASURING ACTUAL FIELD CONDITION LENGTHS AS PART OF THE PRICING PROCESS.

LICENSED PROFESSIONAL ENGINEER 05/21/2026 JEFF MULLIKIN P23384 ARIZONA

| | | | | | |
|---|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR AE DESIGN DENVER, CO | DESIGNED: JR/JM/TW CADD AE DESIGN TECH REVIEW: MRS/ESN DATE: 5/29/2026 | SUB SHEET NO. E400 | TITLE OF SHEET ELECTRICAL SCHEDULES AND DETAILS EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 120 OF 142 | |

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5/21/2026 1:49:06 PM Autodesk Docs://2025-110 NPS EFMO Visitor Center DD-CD/6782.00 - NPS EFMO Effigy Mounds Visitor Center Renovation - Electrical Central - E24.rvt

## Switchboard: MDC

**Location:** MECH 001
**Supply From:** MDP
**Mounting:** NEMA TYPE 1
**Enclosure:** PAD MOUNTED

**Volts:** 120/240 Single
**Phases:** 1
**Wires:** 3

**A.I.C. Rating:** 18K AIC
**Mains Type:** MCB
**Mains Rating:** 400 A
**MCB Rating:** 400 A

**Notes:**

| CKT | Circuit Description | # of Poles | Frame Size | Trip Rating | Load | Remarks |
|---|---|---|---|---|---|---|
| 1 | SPARE | 2 | 60 A | 35 A | 0 VA | |
| 2 | SPARE | 2 | 60 A | 35 A | 0 VA | |
| 3 | SPARE | 2 | 60 A | 60 A | 0 VA | |
| 4 | SPARE | 2 | 60 A | 60 A | 0 VA | |
| 5 | ELEVATOR | 2 | 100 A | 100 A | 15600 VA | |
| 6 | PANEL 'C' | 2 | 100 A | 100 A | 15620 VA | |
| 7 | PANEL 'B' | 2 | 100 A | 100 A | 5860 VA | |
| 8 | PANEL 'A' | 2 | 200 A | 200 A | 30750 VA | |
| 9 | | | | | | |
| 10 | | | | | | |
| | | | | **Total Conn. Load:** | 67830 VA | |
| | | | | **Total Amps:** | 283 A | |

**Legend:**

| Load Classification | Connected Load | Demand Factor | Estimated Demand | Panel Totals | |
|---|---|---|---|---|---|
| E | 19217 VA | 100.00% | 19217 VA | | |
| KE | 2100 VA | 90.00% | 1890 VA | **Total Conn. Load:** | 67830 VA |
| R | 27132 VA | 68.43% | 18566 VA | **Total Est. Demand:** | 62954 VA |
| M | 19381 VA | 120.12% | 23281 VA | **Total Conn.:** | 283 A |
| | | | | **Total Est. Demand:** | 262 A |
| | | | | | |
| | | | | | |

**Notes:**

## PANEL: A

**LOCATION:** MECH 001
**SUPPLY FROM:** MDC
**MOUNTING:** SURFACE
**ENCLOSURE:** TYPE 1

**VOLTS:** 120/240 Single
**PHASES:** 1
**WIRES:** 3

**A.I.C. RATING:** 10K AIC
**MAINS TYPE:** MLO
**MAINS RATING:** 200 A
**MCB RATING:** N/A

**Notes:**
EXISTING PANEL

| CKT | LOAD TYPE | LOAD DESCRIPTION | TRIP | POLES | CB TYPE | A | | B | | CB TYPE | POLES | TRIP | LOAD DESCRIPTION | LOAD TYPE | CKT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | R | RCPT - STORAGE 2, 3 (N2) | 20 | 1 | | 720 | 696 | | | | 1 | 20 | PROJECTOR SCREEN (N2) | M | 2 |
| 3 | R | RCPT - COLLECTIONS (N2) | 20 | 1 | | | | 720 | 1260 | | 1 | 20 | RCPT - AUDITORIUM (N2) | R | 4 |
| 5 | R | RCPT - COLLECTIONS (N2) | 20 | 1 | | 720 | 180 | | | | 1 | 20 | PRJCTR - AUDITORIUM (N2) | R | 6 |
| 7 | E | NITROGEN GENERATOR (N2) | 20 | 1 | | | | 200 | 540 | | 1 | 20 | RCPT - EXTERIOR (N2) | R | 8 |
| 9 | R | RCPT - MECH (N2) | 20 | 1 | | 360 | 540 | | | | 1 | 20 | RCPT - AUDITORIUM (N2) | R | 10 |
| 11 | R | RCPT - FLEX/MTG (N2) | 20 | 1 | | | | 720 | 720 | | 1 | 20 | RCPT - RECEPTION (N2) | R | 12 |
| 13 | R | RCPT - JAN / STAFF RR (N2) | 20 | 1 | | 360 | 900 | | | | 1 | 20 | RCPT - RECEPTION (N2) | R | 14 |
| 15 | E | HAND DRYER - STAFF RR (N2) | 20 | 1 | | | | 900 | 360 | | 1 | 20 | RCPT - BREAK (N2) | R | 16 |
| 17 | M | GD - BREAK (N2) | 20 | 1 | | 1127 | 180 | | | | 1 | 20 | RCPT - BREAK (N2) | R | 18 |
| 19 | E | WH-1 (N2) | 20 | 2 | | | | 1500 | 180 | | 1 | 20 | RCPT - BREAK (N2) | R | 20 |
| 21 | -- | -- | -- | -- | | 1500 | 700 | | | | 1 | 20 | REF - BREAK (N3) | KE | 22 |
| 23 | E | ASD (N3) | 20 | 1 | HC-ON | | | 200 | 1000 | | 1 | 20 | COPIER - OPEN OFFICE (N3) | E | 24 |
| 25 | E | FACP (N3) | 20 | 1 | HC-ON | 200 | 0 | | | | 2 | 40 | SPARE (N2) | -- | 26 |
| 27 | E | UH-2 (N1) | 30 | 2 | | | | 1737 | 0 | | -- | -- | -- | -- | 28 |
| 29 | -- | -- | -- | -- | | 1737 | 0 | | | | 2 | 40 | SPARE (N2) | -- | 30 |
| 31 | R | RCPT - SUPERINT (N3) | 20 | 1 | | | | 1080 | 0 | | -- | -- | -- | -- | 32 |
| 33 | R | RCPT - OFFICE 117 (N3) | 20 | 1 | | 720 | 960 | | | | 2 | 20 | UH-3 (N1) | E | 34 |
| 35 | R | RCPT - ADMIN (N3) | 20 | 1 | | | | 900 | 960 | | -- | -- | -- | -- | 36 |
| 37 | R | RCPT - OPEN OFFICE (N3) | 20 | 1 | | 1440 | 540 | | | | 1 | 20 | RCPT - OPEN OFFICE (N2) | R | 38 |
| 39 | R | RCPT - AV (N3) | 20 | 1 | | | | 360 | 1737 | | 2 | 30 | UH-1 (N1) | E | 40 |
| 41 | R | RCPT - AV (N3) | 20 | 1 | | 360 | 1737 | | | | -- | -- | -- | -- | 42 |
| | | | | | **Total Load:** | 15677 VA | | 15074 VA | | | | | | | |
| | | | | | **Total Amps:** | 131 A | | 126 A | | | | | | | |

**CB TYPE LEGEND**
GFCI: 5mA GROUND FAULT CIRCUIT INTERRUPTER
GFEP: 30mA GROUND FAULT PROTECTION FOR EQUIPMENT
AFCI: ARC FAULT CIRCUIT INTERRUPTER
CAFCI: COMBO ARC FAULT & 5mA GROUND FAULT CIRCUIT INTERRUPTER
HC(ON/OFF): HANDLE CLAMP FOR LOCKING...
HT#: HANDLE TIE WITH GROUPING #
ST: SHUNT TRIP
LOCK: PERMANENTLY LOCKABLE BREAKER

**CIRCUIT PHASE CODE LEGEND**
N1. EXISTING LOAD ON EXISTING CIRCUIT BREAKER.
N2. NEW LOAD ON EXISTING CIRCUIT BREAKER.
N3. NEW LOAD ON NEW CIRCUIT BREAKER. CIRCUIT BREAKER AND AIC RATING TO MATCH EXISTING.

| LOAD TYPE: | LOAD | DEMAND LOAD | PANEL TOTALS | |
|---|---|---|---|---|
| LIGHTING: | | | | |
| RECEPTACLE: | 13860 VA | 11930 VA | **TOTAL CONN. LOAD:** | 30750 VA |
| MOTOR: | 1823 VA | 2105 VA | **TOTAL EST. LOAD:** | 29102 VA |
| EQUIPMENT: | 14367 VA | 14367 VA | **TOTAL CONN.:** | 128 A |
| KITCH EQUIP: | 700 VA | 700 VA | **TOTAL EST. DEMAND:** | 121 A |
| CONTINUOUS: | | | | |
| EXISTING: | | | | |

**NOTES:**

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.

## MECHANICAL EQUIPMENT SCHEDULE

| KEY | | EQUIPMENT DESCRIPTION | LOAD | ELECTRICAL | MOCP/MFS | FEEDER | DISCONNECT | PANEL | CIRCUIT | NOTES |
|---|---|---|---|---|---|---|---|---|---|---|
| AHU | 1 | AIR HANDLING UNIT | 10.9 MCA | 120 V/1-1308 VA | 20A | 2#12, 1#12G, 3/4"C | 30A/1P | D | 3 | |
| AHU | 2 | AIR HANDLING UNIT | 10.9 MCA | 120 V/1-1308 VA | 20A | 2#12, 1#12G, 3/4"C | 30A/1P | D | 7 | |
| AHU | 3 | AIR HANDLING UNIT | 10.9 MCA | 120 V/1-1308 VA | 20A | 2#12, 1#12G, 3/4"C | 30A/1P | D | 11 | |
| AHU | 4 | AIR HANDLING UNIT | 10.9 MCA | 120 V/1-1308 VA | 20A | 2#12, 1#12G, 3/4"C | 30A/1P | D | 15 | |
| AHU | 5 | AIR HANDLING UNIT | 10.9 MCA | 120 V/1-1308 VA | 20A | 2#12, 1#12G, 3/4"C | 30A/1P | D | 19 | |
| AHU | 6 | AIR HANDLING UNIT | 10.9 MCA | 120 V/1-1308 VA | 20A | 2#12, 1#12G, 3/4"C | 30A/1P | D | 23 | |
| AHU | 7 | AIR HANDLING UNIT | 1 MCA | 240 V/1-200 VA | -- | -- | -- | MDP | 10 | 1 |
| AHU | 8 | AIR HANDLING UNIT | 1 MCA | 240 V/1-200 VA | -- | -- | -- | MDP | 11 | 1 |
| CU | 1 | CONDENSING UNIT | 18.5 RLA | 240 V/1-3848 VA | 40A | 2#8, 1#10G, 1"C | 60A/2P | MDP | 4 | |
| CU | 2 | CONDENSING UNIT | 18.5 RLA | 240 V/1-3848 VA | 40A | 2#8, 1#10G, 1"C | 60A/2P | MDP | 5 | |
| CU | 3 | CONDENSING UNIT | 18.5 RLA | 240 V/1-3848 VA | 40A | 2#8, 1#10G, 1"C | 60A/2P | MDP | 6 | |
| CU | 4 | CONDENSING UNIT | 18.5 RLA | 240 V/1-3848 VA | 40A | 2#8, 1#10G, 1"C | 60A/2P | MDP | 7 | |
| CU | 5 | CONDENSING UNIT | 18.5 RLA | 240 V/1-3848 VA | 40A | 2#8, 1#10G, 1"C | 60A/2P | MDP | 8 | |
| CU | 6 | CONDENSING UNIT | 18.5 RLA | 240 V/1-3848 VA | 40A | 2#8, 1#10G, 1"C | 60A/2P | MDP | 9 | |
| CU | 7 | CONDENSING UNIT | 16 MCA | 240 V/1-3328 VA | 20A | 2#12, 1#12G, 3/4"C | 30A/2P | MDP | 10 | 1 |
| CU | 8 | CONDENSING UNIT | 16 MCA | 240 V/1-3328 VA | 20A | 2#12, 1#12G, 3/4"C | 30A/2P | MDP | 11 | 1 |
| CUH | 1 | CEILING UNIT HEATER | 9.8 FLA | 240 V/1-2038 VA | 15A | 2#12, 1#12G, 3/4"C | INTEGRAL | D | 27,29 | |
| CUH | 2 | CEILING UNIT HEATER | 16.7 FLA | 240 V/1-3474 VA | 25A | 2#10, 1#10G, 3/4"C | INTEGRAL | D | 31,33 | |
| CUH | 3 | CEILING UNIT HEATER | 7.5 FLA | 120 V/1-900 VA | 15A | 2#12, 1#12G, 3/4"C | INTEGRAL | D | 35 | |
| CUH | 4 | CEILING UNIT HEATER | 7.5 FLA | 120 V/1-900 VA | 15A | 2#12, 1#12G, 3/4"C | INTEGRAL | D | 37 | |
| CUH | 5 | CEILING UNIT HEATER | 7.5 FLA | 120 V/1-900 VA | 15A | 2#12, 1#12G, 3/4"C | INTEGRAL | D | 39 | |
| DU | 1 | DEHUMIDIFICATION UNIT | 1115W | 120 V/1-1115 VA | 20A | 2#12, 1#12G, 3/4"C | NEMA 5-20R | D | 5 | |
| DU | 2 | DEHUMIDIFICATION UNIT | 1115W | 120 V/1-1115 VA | 20A | 2#12, 1#12G, 3/4"C | NEMA 5-20R | D | 9 | |
| DU | 3 | DEHUMIDIFICATION UNIT | 1115W | 120 V/1-1115 VA | 20A | 2#12, 1#12G, 3/4"C | NEMA 5-20R | D | 13 | |
| DU | 4 | DEHUMIDIFICATION UNIT | 1115W | 120 V/1-1115 VA | 20A | 2#12, 1#12G, 3/4"C | NEMA 5-20R | D | 17 | |
| DU | 5 | DEHUMIDIFICATION UNIT | 1115W | 120 V/1-1115 VA | 20A | 2#12, 1#12G, 3/4"C | NEMA 5-20R | D | 21 | |
| DU | 6 | DEHUMIDIFICATION UNIT | 1115W | 120 V/1-1115 VA | 20A | 2#12, 1#12G, 3/4"C | NEMA 5-20R | D | 25 | |
| EF | 1 | EXHAUST FAN | 4.1 FLA | 120 V/1-492 VA | 15A | 2#12, 1#12G, 3/4"C | 30A/1P | C | 15 | 2 |
| EF | 2 | EXHAUST FAN | 2.5 FLA | 120 V/1-300 VA | 15A | 2#12, 1#12G, 3/4"C | 30A/1P | C | 15 | |
| EF | 3 | EXHAUST FAN | 1.8 FLA | 120 V/1-216 VA | 15A | 2#12, 1#12G, 3/4"C | 30A/1P | C | 17 | |
| ELEV | 1 | ELEVATOR | 15600VA | 240 V/1-15600 VA | 100A | EXISTING | SEE NOTES | MDC | 5 | 3 |
| ERV | 1 | ENERGY RECOVER VENTILATOR | 14.2 MCA | 240 V/1-3266 VA | 20A | 2#12, 1#12G, 3/4"C | 30A/2P | D | 14,16 | |
| HWCP | 1 | HOT WATER CIRCULATION PUMP | 1/12 HP | 120 V/1-50 VA | 20A | 2#12, 1#12G, 3/4"C | 30A/1P | B | 15 | |
| UH | 1 | EXISTING UNIT HEATER | EXISTING | 240 V/1-3474 VA | EXISTING | EXISTING | EXISTING | A | 40,42 | 4 |
| UH | 2 | EXISTING UNIT HEATER | EXISTING | 240 V/1-3474 VA | EXISTING | EXISTING | EXISTING | A | 27,29 | 4 |
| UH | 3 | EXISTING UNIT HEATER | EXISTING | 240 V/1-1920 VA | EXISTING | EXISTING | EXISTING | A | 34,36 | 4 |
| UH | 4 | CEILING UNIT HEATER | 9 FLA | 240 V/1-2160 VA | 15A | 2#12, 1#12G, 3/4"C | INTEGRAL | D | 10,12 | |
| WH | 1 | WATER HEATER | 3 KW | 240 V/1-3000 VA | 20A | 2#12, 1#12G, 3/4"C | 30A/2P | A | 19,21 | |

## MECHANICAL EQUIPMENT GENERAL NOTES

A. REFER TO MECHANICAL PLANS FOR SPECIFIC EQUIPMENT LOCATIONS AND REQUIREMENTS.

B. PRIOR TO ROUGH-IN, COORDINATE ALL MECHANICAL EQUIPMENT POWER AND CONNECTION REQUIREMENTS WITH MECHANICAL CONTRACTOR'S FINAL SHOP DRAWINGS.

C. PROVIDE ALL 120V CONTROL WIRING, REFER TO SPECIFICATIONS FOR FURTHER CONTROL WIRING CLARIFICATION.

D. EXTERIOR DISCONNECT SWITCHES ARE TO BE PROVIDED AS NEMA 3R EQUIPMENT UNLESS OTHERWISE NOTED.

E. PROVIDE WEATHERPROOF 120 VOLT GFCI RECEPTACLES WITHIN 25' OF ALL ROOFTOP HEATING, VENTILATING, AND AIR CONDITIONING EQUIPMENT. CIRCUIT TO SPARE CIRCUIT ON NEAREST 120V PANELBOARD OR AS INDICATED ON PLANS.

F. PROVIDE DUCT DETECTION ON ALL RETURN AIR SYSTEMS OF 2,000 CFM OR GREATER, AND FOR ALL SUPPLY AIR SYSTEMS 15,000 CFM OR GREATER, INCLUDING THOSE SYSTEMS SERVING MULTIPLE FLOORS. PROVIDE ADDITIONAL DUCT DETECTORS AND INSTALL REMOTE INDICATOR LIGHTS AS REQUIRED BY LOCAL AUTHORITY HAVING JURISDICTION.

G. EC TO PROVIDE HAND/OFF/AUTO STARTERS FOR ALL MOTORS WHEN NOT INDICATED AS TO BE PROVIDED BY THE MECHANICAL CONTRACTOR ON THE MECHANICAL PLANS. SIZE OF STARTER TO BE BASED UPON SIZE OF MOTOR HORSEPOWER INDICATED.

## MECHANICAL EQUIPMENT SPECIFIC NOTES

1. OUTDOOR UNIT TO SUPPLY INDOPOR UNIT. EC SHALL PROVIDE EXACT WIRE SIZE, TYPE, AND FINAL CONNECTION REQUIREMENTS TO INDOOR UNIT PER MANUFACTURER'S REQUIREMENTS FOR COMPLETE AND FUNCTIONAL SYSTEM.

2. FAN SHALL BE CONTROLLED BY TIMECLOCK. COORDINATE SCHEDULE WITH BUILDING MANAGEMENT.

3. EC SHALL PROVIDE ELEVATOR CONTROL SWITCH WITH FIRE SAFETY INTERFACE RELAY, VOLTAGE MONITORING RELAY, AND AUXIIARY CONTACTS AS REQUIRED FOR FIRE ALARM SHUNT TRIP OPERATION OF ELEVATOR POWER PER CODE. PROVIDE 100A/2P LOCKABLE ELEVATOR ALL-IN-ONE CONTROL SWITCH (EATON SERIES #ES OR APPROVED EQUAL) WITH FUSING TO MATCH EXISTING ELEVATOR DISCONNECT. PROVIDE LPS-RK FUSES AT ELEVATOR LOCAL DISCONNECT TO LIMIT SHORT CIRCUIT AVAILABLE AT CONTROLLER TO LESS THAN 10,000AIC.

4. EQUIPMENT IS EXISTING TO REMAIN. PROTECT BRANCH CIRCUIT DURING CONSTRUCTION.

LICENSED PROFESSIONAL ENGINEER
05/21/2026
JEFF MULLIKIN
P25384
IOWA

| | | | | | |
|---|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR AE DESIGN DENVER, CO | DESIGNED: JR/JM/TW CADD AE DESIGN TECH REVIEW: MRS/ESN DATE: 5/29/2026 | SUB SHEET NO. E401 | TITLE OF SHEET: ELECTRICAL PANEL SCHEDULES — EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 121 OF 142 | |

5/21/2026 1:49:07 PM Autodesk Docs://2025-110 NPS EFMO Visitor Center DD-CD/6782.00 - NPS EFMO Effigy Mounds Visitor Center Renovation - Electrical Central - E24.rvt

## PANEL: C

**LOCATION:** ELEC 107
**SUPPLY FROM:** MDC
**MOUNTING:** SURFACE
**ENCLOSURE:** TYPE 1

**VOLTS:** 120/240 Single
**PHASES:** 1
**WIRES:** 3

**A.I.C. RATING:** 10K AIC
**MAINS TYPE:** MLO
**MAINS RATING:** 225 A
**MCB RATING:** N/A

**Notes:**
EXISTING PANEL

| CKT | LOAD TYPE | LOAD DESCRIPTION | TRIP | POLES | CB TYPE | A | | B | | CB TYPE | POLES | TRIP | LOAD DESCRIPTION | LOAD TYPE | CKT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | R | EWC (N2) | 20 | 1 | GFCI | 696 | 720 | | | | 1 | 20 | RCPT - EXHIBIT LOBBY (N2) | R | 2 |
| 3 | R | RCPT - EXTERIOR (N2) | 20 | 1 | | | | 540 | 1080 | | 1 | 20 | RCPT - WORKSTATIONS (N2) | R | 4 |
| 5 | R | RCPT - EXHIBIT (N2) | 20 | 1 | | 540 | 900 | | | | 1 | 20 | RCPT - HALL 108 (N2) | R | 6 |
| 7 | R | RCPT - EXHIBIT (N2) | 20 | 1 | | | | 360 | 180 | | 1 | 20 | RCPT - FIBER (N2) | R | 8 |
| 9 | R | RCPT - EXHIBIT (N2) | 20 | 1 | | 720 | 900 | | | | 1 | 20 | RCPT - CHIEF OF INTERP (N2) | R | 10 |
| 11 | R | RCPT - EXHIBIT (N2) | 20 | 1 | | | | 360 | 900 | | 1 | 20 | RCPT - EASTERN NATIONAL... | R | 12 |
| 13 | -- | SPARE (N2) | 20 | 1 | | 0 | 720 | | | | 1 | 20 | RCPT - RESTROOMS (N2) | R | 14 |
| 15 | M | EF-1,2 (N3) | 15 | 1 | | | | 792 | 900 | | 1 | 20 | HAND DRYER - RR (N2) | E | 16 |
| 17 | M | EF-3 (N3) | 15 | 1 | | 216 | 900 | | | | 1 | 20 | HAND DRYER - MENS RR (N3) | E | 18 |
| 19 | M | MOTORIZED DOORS (N3) | 20 | 1 | | | | 200 | 900 | | 1 | 20 | HAND DRYER - MENS RR (N3) | E | 20 |
| 21 | E | REFUGE COMMAND UNIT (N3) | 20 | 1 | | 100 | 900 | | | | 1 | 20 | HAND DRYER - WOMENS RR... | E | 22 |
| 23 | E | FIRE ALARM PANEL (N3) | 20 | 1 | HC-ON | | | 200 | 900 | | 1 | 20 | HAND DRYER - WOMENS RR... | E | 24 |
| 25 | -- | SPARE (N3) | 20 | 1 | | 0 | 696 | | | GFCI | 1 | 20 | EWC (N3) | R | 26 |
| 27 | -- | SPARE (N3) | 20 | 1 | | | | 0 | 200 | | 1 | 20 | EM EXIT DOOR - EXHIBIT (N3) | M | 28 |
| 29 | -- | SPARE (N3) | 20 | 1 | | 0 | 100 | | | | 1 | 20 | MOTORIZED DOOR (N3) | M | 30 |
| | | | | **Total Load:** | | 8108 VA | | 7512 VA | | | | | | | |
| | | | | **Total Amps:** | | 68 A | | 63 A | | | | | | | |

**CB TYPE LEGEND**
GFCI: 5mA GROUND FAULT CIRCUIT INTERRUPTER
GFEP: 30mA GROUND FAULT PROTECTION FOR EQUIPMENT
AFCI: ARC FAULT CIRCUIT INTERRUPTER
CAFCI: COMBO ARC FAULT & 5mA GROUND FAULT CIRCUIT INTERRUPTER
HC(ON/OFF): HANDLE CLAMP LOCKING ON/OFF
HT#: HANDLE TIE WITH GROUPING #
ST: SHUNT TRIP
LOCK: PERMANENTLY LOCKABLE BREAKER

**CIRCUIT PHASE CODE LEGEND**
N1. EXISTING LOAD ON EXISTING CIRCUIT BREAKER.
N2. NEW LOAD ON EXISTING CIRCUIT BREAKER.
N3. NEW LOAD ON NEW CIRCUIT BREAKER. CIRCUIT BREAKER AND AIC RATING TO MATCH EXISTING.

| LOAD TYPE: | LOAD | DEMAND LOAD | PANEL TOTALS | |
|---|---|---|---|---|
| LIGHTING: | | | | |
| RECEPTACLE: | 9312 VA | 9312 VA | **TOTAL CONN. LOAD:** | 15620 VA |
| MOTOR: | 1508 VA | 1631 VA | **TOTAL EST. LOAD:** | 15743 VA |
| EQUIPMENT: | 4800 VA | 4800 VA | **TOTAL CONN.:** | 65 A |
| KITCH EQUIP: | | | **TOTAL EST. DEMAND:** | 66 A |
| CONTINUOUS: | | | | |
| EXISTING: | | | | |

**NOTES:**

## PANEL: B

**LOCATION:** HALL 005
**SUPPLY FROM:** MDC
**MOUNTING:** RECESSED
**ENCLOSURE:** TYPE 1

**VOLTS:** 120/240 Single
**PHASES:** 1
**WIRES:** 3

**A.I.C. RATING:** 10K AIC
**MAINS TYPE:** MLO
**MAINS RATING:** 100 A
**MCB RATING:** N/A

**Notes:**
EXISTING PANEL

| CKT | LOAD TYPE | LOAD DESCRIPTION | TRIP | POLES | CB TYPE | A | | B | | CB TYPE | POLES | TRIP | LOAD DESCRIPTION | LOAD TYPE | CKT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | E | CALL STATION (N2) | 20 | 1 | | 50 | 900 | | | | 1 | 20 | RCPT - VEST 013 (N2) | R | 2 |
| 3 | M | MOTORIZED DOORS (N2) | 20 | 1 | | | | 200 | 1080 | | 1 | 20 | RCPT - WORK ROOM (N2) | R | 4 |
| 5 | R | RCPT - EXTERIOR (N2) | 20 | 1 | | 360 | 700 | | | GFCI | 1 | 20 | REF - WORK ROOM (N3) | KE | 6 |
| 7 | R | RCPT - STORAGE 1 (N2) | 20 | 1 | | | | 720 | 700 | GFCI | 1 | 20 | REF - WORK ROOM (N3) | KE | 8 |
| 9 | -- | SPARE (N2) | 40 | 2 | | 0 | 720 | | | | 1 | 20 | RCPT - WORK ROOM (N2) | R | 10 |
| 11 | -- | -- | -- | -- | | | | 0 | 180 | | 1 | 20 | RCPT - ELEVATOR SHAFT (N3) | R | 12 |
| 13 | M | MOTORIZED DOORS (N2) | 20 | 1 | | 200 | -- | | | | 1 | -- | BUSSED SPACE | -- | 14 |
| 15 | M | HWCP-1 (N2) | 20 | 1 | | | | 50 | -- | | 1 | -- | BUSSED SPACE | -- | 16 |
| 17 | -- | BUSSED SPACE | -- | 1 | | -- | -- | | | | 1 | -- | BUSSED SPACE | -- | 18 |
| 19 | -- | BUSSED SPACE | -- | 1 | | | | -- | -- | | 1 | -- | BUSSED SPACE | -- | 20 |
| | | | | **Total Load:** | | 2930 VA | | 2930 VA | | | | | | | |
| | | | | **Total Amps:** | | 24 A | | 24 A | | | | | | | |

**CB TYPE LEGEND**
GFCI: 5mA GROUND FAULT CIRCUIT INTERRUPTER
GFEP: 30mA GROUND FAULT PROTECTION FOR EQUIPMENT
AFCI: ARC FAULT CIRCUIT INTERRUPTER
CAFCI: COMBO ARC FAULT & 5mA GROUND FAULT CIRCUIT INTERRUPTER
HC(ON/OFF): HANDLE CLAMP FOR LOCKING...
HT#: HANDLE TIE WITH GROUPING #
ST: SHUNT TRIP
LOCK: PERMANENTLY LOCKABLE BREAKER

**CIRCUIT PHASE CODE LEGEND**
N1. EXISTING LOAD ON EXISTING CIRCUIT BREAKER.
N2. NEW LOAD ON EXISTING CIRCUIT BREAKER.
N3. NEW LOAD ON NEW CIRCUIT BREAKER. CIRCUIT BREAKER AND AIC RATING TO MATCH EXISTING.

| LOAD TYPE: | LOAD | DEMAND LOAD | PANEL TOTALS | |
|---|---|---|---|---|
| LIGHTING: | | | | |
| RECEPTACLE: | 3960 VA | 3960 VA | **TOTAL CONN. LOAD:** | 5860 VA |
| MOTOR: | 450 VA | 475 VA | **TOTAL EST. LOAD:** | 5885 VA |
| EQUIPMENT: | 50 VA | 50 VA | **TOTAL CONN.:** | 24 A |
| KITCH EQUIP: | 1400 VA | 1400 VA | **TOTAL EST. DEMAND:** | 25 A |
| CONTINUOUS: | | | | |
| EXISTING: | | | | |

**NOTES:**

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.

## PANEL: D

**LOCATION:** MECH 001
**SUPPLY FROM:** MDP
**MOUNTING:** SURFACE
**ENCLOSURE:** TYPE 1

**VOLTS:** 120/240 Single
**PHASES:** 1
**WIRES:** 3

**A.I.C. RATING:** FULLY RATED TO MIN 18K AIC
**MAINS TYPE:** MLO
**MAINS RATING:** 200 A
**MCB RATING:** N/A

**Notes:**
NEW PANEL

| CKT | LOAD TYPE | LOAD DESCRIPTION | TRIP | POLES | CB TYPE | A | | B | | CB TYPE | POLES | TRIP | LOAD DESCRIPTION | LOAD TYPE | CKT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | L; R | LTS/REC - CRAWLSPACE | 20 | 1 | GFCI | 424 | 725 | | | | 1 | 20 | LTG - EXTERIOR LIGHTING | L | 2 |
| 3 | E | AHU-1, SPLIT SYSTEM INDOOR | 20 | 1 | | | | 1308 | 1194 | | 1 | 20 | LTG - BASEMENT | L | 4 |
| 5 | E | DU-1, DEHUMIDIFIER | 20 | 1 | | 1115 | 1648 | | | | 1 | 20 | LTG - FIRST FLOOR | L | 6 |
| 7 | E | AHU-2, SPLIT SYSTEM INDOOR | 20 | 1 | | | | 1308 | 1256 | | 1 | 20 | LTG - EXHIBIT & RR | L | 8 |
| 9 | E | DU-2, DEHUMIDIFIER | 20 | 1 | | 1115 | 1080 | | | | 2 | 15 | UH-4 | E | 10 |
| 11 | E | AHU-3, SPLIT SYSTEM INDOOR | 20 | 1 | | | | 1308 | 1080 | | -- | -- | -- | -- | 12 |
| 13 | E | DU-3, DEHUMIDIFIER | 20 | 1 | | 1115 | 1633 | | | | 2 | 20 | ERV-1 | E | 14 |
| 15 | E | AHU-4, SPLIT SYSTEM INDOOR | 20 | 1 | | | | 1308 | 1633 | | -- | -- | -- | -- | 16 |
| 17 | E | DU-4, DEHUMIDIFIER | 20 | 1 | | 1115 | 0 | | | | 1 | 20 | SPARE | -- | 18 |
| 19 | E | AHU-5, SPLIT SYSTEM INDOOR | 20 | 1 | | | | 1308 | 0 | | 1 | 20 | SPARE | -- | 20 |
| 21 | E | DU-5, DEHUMIDIFIER | 20 | 1 | | 1115 | 0 | | | | 1 | 20 | SPARE | -- | 22 |
| 23 | E | AHU-6, SPLIT SYSTEM INDOOR | 20 | 1 | | | | 1308 | 0 | | 1 | 20 | SPARE | -- | 24 |
| 25 | E | DU-6, DEHUMIDIFIER | 20 | 1 | | 1115 | 0 | | | | 1 | 20 | SPARE | -- | 26 |
| 27 | E | CUH-1 | 15 | 2 | | | | 1019 | 0 | | 1 | 20 | SPARE | -- | 28 |
| 29 | -- | -- | -- | -- | | 1019 | 0 | | | | 1 | 20 | SPARE | -- | 30 |
| 31 | E | CUH-2 | 25 | 2 | | | | 1737 | 0 | | 1 | 20 | SPARE | -- | 32 |
| 33 | -- | -- | -- | -- | | 1737 | 0 | | | | 1 | 20 | SPARE | -- | 34 |
| 35 | E | CUH-3 | 15 | 1 | | | | 900 | 0 | | 1 | 20 | SPARE | -- | 36 |
| 37 | E | CUH-4 | 15 | 1 | | 900 | 0 | | | | 1 | 20 | SPARE | -- | 38 |
| 39 | E | CUH-5 | 15 | 1 | | | | 900 | 0 | | 1 | 20 | SPARE | -- | 40 |
| 41 | -- | SPARE | 20 | 1 | | 0 | 0 | | | | 1 | 20 | SPARE | -- | 42 |
| | | | | **Total Load:** | | 15856 VA | | 17567 VA | | | | | | | |
| | | | | **Total Amps:** | | 132 A | | 146 A | | | | | | | |

**CB TYPE LEGEND**
GFCI: 5mA GROUND FAULT CIRCUIT INTERRUPTER
GFEP: 30mA GROUND FAULT PROTECTION FOR EQUIPMENT
AFCI: ARC FAULT CIRCUIT INTERRUPTER
CAFCI: COMBO ARC FAULT & 5mA GROUND FAULT CIRCUIT INTERRUPTER
HC(ON/OFF): HANDLE CLAMP FOR LOCKING...
HT#: HANDLE TIE WITH GROUPING #
ST: SHUNT TRIP
LOCK: PERMANENTLY LOCKABLE BREAKER

**CIRCUIT PHASE CODE LEGEND**
N1. EXISTING LOAD ON EXISTING CIRCUIT BREAKER.
N2. NEW LOAD ON EXISTING CIRCUIT BREAKER.
N3. NEW LOAD ON NEW CIRCUIT BREAKER. CIRCUIT BREAKER AND AIC RATING TO MATCH EXISTING.

| LOAD TYPE: | LOAD | DEMAND LOAD | PANEL TOTALS | |
|---|---|---|---|---|
| LIGHTING: | 4887 VA | 6109 VA | | |
| RECEPTACLE: | 360 VA | 360 VA | **TOTAL CONN. LOAD:** | 33423 VA |
| MOTOR: | | | **TOTAL EST. LOAD:** | 34645 VA |
| EQUIPMENT: | 28176 VA | 28176 VA | **TOTAL CONN.:** | 139 A |
| KITCH EQUIP: | | | **TOTAL EST. DEMAND:** | 144 A |
| CONTINUOUS: | | | | |
| EXISTING: | | | | |

**NOTES:**

A/E FIRM:
PRIME:
OTAK, INC.
REDMOND, WA
SUBCONTRACTOR
AE DESIGN
DENVER, CO

DESIGNED: JR/JM/TW
DRAWN: AE DESIGN
TECH REVIEW: MRS/ESN
DATE: 5/29/2026

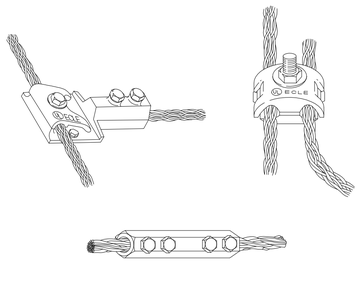
SUB SHEET NO. E402

TITLE OF SHEET
ELECTRICAL PANEL SCHEDULES
EFFIGY MOUNDS NATIONAL MONUMENT

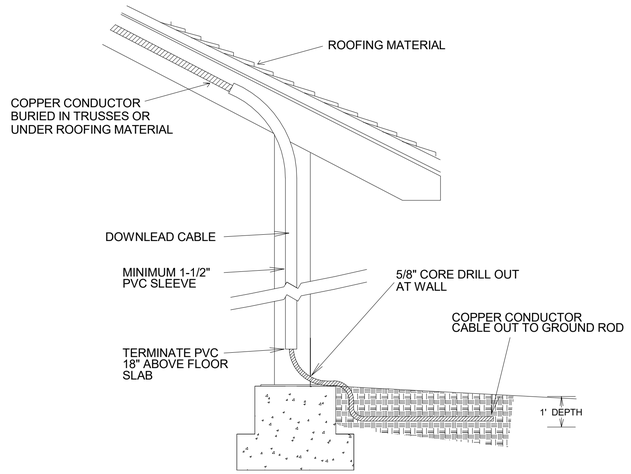
DRAWING NO. 394 194248
PMIS/PKG NO. 318539
SHEET 122 OF 142

5/21/2026 1:49:07 PM Autodesk Docs://2025-110 NPS EFMO Visitor Center DD-CD/6782.00 - NPS EFMO Effigy Mounds Visitor Center Renovation - Electrical Central - E24.rvt

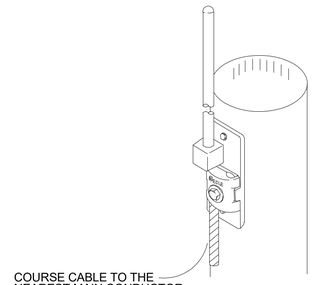
ALL SCOPE ON THIS SHEET SHALL BE IN OPTION E.



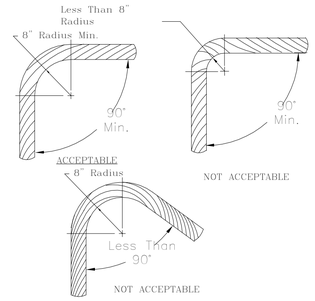
6 E403 TYPICAL CABLE CONNECTORS
SCALE: NONE



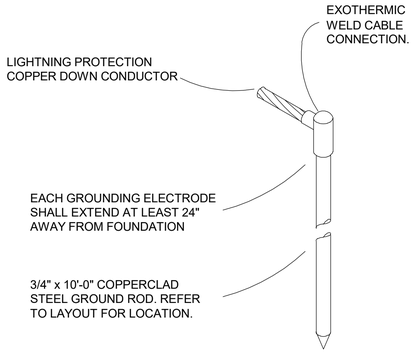

4 E403 EXTERIOR DOWNLEAD DETAIL
SCALE: NONE



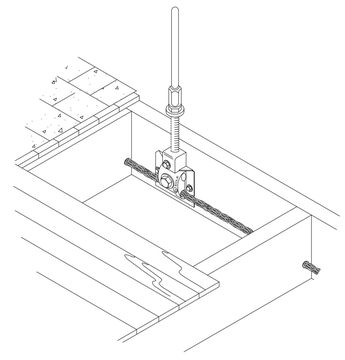

2 E403 FLUE AIR TERMINAL DETAIL
SCALE: NONE




5 E403 TYPICAL CABLE BEND REQUIREMENTS
SCALE: NONE




3 E403 GROUND ROD DETAIL
SCALE: NONE



1 E403 THRU ROOF BASE & AIR TERMINAL
SCALE: NONE

LICENSED PROFESSIONAL ENGINEER
05/21/2026
JEFF MULLIKIN
P25384
REGISTERED
ROYAL

| | | | | |
|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR AE DESIGN DENVER, CO | DESIGNED: JR/JM/TW CADD: AE DESIGN TECH REVIEW: MRS/ESN DATE: 5/29/2026 | SUB SHEET NO. E403 | TITLE OF SHEET: ELECTRICAL OPTION LIGHTNING PROTECTION DETAILS — EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 123 OF 142 |

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# LIGHTING FIXTURE GENERAL NOTES

| | |
|---|---|
| A. | BOF = BOTTOM OF FIXTURE HEIGHT, RFD = RECESSED FIXTURE DEPTH, OFH = OVERALL FIXTURE HEIGHT |
| B. | LUMENS LISTED ARE DELIVERED LUMENS, NOT INITIAL. |
| C. | FOR ALL SPECIFIED LUMINAIRES, THE ELECTRICAL CONTRACTOR SHALL BE RESPONSIBLE FOR PROVIDING ALL MOUNTING HARDWARE, ACCESSORIES, COMPONENTS, LEADER/JUMPER CABLES, WIRE FEED, CONNECTORS, END CAPS, REMOTE POWER SUPPLIES, AND ANY OTHER NECESSARY COMPONENT AS REQUIRED FOR INSTALLING A SECURE AND FULLY FUNCTIONAL SYSTEM. |
| D. | THE CONTRACTOR SHALL VERIFY THE CEILING TYPE BEFORE ORDERING LIGHT FIXTURES TO ENSURE COMPATIBILITY WITH SPECIFIED FIXTURES. NOTIFY CONTRACTING OFFICER OF ANY DISCREPANCIES. |
| E. | ALL FINISH SELECTIONS SHALL BE VERIFIED BY THE CONTRACTING OFFICER AS PART OF THE SUBMITTAL PROCESS. UNLESS OTHERWISE NOTED, EC SHALL ASSUME STANDARD LUMINAIRE FINISH OPTION FOR PRICING. |
| F. | ALL MOUNTING HEIGHTS SHALL BE VERIFIED WITH ARCHITECTURAL ELEVATIONS PRIOR TO ANY ROUGH-IN. |
| G. | SPECIFICATIONS DO NOT INCLUDE FIXTURE QUANTITIES, FINAL DIMENSIONS WITH ARCHITECTURAL FINISHES, WEIGHTS OR GAUGES, FABRICATION PROCESSES, CONSTRUCTION MEANS OR METHODS, SAFETY PRECAUTIONS, ALL OF WHICH ARE THE SOLE RESPONSIBILITY OF THE CONTRACTOR. |
| H. | LIGHTING FIXTURE SCHEDULE DIMMING COLUMN INDICATES DIMMING CAPABILITIES OF THE FIXTURES BASED ON THE SPECIFIED DRIVER. LIGHTING ZONE SCHEDULE INDICATES ASSOCIATED DIMMING PROTOCOL CAPABILITIES/REQUIREMENTS SHOULD THE ZONES UTILIZE DIMMING. INTENT TO UTILIZE DIMMING IN SPECIFIC AREAS IS DEFINED/IDENTIFIED PER ROOM BY AS NOTED IN THE LIGHTING SEQUENCE OF OPERATION SCHEDULE. THOUGH FIXTURES/ZONES MAY BE DIMMING CAPABLE BASED ON THE SPECIFIED HARDWARE, SEQUENCE OF OPERATION SCHEDULE TAKES PRECEDENT AS RELATED TO UTILIZATION/PROGRAMMING NEEDS PER SPACE. |
| I. | FOR LUMINAIRES SPECIFIED WITH 0-10V DIMMING, 0-10V DIMMING DRIVER(S) REQUIRE ADDITIONAL LOW VOLTAGE CONTROL WIRES IN ADDITION TO STANDARD WIRING FOR POWER. |
| J. | EXTERIOR LUMINAIRES SHALL BE COLD WEATHER RATED FOR 0 DEG. F / -18 DEG. C., AND RATED FOR OUTDOOR USE. |
| K. | REMOTE DRIVER(S)/TRANSFORMER(S) SHALL BE LOCATED IN A CONCEALED, ACCESSIBLE, AND VENTILATED LOCATION AS PROPOSED BY THE EC AND APPROVED BY THE CONTRACTING OFFICER. REFER TO THE MANUFACTURER'S INSTALLATION INSTRUCTIONS FOR MINIMUM REQUIRED CLEARANCES FROM ADJACENT MATERIALS, AND WIRING REQUIREMENTS TO PREVENT VOLTAGE DROP. DO NOT EXCEED MANUFACTURER'S RECOMMENDATIONS REGARDING WIRE GAGE AND MAXIMUM WIRE RUN LENGTHS. |
| L. | AFTER THE ISSUANCE OF 100% CONSTRUCTION DOCUMENTS, ON THREE-MONTH INTERVALS, THE ELECTRICAL CONTRACTOR SHALL BE REQUIRED TO NOTIFY AE DESIGN OF ANY LUMINAIRE OR CONTROL PRICING CHANGES DUE TO VARIATIONS IN TARIFF REQUIREMENTS, MANUFACTURER PRICING, MATERIAL COST, OR OTHER CIRCUMSTANCES THAT RESULT IN A 20% OVERALL INCREASE OR DECREASE TO THE 100% CD ISSUANCE BUDGET. REVISED PRICING SHALL BE ISSUED BY THE CONTRACTOR IN THE FORM OF A LINE-ITEM BUDGET IDENTIFYING ANY PRICE CHANGES BY INDIVIDUAL FIXTURE TYPE. BUDGETS CONTAINING ONLY A LUMP SUM DELTA WILL NOT BE REVIEWED. |

# LIGHTING FIXTURE SPECIFIC NOTES

| | |
|---|---|
| 1. | CONFIRM FINISH WITH CONTRACTING OFFICER PRIOR TO PROCUREMENT. |
| 2. | EC SHALL COORDINATE LOCATION OF REMOTE DRIVER WITH CONTRACTING OFFICER PRIOR TO ROUGH-IN. EE SHALL CONFIRM DRIVER WATTAGE. EC SHALL LOCATE ALL REMOTE DRIVERS IN A CONCEALED, DRY, VENTILATED, AND ACCESSIBLE LOCATION. GENERAL LOCATION AND PROXIMITY HAVE BEEN IDENTIFIED IN PLANS. CONTRACTOR SHALL VERIFY FINAL LOCATIONS WITH CONTRACTING OFFICER PRIOR TO ROUGH-IN, AND SHALL BE LOCATED TO MEET ANY DISTANCE LIMITATIONS IDENTIFIED BY THE MANUFACTURER FOR FINAL APPROVED FIXTURES. REFER TO THE MANUFACTURER'S INSTALLATION INSTRUCTIONS FOR MINIMUM REQUIRED CLEARANCES FROM ADJACENT MATERIALS AND WIRING REQUIREMENTS TO PREVENT VOLTAGE DROP. DO NOT EXCEED MANUFACTURER'S RECOMMENDATIONS REGARDING WIRE GAUGE AND MAXIMUM WIRE RUN LENGTHS. |
| 3. | EC SHALL REFER TO LIGHTING PLANS TO VERIFY FIXTURE LENGTHS PRIOR TO PROCUREMENT. SHOP DRAWINGS SHALL BE PROVIDED FOR TRACK SYSTEM FOR REVIEW AND APPROVAL PRIOR TO PROCURMENT. SHOP DRAWINGS SHALL INCLUDE ALL NECESSARY COMPONENTS FOR A COMPLETE SYSTEM. |
| 4. | EC SHALL INSTALL EXIT SIGN CENTERED ABOVE THE DOOR SUCH THAT THE BOF IS 3"-6" ABOVE TOP OF DOOR FRAME. ALL SIGNS SHALL BE MOUNTED AT A CONSISTENT DISTANCE FROM TOP OF DOOR FRAME. |
| 5. | SHOP DRAWINGS SHALL BE PROVIDED FOR REVIEW AND APPROVAL PRIOR TO PROCUREMENT. VERIFY FIXTURE MOUNTING HEIGHT WITH CONTRACTING OFFICER PRIOR TO INSTALLATION. |
| 6. | FIXTURES SHALL BE INCLUDED IN OPTION F. |

# OPTION C: LIGHTING FIXTURE SCHEDULE

| TYPE | DESCRIPTION | MANUFACTURER | CATALOG NUMBER | VOLTAGE | LAMP / CCT / CRI | MAX WATTAGE | LUMEN OUTPUT | DIMMING CAPABILITY | FIXTURE FINISH | MOUNTING | BOF/RFD/OFH | NOTES |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| C1A | 11'-4" LONG RECESSED PERIMETER LED, 2" WIDE, 1" REGRESS, FIELD-CUTTABLE FILLER | PRUDENTIAL (OR APPROVED EQUAL) | BPRO2-PER-REG1-LED35-90-LO-11'-TMW-SAL-WTW-SC-UNV-X3-DM01 (OR APPROVED EQUAL) | 120 V | LED / 3500K / 90 | 44 VA | 390 LM/FT | 0-10V / 1% | WHITE | RECESSED CEILING | 0'-8 3/16" RFD | |
| C1B | 8'-4" LONG RECESSED PERIMETER LED, 2" WIDE, 1" REGRESS, FIELD-CUTTABLE FILLER | PRUDENTIAL (OR APPROVED EQUAL) | BPRO2-PER-REG1-LED35-90-LO-8'-TMW-SAL-WTW-SC-UNV-X3-DM01 (OR APPROVED EQUAL) | 120 V | LED / 3500K / 90 | 32 VA | 390 LM/FT | 0-10V / 1% | WHITE | RECESSED CEILING | 0'-8 3/16" RFD | |
| D1 | 4" DIAMETER RECESSED LED DOWNLIGHT | LITHONIA (OR APPROVED EQUAL) | LDN4-10LM-35K-LO4-AR-LSS-MVOLT-UGZ1-90CRI (OR APPROVED EQUAL) | 120 V | LED / 3500K / 90 | 11 VA | 1045 | 0-10V / 1% | CLEAR | RECESSED CEILING | 0'-3 13/16" RFD | |
| D1EM | 4" DIAMETER RECESSED LED DOWNLIGHT, INTEGRAL EM BATTERY | LITHONIA (OR APPROVED EQUAL) | LDN4-10LM-35K-LO4-AR-LSS-MVOLT-UGZ1-EL-90CRI (OR APPROVED EQUAL) | 120 V | LED / 3500K / 90 | 11 VA | 1045 | 0-10V / 1% | CLEAR | RECESSED CEILING | 0'-3 13/16" RFD | |
| D2 | 7" DIAMETER SURFACE MOUNT LED DOWNLIGHT | JUNO (OR APPROVED EQUAL) | JSF-7IN10LM-SWW5-90CRI-MVOLTZT-WH (OR APPROVED EQUAL) | 120 V | LED / 3500K / 90 | 12 VA | 1182 | 0-10V | WHITE | SURFACE CEILING | 0'-0.75" OFH | |
| D3 | 3" DIAMETER RECESSED LED DOWNLIGHT, IC RATED | ACULUX (OR APPROVED EQUAL) | INIT3D-12LM-35K-90CRI-50D-GZ1-MVOLT-ICAT + NT3DP-CD-SF (OR APPROVED EQUAL) | 120 V | LED / 3500K / 90 | 12 VA | 1200 | 0-10V / 1% | CLEAR | RECESSED CEILING | 0'-5.3" RFD | |
| L1-10 | 10' LONG SUSPENDED LINEAR LED, 2" WIDE APERTURE, DIRECT-INDIRECT | NULITE (OR APPROVED EQUAL) | RP24-B-FRF-BW-05-03-H35-UNV-D-11-WH-SRW048-10 (OR APPROVED EQUAL) | 120 V | LED / 3500K / 90 | 60 VA | DOWN: 495 LM/FT UP: 302 LM/FT | 0-10V / 1% | WHITE | SUSPENDED CEILING | 8'-6" BOF | |
| L1-10EM | 10' LONG SUSPENDED LINEAR LED, 2" WIDE APERTURE, DIRECT-INDIRECT, INTEGRAL EM BATTERY | NULITE (OR APPROVED EQUAL) | RP24-B-FRF-BW-05-03-H35-UNV-D-11B10-WH-SRW048-10 (OR APPROVED EQUAL) | 120 V | LED / 3500K / 90 | 60 VA | DOWN: 495 LM/FT UP: 302 LM/FT | 0-10V / 1% | WHITE | SUSPENDED CEILING | 8'-6" BOF | |
| L2-6 | 6' LONG SURFACE MOUNT LINEAR LED | NULITE (OR APPROVED EQUAL) | RR2-D-FRF-05-H35-UNV-D-11-WH-6 (OR APPROVED EQUAL) | 120 V | LED / 3500K / 90 | 24 VA | 495 LM/FT | 0-10V / 1% | WHITE | SURFACE CEILING | 0'-3 3/4" OFH | |
| L2-6EM | 6' LONG SURFACE MOUNT LINEAR LED, INTEGRAL EM BATTERY | NULITE (OR APPROVED EQUAL) | RR2-D-FRF-05-H35-UNV-D-11B10-WH-4 (OR APPROVED EQUAL) | 120 V | LED / 3500K / 90 | 24 VA | 495 LM/FT | 0-10V / 1% | WHITE | SURFACE CEILING | 0'-3 3/4" OFH | |
| P1 | 64.5" LONG DECORATIVE LINEAR PENDANT, 8.75" WIDE, 10" TALL, WAVY STRETCHED FABRIC SHADE | CERNO (OR APPROVED EQUAL) | 07-220-L-A-35-P1 (OR APPROVED EQUAL) | 120 V | LED / 3500K / 90 | 70 VA | 5000 | 0-10V | WHITE, BRUSHED ALUMINUM | SUSPENDED CEILING | 7'-6" BOF | |
| R1 | 2X4 RECESSED LED TROFFER | LITHONIA (OR APPROVED EQUAL) | 2BLT4-30L-ADP-GZ1-LP935 (OR APPROVED EQUAL) | 120 V | LED / 3500K / 90 | 23 VA | 3000 | 0-10V / 1% | WHITE | RECESSED CEILING | 0'-2 3/8" RFD | |
| R1EM | 2X4 RECESSED LED TROFFER, INTEGRAL EM BATTERY | LITHONIA (OR APPROVED EQUAL) | 2BLT4-30L-ADP-GZ1-LP935-EL14L (OR APPROVED EQUAL) | 120 V | LED / 3500K / 90 | 23 VA | 3000 | 0-10V / 1% | WHITE | RECESSED CEILING | 0'-2 3/8" RFD | |
| R2 | 2X2 RECESSED LED TROFFER | LITHONIA (OR APPROVED EQUAL) | 2BLT2-20L-ADP-GZ1-LP935 (OR APPROVED EQUAL) | 120 V | LED / 3500K / 90 | 16 VA | 2000 | 0-10V / 1% | WHITE | RECESSED CEILING | 0'-2 3/8" RFD | |
| R3 | 1X4 RECESSED LED TROFFER | LITHONIA (OR APPROVED EQUAL) | BLT4-30L-ADP-GZ1-LP935 (OR APPROVED EQUAL) | 120 V | LED / 3500K / 90 | 23 VA | 3000 | 0-10V / 1% | WHITE | RECESSED CEILING | 0'-2 3/4" RFD | |
| R3EM | 1X4 RECESSED LED TROFFER, INTEGRAL EM BATTERY | LITHONIA (OR APPROVED EQUAL) | BLT4-30L-ADP-GZ1-LP935-EL14L (OR APPROVED EQUAL) | 120 V | LED / 3500K / 90 | 23 VA | 3000 | 0-10V / 1% | WHITE | RECESSED CEILING | 0'-2 3/4" RFD | |
| S1-2EM | 2' LONG LED STRIP LIGHT, 4" WIDE APERTURE, INTEGRAL EM BATTERY | LITHONIA (OR APPROVED EQUAL) | CLX-L24-2500LM-SEF-FDL-MVOLT-GZ1-35K-80CRI-E10W-WH (OR APPROVED EQUAL) | 120 V | LED / 3500K / 80 | 18 VA | 2500 | 0-10V / 1% | WHITE | SURFACE WALL | 9'-0" BOF | |
| S1-4 | 4' LONG LED STRIP LIGHT, 4" WIDE APERTURE | LITHONIA (OR APPROVED EQUAL) | CLX-L48-5000LM-SEF-FDL-MVOLT-GZ1-35K-80CRI-WH (OR APPROVED EQUAL) | 120 V | LED / 3500K / 80 | 32 VA | 5000 | 0-10V / 1% | WHITE | SURFACE CEILING | 0'-2.72" OFH | |
| S2 | 3" DIAMETER SURFACE MOUNT LED CYLINDER, 8" TALL, 80 DEGREE BEAM, SOLITE LENS | ZANIBONI (OR APPROVED EQUAL) | S2-BONG2-18-35-C-8-S-WS-Z0-NI-D (OR APPROVED EQUAL) | 120 V | LED / 3500K / 90 | 18 VA | 1832 | 0-10V / 1% | WHITE | SURFACE CEILING | 0'-8.15" OFH | 1 |
| S2EM | 3" DIAMETER SURFACE MOUNT LED CYLINDER, 8" TALL, 80 DEGREE BEAM, SOLITE LENS, REMOTE EM INVERTER | ZANIBONI (OR APPROVED EQUAL) | S2-BONG2-18-35-C-8-S-WS-Z0-NI-D (OR APPROVED EQUAL) | 120 V | LED / 3500K / 90 | 18 VA | 1832 | 0-10V / 1% | WHITE | SURFACE CEILING | 0'-8.15" OFH | 1 |
| S3-4 | 4' LONG LED STRIP LIGHT, 4" WIDE APERTURE, CHAIN MOUNT ACCESSORY | LITHONIA (OR APPROVED EQUAL) | CLX-L48-5000LM-SEF-FDL-MVOLT-GZ1-35K-80CRI-WH, HC36-M12 (OR APPROVED EQUAL) | 120 V | LED / 3500K / 80 | 32 VA | 5000 | 0-10V / 1% | WHITE | SUSPENDED CEILING | 8'-6" BOF | |
| S3-4EM | 4' LONG LED STRIP LIGHT, 4" WIDE APERTURE, CHAIN MOUNT ACCESSORY, INTEGRAL EM BATTERY | LITHONIA (OR APPROVED EQUAL) | CLX-L48-5000LM-SEF-FDL-MVOLT-GZ1-35K-80CRI-E10W-WH, HC36-M12 (OR APPROVED EQUAL) | 120 V | LED / 3500K / 80 | 32 VA | 5000 | 0-10V / 1% | WHITE | SUSPENDED CEILING | 8'-6" BOF | |
| S4EM | 4' LONG LED STRIP LIGHT, VAPOR TIGHT, INTEGRAL EM BATTERY | LITHONIA (OR APPROVED EQUAL) | CSVT-L48-5000LM-MVOLT-35K-80CRI-IE7WCP (OR APPROVED EQUAL) | 120 V | LED / 3500K / 80 | 42 VA | 5000 | SWITCHING | GRAY | SURFACE WALL | SEE PLANS | |
| T1 | RECESSED TRACK SYSTEM, LINE VOLTAGE, SINGLE CIRCUIT | JUNO (OR APPROVED EQUAL) | TREC-[NOTE 3]-WH (OR APPROVED EQUAL) | 120 V | LED / 3500K / 90 | 1 VA | -- | -- | WHITE | RECESSED CEILING | 0'-1 1/8" RFD | 3,6 |
| T1A | ADJUSTABLE LED TRACK HEAD, NARROW FLOOD DISTRIBUTION | JUNO (OR APPROVED EQUAL) | T254L-G2-35K-90CRI-PDIM-NFL-WH (OR APPROVED EQUAL) | 120 V | LED / 3500K / 90 | 15 VA | 1569 | PHASE DIMMING | WHITE | TRACK | 0'-7 7/16" OFH | 6 |
| W1 | 24" LONG DECORATIVE LED VANITY LIGHT, DIFFUSE LENS | BROWNLEE (OR APPROVED EQUAL) | 5160-24-WH-H16-EC1-35K-BAC (OR APPROVED EQUAL) | 120 V | LED / 3500K / 90 | 16 VA | 1890 | 0-10V | WHITE | SURFACE WALL | 6'-10" BOF | |
| X1 | LED EXIT SIGN, SINGLE FACE, BACK MOUNT, SELF-CHARGING PHOTOLUMINESCENT BACKUP | EVENLITE (OR APPROVED EQUAL) | AUR-1-BA-1B (OR APPROVED EQUAL) | 120 V | LED / GREEN | 2 VA | -- | -- | ALUMINUM | SURFACE WALL | [NOTE 4] | 4 |
| X2 | LED EXIT SIGN WITH SMALL SCALE INTEGRAL EMERGENCY LIGHTING | EVENLITE (OR APPROVED EQUAL) | TCX1COM-G-U-W | 120 V | LED / GREEN | 3 VA | 200 | -- | WHITE | SURFACE WALL | [NOTE 4] | 4 |
| ED1 | EXTERIOR 6" DIAMETER RECESSED LED DOWNLIGHT | LITHONIA (OR APPROVED EQUAL) | LBR6-NCH-10LM-30K-AR-TRBL-LSS-MWD-MVOLT-UGZ (OR APPROVED EQUAL) | 120 V | LED / 3000K / 80 | 13 VA | 1270 | 0-10V / 10% | BLACK | RECESSED CEILING | 0'-3 7/8" RFD | 1 |
| ED1EM | EXTERIOR 6" DIAMETER RECESSED LED DOWNLIGHT, INTEGRAL EM BATTERY | LITHONIA (OR APPROVED EQUAL) | LBR6-NCH-10LM-30K-AR-TRBL-LSS-MWD-MVOLT-UGZ-E10WCP (OR APPROVED EQUAL) | 120 V | LED / 3000K / 80 | 13 VA | 1270 | 0-10V / 10% | BLACK | RECESSED CEILING | 0'-3 7/8" RFD | 1 |
| ED2 | EXTERIOR 6" DIAMETER RECESSED LED DOWNLIGHT, RETROFIT | LITHONIA (OR APPROVED EQUAL) | LBR6-10LM-30K-AR-TRBL-LSS-MWD-MVOLT-UGZ (OR APPROVED EQUAL) | 120 V | LED / 3000K / 80 | 13 VA | 1270 | 0-10V / 10% | BLACK | RECESSED CEILING | 0'-4 3/16" RFD | 1 |
| ED2EM | EXTERIOR 6" DIAMETER RECESSED LED DOWNLIGHT, RETROFIT, INTEGRAL EM BATTERY | LITHONIA (OR APPROVED EQUAL) | LBR6-10LM-30K-AR-TRBL-LSS-MWD-MVOLT-UGZ-E10WCP (OR APPROVED EQUAL) | 120 V | LED / 3000K / 80 | 13 VA | 1270 | 0-10V / 10% | BLACK | RECESSED CEILING | 0'-4 3/16" RFD | 1 |
| EW1EM | EXTERIOR LED WALL PACK AREA LIGHT, INTEGRAL EM BATTERY | LITHONIA (OR APPROVED EQUAL) | WDGE2-LED-P1-30K-80CRI-VW-MVOLT-SRM-DMG-E10WH-DBLXD (OR APPROVED EQUAL) | 120 V | LED / 3000K / 80 | 10 VA | 1241 | 0-10V | BLACK | SURFACE WALL | 10'-0" BOF | |

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.

LICENSED PROFESSIONAL ENGINEER — JEFF MULLIKIN — P25384 — 05/21/2026 — REGISTERED — STATE OF WASHINGTON

A/E FIRM:
PRIME:
OTAK, INC.
REDMOND, WA
SUBCONTRACTOR
AE DESIGN
DENVER, CO

DESIGNED: JR/JM/TW
DRAWN: AE DESIGN
TECH REVIEW: MRS/ESN
DATE: 5/29/2026

SUB SHEET NO. E500

TITLE OF SHEET
ELECTRICAL LIGHTING SCHEDULES
EFFIGY MOUNDS NATIONAL MONUMENT

DRAWING NO. 394 194248
PMIS/PKG NO. 318539
SHEET 124 OF 142

5/21/2026 1:49:08 PM Autodesk Docs://2025-110 NPS EFMO Visitor Center DD-CD/6782.00 - NPS EFMO Effigy Mounds Visitor Center Renovation - Electrical Central - E24.rvt

# OPTION C: LIGHTING SEQUENCE OF OPERATION

| CONTROL SEQUENCE | ON | OFF | SENSOR TYPE | TIME OUT | DIMMING | CONTROLLED RECEPTACLE | DAYLIGHT HARVESTING | TARGET ILLUMINANCE (FC) | NOTES |
|---|---|---|---|---|---|---|---|---|---|
| EM1 | MANUAL ON TIMER SWITCH, 2 HR MAXIMUM OVERRIDE | AUTOMATIC OFF VIA TIMER SWITCH, MANUAL OVERRIDE | NONE | 30 MIN, 1 HR, 2 HR | SWITCHING | NO | NO | -- | |
| ET1 | TIMECLOCK AUTOMATIC ON 30 MINUTES PRIOR TO SUNSET | TIMECLOCK AUTOMATIC OFF 30 MINUTES AFTER SUNRISE, AUTOMATIC DIM TO 30% BETWEEN MIDNIGHT AND 6AM | NONE | N/A | DIMMING | NO | NO | -- | |
| M1 | MANUAL ON | MANUAL OFF | NONE | N/A | SWITCHING | NO | NO | -- | MANUAL CONTROL HAS BEEN PROVIDED FOR SAFETY OF THE OCCUPANTS RELATED TO MAINTENANCE OF EQUIPMENT |
| O1 | AUTOMATIC ON | AUTOMATIC OFF | OCCUPANCY | 10 MINUTES | SWITCHING | NO | NO | -- | |
| OD1 | AUTOMATIC ON | AUTOMATIC OFF | OCCUPANCY | 10 MINUTES | DIMMING | NO | NO | -- | |
| TO1 | TIMECLOCK AUTOMATIC ON 30 MINUTES PRIOR TO BUSINESS HOURS | OCCUPANCY SENSORS BECOME ENABLED 30 MINUTES PRIOR TO CLOSE OF BUSINESS AND BECOME DISABLED 30 MINUTES PRIOR TO COMMENCE OF BUSINESS | OCCUPANCY | 10 MINUTES | DIMMING | NO | NO | -- | OCCUPANCY SENSOR FUNCTIONALITY DURING TIMECLOCK OFF HOURS ONLY |
| TO2 | AUTOMATIC ON TO 100% OUTPUT UPON OCCUPANCY | AUTOMATIC DIM TO 50% OUTPUT DURING BUSINESS HOURS UPON VACANCY, AUTOMATIC OFF AFTER BUSINESS HOURS | OCCUPANCY | 10 MINUTES | DIMMING | NO | NO | -- | |
| TP1 | TIMECLOCK AUTOMATIC ON 30 MINUTES PRIOR TO BUSINESS HOURS, OVERRIDE WITH PHOTOCELL BELOW TARGET ILLUMINANCE | TIMECLOCK AUTOMATIC OFF 30 MINUTES AFTER CLOSE OF BUSINESS, OVERRIDE WITH PHOTOCELL ABOVE TARGET ILLUMINANCE | PHOTOCELL | N/A | DIMMING | NO | YES | 30 FC | PHOTOCELL OVERRIDE FUNCTIONALITY DURING TIMECLOCK ON HOURS ONLY |
| V1 | MANUAL ON | AUTOMATIC OFF | VACANCY | 10 MINUTES | SWITCHING | NO | NO | -- | |
| VC1 | MANUAL ON FOR LIGHT FIXTURES, AUTOMATIC ON FOR CONTROLLED RECEPTACLES | AUTOMATIC OFF | VACANCY | 10 MINUTES | DIMMING | YES | NO | -- | |

# OPTION C: LIGHTING ZONE SCHEDULE

| ZONE ID | ZONE DESCRIPTION | DIMMING / SWITCHING | VOLTAGE | PANEL-CIRCUIT | ZONE LOAD | CONTROL SEQUENCE |
|---|---|---|---|---|---|---|
| ez01 | CANOPY DOWNLIGHTS | 0-10V / 10% | 120 V | D-2 | 247 VA | ET1 |
| ez02 | EMERGENCY WALL PACKS | 0-10V | 120 V | D-2 | 30 VA | ET1 |
| z1.01 | VESTIBULE CYLINDERS | 0-10V / 1% | 120 V | D-6 | 36 VA | TO1 |
| z1.02 | HALLWAY DOWNLIGHTS | 0-10V / 1% | 120 V | D-6 | 77 VA | TO2 |
| z1.04 | EXHIBIT AND RETAIL LOBBY DECORATIVE PENDANTS | 0-10V | 120 V | D-6 | 140 VA | TO1 |
| z1.05 | EXHIBIT AND RETAIL LOBBY CYLINDERS | 0-10V / 1% | 120 V | D-6 | 180 VA | TO1 |
| z1.06 | EXHIBIT AND RETAIL LOBBY CYLINDERS, SECONDARY DAYLIGHT ZONE | 0-10V / 1% | 120 V | D-6 | 108 VA | TP1 |
| z1.07 | EXHIBIT AND RETAIL LOBBY CYLINDERS, PRIMARY DAYLIGHT ZONE | 0-10V / 1% | 120 V | D-6 | 90 VA | TP1 |
| z1.08 | EXHIBIT TRACK LIGHTING - OPTION F | PHASE DIMMING | 120 V | D-8 | 75 VA | TO1 |
| z1.09 | EXHIBIT TRACK LIGHTING - OPTION F | PHASE DIMMING | 120 V | D-8 | 120 VA | TO1 |
| z1.10 | EXHIBIT TRACK LIGHTING - OPTION F | PHASE DIMMING | 120 V | D-8 | 45 VA | TO1 |
| z1.11 | EXHIBIT TRACK LIGHTING - OPTION F | PHASE DIMMING | 120 V | D-8 | 135 VA | TO1 |
| z1.12 | EXHIBIT TRACK LIGHTING - OPTION F | PHASE DIMMING | 120 V | D-8 | 120 VA | TO1 |
| z1.13 | EXHIBIT 1X4 TROFFERS | 0-10V / 1% | 120 V | D-8 | 230 VA | TO1 |
| z1.14 | HALLWAY TROFFERS | 0-10V / 1% | 120 V | D-8 | 85 VA | TO2 |
| z1.15 | AUDITORIUM/MULTIPURPOSE ROOM LINEARS | 0-10V / 1% | 120 V | D-6 | 180 VA | OD1 |
| z1.16 | AUDITORIUM/MULTIPURPOSE ROOM DOWNLIGHTS | 0-10V / 1% | 120 V | D-6 | 132 VA | OD1 |
| zb.01 | HALLWAY LINEARS | 0-10V / 1% | 120 V | D-4 | 48 VA | TO2 |
| zb.02 | VESTIBULE DOWNLIGHTS | 0-10V / 1% | 120 V | | 72 VA | |
| zb.03 | VESTIBULE DOWNLIGHTS | 0-10V / 1% | 120 V | D-4 | 12 VA | TO2 |
| zp.01 | PARKING LOT BOLLARDS - E | SWITCHING | 120 V | D-2 | 336 VA | EM1 |
| zp.02 | PARKING LOT BOLLARDS - W | SWITCHING | 120 V | D-2 | 112 VA | EM1 |

# OPTION C: LIGHTING CONTROL DEVICES

| TYPE | DESCRIPTION | DETAILS |
|---|---|---|
| NETWORKED COMPONENTS | | |
| NHUB | NETWORKED LIGHTING CONTROL HUB | E502 / 1 |
| NWD1 | NETWORKED WIRELESS BUTTON STATION WITH MANUAL DIMMER, PROGRAM RELAY ZONES AS INDICATED ON LIGHTING PLAN | |
| NWD2 | NETWORKED WIRELESS BUTTON STATION WITH MANUAL DIMMER, PROGRAM RELAY ZONES AS INDICATED ON LIGHTING PLAN, PROVIDE SCENES FOR 'NORMAL' AND 'FILM'. 'FILM' SCENE SHALL DIM ALL LIGHTS TO 25% OF FULL OUTPUT. | |
| NWO1 | NETWORKED WIRELESS CEILING MOUNTED OCCUPANCY SENSOR | |
| NWP1 | NETWORKED WIRELESS, CEILING MOUNTED, CLOSED LOOP PHOTOCELL/ DAYLIGHT SENSOR | |
| NWU1 | WIRELESS MULTI-BUTTON TIMER SWITCH, BUTTONS TO BE LABELED AND PROGRAMMED FOR 30 MINUTE, 1 HR, 2 HR, AND ALL OFF FOR EXTERIOR BOLLARD LIGHTING | |
| STANDALONE CONTROL SYSTEMS | | |
| WD1 | WIRELESS MANUAL DIMMER | |
| WO1 | CEILING MOUNTED, WIRELESS OCCUPANCY SENSOR | |
| WOS1 | WIRELESS OCCUPANCY SENSOR ON/OFF SWITCH | |
| WS1 | WIRELESS ON/OFF SWITCH | |
| WV1 | CEILING MOUNTED, WIRELESS OCCUPANCY SENSOR SET TO VACANCY MODE | |
| WVS1 | WIRELESS VACANCY SENSOR ON/OFF SWITCH | |

# OPTION C - EMERGENCY INVERTER SCHEDULE

| INVERTER ID | DESCRIPTION | CONTROL ZONE | PANEL-CIRCUIT | CONNECTED LOAD | MAX LOAD | DETAIL |
|---|---|---|---|---|---|---|
| INV1 | 25W MICRO INVERTER. BOD IOTA IIS 25 I | z1.01 | D-6 | 18 VA | 25 VA | E502 / 2 |
| INV2 | 50W MICRO INVERTER. BOD IOTA IIS 50 I | z1.05 | D-6 | 36 VA | 50 VA | E502 / 2 |
| INV3 | 50W MICRO INVERTER. BOD IOTA IIS 50 I | z1.06 | D-6 | 36 VA | 50 VA | E502 / 2 |
| INV4 | 50W MICRO INVERTER. BOD IOTA IIS 50 I | z1.07 | D-6 | 36 VA | 50 VA | E502 / 2 |

# LIGHTING CONTROL NOTES

## STANDALONE LIGHTING CONTROL GENERAL NOTES

S1 APPROVED STANDALONE LIGHTING CONTROLS TO BE PROVIDED BY ONE OF THE FOLLOWING PRE-APPROVED MANUFACTURERS OR APPROVED EQUAL:
- a. LEVITON
- b. nLIGHT/SENSORSWITCH
- c. LUTRON
- d. WAVELINX

## NETWORKED WIRELESS DISTRIBUTED LIGHTING CONTROLS

N1 APPROVED DISTRIBUTED NETWORK BASED LIGHTING CONTROLS TO BE PROVIDED BY ONE OF THE FOLLOWING PRE-APPROVED MANUFACTURERS OR APPROVED EQUAL:
- a. CRESTRON
- b. nLIGHT
- c. LUTRON
- d. WAVELINX

N2 REFER TO ELECTRICAL LIGHTING LAYOUTS FOR LAYOUT OF DEVICES CONNECTED TO DISTRIBUTED LIGHTING CONTROL SYSTEM. DISTRIBUTED COMPONENTS ARE INDICATED IN THE "LIGHTING CONTROL DEVICE" SCHEDULE, THESE COMPONENTS START WITH THE DESIGNATION 'NW'.

N3 DISTRIBUTED LIGHTING CONTROL HEAD-ENDS/HUBS SHALL BE PROVIDED BASED ON THE QUANTITY OF ZONES INDICATED IN THE 'NETWORKED LIGHTING CONTROL ZONE SCHEDULE'. COMPONENTS PROVIDED SHALL BE CAPABLE OF PROVIDING FUNCTIONALITY IN ACCORDANCE WITH 'SEQUENCE OF OPERATIONS' SCHEDULE.

N4 ALL COMPONENTS OF NETWORKED DISTRIBUTED LIGHTING CONTROL SYSTEM REQUIRING LINE VOLTAGE CONNECTIONS SHALL BE FED FROM LOCAL UNSWITCHED LIGHTING CIRCUIT.

N5 ALL REMOTE LOAD CONTROLLERS SUCH AS POWER PACKS AND RELAY DEVICES SHALL BE LOCATED IN A READILY ACCESSIBLE LOCATION.

N6 POWER REQUIREMENTS SHALL BE CONFIRMED WITH APPROVED SUBMITTAL AND CONNECTED TO NEAREST UNSWITCHED LIGHTING CIRCUIT.

N7 DISTRIBUTED LIGHTING CONTROL HEAD-END/HUBS SHALL BE EQUIPPED WITH CAPABILITY FOR PROGRAMMATIC AND ZONING SOFTWARE EXPANSION FOR A MINIMUM OF 20% ADDITIONAL SPARE CAPACITY.

## GENERAL CONTROL NOTES

G1 THE LIGHTING CONTROL SYSTEM CONSISTS OF THE FOLLOWING:
- a. NETWORKED WIRELESS DISTRIBUTED LIGHTING CONTROLS
- b. STAND-ALONE CONTROLS

G2 ALTERNATE MANUFACTURER'S WILL BE REVIEWED ACCORDING TO THE NOTES PROVIDED IN THE LIGHTING FIXTURE SCHEDULE.

G3 ALL WIRING DIAGRAMS WITHIN THESE DRAWINGS ARE PROVIDED TO COMMUNICATE THE DESIGN INTENT. SYSTEM SHALL BE WIRED ACCORDING TO THE APPROVED SHOP DRAWINGS.

G4 ALL STRUCTURED CABLE WIRING SHOWN ON RISER DIAGRAMS IS INTENDED TO BE BY CONTROL MANUFACTURER APPROVED STANDARD STRUCTURED CABLING, UNLESS OTHERWISE NOTED. EC SHALL PROVIDE ALL CABLING WITHIN THE LIGHTING CONTROL SYSTEM, CABLING BETWEEN THE NETWORKED HEAD-END AND THE BUILDINGS COMMUNICATION NETWORK SHALL BE PROVIDED BY THE LOW VOLTAGE CONTRACTOR.

G5 ALL MANUALLY DIMMED LIGHT LOADS SHALL BE CAPABLE OF DIMMING LIGHTS TO OFF SETTING. DIMMING COMPATIBILITY BETWEEN THE CONTROLS AND LIGHT FIXTURES SHALL BE COORDINATED BY THE EC TO ENSURE THAT LIGHTING IS ABLE TO DIM TO LEVEL NOTED ON LIGHTING FIXTURE SCHEDULE.

G6 LIGHTING CONTROL SYSTEM SHALL INCLUDE A MINIMUM OF (4) HOURS OF MANUFACTURER'S REPRESENTATIVE TIME ON SITE FOR SYSTEM CHECK-OUT AND NPS TRAINING. ELECTRICAL CONTRACTOR SHALL VIDEO RECORD TRAINING SESSION AND PROVIDE COPY OF VIDEO TO CONTRACTING OFFICER AS PART OF PROJECT COMPLETION SUBMITTALS.

G7 ALL DIGITAL SWITCHES FOR OVERRIDE CONTROL OF LIGHTING CONTROL SYSTEM(S) SHALL HAVE A MAXIMUM SETTING OF 2 HOURS PER IECC REQUIREMENTS.

G8 FINAL OCCUPANCY AND DAYLIGHT SENSOR LOCATION SHALL BE PROVIDED BY MANUFACTURER AND LOCATED PER APPROVED SHOP DRAWINGS AND DEVICE REQUIREMENTS. LOCATIONS INDICATED IN THESE DRAWINGS SHALL BE REVIEWED AND ALTERED AS NECESSARY FOR CORRECT OPERATION BY MANUFACTURER. IF OPERATIONS OF SENSORS DOES NOT MEET THE INTENT OUTLINED IN THESE DOCUMENTS THE MANUFACTURER REPRESENTATIVE SHALL PROVIDE FIELD RECTIFICATION SERVICES AS NECESSARY IN ORDER TO RECONFIGURE SYSTEM TO MEET OUTINED INTENT.

G9 OCCUPANCY SENSORS SHALL BE COORDINATED SUCH THAT THE MOUNTING IS NOT WITHIN 6' OF AIR RETURN SYSTEMS.

G10 PHOTOCELL SENSORS SHALL BE COORDINATED SUCH THAT THE DEVICE(S) ARE NOT WITHIN PROXIMITY TO INDIRECT LIGHTING OR WHERE SUBJECT TO VEILING REFLECTIONS FROM GLASS OR WATER SURFACES.

G11 WHERE APPLICABLE, WALLSTATIONS WITH MORE THAN ONE GANG SHALL BE CONGREGATED TOGETHER UNDER A SINGLE FACEPLATE, UON.

# LIGHTING CONTROLS NAMING CONVENTION

**SYSTEM TYPE:**
- **N** = NETWORKED
- **R** = ROOM CONTROLLER
- **W** = WIRELESS
- - THE ABSENCE OF LETTERS UNDER 'SYSTEM TYPE' INDICATES A WIRED STANDALONE SYSTEM
- - THE ABSENCE OF 'W' MEANS THE SYSTEM IS WIRED

**SENSOR TYPE:**
- **O** = OCCUPANCY
- **V** = VACANCY
- **P** = PHOTOCELL

**DEVICES:**
- **D** = DIMMER
- **S** = SWITCH
- **H** = HUB
- **U** = UNIQUE DEVICE TYPE

**NUMBERING:**
- **1**,**2**,**3**... = QUANTITY AS REQUIRED FOR DIFFERENT HARDWARE, DEVICE CHARACTERISTICS, OR MOUNTING CONDITIONS.

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.

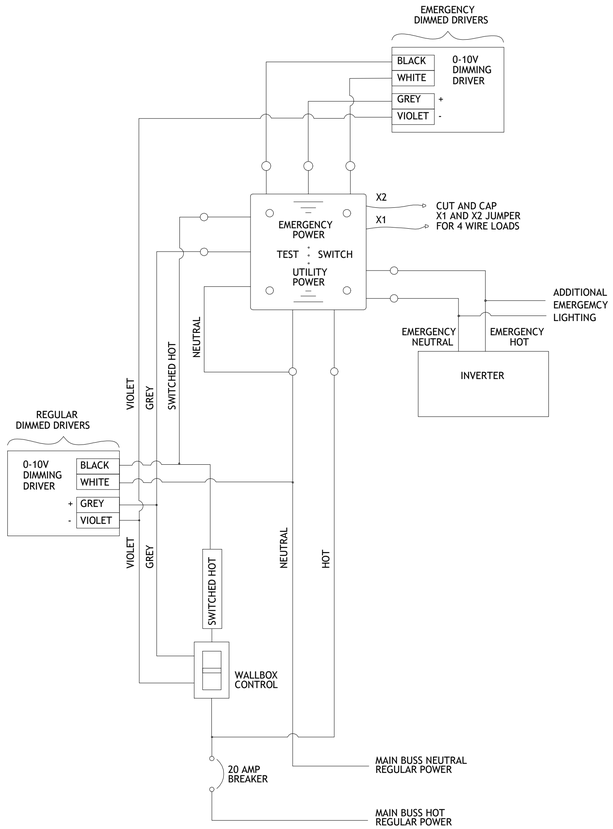
05/21/2026 JEFF MULLIKIN P25384 LICENSED PROFESSIONAL ENGINEER KONA

| | |
|---|---|
| A/E FIRM: | |
| PRIME: | OTAK, INC. |
| | REDMOND, WA |
| SUBCONTRACTOR | AE DESIGN |
| | DENVER, CO |
| DESIGNED: | JR/JM/TW |
| CADD | AE DESIGN |
| TECH REVIEW: | MRS/ESN |
| DATE: | 5/29/2026 |
| SUB SHEET NO. | E501 |

TITLE OF SHEET
ELECTRICAL LIGHTING CONTROLS SCHEDULES
EFFIGY MOUNDS NATIONAL MONUMENT

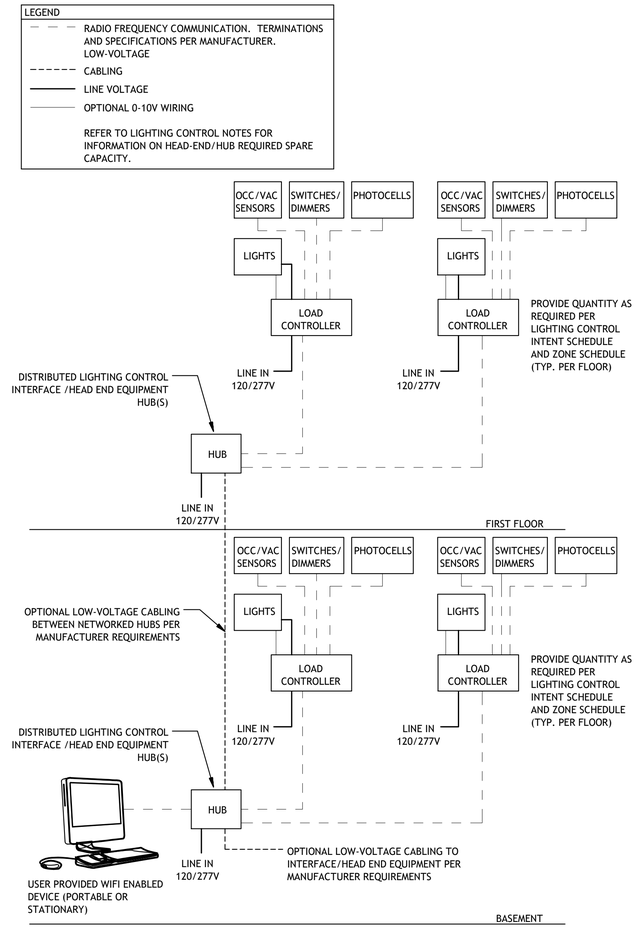
DRAWING NO. 394 194248
PMIS/PKG NO. 318539
SHEET 125 OF 142

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.




| EMERGENCY LIGHTING CONTROL DEVICE GENERAL NOTES |
|---|
| 1. USE WITH MANUFACTURER CONTROL TO DETERMINE THE SUITABILITY OF THE SPECIFIED DIMMER FOR SHUNT TRIP OPERATION. |
| 2. EM SWITCHING DEVICE SHALL BE INSTALLED DOWNSTREAM OF THE EM LIGHTING INVERTER AND POWERED WITH THE SAME CIRCUIT AS THE ZONE IT IS CONTROLLING. |
| 3. INVERTER DEVICE SHALL BE UL924 LISTED. |

2 E502 EMERGENCY CONTROL DEVICE DIAGRAM




1 E502 NETWORKED WIRELESS LIGHTING CONTROLS RISER

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LICENSED PROFESSIONAL ENGINEER 05/21/2026 JEFF MULLIKIN P25384 IOWA

| | |
|---|---|
| A/E FIRM: | |
| PRIME: | OTAK, INC. |
| | REDMOND, WA |
| SUBCONTRACTOR | AE DESIGN |
| | DENVER, CO |
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| TECH REVIEW: | MRS/ESN |
| DATE: | 5/29/2026 |
| SUB SHEET NO. | E502 |
| TITLE OF SHEET | ELECTRICAL LIGHTING CONTROLS DETAILS |
| | EFFIGY MOUNDS NATIONAL MONUMENT |
| DRAWING NO. | 394 194248 |
| PMIS/PKG NO. | 318539 |
| SHEET | 126 OF 142 |

5/21/2026 1:49:08 PM Autodesk Docs://2025-110 NPS EFMO Visitor Center DD-CD/6782.00 - NPS EFMO Effigy Mounds Visitor Center Renovation - Electrical Central - E24.rvt

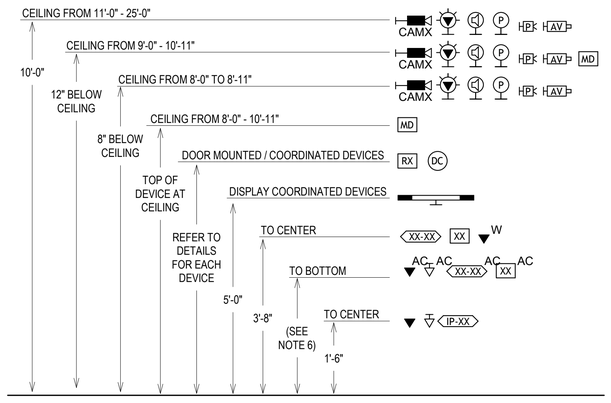
## DISTRIBUTION & RACEWAY

- CONDUIT CONCEALED IN FLOOR OR UNDERGROUND
- CONDUIT EXPOSED OR CONCEALED IN WALL OR CEILING
- RACEWAY UP
- RACEWAY DOWN
- CONDUIT WITH CAPPING
- CONDUIT WITH BUSHING
- CONDUIT WITH CONTINUATION
- LADDER RACK — LENGTH AS INDICATED ON DRAWINGS
- CABLE TRAY — LENGTH AS INDICATED ON DRAWINGS
- MDF OR IDF BOUNDARY
- [XX] PATHWAY/JUNCTION/OUTLET BOX — HH HANDHOLE; PB PULL BOX
- [XX#] (HAND HOLE WILL BE SPECIFIED IN SCHEDULE) (PULL BOX WILL BE SPECIFIED IN SCHEDULE)

PATHWAY/JUNCTION/OUTLET BOX (XX = TYPE, # = IDENTIFIER)

## TELECOMMUNICATIONS LEGEND

- # ## TYPICAL TELECOMMUNICATION SYMBOLS WHERE FIRST (OR SINGLE) '#' DENOTES THE NUMBER OF NETWORK CABLES TO BE PROVIDED AT THE DEVICE, AND THE SECOND '#' DENOTES THE NUMBER OF VOICE CABLES TO BE PROVIDED AT THE DEVICE.
- XXX 'XXX' REFERENCES A CABLE DEFINED IN THE EQUIPMENT SCHEDULES. REFER TO SCHEDULE KEY VALUES FOR CABLE REFERENCE AND ASSOCIATED REQUIREMENTS.
- TELECOMMUNICATION OUTLET
- CABLE TELEVISION (CATV) OUTLET
- FLOOR MOUNTED TELECOMMUNICATION/CATV OUTLET
- FLOOR MOUNTED INTEGRATED POWER / TELECOM OUTLET
- CLG CLG CEILING MOUNTED TELECOMMUNICATION/CATV OUTLET
- WALL MOUNTED WIRELESS ACCESS POINT (WAP) OUTLET
- CEILING MOUNTED WIRELESS ACCESS POINT (WAP) OUTLET




NOTES:

1. HEIGHTS INDICATED ON LEGEND REPRESENT THE CENTERLINE MOUNTING HEIGHT OF THE DEVICE'S BACKBOX/MOUNTING EQUIPMENT, UNLESS OTHERWISE NOTED.
2. AUDIOVIDEO DEVICE SYMBOL SHOWN SHALL REPRESENT ALL AUDIOVISUAL DEVICE SUBSCRIPTS WITHIN THE SYMBOL UNLESS A DEVICE IS SPECIFICALLY CALLED OUT FOR A DIFFERENT HEIGHT ON THE LEGEND OR ON THE PLAN SHEETS.
3. WALL MOUNTED SECURITY DEVICE SYMBOL SHOWN SHALL REPRESENT ALL WALL MOUNTED SECURITY DEVICE SUBSCRIPTS WITHIN THE SYMBOL UNLESS A DEVICE IS SPECIFICALLY CALLED OUT FOR A DIFFERENT HEIGHT ON THE LEGEND OR ON THE PLAN SHEETS.
4. THE E.C. SHALL REFER TO ARCHITECTURAL ELEVATIONS TO COORDINATE ALL COUNTER HEIGHTS. ALL "AC" DEVICES SHALL HAVE BOTTOM OF BACK-BOX MOUNTED 4" ABOVE THE BACK/SIDE SPLASH.

2 T000 TECHNOLOGY DEVICE MOUNTING HEIGHTS

## ABBREVIATIONS AND SYMBOLS

| | |
|---|---|
| AC | ABOVE COUNTER MOUNTED |
| ADA | AMERICANS WITH DISABILITIES ACT |
| AFF | ABOVE FINISHED FLOOR |
| AFG | ABOVE FINISHED GRADE |
| AHJ | AUTHORITY HAVING JURISDICTION |
| ALS | ASSISTIVE LISTENING SYSTEM |
| AV | AUDIOVISUAL |
| AVC | AUDIOVISUAL CONTRACTOR COORDINATED |
| BAS | BUILDING AUTOMATION SYSTEM |
| BGM | BACKGROUND MUSIC |
| C | CONDUIT |
| CATV | COMMUNITY ACCESS TELEVISION |
| CCTV | CLOSED CIRCUIT TELEVISION |
| CH | CHANNEL ROUTED, MOUNTED, TERMINATED |
| CLG | CEILING MOUNTED |
| CPU | CENTRAL PROCESSING UNIT |
| DM | DESK MOUNTED |
| DSP | DIGITAL SIGNAL PROCESSOR |
| DVD | DIGITAL VIDEO DISC/VERSATILE DISC |
| (E) | EXISTING TO REMAIN |
| EC | ELECTRICAL CONTRACTOR COORDINATED |
| ELEV | ELEVATOR CONTRACTOR COORDINATED |
| (ER) | EXISTING TO BE RELOCATED |
| (ERC) | EXISTING TO BE RECABLED |
| FFE | FIXTURES, FURNITURE, AND EQUIPMENT |
| FB | FLOOR BOX MOUNTED |
| FC | FIRE ALARM CONTRACTOR COORDINATED |
| FF | FURNITURE FEEDTHROUGH |
| FM | FREQUENCY MODULATION |
| FPD | FLAT PANEL DISPLAY |
| GAS | GAS SERVICE / METER COORDINATED |
| GC | GENERAL CONTRACTOR COORDINATED |
| GPC | GENERAL PURPOSE COMPUTER |
| IR | INFRARED |
| IT | INFORMATION TECHNOLOGY |
| J | JUNCTION BOX |
| LAN | LOCAL AREA NETWORK |
| LIGH | LIGHTING CONTROL COORDINATED |
| LCD | LIQUID CRYSTAL DISPLAY |
| LTG | LIGHTING |
| MATV | MASTER ANTENNA TELEVISION |
| M | MULLION MOUNTED |
| MC | MECHANICAL CONTRACTOR COORDINATED |
| (N) | NEW |
| OAE | OR APPROVED EQUAL |
| OH | OVERHEAD |
| PA | PUBLIC ADDRESS |
| PC | PERSONAL COMPUTER |
| PDP | PLASMA DISPLAY PANEL |
| PH | PHASE |
| PT | POKE THROUGH MOUNTED |
| POS | POINT OF SALE |
| RCPT | RECEPTACLE |
| (R) | EXISTING TO BE REMOVED |
| (RG) | REMOVED BY GOVERNMENT |
| RF | RADIO FREQUENCY |
| (RL) | RELOCATED LOCATION |
| RU | RACK UNIT (TIA/EIA RACK) |
| SC | SECURITY CONTRACTOR COORDINATED |
| SPC | SPRINKLER CONTROL CONTRACTOR COORDINATED |
| TO | TELECOMMUNICATIONS OUTLET |
| TC | TELECOMMUNICATIONS CONTRACTOR COORDINATED |
| TV | TELEVISION |
| UC | UNDER COUNTER/CABINET MOUNTED |
| UG | UNDERGROUND |
| UHF | ULTRA HIGH FREQUENCY |
| UON | UNLESS OTHERWISE NOTED |
| UPS | UNINTERRUPTIBLE POWER SUPPLY. |
| USB | UNIVERSAL SERIAL BUS |
| VHF | VERY HIGH FREQUENCY |
| VP | VIDEO PROJECTOR |
| W | WALL PHONE |
| WAN | WIDE AREA NETWORK |
| WG | WIREGUARD |
| WP | WEATHERPROOF OR WATERPROOF |
| +XX" | MOUNTING HEIGHT OF 'XX' INCHES |
| (x) | DETAIL NOTE |
| (A) | REVISION (DELTA) TAG |

ALL INFORMATION SHOWN ON THIS SHEET IS COMMON TO ALL OPTIONS.

## PROJECT GENERAL NOTES

1. THE WORK DESCRIBED HEREIN SHALL BE REVIEWED, COORDINATED, AND FACILITATED BY ALL CONTRACTORS; THIS INCLUDES, BUT IS NOT LIMITED TO THE COMMUNICATIONS, SECURITY, LOW VOLTAGE, AV, ELECTRICAL, AND GENERAL CONTRACTORS. FURTHERMORE, ALL WORK SHALL BE COORDINATED WITH THE CONTRACTING OFFICER TO SCHEDULE THE SEQUENCE OF ALL WORK.
2. THESE DRAWINGS ARE DIAGRAMMATIC IN NATURE; NOT ALL EQUIPMENT, PARTS, OR ANCILLARY PIECES ARE SHOWN FOR A COMPLETE SYSTEM. CONTRACTORS ARE RESPONSIBLE FOR FURNISHING, INSTALLING, AND SUCCESSFULLY TESTING FULLY FUNCTIONING AND COMPLETE SYSTEMS, UNLESS OTHERWISE NOTED.
3. DEVICES SHOWN ON PLAN ARE DIAGRAMMATIC. THE PRECISE LOCATION OF DEVICES SHALL BE COORDINATED WITH THE FINAL ARCHITECTURAL DRAWINGS AND APPROVED VENDOR SHOP DRAWINGS PRIOR TO ROUGH-IN. EXACT DEVICE LOCATIONS SHALL ALLOW FOR COORDINATED, PRE-DRYWALL RELOCATION, WITHIN A 10-FOOT RADIUS AT NO CHARGE TO THE CONTRACTING OFFICER.
4. ALL WORK SHOWN HEREIN SHALL BE ASSUMED TO BE RESPONSIBILITY OF THE CONTRACTOR, UNLESS OTHERWISE NOTED. REFER TO RESPONSIBILITY MATRIX FOR FURTHER CLARIFICATION AND DELINEATION OF CONSTRUCTION RESPONSIBILITIES.
5. EACH CONTRACTOR SHALL BE RESPONSIBLE FOR COORDINATING WITH ALL OTHER TRADES AND THEIR ASSOCAITED DRAWINGS/SPECIFICATIONS.
6. CONTRACTORS SHALL MAINTAIN AN UPDATED DOCUMENT SET CONTAINING THE MOST RECENT CONSTRUCTION DOCUMENTS, ALL FIELD AND/OR DOCUMENT CHANGES, AND UPDATED 'AS BUILT' DRAWINGS THROUGHOUT CONSTRUCTION.
7. ALL MATERIAL AND EQUIPMENT SHALL BE ERECTED, INSTALLED, CONNECTED, CLEANED, ADJUSTED, TESTED, CONDITIONED, AND PLACED IN SERVICE IN ACCORDANCE WITH THE MANUFACTURER'S DIRECTIONS AND RECOMMENDATIONS.
8. ALL CUTTING, DRILLING, AND PATCHING OF MASONRY, STEEL, OR IRON WORK BELONGING TO THE BUILDING MUST BE COORDINATED WITH ELECTRICAL, GENERAL, AND STRUCTURAL CONTRACTOR, AS WELL AS THE CONTRACTING OFFICER. UNDER NO CIRCUMSTANCES MAY STRUCTURAL WORK BE CUT UNLESS WRITTEN INSTRUCTION IS PROVIDED BY CONTRACTING OFFICER.
9. ALL MATERIAL, EQUIPMENT, WIRING DEVICES, CABLING, ETC. SHALL BE NEW, UNLESS SPECIFICALLY INDICATED OTHERWISE.
10. PULL BOXES AND HANDHOLES SHALL NOT BE INSTALLED IN PLACE OF CONDUIT BENDS. CONTRACTOR SHALL INSTALL PULL BOXES AND HANDHOLES IN STRAIGHT SECTIONS OF CONDUIT.
11. ALL CONDUITS AND SLEEVES DESIGNATED FOR COMMUNICATIONS USE SHALL BE FIRE RATED TO MATCH OR EXCEED THE PENETRATED MATERIAL. ALL PENETRATIONS OF FIRE-RATED ASSEMBLIES SHALL BE FIRE-STOPPED WITH AN APPROVED SYSTEM AND SHALL COMPLY WITH UL 1479 OR UL 2079. THE F&T RATING OF THE FIRE-STOP SYSTEM SHALL NOT BE LESS THAN THE FIRE-RESISTIVE RATING OF THE ASSEMBLY PENETRATED. THIS SHALL INCLUDE PENETRATIONS INTO AND THROUGH WALLS, FLOORS, CEILINGS, AND ROOFS PER CODE AND AHJ.
12. THE ENTIRE CABLE PLANT SHALL BE TESTED AND CERTIFIED BY CONTRACTOR AS DETERMINED BY BICSI, NEC, AND TIA/EIA STANDARDS AT A MINIMUM, UNLESS OTHERWISE SPECIFIED.
13. ALL TELECOMMUNIATIONS SYSTEMS CONDUITS WHICH ARE INTERIOR, EXPOSED IN THE EXTERIOR, OR WITHIN A TUNNEL SYSTEM SHALL BE METALLIC CONDUIT AS INDICATED IN SPECIFICATIONS. UNDER NO CIRCUMSTANCES SHALL FLEXIBLE CONDUIT BE UTILIZED FOR PATHWAYS. ALL CONDUITS SHALL BE, AT MINIMUM, 1 INCH IN TRADE SIZE, UNLESS SPECIFICALLY AND OTHERWISE NOTED. WHEN CONDUIT IS EXPOSED, COORDINATE WITH CONTRACTING OFFICER FOR CONDUIT COLOR AND/OR PAINTING REQUIREMENTS.
14. ALL TELECOMMUNICATIONS SYSTEMS CONDUITS WHICH ARE UNDERGROUND SHALL BE RIGID POLYVINYL CHLORIDE CONDUIT (PVC), UNLESS OTHERWISE NOTED. REFER TO SPECIFICATIONS FOR MORE INFORMATION.
15. NO MORE THAN TWO (2) 90-DEGREE BENDS (OR A TOTAL OF 180-DEGREES) MAY BE INSTALLED BETWEEN PULLBOXES, HANDHOLES, OR PULL POINTS. A 4 INCH SQUARE, 2-1/8 INCH DEEP BACKBOX SHALL BE PROVIDED AND INSTALLED AS A PULL BOX FOR 1 INCH CONDUIT PATHWAYS, UNLESS OTHERWISE NOTED. FOR ALL OTHER PATHWAYS, REFER TO DRAWINGS AND SPECIFICATIONS FOR PULL BOX SIZE REQUIREMENTS.
16. FOR INTERIOR SPACES, NO MORE THAN 100 FEET OF PATHWAY MAY BE INSTALLED BETWEEN PULLBOXES OR PULL POINTS.
17. FOR EXTERIOR PATHWAY, NO MORE THAN 500 FEET OF PATHWAY MAY BE INSTALLED BETWEEN HANDHOLES OR PULL POINTS.
18. ALL MANUFACTURER CABLE BEND RADIUS REQUIREMENTS SHALL BE OBSERVED AND PROTECTED DURING THE INSTALLATION OF CABLING. FOR 2 INCH (TRADE SIZE) CONDUITS OR LESS, CONDUITS BEND RADIUS SHALL BE A MINIMUM OF 6 TIMES THE CONDUIT DIAMETER. FOR CONDUITS GREATER THAN 2 INCHES (TRADE SIZE), CONDUIT BEND RADIUS SHALL BE A MINIMUM OF 10 TIMES THE CONDUIT DIAMETER. ALL CONDUIT BENDS SHALL BE "SWEEPING" TYPE, NO BEND FITTINGS ARE PERMITTED.
19. FOR LOCATIONS WHERE BOTH POWER AND NETWORK OUTLETS ARE TO BE INSTALLED, NETWORK OUTLETS SHALL BE IMMEDIATELY ADJACENT AND ORTHOGONAL TO THE POWER OUTLET, UNLESS OTHERWISE NOTED. THE CONTRACTOR SHALL ENSURE THAT ALL NETWORK OUTLET LOCATIONS AND POWER OUTLET LOCATIONS ARE COORDINATED TO BE ADJACENT TO EACH OTHER WITH PROPER SPACING PER ARCHITECTURAL ELEVATIONS AND/OR NOTES.
20. BACK-TO-BACK MOUNTED OUTLETS IN WALLS ARE NOT PERMITTED, BOXES SHALL MAINTAIN A MINIMUM OF 12-INCHES OF SEPARATION. IN CASES WHERE OUTLET BOXES IN ADJACENT ROOMS ARE IN THE SAME STUD CAVITY, PROVIDE A LAYER OF EXPANDABLE SPRAY FOAM INSULATION AROUND EACH BOX CAVITY. FOR ALL OUTLETS/BACKBOXES INSTALLED INTO ACOUSTICALLY-RATED PARTITIONS (TYPICALLY OFFICES, CONFERENCE ROOM, CLASSROOMS, PRESENTATION SPACES, ETC.), PROVIDE ACOUSTICAL OUTLET BACKER PUTTY AND/OR PUTTY PADS. ACOUSTICAL RATING SHALL BE MAINTAINED PER ASTM C919 AND ASTM E497, AND BE TESTED TO UL 263 (ASTM E119) AND UL 1479 (ASTM E814) STANDARDS. WHEN INSTALLED INTO FIRE-RATED WALLS, PROVIDE FIRE-RATED PUTTY TO COVER THE BACK OF EACH OUTLET.
21. ALL CABLING SHALL BE SUPPORTED BY DEDICATED INFRASTRUCTURE AND EQUIPMENT. THIS INCLUDES CABLE TRAY, CONDUIT, AND MAUNFACTURED SLEEVES. CABLE SHALL NOT BE SUPPORTED BY OTHER TRADE'S EQUIPMENT OR ARCHITECTURAL ELEMENTS. THIS INCLUDES, BUT IS NOT LIMITED TO, HVAC DUCTS, PIPING, ELECTRICALLY DEDICATED CONDUIT, CEILING TILE/GRID, ETC.
22. CABLE TRAY, CONDUIT, SLEEVES, PENETRATIONS, AND ANY OTHER PATHWAYS USED FOR ROUTING TELECOMMUNICATIONS (NETWORK/VOICE/CATV) CABLING SHALL BE DEDICATED TO TELECOMMUNICATION CABLING. NO OTHER TRADE, SERVICE, OR SYSTEM SHALL UTILIZE SAME PATHWAY.
23. CABLE TRAY SHALL ONLY BE INSTALLED WHERE INDICATED AND ABOVE ACCESSIBLE CEILINGS. EXPOSED OR NON-ACCESSIBLE CABLING IS NOT ACCEPTABLE; CONDUIT PATHWAY SHALL BE PROVIDED AND INSTALLED FOR ROUTING OF CABLING FROM EACH OUTLET BACK TO THE NEAREST CABLE TRAY OR TELECOMMUNICATION SPACE. CONDUIT SHALL BE UTILIZED FOR CABLE ROUTING THROUGH NON-ACCESSIBLE CEILINGS, OPEN CEILING ENVIRONMENTS, AND/OR HIGH CEILING ENVIRONMENTS WHERE THE CABLE IS BELOW THE CEILING, UNLESS OTHERWISE NOTED.
24. ALL INTERIOR CABLING SHALL BE PLENUM RATED UNLESS SPECIFICALLY AND OTHERWISE NOTED.
25. ALL BACKBOX INSTALLATIONS SHALL BE COORDINATED TO ENSURE THAT MUD RING/EXTENSION RING IS MOUNTED FLUSH WITH WALL SURFACE TO ALLOW PROPER MOUNTING OF FACEPLATE(S).
26. FOR ALL OPEN ENDED CONDUIT NOT TERMINATED INTO A BOX OR ENLCOSURE, CONTRACTOR SHALL PROVIDE AND INSTALL A NYLON BUSHING AND CONDUIT END.
27. A PULL STRING SHALL BE PROVIDED FOR ALL CONDUIT PATHWAYS. FOR EXTERIOR PATHWAYS, PULL STRING SHALL BE MULE TAPE WITH A MINIMUM TENSION STRENGTH OF 2200 LBS. FOR INTERIOR PATHWAYS, A STRING OR MULE TAPE WITH A MINIMUM TENSION STRENGTH OF 200 LBS. SHALL BE INSTALLED. ALL PULL STRINGS SHALL BE TIED OFF WITH 3 FEET OF SLACK ON EACH END.

LICENSED PROFESSIONAL ENGINEER — 05/21/2026 — JEFF MULLIKIN — P25384 — WASHINGTON — REGISTERED PROFESSIONAL ENGINEER

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| A/E FIRM: | |
| PRIME: | OTAK, INC. |
| | REDMOND, WA |
| SUBCONTRACTOR | AE DESIGN |
| | DENVER, CO |
| DESIGNED: | JR/JM/TW |
| CADD | AE DESIGN |
| TECH REVIEW: | MRS/ESN |
| DATE: | 5/29/2026 |
| SUB SHEET NO. | T000 |

TITLE OF SHEET: TECHNOLOGY COVER SHEET

EFFIGY MOUNDS NATIONAL MONUMENT

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|---|---|
| DRAWING NO. | 394 194248 |
| PMIS/PKG NO. | 318539 |
| SHEET | 127 OF 142 |

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# MATRIX OF CONSTRUCTION RESPONSIBILITIES

**THE RESPONSIBILITIES LISTED HEREIN ARE PROVIDED AS A RECOMMENDATION AND DO NOT SUPERSEDE OR REPLACE ANY CONTRACTS, OR OTHERWISE DEFINED RESPONSIBILITIES, BETWEEN THE DESIGNATED PARTIES. IN ADDITION, THE INFORMATION IS MEANT TO INDICATE GENERAL RESPONSIBILITY FOR A SCOPE OF WORK AND IN NO WAY DISALLOWS THE RESPONSIBLE PARTY TO SUBCONTRACT THE SCOPE.**

| RESPONSIBLE PARTY | GENERAL CONTRACTOR | | ELECTRICAL CONTRACTOR | | TELECOM CONTRACTOR | | LOW VOLTAGE CONTRACTOR | | AUDIO-VIDEO CONTRACTOR | | SECURITY CONTRACTOR | | GOVERNMENT | | NOTES |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| SCOPE OF WORK | FURNISH | INSTALL | FURNISH | INSTALL | FURNISH | INSTALL | FURNISH | INSTALL | FURNISH | INSTALL | FURNISH | INSTALL | FURNISH | INSTALL | |
| BUDGET OF WORK | CONSTRUCTION | | CONSTRUCTION | | CONSTRUCTION | | CONSTRUCTION | | CONSTRUCTION | | CONSTRUCTION | | FIX-FURN-EQUIP | | |
| **DEMOLITION** | GC | | EC | | TC | | LV | | AV | | SC | | G | | |
| GENERAL | | | | | | | | | | | | | | | 8 |
| **SITE** | GC | | EC | | TC | | LV | | AV | | SC | | G | | |
| INCOMING SERVICE PROVIDER CABLING & COORDINATION | | | | | | | | | | | | | X | X | 1 |
| CAMPUS / LAN CABLING & COORDINATION | | | | | X | X | | | | | | | | | |
| HANDHOLE / MAINTENANCE HOLE | | | X | X | | | | | | | | | | | |
| EXTERIOR CONDUIT PATHWAY / DUCTBANK | | | X | X | | | | | | | | | | | |
| TRENCHING | | | X | X | | | | | | | | | | | |
| **INTERIOR INFRASTRUCTURE** | GC | | EC | | TC | | LV | | AV | | SC | | G | | |
| GROUNDING & BONDING | | | X | X | X | X | | | | | | | | | 3, 4 |
| INTERIOR CONDUIT PATHWAY | | | X | X | | | | | | | | | | | |
| BACKBOX / JUNCTION BOX | | | X | X | | | | | | | | | | | |
| FLOOR BOX / POKE THROUGH | | | X | X | | X | | | | | | | | | 5 |
| SLEEVE / CONDUIT PENETRATIONS | | | X | X | X | X | | | | | | | | | 6, 7 |
| **TELECOMMUNICATIONS** | GC | | EC | | TC | | LV | | AV | | SC | | G | | |
| PLYWOOD BACKBOARD | X | X | | | | | | | | | | | | | |
| LADDER RACK / LADDER RUNWAY / ACCESSORIES | | | | | X | X | | | | | | | | | |
| RACK / FRAME / CABINET (TELECOM) | | | | | X | X | | | | | | | | | |
| WIRE MANAGER | | | | | X | X | | | | | | | | | |
| FIBER PATCH PANEL | | | | | X | X | | | | | | | | | |
| COPPER PATCH PANEL | | | | | X | X | | | | | | | | | |
| POWER DISTRIBUTION UNIT (PDU) | | | | | X | X | | | | | | | | | |
| UNINTERRUPTIBLE POWER SUPPLY (UPS) | | | | | | | | | | | | | X | X | |
| MISCELLANEOUS RACK COMPONENTS (DRAWER, SHELF, ETC.) | | | | | X | X | | | | | | | | | |
| BACKBONE CABLING SYSTEM (NETWORK, VOICE, CATV) | | | | | X | X | | | | | | | | | |
| HORIZONTAL CABLING SYSTEM (NETWORK, VOICE, CATV) | | | | | X | X | | | | | | | | | |
| FACEPLATE / JACK / SURFACE MOUNT BOX | | | | | X | X | | | | | | | | | |
| PATCH CABLE (INTERIOR TO TELECOMMUNICATIONS ROOM) | | | | | X | | | | | | | | | X | |
| PATCH CABLE (END DEVICE / OUTLET) | | | | | X | X | | X | | X | | X | | X | 2 |
| LABELLING | | | | | X | X | | | | | | | | | |
| MEDIA CONVERTERS / COPPER EXTENDERS FOR NETWORK CABLING | | | | | X | X | | | | | | | | | |
| WIRELESS ACCESS POINT (WAP) | | | | | | | | | | | | | X | X | |
| BUILDING NETWORK (LAN) EQUIPMENT (SWITCH, HEADEND, ETC.) | | | | | | | | | | | | | X | X | 9 |
| PERIPHERAL EQUIPMENT (PHONE, PRINTER, PC, ETC.) | | | | | | | | | | | | | X | X | |
| **LOW VOLTAGE (LV)** | GC | | EC | | TC | | LV | | AV | | SC | | G | | |
| PAGING/PUBLIC ADDRESS (HEADEND, END DEVICES, CABLES, ETC.) | | | | | | | X | X | | | | | | | |
| **AUDIO VISUAL (AV)** | GC | | EC | | TC | | LV | | AV | | SC | | G | | |
| RACK / FRAME / CABINET (AV) | | | | | | | | | X | X | | | | | |
| AMPLIFIER / CONTROLLER / PROCESSOR / MATRIX | | | | | | | | | X | X | | | | | |
| MICROPHONE | | | | | | | | | X | X | | | | | |
| FACEPLATE / JACK | | | | | | | | | X | X | | | | | |
| TOUCH PANEL / USER INTERFACE / SCHEDULER | | | | | | | | | X | X | | | | | |
| CABLING (NON-NETWORK) | | | | | | | | | X | X | | | | | |
| SPEAKER | | | | | | | | | X | X | | | | | |
| DISPLAY & MOUNT | | | | | | | | | X | X | | | | | |
| DISPLAY BACKING | X | X | | | | | | | | | | | | | |
| PROJECTOR | | | | | | | | | X | X | | | | | |
| PROJECTOR SCREEN | | | | | | | | | X | X | | | | | |
| AUDIO VIDEO DEDICATED PERSONAL COMPUTER (PC) | | | | | | | | | | | | | X | X | |
| AUDIO VIDEO NETWORK SWITCH(ES) | | | | | | | | | X | X | | | | | |
| AUDIO VIDEO NETWORK CABLING | | | | | | | | | X | X | | | | | 9 |
| BUILDING NETWORK (LAN) CABLING TO AUDIO VIDEO SYSTEM(S) | | | | | X | X | | | | | | | | | 9 |
| **SECURITY - ACCESS CONTROL SYSTEM (ACS)** | GC | | EC | | TC | | LV | | AV | | SC | | G | | |
| ACS HEADEND / CONTROLLER / PANEL | | | | | | | | | | | X | X | | | |
| ACS SOFTWARE, PROGRAMMING, & INTEGRATION | | | | | | | | | | | X | X | | | |
| ACS CABLING & TERMINATIONS (TO PANELS, DEVICES, & HARDWARE) | | | | | | | | | | | X | X | | | |
| INTEGRATION WITH, AND CONNECTION TO, DOOR HARDWARE | | | | | | | | | | | X | X | | | |
| CARD READER / KEYPAD AND LICENSE (AUTHENTICATION DEVICE) | | | | | | | | | | | X | X | | | |
| REQUEST TO EXIT (WHEN NOT INTEGRAL TO DOOR HARDWARE) | | | | | | | | | | | X | X | | | |
| DOOR POSITION SWITCH | | | | | | | | | | | X | X | | | |
| DOOR HARDWARE / COMPONENTS | X | X | | | | | | | | | | | | | |
| BUILDING NETWORK (LAN) CABLING TO ACS HEADEND | | | | | X | X | | | | | | | | | 9 |
| **SECURITY - VIDEO SURVEILLANCE SYSTEM (VSS)** | GC | | EC | | TC | | LV | | AV | | SC | | G | | |
| VSS NETWORK VIDEO RECORDER (NVR) | | | | | | | | | | | X | X | | | |
| VSS SOFTWARE, PROGRAMMING, & INTEGRATION | | | | | | | | | | | X | X | | | |
| VSS DEDICATED NETWORK SWITCH | | | | | | | | | | | X | X | | | 9 |
| CAMERA AND LICENSE | | | | | | | | | | | X | X | | | |
| CABLE TO VIDEO CAMERA | | | | | X | X | | | | | | | | | |
| BUILDING NETWORK (LAN) CABLING TO VSS HEADEND | | | | | X | X | | | | | | | | | 9 |
| **SECURITY - INTRUSION DETECTION SYSTEM (IDS)** | GC | | EC | | TC | | LV | | AV | | SC | | G | | |
| IDS HEADEND / CONTROLLER / PANEL / INTERFACE | | | | | | | | | | | X | X | | | |
| IDS SOFTWARE, PROGRAMMING, & INTEGRATION | | | | | | | | | | | X | X | | | |
| IDS CABLING AND CONNECTION TO DEVICE(S)/PANEL(S) | | | | | | | | | | | X | X | | | |
| MOTION SENSOR | | | | | | | | | | | X | X | | | |
| AUTO-DIALLER & DIAL DESTINATION COORDINATION | | | | | | | | | | | X | X | | | |
| BUILDING NETWORK (LAN) CABLING TO IDS HEADEND | | | | | X | X | | | | | | | | | 9 |

**NOTES:**

1. CONTRACTOR SHALL COORDINATE WITH CONTRACTING OFFICER REGARDING TIMELINE OF INSTALLATION AND REQUIREMENTS FOR INSTALLATION TO ENSURE A TIMELY INSTALLATION.

2. THE PARTY RESPONSIBLE FOR INSTALLING THE END DEVICE (PC, CAMERA, WAP, ETC.) SHALL BE RESPONSIBLE FOR INSTALLING THE END-OF-RUN PATCH CABLE. AFTER INSTALLATION, VERIFICATION OF OPERABILITY IS REQUIRED.

3. THE ELECTRICAL CONTRACTOR SHALL i) EXTEND THE BUILDING GROUND TO EACH TELECOMMUNICATION SPACE ii) PROVIDE AND INSTALL THE BUSBAR(S), GROUNDING CABLES, AND ASSOCIATED EQUIPMENT, iii) AND ENSURE EACH TELECOMMUNICATION SPACE HAS PROPER ACCESS TO BUILDING GROUND THROUGH THE LOCAL BUSBAR AS SHOWN IN THE DRAWINGS.

4. FOR ALL DEVICES, EQUIPMENT, PATHWAY, AND OTHER SUCH MATERIAL REQUIRED TO BE GROUNDED, THE CONTRACTOR/PARTY, WITH WHICH THE DEVICE, EQUIPMENT, PATHWAY OR OTHER SUCH MATERIAL WAS INSTALLED BY, SHALL BE RESPONSIBLE FOR ITS PROPER BONDING AND GROUNDING.

5. THE ELECTRICAL CONTRACTOR SHALL BE RESPONSIBLE FOR THE PROVISION AND INSTALLATION OF ALL FLOOR BOXES AND POKE THROUGHS, AS WELL AS ALL CONDUIT/PATHWAY REQUIREMENTS PERTAINING TO IT, REGARDLESS IF THERE IS POWER CABLING INCLUDED AT DEVICE. THE TELECOM CONTRACTOR SHALL COORDINATE WITH DRAWINGS AND ELECTRICAL CONTRACTOR TO ENSURE LOW VOLTAGE INFRASTRUCTURE AND CABLING REQUIREMENTS ARE MET AND SHALL PROVIDE AND INSTALL ALL CABLING AND FACEPLATE/TERMINATION EQUIPMENT PERTAINING TO DEVICE.

6. FOR ALL PENETRATIONS SHOWN IN DRAWINGS, THE ELECTRICAL CONTRACTOR SHALL PROVIDE AND INSTALL. THE ELECTRICAL CONTRACTOR SHALL ENSURE THE EXTERIOR OF EACH PENETRATION IS SEALED TO EXCLUDE INSECTS, RODENTS, SNAKES, AND OTHER PESTS.

7. ALL PENETRATIONS NOT SHOWN ON DRAWINGS, BUT REQUIRED FOR CABLING INSTALLATION, SHALL BE PROVIDED AND INSTALLED BY THE CABLING INSTALLATION CONTRACTOR/TEAM. FOR EXTERIOR PENETRATIONS, THE CABLE INSTALLATION CONTRACTOR/TEAM SHALL ENSURE THE EXTERIOR OF EACH PENETRATION IS SEALED TO EXCLUDE INSECTS, RODENTS, SNAKES, AND OTHER PESTS.

8. IT IS ACCEPTABLE FOR THE CONTRACTOR TO REUSE AN EXISTING DEVICE WITH CONTRACTING OFFICER APPROVAL. COORDINATE WITH CONTRACTING OFFICER PRIOR TO ANY NEW DEVICE PROCUREMENT TO RECORD ALL DEVICES THAT MAY BE REUSED. FOR ALL DEVICES SCHEDULED FOR REUSE THAT REPLACE AN ITEM SCHEDULED FOR NEW, A CREDIT SHALL BE GIVEN TO CONTRACTING OFFICER FOR ITEM REPLACEMENT.

9. THE BUILDING NETWORK IS THE LOCAL AREA NETWORK (LAN) OF THE BUILDING AND CONNECTS DIRECTLY TO THE INCOMING SERVICE PROVIDER; IT PROVIDES GENERAL ACCESS TO THE WIDE AREA NETWORK (WAN). OTHER SYSTEMS MAY UTILIZE NETWORK EQUIPMENT TO SUPPORT THEIR SPECIFIC NEEDS, HOWEVER, THIS EQUIPMENT IS SEPARATE AND DISTINCT FROM THE BUILDING NETWORK (LAN) AND ASSOCIATED EQUIPMENT.

## SECURITY LEGEND

- CAM# — CAMERA - SINGLE LENSE
- CAM# — CAMERA - DUAL LENSE
- CAM# — CAMERA - TRIPLE LENSE
- CAM# — CAMERA - QUAD LENSE
- MD# — CEILING MOUNTED MOTION DETECTOR - DIRECTIONAL
- MD# — CEILING MOUNTED MOTION DETECTOR - OMNI-DIRECTIONAL
- XX# XX# — SECURITY DEVICES
  - CR CARD READER DEVICE
  - IC INTERCOM (REMOTE)
  - IM INTERCOM (MASTER)
  - KI KEYPAD INTERFACE - INTRUSION SYSTEM
  - PB PANIC BUTTON
  - GB GLASS BREAK
- XXX — SECURITY PANELS
  - ACP ACCESS CONTROL PANEL
  - IDP INTRUSION DETECTION PANEL
  - DCP DOOR CONTROL PANEL
- DOOR SECURITY TAGS
  - C XXX — CONTROLLED DOOR: RE ACCESS CONTROL SCHEDULE AND DIVISION 08 SPECIFICATIONS FOR COORDINATION OF DOOR HARDWARE. (XXX - DOOR NO.)
  - M XXX — MONITORED DOOR: RE ACCESS CONTROL SCHEDULE AND DIVISION 08 SPECIFICATIONS FOR COORDINATION OF DOOR HARDWARE. (XXX - DOOR NO.)
  - S XXX — SCHEDULED DOOR: RE ACCESS CONTROL SCHEDULE AND DIVISION 08 SPECIFICATIONS FOR COORDINATION OF DOOR HARDWARE. (XXX - DOOR NO.)
  - O XXX — OVERHEAD DOOR: RE ACCESS CONTROL SCHEDULE AND DIVISION 08 SPECIFICATIONS FOR COORDINATION OF DOOR HARDWARE. (XXX - DOOR NO.)

ALL INFORMATION SHOWN ON THIS SHEET IS COMMON TO ALL OPTIONS.

## AUDIO-VISUAL LEGEND

- PROJECTION SCREEN - CEILING MOUNTED
- PROJECTION SCREEN - WALL MOUNTED
- DISPLAY - CEILING MOUNTED
- DISPLAY - WALL MOUNTED
- P — VIDEO PROJECTOR
- HP — VIDEO PROJECTOR - WALL MOUNTED SHORT THROW
- AV — AUDIOVISUAL CAMERA - CEILING MOUNTED
- AV — AUDIOVISUAL CAMERA - WALL MOUNTED
- XX-XX XX-XX — AUDIO-VISUAL DEVICE - REFER TO SPECIAL SYMBOLS LEGEND FOR MOUNTING REQUIREMENTS
  - AV AUDIO VIDEO DEVICE
  - BB FLAT PANEL BACK BOX
  - BT BLUETOOTH RECIEVER
  - CP CONTROL PANEL
  - FP FLAT PANEL
  - M DESKTOP MICROPHONE
  - MC CEILING MICROPHONE
  - PB PLENUM BOX
  - PJ PROJECTOR
  - PS PROJECTOR SCREEN
  - PTZ PAN-TILT-ZOOM CAMERA
  - RS ROOM SCHEDULER
  - S SPEAKER
  - SW PROJECTOR SWITCH
  - TP TOUCH PANEL
  - VC VOLUME CONTROL
  - WM WIRELESS MICROPHONE ANTENNA
- LOUDSPEAKER - CEILING MOUNTED
- LOUDSPEAKER - WALL MOUNTED
- PAGING SPEAKER - CEILING MOUNTED
- PAGING SPEAKER - WALL MOUNTED

LICENSED PROFESSIONAL ENGINEER — 05/21/2026 — JEFF MULLIKIN P25384 — IOWA

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|---|---|
| A/E FIRM: | |
| PRIME: | OTAK, INC. |
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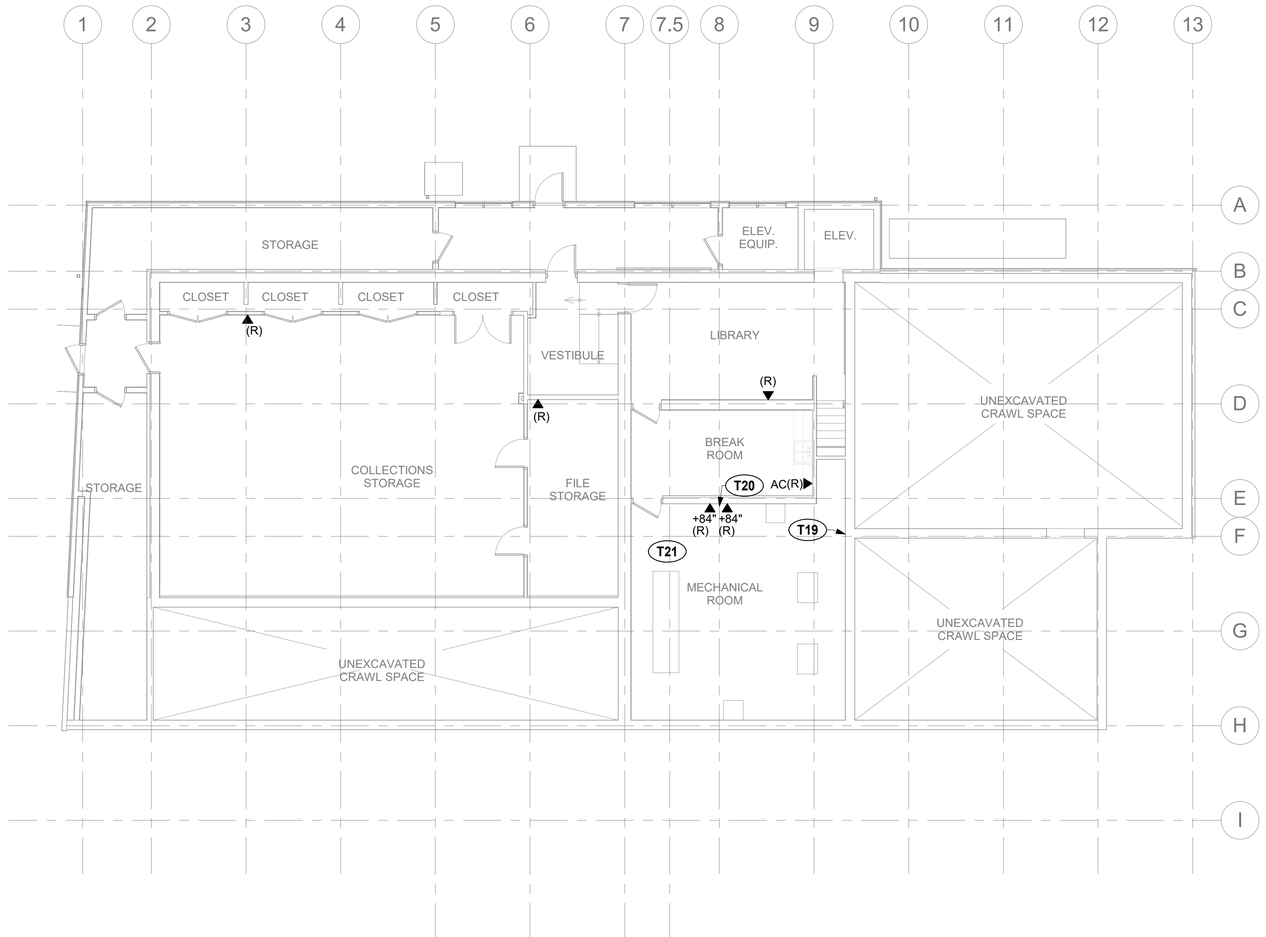
| | DENVER, CO |
| DESIGNED: | JR/JM/TW |
| DRAWN: | AE DESIGN |
| TECH REVIEW: | MRS/ESN |
| DATE: | 5/29/2026 |
| SUB SHEET NO. | T002 |
| TITLE OF SHEET | TECHNOLOGY RESPONSIBILITY MATRIX |
| | EFFIGY MOUNDS NATIONAL MONUMENT |
| DRAWING NO. | 394 194248 |
| PMIS/PKG NO. | 318539 |
| SHEET | 128 OF 142 |




1 TD100 BASEMENT TECHNOLOGY DEMOLITION PLAN

SCALE A

## TECHNOLOGY DEMO GENERAL NOTES

1. CONTRACTOR SHALL REFER TO LEGEND, NOTES, SCHEDULES, DETAILS, AND SPECIFICATIONS FOR MORE INFORMATION.
2. THE BASIS OF THESE DRAWINGS WERE SITE OBSERVATIONS, ORIGINAL BUILDING DRAWINGS AND VARIOUS OTHER SOURCES. EVERY ATTEMPT HAS BEEN MADE TO DOCUMENT THE ACTUAL CONDITIONS. HOWEVER, THE CONTRACTOR SHALL CAREFULLY EXAMINE THE CONTRACT DOCUMENTS AND THOROUGHLY BECOME FAMILIAR WITH THE BUILDING/NPS STANDARDS RELATING TO THE WORK. FAILURE TO DO SO WILL NOT RELIEVE THE CONTRACTOR OF THE OBLIGATIONS OF THE CONTRACT.
3. THE CONTRACTOR SHALL NOTIFY THE CONTRACTING OFFICER OF ANY MATERIALS OR APPARATUS BELIEVED TO BE INADEQUATE, UNSUITABLE, IN VIOLATION OF LAWS, ORDINANCES, RULES OR REGULATIONS OF AUTHORITIES HAVING JURISDICTION. CONTRACTOR IS RESPONSIBLE FOR IDENTIFYING ITEMS TO BE RETURNED TO THE CONTRACTING OFFCIER FOR REUSE. REUSABLE ITEMS DAMAGED OR DISCARDED SHALL BE REPLACED AT NO ADDITIONAL COST TO CONTRACTING OFFICER.
4. THE CONTRACTOR SHALL INCLUDE IN THEIR COST THE REMOVAL OF ALL EXISTING ELECTRICAL AND TELECOMMUNICATIONS DEVICES, CONDUITS, FIXTURES AND EQUIPMENT. DISCARD ALL EQUIPMENT AS REQUIRED, UNLESS NOTED OTHERWISE.
5. TELECOMMUNICATION RACK MOUNTED EQUIPMENT, WIRELESS ACCESS POINTS (WAPs), AND SPECIFICALLY NOTED EQUPMENT SHALL BE PROTECTED AND/OR REMOVED AND RETURNED TO THE CONTRACTING OFFICER. REFER TO INDIVIDUAL REFERENCES FOR MORE INFORMATION. FOR ALL EXISTING SECURITY, TELECOMMUNICATION, AND AUDIO VIDEO EQUIPMENT NOT SPECIFICALLY NOTED OTHERWISE, CABLING AND CONDUIT SHALL BE DEMOLISHED TO ORIGIN, EQUIPMENT SHALL BE RETURNED TO CONTRACTING OFFICER.
6. ALL WORK MUST BE COORDINATED AND SCHEDULED WITH THE CONTRACTING OFFICER AND OCCUPANTS OF THIS BUILDING SO AS TO PROVIDE THE LEAST AMOUNT OF DISRUPTION OF BUILDING ACTIVIES AS POSSIBLE. THE DEMOLITION OF SOME DEVICES MAY INTERRUPT CONNECTIVITY/OPERATION TO DEVICES OUT OF SCOPE. IT IS THE CONTRACTOR'S RESPONSIBILITY TO PROTECT, MAINTAIN, AND/OR RE-ESTABLISH CONNECTIVITY/OPERATION TO ALL DEVICES OUT OF SCOPE.
7. IT IS NOT ACCEPTABLE FOR CABLES, PATHWAY, OR EQUIPMENT TO BE ABANDONDED IN PLACE, UNLESS OTHERWISE NOTED. WHERE DEVICES ARE INDICATED FOR DEMOLITION, CABLING SHALL BE DEMOLISHED BACK TO ORIGIN, DEVICES SHALL BE REMOVED, AND DEDICATED PATHWAYS SHALL BE DEMOLISHED.

## KEYNOTE LEGEND - NPS

| KEY VALUE | KEYNOTE TEXT |
|---|---|
| T19 | CONTRACTOR SHALL DEMOLISH THE EXISTING WALL PANEL, AND ASSOCIATED PATHWAY/CONDUIT, AT THIS LOCATION. IT IS NO LONGER IN USE AND IS A LEGACY TELEPHONE EQUIPMENT HOUSING. IT IS LABELED 'TELEPHONE ALPINE COMM.' IN THE FIELD. |
| T20 | THE OUTLETS AT THIS LOCATION SUPPORT A SWITCH ON A SHELF. THE SHELF SHALL BE DEMOLISHED, THE SWITCH SHALL BE REMOVED BY GOVERNMENT PRIOR TO CONSTRUCTION. |
| T21 | AT THE APPROXIMATE LOCATION INDICATED, THERE IS A CEILING MOUNTED 100-BLOCK WITH SIX (6) CABLES ROUTED THROUGH IT. THE 110-BLOCK AND ALL CABLES SHALL BE DEMOLISHED TO ORIGIN. |

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.

SCALE: A 8 0 8 16 SCALE OF FEET

LICENSED PROFESSIONAL ENGINEER JEFF MULLIKIN P25384 05/21/2026

| | |
|---|---|
| A/E FIRM: | |
| PRIME: | OTAK, INC. |
| | REDMOND, WA |
| SUBCONTRACTOR | AE DESIGN |
| | DENVER, CO |
| DESIGNED: | JR/JM/TW |
| CADD | AE DESIGN |
| TECH REVIEW: | MRS/ESN |
| DATE: | 5/29/2026 |
| SUB SHEET NO. | TD100 |
| TITLE OF SHEET | BASEMENT TECHNOLOGY DEMO PLANS |
| | EFFIGY MOUNDS NATIONAL MONUMENT |
| DRAWING NO. | 394 194248 |
| PMIS/PKG NO. | 318539 |
| SHEET | 129 OF 142 |

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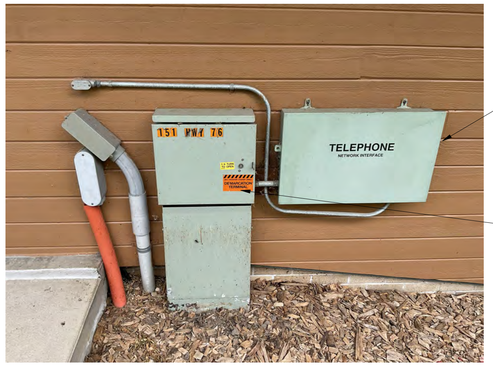
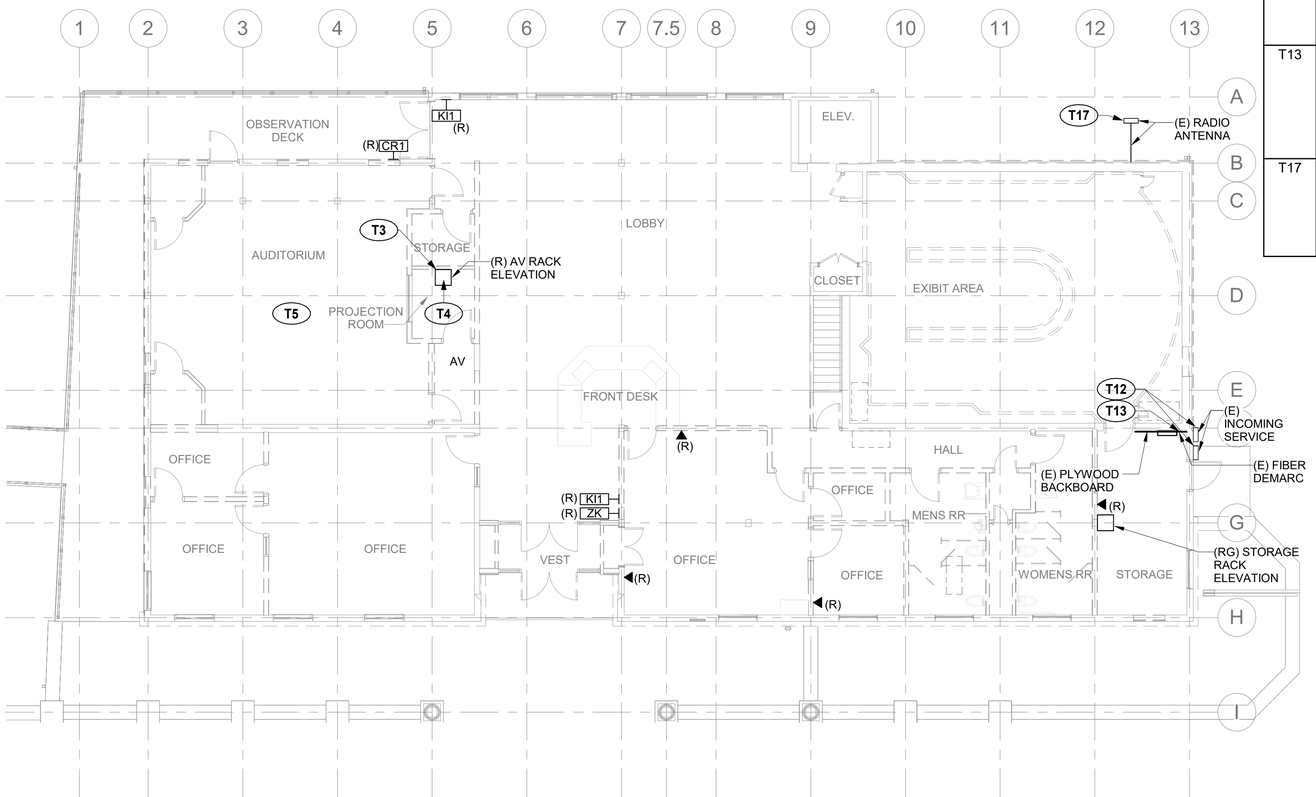



2 TD101 EXTERIOR INCOMING SERVICE EQUIPMENT




1 TD101 FIRST FLOOR TECHNOLOGY DEMOLITION PLAN
SCALE A

## KEYNOTE LEGEND - NPS

| KEY VALUE | KEYNOTE TEXT |
|---|---|
| T3 | AUDIO VIDEO RACK AND ASSOCIATED CABLING SHALL BE REMOVED. CONTRACTOR SHALL RELOCATE ASSISTIVE LISTENING SYSTEM (ALS) DEVICE FROM THE EXISTING AUDIO VIDEO RACK TO THE NEW TELECOMMUNICATIONS RACK. THE EXISTING VHS/DVD PLAYER SHALL BE RETURNED TO THE CONTRACTING OFFICER. |
| T4 | EXISTING AUDIO VIDEO ROOM SHALL BE DEMOLISHED. CONTRACTOR SHALL RETURN ALL REMOVED AUDIO VISUAL DEVICES TO THE CONTRACTING OFFICER |
| T5 | EXISTING AUDITORIUM AUDIO VIDEO SYSTEM SHALL BE REMOVED. CONTRACTOR SHALL RETURN ALL EQUIPMENT, END DEVICES AND CONTROL DEVICES TO THE CONTRACTING OFFICER. |
| T12 | THIS EQUIPMENT SUPPORTS THE INCOMING SERVICE FROM THE PROVIDER AND THE BACKBONE DISTRIBUTION TO OTHER LOCATIONS ON SITE. THIS EQUIPMENT SHALL BE PROTECTED IN PLACE. ALL DEMOLITION WORK ATTRIBUTED TO THE SURROUNDING SIDING AND/OR GRADING SHALL BE COORDINATED TO ENSURE ALL PATHWAY, CABLING, AND ENCLOSURES MAINTAIN OPERATION. IT IS ACCEPTABLE TO DETACH EQUIPMENT FROM WALL FOR THE REMOVAL OF SIDING AND THEN RE-AFFIX THE EQUIPMENT TO THE WALL. COORDINATE WITH CONTRACTING OFFICER PRIOR TO WORK COMMENCEMENT. REFER TO VIEWPORT 2 ON SHEET TD101 FOR EXISTING CONDITION. |
| T13 | EXISTING SERVICE PROVIDER EQUIPMENT AND CABLING SHALL BE PROTECTED IN PLACE. DURING DEMOLITION AND CONSTRUCTION, ENSURE THAT THE FIBER CONNECTION IS MAINTAINED AND ACTIVE BETWEEN THIS LOCATION AND THE MAINTENANCE BUILDING. CONTRACTOR SHALL PROVIDE TEMPORARY POWER AS NEEDED TO ENSURE FIBER CONNECTION IS MAINTAINED DURING CONSTRUCTION. |
| T17 | EXISTING TO REMAIN TOWER AND ANTENNA. CONTRACTOR SHALL PROTECT IN PLACE THE TOWER, THE ANTENNA, AND ANY CABLING ASSOCIATED WITH ANTENNA EQUIPMENT. EXISTING CABLING ROUTES FROM ANTENNA TOWER IN TO THE CEILING SPACE OF THE BUILDING NEAR THE TOWER ANCHOR POINT TO THE BUILDING. |

## TECHNOLOGY DEMO GENERAL NOTES

1. CONTRACTOR SHALL REFER TO LEGEND, NOTES, SCHEDULES, DETAILS, AND SPECIFICATIONS FOR MORE INFORMATION.
2. THE BASIS OF THESE DRAWINGS WERE SITE OBSERVATIONS, ORIGINAL BUILDING DRAWINGS AND VARIOUS OTHER SOURCES. EVERY ATTEMPT HAS BEEN MADE TO DOCUMENT THE ACTUAL CONDITIONS. HOWEVER, THE CONTRACTOR SHALL CAREFULLY EXAMINE THE CONTRACT DOCUMENTS AND THOROUGHLY BECOME FAMILIAR WITH THE BUILDING/NPS STANDARDS RELATING TO THE WORK. FAILURE TO DO SO WILL NOT RELIEVE THE CONTRACTOR OF THE OBLIGATIONS OF THE CONTRACT.
3. THE CONTRACTOR SHALL NOTIFY THE CONTRACTING OFFICER OF ANY MATERIALS OR APPARATUS BELIEVED TO BE INADEQUATE, UNSUITABLE, IN VIOLATION OF LAWS, ORDINANCES, RULES OR REGULATIONS OF AUTHORITIES HAVING JURISDICTION. CONTRACTOR IS RESPONSIBLE FOR IDENTIFYING ITEMS TO BE RETURNED TO THE CONTRACTING OFFCIER FOR REUSE. REUSABLE ITEMS DAMAGED OR DISCARDED SHALL BE REPLACED AT NO ADDITIONAL COST TO CONTRACTING OFFICER.
4. THE CONTRACTOR SHALL INCLUDE IN THEIR COST THE REMOVAL OF ALL EXISTING ELECTRICAL AND TELECOMMUNICATIONS DEVICES, CONDUITS, FIXTURES AND EQUIPMENT. DISCARD ALL EQUIPMENT AS REQUIRED, UNLESS NOTED OTHERWISE.
5. TELECOMMUNICATION RACK MOUNTED EQUIPMENT, WIRELESS ACCESS POINTS (WAPs), AND SPECIFICALLY NOTED EQUPMENT SHALL BE PROTECTED AND/OR REMOVED AND RETURNED TO THE CONTRACTING OFFICER. REFER TO INDIVIDUAL REFERENCES FOR MORE INFORMATION. FOR ALL EXISTING SECURITY, TELECOMMUNICATION, AND AUDIO VIDEO EQUIPMENT NOT SPECIFICALLY NOTED OTHERWISE, CABLING AND CONDUIT SHALL BE DEMOLISHED TO ORIGIN, EQUIPMENT SHALL BE RETURNED TO CONTRACTING OFFICER.
6. ALL WORK MUST BE COORDINATED AND SCHEDULED WITH THE CONTRACTING OFFICER AND OCCUPANTS OF THIS BUILDING SO AS TO PROVIDE THE LEAST AMOUNT OF DISRUPTION OF BUILDING ACTIVIES AS POSSIBLE. THE DEMOLITION OF SOME DEVICES MAY INTERRUPT CONNECTIVITY/OPERATION TO DEVICES OUT OF SCOPE. IT IS THE CONTRACTOR'S RESPONSIBILITY TO PROTECT, MAINTAIN, AND/OR RE-ESTABLISH CONNECTIVITY/OPERATION TO ALL DEVICES OUT OF SCOPE.
7. IT IS NOT ACCEPTABLE FOR CABLES, PATHWAY, OR EQUIPMENT TO BE ABANDONDED IN PLACE, UNLESS OTHERWISE NOTED. WHERE DEVICES ARE INDICATED FOR DEMOLITION, CABLING SHALL BE DEMOLISHED BACK TO ORIGIN, DEVICES SHALL BE REMOVED, AND DEDICATED PATHWAYS SHALL BE DEMOLISHED.

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.

SCALE: A 8 0 8 16
SCALE OF FEET

LICENSED PROFESSIONAL ENGINEER
05/21/2026
JEFF MULLIKIN
P25384
WASHINGTON

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|---|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR AE DESIGN DENVER, CO | DESIGNED: JR/JM/TW CADD: AE DESIGN TECH REVIEW: MRS/ESN DATE: 5/29/2026 | SUB SHEET NO. TD101 | TITLE OF SHEET FIRST FLOOR TECHNOLOGY DEMO PLANS EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 130 OF 142 | |

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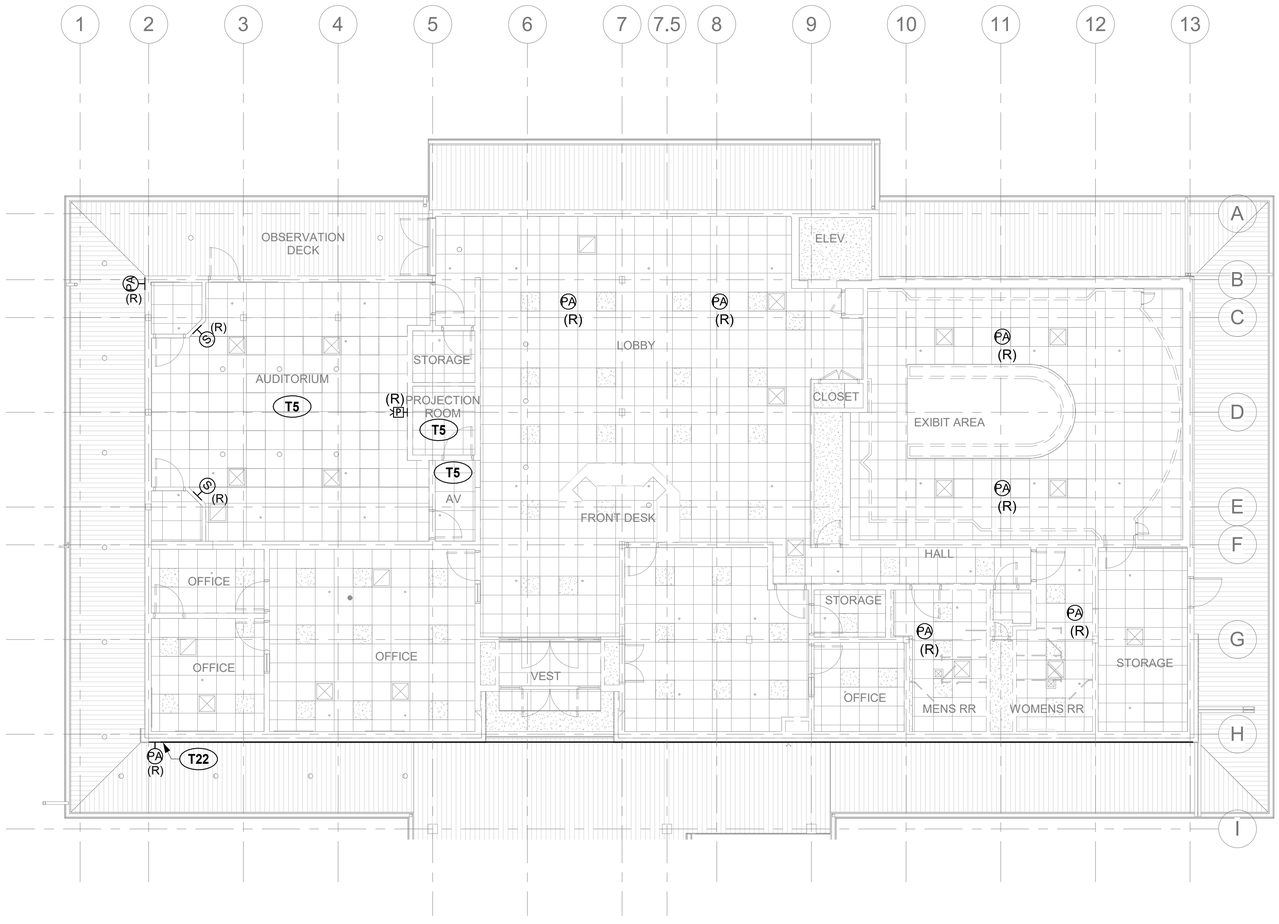


1 TD201 FIRST FLOOR DEMOLITION TECHNOLOGY CEILING PLAN

SCALE A

## TECHNOLOGY DEMO GENERAL NOTES

1. CONTRACTOR SHALL REFER TO LEGEND, NOTES, SCHEDULES, DETAILS, AND SPECIFICATIONS FOR MORE INFORMATION.
2. THE BASIS OF THESE DRAWINGS WERE SITE OBSERVATIONS, ORIGINAL BUILDING DRAWINGS AND VARIOUS OTHER SOURCES. EVERY ATTEMPT HAS BEEN MADE TO DOCUMENT THE ACTUAL CONDITIONS. HOWEVER, THE CONTRACTOR SHALL CAREFULLY EXAMINE THE CONTRACT DOCUMENTS AND THOROUGHLY BECOME FAMILIAR WITH THE BUILDING/NPS STANDARDS RELATING TO THE WORK. FAILURE TO DO SO WILL NOT RELIEVE THE CONTRACTOR OF THE OBLIGATIONS OF THE CONTRACT.
3. THE CONTRACTOR SHALL NOTIFY THE CONTRACTING OFFICER OF ANY MATERIALS OR APPARATUS BELIEVED TO BE INADEQUATE, UNSUITABLE, IN VIOLATION OF LAWS, ORDINANCES, RULES OR REGULATIONS OF AUTHORITIES HAVING JURISDICTION. CONTRACTOR IS RESPONSIBLE FOR IDENTIFYING ITEMS TO BE RETURNED TO THE CONTRACTING OFFCIER FOR REUSE. REUSABLE ITEMS DAMAGED OR DISCARDED SHALL BE REPLACED AT NO ADDITIONAL COST TO CONTRACTING OFFICER.
4. THE CONTRACTOR SHALL INCLUDE IN THEIR COST THE REMOVAL OF ALL EXISTING ELECTRICAL AND TELECOMMUNICATIONS DEVICES, CONDUITS, FIXTURES AND EQUIPMENT. DISCARD ALL EQUIPMENT AS REQUIRED, UNLESS NOTED OTHERWISE.
5. TELECOMMUNICATION RACK MOUNTED EQUIPMENT, WIRELESS ACCESS POINTS (WAPs), AND SPECIFICALLY NOTED EQUPMENT SHALL BE PROTECTED AND/OR REMOVED AND RETURNED TO THE CONTRACTING OFFICER. REFER TO INDIVIDUAL REFERENCES FOR MORE INFORMATION. FOR ALL EXISTING SECURITY, TELECOMMUNICATION, AND AUDIO VIDEO EQUIPMENT NOT SPECIFICALLY NOTED OTHERWISE, CABLING AND CONDUIT SHALL BE DEMOLISHED TO ORIGIN, EQUIPMENT SHALL BE RETURNED TO CONTRACTING OFFICER.
6. ALL WORK MUST BE COORDINATED AND SCHEDULED WITH THE CONTRACTING OFFICER AND OCCUPANTS OF THIS BUILDING SO AS TO PROVIDE THE LEAST AMOUNT OF DISRUPTION OF BUILDING ACTIVIES AS POSSIBLE. THE DEMOLITION OF SOME DEVICES MAY INTERRUPT CONNECTIVITY/OPERATION TO DEVICES OUT OF SCOPE. IT IS THE CONTRACTOR'S RESPONSIBILITY TO PROTECT, MAINTAIN, AND/OR RE-ESTABLISH CONNECTIVITY/OPERATION TO ALL DEVICES OUT OF SCOPE.
7. IT IS NOT ACCEPTABLE FOR CABLES, PATHWAY, OR EQUIPMENT TO BE ABANDONDED IN PLACE, UNLESS OTHERWISE NOTED. WHERE DEVICES ARE INDICATED FOR DEMOLITION, CABLING SHALL BE DEMOLISHED BACK TO ORIGIN, DEVICES SHALL BE REMOVED, AND DEDICATED PATHWAYS SHALL BE DEMOLISHED.

## KEYNOTE LEGEND - NPS

| KEY VALUE | KEYNOTE TEXT |
|---|---|
| T5 | EXISTING AUDITORIUM AUDIO VIDEO SYSTEM SHALL BE REMOVED. CONTRACTOR SHALL RETURN ALL EQUIPMENT, END DEVICES AND CONTROL DEVICES TO THE CONTRACTING OFFICER. |
| T22 | SPEAKER CABLING IS SURFACE MOUNTED WITHIN A SURFACE MOUNTED WIREMOLD. CABLE SHALL BE DEMOLISHED, NEW SPEAKER CABLING SHALL BE REROUTED TO SPEAKER WITHIN BUILDING. COORDINATE WITH RENOVATIONS DRAWINGS FOR MORE INFORMATION. |

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.

SCALE: A 8 0 8 16 SCALE OF FEET

LICENSED PROFESSIONAL ENGINEER 05/21/2026 JEFF MULLIKIN P25384 KONA

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|---|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR AE DESIGN DENVER, CO | DESIGNED: JR/JM/TW DRAWN: AE DESIGN TECH REVIEW: MRS/ESN DATE: 5/29/2026 | SUB SHEET NO. TD201 | TITLE OF SHEET FIRST FLOOR DEMO TECHNOLOGY CEILING PLAN EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 131 OF 142 | |

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| | | KEY | RACK 02 | | |
|---|---|---|---|---|---|
| | 44 | | | 44 | |
| | 43 | | | 43 | |
| | 42 | (R) | ASSISTIVE LISTENING SYSTEM | 42 | |
| VERTICAL WIRE MANAGER (R) | 41 | | | 41 | VERTICAL WIRE MANAGER (R) |
| | 40 | | | 40 | |
| | 39 | | | 39 | |
| | 38 | (R) | DIGITIAL MEDIA PLAYER | 38 | |
| | 37 | | | 37 | |
| | 36 | | | 36 | |
| | 35 | | | 35 | |
| | 34 | | | 34 | |
| | 33 | | | 33 | |
| | 32 | (R) | COMPUTER MONITOR | 32 | |
| | 31 | | | 31 | |
| | 30 | | | 30 | |
| | 29 | | | 29 | |
| | 28 | | | 28 | |
| | 27 | | | 27 | |
| | 26 | | | 26 | |
| | 25 | (R) | SWITCHER | 25 | |
| | 24 | | | 24 | |
| | 23 | (R) | VHS/DVD PLAYER | 23 | |
| | 22 | | | 22 | |
| | 21 | | | 21 | |
| | 20 | (R) | SHELF | 20 | |
| | 19 | | | 19 | |
| | 18 | (R) | STORAGE DRAWER | 18 | |
| | 17 | | | 17 | |
| | 16 | | | 16 | |
| | 15 | (R) | AUDIO VIDEO RECEIVER | 15 | |
| | 14 | | | 14 | |
| | 13 | | | 13 | |
| | 12 | | | 12 | |
| | 11 | | | 11 | |
| | 10 | | | 10 | |
| | 9 | | | 9 | |
| | 8 | | | 8 | |
| | 7 | | | 7 | |
| | 6 | (R) | COMPUTER TOWER \| POWER STRIP \| UPS | 6 | |
| | 5 | | | 5 | |
| | 4 | | | 4 | |
| | 3 | | | 3 | |
| | 2 | | | 2 | |
| | 1 | | | 1 | |
| | RR1 (R) | | | 0 | |

3 TD300 AV RACK - PROJECTION ROOM

| | | KEY | RACK 01 | | |
|---|---|---|---|---|---|
| | 45 | (RG) | HORIZONTAL WIRE MANAGER | 45 | |
| | 44 | | | 44 | |
| | 43 | (RG) | SWITCH | 43 | |
| | 42 | (RG) | CAT6 PATCH PANEL | 42 | |
| | 41 | (RG) | CAT5e PATCH PANEL | 41 | |
| | 40 | (RG) | CAT6 PATCH PANEL | 40 | |
| | 39 | (RG) | CAT6 PATCH PANEL | 39 | |
| | 38 | | | 38 | |
| | 37 | (RG) | HORIZONTAL WIRE MANAGER | 37 | |
| | 36 | | | 36 | |
| | 35 | (RG) | FIBER PANEL | 35 | |
| | 34 | | | 34 | |
| | 33 | (RG) | CAT5e PATCH PANEL | 33 | |
| | 32 | (RG) | PDU | 32 | |
| | 31 | | | 31 | |
| | 30 | | | 30 | |
| | 29 | | | 29 | |
| | 28 | | | 28 | |
| | 27 | | | 27 | |
| | 26 | | | 26 | |
| | 25 | (RG) | UPS | 25 | |
| | 24 | | | 24 | |
| | 23 | | | 23 | |
| | 22 | | | 22 | |
| | 21 | (RG) | SERVER | 21 | |
| | 20 | | | 20 | |
| | 19 | | | 19 | |
| | 18 | | | 18 | |
| | 17 | | | 17 | |
| | 16 | | | 16 | |
| | 15 | (RG) | SHELF | 15 | |
| | 14 | | | 14 | |
| | 13 | | | 13 | |
| | 12 | | | 12 | |
| | 11 | | | 11 | |
| | 10 | (ER) | SERVER | 10 | |
| | 9 | | | 9 | |
| | 8 | | | 8 | |
| | 7 | | | 7 | |
| | 6 | | | 6 | |
| | 5 | (RG) | UPS | 5 | |
| | 4 | | | 4 | |
| | 3 | | | 3 | |
| | 2 | (RG) | SHELF | 2 | |
| | 1 | | | 1 | |
| | RR (RG) | | | | |

4 TD300 SXL - STORAGE - IT RACK (R)

1 TD300 STORAGE 020 WALL ELEVATION

## TECHNOLOGY DEMO GENERAL NOTES

1. CONTRACTOR SHALL REFER TO LEGEND, NOTES, SCHEDULES, DETAILS, AND SPECIFICATIONS FOR MORE INFORMATION.
2. THE BASIS OF THESE DRAWINGS WERE SITE OBSERVATIONS, ORIGINAL BUILDING DRAWINGS AND VARIOUS OTHER SOURCES. EVERY ATTEMPT HAS BEEN MADE TO DOCUMENT THE ACTUAL CONDITIONS. HOWEVER, THE CONTRACTOR SHALL CAREFULLY EXAMINE THE CONTRACT DOCUMENTS AND THOROUGHLY BECOME FAMILIAR WITH THE BUILDING/NPS STANDARDS RELATING TO THE WORK. FAILURE TO DO SO WILL NOT RELIEVE THE CONTRACTOR OF THE OBLIGATIONS OF THE CONTRACT.
3. THE CONTRACTOR SHALL NOTIFY THE CONTRACTING OFFICER OF ANY MATERIALS OR APPARATUS BELIEVED TO BE INADEQUATE, UNSUITABLE, IN VIOLATION OF LAWS, ORDINANCES, RULES OR REGULATIONS OF AUTHORITIES HAVING JURISDICTION. CONTRACTOR IS RESPONSIBLE FOR IDENTIFYING ITEMS TO BE RETURNED TO THE CONTRACTING OFFCIER FOR REUSE. REUSABLE ITEMS DAMAGED OR DISCARDED SHALL BE REPLACED AT NO ADDITIONAL COST TO CONTRACTING OFFICER.
4. THE CONTRACTOR SHALL INCLUDE IN THEIR COST THE REMOVAL OF ALL EXISTING ELECTRICAL AND TELECOMMUNICATIONS DEVICES, CONDUITS, FIXTURES AND EQUIPMENT. DISCARD ALL EQUIPMENT AS REQUIRED, UNLESS NOTED OTHERWISE.
5. TELECOMMUNICATION RACK MOUNTED EQUIPMENT, WIRELESS ACCESS POINTS (WAPs), AND SPECIFICALLY NOTED EQUPMENT SHALL BE PROTECTED AND/OR REMOVED AND RETURNED TO THE CONTRACTING OFFICER. REFER TO INDIVIDUAL REFERENCES FOR MORE INFORMATION. FOR ALL EXISTING SECURITY, TELECOMMUNICATION, AND AUDIO VIDEO EQUIPMENT NOT SPECIFICALLY NOTED OTHERWISE, CABLING AND CONDUIT SHALL BE DEMOLISHED TO ORIGIN, EQUIPMENT SHALL BE RETURNED TO CONTRACTING OFFICER.
6. ALL WORK MUST BE COORDINATED AND SCHEDULED WITH THE CONTRACTING OFFICER AND OCCUPANTS OF THIS BUILDING SO AS TO PROVIDE THE LEAST AMOUNT OF DISRUPTION OF BUILDING ACTIVIES AS POSSIBLE. THE DEMOLITION OF SOME DEVICES MAY INTERRUPT CONNECTIVITY/OPERATION TO DEVICES OUT OF SCOPE. IT IS THE CONTRACTOR'S RESPONSIBILITY TO PROTECT, MAINTAIN, AND/OR RE-ESTABLISH CONNECTIVITY/OPERATION TO ALL DEVICES OUT OF SCOPE.
7. IT IS NOT ACCEPTABLE FOR CABLES, PATHWAY, OR EQUIPMENT TO BE ABANDONDED IN PLACE, UNLESS OTHERWISE NOTED. WHERE DEVICES ARE INDICATED FOR DEMOLITION, CABLING SHALL BE DEMOLISHED BACK TO ORIGIN, DEVICES SHALL BE REMOVED, AND DEDICATED PATHWAYS SHALL BE DEMOLISHED.

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.

SCALE: A 8 0 8 16 SCALE OF FEET

LICENSED PROFESSIONAL ENGINEER 05/21/2026 JEFF MULLIKIN P25384 ROANOKE VA

A/E FIRM:
PRIME:
OTAK, INC.
REDMOND, WA
SUBCONTRACTOR
AE DESIGN
DENVER, CO

DESIGNED: JR/JM/TW
DRAWN: AE DESIGN
TECH REVIEW: MRS/ESN
DATE: 5/29/2026

SUB SHEET NO. TD300

TITLE OF SHEET
ENLARGED TECHNOLOGY DEMO ROOM PLAN
EFFIGY MOUNDS NATIONAL MONUMENT

DRAWING NO. 394 194248
PMIS/PKG NO. 318539
SHEET 132 OF 142

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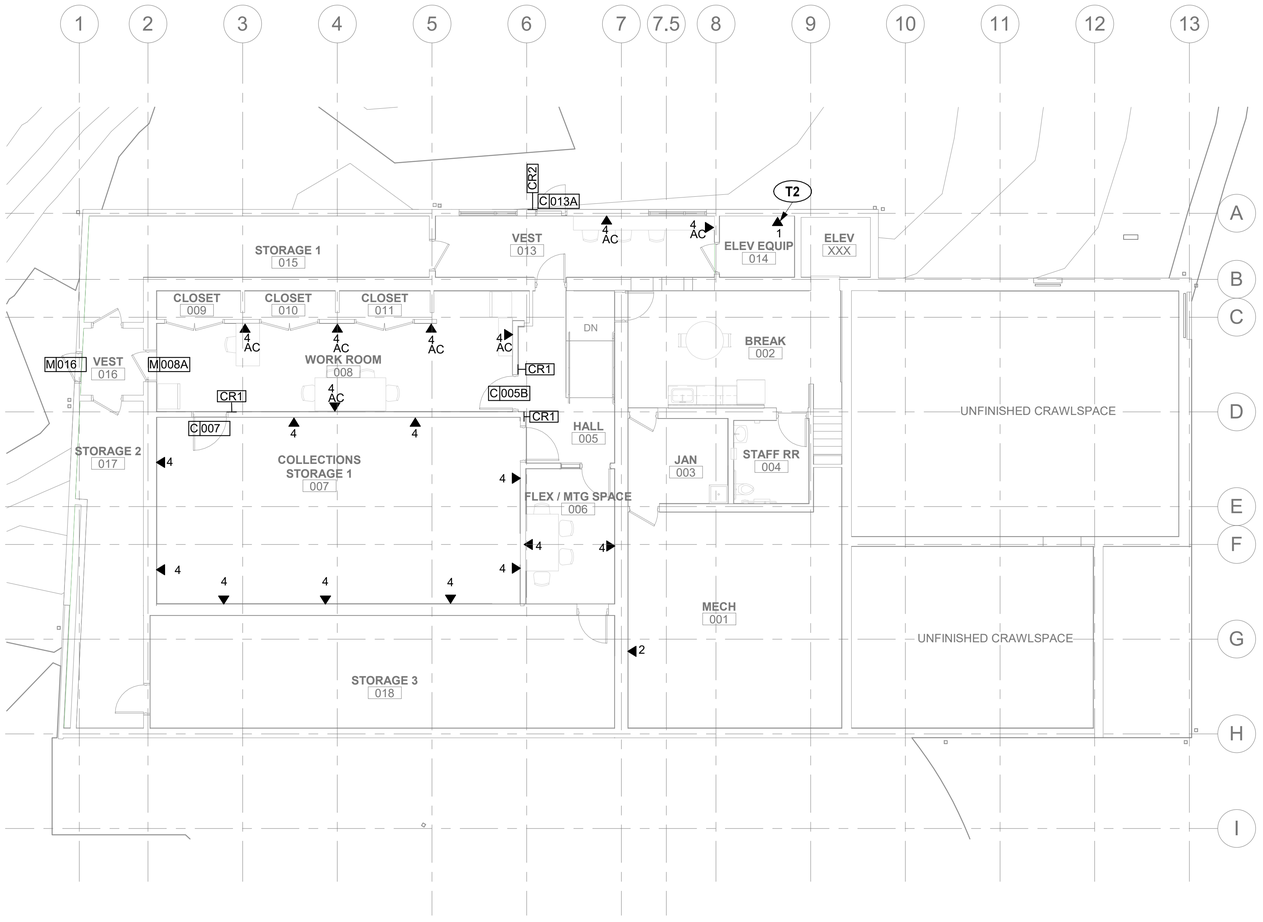


1 T100 BASEMENT TECHNOLOGY PLAN
SCALE A

## TECHNOLOGY GENERAL NOTES

| | |
|---|---|
| 1. | CONTRACTOR SHALL REFER TO LEGEND, NOTES, SCHEDULES, DETAILS, AND SPECIFICATIONS FOR MORE INFORMATION. |
| 2. | VERIFY ALL EXISTING CONDITIONS PRIOR TO BEGINNING WORK. BRING ANY DISCREPANCIES FROM THE DRAWINGS AND NOTES TO THE CONTRACTING OFFICER IMMEDIATELY. |
| 3. | CONTRACTOR IS RESPONSIBLE FOR IDENTIFYING ITEMS TO BE RETURNED TO CONTRACTING OFFICER FOR REUSE. REUSABLE ITEMS DAMAGED OR DISCARDED SHALL BE REPLACED AT NO ADDITIONAL COST TO CONTRACTING OFFICER. |
| 4. | TELECOM BOUNDARIES REPRESENT THE AREA SERVED BY THE CORRESPONDING TELECOM ROOM AND/OR CONSOLIDATION POINT. |
| 5. | SHADING OF AN AREA INDICATES THAT DEVICES IN THE AREA SHALL BE CONSIDERED EXISTING TO REMAIN. CABLING AND/OR PATHWAY PASSING THROUGH THE AREA (THAT SUPPORT DEVICES OUTSIDE THE AREA) SHALL BE DEMOLISHED OR PROTECTED IN PLACE, AS DEFINED PER EACH INDIVIDUAL DEVICE. |

## KEYNOTE LEGEND - NPS

| KEY VALUE | KEYNOTE TEXT |
|---|---|
| T2 | TELECOM SHOWN FOR CONNECTION TO ELEVATOR CONTROL SYSTEM. LOCATION AND QUANTITY SHOWN FOR PRICING PURPOSES ONLY. COORDINATE PRECISE LOCATION OF OUTLETS WITH ELEVATOR EQUIPMENT INTEGRATOR AND APPROVED ELEVATOR SHOP DRAWINGS. |

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.

SCALE: A 8 0 8 16 SCALE OF FEET

LICENSED PROFESSIONAL ENGINEER 05/21/2026 JEFF MULLIKIN P25384 STATE OF WASHINGTON

| | |
|---|---|
| A/E FIRM: | |
| PRIME: | OTAK, INC. REDMOND, WA |
| SUBCONTRACTOR | AE DESIGN DENVER, CO |
| DESIGNED: | JR/JM/TW |
| CADD | AE DESIGN |
| TECH REVIEW: | MRS/ESN |
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| SUB SHEET NO. | T100 |
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| | EFFIGY MOUNDS NATIONAL MONUMENT |
| DRAWING NO. | 394 194248 |
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| SHEET | 133 OF 142 |

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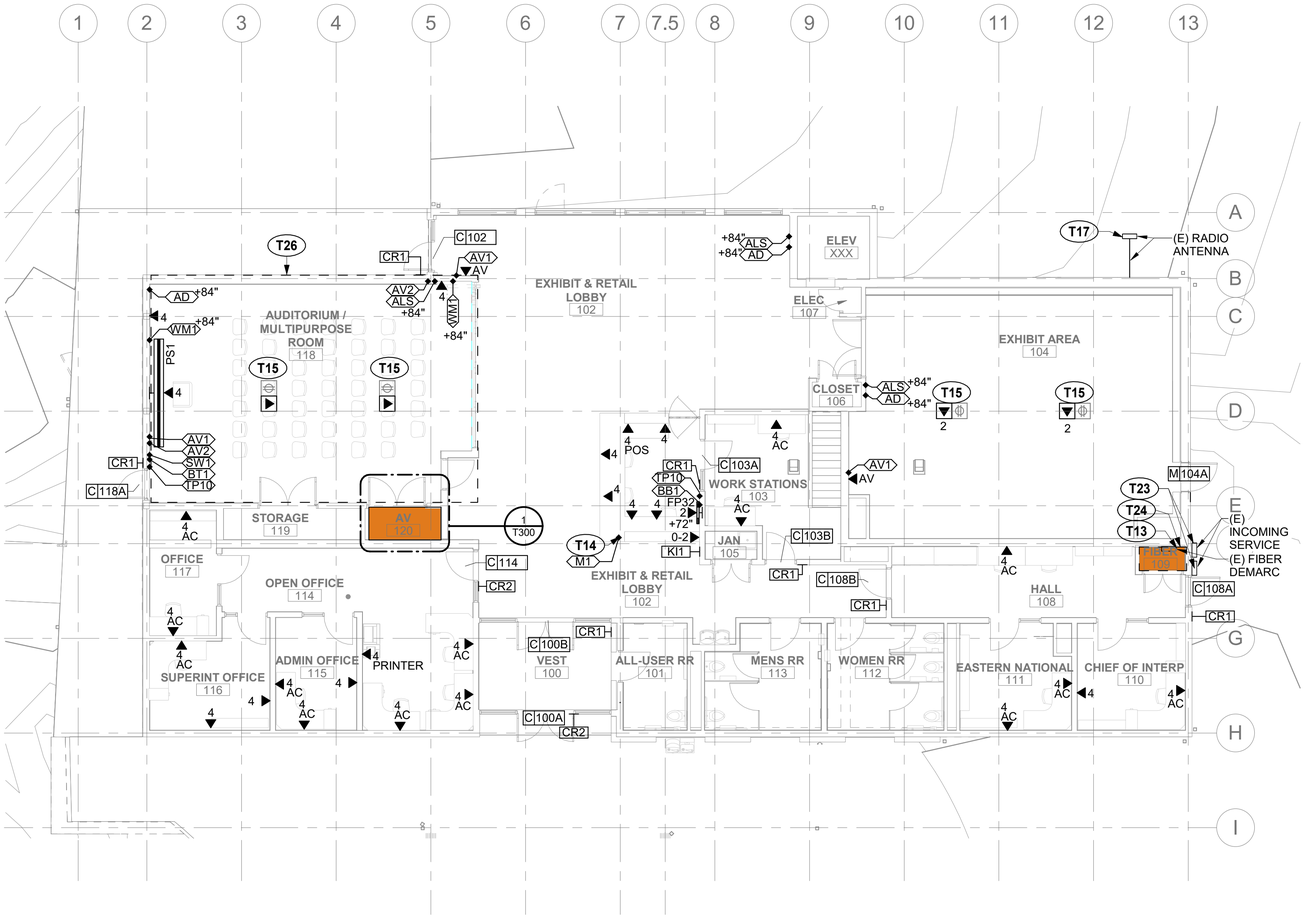



## TECHNOLOGY GENERAL NOTES

| | |
|---|---|
| 1. | CONTRACTOR SHALL REFER TO LEGEND, NOTES, SCHEDULES, DETAILS, AND SPECIFICATIONS FOR MORE INFORMATION. |
| 2. | TELECOM BOUNDARIES REPRESENT THE AREA SERVED BY THE CORRESPONDING TELECOM ROOM AND/OR CONSOLIDATION POINT. |
| 3. | DEVICE LOCATIONS SHOWN SHALL BE COORDINATED WITH ARCHITECTURAL ELEMENTS, FURNITURE, AND OTHER TRADES FOR FINAL INSTALLATION LOCATION. |
| 4. | CABLING SHALL BE ENCLOSED IN CONDUIT IN OPEN CEILING ENVIRONMENTS, NO LOOSE CABLING SHALL BE VISIBLE IN OPEN CEILING ENVIRONMENTS. |
| 5. | ALL CONDUIT PATHWAY ROUTED FROM A BACKBOX, FLOOR BOX, JUNCTION BOX, OR POKE THROUGH SHALL BE ROUTED TO NEAREST ACCESSIBLE CEILING. |
| 6. | NO MORE THAN 100 FEET OF PATHWAY MAY BE INSTALLED BETWEEN PULL POINTS. |
| 7. | ALL CONDUIT BENDS SHALL BE THE "SWEEPING" TYPE AND IN ACCORDANCE WITH THE BEND RADIUS REQUIREMENTS SPECIFIED IN PROJECT GENERAL NOTES. NO "BEND" OR "ELBOW" FITTINGS ARE PERMITTED UNLESS SPECIFICALLY NOTED. |

## KEYNOTE LEGEND - NPS

| KEY VALUE | KEYNOTE TEXT |
|---|---|
| T13 | EXISTING SERVICE PROVIDER EQUIPMENT AND CABLING SHALL BE PROTECTED IN PLACE. DURING DEMOLITION AND CONSTRUCTION, ENSURE THAT THE FIBER CONNECTION IS MAINTAINED AND ACTIVE BETWEEN THIS LOCATION AND THE MAINTENANCE BUILDING. CONTRACTOR SHALL PROVIDE TEMPORARY POWER AS NEEDED TO ENSURE FIBER CONNECTION IS MAINTAINED DURING CONSTRUCTION. |
| T14 | GOOSENECK MICROPHONE SHALL BE INSTALLED AT THE FRONT DESK. |
| T15 | LOCATION OF FLOOR MOUNTED POKE THRU DEVICE FOR POWER AND TELECOMMUNICATION RECEPTACLE INSTALLATION. POKE THRU AND ASSOCIATED CONDUIT RACEWAY FURNISHED AND INSTALLED BY THE ELECTRICAL CONTRACTOR. TELECOM CONTRACTOR SHALL PROVIDE REQUIRED TELECOM DEVICE/JACK/CABLING AS INDICATED ON THESE PLANS. |
| T17 | EXISTING TO REMAIN TOWER AND ANTENNA. CONTRACTOR SHALL PROTECT IN PLACE THE TOWER, THE ANTENNA, AND ANY CABLING ASSOCIATED WITH ANTENNA EQUIPMENT. EXISTING CABLING ROUTES FROM ANTENNA TOWER IN TO THE CEILING SPACE OF THE BUILDING NEAR THE TOWER ANCHOR POINT TO THE BUILDING. |
| T23 | THIS EQUIPMENT SUPPORTS THE INCOMING SERVICE FROM THE PROVIDER AND THE BACKBONE DISTRIBUTION TO OTHER LOCATIONS ON SITE. THIS EQUIPMENT SHALL BE PROTECTED IN PLACE. NEW CONCRETE AND/OR OTHER MATERIAL INSTALLED AT THE GROUND LEVEL OR ON THE WALL SHALL BE COORDINATED TO PROTECT THE EXISTING EQUIPMENT AND ENSURE PROPER ACCESS FOR LONG TERM MAINTENANCE. COORDINATE WITH CONTRACTING OFFICER PRIOR TO WORK COMMENCEMENT. CONTRACTOR SHALL PROVIDE TEMPORARY POWER AS NEEDED TO ENSURE FIBER CONNECTION IS MAINTAINED DURING CONSTRUCTION. |
| T24 | CONTRACTOR SHALL PROVIDE AND INSTALL ONE (1) WALL MOUNTED, 4RU, TELECOM RACK WITH VERTICAL MOUNTING. THE NETWORK EQUIPMENT THAT TAKES IN THE INCOMING SERVICE AND DISTRIBUTES IT TO THE OTHER BUILDINGS SHALL BE RELOCATED FROM THE EXISTING RACK TO THE WALL RACK (FIBER PATCH PANEL AND HEADEND SWITCH/ROUTER) OR BE PROTECTED IN PLACE ON THE EXISTING WALL. COORDINATE WITH CONTRACTING OFFICER ON SITE PRIOR TO DEMOLITION AND RELOCATION. THERE SHALL BE MINIMAL DOWNTIME (LESS THAN 4 HOURS) FOR NETWORK DISTRIBUTION TO OTHER BUILDINGS, ALL DOWNTIME SHALL BE SCHEDULED WITH CONTRACTING OFFICER PRIOR TO WORK. THE INTERNET SERVICE DEMARC AND THE DISTRIBUTION TO OTHER BUILDINGS SHALL REMAIN ACTIVE DURING CONSTRUCTION SO THAT NETWORK CONNECTIVITY AND WIFI AT OTHER BUILDING LOCATIONS REMAIN OPERATIONAL DURING CONSTRUCTION. |
| T26 | **OPTION F:** MAINTAIN <E> ASSISTIVE LISTENING SYSTEM. PROVIDE ALL <N> MEETING ROOM/AUDITORIUM AV EQUIP. INSTALL <N> FOLDING DOOR WITH STRUC SUPPORTS AND <N> WINDOW TREATMENTS. |

1 / T101 FIRST FLOOR TECHNOLOGY PLAN
SCALE (A)

SCALE: (A) 8 0 8 16
SCALE OF FEET

LICENSED PROFESSIONAL ENGINEER 05/21/2026 JEFF MULLIKIN P25384 WASHINGTON

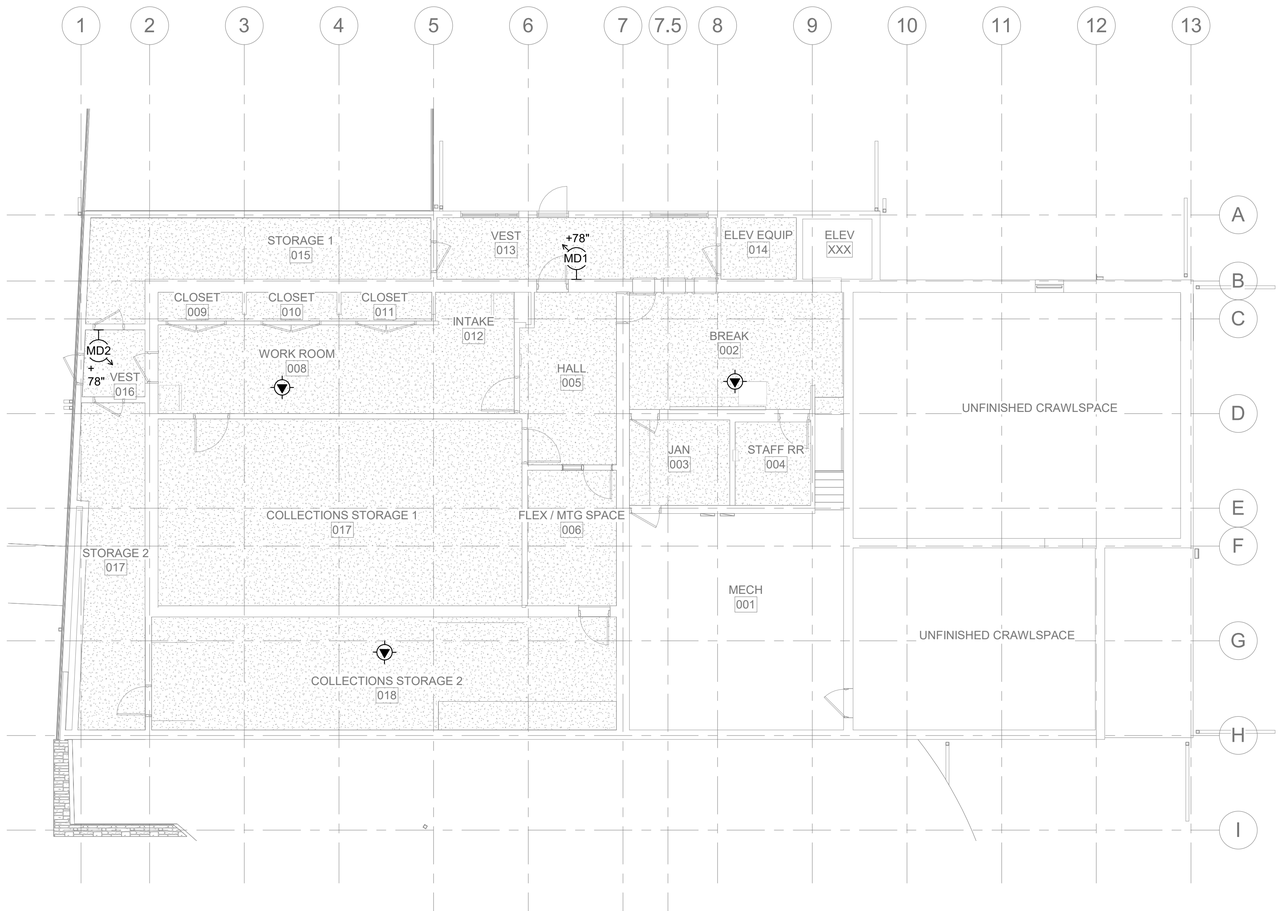
| | | | | | |
|---|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR AE DESIGN DENVER, CO | DESIGNED: JR/JM/TW DRAWN: AE DESIGN TECH REVIEW: MRS/ESN DATE: 5/29/2026 | SUB SHEET NO. T101 | TITLE OF SHEET FIRST FLOOR TECHNOLOGY PLANS EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 134 OF 142 | |

5/21/2026 1:49:12 PM Autodesk Docs://2025-110 NPS EFMO Visitor Center DD-CD/6782.00 - NPS EFMO Effigy Mounds Visitor Center Renovation - Electrical Central - E24.rvt

## TECHNOLOGY GENERAL NOTES

| | |
|---|---|
| 1. | CONTRACTOR SHALL REFER TO LEGEND, NOTES, SCHEDULES, DETAILS, AND SPECIFICATIONS FOR MORE INFORMATION. |
| 2. | TELECOM BOUNDARIES REPRESENT THE AREA SERVED BY THE CORRESPONDING TELECOM ROOM AND/OR CONSOLIDATION POINT. |
| 3. | DEVICE LOCATIONS SHOWN SHALL BE COORDINATED WITH ARCHITECTURAL ELEMENTS, FURNITURE, AND OTHER TRADES FOR FINAL INSTALLATION LOCATION. |
| 4. | CABLING SHALL BE ENCLOSED IN CONDUIT IN OPEN CEILING ENVIRONMENTS, NO LOOSE CABLING SHALL BE VISIBLE IN OPEN CEILING ENVIRONMENTS. |
| 5. | ALL CONDUIT PATHWAY ROUTED FROM A BACKBOX, FLOOR BOX, JUNCTION BOX, OR POKE THROUGH SHALL BE ROUTED TO NEAREST ACCESSIBLE CEILING. |
| 6. | NO MORE THAN 100 FEET OF PATHWAY MAY BE INSTALLED BETWEEN PULL POINTS. |
| 7. | ALL CONDUIT BENDS SHALL BE THE "SWEEPING" TYPE AND IN ACCORDANCE WITH THE BEND RADIUS REQUIREMENTS SPECIFIED IN PROJECT GENERAL NOTES. NO "BEND" OR "ELBOW" FITTINGS ARE PERMITTED UNLESS SPECIFICALLY NOTED. |

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.




1 T200 BASEMENT LEVEL TECHNOLOGY CEILING PLAN
SCALE A

SCALE: A 8 0 8 16
SCALE OF FEET

LICENSED PROFESSIONAL ENGINEER
05/21/2026
JEFF MULLIKIN
P25384
AGENT

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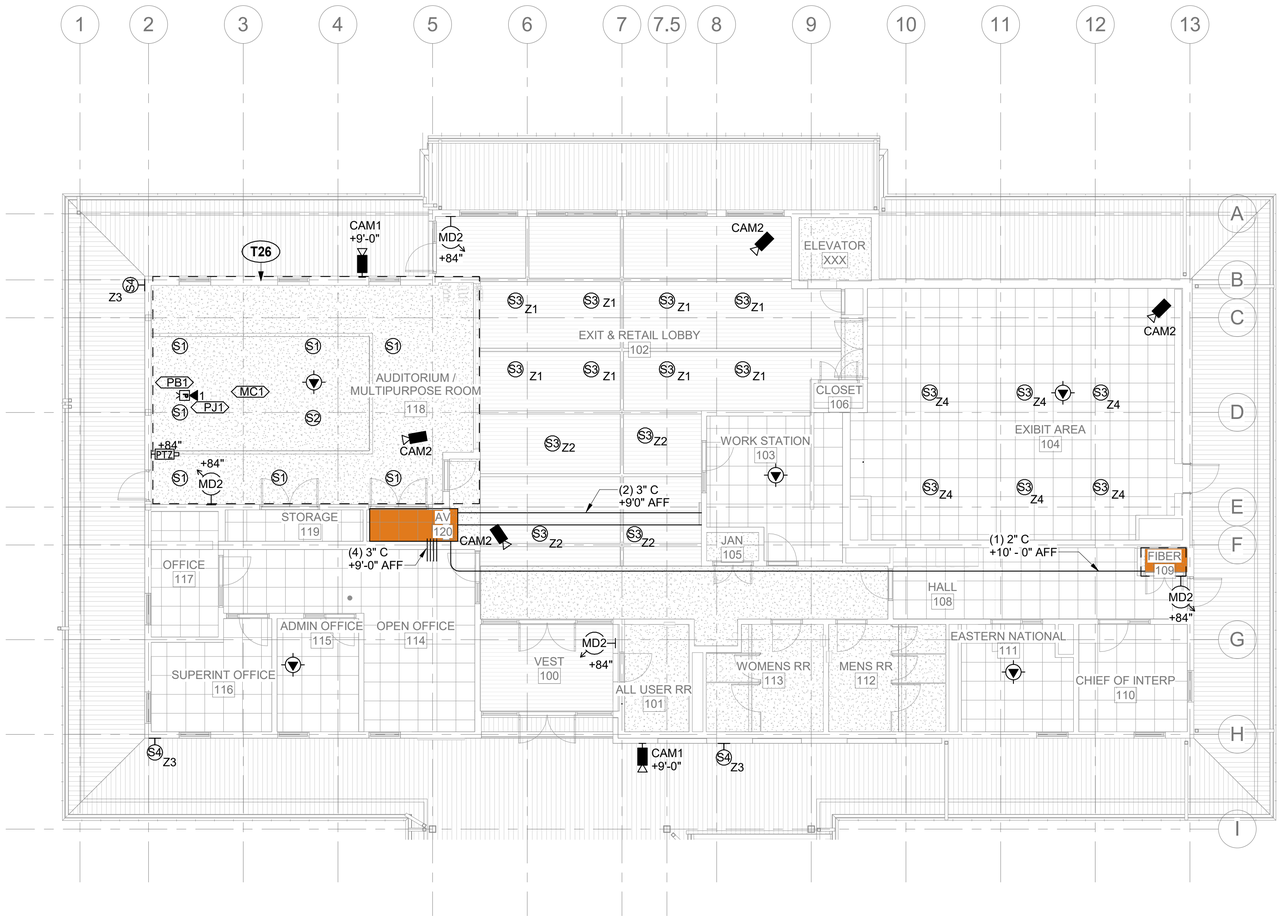
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR AE DESIGN DENVER, CO | DESIGNED: JR/JM/TW CADD AE DESIGN TECH REVIEW: MRS/ESN DATE: 5/29/2026 | SUB SHEET NO. T200 | TITLE OF SHEET BASEMENT TECHNOLOGY CEILING PLAN EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 135 OF 142 | |

5/21/2026 1:49:13 PM Autodesk Docs://2025-110 NPS EFMO Visitor Center DD-CD/6782.00 - NPS EFMO Effigy Mounds Visitor Center Renovation - Electrical Central - E24.rvt




1 T201 FIRST FLOOR TECHNOLOGY CEILING PLAN

SCALE A

## TECHNOLOGY GENERAL NOTES

| | |
|---|---|
| 1. | CONTRACTOR SHALL REFER TO LEGEND, NOTES, SCHEDULES, DETAILS, AND SPECIFICATIONS FOR MORE INFORMATION. |
| 2. | TELECOM BOUNDARIES REPRESENT THE AREA SERVED BY THE CORRESPONDING TELECOM ROOM AND/OR CONSOLIDATION POINT. |
| 3. | DEVICE LOCATIONS SHOWN SHALL BE COORDINATED WITH ARCHITECTURAL ELEMENTS, FURNITURE, AND OTHER TRADES FOR FINAL INSTALLATION LOCATION. |
| 4. | CABLING SHALL BE ENCLOSED IN CONDUIT IN OPEN CEILING ENVIRONMENTS, NO LOOSE CABLING SHALL BE VISIBLE IN OPEN CEILING ENVIRONMENTS. |
| 5. | ALL CONDUIT PATHWAY ROUTED FROM A BACKBOX, FLOOR BOX, JUNCTION BOX, OR POKE THROUGH SHALL BE ROUTED TO NEAREST ACCESSIBLE CEILING. |
| 6. | NO MORE THAN 100 FEET OF PATHWAY MAY BE INSTALLED BETWEEN PULL POINTS. |
| 7. | ALL CONDUIT BENDS SHALL BE THE "SWEEPING" TYPE AND IN ACCORDANCE WITH THE BEND RADIUS REQUIREMENTS SPECIFIED IN PROJECT GENERAL NOTES. NO "BEND" OR "ELBOW" FITTINGS ARE PERMITTED UNLESS SPECIFICALLY NOTED. |

## KEYNOTE LEGEND

| KEY VALUE | KEYNOTE TEXT |
|---|---|
| | |
| T26 | **OPTION F:** MAINTAIN <E> ASSISTIVE LISTENING SYSTEM. PROVIDE ALL <N> MEETING ROOM/AUDITORIUM AV EQUIP. INSTALL <N> FOLDING DOOR WITH STRUC SUPPORTS AND <N> WINDOW TREATMENTS. |

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.

SCALE: A 8 0 8 16 SCALE OF FEET

LICENSED PROFESSIONAL ENGINEER 05/21/2026 JEFF MULLIKIN P25384 KANSAS

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|---|---|---|---|---|---|
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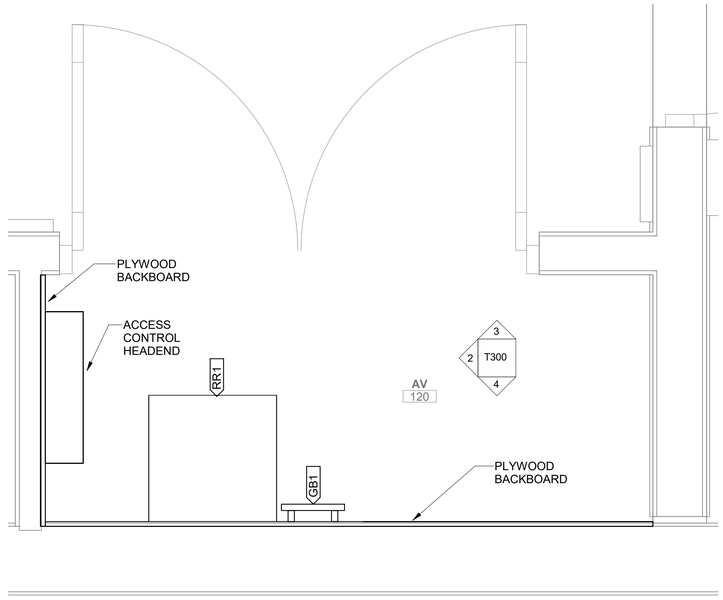
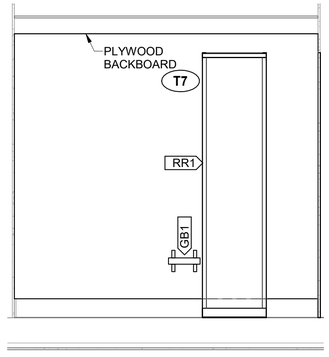
| | KEY | RACK 01 | |
|---|---|---|---|
| 40 | (FP1) | FIBER PANEL | 40 |
| 39 | CP2 | CAT6A PATCH PANEL | 39 |
| 38 | (RL) | SWITCH | 38 |
| 37 | CP4 | CAT6A PATCH PANEL | 37 |
| 36 | | | 36 |
| 35 | | SWITCH | 35 |
| 34 | CP4 | CAT6A PATCH PANEL | 34 |
| 33 | | | 33 |
| 32 | | SWITCH | 32 |
| 31 | CP4 | CAT6A PATCH PANEL | 31 |
| 30 | | | 30 |
| 29 | | SWITCH | 29 |
| 28 | CP4 | CAT6A PATCH PANEL | 28 |
| 27 | | | 27 |
| 26 | | SWITCH | 26 |
| 25 | (CP2) | CAT6A PATCH PANEL | 25 |
| 24 | (SP1) | PDU | 24 |
| 23 | | | 23 |
| 22 | | | 22 |
| 21 | | SERVER | 21 |
| 20 | | | 20 |
| 19 | | | 19 |
| 18 | | | 18 |
| 17 | | | 17 |
| 16 | | | 16 |
| 15 | | | 15 |
| 14 | | | 14 |
| 13 | | | 13 |
| 12 | | RESERVED FOR AV EQUIPMENT | 12 |
| 11 | | | 11 |
| 10 | | | 10 |
| 9 | | | 9 |
| 8 | | | 8 |
| 7 | | | 7 |
| 6 | | | 6 |
| 5 | | | 5 |
| 4 | | | 4 |
| 3 | | RESERVED FOR UPS | 3 |
| 2 | | | 2 |
| 1 | | | 1 |
| | | RR1 | |

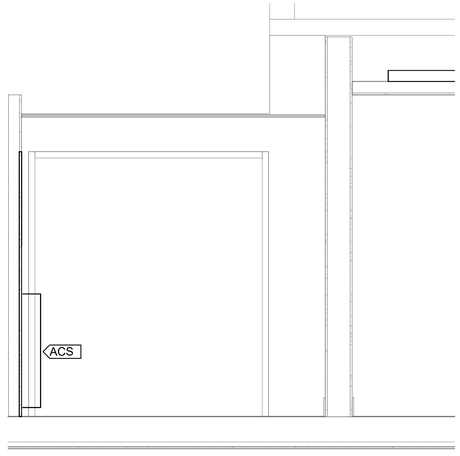
5 / T300 AV 120



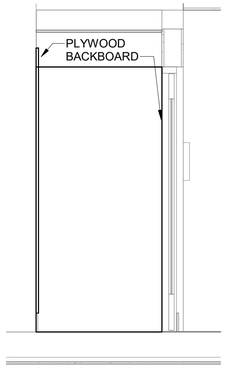

1 / T300 ENLARGED TECHNOLOGY PLAN - AV 120
SCALE (A)




4 / T300 TECHNOLOGY ELEVATION - AV 120 WEST
SCALE (B)




3 / T300 TECHNOLOGY ELEVATION - AV 120 EAST
SCALE (B)




2 / T300 TECHNOLOGY ELEVATION - AV 120 NORTH
SCALE (B)

## TECHNOLOGY GENERAL NOTES

1. CONTRACTOR SHALL REFER TO LEGEND, NOTES, SCHEDULES, DETAILS, AND SPECIFICATIONS FOR MORE INFORMATION.
2. NO EQUIPMENT MAY ENTER OR PASS THROUGH THE TELECOMMUNICATION SPACE UNLESS SPECIFICALLY SUPPORTING THE SPACE. THIS INCLUDES, BUT IS NOT LIMITED TO, HVAC DUCTS, CONDUIT, PLUMBING, ETC.
3. ALL CABLING ROUTED VERTICALLY ON WALLS SHALL BE SUPPORTED WITH CABLE TIES MOUNTED TO WALL MOUNTED VERTICAL LADDER RACK AND/OR D-RINGS.
4. A MINIMUM OF 3 FEET OF CLEARANCE SHALL BE MAINTAINED IN THE FRONT AND BACK OF EACH RACK, AFTER EQUIPMENT IS INSTALLED. CONTRACTOR SHALL ASSUME A MINIMUM OF 24 INCHES OF DEPTH FOR RACK MOUNTED ACTIVE EQUIPMENT IN 2-POST RACKS.
5. ALL WALLS SHALL BE COVERED WITH PLYWOOD, REFER TO SPECIFICATIONS FOR MORE INFORMATION.
6. THE DEDICATED IN-ROOM HVAC THERMOSTAT SHALL BE MOUNTED ADJACENT THE LIGHT SWITCH NEAR ENTRY DOOR.
7. CONTRACTOR SHALL COORDINATE WITH ALL OTHER CONTRACTORS/TRADES TO ENSURE WALL AND RACK SPACE ALLOCATION NEEDS.

## KEYNOTE LEGEND - NPS

| KEY VALUE | KEYNOTE TEXT |
|---|---|
| T7 | CONTRACTOR SHALL PROVIDE MARINE GRADE 1" MARINE GRADE A/B 3/4" PLYWOOD BACKBOARD FOR LENGTHS AS INDICATED MOUNTED TO WALL 0'-6" AFF TO 7'-0" AFF. BACKBOARD SHOULD BE PAINTED PER SPECIFICATIONS WITHOUT COVERING UL RATED STAMP. |

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.

SCALE: (B) 4 0 4 8 SCALE OF FEET

SCALE: (A) 8 0 8 16 SCALE OF FEET

LICENSED PROFESSIONAL ENGINEER
05/21/2026
JEFF MULLIKIN
P25384
WASHINGTON
REGISTERED

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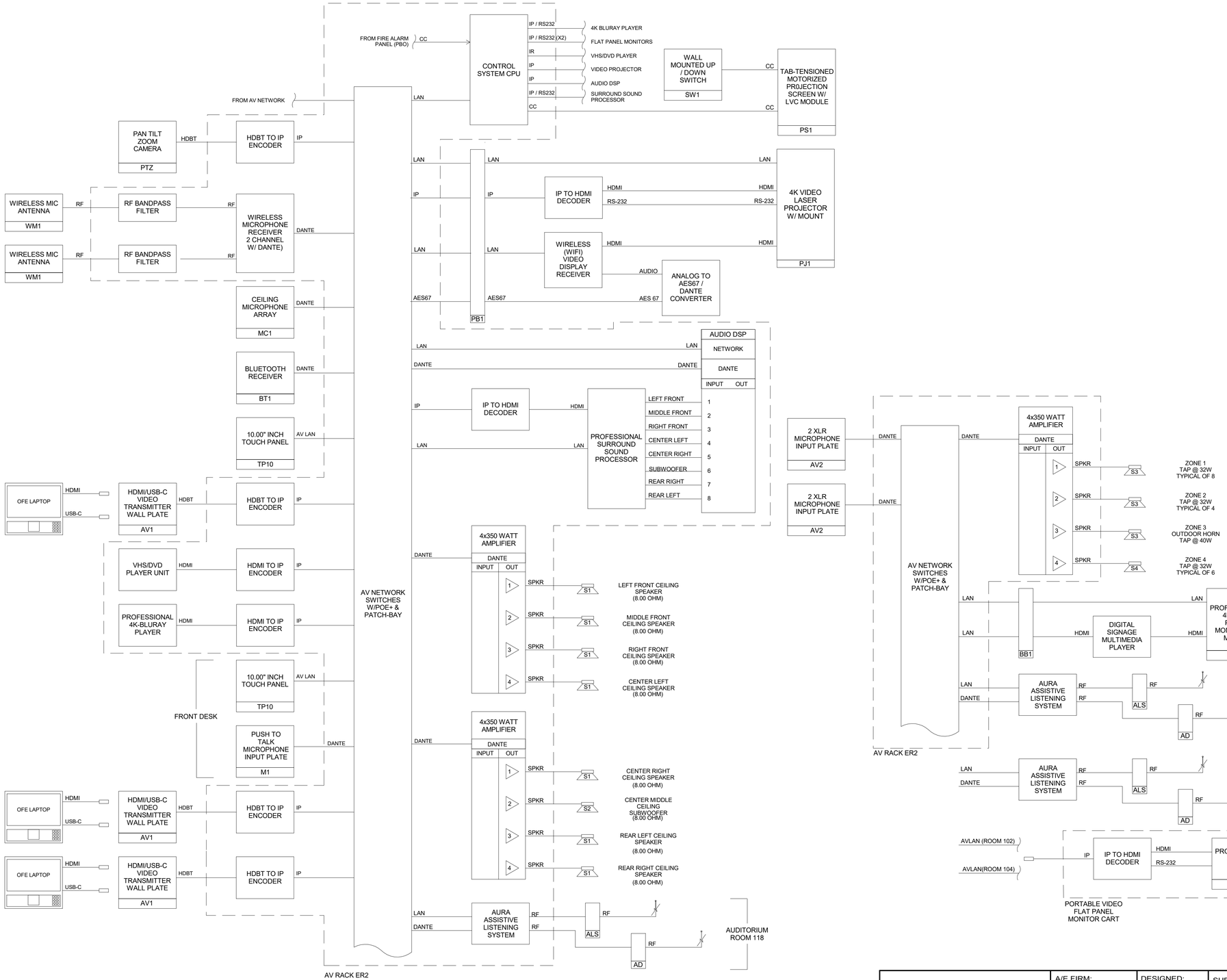
|---|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR AE DESIGN DENVER, CO | DESIGNED: JR/JM/TW DRAWN: AE DESIGN TECH REVIEW: MRS/ESN DATE: 5/29/2026 | SUB SHEET NO. T300 | TITLE OF SHEET: ENLARGED TECHNOLOGY ROOM PLAN — EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 137 OF 142 | |

5/21/2026 1:49:13 PM Autodesk Docs://2025-110 NPS EFMO Visitor Center DD-CD/6782.00 - NPS EFMO Effigy Mounds Visitor Center Renovation - Electrical Central - E24.rvt

5/21/2026 1:49:14 PM Autodesk Docs://2025-110 NPS EFMO Visitor Center DD-CD/6782.00 - NPS EFMO Effigy Mounds Visitor Center Renovation - Electrical Central - E24.rvt




1 T400

## AV ONE-LINE DIAGRAM - AUDITORIUM/MULTIPURPOSE ROOM 118 - OPTION F & EXIT AND RETAIL LOBBY 102

### AV ONE-LINE GENERAL NOTES:

A. ALL DATA PATCH CABLES REQUIRED TO CONNECT AN AV DEVICE TO A DATA NETWORK OUTLET SHALL BE PROVIDED BY THE AVC AND SHALL BE OF EXACTLY THE SAME CATEGORY CLASS ( OR BETTER) AS THE INCOMING CABLES, I.E. A SHIELDED CAT6A INCOMING CABLES TO THE PATCH BAY SHALL REQUIRE A SHIELDED CAT6A RATED PATCH CABLE.

B. AVC SHALL PROVIDE ALL REQUIRED SPECIALIZED/CUSTOM CABLES PROVIDING THE END USER WITH A TURN-KEY SOLUTION.

C. AVC SHALL SUPPLY AND UTILIZE RS-485/422/232 BALUNS FOR LOW VOLTAGE CONTROL APPLICATIONS ETCETERA AS DISTANCE LIMITATIONS REQUIRE.

D. AVC MUST UTILIZE THE 274100 SPECIFICATION SECTION IN CONJUNCTION WITH THESE DRAWINGS TO FULLY UNDERSTAND THE SCOPE OF WORK AND SYSTEM FUNCTIONALITY REQUIRED FOR THIS PROJECT.

E. AVC SHALL VERIFY AND COORDINATE ALL INFRASTRUCTURE REQUIREMENTS, I.E. CONDUIT, PULL, JOB AND BACK BOXES ETCETERA WITH EC PRIOR TO THE COMPLETION OF ELECTRICAL ROUGH-IN PHASE. IMMEDIATELY BRING DISCREPANCIES TO THE ATTENTION OF THE EC AND THE AV CONSULTANT.

F. PROVIDE QUANTITIES AS SHOWN ELSEWHERE ON THE DRAWINGS, AS CALLED FOR IN THE SPECIFICATIONS, AND/OR AS NECESSARY FOR COMPLETE AND FULLY OPERATIONAL SYSTEM.

G. ALL HIGH VOLTAGE CONNECTIONS SHALL BE PERFORMED BY THE BY EC WITH THE LOW VOLTAGE WORK BY AVC.

H. ALL AUDIO CABLING AND TERMINATIONS SHALL UTILIZE THE THREE WIRE "BALANCED" PROFESSIONAL AUDIO INDUSTRY STANDARD WHEREVER POSSIBLE. STEREO AUDIO, THOUGH NOT ALWAYS DETAILED IS REQUIRED.

I. ALL AV EQUIPMENT FURNISHED AND INSTALLED BY AVC UNLESS SPECIFIED OTHERWISE.

J. AVC SHALL INSTALL EACH CEILING LOUDSPEAKER OR LOUDSPEAKER CLUSTER WITH A SEISMIC RESTRAINING SYSTEM PER ALL LOCAL, STATE AND FEDERAL CODES.

K. BASED ON SITE CONDITIONS, THE CONTRACTOR SHALL FIELD VERIFY THE PROJECTION SCREEN IMAGE SIZE AS WELL AD THE PHYSICAL SITE INSTALLATION LOCATIONS PRIOR TO SHOP DRAWING SUBMITTAL, PURCHASE AND INSTALLATION. IN CASES WHEN THE SCREENS ARE PBO THE CONTRACTOR SHALL VERIFY THE VALIDITY OF THE SCREEN SIZE AND IMMEDIATELY CONTACT THE DESIGN TEAM IF VARIANCES OR DEFICIENCIES HAVE BEEN FOUND WHERE THERE IS A MISMATCH IN THE SCREEN ASPECT RATIO AND THE NATIVE PROJECTOR RESOLUTION.

L. EQUIPMENT AND SIGNAL FLOW SHOWN IS OF A CONCEPTUAL NATURE. AVC SHALL PROVIDE ALL NECESSARY INCIDENTAL ELECTRONICS AND EQUIPMENT SUCH THAT A TURN-KEY SOLUTION IS INSTALLED AND READY FOR THE END USERS IMMEDIATE USAGE. ITEMS SUCH AS BALUNS, POWER SUPPLIES, CABLE MANAGEMENT ACCESSORIES, SMALL LOCAL ETHERNET ROUTERS ETCETERA SHALL BE INCLUDED IN THE BASE BID. SUBMIT QUESTIONS FOR CLARIFICATION PRIOR TO YOUR BID RETURN AS MONIES SHALL NOT BE AVAILABLE FOR ITEMS NOT ACCOUNTED FOR IN YOUR BID DOCUMENTATION. NO EXCEPTIONS.

M. AVC SHALL CLOSELY COORDINATE WITH THE TELEDATA CONTRACTOR ALL DATA DROP LOCATIONS AND MOUNTING HEIGHTS AS NECESSARY.

N. AVC SHALL CLOSELY COORDINATE WITH END USERS IT STAFF TO PROCURE THE NECESSARY NETWORK ACCESS, IP ADDRESSES ETCETERA REQUIRED FOR THIS PROJECT.

O. AV CONTRACTOR SHALL PROVIDE A TURN-KEY THE DEDICATED ETHERNET NETWORK FOR THIS PROJECT INCLUDING BUT NOT LIMITED TO SWITCHES,PATCH-BAYS, PATCH CABLES ETCETERA.

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION F UNLESS NOTED OTHERWISE.

SCALE: A — 8 0 8 16 — SCALE OF FEET

LICENSED PROFESSIONAL ENGINEER — 05/21/2026 — JEFF MULLIKIN — P23384 — ARIZONA

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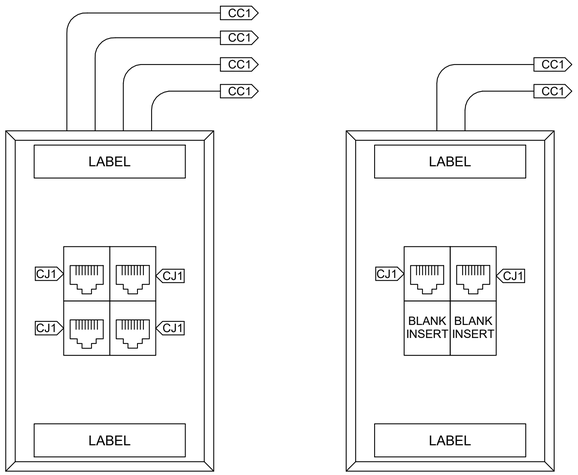
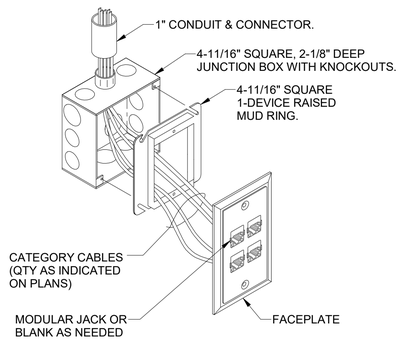
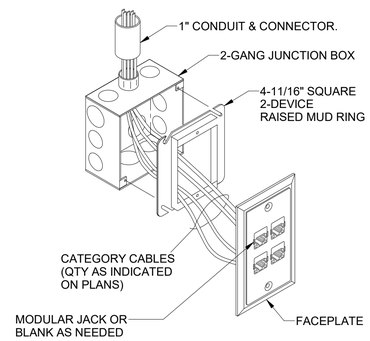
|---|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR AE DESIGN DENVER, CO | DESIGNED: JR/JM/TW DRAWN: AE DESIGN TECH REVIEW: MRS/ESN DATE: 5/29/2026 | SUB SHEET NO. T400 | TITLE OF SHEET: TECHNOLOGY AV ONE-LINES — EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 138 OF 142 | |

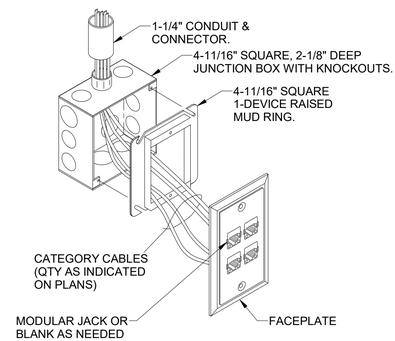
ALL INFORMATION SHOWN ON THIS SHEET IS COMMON TO ALL OPTIONS.




## 2 T500 TELECOMMUNICATION OUTLET FACEPLATE DETAIL

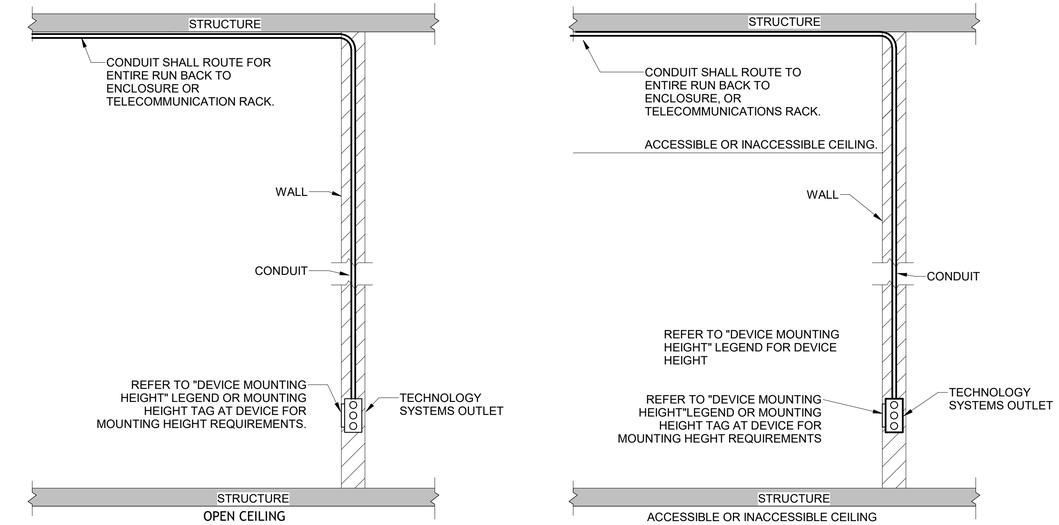


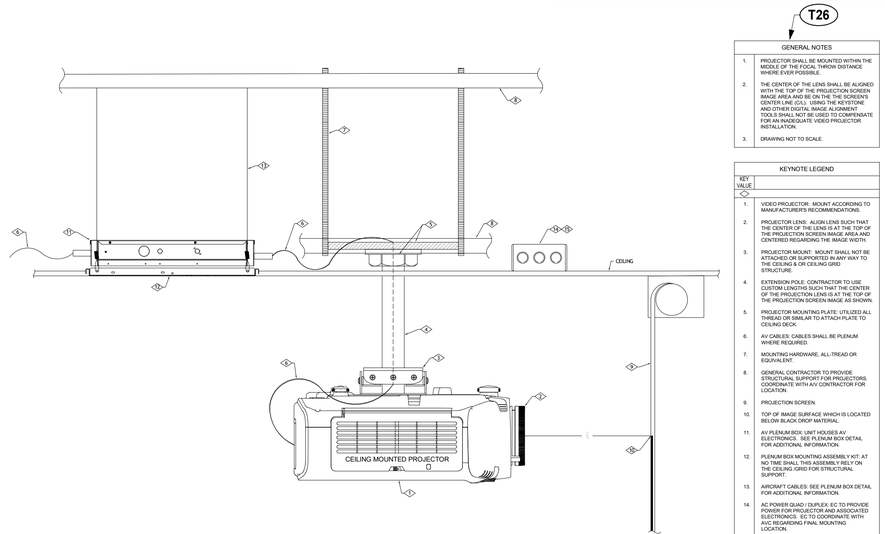




## 1 T500 TECHNOLOGY SYSTEMS OUTLETS AND CONDUIT

| KEYNOTE LEGEND - NPS | |
|---|---|
| KEY VALUE | KEYNOTE TEXT |
| T26 | **OPTION F:** MAINTAIN <E> ASSISTIVE LISTENING SYSTEM. PROVIDE ALL <N> MEETING ROOM/AUDITORIUM AV EQUIP. INSTALL <N> FOLDING DOOR WITH STRUC SUPPORTS AND <N> WINDOW TREATMENTS. |




## 3 T500 TECHNOLOGY SYSTEMS OUTLET AND CONDUIT ROUTING




GENERAL NOTES

1. PROJECTOR SHALL BE MOUNTED WITHIN THE MIDDLE OF THE FOCAL THROW DISTANCE WHERE EVER POSSIBLE.
2. THE CENTER OF THE LENS SHALL BE ALIGNED WITH THE TOP OF THE PROJECTION SCREEN IMAGE AREA AND BE ON THE SCREEN'S CENTER LINE. THE USE OF KEYSTONE AND OTHER DIGITAL IMAGE ALIGNMENT TOOLS SHALL NOT BE USED TO COMPENSATE FOR AN INADEQUATE VIDEO PROJECTOR INSTALLATION.
3. DRAWING NOT TO SCALE.

KEYNOTE LEGEND

| KEY VALUE | |
|---|---|
| 1. | VIDEO PROJECTOR: MOUNT ACCORDING TO MANUFACTURER'S RECOMMENDATIONS. |
| 2. | PROJECTION LENS: ALIGN LENS SUCH THAT THE CENTER OF THE LENS IS AT THE TOP OF THE PROJECTION SCREEN IMAGE AREA AND CENTERED REGARDING THE IMAGE WIDTH. |
| 3. | PROJECTOR MOUNT: MOUNT SHALL NOT BE ATTACHED OR SUPPORTED IN ANY WAY TO THE CEILING & OR CEILING GRID STRUCTURE. |
| 4. | EXTENSION POLE: CONTRACTOR TO USE CUSTOM LENGTHS SUCH THAT THE CENTER OF THE PROJECTION LENS IS AT THE TOP OF THE PROJECTION SCREEN IMAGE AS SHOWN. |
| 5. | PROJECTOR MOUNTING PLATE: UTILIZED ALL THREAD OR SIMILAR TO ATTACH PLATE TO CEILING DECK. |
| 6. | AV CABLES: CABLES SHALL BE PLENUM WHERE REQUIRED. |
| 7. | MOUNTING HARDWARE, ALL-THREAD OR EQUIVALENT. |
| 8. | GENERAL CONTRACTOR TO PROVIDE STRUCTURAL SUPPORT FOR PROJECTORS. COORDINATE WITH AV CONTRACTOR FOR LOCATION. |
| 9. | PROJECTION SCREEN. |
| 10. | TOP OF IMAGE SURFACE WHICH IS LOCATED BELOW BLACK DROP MATERIAL. |
| 11. | AV PLENUM BOX: UNIT HOUSES AV ELECTRONICS. SEE PLENUM BOX DETAIL FOR ADDITIONAL INFORMATION. |
| 12. | PLENUM BOX MOUNTING ASSEMBLY KIT: AT NO TIME SHALL THIS ASSEMBLY RELY ON THE CEILING GRID FOR STRUCTURAL SUPPORT. |
| 13. | AIRCRAFT CABLES: SEE PLENUM BOX DETAIL FOR ADDITIONAL INFORMATION. |
| 14. | AC POWER QUAD / DUPLEX: EC TO PROVIDE POWER FOR PROJECTOR AND ASSOCIATED ELECTRONICS. EC TO COORDINATE WITH AV CONTRACTOR FINAL MOUNTING LOCATION. |
| 15. | DATA PORTS / 2 GANG BACK BOX: EC TO PROVIDE A 2-GANG BACK BOX FOR THE TERMINATION OF THE DATA PORTS. EC & TELE-DATA CONTRACTOR TO COORDINATE WITH AVC FOR FINAL MOUNTING LOCATION. |

## 4 T500 CEILING MOUNTED PROJECTOR

SCALE: A 8 0 8 16
SCALE OF FEET

LICENSED PROFESSIONAL ENGINEER 05/21/2026 JEFF MULLIKIN P25384 COLORADO

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|---|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR AE DESIGN DENVER, CO | DESIGNED: JR/JM/TW CADD: AE DESIGN TECH REVIEW: MRS/ESN DATE: 5/29/2026 | SUB SHEET NO. T500 | TITLE OF SHEET TECHNOLOGY DETAILS - STANDARD EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 139 OF 142 | |

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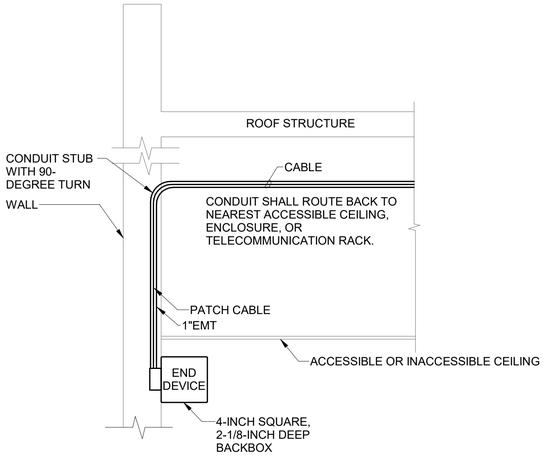
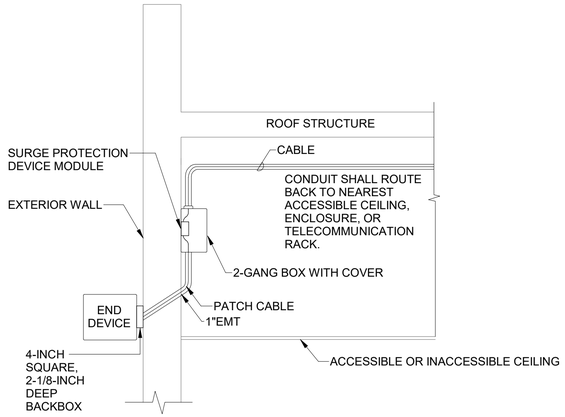
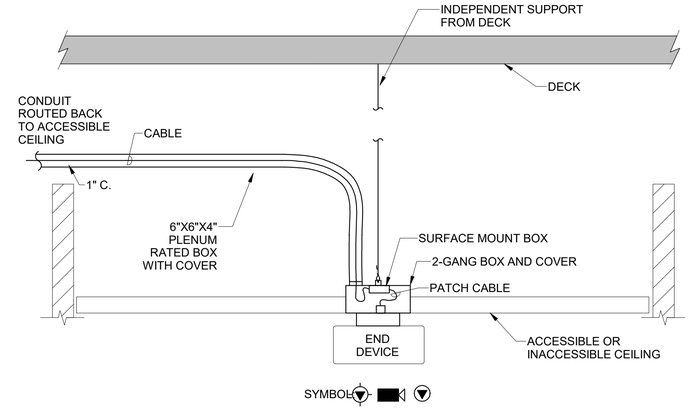
ALL INFORMATION SHOWN ON THIS SHEET IS COMMON TO ALL OPTIONS.



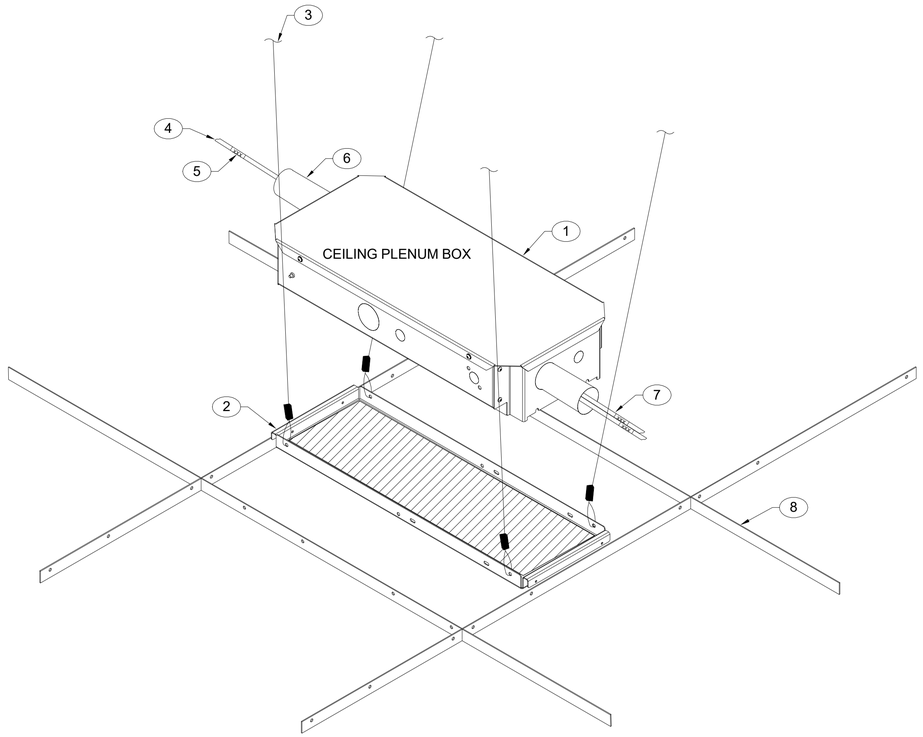

3 T501 INTERIOR WALL MOUNTED DEVICE DETAIL - ACCESSIBLE OR INACCESSIBLE CEILING




2 T501 EXTERIOR WALL MOUNTED DEVICE DETAIL - ACCESSIBLE OR INACCESSIBLE CEILING



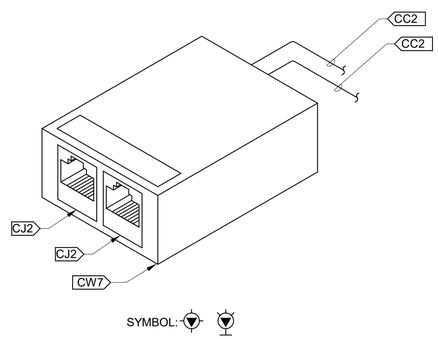

1 T501 CEILING MOUNTED DEVICE DETAIL - ACCESSIBLE OR INACCESSIBLE CEILING



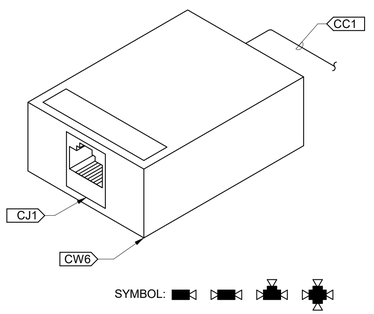

6 T501 CEILING PLENUM BOX

| GENERAL NOTES | |
|---|---|
| 1. | CONTRACTOR SHALL SECURELY ATTACH PLENUM BOX ASSEMBLY TO CEILING DECK STRUCTURE SUCH THAT THE UNIT DOES NOT UTILIZE THE CEILING TILE GRID FOR SUPPORT IN ANY WAY. |
| 2. | PLENUM BOX ASSEMBLY SHALL BE SECURED UTILIZING AIRCRAFT CABLING WITH A MINIMUM OF 5 TIMES THE LOAD WEIGHT CAPACITY OF THE ASSEMBLY. |
| 3. | IN PROJECTS WHERE THIS PLENUM BOX ASSEMBLY IS ASSOCIATED WITH A PROJECTOR MOUNT, THE MOUNTING POLE AND ITS ASSOCIATED HARDWARE SHALL NOT USE THIS DEVICE FOR SUPPORT OR MOUNTING. |
| 4. | CONTRACTOR SHALL INSTALL ALL NECESSARY CONDUIT HOLE PLUGS AS REQUIRED TO MAINTAIN THE AIR PLENUM AT ALL TIMES. NO EXCEPTIONS. |
| 5. | DRAWING NOT TO SCALE. |

| KEYNOTE LEGEND | |
|---|---|
| KEY VALUE | |
| 1. | PLENUM BOX: HOUSES AV ELECTRONICS. |
| 2. | SUSPENDED CEILING MOUNT KIT: UNIT SHALL NOT BE SUPPORTED FROM CEILING GRID STRUCTURE. |
| 3. | AIRCRAFT SUPPORT CABLE: CABLE SHALL BE RATED 5 TIME GREATER THAN MAXIMUM LOAD. |
| 4. | AV CABLING: FROM SOURCE OR EQUIPMENT RACK. All CABLES IN THE PLENUM SPACE SHALL BE PLENUM RATED. |
| 5. | LABELS: PROVIDE WHITE LABELS WITH BLACK TEXT AT BOTH ENDS OF ALL CABLING IN PROJECT. |
| 6. | CONDUIT: EC PROVIDE CONDUIT THAT IS TERMINATED WITHIN THE PLENUM BOX ASSEMBLY AS REQUIRED. |
| 7. | AV CABLING: TO SIGNAL SINK. All CABLES IN THE PLENUM SPACE SHALL BE PLENUM RATED. |




5 T501 2 PORT SURFACE MOUNT BOX




4 T501 1 PORT SURFACE MOUNT BOX

LICENSED PROFESSIONAL ENGINEER 05/21/2026 JEFF MULLIKIN P25384 ARIZONA

| | |
|---|---|
| A/E FIRM: | |
| PRIME: | OTAK, INC. |
| | REDMOND, WA |
| SUBCONTRACTOR | AE DESIGN |
| | DENVER, CO |
| DESIGNED: | JR/JM/TW |
| CADD | AE DESIGN |
| TECH REVIEW: | MRS/ESN |
| DATE: | 5/29/2026 |
| SUB SHEET NO. | T501 |
| TITLE OF SHEET | TECHNOLOGY DETAILS - STANDARD |
| | EFFIGY MOUNDS NATIONAL MONUMENT |
| DRAWING NO. | 394 194248 |
| PMIS/PKG NO. | 318539 |
| SHEET | 140 OF 142 |

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## BACKBONE CABLE MATRIX

| ORIGINATION SPACE | TERMINATION SPACE | CABLE 1 KEY | CABLE 1 QUANTITY | CABLE 1 COLOR | CABLE 2 KEY | CABLE 2 QUANTITY | CABLE 2 COLOR | CABLE 3 KEY | CABLE 3 QUANTITY | CABLE 3 COLOR | COMMENTS |
|---|---|---|---|---|---|---|---|---|---|---|---|
| FIBER 109 | AV 120 | FC1 | 1 | YELLOW | -- | -- | -- | -- | -- | -- | |

## COPPER CONNECTIVITY

| KEY ID | DESCRIPTION | MANUFACTURER | PART NUMBER | COMMENTS |
|---|---|---|---|---|
| CC1 | CAT 6 CABLE, PLENUM, UTP | COMMSCOPE (OR APPROVED EQUAL) | CS34P (OR APPROVED EQUAL) | |
| CC2 | CAT 6A CABLE, PLENUM, UTP | COMMSCOPE (OR APPROVED EQUAL) | CS44P (OR APPROVED EQUAL) | |
| CJ2 | CAT 6A JACK | COMMSCOPE (OR APPROVED EQUAL) | UKJ10G (OR APPROVED EQUAL) | |
| CX3 | CAT 6A PATCH CABLE | COMMSCOPE (OR APPROVED EQUAL) | UC1AAA2 (OR APPROVED EQUAL) | |
| CX4 | CAT 6A PATCH CABLE - PLENUM RATED | COMMSCOPE (OR APPROVED EQUAL) | UC1AA22 (OR APPROVED EQUAL) | |
| CW4 | 4-PORT FACEPLATE, LABELED | COMMSCOPE (OR APPROVED EQUAL) | 2111039-x (OR APPROVED EQUAL) | FINISH SHALL MATCH COLOR OF POWER RECEPTACLE DEVICE FACEPLATES |
| CW6 | SURFACE MOUNT BOX, (BISCUIT), PLENUM, 1 PORT | COMMSCOPE (OR APPROVED EQUAL) | SMB-1P-xxx PLENUM (OR APPROVED EQUAL) | SECURITY CAMERA LOCATIONS |
| CW7 | SURFACE MOUNT BOX, (BISCUIT), PLENUM, 2 PORT | COMMSCOPE (OR APPROVED EQUAL) | SMB-2P-xxx PLENUM (OR APPROVED EQUAL) | WIRELESS ACCESSS POINT LOCATIONS |
| CP2 | CAT 6A PATCH PANEL, 24 PORTS, 1RU, BLACK, FLAT | COMMSCOPE (OR APPROVED EQUAL) | UNP-6A-DM-1U-24 (OR APPROVED EQUAL) | |
| CP4 | CAT 6A PATCH PANEL, 48 PORTS, 2RU, BLACK, FLAT | COMMSCOPE (OR APPROVED EQUAL) | UNP-6A-DM-2U-48 (OR APPROVED EQUAL) | |
| CP5 | MODULAR PATCH PANEL, 48 PORTS, 2RU, BLACK, FLAT | COMMSCOPE (OR APPROVED EQUAL) | CPP-UDDM-KJ-2U-48 (OR APPROVED EQUAL) | |

## FIBER CONNECTIVITY

| KEY ID | DESCRIPTION | MANUFACTURER | PART NUMBER | COMMENTS |
|---|---|---|---|---|
| FC1 | 12 STRAND SM (OS2) FIBER, ISP, PLENUM, TIGHT BUFFERED | CORNING (OR APPROVED EQUAL) | 012E88-33131-29 (OR APPROVED EQUAL) | |
| FP1 | 1 RU FIBER PANEL | CORNING (OR APPROVED EQUAL) | CCH-01U (OR APPROVED EQUAL) | |

## GROUNDING

| KEY ID | DESCRIPTION | MANUFACTURER | PART NUMBER | COMMENTS |
|---|---|---|---|---|
| GB1 | PRIMARY BONDING BUSBAR (PBB) COMPLIANT WITH ANSI/TIA-607 | CPI (OR APPROVED EQUAL) | 40153-020 (OR APPROVED EQUAL) | |
| GS2 | SURGE PROTECTION - CAT6 / CAT6A | CIRCA MAX (OR APPROVED EQUAL) | 604-75RJ (OR APPROVED EQUAL) | |

## HORIZONTAL CABLE MATRIX

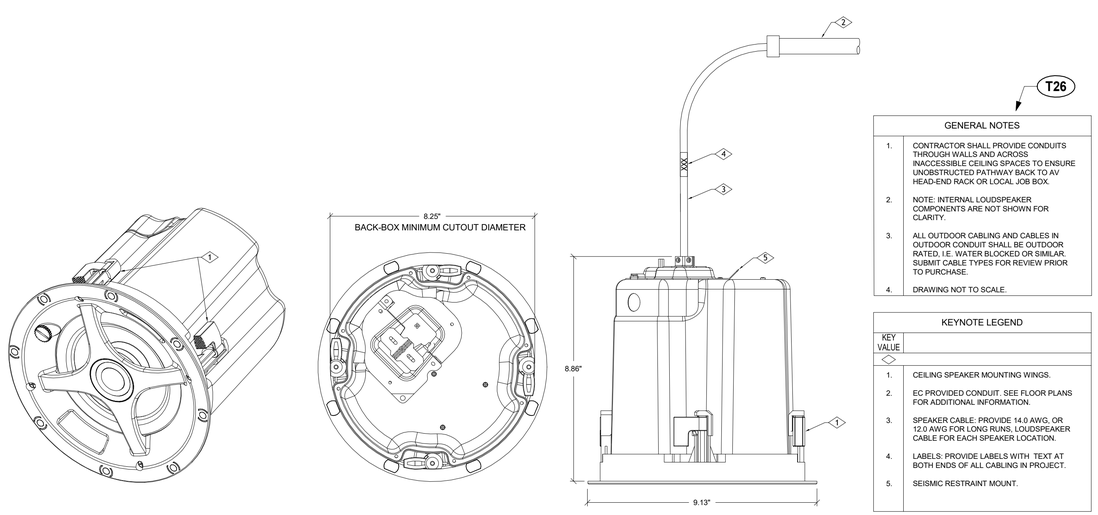
| SYSTEM | KEY ID | CABLE KEY | CABLE COLOR | JACK KEY | JACK COLOR | PATCH CABLE COLOR: IT ROOM | PATCH CABLE COLOR: PREMISE | COMMENTS |
|---|---|---|---|---|---|---|---|---|
| NETWORK | NET | CC1 | BLUE | CJ1 | BLUE | BLUE | BLUE | |
| VOICE | VCE | CC1 | BLUE | CJ1 | BLUE | BLUE | BLUE | |
| WIRELESS ACCESS POINT | WAP | CC2 | GREEN | CJ2 | GREEN | GREEN | GREEN | |
| SECURITY | SEC | CC1 | BLUE | CJ1 | RED | RED | BLUE | |
| PAGING | PAG | CC1 | BLUE | CJ1 | PURPLE | PURPLE | BLUE | |
| AUDIO VIDEO | AVO | CC1 | BLUE | CJ1 | YELLOW | YELLOW | BLUE | |
| CATV | CTV | CC5 | WHITE | CJ5 | STAINLESS | -- | -- | |

## RACKS AND ACCESSORIES

| KEY ID | DESCRIPTION | MANUFACTURER | PART NUMBER | COMMENTS |
|---|---|---|---|---|
| RR1 | WALL MOUNT SWING GATE RACK, EIA. 19-INCH IN WIDTH, BLACK, 40RU | MIDDLE ATLANTIC (OR APPROVED EQUAL) | SR-40-32 (OR APPROVED EQUAL) | |
| RL1 | LADDER RUNWAY, 12-INCH WIDE | CPI (OR APPROVED EQUAL) | 10250-712 (OR APPROVED EQUAL) | |
| RA6 | UPPER TROUGH | CPI (OR APPROVED EQUAL) | 13183-719 (OR APPROVED EQUAL) | |
| RA7 | HORIZONTAL WIRE MANAGER, 1U, 5" DEEP | CPI (OR APPROVED EQUAL) | 30139-719 (OR APPROVED EQUAL) | |
| RA8 | HORIZONTAL WIRE MANAGER, 2U, 5" DEEP | CPI (OR APPROVED EQUAL) | 30130-719 (OR APPROVED EQUAL) | |

## POWER DISTRIBUTION DEVICES

| KEY ID | DESCRIPTION | MANUFACTURER | PART NUMBER | COMMENTS |
|---|---|---|---|---|
| SP1 | PDU, HORIZONTAL, 120V INPUT, (16) 5-20 OUTLETS (1RU) | TRIPPLITE (OR APPROVED EQUAL) | PDUMH20ATS (OR APPROVED EQUAL) | |




RECESSED CEILING SPEAKER

3 T700 RECESSED SPEAKER

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.

## KEYNOTE LEGEND - NPS

| KEY VALUE | KEYNOTE TEXT |
|---|---|
| T26 | **OPTION F:** MAINTAIN <E> ASSISTIVE LISTENING SYSTEM. PROVIDE ALL <N> MEETING ROOM/AUDITORIUM AV EQUIP. INSTALL <N> FOLDING DOOR WITH STRUC SUPPORTS AND <N> WINDOW TREATMENTS. |

LICENSED PROFESSIONAL ENGINEER 05/21/2026 JEFF MULLIKIN P25384 AV KONA, HI

| | | | | | |
|---|---|---|---|---|---|
| A/E FIRM: PRIME: OTAK, INC. REDMOND, WA SUBCONTRACTOR AE DESIGN DENVER, CO | DESIGNED: JR/JM/TW CADD AE DESIGN TECH REVIEW: MRS/ESN DATE: 5/29/2026 | SUB SHEET NO. T700 | TITLE OF SHEET TECHNOLOGY SCHEDULES EFFIGY MOUNDS NATIONAL MONUMENT | DRAWING NO. 394 194248 PMIS/PKG NO. 318539 SHEET 141 OF 142 | |

**SECURITY - VIDEO SURVEILLANCE SYSTEM (VSS)**

| KEY | DESCRIPTION | MANUFACTURER | PART NUMBER | COMMENTS |
|---|---|---|---|---|
| CAM1 | FIXED DOME CAMERA, 5MP, INTERIOR/EXTERIOR RATED | AXIS (OR APPROVED EQUAL) | P3267-LVE (OR APPROVED EQUAL) | PROVIDE MOUNT AS INDICATED ON DRAWINGS AND AS REQUIRED PER LOCATION. |
| CAM2 | FIXED DOME CAMERA, 5MP, INTERIOR RATED | AXIS (OR APPROVED EQUAL) | P3267-LV (OR APPROVED EQUAL) | PROVIDE MOUNT AS INDICATED ON DRAWINGS AND AS REQUIRED PER LOCATION. |

**SECURITY - ACCESS CONTROL SYSTEM (ACS)**

| KEY | DESCRIPTION | MANUFACTURER | PART NUMBER | COMMENTS |
|---|---|---|---|---|
| CR1 | CARD READER - PROXIMITY ONLY | HID (OR APPROVED EQUAL) | SIGNO 40 (OR APPROVED EQUAL) | |
| CR2 | CARD READER - PROXIMITY ONLY, MULLION MOUNT | HID (OR APPROVED EQUAL) | SIGNO 20 (OR APPROVED EQUAL) | |
| CR3 | CARD READER - KEYPAD AND PROXIMITY COMBINATION DEVICE | HID (OR APPROVED EQUAL) | SIGNO 40K (OR APPROVED EQUAL) | |

**SECURITY - INTRUSION DETECTION SYSTEM (IDS)**

| KEY | DESCRIPTION | MANUFACTURER | PART NUMBER | COMMENTS |
|---|---|---|---|---|
| MD1 | MOTION DETECTOR - CEILING MOUNT, 360 DEGREE | BOSCH (OR APPROVED EQUAL) | DS9370 (OR APPROVED EQUAL) | |
| MD2 | MOTION DETECTOR - WALL MOUNT, LONG RANGE | BOSCH (OR APPROVED EQUAL) | DS778 (OR APPROVED EQUAL) | |
| KI1 | KEYPAD INTERFACE | HONEYWELL (OR APPROVED EQUAL) | 6160 ALPHA (OR APPROVED EQUAL) | |

ALL SCOPE ON THIS SHEET SHALL BE IN OPTION C UNLESS NOTED OTHERWISE.

**DOOR COORDINATION SCHEDULE**

| DOOR CLASSIFICATION | | | DOOR TYPE | | | | SECURITY HARDWARE | | | | | | | | | | | | COMMENTS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| # | ENTRANCE LEAF | EXIT LEAF | | | | | | | | | | | | | | | | | |
| DOOR NUMBER | CONTROL (C) / SCHEDULE (S) / MONITOR (M) | CONTROL (C) / SCHEDULE (S) / MONITOR (M) | SINGLE LEAF DOOR | DOUBLE LEAF DOOR | GATE | OVERHEAD DOOR | AUTHENTICATION DEVICE | INTEGRAL AUTHENTICATION DEVICE | INTERCOM | MOTION BASED REQUEST TO EXIT | BUTTON BASED REQUEST TO EXIT | INTEGRAL REQUEST TO EXIT | DOOR POSITION SWITCH | INTEGRAL DOOR POSITION SWITCH | LOCKDOWN RESPONSE ENABLED | REMOTE DOOR RELEASE | LOCAL ALARM | | |
| 005B | C | | X | | | | X | | | | | X | X | | | | | | |
| 005C | C | | X | | | | X | | | | | X | X | | | | | | |
| 007 | C | | X | | | | X | | | | | X | X | | | | | | |
| 008A | M | | X | | | | | | | | | | X | | | | | | |
| 013A | C | | X | | | | X | | | | | X | X | | | | | | |
| 016 | M | | X | | | | | | | | | | X | | | | | | |
| 100A | C, M | | | X | | | X | | | | | X | X | | | | | | DOOR POSITION SWITCH IS NEEDED AT EACH LEAF |
| 100B | C, M | | | X | | | X | | | | | X | X | | | | | | DOOR POSITION SWITCH IS NEEDED AT EACH LEAF |
| 102 | C, M | | | X | | | X | | | | | X | X | | | | | | DOOR POSITION SWITCH IS NEEDED AT EACH LEAF |
| 103A | C, S | | X | | | | X | | | | | X | X | | | | | | |
| 103B | C | | X | | | | X | | | | | X | X | | | | | | |
| 104A | M | | X | | | | | | | | | | X | | | | | | |
| 108A | C | | X | | | | X | | | X | | | X | | | | | | |
| 108B | C | | X | | | | X | | | | | X | X | | | | | | |
| 114 | C | | X | | | | X | | | | | X | X | | | | | | |
| 118A | C | | X | | | | X | | | | | | X | | | | | | |

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LICENSED PROFESSIONAL ENGINEER
05/21/2026
JEFF MULLIKIN
P25384
REGISTERED
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A/E FIRM:
PRIME:
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SUBCONTRACTOR
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TECH REVIEW: MRS/ESN
DATE: 5/29/2026

SUB SHEET NO.
T701

TITLE OF SHEET
TECHNOLOGY SCHEDULES
EFFIGY MOUNDS NATIONAL MONUMENT

DRAWING NO. 394 194248
PMIS/PKG NO. 318539
SHEET 142 OF 142